Seminar on Algebraic Geometry 3

Group Schemes

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Contents

Editorial Notice

We present here a slightly revised edition of the original Seminar, whose purpose and contents are described in the Introduction. The revision consisted mainly of correcting typographical errors, adding a few supplementary remarks or references in footnotes, dividing the work in its present form into three volumes, each with a detailed table of contents and an index of notation, and adding a terminological index at the end of Volume 3. In addition, Exposé $\mathrm{VI_B}$, by J.-E. Bertin, was partially rewritten by the author, especially Sections 5 and 10, so some references to that Exposé differ from the references to the original version. A list of the Exposés in the Seminar appears at the beginning of this volume.

Since the first edition of the present Seminar appeared, all of Chapter IV of the *Éléments de Géométrie Algébrique* has been published, making certain passages of the Seminar unnecessary. We have occasionally indicated in footnotes the relevant references to EGA IV that allow those passages to be bypassed.

For another treatment of algebraic groups that systematically uses the language of schemes, we refer to M. Demazure and P. Gabriel, *Groupes Algébriques* (North-Holland & Masson et Cie). Unlike the present Seminar, that book assumes no knowledge of algebraic geometry, but includes all the required preliminaries from scheme theory; it can therefore serve as an introduction to the study of this Seminar. (It also treats subjects not covered in the Seminar, such as the Dieudonné-style structure theory of commutative affine algebraic groups, in *loc. cit.*, Chapter V.)

Bures-sur-Yvette, March 1970

Introduction

1. The purpose of this Seminar is twofold.

On the one hand, we aim to provide convenient foundations for the theory of group schemes in general. Exposés I–IV will supply the indispensable preliminary exercises in schematic and categorical syntax. To obtain a language that fits geometric intuition without effort, and to avoid circumlocutions that become intolerable in the long run, we always identify a prescheme X over another prescheme S with the functor

$$(\mathbf{Sch})^{\circ}_{/S} \longrightarrow \mathbf{Set}$$

that it represents*. It is therefore necessary to formulate many definitions so that they apply to arbitrary functors, whether representable or not. Moreover, nearly all the functors we shall use will be "sheaves" (for the "faithfully flat quasi-compact topology"). Exposé IV, which treats groups only incidentally, sketches the language of "localization" and sheaves; this language also proves very convenient in questions concerning the representability of functors. Above all, that Exposé will provide the most convenient framework for the questions of passage to a quotient that arise later.

Exposé V gives some general results on the existence of quotients. They are taken up again in Exposé VI_{A} for the quotient of an algebraic group over a field (or, more generally, over an Artinian ring) by a subgroup**. That Exposé and the following Exposé VI_{B} also contain various elementary results specific to algebraic groups over a field that are used routinely later.

Exposé VII studies certain phenomena related to the characteristic of the base field and, in particular, develops with the appropriate generality the correspondence between radicial group schemes of height 1 and restricted p-Lie algebras.

Finally, Exposé XVIII contains the scheme-theoretic generalization of Weil's theorem on the "birational" definition of algebraic groups.

On the other hand, we intend to generalize the Borel–Chevalley structure theory of affine algebraic groups to groups over an arbitrary base prescheme. In fact, while the Seminar notes were being written, it became apparent that the affine hypothesis was unnecessary for many results of the theory. The most complete results are of course obtained for "semisimple

*Such a viewpoint seems to have been considered for the first time eight or nine years ago, in connection with the theory of formal groups, by P. Cartier, who unfortunately did not take the trouble to state it precisely and systematize it as it deserved.

**For a more thorough study of passage to the quotient, especially by reductive groups, see D. Mumford's important work *Geometric Invariant Theory*, Ergebnisse der Mathematik, vol. 34, Springer, 1965. We note that on one important point the terminology of that book does not agree with ours: over a field of characteristic $p > 0$, the groups Mumford calls "reductive"[(1)] are the smooth groups of multiplicative type in the sense of the Seminar (cf. Exposé IX). One may reasonably suppose that Mumford adopted this meaning of "reductive," which loses its significance over a base that is not a field, provisionally and as a kind of stopgap (and this is approximately what Mumford explains, for other reasons, as early as the second paragraph of his preface). [(1)] *Editor's note:* See the remarks added at the end of this Introduction.

group schemes," or more generally "reductive group schemes," with which we shall be exclusively concerned beginning with Exposé XIX. Chevalley himself had already constructed the "Tôhoku groups" over the ring of integers, and that construction will be taken up in the present Seminar. The principal uniqueness theorem[(2)] gives a simple characterization of the "twisted" variants of these Tôhoku groups over a base prescheme S: they are the affine, smooth group schemes over S whose geometric fibers are connected semisimple groups in the usual sense[∗∗∗(2)].

2.

As in the familiar theory over an algebraically closed field, the subtori of the group schemes under consideration play a crucial role. Thus the preliminary study of tori and, more generally, of "group schemes of multiplicative type"—both intrinsically and as multiplicative-type subgroups of a given group—occupies a fairly large part of this Seminar (Exposés VIII–XII). Their remarkable rigidity (in some respects even greater than that of abelian schemes or semisimple group schemes) makes them highly effective tools for studying certain more general groups.

3.

Beginning with Exposé XII (apart from Exposé XVIII, already mentioned), we shall routinely use the theory of affine algebraic groups over an algebraically closed field, which the reader will find in the 1956 Chevalley Seminar, especially in Exposés IV–IX of that Seminar. We shall also use, though to a lesser extent, the later Exposés of the Chevalley Seminar devoted to the structure of semisimple algebraic groups. Indeed, we shall redevelop Chevalley's theory directly within the framework of schemes; it will be seen that this approach makes the exposition simpler and more precise, even over a base field.

4.

The principal purpose of this Seminar is, of course, to develop techniques that apply to the study of group schemes over an arbitrary base, that is, essentially, to the study of families of algebraic groups. The infinitesimal properties of such families, and in particular the case of an Artinian base scheme, therefore play an important role. These properties arise even in the study of algebraic groups over a field k when the field is not perfect, especially when applying descent in the non-Galois case. Among the new results obtained in this setting, let us mention the existence of maximal tori and Cartan subgroups in an arbitrary smooth algebraic group, the rationality of the variety of maximal tori, and various related results (Exposé XIV[(3)]); another example is the correspondence between the "forms" of a semisimple group and homogeneous principal bundles under a suitable semisimple algebraic group (generally not connected) (Exposé XXIV, 1.16–1.20). In general, one may say that the methods required for work over a nonperfect base field are essentially those used over arbitrary base preschemes and thus lie outside the framework of classical algebraic geometry.

[(2)] *Editor's note:* This refers to Corollary XXIII.5.6, which follows easily from the uniqueness theorem for split reductive groups (cf. XXIII, Theorem 4.1 and Corollary 5.2), since every semisimple S-group is a "twisted form" (for the *é*tale topology) of a Tôhoku group (cf. XXII, Corollary 2.3).

[∗∗∗] This is the essential result of M. Demazure's thesis, *Schémas en groupes réductifs*, Bull. Soc. Math. France **93** (1965), 369–413.

[(3)] *Editor's note:* In particular, Theorems 1.1 and 6.1.

5.

It did not seem useful to indicate, at the beginning of the written Exposés, the date or dates of the corresponding oral presentations in the Seminar. We shall merely say that the order of the mimeographed Exposés (I–XXVI) does indeed agree with the order of the oral presentations. Moreover, the final written text was sometimes prepared considerably later than the oral presentation and often differs from it quite substantially: the written text is generally more detailed and more complete (as in Exposés IV and $\mathrm{VII_B}$), or even appreciably more general (as in Exposés XII and $\mathrm{VII_B}$), than the oral presentation. Other written Exposés correspond to no oral presentation ($\mathrm{VI_B}$, $\mathrm{VII_A}$, XV, XVI, XVII, and most of XXVI); they were written and inserted into the mimeographed Seminar either to provide convenient references for various other Exposés (notably in the case of $\mathrm{VI_B}$) or because they form a natural continuation of the concepts and techniques already developed. Consequently, Exposés $\mathrm{VII_A}$, $\mathrm{VII_B}$, XV, XVI, and XVII need not be read in order to study the rest of the Seminar, particularly the part devoted to reductive group schemes.

6.

From scheme theory we shall use primarily the general language of schemes set out in EGA I; the notions of flat, *é*tale, and smooth morphisms set out in SGA 1, I–V; and the theory of faithfully flat descent in SGA 1, VIII. As far as possible, we have avoided imposing unnecessary Noetherian hypotheses. In return, this has required us to replace the usual hypothesis "of finite type" with the hypothesis "of finite presentation." For the notion of a morphism of finite presentation, the reader should consult EGA IV, 1.4 and 1.6. The results of SGA 1, I–IV, which are most often stated in the Noetherian setting, will be developed in general in EGA IV[* * **]; standard methods for reducing certain kinds of statements involving finite-presentation hypotheses to the Noetherian case will also be developed there in detail (EGA IV, Sections 8, 9, and 11). A reader who does not wish to assume these results from EGA IV may simplify certain statements or their proofs by assuming the base prescheme to be locally Noetherian. This does, however, lead to difficulties when a proof proceeds by descent from $\widehat{A}$ to A, where $\widehat{A}$ is the completion of a Noetherian local ring A, because that method introduces the generally non-Noetherian ring

$$\widehat{A} \otimes_A \widehat{A}.$$

7.

References follow the usual decimal system: the reference 5.7.11 denotes the proposition (or lemma, definition, and so forth) with that number in the same Exposé; in a reference such as XVII 7.8, the Roman numeral denotes the Exposé. We use the following abbreviations for our standard references:

Bible Chevalley Seminar, *Groupes de Lie algébriques*, 1956/58.

EGA X**,** $x.y.z$ J. Dieudonné and A. Grothendieck, *Éléments de Géométrie Algébrique*, Chapter X, statement $x.y.z$ (or subsection $x.y$, and so forth).

SGA n**,** X $y.z$ *Séminaire de Géométrie Algébrique du Bois-Marie*, year n, statement $y.z$ of Exposé X.

[* * **]Since this Introduction was written, all four parts (Sections 1–21) of EGA IV have appeared.

TDTE A. Grothendieck, *Techniques de descente et Théorèmes d'existence en Géométrie Algébrique*, Exposés delivered in the Bourbaki Seminar between 1959 and 1962.

[1] *Editor's note.* Concerning quotients by a reductive group, the situation has evolved considerably since A. Grothendieck wrote Note ($**$). For an affine algebraic group over an arbitrary field k, the appropriate notion, introduced by Mumford, is that of a "geometrically reductive" group. (On the other hand, G is said to be "linearly reductive" if every rational representation of G is completely reducible; as noted in Note ($**$), however, that condition is too restrictive when $\mathrm{char}(k) > 0$.) By results of M. Nagata and W. J. Haboush ([Na64], [NM64], [Ha75]), the geometrically reductive k-groups are precisely the reductive k-groups in the sense of the present Seminar. For all of this, see the later editions of Mumford's book ([MF82]). The notion of geometric reductivity over an arbitrary base, together with its consequences for passage to a quotient, was extended to this setting by C. S. Seshadri ([Se77]); additions due to M. Raynaud appear in the article [CTS79] by J.-L. Colliot-Thélène and J.-J. Sansuc. Finally, for more recent developments on passage to the quotient by an algebraic group or, more generally, by a groupoid (cf. Exposé V of the present Seminar), we refer to the articles [Ko97] and [KM97] by J. Kollár and by S. Keel and S. Mori.

Bibliography

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[KM97] S. Keel and S. Mori, *Quotient by groupoids*, Ann. of Math. **145** (1997), no. 1, 193–213.

[Ko97] J. Kollár, *Quotient spaces modulo algebraic groups*, Ann. of Math. **145** (1997), no. 1, 33–79.

[MF82] D. Mumford and J. Fogarty, *Geometric invariant theory*, 2nd ed., Springer-Verlag, 1982; 3rd ed., with F. Kirwan, 1994.

[Na64] M. Nagata, *Invariants of a group in an affine ring*, J. Math. Kyoto Univ. **3** (1964), no. 3, 369–377.

[NM64] M. Nagata and T. Miyata, *Note on semi-reductive groups*, J. Math. Kyoto Univ. **3** (1964), no. 3, 379–382.

[Se77] C. S. Seshadri, *Geometric reductivity over an arbitrary base*, Adv. Math. **26** (1977), no. 3, 225–274.

Exposé I
Algebraic Structures. Group Cohomology

by M. Demazure

This exposé consists of two parts: the first collects a number of general definitions and establishes notation that will often be used in the sequel; the second treats group cohomology and culminates in Theorem 5.3.3, on the vanishing of the cohomology of diagonalizable groups. [1]

We choose once and for all a Universe.[2] All definitions and constructions will be relative to this Universe. We shall systematically allow ourselves the following abuse of language. To define a functor $f : \mathscr{C} \to \mathscr{C}'$, we shall be content to define the object $f(S)$ of $\mathscr{C}'$ for every object S of $\mathscr{C}$, whenever there is no ambiguity about how to define $f(h)$ for an arrow h of $\mathscr{C}$. In practice, we shall say: let $f : \mathscr{C} \to \mathscr{C}'$ be the functor defined by $f(S) = \cdots$.

1. Generalities

1.1.

Let $\mathscr{C}$ be a category. We denote by $\widehat{\mathscr{C}}$ the category

$$\mathrm{Hom}\big(\mathscr{C}^\circ, \mathbf{Set}\big)$$

of contravariant functors from $\mathscr{C}$ to the category **Set** of sets.[3] There is a canonical functor

$$\mathbf{h} : \mathscr{C} \longrightarrow \widehat{\mathscr{C}}$$

which associates with every $X \in \mathrm{Ob}(\mathscr{C})$ the functor $\mathbf{h}_X$ such that

$$\mathbf{h}_X(S) = \mathrm{Hom}(S, X).$$

For every functor $\mathbf{F} \in \mathrm{Ob}(\widehat{\mathscr{C}})$, one has (see, for example, EGA 0_{III}, 8.1.4) a bijection

$$\mathrm{Hom}(\mathbf{h}_X, \mathbf{F}) \xrightarrow{\sim} \mathbf{F}(X).$$

[4]

[1] Editor's note: version of October 13, 2024.

[2] Editor's note: see SGA 4, Exposé I, §0 and the Appendix; see also the discussion in [DG70], p. xxvi.

[3] Editor's note: this is called the category of presheaves on $\mathscr{C}$; see IV, 4.3.1.

[4] Editor's note: this result is often called the Yoneda lemma; we use that terminology in other editor's notes.

In particular, for every pair X, Y of objects of $\mathscr{C}$, the canonical map

$$\mathrm{Hom}_{\mathscr{C}}(X,Y) \xrightarrow{\sim} \mathrm{Hom}_{\widehat{\mathscr{C}}}(\mathbf{h}_X, \mathbf{h}_Y)$$

is bijective; that is, the functor $\mathbf{h}$ is fully faithful. It therefore defines an isomorphism of $\mathscr{C}$ onto a full subcategory of $\widehat{\mathscr{C}}$, and an equivalence of $\mathscr{C}$ with the full subcategory of $\widehat{\mathscr{C}}$ formed by the representable functors (that is, the functors isomorphic to one of the form $\mathbf{h}_X$). In the sequel we shall often identify X with $\mathbf{h}_X$. The following paragraphs are intended to show that this identification can be made safely.

Remark 1.1.1.[5] We sometimes need the following variant. Let $\mathscr{D}$ be a full subcategory of $\mathscr{C}$, and let $X, Y \in \mathrm{Ob}(\mathscr{D})$. Denote by $\mathbf{h}'_X$ and $\mathbf{h}'_Y$ the restrictions of $\mathbf{h}_X$ and $\mathbf{h}_Y$ to $\mathscr{D}$. Then

$$\mathrm{Hom}_{\mathscr{C}}(X,Y) = \mathrm{Hom}_{\mathscr{D}}(X,Y) = \mathrm{Hom}_{\widehat{\mathscr{D}}}(\mathbf{h}'_X, \mathbf{h}'_Y).$$

Thus, to give a morphism $X \to Y$ "is the same thing" as to give, for every $T \in \mathrm{Ob}(\mathscr{D})$, a map

$$\phi(T) : \mathrm{Hom}(T, X) \longrightarrow \mathrm{Hom}(T, Y)$$

functorially in T.

1.2.

We shall say that $\mathbf{F}$ is a *subobject* (or a subfunctor) of $\mathbf{G}$ if $\mathbf{F}(S)$ is a subset of $\mathbf{G}(S)$ for every S.[6]

In $\widehat{\mathscr{C}}$, arbitrary inverse limits exist and are computed pointwise:

$$\left(\varprojlim_i \mathbf{F}_i\right)(S) = \varprojlim_i \mathbf{F}_i(S).$$

[7] In particular, fiber products are defined by

$$(\mathbf{F} \times_{\mathbf{G}} \mathbf{F}')(S) = \mathbf{F}(S) \times_{\mathbf{G}(S)} \mathbf{F}'(S).$$

[8]

As a final object of $\widehat{\mathscr{C}}$, we choose the functor $\underline{\mathbf{e}}$ such that $\underline{\mathbf{e}}(S) = \{\varnothing\}$.[9] Every $\mathbf{F} \in \mathrm{Ob}(\widehat{\mathscr{C}})$ has a unique morphism to $\underline{\mathbf{e}}$, and we set

$$\mathbf{F} \times \mathbf{F}' = \mathbf{F} \times_{\underline{\mathbf{e}}} \mathbf{F}'.$$

The functor $\mathbf{h}$ commutes with inverse limits. In particular, for $X \times X'$ to exist, where $X, X' \in \mathrm{Ob}(\mathscr{C})$, respectively for $\mathscr{C}$ to admit a final object e, it is necessary and sufficient that $\mathbf{h}_X \times \mathbf{h}_{X'}$, respectively $\underline{\mathbf{e}}$, be representable; in that case

$$\mathbf{h}_X \times \mathbf{h}_{X'} \simeq \mathbf{h}_{X \times X'}, \qquad \mathbf{h}_e \simeq \underline{\mathbf{e}}.$$

[5] Editor's note: this remark has been added.

[6] Editor's note: the order has been changed so that fiber products are introduced before monomorphisms; see editor's note (9).

[7] Editor's note: similarly, arbitrary direct limits exist and are computed pointwise: $(\varinjlim_i \mathbf{F}_i)(S) = \varinjlim_i \mathbf{F}_i(S)$. In general, however, the functor $\mathbf{h}$ does not commute with direct limits.

[8] Editor's note: in particular, the equalizer of a pair of morphisms $u, v : \mathbf{F} \rightrightarrows \mathbf{G}$ is the subfunctor $\mathrm{Ker}(u,v)$ of $\mathbf{F}$ defined by $\mathrm{Ker}(u,v)(S) = \{x \in \mathbf{F}(S) \mid u(x) = v(x)\}$.

[9] Editor's note: $\{\varnothing\}$, the power set of the empty set, is a one-element set.

A monomorphism in $\widehat{\mathscr{C}}$ is precisely a morphism $\mathbf{F} \to \mathbf{G}$ such that, for every $S \in \mathrm{Ob}(\mathscr{C})$, the corresponding map of sets $\mathbf{F}(S) \to \mathbf{G}(S)$ is injective.[10]

The functor Γ. For every $\mathbf{F} \in \mathrm{Ob}(\widehat{\mathscr{C}})$, set

$$\Gamma(\mathbf{F}) = \mathrm{Hom}(\underline{\mathbf{e}}, \mathbf{F}).$$

An element of $\Gamma(\mathbf{F})$ is therefore a family $(\gamma_S)_{S \in \mathrm{Ob}(\mathscr{C})}$, with $\gamma_S \in \mathbf{F}(S)$, such that for every arrow $f : S' \to S''$ of $\mathscr{C}$,

$$\mathbf{F}(f)(\gamma_{S''}) = \gamma_{S'}.$$

For $X \in \mathrm{Ob}(\mathscr{C})$, set $\Gamma(X) = \Gamma(\mathbf{h}_X)$. If $\mathscr{C}$ has a final object e, then

$$\Gamma(X) \simeq \mathrm{Hom}(e, X).$$

1.3.

Let $S \in \mathrm{Ob}(\mathscr{C})$. We denote by $\mathscr{C}_{/S}$ the category of objects of $\mathscr{C}$ over S: its objects are the arrows $f : T \to S$ of $\mathscr{C}$, and $\mathrm{Hom}(f, f')$ is the subset of $\mathrm{Hom}(T, T')$ formed by the arrows u such that $f = f' \circ u$. If $\mathscr{C}$ has a final object e, then $\mathscr{C}_{/e}$ is isomorphic to $\mathscr{C}$. The category $\mathscr{C}_{/S}$ has a final object, namely the identity arrow $S \to S$.

If $f : T \to S$ is an object of $\mathscr{C}_{/S}$, one may form the category $(\mathscr{C}_{/S})_{/f}$, which by abuse of language we denote by $(\mathscr{C}_{/S})_{/T}$, and there is a canonical isomorphism

$$\mathscr{C}_{/T} \simeq (\mathscr{C}_{/S})_{/T}.$$

This construction also applies to $\widehat{\mathscr{C}}$; in particular, one obtains the category $\widehat{\mathscr{C}}_{/\mathbf{h}_S}$. On the other hand, one may form the category $\widehat{\mathscr{C}_{/S}}$.

If $f : T \to S$ is an object of $\mathscr{C}_{/S}$, then $\Gamma(f)$ is identified with the set $\Gamma(T/S)$ of sections of T over S, that is, arrows $S \to T$ that are right inverses to f. Moreover, $\mathbf{h}_f : \mathbf{h}_T \to \mathbf{h}_S$ is then an object of $\widehat{\mathscr{C}}_{/\mathbf{h}_S}$, and

$$\Gamma(\mathbf{h}_f) \simeq \Gamma(\mathbf{h}_T/\mathbf{h}_S) \simeq \Gamma(T/S) \simeq \Gamma(f).$$

1.4.

We now define an equivalence between the categories $\widehat{\mathscr{C}_{/S}}$ and $\widehat{\mathscr{C}}_{/\mathbf{h}_S}$. In other words, we shall prove that to give a functor on the category of objects of $\mathscr{C}$ over S is "the same thing" as to give a functor on $\mathscr{C}$ equipped with a morphism to $\mathbf{h}_S$.

(i) Construction of $\alpha_S : \widehat{\mathscr{C}}_{/\mathbf{h}_S} \to \widehat{\mathscr{C}_{/S}}$.

First let $H : \mathbf{F} \to \mathbf{h}_S$ be an object of $\widehat{\mathscr{C}}_{/\mathbf{h}_S}$. We must define a functor $\alpha_S(H)$ on $\mathscr{C}_{/S}$. For an object $f : T \to S$ of $\mathscr{C}_{/S}$, define $\alpha_S(H)(f)$ to be the inverse image of $f \in \mathbf{h}_S(T)$ under the map

$$H(T) : \mathbf{F}(T) \longrightarrow \mathbf{h}_S(T).$$

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[10]Editor's note: if $\mathbf{F}(S) \to \mathbf{G}(S)$ is injective for every S, it is clear that $\mathbf{F} \to \mathbf{G}$ is a monomorphism. The converse follows by considering $\mathbf{F} \times_{\mathbf{G}} \mathbf{F} \rightrightarrows \mathbf{F} \to \mathbf{G}$. Thus $\mathbf{F} \to \mathbf{G}$ is a monomorphism if and only if the diagonal morphism $\mathbf{F} \to \mathbf{F} \times_{\mathbf{G}} \mathbf{F}$ is an isomorphism (see EGA I, 5.3.8). Similarly, a pointwise surjective morphism is an epimorphism, and the converse follows by considering the pushout $\mathbf{G} \amalg_{\mathbf{F}} \mathbf{G}$; see the proof of Lemma 4.4.4 in Exposé IV.

[11]Editor's note: for example, if $\mathbf{F} = \mathbf{h}_X$, then H corresponds to a morphism $h : X \to S$, and $H(T) : \mathrm{Hom}_{\mathscr{C}}(T, X) \to \mathrm{Hom}_{\mathscr{C}}(T, S)$ is the map $g \mapsto h \circ g$. Thus $\alpha_S(\mathbf{h}_X) = \mathrm{Hom}_{\mathscr{C}_{/S}}(-, X)$.

Next, let $u : f \to f'$ be an arrow of $\mathscr{C}_{/S}$. Then $\mathbf{F}(u) : \mathbf{F}(T') \to \mathbf{F}(T)$ induces a map from $\alpha_S(H)(f')$ to $\alpha_S(H)(f)$, denoted $\alpha_S(H)(u)$. It is immediate that the assignments

$$f \longmapsto \alpha_S(H)(f), \qquad u \longmapsto \alpha_S(H)(u)$$

define a functor on $\mathscr{C}_{/S}$, hence an object $\alpha_S(H)$ of $\widehat{\mathscr{C}_{/S}}$.

Finally, let $H : \mathbf{F} \to \mathbf{h}_S$ and $H' : \mathbf{F}' \to \mathbf{h}_S$ be two objects of $\widehat{\mathscr{C}}_{/\mathbf{h}_S}$, and let $U : H \to H'$ be a morphism in that category:

$$\begin{tikzcd}
\mathbf{F} \arrow[rr, "U"] \arrow[dr, "H"'] & & \mathbf{F}' \arrow[dl, "H'"] \\
& \mathbf{h}_S &
\end{tikzcd}.$$

For every $f : T \to S$, the map $U(T) : \mathbf{F}(T) \to \mathbf{F}'(T)$ induces a map

$$\alpha_S(U)(f) : \alpha_S(H)(f) \longrightarrow \alpha_S(H')(f),$$

which defines a morphism of functors

$$\alpha_S(U) : \alpha_S(H) \longrightarrow \alpha_S(H').$$

It is easy to verify that the assignments

$$H \longmapsto \alpha_S(H), \qquad U \longmapsto \alpha_S(U)$$

define a functor

$$\alpha_S : \widehat{\mathscr{C}}_{/\mathbf{h}_S} \longrightarrow \widehat{\mathscr{C}_{/S}}.$$

(ii) Proposition 1.4.1. *The functor $\alpha_S : \widehat{\mathscr{C}}_{/\mathbf{h}_S} \to \widehat{\mathscr{C}_{/S}}$ is an equivalence of categories.*

We indicate only the principle of the construction of a quasi-inverse functor

$$\beta_S : \widehat{\mathscr{C}_{/S}} \longrightarrow \widehat{\mathscr{C}}_{/\mathbf{h}_S}.$$

Let $\mathbf{G}$ be a functor on $\mathscr{C}_{/S}$. For every object T of $\mathscr{C}$, set

$$\beta_S(\mathbf{G})(T) = \coprod_{f \in \mathrm{Hom}(T,S)} \mathbf{G}(f), \qquad \mathrm{Hom}(T,S) = \mathbf{h}_S(T).$$

This defines a functor $\beta_S(\mathbf{G})$ on $\mathscr{C}$, equipped with the evident projection to $\mathbf{h}_S$.

1.5.

The equivalence α_S commutes with the functors Γ. In other words, if $H : \mathbf{F} \to \mathbf{h}_S$ is an object of $\widehat{\mathscr{C}}_{/\mathbf{h}_S}$ and $\alpha_S(H)$ is the corresponding object of $\widehat{\mathscr{C}_{/S}}$, then

$$\Gamma(\alpha_S(H)) \simeq \Gamma(H) \simeq \Gamma(\mathbf{F}/\mathbf{h}_S).$$

The equivalence α_S also commutes with the functors $\mathbf{h}$. That is, if $f : T \to S$ is an object of $\mathscr{C}_{/S}$, then $\mathbf{h}_f : \mathbf{h}_T \to \mathbf{h}_S$ is an object of $\widehat{\mathscr{C}}_{/\mathbf{h}_S}$, and its image under α_S is precisely $\mathbf{h}_{\mathscr{C}_{/S}}(f)$, where

$$\mathbf{h}_{\mathscr{C}_{/S}} : \mathscr{C}_{/S} \longrightarrow \widehat{\mathscr{C}_{/S}}$$

is the canonical functor.[12] Consequently:

Proposition 1.5.1. *Let $H : \mathbf{F} \to \mathbf{h}_S$ be an object of $\widehat{\mathscr{C}}_{/\mathbf{h}_S}$. The functor*

$$\alpha_S(H) : (\mathscr{C}_{/S})^{\circ} \longrightarrow \mathbf{Set}$$

is representable if and only if $\mathbf{F} : \mathscr{C}^{\circ} \to \mathbf{Set}$ is representable. If $\mathbf{F} \simeq \mathbf{h}_T$, then $\alpha_S(H)$ is represented by the object $T \to S$ of $\mathscr{C}_{/S}$.

The equivalence α_S is transitive in S. If $f : T \to S$ is an object of $\mathscr{C}_{/S}$, there is a commutative diagram of equivalences:

$$\begin{tikzcd}
(\widehat{\mathscr{C}}_{/\mathbf{h}_S})_{/\mathbf{h}_T} \arrow[rr, "\alpha_S/\mathbf{h}_T"] \arrow[dr, "\cong"'] & & (\widehat{\mathscr{C}_{/S}})_{/\mathbf{h}_T} \arrow[rr, "\alpha_f"] & & \widehat{(\mathscr{C}_{/S})_{/T}} \arrow[dl, "\cong"] \\
& \widehat{\mathscr{C}}_{/\mathbf{h}_T} \arrow[rr, "\alpha_T"] & & \widehat{\mathscr{C}_{/T}} &
\end{tikzcd},$$

Here $\alpha_{S/\mathbf{h}_T}$ denotes, provisionally, the restriction (see 1.6) of the functor α_S to the objects over $\mathbf{h}_T$.

1.6. Base change in a functor

For every $S \in \mathrm{Ob}(\mathscr{C})$, there is a canonical functor

$$i_S : \mathscr{C}_{/S} \longrightarrow \mathscr{C}$$

defined by $i_S(f) = T$ when f is the arrow $T \to S$. If $f : T \to S$ is an object of $\mathscr{C}_{/S}$, we denote by $i_{T/S} = i_f$ the functor

$$i_{T/S} : (\mathscr{C}_{/S})_{/T} \longrightarrow \mathscr{C}_{/S},$$

and there is a commutative diagram:

$$\begin{tikzcd}
(\mathscr{C}_{/S})_{/T} \arrow[r, "i_{T/S}"] \arrow[dr, "\cong"'] & \mathscr{C}_{/S} \arrow[r, "i_S"] & \mathscr{C} \\
& \mathscr{C}_{/T} \arrow[ur, "i_T"'] &
\end{tikzcd},$$

Thus, identifying $(\mathscr{C}_{/S})_{/T}$ with $\mathscr{C}_{/T}$, as we shall henceforth do,

$$i_S \circ i_{T/S} = i_T.$$

Likewise, if $\mathscr{C}$ is identified with $\mathscr{C}_{/e}$ when $\mathscr{C}$ has a final object e, then $i_{S/e} : \mathscr{C}_{/S} \to \mathscr{C}_{/e}$ is identified with i_S.

For $X \in \mathrm{Ob}(\mathscr{C})$ (respectively, $Y \in \mathrm{Ob}(\mathscr{C}_{/S})$), let $p_S(X)$ (respectively, $p_{T/S}(Y)$) be the object of $\mathscr{C}_{/S}$ (respectively, of $\mathscr{C}_{/T}$), when it exists, defined by $X \times S$ (respectively, $Y \times_S T$) equipped with its second projection:

[12] Editor's note: see editor's note (10).

$$\begin{array}{ccc} \mathrm{X} \times \mathrm{S} & & \mathrm{Y} \times_{\mathrm{S}} \mathrm{T} \\ {\scriptstyle p_{\mathrm{S}}(\mathrm{X})}\big\downarrow & \text{resp.} & \big\downarrow{\scriptstyle p_{\mathrm{T/S}}(\mathrm{Y})} \\ \mathrm{S} & & \mathrm{T} \end{array} .$$

The partially defined functor p_S (respectively, $p_{T/S}$) is called the *base-change functor*. By the definition of a product (respectively, a fiber product), it is right adjoint to i_S (respectively, to $i_{T/S}$). [13] We also write

$$p_S(X) = X_S, \qquad p_{T/S}(Y) = Y_T.$$

The functor i_S defines a restriction functor

$$i_S^* : \widehat{\mathscr{C}} \longrightarrow \widehat{\mathscr{C}_{/S}}.$$

We write

$$\mathbf{F}_S = i_S^*(\mathbf{F}) = \mathbf{F} \circ i_S.$$

Plainly,

$$i_{T/S}^* \circ i_S^* = i_T^*,$$

that is, for every functor $\mathbf{F} \in \mathrm{Ob}(\widehat{\mathscr{C}})$,

$$(\mathbf{F}_S)_T = \mathbf{F}_T.$$

The notation is justified by the following result.

Proposition 1.6.1. *The functor*

$$(\mathbf{h}_X)_S : (\mathscr{C}_{/S})^\circ \longrightarrow \mathbf{Set}$$

is representable if and only if the product $X \times S$ exists. In that case

$$(\mathbf{h}_X)_S \simeq \mathbf{h}_{X_S}.$$

This shows that $\mathbf{F}_S$ has two interpretations: it is the restriction of $\mathbf{F}$ to $\mathscr{C}_{/S}$, and it is the functor obtained by the base change $e \leftarrow S$. This leads to the following notation:

$$\begin{array}{ccccc} \mathbf{F} & \text{———} & \mathbf{F}_{\mathrm{S}} & \text{———} & \mathbf{F}_{\mathrm{T}} \\ \Big| & & \Big| & & \Big| \\ e & \longleftarrow & \mathrm{S} & \longleftarrow & \mathrm{T} \end{array}$$

[13]Editor's note: that is, if $g : U \to S$ (respectively, $h : V \to T$) is an object of $\mathscr{C}_{/S}$ (respectively, of $\mathscr{C}_{/T}$), then $i_S(g) = U$, while $i_{T/S}(h)$ is the object $f \circ h : V \to S$ of $\mathscr{C}_{/S}$, and

$$\mathrm{Hom}_{\mathscr{C}_{/S}}(U, X \times S) \simeq \mathrm{Hom}_{\mathscr{C}}(U, X),$$

respectively,

$$\mathrm{Hom}_{\mathscr{C}_{/T}}(V, X \times_S T) \simeq \mathrm{Hom}_{\mathscr{C}_{/S}}(V, X).$$

This diagram reflects both of the preceding interpretations. Notice that

$$\Gamma(\mathbf{F}_S) \simeq \mathrm{Hom}(\mathbf{h}_S, \mathbf{F}) \simeq \mathbf{F}(S),$$

and, in particular,

$$\Gamma(X_S) \simeq \mathrm{Hom}(S, X).$$

1.7.0.

Let $\mathbf{E}$ be an object of $\widehat{\mathscr{C}}$. [14] Consider the category $\mathscr{C}_{/\mathbf{E}}$ of *objects of* $\mathscr{C}$ over $\mathbf{E}$. Its objects are pairs (V, ρ) formed by an object V of $\mathscr{C}$ and a morphism $\rho : \mathbf{h}_V \to \mathbf{E}$ in $\widehat{\mathscr{C}}$, that is, $\rho \in \mathbf{E}(V)$. A morphism from (V, ρ) to (V', ρ') is a morphism $f : V \to V'$ in $\mathscr{C}$ such that

$$\rho' \circ f = \rho, \qquad \text{that is,} \qquad \mathbf{E}(f)(\rho') = \rho.$$

Let $\mathbf{L}$ be the functor

$$\mathbf{L} = \varinjlim_{(V,\rho)\in\mathscr{C}_{/\mathbf{E}}} \mathbf{h}_V.$$

Thus, for every $S \in \mathrm{Ob}(\mathscr{C})$,

$$\mathbf{L}(S) = \varinjlim_{(V,\rho)\in\mathscr{C}_{/\mathbf{E}}} \mathbf{h}_V(S)$$

is the set of equivalence classes of triples (V, ρ, v), where $v : S \to V$ is a morphism in $\mathscr{C}$, and where (V, ρ, v) is identified with $(V', \rho', f \circ v)$ for every morphism $f : V \to V'$ in $\mathscr{C}$ such that $\rho' \circ f = \rho$.

The map $\phi_{\mathbf{E}}(S)$ which associates with the class of (V, ρ, v) the element $\rho \circ v$ of $\mathbf{E}(S)$ is well defined and gives a morphism of functors

$$\phi_{\mathbf{E}} : \varinjlim_{(V,\rho)\in\mathscr{C}_{/\mathbf{E}}} \mathbf{h}_V \longrightarrow \mathbf{E}.$$

Lemma. *The morphism $\phi_{\mathbf{E}}$ is an isomorphism.*

Indeed, let $S \in \mathrm{Ob}(\mathscr{C})$. Every $x \in \mathbf{E}(S)$ is the image under $\phi_{\mathbf{E}}(S)$ of the triple (S, x, id_S); hence $\phi_{\mathbf{E}}(S)$ is surjective. Conversely, let $\ell_1 = (V_1, \rho_1, v_1)$ and $\ell_2 = (V_2, \rho_2, v_2)$ be two elements of $\mathbf{L}(S)$ having the same image in $\mathbf{E}(S)$, and set

$$\gamma = \rho_1 \circ v_1 = \rho_2 \circ v_2.$$

Then ℓ_1 and ℓ_2 are both equal in $\mathbf{L}(S)$ to the class of $(S, \gamma, \mathrm{id}_S)$. Thus $\phi_{\mathbf{E}}(S)$ is injective.

Corollary. *For every object $\mathbf{F}$ of $\widehat{\mathscr{C}}$,*

$$\mathrm{Hom}(\mathbf{E}, \mathbf{F}) = \varprojlim_{(V,\rho)\in\mathscr{C}_{/\mathbf{E}}} \mathbf{F}(V).$$

[14] Editor's note: the lemma below has been added (see SGA 4, I.3.4); it is used in the proof of 1.7.1 and will be useful repeatedly in the sequel.

1.7. The objects $\underline{\mathrm{Hom}}$, $\underline{\mathrm{Isom}}$, etc.

Let $\mathbf{F}$ and $\mathbf{G}$ be two objects of $\widehat{\mathscr{C}}$. We define another object of $\widehat{\mathscr{C}}$ by

$$\begin{aligned}\underline{\mathrm{Hom}}(\mathbf{F},\mathbf{G})(S) &= \mathrm{Hom}_{\widehat{\mathscr{C}_{/S}}}(\mathbf{F}_S,\mathbf{G}_S)\\ &\simeq \mathrm{Hom}_{\widehat{\mathscr{C}}_{/\mathbf{h}_S}}(\mathbf{F}\times\mathbf{h}_S,\mathbf{G}\times\mathbf{h}_S)\\ &\simeq \mathrm{Hom}_{\widehat{\mathscr{C}}}(\mathbf{F}\times\mathbf{h}_S,\mathbf{G}).\end{aligned}$$

The object $\underline{\mathrm{Hom}}(\mathbf{F},\mathbf{G})$ has the following properties:

(i) $\underline{\mathrm{Hom}}(\underline{\mathbf{e}},\mathbf{G})\simeq\mathbf{G}$. [15]

(ii) The formation of $\underline{\mathrm{Hom}}$ commutes with base extension:

$$\underline{\mathrm{Hom}}(\mathbf{F}_S,\mathbf{G}_S)\simeq\underline{\mathrm{Hom}}(\mathbf{F},\mathbf{G})_S.$$

(iii) $(\mathbf{F},\mathbf{G})\mapsto\underline{\mathrm{Hom}}(\mathbf{F},\mathbf{G})$ is a bifunctor, contravariant in $\mathbf{F}$ and covariant in $\mathbf{G}$.

These three properties follow immediately from the definitions.

We shall prove that, for every object $\mathbf{E}$ of $\widehat{\mathscr{C}}$,

$$\mathrm{Hom}\big(\mathbf{E},\underline{\mathrm{Hom}}(\mathbf{F},\mathbf{G})\big)\simeq\mathrm{Hom}(\mathbf{E}\times\mathbf{F},\mathbf{G}).$$

Let $\phi:\mathbf{E}\times\mathbf{F}\to\mathbf{G}$. We must associate with it a morphism from $\mathbf{E}$ to $\underline{\mathrm{Hom}}(\mathbf{F},\mathbf{G})$. For an arrow $S'\to S$ of $\mathscr{C}$, there are maps

$$\mathbf{E}(S)\times\mathbf{F}(S')\longrightarrow\mathbf{E}(S')\times\mathbf{F}(S')\xrightarrow{\phi(S')}\mathbf{G}(S').$$

Every element e of $\mathbf{E}(S)$ therefore defines, for each $S'\to S$, a map $\mathbf{F}(S')\to\mathbf{G}(S')$ functorial in S'; this is an element $\theta_\phi(e)$ of $\underline{\mathrm{Hom}}(\mathbf{F},\mathbf{G})(S)$. We have thus obtained a map

$$\phi\longmapsto\theta_\phi,\qquad \mathrm{Hom}(\mathbf{E}\times\mathbf{F},\mathbf{G})\longrightarrow\mathrm{Hom}\big(\mathbf{E},\underline{\mathrm{Hom}}(\mathbf{F},\mathbf{G})\big),$$

which is functorial in $\mathbf{E}$.

Proposition 1.7.1. [16] *Let* $\mathbf{E},\mathbf{F},\mathbf{G}\in\mathrm{Ob}(\widehat{\mathscr{C}})$.

(a) The map $\phi\mapsto\theta_\phi$ *is a bijection:*

$$\mathrm{Hom}(\mathbf{E}\times\mathbf{F},\mathbf{G})\xrightarrow{\sim}\mathrm{Hom}\big(\mathbf{E},\underline{\mathrm{Hom}}(\mathbf{F},\mathbf{G})\big).$$

(b) Moreover, there is an isomorphism of functors

$$\underline{\mathrm{Hom}}\big(\mathbf{E},\underline{\mathrm{Hom}}(\mathbf{F},\mathbf{G})\big)\simeq\underline{\mathrm{Hom}}(\mathbf{E}\times\mathbf{F},\mathbf{G}).$$

[15] Editor's note: if $\mathbf{E}$ is a third object of $\widehat{\mathscr{C}}$, then

$$\underline{\mathrm{Hom}}(\mathbf{E},\mathbf{F}\times\mathbf{G})\simeq\underline{\mathrm{Hom}}(\mathbf{E},\mathbf{F})\times\underline{\mathrm{Hom}}(\mathbf{E},\mathbf{G}).$$

[16] Editor's note: part (b), which will be useful in II.1 and II.3.11, has been added. A second, more direct proof of part (a) has also been sketched.

For (a), regard both sides as functors of $\mathbf{E}$. The asserted result is true when $\mathbf{E} = \mathbf{h}_X$, since it is then precisely the definition of the functor $\underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})$. Moreover, both sides, as functors of $\mathbf{E}$, take direct limits to inverse limits. Finally, by Lemma 1.7.0, every object $\mathbf{E}$ of $\widehat{\mathscr{C}}$ is isomorphic to the direct limit of the $\mathbf{h}_X$, as X ranges over $\mathscr{C}_{/\mathbf{E}}$. This proves (a).

Let us also sketch a direct proof of (a). To every

$$\theta \in \mathrm{Hom}\big(\mathbf{E}, \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})\big)$$

associate $\phi_\theta \in \mathrm{Hom}(\mathbf{E} \times \mathbf{F}, \mathbf{G})$ as follows. For every $S \in \mathrm{Ob}(\mathscr{C})$, there is a map

$$\theta(S) : \mathbf{E}(S) \longrightarrow \mathrm{Hom}(\mathbf{F} \times S, \mathbf{G}),$$

functorial in S. If $(e, f) \in \mathbf{E}(S) \times \mathbf{F}(S)$, then f is a morphism $S \to \mathbf{F}$, so $f \times \mathrm{id}_S$ is a morphism $S \to \mathbf{F} \times S$. On the other hand, $\theta(S)(e)$ is a morphism $\mathbf{F} \times S \to \mathbf{G}$. Composition therefore gives a morphism

$$\theta(S)(e) \circ (f \times \mathrm{id}_S) : S \longrightarrow \mathbf{G},$$

that is, an element $\phi_\theta(S)(e, f)$ of $\mathbf{G}(S)$. It is easy to check that $S \mapsto \phi_\theta(S)$ is functorial in S, and therefore defines a morphism ϕ_θ from $\mathbf{E} \times \mathbf{F}$ to $\mathbf{G}$. We leave to the reader the verification that $\theta \mapsto \phi_\theta$ and $\phi \mapsto \theta_\phi$ are mutually inverse bijections.

For (b), if $S \in \mathrm{Ob}(\mathscr{C})$, then by 1.7(ii) and part (a), applied to $\mathscr{C}_{/S}$,

$$\begin{aligned}
&\underline{\mathrm{Hom}}\big(\mathbf{E}, \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})\big)(S) \\
&\quad \simeq \mathrm{Hom}_S\big(\mathbf{E}_S, \underline{\mathrm{Hom}}_S(\mathbf{F}_S, \mathbf{G}_S)\big) \\
&\quad \simeq \mathrm{Hom}_S(\mathbf{E}_S \times_S \mathbf{F}_S, \mathbf{G}_S) \\
&\quad \simeq \mathrm{Hom}(\mathbf{E} \times \mathbf{F} \times S, \mathbf{G}) \\
&\quad \simeq \underline{\mathrm{Hom}}(\mathbf{E} \times \mathbf{F}, \mathbf{G})(S),
\end{aligned}$$

and these isomorphisms are functorial in S.

Corollary 1.7.2. *There are isomorphisms*

$$\mathrm{Hom}\big(\mathbf{E}, \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})\big) \simeq \mathrm{Hom}\big(\mathbf{F}, \underline{\mathrm{Hom}}(\mathbf{E}, \mathbf{G})\big),$$

and

$$\underline{\mathrm{Hom}}\big(\mathbf{E}, \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})\big) \simeq \underline{\mathrm{Hom}}\big(\mathbf{F}, \underline{\mathrm{Hom}}(\mathbf{E}, \mathbf{G})\big).$$

In particular, taking $\mathbf{E} = \underline{\mathbf{e}}$ and using $\underline{\mathrm{Hom}}(\underline{\mathbf{e}}, \mathbf{G}) \simeq \mathbf{G}$, we obtain

$$\Gamma\big(\underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})\big) \simeq \mathrm{Hom}(\mathbf{F}, \mathbf{G}).$$

Composition supplies functorial morphisms

$$\underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G}) \times \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{H}) \longrightarrow \underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{H}).$$

If $\mathbf{F}$ and $\mathbf{G}$ are two objects of $\widehat{\mathscr{C}}$, let $\mathrm{Isom}(\mathbf{F}, \mathbf{G})$ denote the subset of $\mathrm{Hom}(\mathbf{F}, \mathbf{G})$ formed by the isomorphisms from $\mathbf{F}$ onto $\mathbf{G}$. Define a subobject $\underline{\mathrm{Isom}}(\mathbf{F}, \mathbf{G})$ of $\underline{\mathrm{Hom}}(\mathbf{F}, \mathbf{G})$ by

$$\underline{\mathrm{Isom}}(\mathbf{F}, \mathbf{G})(S) = \mathrm{Isom}(\mathbf{F}_S, \mathbf{G}_S).$$

Then

$$\Gamma\big(\underline{\mathrm{Isom}}(\mathbf{F}, \mathbf{G})\big) \simeq \mathrm{Isom}(\mathbf{F}, \mathbf{G}), \qquad \underline{\mathrm{Isom}}(\mathbf{F}, \mathbf{G}) \simeq \underline{\mathrm{Isom}}(\mathbf{G}, \mathbf{F}).$$

In the special case $\mathbf{F} = \mathbf{G}$, set

$$\begin{aligned} \underline{\mathrm{End}}(\mathbf{F}) &= \underline{\mathrm{Hom}}(\mathbf{F},\mathbf{F}), & \mathrm{End}(\mathbf{F}) &= \mathrm{Hom}(\mathbf{F},\mathbf{F}) \simeq \Gamma\big(\underline{\mathrm{End}}(\mathbf{F})\big),\\ \underline{\mathrm{Aut}}(\mathbf{F}) &= \underline{\mathrm{Isom}}(\mathbf{F},\mathbf{F}), & \mathrm{Aut}(\mathbf{F}) &= \mathrm{Isom}(\mathbf{F},\mathbf{F}) \simeq \Gamma\big(\underline{\mathrm{Aut}}(\mathbf{F})\big). \end{aligned}$$

Formation of the objects $\underline{\mathrm{Hom}}$, $\underline{\mathrm{Isom}}$, $\underline{\mathrm{Aut}}$, and $\underline{\mathrm{End}}$ commutes with base change.

Remark 1.7.3. [17] An object isomorphic to $\underline{\mathrm{Isom}}(\mathbf{F},\mathbf{G})$ may also be constructed as follows. There is a morphism

$$\underline{\mathrm{Hom}}(\mathbf{F},\mathbf{G}) \times \underline{\mathrm{Hom}}(\mathbf{G},\mathbf{F}) \longrightarrow \underline{\mathrm{End}}(\mathbf{F}).$$

Interchanging $\mathbf{F}$ and $\mathbf{G}$ gives a morphism

$$\underline{\mathrm{Hom}}(\mathbf{F},\mathbf{G}) \times \underline{\mathrm{Hom}}(\mathbf{G},\mathbf{F}) \longrightarrow \underline{\mathrm{End}}(\mathbf{F}) \times \underline{\mathrm{End}}(\mathbf{G}).$$

On the other hand, the identity morphism of $\mathbf{F}$ is an element of $\mathrm{End}(\mathbf{F})$, and hence defines a morphism $\underline{\mathbf{e}} \to \underline{\mathrm{End}}(\mathbf{F})$. Doing the same for $\mathbf{G}$ and taking the product gives a morphism

$$\underline{\mathbf{e}} \longrightarrow \underline{\mathrm{End}}(\mathbf{F}) \times \underline{\mathrm{End}}(\mathbf{G}).$$

It is then immediate that the fiber product of $\underline{\mathbf{e}}$ and

$$\underline{\mathrm{Hom}}(\mathbf{F},\mathbf{G}) \times \underline{\mathrm{Hom}}(\mathbf{G},\mathbf{F})$$

over $\underline{\mathrm{End}}(\mathbf{F}) \times \underline{\mathrm{End}}(\mathbf{G})$ is isomorphic to $\underline{\mathrm{Isom}}(\mathbf{F},\mathbf{G})$.

All these definitions apply, in particular, when $\mathbf{F} = \mathbf{h}_X$ and $\mathbf{G} = \mathbf{h}_Y$. If $\underline{\mathrm{Hom}}(\mathbf{h}_X, \mathbf{h}_Y)$ is representable by an object of $\mathscr{C}$, we denote that object by $\underline{\mathrm{Hom}}(X,Y)$. It has the following property: if $Z \times X$ exists, then

$$\mathrm{Hom}\big(Z, \underline{\mathrm{Hom}}(X,Y)\big) \simeq \mathrm{Hom}(Z \times X, Y).$$

This property characterizes it when products exist in $\mathscr{C}$.

Similarly, one defines, when they exist, the objects

$$\underline{\mathrm{Isom}}(X,Y), \qquad \underline{\mathrm{End}}(X), \qquad \underline{\mathrm{Aut}}(X).$$

We merely note that the construction above shows that $\underline{\mathrm{Isom}}(X,Y)$ exists whenever fiber products exist in $\mathscr{C}$ and the objects

$$\underline{\mathrm{Hom}}(X,Y), \quad \underline{\mathrm{Hom}}(Y,X), \quad \underline{\mathrm{End}}(X), \quad \text{and} \quad \underline{\mathrm{End}}(Y)$$

exist.

Everything above also applies to categories of the form $\mathscr{C}_{/S}$. The corresponding objects will be denoted as explicitly as possible by appropriate symbols. For example, if T and T' are two objects of $\mathscr{C}$ over S, we denote by $\underline{\mathrm{Hom}}_S(T,T')$ the object

$$\underline{\mathrm{Hom}}_{\mathscr{C}_{/S}}(T/S, T'/S).$$

[17] Editor's note: the numbering 1.7.3 has been added for later references.

1.8. Constant objects

Let $\mathscr{C}$ be a category in which direct sums and fiber products exist and in which direct sums commute with base change (for example, the category of schemes).[18] For every set E and every object S of $\mathscr{C}$, set

$$E_S = \big\{\text{the direct sum of a family } (S_i)_{i\in E} \text{ of objects} \quad \text{of } \mathscr{C}\text{, all isomorphic to } S. \tag{1.8.1}$$

This object is characterized by the formula

$$\mathrm{Hom}_{\mathscr{C}}(E_S, T) = \mathrm{Hom}_{\mathbf{Set}}\big(E, \mathrm{Hom}_{\mathscr{C}}(S, T)\big), \tag{1.8.2}$$

for every $T \in \mathrm{Ob}(\mathscr{C})$.

The object E_S has a canonical projection to S, so that $E \mapsto E_S$ is in fact a functor from **Set** to $\mathscr{C}_{/S}$.[19] If $S' \to S$ is an arrow of $\mathscr{C}$, then, because direct sums commute with base change,

$$E_{S'} = (E_S)_{S'}.$$

In particular, if $\mathscr{C}$ has a final object e, then

$$E_S = (E_e)_S.$$

The functor $E \mapsto E_S/S$, from **Set** to $\mathscr{C}_{/S}$, commutes with finite products. It is enough to prove that

$$E_S \times_S F_S = (E \times F)_S. \tag{$\times$}$$

Indeed, by the results of 1.7 applied to $\mathscr{C}_{/S}$, for every $T \in \mathrm{Ob}(\mathscr{C}_{/S})$ there are natural isomorphisms (all unspecified Hom's being taken in $\mathscr{C}_{/S}$)

$$\begin{aligned}\mathrm{Hom}((E\times F)_S, T) &\simeq \mathrm{Hom}_{\mathbf{Set}}\big(E\times F, \mathrm{Hom}(S,T)\big)\\ &\simeq \mathrm{Hom}_{\mathbf{Set}}\big(E, \mathrm{Hom}_{\mathbf{Set}}(F, \mathrm{Hom}(S,T))\big)\\ &\simeq \mathrm{Hom}_{\mathbf{Set}}\big(E, \mathrm{Hom}(F_S, T)\big),\end{aligned}$$

and

$$\begin{aligned}\mathrm{Hom}(E_S \times_S F_S, T) &\simeq \mathrm{Hom}\big(E_S, \underline{\mathrm{Hom}}(F_S, T)\big)\\ &\simeq \mathrm{Hom}_{\mathbf{Set}}\big(E, \mathrm{Hom}(S, \underline{\mathrm{Hom}}(F_S, T))\big).\end{aligned}$$

But

$$\mathrm{Hom}\big(S, \underline{\mathrm{Hom}}(F_S, T)\big) \simeq \mathrm{Hom}(F_S, T),$$

which proves ($\times$).

Suppose that, with $\varnothing$ denoting an initial object of $\mathscr{C}$, the following diagram is cartesian:

$$\begin{tikzcd} \varnothing \arrow[r] \arrow[d] & \mathrm{S} \arrow[d] \\ \mathrm{S} \arrow[r] & \mathrm{S} \coprod \mathrm{S} \end{tikzcd}$$

(*)

[18]Editor's note: the former terminology "preschemes/schemes" has been replaced by the current terminology "schemes/separated schemes."

[19]Editor's note: in the next three paragraphs, the order of the sentences has been changed and a few details have been added concerning the role of condition ($*$) below.

This is the case for the category of schemes.[20] Then the functor $E \mapsto E_S$ commutes with finite inverse limits.

Indeed, in view of $(\times)$, it is enough to show that $E \mapsto E_S$ commutes with fiber products. Let $u : E \to G$ and $v : F \to G$ be two maps of sets. Since direct sums in $\mathscr{C}$ commute with base change, one has

$$F_S \times_{G_S} E_S \simeq \coprod_{f \in F} \big(S_f \times_{G_S} E_S\big) \simeq \coprod_{\substack{f \in F \\ x \in E}} \big(S_f \times_{G_S} S_x\big). \tag{1}$$

If $v(f) \neq u(x)$, there is a morphism in $\mathscr{C}_{/S}$

$$S_f \times_{G_S} S_x \longrightarrow S_f \times_{S_{v(f)} \coprod S_{u(x)}} S_x.$$

By condition $(*)$, the term on the right is $\varnothing$. Hence

$$S_f \times_{G_S} S_x \simeq \varnothing \qquad \text{if } v(f) \neq u(x). \tag{2}$$

On the other hand, if $v(f) = u(x)$, there is a morphism in $\mathscr{C}_{/S}$

$$S \longrightarrow S_f \times_{G_S} S_x.$$

Since $S \xrightarrow{\mathrm{id}} S$ is the final object of $\mathscr{C}_{/S}$, it follows that

$$S_f \times_{G_S} S_x \simeq S \qquad \text{if } v(f) = u(x). \tag{3}$$

Combining (1), (2), and (3), we obtain an isomorphism functorial in S:

$$F_S \times_{G_S} E_S \simeq \coprod_{F \times_G E} S = (F \times_G E)_S.$$

An object of the form E_S is called a *constant object*. There is a morphism, functorial in E,

$$E \longrightarrow \Gamma(E_S/S),$$

which associates with each $i \in E$ the section of E_S over S defined by the isomorphism from S onto S_i. Suppose condition $(*)$ is satisfied for every object S of $\mathscr{C}$. Then $E \to \Gamma(E_S/S)$ is a monomorphism whenever $S \not\simeq \varnothing$.

If $\mathscr{C}$ is the category of schemes, then $\Gamma(E_S/S)$ is identified with the locally constant maps from the topological space S to the set E, the preceding map associating with each element of E the corresponding constant map. It follows that E_S may equivalently be defined as representing the functor that assigns to every S' over S the set of locally constant functions from the topological space S' to E.[21]

2. Algebraic structures

Given a species of algebraic structure in the category of sets, we shall extend it to the category $\mathscr{C}$. We first treat the example of groups.

[20] Editor's note: Condition $(*)$ is not satisfied if $\mathscr{C}$ is the category whose arrows are $A \to B$, id_A, and id_B. In that case $B \coprod B = B$ and $B \times_B B = B \not\simeq A$.

[21] Editor's note: the diagonal morphism $E_S \to E_S \times_S E_S = (E \times E)_S$ is also a closed immersion; that is, E_S is separated over S.

2.1. Group structures

We retain the notation of the preceding paragraph.

Definition 2.1.1. Let $\mathbf{G} \in \mathrm{Ob}(\widehat{\mathscr{C}})$. A *$\widehat{\mathscr{C}}$-group structure* on $\mathbf{G}$ is the datum, for every $S \in \mathrm{Ob}(\mathscr{C})$, of a group structure on the set $\mathbf{G}(S)$, such that for every arrow $f : S' \to S''$ of $\mathscr{C}$, the map

$$\mathbf{G}(f) : \mathbf{G}(S'') \longrightarrow \mathbf{G}(S')$$

is a group homomorphism. If $\mathbf{G}$ and $\mathbf{H}$ are two $\widehat{\mathscr{C}}$-groups, a *morphism of $\widehat{\mathscr{C}}$-groups* from $\mathbf{G}$ to $\mathbf{H}$ is a morphism $u \in \mathrm{Hom}(\mathbf{G}, \mathbf{H})$ such that $u(S) : \mathbf{G}(S) \to \mathbf{H}(S)$ is a group homomorphism for every $S \in \mathrm{Ob}(\mathscr{C})$.

We denote by $\mathrm{Hom}_{\widehat{\mathscr{C}}\text{-Gr}}(\mathbf{G}, \mathbf{H})$ the set of these morphisms and by $(\widehat{\mathscr{C}}\text{-Gr})$ their category.

Examples. For $\mathbf{E} \in \mathrm{Ob}(\widehat{\mathscr{C}})$, the object $\underline{\mathrm{Aut}}(\mathbf{E})$ has an evident $\widehat{\mathscr{C}}$-group structure. The final object $\underline{\mathbf{e}}$ has a unique $\widehat{\mathscr{C}}$-group structure, making it a final object of $(\widehat{\mathscr{C}}\text{-Gr})$.

For every $S \in \mathrm{Ob}(\mathscr{C})$, let $e_{\mathbf{G}}(S)$ be the identity element of $\mathbf{G}(S)$. The family of elements $e_{\mathbf{G}}(S)$ defines

$$e_{\mathbf{G}} \in \Gamma(\mathbf{G}) = \mathrm{Hom}(\underline{\mathbf{e}}, \mathbf{G}).$$

It is a morphism of $\widehat{\mathscr{C}}$-groups $\underline{\mathbf{e}} \to \mathbf{G}$, called the *unit section* of $\mathbf{G}$.

To give a $\widehat{\mathscr{C}}$-group structure on $\mathbf{G}$ is equivalently to give a composition law on $\mathbf{G}$, that is, a $\widehat{\mathscr{C}}$-morphism

$$\pi_{\mathbf{G}} : \mathbf{G} \times \mathbf{G} \longrightarrow \mathbf{G}$$

such that $\pi_{\mathbf{G}}(S)$ equips $\mathbf{G}(S)$ with a group structure for every $S \in \mathrm{Ob}(\mathscr{C})$.

Similarly, $f : \mathbf{G} \to \mathbf{H}$ is a morphism of $\widehat{\mathscr{C}}$-groups if and only if the following diagram is commutative:

$$\begin{tikzcd}
\mathbf{G} \times \mathbf{G} \arrow[r, "\pi_{\mathbf{G}}"] \arrow[d, "{(f,f)}"'] & \mathbf{G} \arrow[d, "f"'] \\
\mathbf{H} \times \mathbf{H} \arrow[r, "\pi_{\mathbf{H}}"] & \mathbf{H}
\end{tikzcd}$$

A subobject $\mathbf{H}$ of $\mathbf{G}$ such that $\mathbf{H}(S)$ is a subgroup of $\mathbf{G}(S)$ for every S has the evident $\widehat{\mathscr{C}}$-group structure induced by that of $\mathbf{G}$. It is the unique structure for which the monomorphism $\mathbf{H} \to \mathbf{G}$ is a morphism of $\widehat{\mathscr{C}}$-groups. With this structure, $\mathbf{H}$ is called a *sub-$\widehat{\mathscr{C}}$-group* of $\mathbf{G}$.

If $\mathbf{G}$ and $\mathbf{H}$ are two $\widehat{\mathscr{C}}$-groups, then $\mathbf{G} \times \mathbf{H}$ has the evident product structure: for every S, the set $\mathbf{G}(S) \times \mathbf{H}(S)$ is given the product of the prescribed group structures. The resulting object is called the *product $\widehat{\mathscr{C}}$-group* of $\mathbf{G}$ and $\mathbf{H}$; it is their product in the category of $\widehat{\mathscr{C}}$-groups.

If $\mathbf{G}$ is a $\widehat{\mathscr{C}}$-group, then $\mathbf{G}_S$ is a $\widehat{\mathscr{C}_{/S}}$-group for every $S \in \mathrm{Ob}(\mathscr{C})$. For two $\widehat{\mathscr{C}}$-groups $\mathbf{G}$ and $\mathbf{H}$, define the object $\underline{\mathrm{Hom}}_{\widehat{\mathscr{C}}\text{-Gr}}(\mathbf{G}, \mathbf{H})$ of $\widehat{\mathscr{C}}$ by

$$\underline{\mathrm{Hom}}_{\widehat{\mathscr{C}}\text{-Gr}}(\mathbf{G}, \mathbf{H})(S) = \mathrm{Hom}_{\widehat{\mathscr{C}_{/S}}\text{-Gr}}(\mathbf{G}_S, \mathbf{H}_S).$$

Notice that $\underline{\mathrm{Hom}}_{\widehat{\mathscr{C}}\text{-Gr}}$ is not in general a $\widehat{\mathscr{C}}$-group, still less the internal $\underline{\mathrm{Hom}}$ object in the category $(\widehat{\mathscr{C}}\text{-Gr})$.

One similarly defines

$$\underline{\mathrm{Isom}}_{\widehat{\mathscr{C}}\text{-Gr}}(\mathbf{G}, \mathbf{H}), \qquad \underline{\mathrm{End}}_{\widehat{\mathscr{C}}\text{-Gr}}(\mathbf{G}), \qquad \underline{\mathrm{Aut}}_{\widehat{\mathscr{C}}\text{-Gr}}(\mathbf{G}).$$

Definition 2.1.2. Let $G \in \mathrm{Ob}(\mathscr{C})$. A *$\mathscr{C}$-group structure* on G is a $\widehat{\mathscr{C}}$-group structure on $\mathbf{h}_G \in \mathrm{Ob}(\widehat{\mathscr{C}})$. A morphism from the $\mathscr{C}$-group G to the $\mathscr{C}$-group H is an element

$$u \in \mathrm{Hom}(G, H) \simeq \mathrm{Hom}(\mathbf{h}_G, \mathbf{h}_H)$$

that defines a morphism of $\widehat{\mathscr{C}}$-groups from $\mathbf{h}_G$ to $\mathbf{h}_H$.

We denote the category of $\mathscr{C}$-groups by ($\mathscr{C}$-Gr). There is a cartesian square in **Cat**:

$$\begin{tikzcd}
(\mathscr{C}\text{-Gr.}) \arrow[r] \arrow[d] & (\widehat{\mathscr{C}}\text{-Gr.}) \arrow[d] \\
\mathscr{C} \arrow[r, "\mathbf{h}"] & \widehat{\mathscr{C}}
\end{tikzcd} .$$

All the preceding definitions and constructions therefore carry over at once to ($\mathscr{C}$-Gr) whenever the functors they use (products, internal $\underline{\mathrm{Hom}}$ objects, and so on) are representable. They also apply to the categories $\mathscr{C}_{/S}$. In that case we write $\underline{\mathrm{Hom}}_{S\text{-Gr}}$ for $\underline{\mathrm{Hom}}_{\mathscr{C}_{/S}\text{-Gr}}$, and similarly for the other objects.

2.2.

More generally, suppose (T) is a species of structure on n base sets defined by finite inverse limits—for example, by commutative diagrams constructed from cartesian products, as in the structures of a monoid, a group, a set with operators, a module over a ring, or a Lie algebra over a ring. The preceding construction defines the notion of a "structure of species (T)" on n objects $\mathbf{F}_1, \ldots, \mathbf{F}_n$ of $\widehat{\mathscr{C}}$. Such a structure consists, for every $S \in \mathrm{Ob}(\mathscr{C})$, of a structure of species (T) on the sets $\mathbf{F}_1(S), \ldots, \mathbf{F}_n(S)$, in such a way that for every arrow $S' \to S''$ of $\mathscr{C}$, the family of maps

$$\big(\mathbf{F}_i(S'')\big)_i \longrightarrow \big(\mathbf{F}_i(S')\big)_i$$

is a polyhomomorphism for the species (T).

Morphisms of structures of species (T) are defined similarly, giving a category

$$(\widehat{\mathscr{C}} \times \cdots \times \widehat{\mathscr{C}})^{(T)}.$$

The fully faithful functor $\mathbf{h} \times \cdots \times \mathbf{h}$ then defines, by inverse image, the category

$$(\mathscr{C} \times \cdots \times \mathscr{C})^{(T)}.$$

Since it commutes with inverse limits, all the functorial properties, notions, and notation introduced in $\widehat{\mathscr{C}}$ carry over to this category.

Now suppose fiber products exist in $\mathscr{C}$, and let (T) be a species of algebraic structure specified by certain morphisms between cartesian products, subject to axioms consisting of commutative diagrams formed with those morphisms. A structure of species (T) on a family of objects of $\mathscr{C}$ is then defined by corresponding morphisms between cartesian products satisfying the corresponding commutativity conditions. Consequently, if $\mathscr{C}$ and $\mathscr{C}'$

are categories with products and $f : \mathscr{C} \to \mathscr{C}'$ is a functor commuting with products, then a structure of species (T) on a family (F_i) of objects of $\mathscr{C}$ induces one on the family $(f(F_i))$ in $\mathscr{C}'$. Thus every $\mathscr{C}$-group becomes a $\mathscr{C}'$-group, every pair consisting of a $\mathscr{C}$-ring and a module over it becomes the analogous pair in $\mathscr{C}'$, and so on.

In particular, let $\mathscr{C}$ satisfy the conditions of 1.8.[22] The functor $E \mapsto E_S$ defined there commutes with finite inverse limits. It therefore carries groups to S-groups (that is, $\mathscr{C}_{/S}$-groups), rings to S-rings, and so forth.

Remark. It is worth observing that the preceding construction, when applied to $\widehat{\mathscr{C}}$, gives back precisely the notions already defined there. In other words, to give an object of $\widehat{\mathscr{C}}$ a structure of species (T) when it is viewed as a functor on $\mathscr{C}$ is the same as to give the representable functor on $\widehat{\mathscr{C}}$ defined by that object a structure of species (T).[23]

We shall treat two further special cases of this construction: structures with operator groups and modules.

2.3. Structures with operator groups

Definition 2.3.1. Let $\mathbf{E} \in \mathrm{Ob}(\widehat{\mathscr{C}})$ and $\mathbf{G} \in \mathrm{Ob}(\widehat{\mathscr{C}}\text{-}\mathbf{Gr})$. A structure on $\mathbf{E}$ of an object with $\widehat{\mathscr{C}}$-operator group $\mathbf{G}$ (or of a $\mathbf{G}$-object) consists in giving $\mathbf{E}(S)$, for every $S \in \mathrm{Ob}(\mathscr{C})$, the structure of a set with operator group $\mathbf{G}(S)$, in such a way that, for every arrow $S' \to S''$ in $\mathscr{C}$, the map of sets $\mathbf{E}(S'') \to \mathbf{E}(S')$ is compatible with the homomorphism of operator groups $\mathbf{G}(S'') \to \mathbf{G}(S')$.

As usual, this is equivalent to giving a morphism

$$\mu : \mathbf{G} \times \mathbf{E} \longrightarrow \mathbf{E}$$

which, for every S, equips $\mathbf{E}(S)$ with the structure of a set on which $\mathbf{G}(S)$ acts. But

$$\mathrm{Hom}(\mathbf{G} \times \mathbf{E}, \mathbf{E}) \simeq \mathrm{Hom}(\mathbf{G}, \underline{\mathrm{End}}(\mathbf{E})),$$

so μ defines a morphism $\mathbf{G} \to \underline{\mathrm{End}}(\mathbf{E})$. It is immediate that this morphism maps $\mathbf{G}$ into $\underline{\mathrm{Aut}}(\mathbf{E})$ and is a morphism of $\widehat{\mathscr{C}}$-groups. Consequently, giving $\mathbf{E}$ the structure of an object with $\widehat{\mathscr{C}}$-operator group $\mathbf{G}$ is equivalent to giving a morphism of $\widehat{\mathscr{C}}$-groups

$$\rho : \mathbf{G} \longrightarrow \underline{\mathrm{Aut}}(\mathbf{E}).$$

In particular, every element $g \in \mathbf{G}(S)$ defines an automorphism $\rho(g)$ of the functor $\mathbf{E}_S$, that is, an automorphism of $\mathbf{E} \times \mathbf{h}_S$ commuting with the projection $\mathbf{E} \times \mathbf{h}_S \to \mathbf{h}_S$, and hence, in particular, an automorphism of the set $\mathbf{E}(S')$ for every $S' \to S$.

Definition 2.3.2. We denote by $\mathbf{E}^{\mathbf{G}}$ the subobject of $\mathbf{E}$ defined as follows:

$$\mathbf{E}^{\mathbf{G}}(S) = \big\{x \in \mathbf{E}(S) \mid x_{S'} \text{ is invariant under } \mathbf{G}(S') \text{ for every } S' \to S\big\},$$

where $x_{S'}$ denotes the image of x under $\mathbf{E}(S) \to \mathbf{E}(S')$.

[22] Editor's note: including condition $(*)$.

[23] Editor's note: for group structures, for example, let $\mathbf{G} \in \mathrm{Ob}(\widehat{\mathscr{C}})$. If the functor

$$\widehat{\mathscr{C}} \longrightarrow \mathbf{Set}, \qquad \mathbf{F} \longmapsto \mathrm{Hom}_{\widehat{\mathscr{C}}}(\mathbf{F}, \mathbf{G})$$

has a group structure, then so does its restriction to $\mathscr{C}$, namely $X \mapsto \mathbf{G}(X)$. Conversely, if $\mathbf{G}$ is a $\widehat{\mathscr{C}}$-group, its multiplication morphism $\pi_{\mathbf{G}} : \mathbf{G} \times \mathbf{G} \to \mathbf{G}$ induces, for every $\mathbf{F} \in \mathrm{Ob}(\widehat{\mathscr{C}})$, a group structure on $\mathrm{Hom}_{\widehat{\mathscr{C}}}(\mathbf{F}, \mathbf{G})$, functorially in $\mathbf{F}$.

Thus $\mathbf{E}^{\mathbf{G}}$, the "subobject of invariants of $\mathbf{G}$," is the largest subobject of $\mathbf{E}$ on which $\mathbf{G}$ acts trivially.

Definition 2.3.3. Let $\mathbf{F}$ be a subobject of $\mathbf{E}$. We denote by $\underline{\mathrm{Norm}}_{\mathbf{G}}\mathbf{F}$ and $\underline{\mathrm{Centr}}_{\mathbf{G}}\mathbf{F}$ the sub-$\widehat{\mathscr{C}}$-groups of $\mathbf{G}$ defined by

$$\begin{aligned}(\underline{\mathrm{Norm}}_{\mathbf{G}}\mathbf{F})(S) &= \{g \in \mathbf{G}(S) \mid \rho(g)\mathbf{F}_S = \mathbf{F}_S\} \\ &= \{g \in \mathbf{G}(S) \mid \rho(g)\mathbf{F}(S') = \mathbf{F}(S') \text{ for every } S' \to S\}, \\ (\underline{\mathrm{Centr}}_{\mathbf{G}}\mathbf{F})(S) &= \{g \in \mathbf{G}(S) \mid \rho(g)|_{\mathbf{F}_S} = \mathrm{id}\} \\ &= \{g \in \mathbf{G}(S) \mid \rho(g)|_{\mathbf{F}(S')} = \mathrm{id} \text{ for every } S' \to S\}.\end{aligned}$$

Here the vertical bar after $\rho(g)$ denotes restriction. [24]

Scholium 2.3.3.1. [25] In particular, let $x \in \Gamma(\mathbf{E})$, that is (cf. 1.2), a collection of elements $x_S \in \mathbf{E}(S)$, $S \in \mathrm{Ob}(\mathscr{C})$, such that for every arrow $f : S' \to S$ one has $\mathbf{E}(f)(x_S) = x_{S'}$. If $\mathscr{C}$ has a final object S_0, then $\Gamma(\mathbf{E}) = \mathbf{E}(S_0)$. The element x then defines a subfunctor of $\mathbf{E}$, denoted by $\mathbf{x}$, and

$$\underline{\mathrm{Norm}}_{\mathbf{G}}\mathbf{x} = \underline{\mathrm{Centr}}_{\mathbf{G}}\mathbf{x}.$$

We denote this functor by $\underline{\mathrm{Stab}}_{\mathbf{G}}(x)$ and call it the *stabilizer* of x; thus, for every $S \in \mathrm{Ob}(\mathscr{C})$,

$$\underline{\mathrm{Stab}}_{\mathbf{G}}(x)(S) = \{g \in \mathbf{G}(S) \mid \rho(g)x_S = x_S\}.$$

Suppose that fiber products exist in $\mathscr{C}$. If $\mathbf{G} = \mathbf{h}_G$ (respectively, $\mathbf{E} = \mathbf{h}_E$), where G is a $\mathscr{C}$-group (respectively, $E \in \mathrm{Ob}(\mathscr{C})$), and if $\mathscr{C}$ has a final object S_0, so that x is a morphism $S_0 \to E$, then $\underline{\mathrm{Stab}}_{\mathbf{G}}(x)$ is represented by the fiber product $G \times_E S_0$, where the morphism $G \to E$ is the composite of

$$\mathrm{id}_G \times x : G = G \times S_0 \longrightarrow G \times E$$

and

$$\mu : G \times E \longrightarrow E.$$

Remark 2.3.3.2.[25] The formation of $\mathbf{E}^{\mathbf{G}}$, $\underline{\mathrm{Norm}}_{\mathbf{G}}\mathbf{F}$, and $\underline{\mathrm{Centr}}_{\mathbf{G}}\mathbf{F}$ commutes with base change; that is, for every $S \in \mathrm{Ob}(\mathscr{C})$,

$$(\mathbf{E}^{\mathbf{G}})_S = (\mathbf{E}_S)^{\mathbf{G}_S}, \qquad (\underline{\mathrm{Norm}}_{\mathbf{G}}\mathbf{F})_S \simeq \underline{\mathrm{Norm}}_{\mathbf{G}_S}\mathbf{F}_S, \qquad (\underline{\mathrm{Centr}}_{\mathbf{G}}\mathbf{F})_S \simeq \underline{\mathrm{Centr}}_{\mathbf{G}_S}\mathbf{F}_S.$$

Definition 2.3.4. If G is a $\mathscr{C}$-group and $\mathbf{E}$ is an object of $\widehat{\mathscr{C}}$ (respectively, E is an object of $\mathscr{C}$), a G-object structure on $\mathbf{E}$ (respectively, on E) is an $\mathbf{h}_G$-object structure on $\mathbf{E}$ (respectively, on $\mathbf{h}_E$).

In view of this definition, all the notions and notation defined above carry over to $\mathscr{C}$ whenever they involve only representable functors. For example, if $\underline{\mathrm{Norm}}_{\mathbf{h}_G}(\mathbf{h}_F)$ is representable, then there is a unique subobject of G representing it; it is then a sub-$\mathscr{C}$-group of G, and is denoted by $\mathrm{Norm}_G(F)$, and so forth.

Definition 2.3.5.

[24] Editor's note: the original has been corrected by replacing the inclusion $\rho(g)\mathbf{F}_S \subset \mathbf{F}_S$ with an equality, so as to ensure that $\underline{\mathrm{Norm}}_{\mathbf{G}}(\mathbf{F})$ is indeed a group. See $\mathrm{VI_B}$, 6.4 for conditions under which the "transporter" coincides with the "strict transporter."

[25] Editor's note: Scholium 2.3.3.1 and Remark 2.3.3.2 have been added.

(a) We say that the $\widehat{\mathscr{C}}$-group $\mathbf{G}$ acts on the $\widehat{\mathscr{C}}$-group $\mathbf{H}$ if $\mathbf{H}$ is equipped with a $\mathbf{G}$-object structure such that, for every $g \in \mathbf{G}(S)$, the automorphism of $\mathbf{H}(S)$ defined by g is a group automorphism.

Equivalently, for every $g \in \mathbf{G}(S)$, the automorphism $\rho(g)$ of $\mathbf{H}_S$ is an automorphism of $\widehat{\mathscr{C}}_{/S}$-groups; or equivalently, the morphism of $\widehat{\mathscr{C}}$-groups $\mathbf{G} \to \underline{\mathrm{Aut}}(\mathbf{H})$ maps $\mathbf{G}$ into

$$\underline{\mathrm{Aut}}_{\widehat{\mathscr{C}}\text{-}\mathbf{Gr}}(\mathbf{H}).$$

(b) In the situation above, there is a unique $\widehat{\mathscr{C}}$-group structure on $\mathbf{H} \times \mathbf{G}$ such that, for every S, $(\mathbf{H} \times \mathbf{G})(S)$ is the semidirect product of the groups $\mathbf{H}(S)$ and $\mathbf{G}(S)$ for the given action of $\mathbf{G}(S)$ on $\mathbf{H}(S)$. We denote this $\widehat{\mathscr{C}}$-group by $\mathbf{H} \cdot \mathbf{G}$ and call it the *semidirect product of* $\mathbf{H}$ *by* $\mathbf{G}$. Thus, by definition,

$$(\mathbf{H} \cdot \mathbf{G})(S) = \mathbf{H}(S) \cdot \mathbf{G}(S).$$

Let $\mathbf{G}$ be a $\widehat{\mathscr{C}}$-group. For every arrow $S' \to S$ in $\mathscr{C}$ and every $g \in \mathbf{G}(S)$, let $\mathrm{Int}(g)$ be the automorphism of $\mathbf{G}(S')$ defined by

$$\mathrm{Int}(g)h = ghg^{-1}.$$

This definition extends to a morphism of $\widehat{\mathscr{C}}$-groups

$$\mathrm{Int} : \mathbf{G} \longrightarrow \underline{\mathrm{Aut}}_{\widehat{\mathscr{C}}\text{-}\mathbf{Gr}}(\mathbf{G}) \subset \underline{\mathrm{Aut}}(\mathbf{G}).$$

Definition 2.3.3 therefore applies, and for every subobject $\mathbf{E}$ of $\mathbf{G}$ one has the sub-$\widehat{\mathscr{C}}$-groups of $\mathbf{G}$

$$\underline{\mathrm{Norm}}_{\mathbf{G}}(\mathbf{E}) \qquad \text{and} \qquad \underline{\mathrm{Centr}}_{\mathbf{G}}(\mathbf{E}).$$

Definition 2.3.6. The *center* of $\mathbf{G}$, denoted $\underline{\mathrm{Centr}}(\mathbf{G})$, is the sub-$\widehat{\mathscr{C}}$-group $\underline{\mathrm{Centr}}_{\mathbf{G}}(\mathbf{G})$ of $\mathbf{G}$. We say that $\mathbf{G}$ is commutative if $\underline{\mathrm{Centr}}(\mathbf{G}) = \mathbf{G}$, or, equivalently, if $\mathbf{G}(S)$ is commutative for every S.

We say that the sub-$\widehat{\mathscr{C}}$-group $\mathbf{H}$ of $\mathbf{G}$ is invariant in $\mathbf{G}$ if $\underline{\mathrm{Norm}}_{\mathbf{G}}(\mathbf{H}) = \mathbf{G}$, or, equivalently, if $\mathbf{H}(S)$ is invariant in $\mathbf{G}(S)$ for every S. [26]

Definition 2.3.6.1. [27] Let $f : \mathbf{G} \to \mathbf{G}'$ be a morphism of $\widehat{\mathscr{C}}$-groups. The *kernel* of f, denoted $\mathrm{Ker}\, f$, is the sub-$\widehat{\mathscr{C}}$-group of $\mathbf{G}$ defined by

$$(\mathrm{Ker}\, f)(S) = \{x \in \mathbf{G}(S) \mid f(S)(x) = 1\} = \mathrm{Ker}\, f(S)$$

for every $S \in \mathrm{Ob}(\mathscr{C})$; it is an invariant sub-$\widehat{\mathscr{C}}$-group.

Notice that if $\mathbf{G} = \mathbf{h}_G$ and $\mathbf{G}' = \mathbf{h}_{G'}$, if $\mathscr{C}$ has a final object S_0, and if fiber products exist in $\mathscr{C}$, then $\mathrm{Ker}(f)$ is represented by $S_0 \times_{G'} G$.

Definition 2.3.6.2.[27] Let $\mathbf{E} \in \widehat{\mathscr{C}}$, and let $\mathbf{G}$ be a $\widehat{\mathscr{C}}$-group acting on $\mathbf{E}$. We say that the action of $\mathbf{G}$ on $\mathbf{E}$ is *faithful* if the kernel of the morphism $\mathbf{G} \to \underline{\mathrm{Aut}}(\mathbf{E})$ is trivial; that is, if, for every $S \in \mathrm{Ob}(\mathscr{C})$ and $g \in \mathbf{G}(S)$, the condition

$$g_{S'} \cdot x = x \qquad \text{for every } S' \to S \text{ and every } x \in \mathbf{E}(S')$$

implies $g = 1$.

[26]Editor's note: in addition, $\mathbf{H}$ is said to be *central* in $\mathbf{G}$ if $\underline{\mathrm{Centr}}_{\mathbf{G}}(\mathbf{H}) = \mathbf{G}$, or, equivalently, if $\mathbf{H}(S)$ is central in $\mathbf{G}(S)$ for every S.

[27]Editor's note: Definitions 2.3.6.1 and 2.3.6.2 have been added.

Many definitions and propositions from elementary group theory carry over readily. We record only the following one, which will be useful to us.

Proposition 2.3.7. *Let $f : \mathbf{W} \to \mathbf{G}$ be a morphism of $\widehat{\mathscr{C}}$-groups. Put $\mathbf{H}(S) = \operatorname{Ker} f(S)$. Let $u : \mathbf{G} \to \mathbf{W}$ be a morphism of $\widehat{\mathscr{C}}$-groups that is a section of f (and is therefore necessarily a monomorphism). Then $\mathbf{W}$ is identified with the semidirect product of $\mathbf{H}$ by $\mathbf{G}$ for the action of $\mathbf{G}$ on $\mathbf{H}$ defined by*

$$(g, h) \longmapsto \operatorname{Int}(u(g))h$$

for $g \in \mathbf{G}(S)$, $h \in \mathbf{H}(S)$, and $S \in \operatorname{Ob}(\mathscr{C})$.

All these definitions and propositions carry over, as usual, to $\mathscr{C}$. In particular, the semidirect product of two $\mathscr{C}$-groups H and G, with G acting on H, is defined whenever the Cartesian product $H \times G$ exists, and the analog of Proposition 2.3.7 takes the following form.

Proposition 2.3.8. *Let*

$$H \xrightarrow{i} W \xrightarrow{f} G$$

be a sequence of morphisms of $\mathscr{C}$-groups such that, for every $S \in \operatorname{Ob}(\mathscr{C})$, the pair $(H(S), i(S))$ is a kernel of $f(S) : W(S) \to G(S)$. Let $u : G \to W$ be a morphism of $\mathscr{C}$-groups that is a section of f. Then W is identified with the semidirect product of H by G for the action of G on H such that, if $S \in \operatorname{Ob}(\mathscr{C})$, $g \in G(S)$, and $h \in H(S)$, then

$$\operatorname{Int}(u(g))i(h) = i({}^g h).$$

3. The category of O-modules, the category of G-O-modules

Definition 3.1. Let $\mathbf{O}$ and $\mathbf{F}$ be two objects of $\widehat{\mathscr{C}}$. We say that $\mathbf{F}$ is a $\widehat{\mathscr{C}}$-module over the $\widehat{\mathscr{C}}$-ring $\mathbf{O}$, or, for short, an $\mathbf{O}$-module, if, for every $S \in \operatorname{Ob}(\mathscr{C})$, the set $\mathbf{O}(S)$ has been endowed with a ring structure and $\mathbf{F}(S)$ with a module structure over that ring, in such a way that for every arrow $S' \to S''$ in $\mathscr{C}$, the map $\mathbf{O}(S'') \to \mathbf{O}(S')$ is a ring homomorphism and $\mathbf{F}(S'') \to \mathbf{F}(S')$ is a homomorphism of abelian groups compatible with that ring homomorphism.

Once $\mathbf{O}$ is fixed, one defines in the usual way a morphism between two $\mathbf{O}$-modules $\mathbf{F}$ and $\mathbf{F}'$, hence the abelian group $\operatorname{Hom}_{\mathbf{O}}(\mathbf{F}, \mathbf{F}')$, and the category of $\mathbf{O}$-modules, denoted ($\mathbf{O}$-Mod.).

Lemma 3.1.1.[28] *The category ($\mathbf{O}$-Mod.) has the structure of an abelian category, defined objectwise. Moreover, ($\mathbf{O}$-Mod.) satisfies axiom* (AB 5) *(see* [Gr57, 1.5]*); that is, arbitrary direct sums exist, and if $\mathbf{M}$ is an $\mathbf{O}$-module, $\mathbf{N}$ a submodule, and $(\mathbf{F}_i)_{i\in I}$ an increasing filtered family of submodules of $\mathbf{M}$, then*

$$\bigcup_{i\in I}(\mathbf{F}_i \cap \mathbf{N}) = \left(\bigcup_{i\in I}\mathbf{F}_i\right) \cap \mathbf{N}.$$

Indeed, let $f : \mathbf{F} \to \mathbf{F}'$ be a morphism of $\mathbf{O}$-modules. Define the $\mathbf{O}$-modules $\operatorname{Ker} f$ (respectively, $\operatorname{Im} f$ and $\operatorname{Coker} f$) by setting, for every $S \in \operatorname{Ob}(\mathscr{C})$,

$$(\operatorname{Ker} f)(S) = \operatorname{Ker} f(S)$$

(respectively, the analogous formulas). Then $\operatorname{Ker} f$ (respectively, $\operatorname{Coker} f$) is a kernel (respectively, a cokernel) of f, and there is an isomorphism of $\mathbf{O}$-modules

$$\mathbf{F}/\operatorname{Ker} f \xrightarrow{\sim} \operatorname{Im} f.$$

This proves that ($\mathbf{O}$-Mod.) is an abelian category.

Arbitrary direct sums exist and are defined objectwise. Finally, if $\mathbf{M}$ is an $\mathbf{O}$-module, $\mathbf{N}$ a submodule, and $(\mathbf{F}_i)_{i\in I}$ an increasing filtered family of submodules of $\mathbf{M}$, then the inclusion

$$\bigcup_{i\in I}(\mathbf{F}_i \cap \mathbf{N}) \subset \left(\bigcup_{i\in I}\mathbf{F}_i\right) \cap \mathbf{N}$$

is an equality. Indeed, if $S \in \operatorname{Ob}(\mathscr{C})$ and

$$x \in \mathbf{N}(S) \cap \bigcup_i \mathbf{F}_i(S),$$

there is an $i \in I$ such that $x \in \mathbf{N}(S) \cap \mathbf{F}_i(S)$.

Proposition 3.1.2.[28] *Suppose that the category $\mathscr{C}$ is small, that is, that* $\operatorname{Ob}(\mathscr{C})$ *is a set. Then* ($\mathbf{O}$*-Mod.*) *has a generator, and consequently has enough injective objects.*

Proof. For every object S of $\mathscr{C}$, define the $\mathbf{O}$-module $\mathbf{O}_S$ as follows. For every object S', $\mathbf{O}_S(S')$ is a direct sum of copies of $\mathbf{O}(S')$ indexed by $\operatorname{Hom}(S', S)$, that is,

$$\mathbf{O}_S(S') = \bigoplus_{g:S'\to S} \mathbf{O}(S')g.$$

For every morphism $f : S'' \to S'$, writing f^* for the ring homomorphism $\mathbf{O}(S') \to \mathbf{O}(S'')$, the morphism

$$\mathbf{O}_S(f) : \mathbf{O}_S(S') \longrightarrow \mathbf{O}_S(S'')$$

sends each element ag of $\mathbf{O}_S(S')$, where $a \in \mathbf{O}(S')$ and $g : S' \to S$, to the element $f^*(a)(g\circ f)$ of $\mathbf{O}_S(S'')$. It is readily checked that this defines a functor in $\mathbf{O}$-modules.

Since $\operatorname{Ob}(\mathscr{C})$ is a set by hypothesis, we may form the direct sum

$$\mathbf{G} = \bigoplus_{S\in\operatorname{Ob}(\mathscr{C})} \mathbf{O}_S.$$

Lemma 3.1.2.1. *Let $\mathbf{F}$ be an $\mathbf{O}$-module. For every $S \in \operatorname{Ob}(\mathscr{C})$, every element $x \in \mathbf{F}(S)$ defines a unique morphism of $\mathbf{O}$-modules*

$$\mathbf{x} : \mathbf{O}_S \longrightarrow \mathbf{F}$$

such that $\mathbf{x}(1\,\mathrm{id}_S) = x$.

Proof. Let $\phi : \mathbf{O}_S \to \mathbf{F}$ be a morphism of $\mathbf{O}$-modules such that $\phi(1\,\mathrm{id}_S) = x$. Let $g : S' \to S$ and $a \in \mathbf{O}(S')$. The $\mathbf{O}(S')$-linearity and the commutativity of the diagram

$$\begin{tikzcd}
\mathbf{O}_S(S) \arrow[r, "\psi"] \arrow[d, "\mathbf{O}_S(g)"'] & \mathbf{F}(S) \arrow[d, "\mathbf{F}(g)"] \\
\mathbf{O}_S(S') \arrow[r, "\phi"] & \mathbf{F}(S')
\end{tikzcd}$$

[28]Editor's note: Statements 3.1.1 and 3.1.2 have been added to show that the category ($\mathbf{O}$-Mod.) is abelian and satisfies axiom (AB 5), and moreover has enough injective objects when the category $\mathscr{C}$ is small. An error in 3.1.2, pointed out in 2016 by Linyuan Liu, is corrected here.

imply that

$$\phi(ag) = a\phi(1g) = a\mathbf{F}(g)\big(\phi(1\,\mathrm{id}_S)\big) = a\mathbf{F}(g)(x).$$

Thus ϕ, if it exists, is determined by the preceding equality. Conversely, if ϕ is defined in this way, then for every $f : S'' \to S'$ one has

$$\begin{aligned}\phi \circ \mathbf{O}_S(f)(ag) &= \phi\big(f^*(a)(g \circ f)\big)\\ &= f^*(a)\mathbf{F}(g \circ f)(x)\\ &= \mathbf{F}(f)\big(a\mathbf{F}(g)(x)\big)\\ &= \mathbf{F}(f) \circ \phi(ag),\end{aligned}$$

that is, the diagram

$$\begin{tikzcd} \mathbf{O}_S(S') \arrow[r, "\phi"] \arrow[d, "\mathbf{O}_S(f)"'] & \mathrm{F}(S') \arrow[d, "\mathbf{F}(f)"] \\ \mathbf{O}_S(S'') \arrow[r, "\phi"] & \mathrm{F}(S'') \end{tikzcd}$$

is commutative. This proves the lemma.

Now let $\mathbf{F}$ be an $\mathbf{O}$-module. For each $S \in \mathrm{Ob}(\mathscr{C})$, let $F_0(S)$ be a set of generators of the $\mathbf{O}(S)$-module $\mathbf{F}(S)$, and let $\mathbf{L}_S$ be the direct sum, indexed by $F_0(S)$, of copies of $\mathbf{O}_S$. By the lemma, we obtain a morphism of $\mathbf{O}$-modules

$$\mathbf{L}_S \longrightarrow \mathbf{F}$$

such that the morphism $\mathbf{L}_S(S) \to \mathbf{F}(S)$ is surjective, and hence is an epimorphism in the category of $\mathbf{O}(S)$-modules.

Set

$$I = \coprod_{S \in \mathrm{Ob}(\mathscr{C})} F_0(S).$$

We then obtain a morphism of $\mathbf{O}$-modules

$$\mathbf{G}^{\oplus I} \longrightarrow \bigoplus_{S \in \mathrm{Ob}(\mathscr{C})} \mathbf{L}_S \longrightarrow \mathbf{F}$$

which is an epimorphism objectwise, and hence an epimorphism of ($\mathbf{O}$-Mod.). This shows that $\mathbf{G}$ is a generator of ($\mathbf{O}$-Mod.) (see [Gr57, 1.9.1]). Since ($\mathbf{O}$-Mod.) satisfies (AB 5), it follows from [Gr57, 1.10.1] that ($\mathbf{O}$-Mod.) has enough injective objects.

Remark 3.1.3. If $\mathbf{O}_0$ is the $\widehat{\mathscr{C}}$-ring defined by

$$\mathbf{O}_0(S) = \mathbb{Z}$$

(which must not be confused with the functor associated with the constant object $\mathbb{Z}$), then the category of $\mathbf{O}_0$-modules is isomorphic to the category of commutative $\widehat{\mathscr{C}}$-groups.

Definition 3.1.4. Notice that if $\mathbf{F}$ is an $\mathbf{O}$-module, then for every $S \in \mathrm{Ob}(\mathscr{C})$, $\mathbf{F}_S$ is an $\mathbf{O}_S$-module. We may therefore define an abelian $\widehat{\mathscr{C}}$-group $\underline{\mathrm{Hom}}_{\mathbf{O}}(\mathbf{F}, \mathbf{F}')$ by

$$\underline{\mathrm{Hom}}_{\mathbf{O}}(\mathbf{F}, \mathbf{F}')(S) = \mathrm{Hom}_{\mathbf{O}_S}(\mathbf{F}_S, \mathbf{F}'_S).$$

In the same way one defines the objects

$$\underline{\mathrm{Isom}}_{\mathbf{O}}(\mathbf{F},\mathbf{F}'), \qquad \underline{\mathrm{End}}_{\mathbf{O}}(\mathbf{F}), \qquad \text{and} \qquad \underline{\mathrm{Aut}}_{\mathbf{O}}(\mathbf{F}),$$

the last of which is a $\widehat{\mathscr{C}}$-group under the structure induced by composition of automorphisms.

Definition 3.2. Let $\mathbf{O}$ be a $\widehat{\mathscr{C}}$-ring, $\mathbf{F}$ an $\mathbf{O}$-module, and $\mathbf{G}$ a $\widehat{\mathscr{C}}$-group. A $\mathbf{G}$-$\mathbf{O}$-module structure on $\mathbf{F}$ is a $\mathbf{G}$-object structure such that, for every $S \in \mathrm{Ob}(\mathscr{C})$ and every $g \in \mathbf{G}(S)$, the automorphism of $\mathbf{F}(S)$ defined by g is an automorphism of its $\mathbf{O}(S)$-module structure.

Equivalently, the morphism of $\widehat{\mathscr{C}}$-groups

$$\rho : \mathbf{G} \longrightarrow \underline{\mathrm{Aut}}(\mathbf{F})$$

defined in 2.3 maps $\mathbf{G}$ into the $\widehat{\mathscr{C}}$-subgroup $\underline{\mathrm{Aut}}_{\mathbf{O}}(\mathbf{F})$ of $\underline{\mathrm{Aut}}(\mathbf{F})$. Thus, giving a $\mathbf{G}$-$\mathbf{O}$-module structure on the $\mathbf{O}$-module $\mathbf{F}$ is the same as giving a morphism of $\widehat{\mathscr{C}}$-groups

$$\rho : \mathbf{G} \longrightarrow \underline{\mathrm{Aut}}_{\mathbf{O}}(\mathbf{F}).$$

One defines naturally the abelian group $\mathrm{Hom}_{\mathbf{G}\text{-}\mathbf{O}}(\mathbf{F},\mathbf{F}')$, and hence the additive category of $\mathbf{G}$-$\mathbf{O}$-modules, denoted ($\mathbf{G}$-$\mathbf{O}$-Mod.).

Remark 3.2.1. The reader will notice that ($\mathbf{G}$-$\mathbf{O}$-Mod.) may also be defined as the category of $\mathbf{O}[\mathbf{G}]$-modules, where $\mathbf{O}[\mathbf{G}]$ is the algebra of the $\widehat{\mathscr{C}}$-group $\mathbf{G}$ over the $\widehat{\mathscr{C}}$-ring $\mathbf{O}$, whose definition is clear. [29] Consequently, by 3.1.1 and 3.1.2, ($\mathbf{G}$-$\mathbf{O}$-Mod.) is an abelian category satisfying axiom (AB 5); moreover, if $\mathscr{C}$ is small, ($\mathbf{G}$-$\mathbf{O}$-Mod.) has enough injective objects.

All the constructions of this section specialize immediately to the case where $\mathbf{G}$ (or $\mathbf{O}$, or both) is representable by an object of $\mathscr{C}$, which is thereby endowed with the corresponding algebraic structure.

We have treated briefly the principal algebraic structures that occur later in this seminar. For the others (the structure of an $\mathbf{O}$-Lie algebra, for example), we believe that the reader has now seen enough examples in this section to make the general mechanism sketched in 2.2 work in each particular case.

We shall now apply what we have just done to the category of schemes, denoted (**Sch**), and more generally to the categories $(\mathbf{Sch})_{/S}$, also denoted $(\mathbf{Sch}_{/S})$.

4. Algebraic structures in the category of schemes

Whenever there is no ambiguity, we shall permit ourselves the following abuses of language. We shall denote by T the object $T \xrightarrow{f} S$ of $\mathscr{C}_{/S}$, with the datum of f (the structural morphism of T) understood, and we shall identify $\mathscr{C}$ with a subcategory of $\widehat{\mathscr{C}}$. In view of the compatibilities stated in the preceding sections, these identifications may be made without danger.

We shall also simplify terminology according to the following pattern: a (**Sch**)-group will also be called a *group scheme*; a $(\mathbf{Sch})_{/S}$-group will be called a *group scheme over S*, an *S-group scheme*, or an *S-group*; and it will be called an *A-group* when S is the spectrum of the ring A.

[29] Editor's note: the sentence which follows has been added; it will be used in Section 5.

4.1. Constant schemes

The category of schemes satisfies the conditions required in 1.8. Constant objects may therefore be defined in it. For every set E, there is a scheme $E_{\mathbb{Z}}$, and for every scheme S, an S-scheme

$$E_S = (E_{\mathbb{Z}})_S.$$

Recall (cf. *loc. cit.*) that for every S-scheme T,

$$\mathrm{Hom}_S(T, E_S)$$

is the set of locally constant maps from the topological space underlying T to E.

The functor $E \mapsto E_S$ commutes with finite inverse limits. In particular, if G is a group, then G_S is an S-group scheme; if O is a ring, then O_S is an S-ring scheme; and similarly for other algebraic structures.

4.2. Affine S-groups

Let us recall some facts about affine S-schemes (EGA II, Section 1). An S-scheme T is said to be *affine over* S if the inverse image of every affine open subset of S is affine. The $\mathcal{O}_S$-algebra $f_*\mathcal{O}_T$, denoted by $\mathscr{A}(T)$, is then quasi-coherent, where f denotes the structural morphism of T. Conversely, to every quasi-coherent $\mathcal{O}_S$-algebra $\mathscr{A}$, one can associate an S-scheme affine over S, denoted $\mathrm{Spec}(\mathscr{A})$. The functors

$$T \longmapsto \mathscr{A}(T), \qquad \mathscr{A} \longmapsto \mathrm{Spec}(\mathscr{A})$$

are quasi-inverse equivalences between the category of S-schemes affine over S and the category opposite to that of quasi-coherent $\mathcal{O}_S$-algebras.

It follows that giving an algebraic structure on an S-scheme T affine over S is equivalent to giving the corresponding structure on $\mathscr{A}(T)$ in the category opposite to that of quasi-coherent $\mathcal{O}_S$-algebras. In particular, if G is an S-group affine over S, then $\mathscr{A}(G)$ is endowed with the structure of an augmented $\mathcal{O}_S$-bialgebra: there are two morphisms of $\mathcal{O}_S$-algebras

$$\Delta : \mathscr{A}(G) \longrightarrow \mathscr{A}(G) \otimes_{\mathcal{O}_S} \mathscr{A}(G), \qquad \varepsilon : \mathscr{A}(G) \longrightarrow \mathcal{O}_S,$$

corresponding to the morphisms of S-schemes

$$\pi : G \times G \longrightarrow G, \qquad e_G : S \longrightarrow G.$$

The maps Δ and ε satisfy the following conditions, which express that G is an S-monoid. [30]

(HA 1) Δ is coassociative: the following diagram is commutative.

[30] Editor's note: of course, the inversion morphism $G \to G$ induces a morphism of $\mathcal{O}_S$-algebras $\tau : \mathscr{A}(G) \to \mathscr{A}(G)$ which, together with Δ and ε, makes $\mathscr{A}(G)$ into a Hopf $\mathcal{O}_S$-algebra.

$$\begin{tikzcd}
 & \mathscr{A}(\mathrm{G}) \otimes_{\mathscr{O}_\mathrm{S}} \mathscr{A}(\mathrm{G}) \arrow[dr, "\mathrm{Id} \otimes \Delta"] & \\
\mathscr{A}(\mathrm{G}) \arrow[ur, "\Delta"] \arrow[dr, "\Delta"'] & & \mathscr{A}(\mathrm{G}) \otimes_{\mathscr{O}_\mathrm{S}} \mathscr{A}(\mathrm{G}) \otimes_{\mathscr{O}_\mathrm{S}} \mathscr{A}(\mathrm{G}) \\
 & \mathscr{A}(\mathrm{G}) \otimes_{\mathscr{O}_\mathrm{S}} \mathscr{A}(\mathrm{G}) \arrow[ur, "\Delta \otimes \mathrm{Id}"'] & .
\end{tikzcd}$$

Figure 1: Coassociativity of the diagonal map Δ.

Temporary Loop-1 image; source locator: Polo–Gille Exp1-13oct24.pdf, component-local p. 22; combined-reader p. 32.

(HA 2) Δ is compatible with ε: the following two composites are the identity:

$$\mathscr{A}(G) \xrightarrow{\Delta} \mathscr{A}(G) \otimes_{\mathcal{O}_S} \mathscr{A}(G) \xrightarrow{\mathrm{Id} \otimes \varepsilon} \mathscr{A}(G) \otimes_{\mathcal{O}_S} \mathcal{O}_S \xrightarrow{\sim} \mathscr{A}(G),$$

$$\mathscr{A}(G) \xrightarrow{\Delta} \mathscr{A}(G) \otimes_{\mathcal{O}_S} \mathscr{A}(G) \xrightarrow{\varepsilon \otimes \mathrm{Id}} \mathcal{O}_S \otimes_{\mathcal{O}_S} \mathscr{A}(G) \xrightarrow{\sim} \mathscr{A}(G).$$

Let us take this opportunity to note once again that it follows from the definition of an S-group structure that, in order to give such a structure on an S-scheme G affine over S, it is not necessary to verify anything on $\mathscr{A}(G)$; one need only endow every $G(S')$, for S' over S, with a group structure functorial in S'. This observation applies *mutatis mutandis* to morphisms.

4.3. The groups $\mathbb{G}_a$ and $\mathbb{G}_m$. The ring O

4.3.1.

Let $\mathbb{G}_a$ be the functor from $(\mathbf{Sch})^\circ$ to $\mathbf{Set}$ defined by

$$\mathbb{G}_a(S) = \Gamma(S, \mathcal{O}_S),$$

endowed with the $\widehat{(\mathbf{Sch})}$-group structure defined by the additive-group structure of the ring $\Gamma(S, \mathcal{O}_S)$. It is representable by an affine scheme, which we denote by $\mathbb{G}_a$, and which is therefore a group scheme:

$$\mathbb{G}_a = \operatorname{Spec} \mathbb{Z}[T].$$

Indeed,

$$\mathbb{G}_a(S) = \operatorname{Hom}(S, \mathbb{G}_a) = \operatorname{Hom}_{\mathrm{Alg.}}\big(\mathbb{Z}[T], \Gamma(S, \mathcal{O}_S)\big) \simeq \Gamma(S, \mathcal{O}_S).$$

For every scheme S, one therefore has an S-group affine over S, denoted $\mathbb{G}_{a,S}$, corresponding to the $\mathcal{O}_S$-bialgebra $\mathcal{O}_S[T]$, whose diagonal map and augmentation are defined by

$$\Delta(T) = T \otimes 1 + 1 \otimes T, \qquad \varepsilon(T) = 0.$$

4.3.2.

Let $\mathbb{G}_m$ be the functor from $(\mathbf{Sch})^\circ$ to $\mathbf{Set}$ defined by

$$\mathbb{G}_m(S) = \Gamma(S, \mathcal{O}_S)^\times,$$

where $\Gamma(S,\mathcal{O}_S)^\times$ denotes the multiplicative group of invertible elements of the ring $\Gamma(S,\mathcal{O}_S)$, endowed with its natural $\widehat{(\mathbf{Sch})}$-group structure. It is representable by an affine scheme denoted by $\mathbb{G}_m$:

$$\mathbb{G}_m = \operatorname{Spec}\mathbb{Z}[T,T^{-1}] = \operatorname{Spec}\mathbb{Z}[\mathbb{Z}],$$

where $\mathbb{Z}[\mathbb{Z}]$ denotes the group algebra of $\mathbb{Z}$ over the ring $\mathbb{Z}$. Indeed,

$$\mathbb{G}_m(S) = \operatorname{Hom}_{\mathrm{Alg.}}\big(\mathbb{Z}[T,T^{-1}],\Gamma(S,\mathcal{O}_S)\big) \simeq \Gamma(S,\mathcal{O}_S)^\times.$$

For every scheme S, one therefore has an S-group affine over S, denoted $\mathbb{G}_{m,S}$, corresponding to the $\mathcal{O}_S$-algebra $\mathcal{O}_S[\mathbb{Z}]$, whose diagonal map and augmentation are defined by

$$\Delta(x) = x\otimes x, \qquad \varepsilon(x) = 1, \qquad x\in\mathbb{Z}.$$

4.3.3.

Let $\mathbf{O}$ be the functor $\mathbb{G}_a$ endowed with its $\widehat{(\mathbf{Sch})}$-ring structure. It is represented by the scheme $\operatorname{Spec}\mathbb{Z}[T]$, which we denote by $\mathbf{O}$ when it is regarded as endowed with its ring-scheme structure.

For every scheme S,

$$\mathbf{O}_S = S\times_{\operatorname{Spec}\mathbb{Z}}\operatorname{Spec}\mathbb{Z}[T] = \operatorname{Spec}(\mathcal{O}_S[T])$$

is therefore an S-ring scheme affine over S. (Note: in EGA II, 1.7.13, $\mathbf{O}_S$ is denoted by $S[T]$.)

4.3.3.1. [31] For every object X of $\widehat{(\mathbf{Sch})}$,

$$\mathbf{O}(X) = \operatorname{Hom}(X,\mathbf{O})$$

has a ring structure, functorial in X. In particular, if X' is a scheme and morphisms $x : X'\to X$ and $f : X\to\mathbf{O}$ are given (that is, $f\in\mathbf{O}(X)$), then

$$f(x) = f\circ x\in\mathbf{O}(X') = \Gamma(X',\mathcal{O}_{X'}).$$

Definition. Let $\pi : M\to X$ be a morphism in $\widehat{(\mathbf{Sch})}$, and set

$$\mathbf{O}_X = \mathbf{O}\times X.$$

We say that M is an $\mathbf{O}_X$-module if, for every X-scheme X', the set

$$\operatorname{Hom}_X(X',M)$$

has been endowed with an $\mathbf{O}(X')$-module structure, functorial in X'.

This is equivalent to giving an X-abelian-group law

$$\mu : M\times_X M\longrightarrow M$$

and an *external law*

$$\mathbf{O}\times M = \mathbf{O}_X\times_X M\longrightarrow M, \qquad (f,m)\longmapsto f\cdot m,$$

which is an X-morphism, that is, $\pi(f\cdot m) = \pi(m)$, and which, for every $x\in X(S)$, endows

$$M(x) = \{m\in M(S)\mid \pi(m) = x\}$$

with an $\mathbf{O}(S)$-module structure.

In that case, for every $Y\in\operatorname{Ob}\big(\widehat{(\mathbf{Sch})}_{/X}\big)$, not necessarily representable,

$$\operatorname{Hom}_X(Y,M) = \Gamma(M_Y/Y)$$

is an $\mathbf{O}(Y)$-module, functorially in Y.

[31] Editor's note: this paragraph has been added; it will be useful later (cf. II, 1.3).

4.4. Diagonalizable groups

4.4.1. The construction of $\mathbb{G}_m$ generalizes as follows. Let M be a commutative group, and let $M_{\mathbb{Z}}$ be the group scheme associated with it by the procedure of 4.1. Consider the functor defined by

$$\mathbf{D}(M)(S) = \mathrm{Hom}_{\mathrm{groups}}(M, \mathbb{G}_m(S)) \simeq \underline{\mathrm{Hom}}_{S\text{-}\mathbf{Gr}}(M_S, \mathbb{G}_{m,S}).$$

This is a commutative $\widehat{(\mathbf{Sch})}$-group represented by a group scheme denoted $D(M)$; thus, by definition,

$$D(M) \simeq \underline{\mathrm{Hom}}_{(\mathbf{Sch})\text{-}\mathbf{Gr}}(M_{\mathbb{Z}}, \mathbb{G}_m).$$

Indeed, set

$$D(M) = \operatorname{Spec} \mathbb{Z}[M],$$

where $\mathbb{Z}[M]$ is the group algebra of M over $\mathbb{Z}$. Then

$$D(M)(S) = \mathrm{Hom}_{\mathrm{Alg.}}\big(\mathbb{Z}[M], \Gamma(S, \mathcal{O}_S)\big) \simeq \mathrm{Hom}_{\mathrm{groups}}\big(M, \Gamma(S, \mathcal{O}_S)^{\times}\big)$$

by the very definition of the algebra $\mathbb{Z}[M]$.

4.4.2. For every scheme S, one therefore has an S-group scheme affine over S,

$$D_S(M) = D(M)_S = \big(\underline{\mathrm{Hom}}_{(\mathbf{Sch})\text{-}\mathbf{Gr}}(M_{\mathbb{Z}}, \mathbb{G}_m)\big)_S = \underline{\mathrm{Hom}}_{S\text{-}\mathbf{Gr}}(M_S, \mathbb{G}_{m,S}).$$

It is associated with the $\mathcal{O}_S$-bialgebra $\mathcal{O}_S[M]$, endowed with the diagonal map and augmentation defined by

$$\Delta(x) = x \otimes x, \qquad \varepsilon(x) = 1, \qquad x \in M.$$

4.4.3. If $f : M \to N$ is a homomorphism of commutative groups, one defines in the evident way a morphism of S-groups

$$D_S(f) : D_S(N) \longrightarrow D_S(M),$$

and hence a functor

$$D_S : M \longmapsto D_S(M)$$

from the category of abelian groups to the category of S-groups affine over S. It may also be defined as the composite of the functor $M \mapsto M_S$ and the functor

$$M_S \longmapsto \underline{\mathrm{Hom}}_{S\text{-}\mathbf{Gr}}(M_S, \mathbb{G}_{m,S}).$$

This functor commutes with base extension.

An S-group isomorphic to a group of the form $D_S(M)$ is said to be *diagonalizable.* Notice that the elements of M are interpreted as certain characters of $D_S(M)$, that is, as certain elements of

$$\mathrm{Hom}_{S\text{-}\mathbf{Gr}}(D_S(M), \mathbb{G}_{m,S}).$$

(Compare VIII, 1.)

4.4.4. Let us give some examples of diagonalizable groups. First,

$$D(\mathbb{Z}) = \mathbb{G}_m, \qquad D(\mathbb{Z}^r) = (\mathbb{G}_m)^r.$$

Set

$$\boldsymbol{\mu}_n = D(\mathbb{Z}/n\mathbb{Z}),$$

and call it the *group of n-th roots of unity.* Indeed,

$$\boldsymbol{\mu}_n(S) = \mathrm{Hom}_{\mathrm{groups}}\big(\mathbb{Z}/n\mathbb{Z}, \Gamma(S, \mathcal{O}_S)^{\times}\big) = \{f \in \Gamma(S, \mathcal{O}_S) \mid f^n = 1\}.$$

The S-group $\boldsymbol{\mu}_{n,S}$ corresponds to the $\mathcal{O}_S$-algebra $\mathcal{O}_S[T]/(T^n - 1)$. Suppose, in particular, that S is the spectrum of a field k of characteristic $p = n$. On setting $T - 1 = s$, one obtains

$$k[T]/(T^p - 1) = k[s]/(s^p),$$

which shows that the underlying topological space of $\boldsymbol{\mu}_{p,k}$ consists of one point, whose local ring is the Artinian k-algebra $k[s]/(s^p)$. In the same vein, note that the S-schemes $\mathbb{G}_{a,S}$, $\mathbb{G}_{m,S}$, and $\mathbf{O}_S$ are smooth over S, and that $D_S(M)$ is

flat over S. Moreover, it is formally smooth (respectively, smooth) over S if and only if no residue characteristic of S divides the torsion of M (respectively, and if in addition M is of finite type); see Exposé VIII, 2.1.

4.5. Other examples of groups

The preceding procedure applies to the "classical groups" (linear groups GL_n, symplectic groups Sp_n, orthogonal groups O_n, and so forth). For example, one defines GL_n as representing the functor $\mathbf{GL}_n$ such that

$$\mathbf{GL}_n(S) = \mathrm{GL}\big(n, \Gamma(S, \mathcal{O}_S)\big) = \mathrm{Aut}_{\mathcal{O}_S}(\mathcal{O}_S^n).$$

It may be constructed, for example, as the open subscheme of $\mathrm{Spec}\,\mathbb{Z}[T_{ij}]$, $1 \le i, j \le n$, defined by the function

$$f = \det\big((T_{ij})_{i,j=1}^n\big),$$

that is, as $\mathrm{Spec}\,\mathbb{Z}[T_{ij}, f^{-1}]$.

4.6. Functor-modules in the category of schemes

We shall associate with every $\mathcal{O}_S$-module on a scheme S a $\mathbf{O}_S$-module, where $\mathbf{O}_S$ denotes the functor-ring introduced in 4.3.3. This can be done in two different ways, as follows.

Definition 4.6.1. Let S be a scheme. For every $\mathcal{O}_S$-module $\mathscr{F}$, denote by $\mathbf{V}(\mathscr{F})$ and $\mathbf{W}(\mathscr{F})$ the contravariant functors on $(\mathbf{Sch})_{/S}$ defined by

$$\begin{aligned} \mathbf{V}(\mathscr{F})(S') &= \mathrm{Hom}_{\mathcal{O}_{S'}}(\mathscr{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}, \mathcal{O}_{S'}), \\ \mathbf{W}(\mathscr{F})(S') &= \Gamma(S', \mathscr{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}), \end{aligned}$$

where $\mathscr{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$ denotes the inverse image on S' of the $\mathcal{O}_S$-module $\mathscr{F}$.

Then $\mathbf{V}(\mathscr{F})$ and $\mathbf{W}(\mathscr{F})$ are endowed in the evident way with $\mathbf{O}_S$-module structures (recall that $\mathbf{O}_S(S') = \Gamma(S', \mathcal{O}_{S'}) = \mathbf{W}(\mathcal{O}_S)(S')$). Thus one has in fact defined two functors $\mathbf{V}$ and $\mathbf{W}$ from the category of $\mathcal{O}_S$-modules to the category of $\mathbf{O}_S$-modules, with $\mathbf{V}$ contravariant and $\mathbf{W}$ covariant.

For the remainder of this paragraph, we restrict to the case in which the $\mathcal{O}_S$-modules in question are quasi-coherent. Thus we regard $\mathbf{V}$ and $\mathbf{W}$ as functors

$$\mathbf{V} : (\mathcal{O}_S\text{-}\mathbf{Mod.q.c.})^{\circ} \longrightarrow (\mathbf{O}_S\text{-}\mathbf{Mod.}), \qquad \mathbf{W} : (\mathcal{O}_S\text{-}\mathbf{Mod.q.c.}) \longrightarrow (\mathbf{O}_S\text{-}\mathbf{Mod.}).$$

Remark 4.6.1.1. [32] The reader will notice that, in what follows, every proposition involving only the functor $\mathbf{W}$ remains valid for arbitrary $\mathcal{O}_S$-modules, not necessarily quasi-coherent.

Proposition 4.6.2.

[32]Editor's note: the numbering 4.6.1.1 has been added for later references.

(i) The functors $\mathbf{V}$ and $\mathbf{W}$ commute with base extension: if S' lies over S and $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_S$-module, then

$$\mathbf{V}(\mathscr{F} \otimes \mathscr{O}_{S'}) \simeq \mathbf{V}(\mathscr{F})_{S'}, \qquad \mathbf{W}(\mathscr{F} \otimes \mathscr{O}_{S'}) \simeq \mathbf{W}(\mathscr{F})_{S'}.$$

(ii) The functors $\mathbf{V}$ and $\mathbf{W}$ are fully faithful: the canonical maps

$$\mathrm{Hom}_{\mathscr{O}_S}(\mathscr{F}, \mathscr{F}') \longrightarrow \mathrm{Hom}_{\mathbf{O}_S}(\mathbf{V}(\mathscr{F}'), \mathbf{V}(\mathscr{F})),$$

$$\mathrm{Hom}_{\mathscr{O}_S}(\mathscr{F}, \mathscr{F}') \longrightarrow \mathrm{Hom}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}), \mathbf{W}(\mathscr{F}'))$$

are bijective.

(iii) The functors $\mathbf{V}$ and $\mathbf{W}$ are additive:

$$\mathbf{V}(\mathscr{F} \oplus \mathscr{F}') \simeq \mathbf{V}(\mathscr{F}) \times_S \mathbf{V}(\mathscr{F}'), \qquad \mathbf{W}(\mathscr{F} \oplus \mathscr{F}') \simeq \mathbf{W}(\mathscr{F}) \times_S \mathbf{W}(\mathscr{F}').$$

Parts (i) and (iii) are immediate from the definitions. For (ii), take the S' to be open subschemes of S. We leave the proof to the reader (for $\mathbf{V}$, use EGA II, 1.7.14).

Recall (cf. 3.1.4) that if $\mathbf{F}$ and $\mathbf{F}'$ are $\mathbf{O}_S$-modules, then $\underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{F}, \mathbf{F}')$ denotes the S-functor in abelian groups that assigns to every $S' \to S$ the group

$$\mathrm{Hom}_{\mathbf{O}_{S'}}(\mathbf{F}_{S'}, \mathbf{F}'_{S'}).$$

Proposition 4.6.3. *There are canonical morphisms in* $(\mathbf{O}_S\text{-}\mathbf{Mod}.)$*:* [33]

$$\begin{tikzcd}
\underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}), \mathbf{W}(\mathscr{F}')) \arrow[rr, equal] & & \underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{V}(\mathscr{F}'), \mathbf{V}(\mathscr{F})) \\
& \mathbf{W}(\mathscr{H}om_{\mathscr{O}_S}(\mathscr{F}, \mathscr{F}')) \arrow[ul] \arrow[ur] &
\end{tikzcd}.$$

Figure 2: Canonical morphisms of Proposition 4.6.3.

Temporary Loop-1 image; source locator: Polo–Gille Exp1-13oct24.pdf, component-local p. 26; combined-reader p. 36.

This follows immediately from 4.6.2(i) and (ii).

Notation 4.6.3.1. Let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_S$-module. By EGA II, 1.7.8, the S-functor $\mathbf{V}(\mathscr{F})$ is represented by an S-scheme affine over S, denoted $\mathbb{V}(\mathscr{F})$, and called the *vector fibration* [34] defined by $\mathscr{F}$:

$$\mathbb{V}(\mathscr{F}) = \mathrm{Spec}(\mathcal{S}(\mathscr{F})),$$

where $\mathcal{S}(\mathscr{F})$ denotes the symmetric algebra of the $\mathscr{O}_S$-module $\mathscr{F}$. [35]

Proposition 4.6.4. *Let $\mathscr{F}$ and $\mathscr{F}'$ be two quasi-coherent $\mathscr{O}_S$-modules, and let $\mathscr{A}$ be a quasi-coherent $\mathscr{O}_S$-algebra. There is a functorial isomorphism*

$$\mathrm{Hom}_S\big(\mathrm{Spec}(\mathscr{A}), \underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}'), \mathbf{W}(\mathscr{F}))\big) \xrightarrow{\sim} \mathrm{Hom}_{\mathscr{O}_S}(\mathscr{F}', \mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{A}).$$

[33] Editor's note: the isomorphism $\underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}), \mathbf{W}(\mathscr{F}')) \simeq \underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{V}(\mathscr{F}'), \mathbf{V}(\mathscr{F}))$ has been added.

[34] Editor's note: "vector bundle" has been replaced by "vector fibration." Current usage calls a vector fibration that is locally trivial of rank r a "vector bundle of rank r," that is, one whose sheaf of sections is locally isomorphic to $\mathscr{O}_S^{\oplus r}$.

[35] Editor's note: the articles [Ni04] and [Ni02], respectively, show that if S is Noetherian and $\mathscr{F}$ is a coherent $\mathscr{O}_S$-module, then $\mathbf{W}(\mathscr{F})$, respectively the S-group assigning $\mathrm{Aut}_{\mathscr{O}_T}(\mathscr{F} \otimes \mathscr{O}_T)$ to every $T \to S$, is representable if and only if $\mathscr{F}$ is locally free.

Indeed, put $X = \mathrm{Spec}(\mathscr{A})$. The left-hand side is canonically isomorphic to

$$\underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}'), \mathbf{W}(\mathscr{F}))(X),$$

that is, by definition, to

$$\begin{aligned} &\mathrm{Hom}_{\mathbf{O}_X}(\mathbf{W}(\mathscr{F}')_X, \mathbf{W}(\mathscr{F})_X) \\ &\qquad \simeq \mathrm{Hom}_{\mathbf{O}_X}\big(\mathbf{W}(\mathscr{F}' \otimes \mathcal{O}_X), \mathbf{W}(\mathscr{F} \otimes \mathcal{O}_X)\big) \end{aligned}$$

(cf. 4.6.2(i)). By 4.6.2(ii), this may also be written as

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}' \otimes \mathcal{O}_X, \mathscr{F} \otimes \mathcal{O}_X) = \mathrm{Hom}_{\mathcal{O}_S}(\mathscr{F}', \pi_*\pi^*(\mathscr{F})),$$

where $\pi : X \to S$ is the structural morphism. But EGA II, 1.4.7 gives

$$\pi_*\pi^*(\mathscr{F}) \simeq \mathscr{F} \otimes \mathscr{A},$$

which completes the proof.

Corollary 4.6.4.1. *There is a canonical isomorphism*

$$\mathbf{W}(\mathscr{F} \otimes \mathscr{A}) \simeq \underline{\mathrm{Hom}}_S(\mathrm{Spec}(\mathscr{A}), \mathbf{W}(\mathscr{F})).$$

Indeed, [36] let $f : S' \to S$ be an S-scheme and put $X' = X \times_S S'$. There is a Cartesian square

$$\begin{tikzcd} X' \arrow[r, "f'"] \arrow[d, "\pi'"'] & X \arrow[d, "\pi"] \\ S' \arrow[r, "f"] & S \end{tikzcd}$$

Figure 3: The Cartesian base-change square used in Corollary 4.6.4.1.

Temporary Loop-1 image; source locator: Polo–Gille Exp1-13oct24.pdf, component-local p. 27; combined-reader p. 37.

By EGA II, 1.5.2, X' is affine over S' and $\pi'_*\mathcal{O}_{X'} = f^*(\mathscr{A})$. Hence

$$\begin{aligned} &\underline{\mathrm{Hom}}_S(\mathrm{Spec}(\mathscr{A}), \mathbf{W}(\mathscr{F}))(S') \\ &\qquad = \mathrm{Hom}_{S'}\big(\mathrm{Spec}(f^*\mathscr{A}), \mathbf{W}(f^*\mathscr{F})\big). \end{aligned}$$

By 4.6.4, applied to $f^*\mathscr{F}$, $\mathscr{F}' = \mathcal{O}_{S'}$, and $f^*\mathscr{A}$, this is equal to

$$\begin{aligned} \Gamma(S', f^*\mathscr{F} \otimes f^*\mathscr{A}) &= \Gamma(S', f^*(\mathscr{F} \otimes \mathscr{A})) \\ &= \mathbf{W}(\mathscr{F} \otimes \mathscr{A})(S'). \end{aligned}$$

Proposition 4.6.5. *If $\mathscr{F}$ and $\mathscr{F}'$ are two $\mathcal{O}_S$-modules locally free of finite type, the morphisms of 4.6.3 are isomorphisms.*

Indeed, for every $S' \to S$,

$$\begin{aligned} &\mathbf{W}\big(\mathscr{H}om_{\mathcal{O}_S}(\mathscr{F}, \mathscr{F}')\big)(S') \\ &\quad = \Gamma\big(S', \mathscr{H}om_{\mathcal{O}_S}(\mathscr{F}, \mathscr{F}') \otimes \mathcal{O}_{S'}\big) \\ &\quad = \mathrm{Hom}_{\mathcal{O}_{S'}}(\mathscr{F} \otimes \mathcal{O}_{S'}, \mathscr{F}' \otimes \mathcal{O}_{S'}). \end{aligned}$$

[36] Editor's note: the original has been expanded in what follows.

The last member is isomorphic both to

$$\underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}), \mathbf{W}(\mathscr{F}'))(S')$$

and to

$$\underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{V}(\mathscr{F}'), \mathbf{V}(\mathscr{F}))(S')$$

by 4.6.2(i) and (ii).

Corollary 4.6.5.1. *Let $\mathscr{F}$ be an $\mathcal{O}_S$-module locally free of finite type. Put $\mathscr{F}^\vee = \mathscr{H}om_{\mathcal{O}_S}(\mathscr{F}, \mathcal{O}_S)$. There are canonical isomorphisms*

$$\mathbf{W}(\mathscr{F}^\vee) \simeq \underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}), \mathbf{O}_S) \simeq \mathbf{V}(\mathscr{F}),$$

$$\mathbf{V}(\mathscr{F}^\vee) \simeq \underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{V}(\mathscr{F}), \mathbf{O}_S) \simeq \mathbf{W}(\mathscr{F}).$$

Finally, we have the following proposition.

Proposition 4.6.6. *Let $f : \mathscr{F} \to \mathscr{F}'$ be a morphism of $\mathcal{O}_S$-modules locally free of finite rank. The morphism*

$$\mathbf{W}(f) : \mathbf{W}(\mathscr{F}) \longrightarrow \mathbf{W}(\mathscr{F}')$$

is a monomorphism if and only if f identifies $\mathscr{F}$ locally with a direct summand of $\mathscr{F}'$.

The direct assertion is essentially contained in EGA 0_{I}, 5.5.5. [37] Conversely, if $\mathscr{F}$ is locally a direct summand of $\mathscr{F}'$, then for every $\pi : S' \to S$, the module $\pi^*\mathscr{F}$ is a submodule of $\pi^*\mathscr{F}'$, since this holds locally as a direct summand. Thus

$$\mathbf{W}(\mathscr{F})(S') = \Gamma(S', \pi^*\mathscr{F})$$

is a submodule of

$$\mathbf{W}(\mathscr{F}')(S') = \Gamma(S', \pi^*\mathscr{F}').$$

4.7. The category of G-$\mathcal{O}_S$-modules

Let G be an S-group and $\mathscr{F}$ an $\mathcal{O}_S$-module; $\mathbf{W}(\mathscr{F})$ has a $\mathbf{O}_S$-module structure.

Definition 4.7.1. A *G-$\mathcal{O}_S$-module structure* on $\mathscr{F}$ is an $\mathbf{h}_G$-$\mathbf{O}_S$-module structure on $\mathbf{W}(\mathscr{F})$ (cf. 3.2). A morphism of G-$\mathcal{O}_S$-modules is, by definition, a morphism of the associated $\mathbf{h}_G$-$\mathbf{O}_S$-modules. This gives the category (G-$\mathcal{O}_S$-**Mod**.); we denote by (G-$\mathcal{O}_S$-**Mod.q.c.**) its full subcategory consisting of those G-$\mathcal{O}_S$-modules that are quasi-coherent as $\mathcal{O}_S$-modules.

By 3.2, giving a G-$\mathcal{O}_S$-module structure on $\mathscr{F}$ is therefore equivalent to giving a morphism of $\widehat{(\mathbf{Sch})}_{/S}$-groups

$$\rho : \mathbf{h}_G \longrightarrow \underline{\mathrm{Aut}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F})).$$

Remark 4.7.1.0. [38] Since 4.6.2 gives an anti-isomorphism of S-functors in groups

$$(\dagger) \qquad \underline{\mathrm{Aut}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F})) \simeq \underline{\mathrm{Aut}}_{\mathbf{O}_S}(\mathbf{V}(\mathscr{F})),$$

it is equivalent to give an $\mathbf{h}_G$-$\mathbf{O}_S$-module structure on $\mathbf{W}(\mathscr{F})$ or on $\mathbf{V}(\mathscr{F})$. Indeed, let

$$\rho : \mathbf{h}_G \longrightarrow \underline{\mathrm{Aut}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}))$$

be as above. For every $T \to S$ and $g \in G(T)$, denote by $\rho^*(g)$ the image of $\rho(g)$ under the anti-isomorphism (†). Then

$$\rho^*(gh) = \rho^*(h) \circ \rho^*(g),$$

[37] Editor's note: the following sentence has been expanded.
[38] Editor's note: this remark has been added from [DG], II, Section 2, 1.1.

so ρ^* defines a *right* $\mathbf{h}_G$-$\mathbf{O}_S$-module structure on $\mathbf{V}(\mathscr{F})$. Setting

$$\rho^\vee(g) = \rho^*(g^{-1}),$$

one obtains an $\mathbf{h}_G$-$\mathbf{O}_S$-module structure on $\mathbf{V}(\mathscr{F})$, whose datum is equivalent to that of ρ.

Remark 4.7.1.1. One may say that the categories just defined have been constructed by the following Cartesian squares.

$$\begin{tikzcd}
(\mathrm{G}\text{-}\mathscr{O}_{\mathrm{S}}\text{-Mod.q.c.}) \arrow[r, hook] \arrow[d] & (\mathrm{G}\text{-}\mathscr{O}_{\mathrm{S}}\text{-Mod.}) \arrow[r] \arrow[d] & (\mathbf{h}_{\mathrm{G}}\text{-}\mathbf{O}_{\mathrm{S}}\text{-Mod.}) \arrow[d, "\text{oubli}"] \\
(\mathscr{O}_{\mathrm{S}}\text{-Mod.q.c.}) \arrow[r, hook] & (\mathscr{O}_{\mathrm{S}}\text{-Mod.}) \arrow[r, "\mathbf{W}"] & (\mathbf{O}_{\mathrm{S}}\text{-Mod.}).
\end{tikzcd}$$

Figure 4: Cartesian squares defining the categories of G-$\mathcal{O}_S$-modules.

Temporary Loop-1 image; source locator: Polo–Gille Exp1-13oct24.pdf, component-local p. 28; combined-reader p. 38.

The categories ($\mathcal{O}_S$-**Mod.**) and ($\mathbf{O}_S$-**Mod.**) are abelian, but one should remember that in general the functor $\mathbf{W}$ is neither left nor right exact. [39]

Remark 4.7.1.2. [40] Let $\mathscr{F}$ be a G-$\mathcal{O}_S$-module. The subsheaf of invariants $\mathscr{F}^G$ is defined as follows: for every open subset U of S,

$$\mathscr{F}^G(U) = \mathbf{W}(\mathscr{F})^G(U) = \big\{x \in \mathscr{F}(U) \mid g \cdot x_{S'} = x_{S'} \text{ for every } S' \xrightarrow{f} U,\ g \in G(S')\big\},$$

where $x_{S'}$ denotes the image of x in

$$\Gamma(S', f^*\mathscr{F}) = \Gamma(U, f_*f^*\mathscr{F}).$$

One should note that the natural morphism

$$\mathbf{W}(\mathscr{F}^G) \longrightarrow \mathbf{W}(\mathscr{F})^G$$

is not an isomorphism in general. For example, let $S = \mathrm{Spec}(\mathbb{Z})$, and let G be the constant group $\mathbb{Z}/2\mathbb{Z} = \{1, \tau\}$ acting on $\mathscr{F} = \mathcal{O}_S$ by $\tau \cdot 1 = -1$. Then $\mathscr{F}^G = 0$, whereas, if R is an $\mathbb{F}_2$-algebra,

$$\mathbf{W}(\mathscr{F})^G(\mathrm{Spec}(R)) = R.$$

4.7.2. From now until the end of 4.7, suppose that G is affine over S. [41] By 4.6.4, giving a morphism of S-functors

$$\rho : \mathbf{h}_G \longrightarrow \underline{\mathrm{End}}_{\mathbf{O}_S}(\mathbf{W}(\mathscr{F}))$$

is equivalent to giving a morphism of $\mathcal{O}_S$-modules

$$\mu : \mathscr{F} \longrightarrow \mathscr{F} \otimes_{\mathcal{O}_S} \mathscr{A}(G).$$

[39] Editor's note: the original has been corrected by deleting the assertion that the category (G-$\mathcal{O}_S$-**Mod.**) is abelian; see 4.7.2.1 below.

[40] Editor's note: this remark has been added.

[41] Editor's note: see VI_{B}, Sections 11.1–11.6 for the extension of the results of 4.7.2 to the case where G is not necessarily affine but G and $\mathscr{F}$ are assumed flat over S.

Saying that ρ is a morphism of $\widehat{(\mathbf{Sch})}_{/S}$-groups is then equivalent to saying that μ satisfies the following axioms.

(CM 1) The following diagram is commutative.

$$\begin{tikzcd}
\mathscr{F} \arrow[rr, "\mu"] \arrow[d, "\mu"'] & & \mathscr{F} \otimes_{\mathscr{O}_{\mathrm{S}}} \mathscr{A}(\mathrm{G}) \arrow[d, "\mathrm{id} \otimes \Delta"] \\
\mathscr{F} \otimes_{\mathscr{O}_{\mathrm{S}}} \mathscr{A}(\mathrm{G}) \arrow[rr, "\mu \otimes \mathrm{id}"] & & \mathscr{F} \otimes_{\mathscr{O}_{\mathrm{S}}} \mathscr{A}(\mathrm{G}) \otimes_{\mathscr{O}_{\mathrm{S}}} \mathscr{A}(\mathrm{G}) \ .
\end{tikzcd}$$

Figure 5: The coassociativity axiom for the coaction μ.

Temporary Loop-1 image; source locator: Polo–Gille Exp1-13oct24.pdf, component-local p. 29; combined-reader p. 39.

(CM 2) The following composite is the identity.

$$\mathscr{F} \xrightarrow{\mu} \mathscr{F} \otimes \mathscr{A}(\mathrm{G}) \xrightarrow{\mathrm{id} \otimes \varepsilon} \mathscr{F} \otimes \mathscr{O}_{\mathrm{S}} \xrightarrow{\sim} \mathscr{F}.$$

Figure 6: The counit axiom for the coaction μ.

Temporary Loop-1 image; source locator: Polo–Gille Exp1-13oct24.pdf, component-local p. 29; combined-reader p. 39.

These axioms (CM 1) and (CM 2) are those of a right *comodule* structure over the bialgebra $\mathscr{A}(G)$. [42]

Put $\mathscr{A} = \mathscr{A}(G)$. If $\mathscr{F}$ and $\mathscr{F}'$ are $\mathscr{A}$-comodules, a morphism of comodules $f : \mathscr{F} \to \mathscr{F}'$ is a morphism of $\mathcal{O}_S$-modules such that the following diagram is
commutative.

$$\begin{tikzcd}
\mathscr{F} \arrow[r, "f"] \arrow[d, "\mu_{\mathscr{F}}"'] & \mathscr{F}' \arrow[d, "\mu_{\mathscr{F}'}"] \\
\mathscr{F} \otimes \mathscr{A} \arrow[r, "f \otimes \mathrm{id}"] & \mathscr{F}' \otimes \mathscr{A} .
\end{tikzcd}$$

Figure 7: Compatibility required of a morphism of comodules.

Temporary Loop-1 image; source locator: Polo–Gille Exp1-13oct24.pdf, component-local p. 30; combined-reader p. 40.

This gives the category ($\mathscr{A}$-**Comod.**); we denote by ($\mathscr{A}$-**Comod.q.c.**) its full subcategory consisting of those $\mathscr{A}$-comodules that are quasi-coherent as $\mathcal{O}_S$-modules. We have therefore proved the following statement.

Proposition 4.7.2. *Let G be an S-group affine over S. There are equivalences of categories*

$$(G\text{-}\mathcal{O}_S\text{-}\mathbf{Mod.}) \simeq (\mathscr{A}(G)\text{-}\mathbf{Comod.}),$$

$$(G\text{-}\mathcal{O}_S\text{-}\mathbf{Mod.q.c.}) \simeq (\mathscr{A}(G)\text{-}\mathbf{Comod.q.c.}).$$

[42] Editor's note: left G-$\mathcal{O}_S$-modules correspond naturally to right $\mathscr{A}(G)$-comodules.

If, in addition, $S = \operatorname{Spec}(\Lambda)$ is affine and one sets $\Lambda[G] = \Gamma(S, \mathscr{A}(G))$, there is an equivalence of categories

$$(\mathscr{A}(G)\text{-}\mathbf{Comod.q.c.}) \simeq (\Lambda[G]\text{-}\mathbf{Comod.}).$$

[43]

Suppose, moreover, that $\mathscr{A} = \mathscr{A}(G)$ is a flat $\mathcal{O}_S$-module. [44] Let $\mathscr{E}$ be an $\mathscr{A}$-comodule and $\mathscr{F}$ an $\mathcal{O}_S$-submodule of $\mathscr{E}$. Since $\mathscr{A}$ is flat over $\mathcal{O}_S$, one may identify $\mathscr{F} \otimes \mathscr{A}$ (respectively, $\mathscr{F} \otimes \mathscr{A} \otimes \mathscr{A}$) with an $\mathcal{O}_S$-submodule of $\mathscr{E} \otimes \mathscr{A}$ (respectively, of $\mathscr{E} \otimes \mathscr{A} \otimes \mathscr{A}$). Suppose that $\mu_{\mathscr{E}}$ maps $\mathscr{F}$ into $\mathscr{F} \otimes \mathscr{A}$. Then its restriction

$$\mu_{\mathscr{F}} : \mathscr{F} \longrightarrow \mathscr{F} \otimes \mathscr{A}$$

endows $\mathscr{F}$ with an $\mathscr{A}$-comodule structure; one says that $\mathscr{F}$ is a *subcomodule* of $\mathscr{E}$. Passing to the quotient, $\mu_{\mathscr{E}}$ defines a morphism of $\mathcal{O}_S$-modules

$$\mathscr{E}/\mathscr{F} \longrightarrow (\mathscr{E}/\mathscr{F}) \otimes \mathscr{A},$$

which endows $\mathscr{E}/\mathscr{F}$ with an $\mathscr{A}$-comodule structure. If $f : \mathscr{E} \to \mathscr{E}'$ is a morphism of $\mathscr{A}$-comodules, then $\operatorname{Ker} f$ (respectively, $\operatorname{Im} f$) is an $\mathscr{A}$-subcomodule of $\mathscr{E}$ (respectively, of $\mathscr{E}'$), and f induces an isomorphism of $\mathscr{A}$-comodules

$$\mathscr{E}/\operatorname{Ker} f \xrightarrow{\sim} \operatorname{Im} f.$$

Moreover, if $\mathscr{E}$ and $\mathscr{E}'$ are quasi-coherent $\mathcal{O}_S$-modules, then so are $\operatorname{Ker} f$ and $\operatorname{Im} f$. Consequently, ($\mathscr{A}$-**Comod.**) and ($\mathscr{A}$-**Comod.q.c.**) are abelian categories.

Corollary 4.7.2.1. *Suppose that G is affine and flat over S. Then the category* (G-$\mathcal{O}_S$-**Mod.q.c.**) *(respectively,* (G-$\mathcal{O}_S$-**Mod.**)*), equivalent to the category of $\mathscr{A}(G)$-comodules that are quasi-coherent over $\mathcal{O}_S$ (respectively, of all $\mathscr{A}(G)$-comodules), is abelian.*

4.7.3. Suppose now that G is a *diagonalizable group*, that is, that $\mathscr{A}(G)$ is the algebra of a commutative group M over the sheaf of rings $\mathcal{O}_S$. If $\mathscr{F}$ is an $\mathcal{O}_S$-module, then

$$\mathscr{F} \otimes \mathscr{A}(G) = \coprod_{m \in M} \mathscr{F} \otimes m\mathcal{O}_S.$$

Giving a morphism of $\mathcal{O}_S$-modules

$$\mu : \mathscr{F} \longrightarrow \mathscr{F} \otimes \mathscr{A}(G)$$

is therefore equivalent to giving $\mathcal{O}_S$-endomorphisms $(\mu_m)_{m \in M}$ of $\mathscr{F}$ such that, for every section x of $\mathscr{F}$ on an open subset of S, the family $(\mu_m(x))$ is a section of the direct sum $\coprod_{m \in M} \mathscr{F}$. This means that on every sufficiently small open subset, only finitely many restrictions of the $\mu_m(x)$ are nonzero.

For μ defined by

$$\mu(x) = \sum_{m \in M} \mu_m(x) \otimes m,$$

condition (CM 1), respectively (CM 2), holds if and only if

$$\mu_m \circ \mu_{m'} = \delta_{mm'}\mu_m, \qquad \left(\text{respectively, } \sum_{m \in M} \mu_m = \operatorname{Id}_{\mathscr{F}}\right).$$

[43] Editor's note: the preceding sentence has been added.

[44] Editor's note: the following paragraph has been added from [Se68, Section 1.3].

Thus the μ_m are pairwise orthogonal projectors whose sum is the identity. We have proved the following result.

Proposition 4.7.3. *If $G = D_S(M)$ is a diagonalizable S-group, the category of quasi-coherent G-$\mathscr{O}_S$-modules (respectively, of all G-$\mathscr{O}_S$-modules) is equivalent to the category of quasi-coherent $\mathscr{O}_S$-modules (respectively, of all $\mathscr{O}_S$-modules) graded by M.*

Remark. If $\mathscr{F}$ has an $\mathscr{O}_S$-algebra structure preserved by the operations of G, then the grading of $\mathscr{F}$ is an algebra grading. More precisely:

Corollary 4.7.3.1. *The functor $\mathscr{A} \mapsto \operatorname{Spec}(\mathscr{A})$ induces an equivalence between the category of quasi-coherent $\mathscr{O}_S$-algebras graded by M and the category opposite to that of S-schemes affine over S with S-operator group $G = D_S(M)$.*

Proposition 4.7.4. *Let G be a diagonalizable S-group. If*

$$0 \longrightarrow \mathscr{F}_1 \longrightarrow \mathscr{F}_2 \longrightarrow \mathscr{F}_3 \longrightarrow 0$$

is an exact sequence of quasi-coherent G-$\mathscr{O}_S$-modules that splits as a sequence of $\mathscr{O}_S$-modules, then it also splits as a sequence of G-$\mathscr{O}_S$-modules.

Indeed, if $G = D_S(M)$, each $\mathscr{F}_i$ is graded by the $(\mathscr{F}_i)_m$, and for every $m \in M$ the sequence

$$0 \longrightarrow (\mathscr{F}_1)_m \longrightarrow (\mathscr{F}_2)_m \longrightarrow (\mathscr{F}_3)_m \longrightarrow 0$$

of $\mathscr{O}_S$-modules is split. The preceding proposition then gives the result.

5. Group cohomology

5.1. The standard complex

Let $\mathscr{C}$ be a category, $\mathbf{G}$ a $\widehat{\mathscr{C}}$-group, $\mathbf{O}$ a $\widehat{\mathscr{C}}$-ring, and $\mathbf{F}$ a $\mathbf{G}$-$\mathbf{O}$-module.[45] For $n \geq 0$, set

$$C^n(\mathbf{G}, \mathbf{F}) = \operatorname{Hom}(\mathbf{G}^n, \mathbf{F}), \qquad \underline{C}^n(\mathbf{G}, \mathbf{F}) = \underline{\operatorname{Hom}}(\mathbf{G}^n, \mathbf{F}),$$

where $\mathbf{G}^0$ is the final object $\underline{\mathbf{e}}$. Then $\underline{C}^n(\mathbf{G}, \mathbf{F})$ (respectively, $C^n(\mathbf{G}, \mathbf{F})$) is endowed in the evident way with an $\mathbf{O}$-module structure (respectively, a $\Gamma(\mathbf{O})$-module structure), and

$$C^n(\mathbf{G}, \mathbf{F}) \simeq \Gamma\big(\underline{C}^n(\mathbf{G}, \mathbf{F})\big), \qquad \underline{C}^n(\mathbf{G}, \mathbf{F})(S) = C^n(\mathbf{G}_S, \mathbf{F}_S).$$

To give an element of $C^n(\mathbf{G}, \mathbf{F})$ is to give, for every $S \in \operatorname{Ob}(\mathscr{C})$, an n-cochain of $\mathbf{G}(S)$ with values in $\mathbf{F}(S)$, functorially in S. Recall that the boundary operator

$$\partial : C^n\big(\mathbf{G}(S), \mathbf{F}(S)\big) \longrightarrow C^{n+1}\big(\mathbf{G}(S), \mathbf{F}(S)\big)$$

is given by

$$\begin{aligned} \partial f(g_1, \dots, g_{n+1}) = {} & g_1 f(g_2, \dots, g_{n+1}) \\ & + \sum_{i=1}^{n} (-1)^i f(g_1, \dots, g_i g_{i+1}, \dots, g_{n+1}) \\ & + (-1)^{n+1} f(g_1, \dots, g_n). \end{aligned}$$

[45] Editor's note: This complex is often called the Hochschild complex; see, for example, Section II.3 of [DG70].

It is functorial in S, and hence defines a homomorphism

$$\partial : C^n(\mathbf{G}, \mathbf{F}) \longrightarrow C^{n+1}(\mathbf{G}, \mathbf{F})$$

such that $\partial \circ \partial = 0$. We have thus defined a complex of abelian groups (indeed, of $\Gamma(\mathbf{O})$-modules), denoted $C^*(\mathbf{G}, \mathbf{F})$. In the same way one defines the complex of $\mathbf{O}$-modules $\underline{C}^*(\mathbf{G}, \mathbf{F})$, and

$$C^*(\mathbf{G}, \mathbf{F}) = \Gamma\big(\underline{C}^*(\mathbf{G}, \mathbf{F})\big).$$

We denote by $H^n(\mathbf{G}, \mathbf{F})$ (respectively, $\underline{H}^n(\mathbf{G}, \mathbf{F})$) the groups (respectively, the $\widehat{\mathscr{C}}$-groups) of cohomology of $C^*(\mathbf{G}, \mathbf{F})$ (respectively, $\underline{C}^*(\mathbf{G}, \mathbf{F})$). In particular,

$$\underline{H}^0(\mathbf{G}, \mathbf{F}) = \mathbf{F}^{\mathbf{G}}, \qquad H^0(\mathbf{G}, \mathbf{F}) = \Gamma(\mathbf{F}^{\mathbf{G}}).$$

Remark 5.1.1. The "set-theoretic" description of ∂ given above is convenient for checking that $\partial \circ \partial = 0$. One may also define ∂ in terms of the multiplication $m : \mathbf{G} \times \mathbf{G} \to \mathbf{G}$ and the action $\mu : \mathbf{G} \times \mathbf{F} \to \mathbf{F}$ as follows: for every $f \in C^n(\mathbf{G}, \mathbf{F})$,

$$\partial f = \mu \circ (\mathrm{id}_{\mathbf{G}} \times f) + \sum_{i=1}^{n} (-1)^i f \circ \big(\mathrm{id}_{\mathbf{G}^{i-1}} \times m \times \mathrm{id}_{\mathbf{G}^{n-i}}\big) + (-1)^{n+1} f \circ \mathrm{pr}_{[1,n]},$$

where $\mathrm{pr}_{[1,n]}$ denotes the projection of $\mathbf{G}^{n+1} = \mathbf{G}^n \times \mathbf{G}$ onto $\mathbf{G}^n$. Similarly, for every $S \in \mathrm{Ob}(\mathscr{C})$ and $f \in \underline{C}^n(\mathbf{G}, \mathbf{F})(S) = C^n(\mathbf{G}_S, \mathbf{F}_S)$,

$$\partial f = \mu_S \circ (\mathrm{id}_{\mathbf{G}_S} \times f) + \sum_{i=1}^{n} (-1)^i f \circ \big(\mathrm{id}_{\mathbf{G}_S^{i-1}} \times m_S \times \mathrm{id}_{\mathbf{G}_S^{n-i}}\big) + (-1)^{n+1} f \circ \mathrm{pr}_{[1,n]},$$

where m_S and μ_S are obtained from m and μ by base change. [46]

5.2.

Recall (see Section 3) that ($\mathbf{G}$-$\mathbf{O}$-Mod.) is an abelian category, with its structure defined objectwise. Thus

$$0 \longrightarrow \mathbf{F}' \longrightarrow \mathbf{F} \longrightarrow \mathbf{F}'' \longrightarrow 0$$

is an exact sequence of $\mathbf{G}$-$\mathbf{O}$-modules if and only if the sequence of abelian groups

$$0 \longrightarrow \mathbf{F}'(S) \longrightarrow \mathbf{F}(S) \longrightarrow \mathbf{F}''(S) \longrightarrow 0$$

is exact for every $S \in \mathrm{Ob}(\mathscr{C})$. [47]

Suppose that $\mathscr{C}$ is small. Then, by 3.2.1, ($\mathbf{G}$-$\mathbf{O}$-Mod.) has enough injective objects, so the derived functors of the left exact functors $\underline{H}^0$ and H^0 are defined. We shall now show that the functors $\underline{H}^n$ (respectively, H^n) are indeed the derived functors of $\underline{H}^0$ (respectively, H^0). [48]

Definition 5.2.0. For every $\mathbf{O}$-module $\mathbf{P}$, let $E(\mathbf{P})$ denote the object $\underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P})$ of $\widehat{\mathscr{C}}$, endowed with the following $\mathbf{G}$-$\mathbf{O}$-module structure. For every $S \in \mathrm{Ob}(\mathscr{C})$,

$$\underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P})(S) = \mathrm{Hom}_S(\mathbf{G}_S, \mathbf{P}_S),$$

[46] Editor's note: This remark has been added.
[47] Editor's note: The preceding reminder has been added.
[48] Editor's note: The preceding sentence has been added.

and $g \in \mathbf{G}(S)$ and $a \in \mathbf{O}(S)$ act on $\phi \in \mathrm{Hom}_S(\mathbf{G}_S, \mathbf{P}_S)$ by

$$(g \cdot \phi)(h) = \phi(hg), \qquad (a \cdot \phi)(h) = a\phi(h),$$

for every $h \in \mathbf{G}(S')$, $S' \to S$. Moreover, for every $\phi \in \mathrm{Hom}_S(\mathbf{G}_S, \mathbf{P}_S)$, set

$$\varepsilon(\phi) = \phi(1) \in \mathbf{P}(S),$$

where 1 denotes the unit element of $\mathbf{G}(S)$. This defines a functor

$$E : (\mathbf{O}\text{-Mod.}) \longrightarrow (\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$$

and a natural transformation $\varepsilon : E \to \mathrm{Id}$, where Id denotes the identity functor of ($\mathbf{O}$-Mod.).
[49]

Remark 5.2.0.1. In what follows, let $\mathbf{G}_1$ and $\mathbf{G}_2$ denote two copies of $\mathbf{G}$. The morphism

$$\mathbf{G}_1 \times E(\mathbf{P}) \longrightarrow E(\mathbf{P}), \qquad (g_1, \phi) \longmapsto \big(g_2 \longmapsto \phi(g_2 g_1)\big)$$

corresponds, via the isomorphisms

$$\begin{aligned} \mathrm{Hom}\big(\mathbf{G}_1 \times E(\mathbf{P}), E(\mathbf{P})\big) &\simeq \mathrm{Hom}\Big(E(\mathbf{P}), \underline{\mathrm{Hom}}\big(\mathbf{G}_1, \underline{\mathrm{Hom}}(\mathbf{G}_2, \mathbf{P})\big)\Big) \\ &\simeq \mathrm{Hom}\Big(E(\mathbf{P}), \underline{\mathrm{Hom}}(\mathbf{G}_2 \times \mathbf{G}_1, \mathbf{P})\Big), \end{aligned}$$

to the morphism

$$\phi \longmapsto \big((g_2, g_1) \longmapsto \phi(g_2 g_1)\big),$$

that is, to the morphism

$$\underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P}) \longrightarrow \underline{\mathrm{Hom}}(\mathbf{G}_2 \times \mathbf{G}_1, \mathbf{P})$$

induced by the multiplication

$$\mu_{\mathbf{G}} : \mathbf{G} \times \mathbf{G} \longrightarrow \mathbf{G}, \qquad (g_2, g_1) \longmapsto g_2 g_1.$$

Lemma 5.2.0.2. *(i) The functor E is right adjoint to the forgetful functor*

$$(\mathbf{G}\text{-}\mathbf{O}\text{-}\mathit{Mod.}) \longrightarrow (\mathbf{O}\text{-}\mathit{Mod.}).$$

More precisely, $\varepsilon : E \to \mathrm{Id}$ induces, for every $\mathbf{M} \in (\mathbf{G}\text{-}\mathbf{O}\text{-}\mathit{Mod.})$ and $\mathbf{P} \in \mathrm{Ob}(\mathbf{O}\text{-}\mathit{Mod.})$, a bijection

$$\mathrm{Hom}_{\mathbf{G}\text{-}\mathbf{O}\text{-Mod.}}\big(\mathbf{M}, E(\mathbf{P})\big) \xrightarrow{\ \sim\ } \mathrm{Hom}_{\mathbf{O}\text{-Mod.}}(\mathbf{M}, \mathbf{P})$$

functorial in $\mathbf{M}$ and $\mathbf{P}$.

(ii) Consequently, if $\mathbf{I}$ is an injective object of ($\mathbf{O}$-Mod.), then $E(\mathbf{I})$ is an injective object of ($\mathbf{G}$-$\mathbf{O}$-Mod.).

Proof. To every $\mathbf{O}$-morphism $f : \mathbf{M} \to \mathbf{P}$, associate the element ϕ_f of $\mathrm{Hom}_{\mathbf{O}}(\mathbf{M}, E(\mathbf{P}))$ defined as follows. For every $S \in \mathrm{Ob}(\mathscr{C})$ and $m \in \mathbf{M}(S)$, the element $\phi_f(m)$ of $\mathrm{Hom}_S(\mathbf{G}_S, \mathbf{P}_S)$ is defined, for every $g \in \mathbf{G}(S')$, $S' \to S$, by

$$\phi_f(m)(g) = f(gm) \in \mathbf{P}(S').$$

[49] Editor's note: The original has been modified so as to introduce 5.2.0.1 and 5.2.0.2, which will be useful in the proof of Theorem 5.3.1.

Then, for every $h \in \mathbf{G}(S)$, one has $\phi_f(hm) = h \cdot f(m)$; that is, ϕ_f is an element of

$$\operatorname{Hom}_{\mathbf{G}\text{-}\mathbf{O}\text{-Mod.}}\big(\mathbf{M}, E(\mathbf{P})\big).$$

If $\phi \in \operatorname{Hom}_{\mathbf{G}\text{-}\mathbf{O}\text{-Mod.}}(\mathbf{M}, E(\mathbf{P}))$, and if for every $m \in \mathbf{M}(S)$ we write $f(m) = \phi(m)(1)$, then

$$\phi_f(m)(g) = f(gm) = \phi(gm)(1) = (g \cdot \phi(m))(1) = \phi(m)(g);$$

thus $\phi_f = \phi$. Conversely, it is clear that $\phi_f(m)(1) = f(m)$. This proves (i), and (ii) follows at once.

Definition 5.2.0.3. Let $\mathbf{M}$ be a $\mathbf{G}$-$\mathbf{O}$-module. The identity map of $\mathbf{M}$, considered as an $\mathbf{O}$-module, corresponds under the adjunction to the morphism of $\mathbf{G}$-$\mathbf{O}$-modules

$$\mu_{\mathbf{M}} : \mathbf{M} \longrightarrow E(\mathbf{M})$$

such that, for every $S \in \operatorname{Ob}(\mathscr{C})$ and $m \in \mathbf{M}(S)$, $\mu_{\mathbf{M}}(m)$ is the morphism $\mathbf{G}_S \to \mathbf{M}_S$ defined as follows: for every $S' \to S$ and $g \in \mathbf{G}(S')$,

$$\mu_{\mathbf{M}}(m)(g) = g \cdot m_{S'} \in \mathbf{M}(S').$$

Notice that $\mu_{\mathbf{M}}$ is a monomorphism. Indeed, $\varepsilon_{\mathbf{M}} : E(\mathbf{M}) \to \mathbf{M}$ is a morphism of $\mathbf{O}$-modules such that $\varepsilon_{\mathbf{M}} \circ \mu_{\mathbf{M}} = \mathrm{id}_{\mathbf{M}}$. Consequently, as an $\mathbf{O}$-module, $\mathbf{M}$ is a direct summand of $E(\mathbf{M})$.

Proposition 5.2.1. *Suppose that $\mathscr{C}$ is small, that finite products exist in $\mathscr{C}$, and that $\mathbf{G}$ is representable. Then the functors $\underline{H}^n(\mathbf{G}, -)$ (respectively, $H^n(\mathbf{G}, -)$) are the derived functors of the left exact functor $\underline{H}^0(\mathbf{G}, -)$ (respectively, $H^0(\mathbf{G}, -)$) on the category of $\mathbf{G}$-$\mathbf{O}$-modules.*

By the well-known general results, [50] it suffices to verify that the $\underline{H}^n(\mathbf{G}, -)$ (respectively, $H^n(\mathbf{G}, -)$) form an effaceable cohomological functor in degrees > 0. Let

$$0 \longrightarrow \mathbf{F}' \longrightarrow \mathbf{F} \longrightarrow \mathbf{F}'' \longrightarrow 0$$

be an exact sequence of $\mathbf{G}$-$\mathbf{O}$-modules, and let $S \in \operatorname{Ob}(\mathscr{C})$. By hypothesis, $\mathbf{G}$ is represented by an object $G \in \operatorname{Ob}(\mathscr{C})$, and finite products exist in $\mathscr{C}$; in particular, $\mathscr{C}$ has a final object e. Hence every $\mathbf{G}^n \times \mathbf{h}_S$ is represented by $G^n \times S$ (with $G^0 = e$), and the sequence

$$0 \longrightarrow \mathbf{F}'(G^n \times S) \longrightarrow \mathbf{F}(G^n \times S) \longrightarrow \mathbf{F}''(G^n \times S) \longrightarrow 0$$

is exact. This shows that the sequence of $\mathbf{O}$-modules

$$0 \longrightarrow \underline{C}^n(\mathbf{h}_G, \mathbf{F}') \longrightarrow \underline{C}^n(\mathbf{h}_G, \mathbf{F}) \longrightarrow \underline{C}^n(\mathbf{h}_G, \mathbf{F}'') \longrightarrow 0$$

is exact. It follows that $\underline{C}^*(\mathbf{G}, -)$, regarded as a functor on ($\mathbf{G}$-$\mathbf{O}$-Mod.) with values in the category of complexes of ($\mathbf{O}$-Mod.), is exact. Thus the functors $\underline{H}^n(\mathbf{G}, -)$ are cohomological.

Since Γ is exact, the same is true of the functors $H^n(\mathbf{G}, -)$. It remains only to prove the following.

Lemma 5.2.2. *For every $\mathbf{P} \in \operatorname{Ob}(\mathbf{O}\text{-Mod.})$, one has*

$$\underline{H}^n\big(\mathbf{G}, \underline{\operatorname{Hom}}(\mathbf{G}, \mathbf{P})\big) = 0, \qquad H^n\big(\mathbf{G}, \underline{\operatorname{Hom}}(\mathbf{G}, \mathbf{P})\big) = 0, \qquad n > 0.$$

It is enough to show that $\underline{C}^*(\mathbf{G}, \underline{\operatorname{Hom}}(\mathbf{G}, \mathbf{P}))$ and $C^*(\mathbf{G}, \underline{\operatorname{Hom}}(\mathbf{G}, \mathbf{P}))$ are homotopically trivial in degrees > 0. It is enough, in fact, to do this for the second complex, since the

[50] Editor's note: See [Gr57, 2.2.1] and [Gr57, 2.3]. The original has also been expanded in what follows.

corresponding result for the first follows by base change. [51] For every $n \geq 0$, define a morphism

$$\sigma : C^{n+1}\big(\mathbf{G}, \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P})\big) \longrightarrow C^{n}\big(\mathbf{G}, \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P})\big)$$

as follows. Let $f \in C^{n+1}(\mathbf{G}, \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P}))$. For every $S \in \mathrm{Ob}(\mathscr{C})$ and $g_1, \ldots, g_n \in \mathbf{G}(S)$, the element $\sigma(f)(g_1, \ldots, g_n)$ of $\mathrm{Hom}_S(\mathbf{G}_S, \mathbf{P}_S)$ is defined, for every $S' \to S$ and $x \in \mathbf{G}(S')$, by

$$\sigma(f)(g_1, \ldots, g_n)(x) = f(x, g_1, \ldots, g_n)(e) \in \mathbf{P}(S'),$$

where e denotes the unit element of $\mathbf{G}(S')$. Then σ is a homotopy operator in degrees > 0. Indeed, for every $g_1, \ldots, g_{n+1} \in \mathbf{G}(S)$ and $x \in \mathbf{G}(S')$, on the one hand,

$$\begin{aligned} \partial\sigma(f)(g_1, \ldots, g_{n+1})(x) = f(&xg_1, g_2, \ldots, g_{n+1})(e) \\ &+ \sum_{i=1}^{n}(-1)^i f(x, g_1, \ldots, g_i g_{i+1}, \ldots, g_{n+1})(e) \\ &+ (-1)^{n+1} f(x, g_1, \ldots, g_n)(e), \end{aligned}$$

and, on the other hand,

$$\begin{aligned} \sigma(\partial f)(g_1, \ldots, g_{n+1})(x) = \big(&x f(g_1, g_2, \ldots, g_{n+1})\big)(e) - f(xg_1, g_2, \ldots, g_{n+1})(e) \\ &+ \sum_{i=1}^{n}(-1)^{i+1} f(x, g_1, \ldots, g_i g_{i+1}, \ldots, g_{n+1}) \\ &+ (-1)^{n+2} f(x, g_1, \ldots, g_n)(e). \end{aligned}$$

Consequently,

$$\big(\partial\sigma(f) + \sigma(\partial f)\big)(g_1, \ldots, g_{n+1})(x) = f(g_1, \ldots, g_{n+1})(x);$$

that is, $\partial\sigma + \sigma\partial$ is the identity map of $C^{n+1}(\mathbf{G}, \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{P}))$ for every $n \geq 0$.

Remark 5.2.3. The hypothesis that $\mathscr{C}$ be small is used only to ensure the existence of the derived functors $R^n\underline{H}^0$ and R^nH^0. In every case, what precedes shows that the functors $\underline{H}^n(\mathbf{G}, -)$ (respectively, $H^n(\mathbf{G}, -)$) form an effaceable cohomological functor in degrees > 0, and hence are the (right) satellite functors of the left exact functor $\underline{H}^0(\mathbf{G}, -)$ (respectively, $H^0(\mathbf{G}, -)$), in the sense of [Gr57, 2.2]. If ($\mathbf{G}$-$\mathbf{O}$-Mod.) has enough injective objects (as it does when $\mathscr{C}$ is small), they coincide with the derived functors (*loc. cit.*, 2.3). [52]

5.3. Cohomology of G-$\mathscr{O}_S$-modules

Let S be a scheme, G an S-group, and $\mathscr{F}$ a quasi-coherent G-$\mathscr{O}_S$-module. The cohomology groups of G with values in $\mathscr{F}$ are defined by

$$H^n(G, \mathscr{F}) = H^n\big(\mathbf{h}_G, \mathbf{W}(\mathscr{F})\big)$$

(for the notation, see 4.6).

Suppose that G is affine over S. In view of Proposition 4.6.4, this cohomology is computed as follows: $H^n(G, \mathscr{F})$ is the n-th cohomology group of the complex $C^*(G, \mathscr{F})$, whose n-th term is

$$C^n(G, \mathscr{F}) = \Gamma\Big(S, \mathscr{F} \otimes \underbrace{\mathscr{A}(G) \otimes \cdots \otimes \mathscr{A}(G)}_{n \text{ times}}\Big).$$

[51] Editor's note: The original has been expanded in what follows.
[52] Editor's note: This remark has been added.

If f (respectively, a_i) is a section of $\mathscr{F}$ (respectively, of $\mathscr{A}(G)$) over an open subset of S, then

$$\begin{aligned}\partial(f \otimes a_1 \otimes \cdots \otimes a_n) = \mu_{\mathscr{F}}(f) \otimes a_1 \otimes a_2 \otimes \cdots \otimes a_n \\ + \sum_{i=1}^{n} (-1)^i f \otimes a_1 \otimes \cdots \otimes \Delta a_i \otimes \cdots \otimes a_n \\ + (-1)^{n+1} f \otimes a_1 \otimes a_2 \otimes \cdots \otimes a_n \otimes 1,\end{aligned}$$

where

$$\Delta : \mathscr{A}(G) \longrightarrow \mathscr{A}(G) \otimes \mathscr{A}(G), \qquad \mu_{\mathscr{F}} : \mathscr{F} \longrightarrow \mathscr{F} \otimes \mathscr{A}(G)$$

describe the coalgebra structure of $\mathscr{A}(G)$ and the comodule structure of $\mathscr{F}$. Notice in passing that the cohomology of G with values in $\mathscr{F}$ therefore depends only on the comodule structure of $\mathscr{F}$, and in particular only on the S-monoid structure of G.

In particular,

$$H^0(G, \mathscr{F}) = \Gamma(S, \mathscr{F}^G),$$

where $\mathscr{F}^G$, the *sheaf of invariants of* $\mathscr{F}$, is the sheaf whose sections over an open subset U of S are those sections of $\mathscr{F}$ over U whose inverse image in every S' over U is invariant under $G(S')$ (see 4.7.1.2).

Theorem 5.3.1. *Let S be an affine scheme and G an S-group that is affine and flat over S. The functors $H^n(G, -)$ are the derived functors of $H^0(G, -)$ on the category of quasi-coherent G-$\mathscr{O}_S$-modules.*

Proof. Since G is affine and flat over S, 4.7.2.1 shows that the category (G-$\mathscr{O}_S$-Mod.q.c.) is equivalent to the category ($\mathscr{A}(G)$-Comod.q.c.) of quasi-coherent $\mathscr{A}(G)$-comodules over $\mathscr{O}_S$, and is therefore abelian. Moreover, since $\mathscr{A}(G)$ is a flat $\mathscr{O}_S$-module, each functor

$$\mathscr{F} \longmapsto \mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{A}(G)^{\otimes n}$$

is exact; since S is affine, it follows that $C^*(G, -)$ is an exact functor on (G-$\mathscr{O}_S$-Mod.q.c.). [53]

Let Δ (respectively, η) denote the comultiplication (respectively, the augmentation) of $\mathscr{A}(G)$. For every quasi-coherent $\mathscr{O}_S$-module $\mathscr{P}$, let

$$\mathrm{Ind}(\mathscr{P}) = \mathscr{P} \otimes_{\mathscr{O}_S} \mathscr{A}(G),$$

endowed with the $\mathscr{A}(G)$-comodule structure defined by

$$\mathrm{id}_{\mathscr{P}} \otimes \Delta : \mathscr{P} \otimes_{\mathscr{O}_S} \mathscr{A}(G) \longrightarrow \mathscr{P} \otimes_{\mathscr{O}_S} \mathscr{A}(G) \otimes_{\mathscr{O}_S} \mathscr{A}(G).$$

This defines a functor

$$\mathrm{Ind} : (\mathscr{O}_S\text{-Mod.q.c.}) \longrightarrow (G\text{-}\mathscr{O}_S\text{-Mod.q.c.}).$$

It follows from 4.6.4.1, 5.2.0, and 5.2.0.1 that there is an isomorphism of $\mathbf{G}$-$\mathbf{O}_S$-modules

$$\mathbf{W}\big(\mathrm{Ind}(\mathscr{P})\big) \simeq E\big(\mathbf{W}(\mathscr{P})\big) = \underline{\mathrm{Hom}}\big(\mathbf{G}, \mathbf{W}(\mathscr{P})\big). \tag{*}$$

[53] Editor's note: The original has been modified to show that the category (G-$\mathscr{O}_S$-Mod.q.c.) is abelian and has enough injective objects. Compare [Ja03], Part I, 3.3–3.4, 3.9, 4.2, and 4.14–4.16 (bearing in mind that "k-group scheme" there means "affine k-group scheme"; see *loc. cit.*, 2.1).

Under this identification, the morphism $\varepsilon : E(\mathbf{W}(\mathscr{P})) \to \mathbf{W}(\mathscr{P})$ corresponds to the morphism of $\mathscr{O}_S$-modules

$$\mathrm{id}_{\mathscr{P}} \otimes \eta : \mathrm{Ind}(\mathscr{P}) \longrightarrow \mathscr{P}.$$

We have already used the fact that the functor

$$\mathbf{W} : (\mathscr{O}_S\text{-Mod.}) \longrightarrow (\mathbf{O}_S\text{-Mod.})$$

is fully faithful. By Definition 4.7.1, the same is true of its restriction to (G-$\mathscr{O}_S$-Mod.): if $\mathscr{M}$ and $\mathscr{M}'$ are G-$\mathscr{O}_S$-modules, there is a functorial isomorphism

$$\mathrm{Hom}_{G\text{-}\mathscr{O}_S\text{-Mod.}}(\mathscr{M}, \mathscr{M}') \simeq \mathrm{Hom}_{\mathbf{G}\text{-}\mathbf{O}_S\text{-Mod.}}\big(\mathbf{W}(\mathscr{M}), \mathbf{W}(\mathscr{M}')\big).$$

Consequently, Lemma 5.2.0.2 gives the following result.

Corollary 5.3.1.1. *(i) The functor* Ind *is right adjoint to the forgetful functor*

$$(G\text{-}\mathscr{O}_S\text{-Mod.q.c.}) \longrightarrow (\mathscr{O}_S\text{-Mod.q.c.}).$$

More precisely, the map $\mathrm{id}_{\mathscr{P}} \otimes \eta : \mathrm{Ind}(\mathscr{P}) \to \mathscr{P}$ *induces, for every object* $\mathscr{M}$ *of* (G-$\mathscr{O}_S$*-Mod.q.c.*)*, a bijection*

$$\mathrm{Hom}_{G\text{-}\mathscr{O}_S\text{-Mod.}}\big(\mathscr{M}, \mathrm{Ind}(\mathscr{P})\big) \xrightarrow{\sim} \mathrm{Hom}_{\mathscr{O}_S}(\mathscr{M}, \mathscr{P}).$$

(ii) Consequently, if $\mathscr{I}$ *is an injective object of* ($\mathscr{O}_S$*-Mod.q.c.*)*, then* $\mathrm{Ind}(\mathscr{I})$ *is an injective object of* (G-$\mathscr{O}_S$*-Mod.q.c.*).

Let $\mathscr{F}$ be a G-$\mathscr{O}_S$-module, and let $\mu_{\mathscr{F}} : \mathscr{F} \to \mathrm{Ind}(\mathscr{F})$ be the map defining its $\mathscr{A}(G)$-comodule structure. It follows from 5.2.0.3 (or directly from axioms (CM 1) and (CM 2) of 4.7.2) that $\mu_{\mathscr{F}}$ is a morphism of G-$\mathscr{O}_S$-modules, and that

$$(\mathrm{id}_{\mathscr{F}} \otimes \eta) \circ \mu_{\mathscr{F}} = \mathrm{id}_{\mathscr{F}}\,.$$

Thus $\mathscr{F}$ is a direct summand of $\mathrm{Ind}(\mathscr{F})$ as an $\mathscr{O}_S$-module; in particular, $\mu_{\mathscr{F}}$ is a monomorphism. By (*) and 5.2.2,

$$H^n\big(\mathbf{G}, \mathbf{W}(\mathrm{Ind}(\mathscr{F}))\big) \simeq H^n\Big(\mathbf{G}, \underline{\mathrm{Hom}}_S\big(\mathbf{G}, \mathbf{W}(\mathscr{F})\big)\Big) = 0 \qquad (n > 0).$$

Therefore $H^n(G, -)$ is effaceable for $n > 0$.

Finally, since S is affine, ($\mathscr{O}_S$-Mod.q.c.) has enough injective objects. Let $\mathscr{F} \hookrightarrow \mathscr{I}$ be a monomorphism of $\mathscr{O}_S$-modules, where $\mathscr{I}$ is injective in ($\mathscr{O}_S$-Mod.q.c.). Since $\mathscr{A}(G)$ is flat over $\mathscr{O}_S$, $\mathrm{Ind}(\mathscr{F})$ is a G-$\mathscr{O}_S$-submodule of $\mathrm{Ind}(\mathscr{I})$. Hence:

Corollary 5.3.1.2. *The abelian category* (G-$\mathscr{O}_S$*-Mod.q.c.*) *has enough injective objects.*

Taking account of [Gr57, 2.2.1 and 2.3], already used in the proof of 5.2.1, completes the proof of Theorem 5.3.1.

Remark 5.3.1.3. Corollary 5.3.1.1 may also be proved by the following calculation. To every morphism of G-$\mathscr{O}_S$-modules

$$\phi : \mathscr{M} \longrightarrow \mathscr{P} \otimes_{\mathscr{O}_S} \mathscr{A}(G)$$

associate the $\mathscr{O}_S$-morphism

$$(\mathrm{id}_{\mathscr{P}} \otimes \eta) \circ \phi : \mathscr{M} \longrightarrow \mathscr{P}.$$

Conversely, to every $\mathscr{O}_S$-morphism $f : \mathscr{M} \to \mathscr{P}$, associate the morphism of G-$\mathscr{O}_S$-modules

$$(f \otimes \mathrm{id}_{\mathscr{A}(G)}) \circ \mu_{\mathscr{M}} : \mathscr{M} \longrightarrow \mathscr{P} \otimes_{\mathscr{O}_S} \mathscr{A}(G).$$

One sees at once that

$$(\mathrm{id}_{\mathscr{P}} \otimes \eta) \circ (f \otimes \mathrm{id}_{\mathscr{A}(G)}) \circ \mu_{\mathscr{M}} = (f \otimes \mathrm{id}_{\mathscr{O}_S}) \circ (\mathrm{id}_{\mathscr{P}} \otimes \eta) \circ \mu_{\mathscr{M}} = f.$$

On the other hand, for every ϕ, the following diagram is commutative.

$$\begin{tikzcd}
\mathscr{M} \arrow[r, "\varphi"] \arrow[d, "\mu_{\mathscr{M}}"'] & \mathscr{P} \otimes_{\mathscr{O}_{\mathrm{S}}} \mathscr{A}(\mathrm{G}) \arrow[d, "\mathrm{id}_{\mathscr{P}} \otimes \Delta"] \\
\mathscr{M} \otimes_{\mathscr{O}_{\mathrm{S}}} \mathscr{A}(\mathrm{G}) \arrow[r, "\phi \otimes \mathrm{id}_{\mathscr{A}(\mathrm{G})}"] & \mathscr{P} \otimes_{\mathscr{O}_{\mathrm{S}}} \mathscr{A}(\mathrm{G}) \otimes_{\mathscr{O}_{\mathrm{S}}} \mathscr{A}(\mathrm{G}) .
\end{tikzcd}$$

Figure 8: The adjunction square for ϕ, the comodule map $\mu_{\mathscr{M}}$, and the comultiplication Δ.

Temporary Loop-1 image; source locator: Polo–Gille Exp1-13oct24.pdf, component-local/printed p. 38; combined-reader p. 48.

It follows that

$$\begin{aligned}
&\Big(\big((\mathrm{id}_{\mathscr{P}} \otimes \eta) \circ \phi\big) \otimes \mathrm{id}_{\mathscr{A}(G)}\Big) \circ \mu_{\mathscr{M}} \\
&\qquad = (\mathrm{id}_{\mathscr{P}} \otimes \eta \otimes \mathrm{id}_{\mathscr{A}(G)}) \circ (\phi \otimes \mathrm{id}_{\mathscr{A}(G)}) \circ \mu_{\mathscr{M}} \\
&\qquad = (\mathrm{id}_{\mathscr{P}} \otimes \eta \otimes \mathrm{id}_{\mathscr{A}(G)}) \circ (\mathrm{id}_{\mathscr{P}} \otimes \Delta) \circ \phi = \phi.
\end{aligned}$$

This proves Corollary 5.3.1.1(i), and (ii) follows.

Let $\mathscr{F}$ be a G-$\mathscr{O}_S$-module. We saw above that axiom (CM 2) of 4.7.2 shows that, as an $\mathscr{O}_S$-module, $\mathscr{F}$ is a direct summand of $E(\mathscr{F})$. This gives:

Proposition 5.3.2. *Let S be an affine scheme and G an S-group that is affine and flat. Suppose that every exact sequence*

$$0 \longrightarrow \mathscr{F}_1 \longrightarrow \mathscr{F}_2 \longrightarrow \mathscr{F}_3 \longrightarrow 0$$

of quasi-coherent G-$\mathscr{O}_S$-modules which splits as a sequence of $\mathscr{O}_S$-modules also splits as a sequence of G-$\mathscr{O}_S$-modules. Then the functors $H^n(G,-)$ vanish for $n > 0$ (or, equivalently, the functor $H^0(G,-)$ is exact).

Indeed, by hypothesis, the sequence of G-$\mathscr{O}_S$-modules

$$0 \longrightarrow \mathscr{F} \longrightarrow E(\mathscr{F}) \longrightarrow E(\mathscr{F})/\mathscr{F} \longrightarrow 0$$

is split. Thus $\mathscr{F}$ is a direct summand of $E(\mathscr{F})$ as a G-$\mathscr{O}_S$-module, while the cohomology of the latter vanishes.

Combining this with Proposition 4.7.4 immediately gives:

Theorem 5.3.3. *Let S be an affine scheme and G a diagonalizable S-group. For every quasi-coherent G-$\mathscr{O}_S$-module $\mathscr{F}$, one has*

$$H^n(G, \mathscr{F}) = 0 \qquad (n > 0).$$

Remark. Proposition 5.3.2 remains valid when G is not necessarily flat over S; the proof then uses relative cohomology. [54]

[54] Editor's note: The editors have not attempted to develop this remark.

6. G-equivariant objects and modules

Let $\mathscr{C}$ be a category having a final object and in which fiber products exist.[55] Let G be a $\widehat{\mathscr{C}}$-group, $\pi : M \to X$ a morphism in $\widehat{\mathscr{C}}$, and $\lambda = \lambda_X : G \times X \to X$ an action of G on X. In what follows, $Y \times_f M$ denotes the fiber product of $\pi : M \to X$ and an X-functor $f : Y \to X$.

For every $U \in \mathrm{Ob}(\mathscr{C})$ and $x \in X(U)$, write $M_x = U \times_x M$; thus, for every $\phi : U' \to U$,

$$M_x(U') = \{m \in M(U') \mid \pi(m) = x_{U'} = \phi^*(x)\}.$$

Finally, if $g \in G(U)$, we also write gx for the element $\lambda(g, x)$ of $X(U)$.

Definition 6.1. *(a) We say that M is a G-equivariant X-object if an action $\Lambda : G \times M \to M$ of G on M, lifting λ, has been given; that is, if the following square commutes.*

$$\begin{tikzcd}
\mathbf{G} \times \mathbf{M} \arrow[r, "\Lambda"] \arrow[d, "\mathrm{id}_{\mathbf{G}} \times \pi"'] & \mathbf{M} \arrow[d, "\pi"] \\
\mathbf{G} \times \mathbf{X} \arrow[r, "\lambda"] & \mathbf{X}
\end{tikzcd}$$

Figure 9: The action of G on M lifts its action on X.

Temporary Loop-1 image; source locator: Polo–Gille Exp1-13oct24.pdf, component-local p. 39; combined-reader p. 49.

Equivalently, for every morphism $(g, x) : U \to G \times X$, one is given maps

$$\Lambda_x^U(g) : M_x(U) \longrightarrow M_{gx}(U), \qquad m \longmapsto g \cdot m,$$

which satisfy $1 \cdot m = m$ and $g \cdot (h \cdot m) = (gh) \cdot m$, and which are functorial in the $(G \times X)$-object U. Equivalently again, one is given morphisms of U-objects

$$\Lambda_x(g) : M_x \longrightarrow M_{gx}$$

satisfying $\Lambda_x(1) = \mathrm{id}$ and $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$.

(b) Now let $\mathbf{O}$ be a $\widehat{\mathscr{C}}$-ring and put $\mathbf{O}_X = \mathbf{O} \times X$. Under the hypotheses of (a), we say that M is a G-equivariant $\mathbf{O}_X$-module if it is an $\mathbf{O}_X$-module (cf. Definition 4.3.3.1, which is valid for every ring functor on a category $\mathscr{C}$) and if the action Λ is compatible with the $\mathbf{O}_X$-module structure on M. Thus, for every $(g, x) \in G(U) \times X(U)$, $U \in \mathrm{Ob}(\mathscr{C})$, the map

$$\Lambda_x(g) : M_x \longrightarrow M_{gx}, \qquad m \longmapsto g \cdot m,$$

is a morphism of $\mathbf{O}_U$-modules.

Remark 6.2. (i) In (a), the conditions $\Lambda_x(1) = \mathrm{id}$ and $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$ plainly imply that each $\Lambda_x(g)$ is an isomorphism, with inverse $\Lambda_{gx}(g^{-1})$. Conversely, if each $\Lambda_x(g)$ is an isomorphism, applying the second condition with $h = 1$ gives $\Lambda_x(1) = \mathrm{id}$.

[55] Editor's note: This section has been added.

(ii) Let M be an $\mathbf{O}_X$-module. Giving a morphism $\Lambda : G \times M \to M$ which makes the diagram of Definition 6.1 commute and for which every morphism

$$\Lambda_x(g) : M_x \longrightarrow M_{gx}, \qquad m \longmapsto g \cdot m,$$

is an isomorphism of $\mathbf{O}_U$-modules is equivalent to giving an isomorphism of $\mathbf{O}_{G\times X}$-modules

$$\theta : G \times M = (G \times X) \times_{\mathrm{pr}_X} M \xrightarrow{\sim} (G \times X) \times_\lambda M, \qquad (g, x, m) \longmapsto (g, x, g \cdot m).$$

Since each $\Lambda_x(g)$ is assumed to be an isomorphism, the equality $\Lambda_x(1) = \mathrm{id}$ follows from $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$ with $h = 1$. Consequently, Λ is an action of G on M, that is, $g \cdot (h \cdot m) = (gh) \cdot m$, if and only if the following diagram of $(G \times G \times X)$-isomorphisms commutes. Here μ denotes multiplication in G, and $f^*(\theta)$ is obtained from θ by base change along $f : G \times G \times X \to G \times X$.

$$\begin{tikzcd}
(\mathbf{G}\times\mathbf{G}\times\mathbf{X}) \underset{\mathrm{pr}_{\mathbf{X}}\circ\mathrm{pr}_{2,3}}{\times} \mathbf{M} \arrow[r, "\mathrm{pr}_{2,3}^*(\theta)", "\sim"'] \arrow[d, equal] & (\mathbf{G}\times\mathbf{G}\times\mathbf{X}) \underset{\lambda\circ\mathrm{pr}_{2,3}}{\times} \mathbf{M} \arrow[d, equal] \\
(\mathbf{G}\times\mathbf{G}\times\mathbf{X}) \underset{\mathrm{pr}_{\mathbf{X}}\circ(\mu\times\mathrm{id}_{\mathbf{X}})}{\times} \mathbf{M} \arrow[d, "(\mu\times\mathrm{id}_{\mathbf{X}})^*(\theta)"', "\wr"] & (\mathbf{G}\times\mathbf{G}\times\mathbf{X}) \underset{\mathrm{pr}_{\mathbf{X}}\circ(\mathrm{id}_{\mathbf{G}}\times\lambda)}{\times} \mathbf{M} \arrow[d, "(\mathrm{id}_{\mathbf{G}}\times\lambda)^*(\theta)", "\wr"'] \\
(\mathbf{G}\times\mathbf{G}\times\mathbf{X}) \underset{\lambda\circ(\mu\times\mathrm{id}_{\mathbf{X}})}{\times} \mathbf{M} \arrow[r, equal] & (\mathbf{G}\times\mathbf{G}\times\mathbf{X}) \underset{\lambda\circ(\mathrm{id}_{\mathbf{G}}\times\lambda)}{\times} \mathbf{M}\ .
\end{tikzcd}$$

Figure 10: The cocycle condition for the isomorphism θ.

Temporary Loop-1 image; source locator: Polo–Gille Exp1-13oct24.pdf, component-local p. 40; combined-reader p. 50.

Thus, giving a G-equivariant $\mathbf{O}_X$-module structure on M is equivalent to giving an isomorphism θ of $\mathbf{O}_{G\times X}$-modules as above for which this diagram commutes.

(iii) Everything above extends to the case in which G is merely a $\widehat{\mathscr{C}}$-monoid. In that case, giving an action $\Lambda : G \times M \to M$ lifting λ, such that every $\Lambda_x(g) : M_x \to M_{gx}$ is a morphism of $\mathbf{O}_U$-modules, is equivalent to giving a morphism θ of $\mathbf{O}_{G\times X}$-modules as in (ii), for which the diagram above commutes without the isomorphism signs under the arrows and which satisfies

$$p_M \circ \theta \circ (\varepsilon_G \times \mathrm{id}_M) = \mathrm{id}_M,$$

where ε_G is the unit section of G and p_M is the projection onto M.

6.3. G-equivariant morphisms

Let Y be a second object of $\widehat{\mathscr{C}}$, endowed with an action $\lambda_Y : G \times Y \to Y$, and let N be a second G-equivariant $\mathbf{O}_X$-module. A morphism $f : Y \to X$ in $\widehat{\mathscr{C}}$ (respectively, a morphism of $\mathbf{O}_X$-modules $\phi : M \to N$) is called *G-equivariant* if it commutes with the action of G. In set-theoretic notation, this says

$$f(g \cdot y) = g \cdot f(y) \qquad \big(\text{respectively, } \phi(g \cdot m) = g \cdot \phi(m)\big),$$

or, equivalently,

$$f \circ \lambda_Y = \lambda_X \circ (\mathrm{id}_G \times f), \qquad \phi \circ \Lambda_M = \Lambda_N \circ (\mathrm{id}_G \times \phi).$$

The following lemma is immediate.

Lemma 6.3.1. *Let $f : Y \to X$ be a G-equivariant morphism and let M be a G-equivariant $\mathbf{O}_X$-module. Then the inverse image $f^*(M) = Y \times_f M$ is a G-equivariant $\mathbf{O}_Y$-module.*

On the other hand, G acts on $\mathrm{Hom}_{\widehat{\mathscr{C}}}(Y, X)$. Indeed, let $T \in \mathrm{Ob}(\mathscr{C})$, $g \in G(T)$, and let ϕ be a T-morphism $Y_T \to X_T$. Then g defines automorphisms $\lambda_Y(g)$ and $\lambda_X(g)$ of Y_T and X_T, and we write $g \cdot \phi$, or $g\phi g^{-1}$, for the morphism

$$\lambda_X(g) \circ \phi \circ \lambda_Y(g^{-1}).$$

This defines an action of $G(T)$ on $\mathrm{Hom}_T(Y_T, X_T)$, functorially in T.

Definition 6.3.2. *If $\phi : Y \to X$ is an arbitrary morphism in $\widehat{\mathscr{C}}$, we may therefore consider its stabilizer* $\mathrm{Stab}_G(\phi)$ *(cf. 2.3.3.1). For every $T \in \mathrm{Ob}(\mathscr{C})$, $\mathrm{Stab}_G(\phi)(T)$ is the subgroup of $G(T)$ formed by those g for which $g \circ \phi_T = \phi_T \circ g$, that is, for which the following diagram commutes.*

$$\begin{tikzcd} Y_T \arrow[r, "\phi_T"] \arrow[d, "g"'] & X_T \arrow[d, "g"] \\ Y_T \arrow[r, "\phi_T"] & X_T \end{tikzcd}$$

Figure 11: The defining square for $\mathrm{Stab}_G(\phi)$.

Temporary Loop-1 image; source locator: Polo–Gille Exp1-13oct24.pdf, component-local p. 41; combined-reader p. 51.

Then $\phi : Y \to X$ is equivariant under $\mathrm{Stab}_G(\phi)$.

6.4. Global sections

Let M be a G-equivariant $\mathbf{O}_X$-module. Let S_0 denote the final object of $\mathscr{C}$, and denote by $\prod_{X/S_0} M$ the *functor of sections of M over X* (cf. Exposé II, 1.1): to every $T \in \mathrm{Ob}(\mathscr{C})$ it assigns

$$\mathrm{Hom}_X(X_T, M) = \mathrm{Hom}_{X_T}(X_T, M_T) = \Gamma(M_T/X_T).$$

Recall also that every morphism $g : Z \to Y$ of $\widehat{\mathscr{C}}$-objects over X induces a morphism of abelian groups

$$M(g) : M(Y) \longrightarrow M(Z),$$

compatible with the ring homomorphism $g^* : \mathbf{O}(Y) \to \mathbf{O}(Z)$. In particular, when $Z = Y$, so that g is an X-endomorphism of Y, one obtains a morphism of abelian groups

$$M(g) : M(Y) \longrightarrow M(Y)$$

which is not generally a morphism of $\mathbf{O}(Y)$-modules, but which satisfies

$$M(g)(\alpha \cdot m) = g^*(\alpha) \cdot M(g)(m)$$

for every $m \in M(Y)$ and $\alpha \in \mathbf{O}(Y)$. In what follows, we simply write g^* instead of $M(g)$.

Let $T \in \mathrm{Ob}(\mathscr{C})$, put $X_T = X \times T$, and let $\mathrm{pr}_X : X_T \to X$ be the projection. For $\alpha \in \mathbf{O}(X_T)$ and $g \in G(T)$, set

$$g(\alpha) = (g^{-1})^*(\alpha).$$

Set-theoretically, this element of $\mathbf{O}(X_T)$ is defined, for every $S \in \mathrm{Ob}(\mathscr{C})$, $x \in X(S)$, and $t \in T(S)$, by

$$g(\alpha)(x, t) = \alpha(g^{-1}x, t).$$

This gives a left action of $G(T)$ by ring automorphisms on $\mathbf{O}(X_T)$, functorially in T, and $g(\alpha) = \alpha$ when α is the image in $\mathbf{O}(X_T)$ of an element of $\mathbf{O}(T)$.

Let ϕ now be the identity morphism of X_T (cf. 6.3 and the generalization below to a morphism $\phi : Y \to X$). Denote $\mathrm{Hom}_X(X_T, M)$ by $M(\phi g^{-1})$ or $M(\phi)$ according as X_T is regarded as an X-object through $\mathrm{pr}_X \circ \lambda(g^{-1})_T$ or through pr_X, respectively.

Then $\lambda(g^{-1})_T$ is an X-morphism between these two X-objects. Hence there is a morphism of abelian groups

$$(g^{-1})^* : M(\phi) \longrightarrow M(\phi g^{-1}), \qquad m \longmapsto m \circ \lambda(g^{-1})_T,$$

and

$$(g^{-1})^*(\alpha \cdot m) = (g\alpha) \cdot (g^{-1})^*(m)$$

for $m \in M(\phi)$ and $\alpha \in \mathbf{O}(X_T)$. If m is a section of M_T over X_T, then $(g^{-1})^*(m)$ is the section defined set-theoretically by $(x, t) \mapsto m(g^{-1}x, t)$. In particular, $(g^{-1})^*$ is a morphism of $\mathbf{O}(T)$-modules.

The functoriality of the $\mathbf{O}(X_T)$-module morphisms $\Lambda_x(g)$ gives the following commutative diagram.

$$\begin{tikzcd}
\mathbf{M}(\phi) \arrow[r, "(g^{-1})^*"] \arrow[d, "\Lambda_\phi(g)"'] & \mathbf{M}(\phi g^{-1}) \arrow[d, "\Lambda_{\phi g^{-1}}(g)"] \\
\mathbf{M}(g\phi) \arrow[r, "(g^{-1})^*"] & \mathbf{M}(g\phi g^{-1})
\end{tikzcd}$$

Figure 12: Naturality of the maps $\Lambda_x(g)$ on sections.

Temporary Loop-1 image; source locator: Polo–Gille Exp1-13oct24.pdf, component-local p. 42; combined-reader p. 52.

Moreover, $g\phi g^{-1} = \phi$, because ϕ is the identity. Writing

$$A(g) = (g^{-1})^* \circ \Lambda_\phi(g),$$

we obtain a morphism of abelian groups

$$A(g) : M(\phi) \longrightarrow M(\phi)$$

which is compatible with the action of $G(T)$ on $\mathbf{O}(X_T)$, namely

$$A(g)(\alpha \cdot m) = (g\alpha) \cdot A(g)(m).$$

Finally, if h is a second element of $G(T)$, functoriality of M and of the morphisms $\Lambda_x(g)$ gives the following commutative diagram.

$$\begin{tikzcd}
\mathbf{M}(\phi) \arrow[d, "\Lambda_\phi(g)"'] \arrow[dr, "\Lambda_\phi(hg)"] & & \\
\mathbf{M}(g\phi) \arrow[r, "\Lambda_\phi(h)"] \arrow[d, "(g^{-1})^*"'] & \mathbf{M}(hg\phi) \arrow[d, "(g^{-1})^*"'] \arrow[dr, "((hg)^{-1})^*"] & \\
\mathbf{M}(\phi) \arrow[r, "\Lambda_\phi(h)"] & \mathbf{M}(h\phi) \arrow[r, "(h^{-1})^*"] & \mathbf{M}(\phi)
\end{tikzcd}$$

Figure 13: Composition of the induced action on global sections.

Temporary Loop-1 image; source locator: Polo–Gille Exp1-13oct24.pdf, component-local p. 42; combined-reader p. 52.

It follows that $A(h) \circ A(g) = A(hg)$, and hence:

Proposition 6.4.1. *For every T, the $\mathbf{O}(X_T)$-module*

$$\left(\prod_{X/S_0} M \right) (T) = \Gamma(M_T/X_T)$$

is endowed, functorially in T, with an action of $G(T)$ compatible with the action of $G(T)$ on $\mathbf{O}(X_T)$; that is,

$$A(g)(\alpha \cdot m) = g(\alpha) \cdot A(g)(m).$$

Since $g(\alpha) = \alpha$ for $\alpha \in \mathbf{O}(T)$, it follows in particular that $\prod_{X/S_0} M$ is a G-$\mathbf{O}_{S_0}$-module.

More generally, let Y be a second G-equivariant $\widehat{\mathscr{C}}$-object, let $\phi : Y \to X$ be a morphism in $\widehat{\mathscr{C}}$, and put $H = \mathrm{Stab}_G(\phi)$, as in 6.3. Then

$$M_\phi = Y \times_\phi M$$

is an H-equivariant $\mathbf{O}_Y$-module. We obtain:

Corollary 6.4.2. *The functor $\prod_{Y/S_0} M_\phi$ is a $\mathrm{Stab}_G(\phi)$-$\mathbf{O}_{S_0}$-module.*

6.5. G-equivariant $\mathcal{O}_X$-modules

Apply the preceding discussion in the following situation. Let S be a scheme, let G be an S-group scheme acting on an S-scheme X, and let $\mathcal{F}$ be an $\mathcal{O}_X$-module, not necessarily quasi-coherent.

Definition 6.5.1. *We say that $\mathcal{F}$ is a G-equivariant $\mathcal{O}_X$-module if the $\mathbf{O}_X$-module $M = W(\mathcal{F})$ is G-equivariant. Thus, for every morphism $(g,x) : U \to G \times_S X$, one is given isomorphisms of $\mathcal{O}_U$-modules*

$$\Lambda_x(g) : W\big(x^*(\mathcal{F})\big) \xrightarrow{\sim} W\big((gx)^*(\mathcal{F})\big),$$

functorial in U and satisfying $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$. Because W is fully faithful (cf. 4.6.1.1 and 4.6.2), these yield isomorphisms of $\mathcal{O}_U$-modules

$$\Lambda_x(g) : x^*(\mathcal{F}) \xrightarrow{\sim} (gx)^*(\mathcal{F}),$$

where gx denotes $\lambda \circ (g,x) : U \to X$. Applying this to the identity morphism of $G \times_S X$ gives an isomorphism of $\mathcal{O}_{G\times_S X}$-modules

$$(\star) \qquad \theta : \mathrm{pr}_X^*(\mathcal{F}) \xrightarrow{\sim} \lambda^*(\mathcal{F})$$

such that the following diagram $(\star\star)$ of morphisms of sheaves on $G \times_S G \times_S X$ commutes.

$$\begin{tikzcd}
\mathrm{pr}_{2,3}^* \circ \mathrm{pr}_{\mathrm{X}}^*(\mathscr{F}) \arrow[r, "\mathrm{pr}_{2,3}^*(\theta)", "\sim"'] \arrow[d, equal] & \mathrm{pr}_{2,3}^* \circ \lambda^*(\mathscr{F}) \arrow[r, equal] \arrow[d, phantom, "(\star\star)"] & (\mathrm{id}_{\mathrm{G}} \times\lambda)^* \circ \mathrm{pr}_{\mathrm{X}}^*(\mathscr{F}) \arrow[d, "(\mathrm{id}_{\mathrm{G}} \times\lambda)^*(\theta)", "\wr"'] \\
(\mu \times \mathrm{id}_{\mathrm{X}})^* \circ \mathrm{pr}_{\mathrm{X}}^*(\mathscr{F}) \arrow[r, "(\mu\times\mathrm{id}_{\mathrm{X}})^*(\theta)", "\sim"'] & (\mu \times \mathrm{id}_{\mathrm{X}})^* \circ \lambda^*(\mathscr{F}) \arrow[r, equal] & (\mathrm{id}_{\mathrm{G}} \times\lambda)^* \circ \lambda^*(\mathscr{F}) .
\end{tikzcd}$$

Figure 14: The cocycle condition $(\star\star)$ for a G-equivariant $\mathcal{O}_X$-module.

Temporary Loop-1 image; source locator: Polo–Gille Exp1-13oct24.pdf, component-local p. 43; combined-reader p. 53.

If $\mathcal{E}$ is a second G-equivariant $\mathcal{O}_X$-module, a morphism $\phi : \mathcal{E} \to \mathcal{F}$ of $\mathcal{O}_X$-modules is called *G-equivariant* if $W(\phi) : W(\mathcal{E}) \to W(\mathcal{F})$ is G-equivariant. With the notation above, this is equivalent to

$$\theta_{\mathcal{F}} \circ \mathrm{pr}_X^*(\phi) = \lambda^*(\phi) \circ \theta_{\mathcal{E}}.$$

Remark 6.5.2. If $f : Y \to X$ is G-equivariant, Lemma 6.3.1 shows that $f^*(\mathcal{F})$ is a G-equivariant $\mathcal{O}_Y$-module.

Remark 6.5.3. Suppose in addition that $\mathcal{F}$ is quasi-coherent. Giving a G-equivariant $\mathcal{O}_X$-module structure on $M = W(\mathcal{F})$ is then equivalent to giving such a structure on $V(\mathcal{F})$, and this in turn is equivalent to giving an action of G on the vector fibration $V(\mathcal{F})$, compatible with the action on X and linear on the fibers.

Indeed, let

$$\varphi : \lambda^*(\mathcal{F}) \xrightarrow{\sim} \mathrm{pr}_X^*(\mathcal{F})$$

be the inverse of θ. For every morphism $(g,x) : U \to G \times_S X$, the isomorphism of $\mathcal{O}_U$-modules

$$\varphi_x(g) = \Lambda_x(g)^{-1} = \Lambda_{gx}(g^{-1})$$

from $(gx)^*(\mathcal{F})$ to $x^*(\mathcal{F})$ induces an isomorphism of $\mathcal{O}_U$-modules

$$V\big((gx)^*(\mathcal{F})\big) \xrightarrow{\sim} V\big(x^*(\mathcal{F})\big).$$

Denote the latter by ${}^t\varphi_x(g)$. Then

$${}^t\varphi_{gx}(h) \circ {}^t\varphi_x(g) = {}^t\big(\Lambda_{gx}(h) \circ \varphi_x(g)\big)^{-1} = \Lambda_x(hg)^{-1} = {}^t\varphi_x(hg).$$

For every X-scheme $f : Y \to X$, one has

$$V(\mathcal{F}) \times_f Y \simeq V\big(f^*(\mathcal{F})\big) \qquad \text{(cf. 4.6.2)}.$$

Therefore the isomorphism

$$\begin{aligned} {}^t\varphi : \quad G \times_S V(\mathcal{F}) &= (G \times_S X) \times_{\mathrm{pr}_X} V(\mathcal{F}) = V\big(\mathrm{pr}_X^*(\mathcal{F})\big) \\ &\xrightarrow{\sim} V\big(\lambda^*(\mathcal{F})\big) = (G \times_S X) \times_\lambda V(\mathcal{F}) \end{aligned}$$

endows $V(\mathcal{F})$ with a G-equivariant $\mathcal{O}_X$-module structure. Identifying every functor $V(-)$ with the vector fibration which represents it, and composing ${}^t\varphi$ with the projection onto $V(\mathcal{F})$, gives an action of G on $V(\mathcal{F})$, compatible with the action on X and linear on the fibers.

This recovers the definition in [GIT, Chapter 1, Section 3], except that Mumford there considers a locally free $\mathcal{O}_X$-module $\mathcal{E}$ of finite rank and defines an action of G on $V(\mathcal{E}) = W(\mathcal{E}^\vee)$. Indeed, the preceding diagram ($\star\star$), with θ replaced by φ and the directions of the arrows reversed, is exactly the diagram in *loc. cit.*, Definition 1.6, and the isomorphism ${}^t\varphi$ above coincides with the isomorphism Φ of *loc. cit.*, p. 31.

Remark 6.5.4. Consider in particular the case $X = S$, with the trivial action of G. A G-equivariant $\mathcal{O}_S$-module $\mathcal{F}$ is then the same thing as a G-$\mathcal{O}_S$-module (cf. 4.7.1). If $f : G \to S$ denotes the morphism which here equals both pr_X and λ, the isomorphism

$$(\star) \qquad \theta : f^*(\mathcal{F}) \xrightarrow{\sim} f^*(\mathcal{F})$$

is an element of

$$\begin{aligned} \mathrm{Hom}_{\mathcal{O}_G}\big(f^*(\mathcal{F}), f^*(\mathcal{F})\big) &= \mathrm{Hom}_{\mathcal{O}_G}\big(W(f^*(\mathcal{F})), W(f^*(\mathcal{F}))\big) \\ &= \mathrm{End}_{\mathcal{O}_S}\big(W(\mathcal{F})\big)(G). \end{aligned}$$

It is precisely the morphism

$$\rho : G \longrightarrow \mathrm{End}_{\mathcal{O}_S}\big(W(\mathcal{F})\big)$$

which defines the action of G on $W(\mathcal{F})$. Moreover, θ corresponds by adjunction to the morphism of $\mathcal{O}_S$-modules

$$f_*(\theta) \circ \tau : \mathcal{F} \longrightarrow f_*f^*(\mathcal{F}),$$

where $\tau : \mathcal{F} \to f_*f^*(\mathcal{F})$ is the unit morphism of the adjunction. This will be used in Exposé $\mathrm{VI_B}$, 11.10 bis.

6.6. The functors $\prod_{X/S} W(\mathcal{F})$ and $W(p_*(\mathcal{F}))$

Let S be a scheme, let G be an S-group scheme acting on S-schemes X and Y through

$$\lambda : G \times_S X \to X, \qquad \mu : G \times_S Y \to Y,$$

and let $p : X \to Y$ be a G-equivariant morphism. Suppose that $p : X \to Y$ and $\nu_Y : Y \to S$ are quasi-compact and quasi-separated and that $\pi : G \to S$ is flat.

The projection $\mathrm{pr}_Y : G \times_S Y \to Y$ is then flat, as is μ, since μ is the composite of pr_Y and the automorphism $(g, y) \mapsto (g, gy)$. For a variable S-scheme $f : T \to S$, denote by $p_T : X_T \to T$ and $f_X : X_T \to X$ the morphisms obtained from p and f by base change.

Let $\mathcal{F}$ be a quasi-coherent G-equivariant $\mathcal{O}_X$-module, and let

$$\theta : \mathrm{pr}_X^*(\mathcal{F}) \xrightarrow{\sim} \lambda^*(\mathcal{F})$$

be the isomorphism of 6.5.1 ($\star$). Since p is quasi-compact and quasi-separated, $p_*(\mathcal{F})$ is quasi-coherent (cf. EGA I, 9.2.1).

Lemma 6.6.1. *The quasi-coherent $\mathcal{O}_Y$-module $p_*(\mathcal{F})$ is G-equivariant.*

Indeed, one has the following two cartesian squares.

$$\begin{tikzcd}
G \times_S X \arrow[r, "\mathrm{pr}_X"] \arrow[d, "q"'] & X \arrow[d, "p"] & G \times_S X \arrow[l, "\lambda"'] \arrow[d, "q"] \\
G \times_S Y \arrow[r, "\mathrm{pr}_Y"] & Y & G \times_S Y. \arrow[l, "\mu"']
\end{tikzcd}$$

Figure 15: The two cartesian base-change squares used in Lemma 6.6.1.

Temporary Loop-1 image; source locator: Polo–Gille Exp1-13oct24.pdf, component-local p. 45; combined-reader p. 55.

Since p is quasi-compact and quasi-separated and pr_Y and μ are flat, EGA III, 1.4.15, completed by EGA IV_1, 1.7.21, gives

$$q_* \mathrm{pr}_X^*(\mathcal{F}) = \mathrm{pr}_Y^* p_*(\mathcal{F}), \qquad q_* \lambda^*(\mathcal{F}) = \mu^* p_*(\mathcal{F}).$$

Consequently, θ induces an isomorphism

$$\theta_Y : \mathrm{pr}_Y^* p_*(\mathcal{F}) \xrightarrow{\sim} \mu^* p_*(\mathcal{F}).$$

The same argument shows that θ_Y satisfies the cocycle condition 6.5.1 ($\star\star$), because the morphisms $G \times_S G \times_S Y \to G \times_S Y$ involved are flat. This proves the lemma.

In particular, take $Y = S$ with the trivial action of G. By Remark 6.5.4, $p_*(\mathcal{F})$ is then a G-$\mathcal{O}_S$-module. If, in addition, G is affine over S, and if

$$\mathscr{A}(G) = \pi_*(\mathcal{O}_G),$$

then $p_*(\mathcal{F})$ is an $\mathscr{A}(G)$-comodule, by 4.7.2.

On the other hand, by Proposition 6.4.1, the functor $\prod_{X/S} W(\mathcal{F})$, which assigns to every $f : T \to S$

$$W\big(f_X^*(\mathcal{F})\big)(X_T) = \Gamma\big(X_T, f_X^*(\mathcal{F})\big) = \Gamma\big(T, p_{T*} f_X^*(\mathcal{F})\big),$$

is a G-$\mathcal{O}_S$-module. There is moreover a canonical morphism

$$\tau : W\big(p_*(\mathcal{F})\big) \longrightarrow \prod_{X/S} W(\mathcal{F}),$$

which, for each $f : T \to S$, is given by the canonical morphism

$$\Gamma\big(T, f^* p_*(\mathcal{F})\big) \longrightarrow \Gamma\big(T, p_{T*} f_X^*(\mathcal{F})\big).$$

It is an isomorphism when restricted to the full subcategory of schemes T flat over S, and it is immediate that τ is a morphism of G-$\mathcal{O}_S$-modules. We have therefore proved the following proposition; for part (ii), compare [GIT], p. 32.

Proposition 6.6.2. *Let S be a scheme, let G be an S-group scheme acting on an S-scheme X, and let $\mathcal{F}$ be a quasi-coherent G-equivariant $\mathcal{O}_X$-module. Suppose that $\pi : G \to S$ is flat and $p : X \to S$ is quasi-compact and quasi-separated.*

(i) Then $p_*(\mathcal{F})$ is a quasi-coherent G-$\mathcal{O}_S$-module. The canonical morphism

$$W\big(p_*(\mathcal{F})\big) \longrightarrow \prod_{X/S} W(\mathcal{F})$$

is a morphism of G-$\mathcal{O}_S$-modules, and the two functors coincide on the category of flat S-schemes.

(ii) If, in addition, G is affine over S, and if $\mathscr{A}(G) = \pi_*(\mathcal{O}_G)$, then $p_*(\mathcal{F})$ has the structure of an $\mathscr{A}(G)$-comodule.[56]

6.7. Stabilizers

Let S be a scheme, let G be an S-group scheme acting on an S-scheme X, and let $\mathcal{F}$ be a quasi-coherent G-equivariant $\mathcal{O}_X$-module. Let Y be a second S-scheme with an action of G, possibly the trivial action, let $\phi : Y \to X$ be an S-morphism, not necessarily G-equivariant, and let $\mathrm{Stab}_G(\phi)$ be the stabilizer of ϕ in G (cf. 6.3.2).

Let us point out at once (see Exposé $\mathrm{VI_B}$, Section 6) that $\mathrm{Stab}_G(\phi)$ is representable by a closed subgroup scheme H of G if X is separated over S and Y is essentially free over S (cf. *loc. cit.*, Definition 6.2.1). Indeed, consider

$$r : G \times_S Y \longrightarrow X \times_S X, \qquad r(g,y) = \big(\phi(y), g\phi(g^{-1}y)\big),$$

put $P = G \times_S Y$, and let P' be the inverse image under r of the diagonal $\Delta_{X/S}$. Then (cf. *loc. cit.*, 6.2.4(a))

$$\mathrm{Stab}_G(\phi) = \prod_{P/G} P'.$$

It follows that $\mathrm{Stab}_G(\phi)$ is representable by a closed subgroup scheme H of G if X is separated over S and Y is essentially free over S. The latter condition is automatic if S is the spectrum of a field or if $Y = S$. Under these hypotheses, $\phi^*(\mathcal{F})$ is a quasi-coherent H-equivariant $\mathcal{O}_Y$-module (cf. 6.5.2).

If, in addition, $\pi : G \to S$ and $p : Y \to S$ are quasi-compact and quasi-separated over S, and π is flat, then $p_*\phi^*(\mathcal{F})$ is an H-$\mathcal{O}_S$-module by Proposition 6.6.2. In particular:

Corollary 6.7.1. *Let S be a scheme, let G be an S-group scheme acting on an S-scheme X, and let $\mathcal{F}$ be a quasi-coherent G-equivariant $\mathcal{O}_X$-module. Suppose that $\pi : G \to S$ is flat and $X \to S$ is quasi-compact and separated. Let $\tau : S \to X$ be a section of X over S.*

(i) The stabilizer $H = \mathrm{Stab}_G(\tau)$ is a closed subgroup scheme of G, and $\tau^*(\mathcal{F})$ is a quasi-coherent H-$\mathcal{O}_S$-module.

(ii) If H is moreover affine over S, for example if G is, and if $\mathscr{A}(H) = \pi_*(\mathcal{O}_H)$, then $\tau^*(\mathcal{F})$ is an $\mathscr{A}(H)$-comodule.

[56]Editor's note: The same holds if G is flat, quasi-compact, and quasi-separated over S, and if $p_*(\mathcal{F})$ is a flat $\mathcal{O}_S$-module; cf. Exposé $\mathrm{VI_B}$, 11.6.1(ii).

6.8. G-equivariant sheaves on G

To conclude, we give two results, 6.8.1 and 6.8.6, which will be used in Exposés II and III, in particular in III, 4.25.

Proposition 6.8.1. *Let S be a scheme, let $\pi : G \to S$ be an S-group scheme, and let $\varepsilon : S \to G$ be the unit section. Consider the action of G on itself by left translations. The functors*

$$\mathcal{E} \longmapsto \pi^*(\mathcal{E}), \qquad \mathcal{F} \longmapsto \varepsilon^*(\mathcal{F})$$

induce mutually quasi-inverse equivalences between the category of quasi-coherent $\mathcal{O}_S$-modules and the category of quasi-coherent G-equivariant $\mathcal{O}_G$-modules.

Proof. Let μ denote multiplication in G, and let $\mathrm{pr}_2 : G \times_S G \to G$ be the second projection. Since $\pi \circ \mu = \pi \circ \mathrm{pr}_2$, every quasi-coherent $\mathcal{O}_S$-module $\mathcal{E}$ has a canonical isomorphism

$$\mu^*\pi^*(\mathcal{E}) = \mathrm{pr}_2^*\pi^*(\mathcal{E}).$$

It is immediate that this isomorphism satisfies the cocycle condition 6.5.1 ($\star\star$), so $\pi^*(\mathcal{E})$ is a G-equivariant $\mathcal{O}_G$-module. Since $\varepsilon^*\pi^*(\mathcal{E}) = \mathcal{E}$, the functor $\mathcal{E} \mapsto \pi^*(\mathcal{E})$ is

fully faithful. It remains to show that, for every G-equivariant $\mathcal{O}_G$-module $\mathcal{F}$,

$$\mathcal{F} \simeq \pi^*\varepsilon^*(\mathcal{F}).$$

By hypothesis, there is an isomorphism

$$\theta : \mathrm{pr}_2^*(\mathcal{F}) \xrightarrow{\sim} \mu^*(\mathcal{F}).$$

Pulling θ back along the morphism

$$\tau : G \longrightarrow G \times_S G, \qquad \tau = (\mathrm{id}_G, \varepsilon \circ \pi),$$

gives an isomorphism $\mathcal{F} \xrightarrow{\sim} \pi^*\varepsilon^*(\mathcal{F})$.

Remarks 6.8.2. (a) Consider the action of $G \times_S G$ on G defined by

$$(g_1, g_2) \cdot g = g_1 g g_2^{-1}.$$

The stabilizer of the unit section $\varepsilon : S \to G$ is the diagonal subgroup H of $G \times_S G$. Consequently, if $\mathcal{F}$ is a quasi-coherent $(G \times_S G)$-equivariant $\mathcal{O}_G$-module, then $\mathcal{E} = \varepsilon^*(\mathcal{F})$ has the structure of an H-$\mathcal{O}_S$-module, by 6.5.2.

(b) One can show that $\mathcal{F} \mapsto \varepsilon^*(\mathcal{F})$ is an equivalence between the category of quasi-coherent $(G \times_S G)$-equivariant $\mathcal{O}_G$-modules and the category of quasi-coherent H-$\mathcal{O}_S$-modules. This is a special case of more general descent results (cf. Exposé IV, Section 2, and SGA 1, VIII); see, for example, [Th87], 1.2–1.3.

Remark 6.8.3. Retain the notation of 6.8.1. For every quasi-coherent $\mathcal{O}_S$-module $\mathcal{E}$, let $\pi_*^G\pi^*(\mathcal{E})$ be the $\mathcal{O}_S$-submodule of $\pi_*\pi^*(\mathcal{E})$ whose sections over an open subset V of S are the elements

$$\gamma \in \Gamma\big(\pi^{-1}(V), \pi^*(\mathcal{E})\big)$$

such that $g \cdot \gamma_{S'} = \gamma_{S'}$ for every $S' \to V$ and $g \in G(S')$. Then the natural morphism

$$\mathcal{E} \longrightarrow \pi_*^G\pi^*(\mathcal{E})$$

is an isomorphism. This is immediate if

$$\pi_*\pi^*(\mathcal{E}) = \mathcal{E} \otimes_{\mathcal{O}_S} \pi_*(\mathcal{O}_G),$$

for example if $G \to S$ is affine, or if $G \to S$ is quasi-compact and quasi-separated and $\mathcal{E}$ is flat; the general case is verified without difficulty.

Remark 6.8.4. Let S be a scheme, let H be an S-group scheme acting on an S-scheme X, and let $\mathcal{F}$ be an $\mathcal{O}_X$-module. Suppose H is flat over S, and let $W_P(\mathcal{F})$ denote the restriction of $W(\mathcal{F})$ to the full subcategory of flat S-schemes. By 6.5.1, endowing $\mathcal{F}$ with an H-equivariant module structure is equivalent to giving an isomorphism

$$\theta : \mathrm{pr}_X^*(\mathcal{F}) \longrightarrow \lambda^*(\mathcal{F})$$

of sheaves on $H \times_S X$ which satisfies the cocycle condition $(\star\star)$. Hence, to endow $\mathcal{F}$ with an H-equivariant module structure, it suffices to endow $W_P(\mathcal{F})$ with such a structure.

Recollection 6.8.5. Let X be an S-scheme and Y a closed S-subscheme of X. Let $\mathcal{N}_{Y/X}$ denote the conormal sheaf of the immersion $i : Y \hookrightarrow X$ (cf. EGA IV$_4$, 16.1.2). If $S' \to S$ is flat and $i' : Y' \hookrightarrow X'$ is obtained from i by base change, then *loc. cit.*, 16.2.2(iii), gives

$$\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'} = \mathcal{N}_{Y'/X'}.$$

Proposition 6.8.6. *Let S be a scheme, let X be an S-group scheme, and let Y be an S-subgroup scheme of X. Suppose that Y is flat over S. Then the conormal sheaf $\mathcal{N}_{Y/X}$ is a $(Y \times_S Y)$-equivariant $\mathcal{O}_Y$-module.*

Indeed, $H = Y \times_S Y$ is flat over S. By Remark 6.8.4, it is therefore enough to endow $W_P(\mathcal{N}_{Y/X})$ with an H-equivariant module structure. Let S' be a flat S-scheme, let $Y' \hookrightarrow X'$ be the immersion obtained by base change, and put

$$\mathcal{N}' = \mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'}.$$

By 6.8.5, $\mathcal{N}' = \mathcal{N}_{Y'/X'}$. Every $h \in H(S')$ induces an automorphism of Y', and hence, for every $y \in Y'(S')$, isomorphisms

$$\Gamma\big(S', y^*(\mathcal{N}')\big) \xrightarrow{\sim} \Gamma\big(S', y^*h^*(\mathcal{N}')\big)$$

which endow $W_P(\mathcal{N}_{Y/X})$ with an H-equivariant module structure (cf. 6.1).

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Exposé II
Tangent Bundles – Lie Algebras

by M. Demazure

We propose in this exposé to construct, in the theory of schemes, the analogues of the tangent bundles and Lie algebras of classical theory. It will, however, be useful not to restrict ourselves to schemes proper, but also to consider certain functors on the category of schemes that are not necessarily representable (for example, the functors $\underline{\mathrm{Hom}}$, $\underline{\mathrm{Norm}}$, and so forth). As announced in the preceding exposé (cf. I 1.1), we shall identify a scheme with its associated functor.[0]

On the other hand, the constructions presented below go beyond the setting of scheme theory. They are equally valid, for example, in the theory of analytic spaces with nilpotent elements, apart from a few changes of detail.

Before beginning this construction, we must give a few general definitions that supplement those of I 1.7.

1. The functors $\underline{\mathrm{Hom}}_{Z/S}(X, Y)$

We resume the notation of I 1.1. We identify the category $\mathscr{C}$ with a full subcategory of

$$\widehat{\mathscr{C}} = \mathrm{Hom}(\mathscr{C}^{\circ}, \mathbf{Set})$$

(in particular, we suppress the underlinings that allowed us to distinguish graphically an object of $\widehat{\mathscr{C}}$ from an object of $\mathscr{C}$).[1]

Consider the following situation: four objects of $\widehat{\mathscr{C}}$, denoted by S, X, Y, Z, the first of which is in fact an object of $\mathscr{C}$, with X and Y over Z, and Z over S.

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[0]Editor's note: version of October 14, 2024.

[1]Editor's note: in this edition these are rendered by bold letters $\mathbf{F}, \mathbf{G}, \mathbf{V}, \mathbf{W}$; cf. Exposé I. For functors such as $\underline{\mathrm{Norm}}$ and $\underline{\mathrm{Centr}}$ (cf. 5.2), however, the underlinings of the original have been retained, and one has been added to the functor $\underline{\mathrm{Lie}}$, in order to distinguish the functor $\underline{\mathrm{Lie}}(G/S)$ from the $\Gamma(S, \mathcal{O}_S)$-Lie algebra $\mathrm{Lie}(G/S) = \underline{\mathrm{Lie}}(G/S)(S)$, used, for example, in Exposé $\mathrm{VII_A}$.

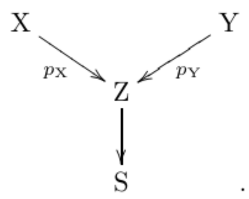

Figure 16: The objects X and Y over Z, and Z over S.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 2; combined-reader p. 60.

Definition 1.1.

We define an object of $\widehat{\mathscr{C}}_{/S}$, denoted $\underline{\mathrm{Hom}}_{Z/S}(X,Y)$, by

$$\underline{\mathrm{Hom}}_{Z/S}(X,Y)(S') = \mathrm{Hom}_{Z_{S'}}(X_{S'}, Y_{S'}) = \mathrm{Hom}_Z(X \times_S S', Y), \tag{1}$$

for every object S' of $\mathscr{C}_{/S}$. One sees at once that $\underline{\mathrm{Hom}}_{Z/S}(X,Y)$ is simply the subobject of $\underline{\mathrm{Hom}}_S(X,Y)$ formed by the morphisms compatible with p_X and p_Y; that is, the kernel of the pair of morphisms

$$\underline{\mathrm{Hom}}_S(X,Y) \rightrightarrows \underline{\mathrm{Hom}}_S(X,Z),$$

the first defined by composition with p_Y, and the second being the constant morphism whose "image" is p_X.

As in I 1.7, for every object T of $\widehat{\mathscr{C}}$ over S, there is a natural bijection

$$\mathrm{Hom}_S\big(T, \underline{\mathrm{Hom}}_{Z/S}(X,Y)\big) \simeq \mathrm{Hom}_Z(X \times_S T, Y). \tag{2}$$

Moreover, by I 1.7.1, if E, F are objects of $\widehat{\mathscr{C}}$ over Z, then

$$\mathrm{Hom}_Z\big(E, \underline{\mathrm{Hom}}_Z(F,Y)\big) \simeq \mathrm{Hom}_Z(E \times_Z F, Y) \simeq \mathrm{Hom}_Z\big(F, \underline{\mathrm{Hom}}_Z(E,Y)\big).$$

Applying this with $E = X$ and $F = Z \times_S T$, one obtains natural bijections, for every object T of $\widehat{\mathscr{C}}_{/S}$,

$$\begin{aligned}\mathrm{Hom}_S\big(T, \underline{\mathrm{Hom}}_{Z/S}(X,Y)\big) &\simeq \mathrm{Hom}_Z(X \times_S T, Y)\\ &\simeq \begin{cases}\mathrm{Hom}_Z\big(Z \times_S T, \underline{\mathrm{Hom}}_Z(X,Y)\big),\\ \mathrm{Hom}_Z\big(X, \underline{\mathrm{Hom}}_Z(Z \times_S T, Y)\big).\end{cases}\end{aligned} \tag{3}$$

These bijections are functorial in T, and hence give isomorphisms of S-functors.[2]

(4)

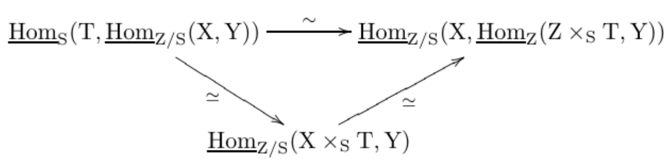

Figure 17: The functorial isomorphisms (4).

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 2; combined-reader p. 60.

[2]Editor's note: points (2), (3), and (4) have been expanded; in particular, (4) will be used in 3.11.

Let us point out two special cases of the definition. If $Z = S$, then

$$\underline{\mathrm{Hom}}_{S/S}(X, Y) = \underline{\mathrm{Hom}}_S(X, Y).$$

On the other hand, when $X = Z$, we set

$$\prod_{Z/S} Y = \underline{\mathrm{Hom}}_{Z/S}(Z, Y), \tag{5}$$

so that, by definition,

$$\left(\prod_{Z/S} Y\right)(S') = \mathrm{Hom}_Z(Z \times_S S', Y) \simeq \Gamma(Y_{S'}/Z_{S'}).$$

The functor $\prod_{Z/S} : \widehat{\mathscr{C}}_{/Z} \to \widehat{\mathscr{C}}_{/S}$ is right adjoint to the base-change functor from S to Z: for every S-functor U,

$$\mathrm{Hom}_S\left(U, \prod_{Z/S} Y\right) = \mathrm{Hom}_Z(U \times_S Z, Y).$$

If $\mathscr{C} = \mathbf{Sch}$ and Z is an S-scheme, the functor $\prod_{Z/S}$ is called the “Weil restriction of scalars.” [3] One also has an isomorphism

$$\underline{\mathrm{Hom}}_{Z/S}(X, Y) \simeq \underline{\mathrm{Hom}}_{X/S}(X, Y \times_Z X) = \prod_{X/S}(Y \times_Z X), \tag{6}$$

which gives in particular, when $Z = S$, an isomorphism

$$\underline{\mathrm{Hom}}_S(X, Y) \simeq \prod_{X/S} Y_X. \tag{7}$$

Remark 1.2.

The functor $Y \mapsto \underline{\mathrm{Hom}}_{Z/S}(X, Y)$ commutes with products in the following sense: there is a functorial isomorphism

$$\underline{\mathrm{Hom}}_{Z/S}(X, Y \times_Z Y') \simeq \underline{\mathrm{Hom}}_{Z/S}(X, Y) \times_S \underline{\mathrm{Hom}}_{Z/S}(X, Y'). \tag{*}$$

It follows that if Y is a Z-group, respectively a Z-ring, and so on, then $\underline{\mathrm{Hom}}_{Z/S}(X, Y)$ is an S-group, respectively an S-ring, and so on.

Remark 1.3.

Let $\pi : M \to Y$ be a Y-functor in $\mathcal{O}_Y$-modules (cf. I 4.3.3.1), and set $H = \underline{\mathrm{Hom}}_{Z/S}(X, Y)$. Then $\underline{\mathrm{Hom}}_{Z/S}(X, M)$ has a natural $\mathcal{O}_H$-module structure; more precisely, for every H' over H, the set

$$\mathrm{Hom}_H\big(H', \underline{\mathrm{Hom}}_{Z/S}(X, M)\big)$$

has a natural $\mathcal{O}(X \times_S H')$-module structure. [4]

Indeed, let $m : M \times_Y M \to M$ and $\lambda : \mathcal{O}_Y \times_Y M \to M$ denote the morphisms defining the structures of a Y-abelian group and an $\mathcal{O}_Y$-module. Let H' be an S-scheme over

[3] Editor’s note: the preceding two sentences have been added.

[4] Editor’s note: this remark has been added; it is useful for Corollary 3.11.1.

$H = \underline{\mathrm{Hom}}_{Z/S}(X, Y)$; equivalently, one has been given a Z-morphism $f : X \times_S H' \to Y$, which makes $X \times_S H'$ a Y-object. Then

$$\mathrm{Hom}_H\big(H', \underline{\mathrm{Hom}}_{Z/S}(X, M)\big)$$

is the set of Z-morphisms $\phi : X \times_S H' \to M$ such that $\pi \circ \phi = f$, that is, the set of Y-morphisms $X \times_S H' \to M$.

If ϕ, ψ are two such morphisms, define $\phi + \psi$ to be the composite Y-morphism

$$X \times_S H' \xrightarrow{\phi\times\psi} M \times_Y M \xrightarrow{m} M.$$

This makes $\underline{\mathrm{Hom}}_{Z/S}(X, M)$ an abelian group over $H = \underline{\mathrm{Hom}}_{Z/S}(X, Y)$. [5]

Similarly, if $a \in \mathcal{O}(X \times_S H')$, that is, if $a : X \times_S H' \to \mathcal{O}_S$ is an S-morphism, define $a\phi$ to be the composite $\lambda \circ (a \times \phi)$, where $a \times \phi$ is the Y-morphism from $X \times_S H'$ to $\mathcal{O}_Y \times_Y M \simeq \mathcal{O}_S \times_S M$ with components a and ϕ. This gives $\mathrm{Hom}_H(H', \underline{\mathrm{Hom}}_{Z/S}(X, M))$ an $\mathcal{O}(X \times_S H')$-module structure, functorial in the H-object H'.

2. The schemes $I_S(\mathcal{M})$

Definition 2.1.

Let S be a scheme and $\mathcal{M}$ a quasi-coherent $\mathcal{O}_S$-module. Denote by $\mathscr{D}_{\mathcal{O}_S}(\mathcal{M})$ the quasi-coherent $\mathcal{O}_S$-algebra $\mathcal{O}_S \oplus \mathcal{M}$, where $\mathcal{M}$ is regarded as an ideal of square zero. Denote by

$$I_S(\mathcal{M}) = \mathrm{Spec}\, \mathscr{D}_{\mathcal{O}_S}(\mathcal{M})$$

the corresponding S-scheme.[6] In particular, put

$$\mathscr{D}_{\mathcal{O}_S} = \mathscr{D}_{\mathcal{O}_S}(\mathcal{O}_S), \qquad I_S = I_S(\mathcal{O}_S),$$

and call these, respectively, the algebra of dual numbers over S and the scheme of dual numbers over S.

The assignment $\mathcal{M} \mapsto I_S(\mathcal{M})$ is a contravariant functor from quasi-coherent $\mathcal{O}_S$-modules to S-schemes. In particular, the morphisms $0 \to \mathcal{M}$ and $\mathcal{M} \to 0$ define, respectively, the structural morphism

$$\rho : I_S(\mathcal{M}) \longrightarrow I_S(0) = S$$

and a section $\varepsilon_{\mathcal{M}}$ of it, called the *zero section.*[7]

2.1.1.

Since $\mathcal{M} \mapsto I_S(\mathcal{M})$ is contravariant, every $a \in \mathrm{End}_{\mathcal{O}_S}(\mathcal{M})$ defines an S-endomorphism a^* of $I_S(\mathcal{M})$, and

$$1^* = \mathrm{id}, \qquad (ab)^* = b^* \circ a^*, \qquad 0^* = \varepsilon_{\mathcal{M}} \circ \rho, \qquad a^* \circ \varepsilon_{\mathcal{M}} = \varepsilon_{\mathcal{M}}.$$

Consequently, $I_S(\mathcal{M})$ carries a right action of the multiplicative monoid of $\mathrm{End}_{\mathcal{O}_S}(\mathcal{M})$, commuting with the S-morphisms $I_S(\mathcal{M}) \to I_S(\mathcal{M}')$ arising from morphisms $\mathcal{M}' \to \mathcal{M}$; in particular, the operators a^* preserve the zero section.[8]

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[5] Editor's note: if M is a Y-group, not necessarily abelian, and one sets $\phi \cdot \psi = m \circ (\phi \times \psi)$, this makes $\underline{\mathrm{Hom}}_{Z/S}(X, M)$ a group over $\underline{\mathrm{Hom}}_{Z/S}(X, Y)$.

[6] Editor's note: the underlying space of $I_S(\mathcal{M})$ is the same as that of S.

[7] Editor's note: compare 3.1.1.

[8] Editor's note: what follows expands the original; the numbering 2.1.1–2.1.3 has also been added.

For every $a \in \mathrm{End}_{\mathcal{O}_S}(\mathcal{M})$, every $f : S' \to S$, and every $m \in I_S(\mathcal{M})(S')$, write $m \cdot a = a^*(m)$. Then

$$m \cdot 1 = m, \qquad (m \cdot a) \cdot b = m \cdot (ab), \qquad m \cdot 0 = \varepsilon_{\mathcal{M}}(\rho(m)),$$

and if $m = \varepsilon_{\mathcal{M}}(f)$, then $m \cdot a = m$. [9]

Remark 2.1.2.

Formation of $I_S(\mathcal{M})$ commutes with base extension: there are canonical isomorphisms

$$I_S(\mathcal{M})_{S'} \simeq I_{S'}(\mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}).$$

For simplicity write $I_{S'}(\mathcal{M}) = I_S(\mathcal{M})_{S'}$. More generally, if X is an S-functor, not necessarily representable, write $I_X(\mathcal{M}) = I_S(\mathcal{M}) \times_S X$.

2.1.3.

By the preceding discussion, the multiplicative monoid of $\mathcal{O}(S')$ acts on the S'-scheme $I_{S'}(\mathcal{M})$, functorially in $\mathcal{M}$. In other words, the S-scheme $I_S(\mathcal{M})$ is an object with an operator-monoid structure $\mathcal{O}_S$, functorial in $\mathcal{M}$. Thus there is a morphism of S-schemes

$$\lambda : I_S(\mathcal{M}) \times_S \mathcal{O}_S \longrightarrow I_S(\mathcal{M}),$$

satisfying the evident conditions. For every S-functor X, base change gives a morphism of X-functors

$$\lambda_X : I_X(\mathcal{M}) \times_S \mathcal{O}_S \longrightarrow I_X(\mathcal{M}),$$

making $I_X(\mathcal{M})$ an object with operator monoid $\mathcal{O}(X)$. Every $a \in \mathcal{O}(X) = \mathrm{Hom}_S(X, \mathcal{O}_S)$ defines the X-endomorphism

$$a^* = \lambda_X \circ \big(\mathrm{id}_{I_X(\mathcal{M})} \times (a \circ \mathrm{pr}_X)\big).$$

Explicitly, if $x \in X(S')$ and $m \in I_S(\mathcal{M})(S') = I_{S'}(\mathcal{M})(S')$, then $a(x) = a \circ x \in \mathcal{O}(S')$ and

$$(m, x) \cdot a = (m \cdot a(x), x).$$

This operation is functorial in $\mathcal{M}$ and preserves the zero section $\varepsilon_{\mathcal{M}} : X \to I_X(\mathcal{M})$; thus $a^* \circ \varepsilon_{\mathcal{M}} = \varepsilon_{\mathcal{M}}$. [10]

The operation is also "functorial in X" in the following sense. If $\pi : Y \to X$ is a morphism of S-functors and $u : \mathcal{O}(X) \to \mathcal{O}(Y)$ the corresponding ring morphism, so that $u(a) = a \circ \pi$, then the following diagram is commutative.

$$\begin{tikzcd}
\mathrm{I}_{\mathrm{X}}(\mathscr{M}) \arrow[r, "a^*"] & \mathrm{I}_{\mathrm{X}}(\mathscr{M}) \\
\mathrm{I}_{\mathrm{Y}}(\mathscr{M}) \arrow[u, "\pi"] \arrow[r, "u(a)^*"] & \mathrm{I}_{\mathrm{Y}}(\mathscr{M}) \arrow[u, "\pi"']
\end{tikzcd} \quad .$$

Figure 18: Functoriality of the operator action under $Y \to X$.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 5; combined-reader p. 63.

[9] Editor's note: later we shall be chiefly interested in $a \in \mathcal{O}(S)$, acting by homotheties on $\mathcal{M}$. For example, if $\mathcal{M}$ is free of rank r, then for every $S' \to S$, $\mathrm{Hom}_S(S', I_S(\mathcal{M}))$ identifies with the set of r-tuples $(\epsilon_1, \dots, \epsilon_r) \in \mathcal{O}(S')^r$ satisfying $\epsilon_i \epsilon_j = 0$ for all i, j, and $(\epsilon_1, \dots, \epsilon_r) \cdot a = (a\epsilon_1, \dots, a\epsilon_r)$ for every $a \in \mathcal{O}(S)$.

[10] Editor's note: this paragraph has been added; it will be useful in 3.4.2 and 4.6.2.

2.2.

Let $\mathcal{M}$ and $\mathcal{N}$ be two quasi-coherent $\mathcal{O}_S$-modules. The commutative diagram below induces the corresponding diagram of S-schemes on the following page.

$$\begin{tikzcd}
 & \mathscr{M} \oplus \mathscr{N} \arrow[dl] \arrow[dr] & \\
\mathscr{M} \arrow[dr] & & \mathscr{N} \arrow[dl] \\
 & 0 &
\end{tikzcd}$$

Figure 19: The direct-sum diagram of $\mathcal{O}_S$-modules.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 5; combined-reader p. 63.

$$\begin{tikzcd}
 & \mathrm{I_S}(\mathscr{M} \oplus \mathscr{N}) & \\
\mathrm{I_S}(\mathscr{M}) \arrow[ur] & & \mathrm{I_S}(\mathscr{N}) \arrow[ul] \\
 & \mathrm{S} \arrow[ul, "\varepsilon_{\mathscr{M}}"] \arrow[ur, "\varepsilon_{\mathscr{N}}"'] \arrow[uu, "\varepsilon_{\mathscr{M}\oplus\mathscr{N}}"'] &
\end{tikzcd} .$$

Figure 20: The induced commutative diagram of S-schemes.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 6; combined-reader p. 64.

Proposition 2.2.

For every S-scheme X, the diagram of functors over S obtained by applying $\underline{\mathrm{Hom}}_S(-, X)$ to the preceding diagram is cartesian.

$$\begin{tikzcd}
 & \underline{\mathrm{Hom}}_{\mathrm{S}}(\mathrm{I_S}(\mathscr{M} \oplus \mathscr{N}), \mathrm{X}) \arrow[dl] \arrow[dr] & \\
\underline{\mathrm{Hom}}_{\mathrm{S}}(\mathrm{I_S}(\mathscr{M}), \mathrm{X}) \arrow[dr] & & \underline{\mathrm{Hom}}_{\mathrm{S}}(\mathrm{I_S}(\mathscr{N}), \mathrm{X}) \arrow[dl] \\
 & \underline{\mathrm{Hom}}_{\mathrm{S}}(\mathrm{S}, \mathrm{X}) = \mathrm{X} &
\end{tikzcd} .$$

Figure 21: The cartesian diagram of $\underline{\mathrm{Hom}}_S$-functors.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 6; combined-reader p. 64.

Indeed, for every $S' \to S$, one must verify that the diagram of sets obtained by evaluating the functors on S' is cartesian. Since formation of $I_S(\mathcal{P})$ commutes with base extension, it is enough to take $S' = S$, and hence to verify that the following diagram of sets is cartesian.

$$\begin{tikzcd}
 & \mathrm{X}(\mathrm{I_S}(\mathscr{M} \oplus \mathscr{N})) \arrow[dl] \arrow[dr] \arrow[dd, "\mathrm{X}(\varepsilon_{\mathscr{M}\oplus\mathscr{N}})"] & \\
\mathrm{X}(\mathrm{I_S}(\mathscr{M})) \arrow[dr, "\mathrm{X}(\varepsilon_{\mathscr{M}})"'] & & \mathrm{X}(\mathrm{I_S}(\mathscr{N})) \arrow[dl, "\mathrm{X}(\varepsilon_{\mathscr{N}})"] \\
 & \mathrm{X}(\mathrm{S}) &
\end{tikzcd} .$$

Figure 22: The cartesian diagram after evaluation on S.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 6; combined-reader p. 64.

If $x \in X(S)$, it follows from SGA 1, III 5.1 that $X(\varepsilon_{\mathcal{M}})^{-1}(x)$ is isomorphic, functorially in $\mathcal{M}$, to

$$\mathrm{Hom}_{\mathcal{O}_S}\big(x^*(\Omega^1_{X/S}), \mathcal{M}\big),$$

where $\Omega^1_{X/S}$ denotes the sheaf of relative differentials of X over S. This latter functor in $\mathcal{M}$ plainly carries a direct sum of $\mathcal{O}_S$-modules to the product of the corresponding sets, which proves the proposition.[11]

Corollary 2.2.1.

Let X be an S-scheme and $\mathcal{M}$ a finite free $\mathcal{O}_S$-module. The functor $\underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), X)$, as a functor over X, is isomorphic to a finite product over X of copies of $\underline{\mathrm{Hom}}_S(I_S, X)$.

Nota 2.2.2.

It follows from the proof of the proposition that $\underline{\mathrm{Hom}}_S(I_S, X)$, as an X-functor, is isomorphic to $\mathbf{V}(\Omega^1_{X/S})$ (I 4.6.1), and is therefore represented by the vector fibration[12]

$$V(\Omega^1_{X/S}).$$

3. The tangent bundle, condition (E)

Throughout this section, unless otherwise stated, the letters $\mathcal{M}, \mathcal{M}', \mathcal{N}, \ldots$ will denote finite free $\mathcal{O}_S$-modules (that is, modules isomorphic to finite direct sums of copies of $\mathcal{O}_S$).

We shall systematically use the identifications established in Exposé I. Thus, by a "functor over S" we shall mean indifferently either a functor equipped with a morphism to $S = h_S$, or a functor on the category of objects over S. We shall use "group functor over S," and similar expressions, in the same way.

Definition 3.1.

Let S be a scheme, let $\mathcal{M}$ be a finite free $\mathcal{O}_S$-module, and let X be a functor over S. The *tangent bundle of X over S, relative to the $\mathcal{O}_S$-module $\mathcal{M}$*, is the S-functor

$$T_{X/S}(\mathcal{M}) = \underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), X).$$

[11] Editor's note: see also the addition 0.1.8 in Exposé III.

[12] Editor's note: cf. editor's note (33) of Exposé I.

In particular, the *tangent bundle of X over S* is [13]

$$T_{X/S} = T_{X/S}(\mathcal{O}_S) = \underline{\mathrm{Hom}}_S(I_S, X).$$

The assignment $\mathcal{M} \mapsto T_{X/S}(\mathcal{M})$ is a covariant functor from finite free $\mathcal{O}_S$-modules to S-functors. In particular, the morphisms $\mathcal{M} \to 0$ and $0 \to \mathcal{M}$ define, respectively, an S-morphism

$$\pi_{\mathcal{M}} : T_{X/S}(\mathcal{M}) \longrightarrow T_{X/S}(0) \simeq X$$

and a section τ_0 of it, called the *zero section.* [14]

Remark 3.1.1.

[15] The projection $\pi_{\mathcal{M}} : T_{X/S}(\mathcal{M}) \to X$ is induced by the zero section $\varepsilon_{\mathcal{M}} : S \to I_S(\mathcal{M})$, whereas the zero section $\tau_0 : X \to T_{X/S}(\mathcal{M})$ is induced by the structural morphism $\rho : I_S(\mathcal{M}) \to S$. More explicitly, let $t \in T_{X/S}(\mathcal{M})(S')$ correspond to the S-morphism $f : S' \times_S I_S(\mathcal{M}) \to X$, and let $x \in X(S')$ correspond to $g : S' \to X$. Then

$$\pi(t) = f \circ (\mathrm{id}_{S'} \times \varepsilon_{\mathcal{M}}), \qquad \tau_0(x) = g \circ (\mathrm{id}_{S'} \times \rho).$$

It follows from 3.1 that $\mathcal{M} \mapsto T_{X/S}(\mathcal{M})$ is a covariant functor from finite free $\mathcal{O}_S$-modules to functors over X. In particular, $\mathcal{O}(S)$ is an operator monoid of the X-functor $T_{X/S}(\mathcal{M})$, compatible with functoriality in $\mathcal{M}$.

Scholium 3.1.2.

[15] The preceding discussion means, in particular, the following. For every S-morphism $\varphi : X' \to X$, put

$$\Sigma(X', \mathcal{M}) = \mathrm{Hom}_X\big(X', T_{X/S}(\mathcal{M})\big).$$

The multiplicative monoid $\mathrm{End}_{\mathcal{O}_S}(\mathcal{M})$ acts on $\Sigma(X', \mathcal{M})$; write the action as $(\lambda, x) \mapsto \lambda * x$. Then

$$\lambda * (\mu * x) = (\lambda\mu) * x, \qquad 1 * x = x, \qquad 0 * x = \tau_0 \circ \varphi.$$

There is likewise an action of $\mathrm{End}_{\mathcal{O}_S}(\mathcal{M} \oplus \mathcal{M})$ on $\Sigma(X', \mathcal{M} \oplus \mathcal{M})$. [16]

Let $m : \mathcal{M} \oplus \mathcal{M} \to \mathcal{M}$ be addition, and let $\delta : \mathcal{M} \to \mathcal{M} \oplus \mathcal{M}$ be the diagonal. Write

$$m_{X'} : \Sigma(X', \mathcal{M} \oplus \mathcal{M}) o \Sigma(X', \mathcal{M}), \qquad \delta_{X'} : \Sigma(X', \mathcal{M}) o \Sigma(X', \mathcal{M} \oplus \mathcal{M})$$

for the induced maps. For $\lambda, \mu \in \mathcal{O}(S)$, let ℓ_λ denote multiplication by λ on $\mathcal{M}$, and let $\ell_{\lambda,\mu}$ denote multiplication by (λ, μ) on $\mathcal{M} \oplus \mathcal{M}$. Since

$$m \circ \ell_{\lambda,\lambda} = \ell_\lambda \circ m, \qquad m \circ \ell_{\lambda,\mu} \circ \delta = \ell_{\lambda+\mu},$$

we have, for every $z \in \Sigma(X', \mathcal{M} \oplus \mathcal{M})$ and $x \in \Sigma(X', \mathcal{M})$,

$$\lambda * m(z) = m((\lambda, \lambda) * z), \qquad m((\lambda, \mu) * \delta(x)) = (\lambda + \mu) * x. \qquad (\dagger)$$

[13] Editor's note: when $S = \mathrm{Spec}(k)$ and X is a k-scheme, one has $T_{X/S}(S') = \mathrm{Hom}_S(S' \otimes_k k[\epsilon], X) = X(S' \otimes_k k[\epsilon])$; this recovers one of the usual definitions of the tangent bundle.

[14] Editor's note: the original order "$0 \to \mathcal{M}$ and $\mathcal{M} \to 0$" has been corrected to "$\mathcal{M} \to 0$ and $0 \to \mathcal{M}$."

[15] Editor's note: paragraphs 3.1.1 and 3.1.2 have been added.

[16] Editor's note: in fact, only the actions of $\mathcal{O}(S)$ (respectively $\mathcal{O}(S) \times \mathcal{O}(S)$) on $\Sigma(X', \mathcal{M})$ (respectively $\Sigma(X', \mathcal{M} \oplus \mathcal{M})$) will be used; see below and the proof of 3.6.

Definition 3.2.

Let $u \in X(S) = \mathrm{Hom}_S(S, X) = \Gamma(X/S)$. The *tangent space of X over S at u, relative to $\mathcal{M}$*, denoted $L^u_{X/S}(\mathcal{M})$, is the S-functor obtained from the X-functor $T_{X/S}(\mathcal{M})$ by pullback along $u : S \to X$.

$$\begin{tikzcd} \mathrm{L}^u_{\mathrm{X/S}}(\mathscr{M}) \arrow[r] \arrow[d] & \mathrm{T}_{\mathrm{X/S}}(\mathscr{M}) \arrow[d, "\pi"] \\ \mathrm{S} \arrow[r, "u"] & \mathrm{X} \end{tikzcd}$$

Figure 23: The cartesian square defining $L^u_{X/S}(\mathcal{M})$.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 8; combined-reader p. 66.

In particular, $L^u_{X/S}(\mathcal{O}_S)$ is denoted $L^u_{X/S}$ and is called the *tangent space of X over S at u*.

Remark 3.2.1.

[17] It follows from 3.1.1 that, for every $t : S' \to S$, $L^u_{X/S}(\mathcal{M})(S')$ is the set of S-morphisms $f : I_{S'}(\mathcal{M}) \to X$ such that $f \circ \varepsilon_{\mathcal{M}} = u \circ t$, where $\varepsilon_{\mathcal{M}} : S' \to I_{S'}(\mathcal{M})$ is the zero section.

Proposition 3.3.

If X is represented by an S-scheme $\widetilde{X}$, then $T_{X/S}(\mathcal{M})$ and $L^u_{X/S}(\mathcal{M})$ are representable. In particular, $T_{X/S}$ and $L^u_{X/S}$ are represented by the vector fibrations

$$V(\Omega^1_{\widetilde{X}/S}) \qquad \text{and} \qquad V(u^*\Omega^1_{\widetilde{X}/S})$$

over $\widetilde{X}$ and S, respectively.

It plainly suffices to prove the assertion for $T_{X/S}(\mathcal{M})$, since the corresponding assertion for $L^u_{X/S}(\mathcal{M})$ follows by pullback. By Corollary 2.2.1, it even suffices to treat $T_{X/S}$, in which case the result is exactly Nota 2.2.2.

Remark 3.3.1.

The proposition gives a particularly simple description of the vector fibration representing $L^u_{X/S}$. The image of the section u of X over S is locally closed, [18] and hence is defined by a quasi-coherent ideal $\mathfrak{m}$ in the scheme induced on an open subscheme of X. The quotient $\mathfrak{m}/\mathfrak{m}^2$ may be regarded as a quasi-coherent module on S; it is this module that defines the vector fibration in question.

For example, let X be an algebraic scheme over a field k, and let u be a k-rational point of X. Let $\mathfrak{m}$ be the maximal ideal of the local ring $\mathcal{O}_{X,u}$, and let $\mathfrak{t}$ be the k-vector space dual to $\mathfrak{m}/\mathfrak{m}^2$; this is the Zariski tangent space of $\mathcal{O}_{X,u}$ at u. In the notation of I 4.6.5.1,

$$L^u_{X/k} = V(\mathfrak{m}/\mathfrak{m}^2) = W(\mathfrak{t}).$$

[17] Editor's note: this remark has been added.
[18] Editor's note: cf. EGA I, 5.3.11.

Returning to the general situation, observe first that $L^u_{X/S}(\mathcal{M})$ is covariant from finite free $\mathcal{O}_S$-modules to functors over S. In particular, $\mathcal{O}(S)$ is a set of operators of the S-functor $L^u_{X/S}(\mathcal{M})$, compatible with functoriality in $\mathcal{M}$. [19]

Proposition 3.4.

Formation of $T_{X/S}(\mathcal{M})$ and $L^u_{X/S}(\mathcal{M})$ commutes with base extension. For every S-scheme S', there are isomorphisms, functorial in $\mathcal{M}$,

$$T_{X_{S'}/S'}(\mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) \xrightarrow{\sim} T_{X/S}(\mathcal{M})_{S'},$$

$$L^{u'}_{X_{S'}/S'}(\mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) \xrightarrow{\sim} L^u_{X/S}(\mathcal{M})_{S'}, \qquad u' = u_{S'}.$$

This follows immediately from the compatibility of Hom with base extension.

Corollary 3.4.1.

The X-functor $T_{X/S}(\mathcal{M})$, respectively the S-functor $L^u_{X/S}(\mathcal{M})$, is naturally an object with operators $\mathcal{O}_X$, respectively $\mathcal{O}_S$, and these structures are functorial in $\mathcal{M}$.

For $L^u_{X/S}(\mathcal{M})$, this is immediate. For each S' over S, the ring $\mathcal{O}(S')$ acts on $\mathcal{M} \otimes \mathcal{O}_{S'}$, hence on

$$L^{u'}_{X_{S'}/S'}(\mathcal{M} \otimes \mathcal{O}_{S'}) = L^u_{X/S}(\mathcal{M})_{S'};$$

this action is functorial in S'.

The case of $T_{X/S}(\mathcal{M})$ requires a little more detail. For an X-object $f : X' \to X$, put $T_{X/S}(\mathcal{M})_{X'} = T_{X/S}(\mathcal{M}) \times_X X'$. We must equip

$$T_{X/S}(\mathcal{M})_{X'}(X') = \mathrm{Hom}_X\big(X', T_{X/S}(\mathcal{M})\big)$$

with an action of $\mathcal{O}(X')$, functorially in X'. Let $X_{X'} = X \times_S X'$, and let f' be the section of $X_{X'}$ over X' defined by f. The following diagram provides the required identification.

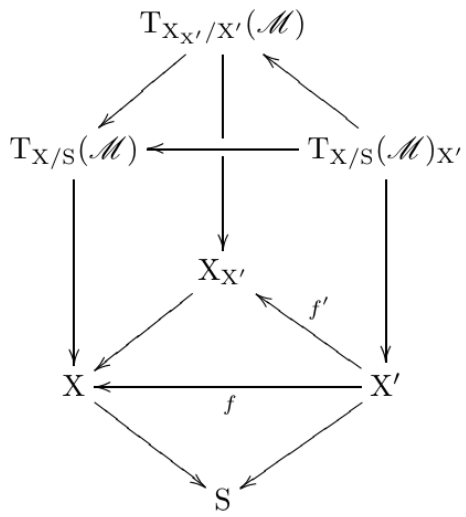

Figure 24: The base-change diagram used to define the $\mathcal{O}_X$-operator structure.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 10; combined-reader p. 68.

[19] Editor's note: cf. 3.1.2.

Together with 3.2.1, the diagram identifies $T_{X/S}(\mathcal{M})_{X'}(X')$ with

$$L^{f'}_{X_{X'}/X'}(\mathcal{M})(X') = \left\{ \ X'\text{-morphisms } \psi : I_{X'}(\mathcal{M}) \to X_{X'} \ \middle| \ \psi \circ \varepsilon_{\mathcal{M}} = f' \ \right\}.$$

Every $a \in \mathcal{O}(X')$ acts through its action on $I_{X'}(\mathcal{M})$: in the notation of 2.1.1,

$$a\psi = \psi \circ a^*, \qquad (a\psi)(x) = \psi(x \cdot a)$$

for every $X'' \to X'$ and every $x \in I_{X'}(\mathcal{M})(X'')$. This construction is plainly functorial in X'. [20] The isomorphisms of Proposition 3.4 are therefore, by construction, isomorphisms for the corresponding $\mathcal{O}_{X_{S'}}$- and $\mathcal{O}_{S'}$-operator structures.

Remark 3.4.2.

[21] The action of $\mathcal{O}_X$ on $T_{X/S}(\mathcal{M})$ can be seen more directly as follows. For every $f : X' \to X$,

$$\mathrm{Hom}_X\big(X', T_{X/S}(\mathcal{M})\big) = \{\varphi \in \mathrm{Hom}_S(I_{X'}(\mathcal{M}), X) \mid \varphi \circ \varepsilon_{\mathcal{M}} = f\}.$$

By 2.1.3, the S-functor $I_{X'}(\mathcal{M})$ carries an action of the monoid $\mathcal{O}(X')$ preserving the zero section. Thus, if a^* denotes the X'-endomorphism defined by $a \in \mathcal{O}(X')$, then

$$a\varphi = \varphi \circ a^*, \qquad (a\varphi)(m, x') = \varphi(m \cdot a(x'), x')$$

for every $S' \to S$ and every $(m, x') \in \mathrm{Hom}_S(S', I_S(\mathcal{M}) \times_S X')$. Notice that $a^* \circ \varepsilon_{\mathcal{M}} = \varepsilon_{\mathcal{M}}$, so that $(a\varphi) \circ \varepsilon_{\mathcal{M}} = f$.

Likewise, for $t : S' \to S$, the set $L^u_{X/S}(\mathcal{M})(S')$ consists of the S-morphisms $\varphi : I_{S'}(\mathcal{M}) \to X$ satisfying $\varphi \circ \varepsilon_{\mathcal{M}} = u \circ t$, and $a\varphi = \varphi \circ a^*$ for $a \in \mathcal{O}(S')$.

Remark 3.4.3.

[21] The observations of 3.1.2 apply equally to the action of $\mathcal{O}_S$ on $L^u_{X/S}(\mathcal{M})$ and to that of $\mathcal{O}_X$ on $T_{X/S}(\mathcal{M})$.

Definition 3.5.

Let S be a scheme and X an S-functor. We say that X *satisfies condition (E) relative to* S if, for every S' over S and every pair of finite free $\mathcal{O}_{S'}$-modules $\mathcal{M}, \mathcal{N}$, the diagram of sets obtained by applying X to diagram $(*)$ of 2.1 is cartesian. [22]

$$\begin{tikzcd}
& \mathrm{X}(\mathrm{I}_{\mathrm{S}'}(\mathscr{M} \oplus \mathscr{N})) \arrow[dl] \arrow[dr] & \\
\mathrm{X}(\mathrm{I}_{\mathrm{S}'}(\mathscr{M})) \arrow[dr] & & \mathrm{X}(\mathrm{I}_{\mathrm{S}'}(\mathscr{N})) \arrow[dl] \\
& \mathrm{X}(\mathrm{S}') & ,
\end{tikzcd}$$

Figure 25: The cartesian diagram in condition (E).

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 11; combined-reader p. 69.

[20] Editor's note: the original has been expanded in what follows.
[21] Editor's note: Remarks 3.4.2 and 3.4.3 have been added.
[22] Editor's note: for examples of S-functors and S-groups that do not satisfy (E), see 6.2 and 6.3.

3.5.1.

[23] Equivalently, the functor $\mathcal{M} \mapsto T_{X/S}(\mathcal{M})$ carries direct sums of finite free $\mathcal{O}_S$-modules to products of X-functors. In that case, the same is true of

$$\mathcal{M} \longmapsto L^u_{X/S}(\mathcal{M}) = S \times_X T_{X/S}(\mathcal{M})$$

for every $u \in \Gamma(X/S)$.

Proposition 2.2 shows that every representable functor satisfies condition (E). We shall sometimes abbreviate "X satisfies condition (E) relative to S" to "X/S satisfies condition (E)."

If X/S satisfies (E), then $\mathcal{M} \mapsto T_{X/S}(\mathcal{M})$ commutes with products and hence carries groups to groups. In particular, $T_{X/S}(\mathcal{M})$ is a commutative X-group; for the same reason, $L^u_{X/S}(\mathcal{M})$ is a commutative S-group.

Proposition 3.6.

If X/S satisfies (E), then the abelian-group structure on $T_{X/S}(\mathcal{M})$, respectively on $L^u_{X/S}(\mathcal{M})$, together with the action of $\mathcal{O}_X$, respectively of $\mathcal{O}_S$, makes it an $\mathcal{O}_X$-module, respectively an $\mathcal{O}_S$-module.

The operator action is functorial in $\mathcal{M}$, and therefore respects the abelian-group structure obtained by functoriality from that of $\mathcal{M}$. [24] Indeed, retaining the notation of 3.1.2, the abelian X-group law on $T_{X/S}(\mathcal{M})$ is the composite

$$T_{X/S}(\mathcal{M}) \times_X T_{X/S}(\mathcal{M}) \simeq T_{X/S}(\mathcal{M} \oplus \mathcal{M}) \xrightarrow{m} T_{X/S}(\mathcal{M}),$$

whereas

$$T_{X/S}(\mathcal{M}) \xrightarrow{\delta} T_{X/S}(\mathcal{M} \oplus \mathcal{M}) \simeq T_{X/S}(\mathcal{M}) \times_X T_{X/S}(\mathcal{M})$$

is the diagonal morphism.

By Remark 3.4.3 and the identities (†) of 3.1.2,

$$\lambda(x+y) = \lambda x + \lambda y, \qquad (\lambda + \mu)x = \lambda x + \mu x$$

for every $f : X' \to X$, every $x, y \in \mathrm{Hom}_X(X', T_{X/S}(\mathcal{M}))$, and every $\lambda, \mu \in \mathcal{O}(X')$.

Remark 3.6.1.

If X is representable, then it satisfies (E), and $T_{X/S}$ and $L^u_{X/S}$ are represented by vector fibrations. The preceding laws are then exactly those induced by the vector-fibration structures (cf. I 4.6). [25]

Proposition 3.4 bis.

If X/S satisfies (E), then $X_{S'}/S'$ satisfies (E), and the isomorphisms of 3.4 respect the $\mathcal{O}_{X_{S'}}$-module and $\mathcal{O}_{S'}$-module structures, respectively.

No further comment is needed.

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[23] Editor's note: what follows has been added.

[24] Editor's note: what follows has been expanded.

[25] Editor's note: explicitly, for every S-morphism $f : X' \to X$, the action of $\mathcal{O}(X')$ on $\mathrm{Hom}_X(X', T_{X/S}(\mathcal{M}))$ corresponds, under the identification

$$\mathrm{Hom}_X(X', T_{X/S}(\mathcal{M})) \simeq \mathrm{Hom}_{\mathcal{O}_{X'}}(f^*\Omega^1_{X/S}, \mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{O}_{X'}),$$

to the natural action on the right. This follows from the proof of SGA 1, III 5.1; see also the addition 0.1.8 in Exposé III.

Proposition 3.7.

The functors $T_{X/S}(\mathcal{M})$ and $L^u_{X/S}(\mathcal{M})$ are functorial in X. Thus, if $f : X \to X'$ is an S-morphism, the following diagrams commute.

$$\begin{tikzcd} \mathrm{T}_{\mathrm{X/S}}(\mathscr{M}) \arrow[r, "\mathrm{T}(f)"] \arrow[d] & \mathrm{T}_{\mathrm{X'/S}}(\mathscr{M}) \arrow[d] \\ \mathrm{X} \arrow[r] & \mathrm{X'} \end{tikzcd} , \qquad \begin{tikzcd} \mathrm{L}^u_{\mathrm{X/S}}(\mathscr{M}) \arrow[rr, "\mathrm{L}(f)"] \arrow[dr] & & \mathrm{L}^{f\circ u}_{\mathrm{X'/S}} \arrow[dl] \\ & \mathrm{S} & \end{tikzcd}$$

Figure 26: Functoriality of tangent bundles and tangent spaces in X.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 12; combined-reader p. 70.

Moreover, if f is a monomorphism, then so are $T(f)$ and $L(f)$. [26] The existence of $T(f)$ and $L(f)$, and the last assertion, follow immediately from the definitions. Commutativity follows from functoriality in $\mathcal{M}$ and from $T_{X/S}(0) = X$.

Remark 3.7.1.

[27] Suppose that X and X' are representable, and let r be the rank of $\mathcal{M}$. By 2.2.2, $T_{X/S}(\mathcal{M})$ is isomorphic to the product over X of r copies of $V(\Omega^1_{X/S})$, and similarly for $T_{X'/S}(\mathcal{M})$. Hence the square in 3.7 is cartesian when f is an open immersion, or more generally when $f^*\Omega^1_{X'/S} = \Omega^1_{X/S}$, for example when f is étale. Under these conditions there is an isomorphism of S-functors

$$L^u_{X/S}(\mathcal{M}) \xrightarrow{\sim} L^{f\circ u}_{X'/S}(\mathcal{M}).$$

In general, the square in 3.7 defines the following morphism of X-functors.

$$\begin{tikzcd} \mathrm{T}_{\mathrm{X/S}}(\mathscr{M}) \arrow[rr] \arrow[dr] & & \mathrm{T}_{\mathrm{X'/S}}(\mathscr{M}) \times_{\mathrm{X'}} \mathrm{X} \arrow[dl] \\ & \mathrm{X} & \end{tikzcd} .$$

Figure 27: The induced morphism over X in the general case.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 13; combined-reader p. 71.

Proposition 3.7 bis.

Let $f : X \to X'$ be an S-morphism. If X and X' satisfy (E) relative to S, then

$$T_{X/S}(\mathcal{M}) \xrightarrow{T(f)} T_{X'/S}(\mathcal{M})_X, \qquad \text{respectively} \qquad L^u_{X/S}(\mathcal{M}) \xrightarrow{L(f)} L^{f\circ u}_{X'/S}(\mathcal{M}),$$

is a morphism of $\mathcal{O}_X$-modules, respectively of $\mathcal{O}_S$-modules.

This follows from Proposition 3.7 by functoriality in $\mathcal{M}$.

[26] Editor's note: the preceding sentence has been added.

[27] Editor's note: the hypothesis that X and X' be representable has been added, and what follows has been expanded.

Proposition 3.8.

For two functors X, Y over S, there are isomorphisms, functorial in $\mathcal{M}$,

$$T_{X/S}(\mathcal{M}) \times_S T_{Y/S}(\mathcal{M}) \xrightarrow{\sim} T_{(X\times_S Y)/S}(\mathcal{M}), \tag{3.8.1}$$

$$L^u_{X/S}(\mathcal{M}) \times_S L^v_{Y/S}(\mathcal{M}) \xrightarrow{\sim} L^{(u,v)}_{(X\times_S Y)/S}(\mathcal{M}). \tag{3.8.2}$$

The first isomorphism follows from 1.2 $(*)$; [28] the second follows from it by base change along $(u, v) : S \to X \times_S Y$.

Remark 3.8.0.

The isomorphism (3.8.1) may also be read as an isomorphism of $X \times_S Y$-functors

$$\big(T_{X/S}(\mathcal{M}) \times_X (X \times_S Y)\big) \underset{X\times_S Y}{\times} \big(T_{Y/S}(\mathcal{M}) \times_Y (X \times_S Y)\big) \xrightarrow{\sim} T_{(X\times_S Y)/S}(\mathcal{M}).$$

Corollary 3.8.1.

If X/S carries an algebraic structure defined by finite cartesian products, then $T_{X/S}(\mathcal{M})$ carries a structure of the same kind, and the projection $T_{X/S}(\mathcal{M}) \to X$ is a morphism for that structure.

Proposition 3.8 bis.

If X/S and Y/S satisfy (E), then $(X \times_S Y)/S$ satisfies (E), and (3.8.1), respectively (3.8.2), is an isomorphism of $\mathcal{O}_{X\times_S Y}$-modules, respectively of $\mathcal{O}_S$-modules.

Proof. [29] By 3.5.1 and (3.8.1), the product $(X \times_S Y)/S$ satisfies (E). Let us prove the assertion about (3.8.1). Given an S-morphism $(x, y) : Z \to X \times_S Y$, Remark 3.4.2 reduces the claim to the $\mathcal{O}(Z)$-linearity of the map

$$\{\varphi \in \mathrm{Hom}_S(I_Z(\mathcal{M}), X) \mid \varphi \circ \varepsilon_{\mathcal{M}} = x\} \times \{\psi \in \mathrm{Hom}_S(I_Z(\mathcal{M}), Y) \mid \psi \circ \varepsilon_{\mathcal{M}} = y\}$$
$$\longrightarrow \{\theta \in \mathrm{Hom}_S(I_Z(\mathcal{M}), X \times_S Y) \mid \theta \circ \varepsilon_{\mathcal{M}} = (x, y)\},$$

which sends (φ, ψ) to $\varphi \times \psi$. This is clear, since for $a \in \mathcal{O}(Z)$,

$$(\varphi \circ a^*) \times (\psi \circ a^*) = (\varphi \times \psi) \circ a^* = a \cdot (\varphi \times \psi).$$

The assertion about (3.8.2) follows similarly from 3.2.1.

3.9.0.

[30] If X is an S-group with unit section $e : S \to X$, write

$$\underline{\mathrm{Lie}}(X/S, \mathcal{M}) = L^e_{X/S}(\mathcal{M}).$$

Thus $\underline{\mathrm{Lie}}(X/S, \mathcal{M})$ is defined by the following cartesian square.

[28] Editor's note: what follows has been expanded.
[29] Editor's note: this proof has been added.
[30] Editor's note: this paragraph has been added to explain the introduction of the notion of an S-H-functor.

$$\begin{tikzcd}
\underline{\mathrm{Lie}}(\mathrm{X}/\mathrm{S}, \mathscr{M}) \arrow[r, "i"] \arrow[d] & \mathrm{T}_{\mathrm{X}/\mathrm{S}}(\mathscr{M}) \arrow[d, "p"] \\
\mathrm{S} \arrow[r, "e"] & \mathrm{X}.
\end{tikzcd}$$

Figure 28: The cartesian square defining $\underline{\mathrm{Lie}}(X/S, \mathcal{M})$.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 14; combined-reader p. 72.

By Corollary 3.8.1, the projection $p : T_{X/S}(\mathcal{M}) \to X$ is a morphism of S-groups. Hence $\underline{\mathrm{Lie}}(X/S, \mathcal{M})$ is an S-group, identified by i with the kernel of p.

If X/S also satisfies (E), Proposition 3.9 will show that this group structure, induced by that of X, agrees with the abelian-group structure induced by functoriality in $\mathcal{M}$ (cf. 3.5.1). In fact, the result holds under the weaker assumption that X be an S-functor in monoids, or more generally an S-functor in H-sets, as defined next.

Definitions 3.9.0.1.

(a) An *H-set* is a set X equipped with a composition law having a two-sided identity, denoted e_X or simply e. [31] If $f : X \to Y$ is a morphism of H-sets, its kernel is $\ker f = f^{-1}(e_Y)$, a sub-H-set of X.

(b) An *H-object* in a category $\mathscr{C}$ is an object X equipped with a morphism $X \times X \to X$ and a section of X over the final object that is a two-sided identity. Every $\mathscr{C}$-monoid, and in particular every $\mathscr{C}$-group, is a $\mathscr{C}$-H-object. An H-object in the category of functors over S will be called an *S-H-functor*.

(c) If X is an S-H-functor (for example, an S-group), and $e : S \to X$ is its identity section, write

$$\underline{\mathrm{Lie}}(X/S, \mathcal{M}) = L^{e}_{X/S}(\mathcal{M}), \qquad \underline{\mathrm{Lie}}(X/S) = \underline{\mathrm{Lie}}(X/S, \mathcal{O}_S).$$

We now spell out the following special case of 3.8.1. [32]

Corollary 3.9.0.2.

If X is an S-H-functor, respectively an S-group, then $T_{X/S}(\mathcal{M})$ and $\underline{\mathrm{Lie}}(X/S, \mathcal{M})$ are likewise S-H-functors, respectively S-groups, and there are morphisms of S-H-functors, respectively of S-groups, as follows.

$$\begin{tikzcd}
\underline{\mathrm{Lie}}(\mathrm{X}/\mathrm{S}, \mathscr{M}) \arrow[r, "i"] & \mathrm{T}_{\mathrm{X}/\mathrm{S}}(\mathscr{M}) \arrow[r, shift left, "p"] & \mathrm{X} \arrow[l, shift left, "s"]
\end{tikzcd} \quad ,$$

Figure 29: The inclusion, projection, and zero section for the tangent functor.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 15; combined-reader p. 73.

Here i identifies $\underline{\mathrm{Lie}}(X/S, \mathcal{M})$ with $\ker p$, and s is a section of p.

[31] Editor's note: this terminology is inspired by the notion of an H-space in topology.
[32] Editor's note: Corollary 3.9.0.2 has been added; it will be used in 4.1.

Proposition 3.9.

Let X be an S-H-functor satisfying (E) relative to S. The S-H-functor structure on $\underline{\mathrm{Lie}}(X/S, \mathcal{M})$ induced by that of X agrees with the S-group structure defined in 3.5.1.

The preceding discussion shows that $\underline{\mathrm{Lie}}(X/S, \mathcal{M})$ is an H-object in the category of $\mathcal{O}_S$-modules. The proposition follows from the next lemma. [33]

Lemma 3.10.

Let $\mathscr{C}$ be a category. Let G be an H-object in the category of $\mathscr{C}$-H-objects. Thus G is a $\mathscr{C}$-H-object, whose composition law we denote $f : G \times G \to G$, together with a morphism of $\mathscr{C}$-H-objects $h : G \times G \to G$. [34] Then $f = h$, and f is commutative.

Evaluating the functors on a variable argument reduces the proof, as usual, to the case in which $\mathscr{C}$ is the category of sets. We then have a set G and two maps $f, h : G \times G \to G$ satisfying

$$h(f(x, y), f(z, t)) = f(h(x, z), h(y, t)).$$

Let $e, u \in G$ be the respective identities for f and h, so that

$$f(e, x) = f(x, e) = x, \qquad h(u, x) = h(x, u) = x.$$

First,

$$h(f(u, y), f(x, u)) = f(x, y) = h(f(x, u), f(u, y)).$$

Putting $y = e$, respectively $x = e$, gives

$$\begin{aligned} x = f(x, e) = h(f(u, e), f(x, u)) = h(u, f(x, u)) = f(x, u), \\ y = f(e, y) = h(f(e, u), f(u, y)) = h(u, f(u, y)) = f(u, y). \end{aligned}$$

Substituting these identities into the original relation yields

$$h(y, x) = f(x, y) = h(x, y).$$

This proves the lemma and Proposition 3.9. The following corollaries now follow from 3.9.

Corollary 3.9.1.

If X is an S-H-functor satisfying (E) relative to S, every element of $X(I_S(\mathcal{M}))$ that maps to the identity element of $X(S)$ is invertible.

Corollary 3.9.2.

If X is an S-monoid satisfying (E) relative to S, an element of $X(I_S(\mathcal{M}))$ is invertible if and only if its image in $X(S)$ is invertible.

Corollary 3.9.3.

If X is an S-group satisfying (E) relative to S, the two S-group laws on $\underline{\mathrm{Lie}}(X/S, \mathcal{M})$ coincide.

[33] Editor's note: this is inspired by the standard proof that the fundamental group of an H-space is abelian.
[34] Editor's note: $G \times G$ is equipped with the law $(G \times G) \times (G \times G) \to G \times G$, $((x, z), (y, t)) \mapsto (f(x, y), f(z, t))$.

Corollary 3.9.4.

[35] Let G be an S-group satisfying (E) relative to S. For $n \in \mathbb{Z}$, let $n_G : G \to G$ be the morphism of S-functors defined by $g \mapsto g^n$. Then the derived morphism

$$L(n_G) : \underline{\mathrm{Lie}}(G/S) \longrightarrow \underline{\mathrm{Lie}}(G/S)$$

is multiplication by n; that is, it sends $x \in \underline{\mathrm{Lie}}(G/S)(S')$ to nx.

The map n_G is not generally a group homomorphism, but it preserves the identity section $e : S \to G$, so its derivative indeed carries $\underline{\mathrm{Lie}}(G/S) = L^e_{G/S}$ to itself. If i denotes the inclusion $\underline{\mathrm{Lie}}(G/S) \hookrightarrow T_{G/S}$, then $L(n_G)$ is characterized by

$$i(L(n_G)(x)) = i(x)^n.$$

By 3.9, the group laws on $\underline{\mathrm{Lie}}(G/S)$ coming from (E) and from the law on G agree. Hence $i(x)^n = i(nx)$, and therefore $L(n_G)(x) = nx$.

Before drawing further consequences from Proposition 3.9, we prove one more functoriality result.

Proposition 3.11.

In the situation of Section 1, there is an isomorphism, functorial in $\mathcal{M}$,

$$T_{\underline{\mathrm{Hom}}_{Z/S}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{Z/S}\big(X, T_{Y/Z}(\mathcal{M})\big).$$

Indeed, by definition,

$$T_{\underline{\mathrm{Hom}}_{Z/S}(X,Y)/S}(\mathcal{M}) = \underline{\mathrm{Hom}}_S\big(I_S(\mathcal{M}), \underline{\mathrm{Hom}}_{Z/S}(X,Y)\big).$$

Applying the isomorphism (4) of 1.1 with $T = I_S(\mathcal{M})$ gives

$$\underline{\mathrm{Hom}}_S\big(I_S(\mathcal{M}), \underline{\mathrm{Hom}}_{Z/S}(X,Y)\big) \simeq \underline{\mathrm{Hom}}_{Z/S}\big(X, \underline{\mathrm{Hom}}_Z(Z \times_S I_S(\mathcal{M}), Y)\big).$$

Since $Z \times_S I_S(\mathcal{M}) \simeq I_Z(\mathcal{M})$, this is

$$\underline{\mathrm{Hom}}_{Z/S}\big(X, \underline{\mathrm{Hom}}_Z(I_Z(\mathcal{M}), Y)\big) = \underline{\mathrm{Hom}}_{Z/S}\big(X, T_{Y/Z}(\mathcal{M})\big).$$

Corollary 3.11.1.

If Y/Z satisfies (E), then $\underline{\mathrm{Hom}}_{Z/S}(X,Y)/S$ satisfies (E), and the isomorphism of 3.11 respects the module structures over $\underline{\mathrm{Hom}}_{Z/S}(X,Y)$. [36]

Proof. Let $\mathcal{M}, \mathcal{N}$ be finite free $\mathcal{O}_S$-modules. If Y/Z satisfies (E), then

$$T_{Y/Z}(\mathcal{M} \oplus \mathcal{N}) \simeq T_{Y/Z}(\mathcal{M}) \times_Y T_{Y/Z}(\mathcal{N}).$$

The right-hand side is a subfunctor of $T_{Y/Z}(\mathcal{M}) \times_S T_{Y/Z}(\mathcal{N})$, and 1.2 $(*)$ therefore gives

$$\underline{\mathrm{Hom}}_{Z/S}\big(X, T_{Y/Z}(\mathcal{M} \oplus \mathcal{N})\big) \simeq \underline{\mathrm{Hom}}_{Z/S}\big(X, T_{Y/Z}(\mathcal{M})\big) \underset{\underline{\mathrm{Hom}}_{Z/S}(X,Y)}{\times} \underline{\mathrm{Hom}}_{Z/S}\big(X, T_{Y/Z}(\mathcal{N})\big).$$

[35] Editor's note: this corollary has been added; it amplifies Remark 4.1.1.2 below.

[36] Editor's note: writing $H = \underline{\mathrm{Hom}}_{Z/S}(X,Y)$, this implies in particular that $T_{H/S}(\mathcal{M})$ is naturally a module over $\prod_{X \times_S H/H} \mathcal{O}_H$. This result appeared somewhat surprising to the editors, so the proof has been given in detail.

Together with 3.11, this says that $\underline{\mathrm{Hom}}_{Z/S}(X, Y)$ satisfies (E), proving the first assertion.

For the second, put $H = \underline{\mathrm{Hom}}_{Z/S}(X, Y)$, and let $\Delta : H' \to H$ be an S-morphism. Equivalently, let $\delta : H' \times_S X \to Y$ be the corresponding Z-morphism; it makes $H' \times_S X$ a Y-object. On the one hand, we have the following commutative diagram.

$$\begin{tikzcd}
\mathrm{Hom_H}(\mathrm{H'}, \underline{\mathrm{Hom}}_{\mathrm{Z/S}}(\mathrm{X}, \mathrm{T_{Y/Z}}(\mathscr{M}))) \arrow[r, hook] \arrow[d, equal] & \mathrm{Hom_S}(\mathrm{H'}, \underline{\mathrm{Hom}}_{\mathrm{Z/S}}(\mathrm{X}, \mathrm{T_{Y/Z}}(\mathscr{M}))) \arrow[d, equal] \\
\mathrm{Hom_Y}(\mathrm{H'} \times_{\mathrm{S}} \mathrm{X}, \mathrm{T_{Y/Z}}(\mathscr{M})) \arrow[r, hook] \arrow[d, equal] & \mathrm{Hom_Z}(\mathrm{H'} \times_{\mathrm{S}} \mathrm{X}, \mathrm{T_{Y/Z}}(\mathscr{M})) \arrow[d, equal] \\
\{\psi \in \mathrm{Hom_Z}(\mathrm{I}_{\mathrm{H'}\times_{\mathrm{S}}\mathrm{X}}(\mathscr{M}), \mathrm{Y}) \mid \psi \circ \varepsilon_{\mathscr{M}} = \delta\} \arrow[r, hook] & \mathrm{Hom_Z}(\mathrm{I}_{\mathrm{H'}\times_{\mathrm{S}}\mathrm{X}}(\mathscr{M}), \mathrm{Y}) .
\end{tikzcd}$$

Figure 30: The first comparison diagram in the proof of Corollary 3.11.1.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 17; combined-reader p. 75.

By 1.3, the action of $\alpha \in \mathcal{O}(H' \times_S X)$ on $\Psi \in \mathrm{Hom}_Y(H' \times_S X, T_{Y/Z}(\mathcal{M}))$ is given, for every $U \to S$ and every $(h, x) \in \mathrm{Hom}_S(U, H' \times_S X)$, by

$$(\alpha\Psi)(h, x) = \alpha(h, x)\Psi(h, x),$$

where U is viewed over Y through $\delta \circ (h, x)$, and $\alpha(h, x) \in \mathcal{O}(U)$ acts through the $\mathcal{O}_Y$-module structure on $T_{Y/Z}(\mathcal{M})$. By 3.4.2, under the identification

$$\mathrm{Hom}_Y(H' \times_S X, T_{Y/Z}(\mathcal{M})) = \{\psi \in \mathrm{Hom}_Z(I_{H'\times_S X}(\mathcal{M}), Y) \mid \psi \circ \varepsilon_{\mathcal{M}} = \delta\},$$

this action is

$$(\alpha\psi)(m, h, x) = \psi(m \cdot \alpha(h, x), h, x) \tag{1}$$

for every $(m, h, x) \in \mathrm{Hom}_S(U, I_S(\mathcal{M}) \times_S H' \times_S X)$.

On the other hand, consider $T_{H/S}(\mathcal{M}) = \underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), H)$. We have a second commutative diagram.

$$\begin{tikzcd}
\mathrm{Hom_H}(\mathrm{H'}, \mathrm{T_{H/S}}(\mathscr{M})) \arrow[r, hook] \arrow[d, equal] & \mathrm{Hom_S}(\mathrm{H'}, \mathrm{T_{H/S}}(\mathscr{M})) \arrow[d, equal] \\
\{\Phi \in \mathrm{Hom_S}(\mathrm{I_{H'}}(\mathscr{M}), \mathrm{H}) \mid \Phi \circ \varepsilon_{\mathscr{M}} = \Delta\} \arrow[r, hook] \arrow[d, equal, "(*)"'] & \mathrm{Hom_S}(\mathrm{I_{H'}}(\mathscr{M}), \mathrm{H}) \arrow[d, equal] \\
\{\phi \in \mathrm{Hom_Z}(\mathrm{I}_{\mathrm{H'}\times_{\mathrm{S}}\mathrm{X}}(\mathscr{M}), \mathrm{Y}) \mid \phi \circ \varepsilon_{\mathscr{M}} = \delta\} \arrow[r, hook] & \mathrm{Hom_Z}(\mathrm{I}_{\mathrm{H'}\times_{\mathrm{S}}\mathrm{X}}(\mathscr{M}), \mathrm{Y}) ,
\end{tikzcd}$$

Figure 31: The second comparison diagram in the proof of Corollary 3.11.1.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 18; combined-reader p. 76.

The bijection $(*)$ in the diagram is described as follows (cf. 1.1(2) and I 1.7.1). For every $U \to S$ and $(m, h, x) \in \mathrm{Hom}_S(U, I_S(\mathcal{M}) \times_S H' \times_S X)$, where U lies over Z through $U \xrightarrow{x} X \to Z$, one has $\Phi(m, h) \in \mathrm{Hom}_Z(X \times_S U, Y)$ and

$$\varphi(m, h, x) = \Phi(m, h) \circ (x \times \mathrm{id}_U) \in \mathrm{Hom}_Z(U, Y). \tag{†}$$

By 3.4.2, with X replaced by H and X' by H', the action of $a \in \mathcal{O}(H')$ on $\Phi \in \mathrm{Hom}_S(I_{H'}(\mathcal{M}), H)$ is

$$(a\Phi)(m,h) = \Phi(m \cdot a(h), h).$$

Consequently, if φ, respectively $a\varphi$, corresponds to Φ, respectively $a\Phi$, then (†) gives

$$(a\varphi)(m,h,x) = \Phi(m \cdot a(h), h) \circ (x \times \mathrm{id}_U) = \varphi(m \cdot a(h), h, x). \tag{2}$$

Together with (1), this proves that the isomorphism

$$T_{H/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{Z/S}(X, T_{Y/Z}(\mathcal{M}))$$

of 3.11.1 is an isomorphism of $\mathcal{O}(H)$-modules. Moreover, for every $H' \to H$, the $\mathcal{O}(H')$-module structure on $\mathrm{Hom}_H(H', T_{H/S}(\mathcal{M}))$ extends, functorially in H', to an $\mathcal{O}(H' \times_S X)$-module structure.

Taking $Z = S$ yields the next corollary.

Corollary 3.11.2.

There is an isomorphism, functorial in $\mathcal{M}$,

$$T_{\underline{\mathrm{Hom}}_S(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_S(X, T_{Y/S}(\mathcal{M})).$$

If Y/S satisfies (E), then $\underline{\mathrm{Hom}}_S(X,Y)/S$ satisfies (E), and this isomorphism respects the module structures over $\underline{\mathrm{Hom}}_S(X,Y)$.

Let $u : X \to Y$ be an S-morphism, identified with the constant morphism $u : S \to \underline{\mathrm{Hom}}_S(X,Y)$ such that $u(f) = u$ for every $f : S' \to S$. The fiber product of u with

$$\underline{\mathrm{Hom}}_S(X, T_{Y/S}(\mathcal{M})) \longrightarrow \underline{\mathrm{Hom}}_S(X,Y)$$

identifies with $\underline{\mathrm{Hom}}_{Y/S}(X, T_{Y/S}(\mathcal{M}))$, where X lies over Y through u. Hence the definition of the tangent space and Corollary 3.11.2 give the following result. [37]

Corollary 3.11.3.

For an S-morphism $u : X \to Y$, there is an isomorphism, functorial in $\mathcal{M}$,

$$L^u_{\underline{\mathrm{Hom}}_S(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{Y/S}(X, T_{Y/S}(\mathcal{M})),$$

where on the right X lies over Y through u. If Y/S satisfies (E), this is an isomorphism of $\mathcal{O}_S$-modules. [38]

In particular, for $Y = X$, the functor $\underline{\mathrm{End}}_S(X)$ is an S-functor in monoids, hence an S-H-functor. Recall that $\underline{\mathrm{Lie}}(\underline{\mathrm{End}}_S(X)/S, \mathcal{M})$ denotes $L^e_{\underline{\mathrm{End}}_S(X)/S}(\mathcal{M})$, with e the identity section.

Corollary 3.11.4.

There is an isomorphism, functorial in $\mathcal{M}$,

$$\underline{\mathrm{Lie}}(\underline{\mathrm{End}}_S(X)/S, \mathcal{M}) \xrightarrow{\sim} \prod_{X/S} T_{X/S}(\mathcal{M}).$$

If X/S satisfies (E), this is an isomorphism of $\mathcal{O}_S$-modules. [38]

[37] Editor's note: the preceding sentences have been added.
[38] Editor's note: the preceding sentence has been added.

Remark 3.11.5.

[39] Suppose that X/S satisfies (E). Then

$$\prod_{X/S} T_{X/S}(\mathcal{M}) = \underline{\mathrm{Hom}}_{X/S}(X, T_{X/S}(\mathcal{M}))$$

is a module over $\prod_{X/S} \mathcal{O}_X$. Explicitly, for every $S' \to S$,

$$\underline{\mathrm{Hom}}_{X/S}(X, T_{X/S}(\mathcal{M}))(S')$$
$$= \{\psi \in \mathrm{Hom}_X(I_{S'}(\mathcal{M}) \times_S X, X) \mid \psi \circ (\varepsilon_{\mathcal{M}} \times \mathrm{id}_X) = \mathrm{pr}_X\}$$

is functorially an $\mathcal{O}(X \times_S S')$-module. This follows either from 3.6 and the properties of $\prod_{X/S}$ (cf. 1.2), or from the proof of 3.11.1.

We now give a geometric interpretation of the tangent bundle. Let U be an S-functor. [40] By I 1.7.2, there are isomorphisms, functorial in $\mathcal{M}$,

$$\begin{aligned} T_{X/S}(\mathcal{M})(U) &= \mathrm{Hom}_S\big(U, \underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), X)\big) \\ &\simeq \mathrm{Hom}_S\big(I_S(\mathcal{M}), \underline{\mathrm{Hom}}_S(U, X)\big) \\ &\simeq \mathrm{Hom}_{I_S(\mathcal{M})}\big(U_{I_S(\mathcal{M})}, X_{I_S(\mathcal{M})}\big). \end{aligned}$$

In particular, the morphism $\mathcal{M} \to 0$ gives the following commutative diagram.

$$\begin{tikzcd} \mathrm{Hom}_S(U, T_{X/S}(\mathscr{M})) \arrow[r, "\sim"] \arrow[d, "\pi_{\mathscr{M}}"'] & \mathrm{Hom}_{I_S(\mathscr{M})}(U_{I_S(\mathscr{M})}, X_{I_S(\mathscr{M})}) \arrow[d] \\ \mathrm{Hom}_S(U, X) \arrow[r, "\text{identité}"] & \mathrm{Hom}_S(U, X) \end{tikzcd} ,$$

Figure 32: The geometric interpretation of the projection $\pi_{\mathcal{M}}$.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 19; combined-reader p. 77.

The second vertical arrow is obtained by base change along $\varepsilon_{\mathcal{M}} : S \to I_S(\mathcal{M})$. [41]

Proposition 3.12.

Let $h_0 : U \to X$ be an S-morphism. Then $\mathrm{Hom}_X(U, T_{X/S}(\mathcal{M}))$ identifies with the set of $I_S(\mathcal{M})$-morphisms

$$U_{I_S(\mathcal{M})} \longrightarrow X_{I_S(\mathcal{M})}$$

whose restriction to U is h_0, where U is regarded as a subobject of $U \times_S I_S(\mathcal{M})$ through $\mathrm{id}_U \times_S \varepsilon_{\mathcal{M}}$.

Taking $U = X$ and $h_0 = \mathrm{id}_X$ gives the following. [42]

[39] Editor's note: this remark has been added.

[40] Editor's note: the original has been simplified in what follows.

[41] Editor's note: via

$$\mathrm{Hom}_{I_S(\mathcal{M})}(U_{I_S(\mathcal{M})}, X_{I_S(\mathcal{M})}) \simeq \mathrm{Hom}_S(U \times_S I_S(\mathcal{M}), X),$$

this is equivalent to Remark 3.1.1. Likewise, 3.12 is equivalent to the first part of Remark 3.4.2.

[42] Editor's note: the original has been expanded in what follows.

Corollary 3.12.1.

The set $\Gamma(T_{X/S}(\mathcal{M})/X)$ identifies with the set of $I_S(\mathcal{M})$-endomorphisms ϕ of $X_{I_S(\mathcal{M})}$ inducing the identity on X; equivalently, the following diagram commutes. [43]

$$\begin{tikzcd} \mathrm{I}_{\mathrm{X}}(\mathscr{M}) \arrow[r, "\phi"] & \mathrm{I}_{\mathrm{X}}(\mathscr{M}) \\ \mathrm{X} \arrow[u, "\varepsilon_{\mathscr{M}}"] \arrow[r, "\mathrm{id}"] & \mathrm{X} \arrow[u, "\varepsilon_{\mathscr{M}}"'] \end{tikzcd}$$

Figure 33: An infinitesimal endomorphism inducing the identity on X.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 20; combined-reader p. 78.

On the other hand, by 3.11.2,

$$\Gamma(T_{X/S}(\mathcal{M})/X) \simeq \underline{\mathrm{Lie}}(\underline{\mathrm{End}}_S(X)/S, \mathcal{M})(S).$$

If X/S satisfies (E), then $\underline{\mathrm{End}}_S(X)/S$ satisfies (E), so the latter Lie functor is an $\mathcal{O}_S$-module by 3.6 (indeed, by 3.11.5, it is a $\prod_{X/S} \mathcal{O}_X$-module). Applying 3.9 gives the next proposition. [44]

Proposition 3.13.

If X/S satisfies (E), the abelian group $\Gamma(T_{X/S}(\mathcal{M})/X)$ identifies with the set of $I_S(\mathcal{M})$-endomorphisms of $X_{I_S(\mathcal{M})}$ that induce the identity on X. Consequently, every such endomorphism is an automorphism.

Corollary 3.13.1.

Let $u : X \to Y$ be an S-isomorphism, with Y/S satisfying (E). Every $I_S(\mathcal{M})$-morphism $X_{I_S(\mathcal{M})} \to Y_{I_S(\mathcal{M})}$ extending u is an isomorphism.

Corollary 3.13.2.

If Y/S satisfies (E), the monomorphism

$$\underline{\mathrm{Isom}}_S(X,Y) \longrightarrow \underline{\mathrm{Hom}}_S(X,Y)$$

induces, for every $u \in \underline{\mathrm{Isom}}_S(X,Y)$, an isomorphism

$$L^u_{\underline{\mathrm{Isom}}_S(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} L^u_{\underline{\mathrm{Hom}}_S(X,Y)/S}(\mathcal{M}).$$

Proof. [45] For every $S' \to S$, we must show that this map is a bijection on S'-points. By base change (cf. 3.4), it suffices to take $S' = S$. The right-hand side then parametrizes the $I_S(\mathcal{M})$-morphisms $X_{I_S(\mathcal{M})} \to Y_{I_S(\mathcal{M})}$ extending u, whereas the left-hand side parametrizes those that are isomorphisms. The assertion is therefore Corollary 3.13.1.

[43] Editor's note: when $\mathcal{M} = \mathcal{O}_X$, these are called *infinitesimal endomorphisms* of X; see 6.2 at the end of this exposé.

[44] Editor's note: the original has been expanded in what follows.

[45] Editor's note: the following proof has been added.

Corollary 3.13.3.

If X/S satisfies (E), the monomorphism $\underline{\mathrm{Aut}}_S(X) \to \underline{\mathrm{End}}_S(X)$ induces, for every $u \in \underline{\mathrm{Aut}}_S(X)$, an isomorphism

$$L^u_{\underline{\mathrm{Aut}}_S(X)/S}(\mathcal{M}) \xrightarrow{\sim} L^u_{\underline{\mathrm{End}}_S(X)/S}(\mathcal{M}).$$

In particular,

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_S(X)/S, \mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Lie}}(\underline{\mathrm{End}}_S(X)/S, \mathcal{M}) \xrightarrow{\sim} \prod_{X/S} T_{X/S}(\mathcal{M}),$$

so that $\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_S(X)/S, \mathcal{M})$ is a $\prod_{X/S} \mathcal{O}_X$-module. [46]

3.14.

Finally, suppose that X is representable. [47] By 2.2.2, the X-functor $T_{X/S}$ is represented by $V(\Omega^1_{X/S})$, whence bijections

$$\Gamma(T_{X/S}/X) \simeq \mathrm{Hom}_X(\Omega^1_{X/S}, \mathcal{O}_X) \simeq \mathrm{Der}_{\mathcal{O}_S}(\mathcal{O}_X).$$

This also follows from the preceding results. By 3.13, $\Gamma(T_{X/S}/X)$ identifies with the set of infinitesimal endomorphisms of X, that is, the I_S-endomorphisms of X_{I_S} that induce the identity on X. The schemes X and X_{I_S} have the same underlying topological space, and their structure sheaves are $\mathcal{O}_X$ and

$$\mathscr{D}_{\mathcal{O}_X} = \mathcal{O}_X \oplus \mathcal{M}, \qquad \mathcal{M} = \mathcal{O}_X,$$

where $\mathcal{M}$ is a square-zero ideal. Let $\pi : \mathscr{D}_{\mathcal{O}_X} \to \mathcal{O}_X$ be the morphism of $\mathcal{O}_X$-algebras vanishing on $\mathcal{M}$. Giving an infinitesimal endomorphism of X is equivalent to giving a morphism of $\mathcal{O}_S$-algebras $\phi : \mathcal{O}_X \to \mathscr{D}_{\mathcal{O}_X}$ satisfying $\pi \circ \phi = \mathrm{id}_{\mathcal{O}_X}$, hence to giving an $\mathcal{O}_S$-derivation of $\mathcal{O}_X$.

If $D, D' \in \mathrm{Der}_{\mathcal{O}_S}(\mathcal{O}_X)$, and ϕ_D denotes the infinitesimal endomorphism corresponding to D, then

$$\phi_{D+D'} = \phi_D \circ \phi_{D'}.$$

Thus the identification

$$\{\text{infinitesimal endomorphisms of } X\} \simeq \mathrm{Der}_{\mathcal{O}_S}(\mathcal{O}_X)$$

is an isomorphism of abelian groups. In view of 3.13 and 3.11.5, we have therefore constructed an isomorphism of abelian groups, indeed of $\mathcal{O}(X)$-modules,

$$\Gamma(T_{X/S}/X) \xrightarrow{\sim} \mathrm{Der}_{\mathcal{O}_S}(\mathcal{O}_X),$$

recovering the classical interpretation of tangent vector fields as derivations of the structure sheaf. [48] Finally,

$$\Gamma(T_{X/S}/X) = H^0(X, \mathfrak{g}_{X/S}),$$

where $\mathfrak{g}_{X/S}$ is the dual of $\Omega^1_{X/S}$.

[46] Editor's note: the preceding assertion has been added.

[47] Editor's note: the reminder of 2.2.2 has been added, and the sequel has been expanded.

[48] Editor's note: this isomorphism is also functorial in S. If $S' \to S$ and $X' = X_{S'}$, then

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_S(X)/S)(S') \simeq \Gamma(T_{X'/S'}/X') \simeq \mathrm{Der}_{\mathcal{O}_{S'}}(\mathcal{O}_{X'}).$$

4. Tangent Space to a Group — Lie Algebras

4.1.

Let G be a group functor over S. By 3.9.0.2, $T_{G/S}(\mathcal{M})$ and $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$ are endowed with group structures over S, and there are group morphisms

$$\begin{tikzcd} \underline{\mathrm{Lie}}(\mathrm{G/S}, \mathscr{M}) \arrow[r, "i"] & \mathrm{T}_{\mathrm{G/S}}(\mathscr{M}) \arrow[r, shift left, "p"] & \mathrm{G} \arrow[l, shift left, "s"] \end{tikzcd} \quad ;$$

Figure 34: The split tangent sequence of the group functor G.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 22; combined-reader p. 80; printed p. 70.

By definition, i identifies $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$ with the kernel of p, and s is a section of p. It follows from I 2.3.7 that this sequence identifies $T_{G/S}(\mathcal{M})$ with the semidirect product of G by $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$.

Definition 4.1.A.

The corresponding action of G on $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$ is denoted

$$\mathrm{Ad} : G \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}\big(\underline{\mathrm{Lie}}(G/S, \mathcal{M})\big)$$

and is called the *adjoint representation* of G, relative to $\mathcal{M}$. Thus, by definition, for $x \in G(S')$ and $X \in \underline{\mathrm{Lie}}(G/S, \mathcal{M})(S')$,

$$\mathrm{Ad}(x)X = i^{-1}\big(s(x)i(X)s(x)^{-1}\big).$$

If G is commutative, then $T_{G/S}(\mathcal{M})$ is commutative as well and $\mathrm{Ad}(x)X = X$. [49]

Definition 4.1.B.

Let G, H be group functors over S, and let $f : G \to H$ be a group morphism. Functoriality gives a morphism of exact sequences compatible with the canonical sections.

$$\begin{tikzcd} 1 \arrow[r] & \underline{\mathrm{Lie}}(\mathrm{G/S}, \mathscr{M}) \arrow[r] \arrow[d, "\mathrm{L}(f)"'] & \mathrm{T}_{\mathrm{G/S}}(\mathscr{M}) \arrow[r] \arrow[d, "\mathrm{T}(f)"'] & \mathrm{G} \arrow[r] \arrow[d, "f"'] & 1 \\ 1 \arrow[r] & \underline{\mathrm{Lie}}(\mathrm{H/S}, \mathscr{M}) \arrow[r] & \mathrm{T}_{\mathrm{H/S}}(\mathscr{M}) \arrow[r] & \mathrm{H} \arrow[r] & 1 \end{tikzcd} \quad ;$$

Figure 35: Functoriality of the split tangent sequence.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 22; combined-reader p. 80; printed p. 70.

The morphism $L(f)$, also denoted $\underline{\mathrm{Lie}}(f)$, is called the *derived morphism* of f. [50]

[49] Editor's note: the numbering 4.1.A and 4.1.B has been added in order to bring these definitions into relief.

[50] Editor's note: if f is a monomorphism, then so are $L(f)$ and $T(f)$; cf. 3.7.

Remark 4.1.C.

If G/S and H/S satisfy condition E, then $L(f)$ respects the $\mathcal{O}_S$-module structures obtained from "functoriality in $\mathcal{M}$" (cf. 3.6). [51]

Proposition 4.1.1.

Let $g \in G(S)$. Then

$$\mathrm{Ad}(g) : \underline{\mathrm{Lie}}(G/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}(G/S, \mathcal{M})$$

is the derived morphism of $\mathrm{Int}(g) : G \to G$.

Indeed, $\mathrm{Ad}(g)X = i^{-1}(\mathrm{Int}(g)i(X))$, which is precisely $L(\mathrm{Int}(g))(X)$ by the definition of the derived morphism.

Suppose that G/S satisfies E. By Proposition 3.9, the group structure just defined on $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$ is exactly the one induced by its $\mathcal{O}_S$-module structure (which is defined by (E)). Proposition 4.1.1 and the functoriality of the $\mathcal{O}_S$-action (cf. 3.6) therefore yield the following corollary.

Corollary 4.1.1.1.

Suppose that G/S satisfies E. Then Ad maps G into the subgroup

$$\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}\big(\underline{\mathrm{Lie}}(G/S, \mathcal{M})\big)$$

of $\underline{\mathrm{Aut}}_{S\text{-gr.}}(\underline{\mathrm{Lie}}(G/S, \mathcal{M}))$. In other words, for every $g \in G(S')$, the automorphism $\mathrm{Ad}(g)$ respects the $\mathcal{O}_{S'}$-module structure on $\underline{\mathrm{Lie}}(G'/S', \mathcal{M})$, where $G' = G \times_S S'$. Thus Ad is a *linear representation* (cf. I 3.2) of G on the $\mathcal{O}_S$-module $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$.

Remarks 4.1.1.2.

a) The functor G/S satisfies E if and only if, for every pair $(\mathcal{M}, \mathcal{N})$ of finite free $\mathcal{O}_S$-modules, the following diagram, obtained by applying $\underline{\mathrm{Lie}}(G/S, -)$ to diagram $(*)$ of 2.1, is cartesian.

$$\begin{tikzcd}
& \underline{\mathrm{Lie}}(\mathrm{G/S}, \mathscr{M} \oplus \mathscr{N}) \arrow[dl] \arrow[dr] & \\
\underline{\mathrm{Lie}}(\mathrm{G/S}, \mathscr{M}) \arrow[dr] & & \underline{\mathrm{Lie}}(\mathrm{G/S}, \mathscr{N}) \arrow[dl] \\
& \underline{\mathrm{Lie}}(\mathrm{G/S}, 0) = \mathrm{S} & ,
\end{tikzcd}$$

Figure 36: The cartesian criterion for condition E.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 23; combined-reader p. 81; printed p. 71.

b) Suppose that G/S satisfies E. The derived morphism of the group law $\pi : G \times_S G \to G$ is the addition law on $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$. Notice that π is not a group morphism, but $\pi(e, e) = e$; hence the derived morphism $L(\pi)$ maps

$$T^{(e,e)}_{(G \times_S G)/S}(\mathcal{M}) = \underline{\mathrm{Lie}}(G/S, \mathcal{M}) \times_S \underline{\mathrm{Lie}}(G/S, \mathcal{M})$$

to $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$; cf. 3.7 and 3.8. Likewise, for $n \in \mathbb{Z}$, let $n_G : G \to G$ denote the S-functor morphism $g \mapsto g^n$. Then $L(n_G)$ is multiplication by n on $\underline{\mathrm{Lie}}(G/S)$; cf. 3.9.4.

[51] Editor's note: this remark has been added.

4.1.2.0.

Consider the S-functor $\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M}))$.[52] For every $S' \to S$, one has $T_{G/S}(\mathcal{M})_{S'} \simeq T_{G_{S'}/S'}(\mathcal{M})$ (cf. 3.4), and therefore

$$\begin{aligned}&\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M}))(S')\\ &\qquad \simeq \mathrm{Hom}_{G_{S'}}\big(G_{S'}, T_{G_{S'}/S'}(\mathcal{M})\big) = \Gamma\big(T_{G_{S'}/S'}(\mathcal{M})/G_{S'}\big).\end{aligned}$$

First note the isomorphism, functorial in S',

$$\mathrm{Hom}_{S'}\big(G_{S'}, \underline{\mathrm{Lie}}(G_{S'}/S', \mathcal{M})\big) \xrightarrow{\sim} \Gamma\big(T_{G_{S'}/S'}(\mathcal{M})/G_{S'}\big), \tag{*}$$

which sends $f : G_{S'} \to \underline{\mathrm{Lie}}(G_{S'}/S', \mathcal{M})$ to the section $s_f : G_{S'} \to T_{G_{S'}/S'}(\mathcal{M})$ defined, for every $S'' \to S'$ and $g \in G(S'')$, by

$$s_f(g) = i(f(g))s(g).$$

Let h be an automorphism of the functor $G_{S'}$ over S', not necessarily preserving the group structure. To every section τ of $T_{G_{S'}/S'}(\mathcal{M})$ one associates, by transport of structure, a section $h(\tau)$: it is the unique section that makes the following diagram commute.

$$\begin{tikzcd} \mathrm{G}_{\mathrm{S}'} \arrow[r, "\tau"] \arrow[d, "h"'] & \mathrm{T}_{\mathrm{G}_{\mathrm{S}'}/\mathrm{S}'}(\mathscr{M}) \arrow[d, "\mathrm{T}(h)"] \\ \mathrm{G}_{\mathrm{S}'} \arrow[r, "h(\tau)"] & \mathrm{T}_{\mathrm{G}_{\mathrm{S}'}/\mathrm{S}'}(\mathscr{M}) \end{tikzcd} .$$

Figure 37: Transport of a tangent section by an automorphism h.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 24; combined-reader p. 82; printed p. 72.

In particular, take $h = t_x$, right translation by an element $x \in G(S')$; thus $h(g) = t_x(g) = g \cdot x$ for all $g \in G(S'')$, $S'' \to S'$. One then has

$$t_x(s_f) = s_{t_x(f)},$$

where $t_x(f) : G_{S'} \to \underline{\mathrm{Lie}}(G_{S'}/S', \mathcal{M})$ is given by

$$t_x(f)(g) = f(g \cdot x^{-1}).$$

Let G act by right translations on

$$\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M})) \quad \text{and} \quad \underline{\mathrm{Hom}}_S(G, \underline{\mathrm{Lie}}(G/S, \mathcal{M}))$$

as follows. For $S' \to S$ and $x \in G(S')$, let

$$\begin{aligned}&\sigma \in \Gamma(T_{G_{S'}/S'}(\mathcal{M})/G_{S'}),\\ &f \in \mathrm{Hom}_{S'}(G_{S'}, \underline{\mathrm{Lie}}(G_{S'}/S', \mathcal{M})).\end{aligned}$$

Set

$$(\sigma \cdot x)(g) = \sigma(g \cdot x^{-1}) \cdot s(x), \qquad (f \cdot x)(g) = f(g \cdot x^{-1})$$

[52] Editor's note: the original has been expanded in what follows to display the action of G on the functors $\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M}))$ and $\underline{\mathrm{Hom}}_S(G, \underline{\mathrm{Lie}}(G/S, \mathcal{M}))$.

for every $g \in G(S'')$, $S'' \to S'$. The isomorphism $(*)$ respects these G-actions.

Consequently, under this isomorphism, the elements of

$$\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M}))^G(S')$$

correspond to the constant morphisms from $G_{S'}$ to $\underline{\mathrm{Lie}}(G_{S'}/S', \mathcal{M})$, that is, to the morphisms factoring through $G_{S'} \to S'$; equivalently, they correspond to the elements of $\underline{\mathrm{Lie}}(G/S, \mathcal{M})(S')$.

Terminology. The elements of $\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M}))^G(S')$ are called the sections of $T_{G_{S'}/S'}(\mathcal{M})$ invariant under right translation.

Proposition 4.1.2.

The map

$$\underline{\mathrm{Lie}}(G/S, \mathcal{M})(S) \longrightarrow \Gamma(T_{G/S}(\mathcal{M})/G), \qquad X \longmapsto (x \mapsto X \cdot x),$$

is a bijection onto the set of sections invariant under right translation. [53]

Likewise, let G act on the right on $\underline{\mathrm{End}}_{I_S(\mathcal{M})/S}(G_{I_S(\mathcal{M})})$ by

$$(u \cdot x)(g) = u(g \cdot x^{-1}) \cdot x,$$

for $S' \to S$, $x \in G(S')$, $u \in \mathrm{End}_{I_{S'}(\mathcal{M})}(G_{I_{S'}(\mathcal{M})})$, and $g \in G(S'')$, $S'' \to I_{S'}(\mathcal{M})$. The morphism of 3.12.1

$$\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M})) \longrightarrow \underline{\mathrm{End}}_{I_S(\mathcal{M})/S}(G_{I_S(\mathcal{M})})$$

respects the G-actions. For each $S' \to S$, it therefore induces a bijection between $\Gamma(T_{G_{S'}/S'}(\mathcal{M})/G_{S'})$ and the set of $I_{S'}(\mathcal{M})$-endomorphisms u of $G_{I_{S'}(\mathcal{M})}$ that induce the identity on G and "commute with right translations," meaning that $u_{S''} \cdot x = u_{S''}$ for every $S'' \to S'$ and $x \in G(S'')$.

Proposition 4.1.3.

There is a bijection, functorial in G, between $\underline{\mathrm{Lie}}(G/S, \mathcal{M})(S)$ and the set of $I_S(\mathcal{M})$-endomorphisms of $G_{I_S(\mathcal{M})}$ that induce the identity on G and commute with the right translations of G, in the sense just specified.

Taking 3.13 into account gives the following theorem.

Theorem 4.1.4.

Let G be an S-functor in groups, and suppose that G/S satisfies E. Then $\underline{\mathrm{Lie}}(G/S, \mathcal{M})(S)$ is identified, functorially in G, with the group of $I_S(\mathcal{M})$-automorphisms of $G_{I_S(\mathcal{M})}$ that induce the identity on G and commute with the right translations of G.

For $\mathcal{M} = \mathcal{O}_S$, this recovers one of the classical definitions of the Lie algebra of a group.

[53] Statements 4.1.2, 4.1.3, and 4.1.4 can be obtained more simply by observing that the automorphisms of G invariant under right translations are the left translations. Editor's note: see, for example, the proofs of Propositions 4.6 and 2.5 of [DG70], §II.4.

4.2.0.

Before proceeding, let us establish further consequences of 3.11.[54] Let X, Y be over Z, with Z over S, as in Section 1. As was seen in 3.11, the isomorphisms 1.1(4)

$$\begin{tikzcd}
\underline{\mathrm{Hom}}_{\mathrm{S}}(\mathrm{I}_{\mathrm{S}}(\mathscr{M}), \underline{\mathrm{Hom}}_{\mathrm{Z/S}}(\mathrm{X},\mathrm{Y})) \arrow[rr, "\sim"] \arrow[dr, "\simeq"'] & & \underline{\mathrm{Hom}}_{\mathrm{Z/S}}(\mathrm{X}, \underline{\mathrm{Hom}}_{\mathrm{Z}}(\mathrm{I}_{\mathrm{Z}}(\mathscr{M}),\mathrm{Y})) \\
& \underline{\mathrm{Hom}}_{\mathrm{Z/S}}(\mathrm{X} \times_{\mathrm{S}} \mathrm{I}_{\mathrm{S}}(\mathscr{M}), \mathrm{Y}) \arrow[ur, "\simeq"'] &
\end{tikzcd}$$

Figure 38: The adjunction isomorphisms of 1.1(4).

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 25; combined-reader p. 83; printed p. 73.

induce the following isomorphism θ.

$$\begin{tikzcd}
\mathrm{T}_{\underline{\mathrm{Hom}}_{\mathrm{Z/S}}(\mathrm{X},\mathrm{Y})}(\mathscr{M}) \arrow[rr, "\theta", "\sim"'] \arrow[dr, "\simeq"'] & & \underline{\mathrm{Hom}}_{\mathrm{Z/S}}(\mathrm{X}, \mathrm{T}_{\mathrm{Y/Z}}(\mathscr{M})) \\
& \underline{\mathrm{Hom}}_{\mathrm{Z/S}}(\mathrm{X} \times_{\mathrm{S}} \mathrm{I}_{\mathrm{S}}(\mathscr{M}), \mathrm{Y}) \arrow[ur, "\simeq"'] & .
\end{tikzcd}$$

Figure 39: The induced tangent-space isomorphism θ.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 25; combined-reader p. 83; printed p. 73.

By 1.3, if Y is a Z-group, then $\underline{\mathrm{Hom}}_Z(V, Y)$ is a Z-group for every $V \to Z$, in particular for $V = I_Z(\mathcal{M})$. Explicitly, if $Z'' \to Z' \to Z$ and $\varphi, \psi \in \mathrm{Hom}_Z(V_{Z'}, Y)(Z')$, their product is defined by

$$(\varphi \cdot \psi)(v) = \varphi(v)\psi(v) \qquad (v \in V_{Z'}(Z'')).$$

Suppose now that X and Y are Z-groups.

Definition 4.2.0.1.

We denote by $\underline{\mathrm{Hom}}_{(Z/S)\text{-gr.}}(X, Y)$ the subfunctor of $\underline{\mathrm{Hom}}_{Z/S}(X, Y)$ defined, for every $S' \to S$, by

$$\underline{\mathrm{Hom}}_{(Z/S)\text{-gr.}}(X, Y)(S') = \mathrm{Hom}_{Z_{S'}\text{-gr.}}(X_{S'}, Y_{S'}). \tag{3}$$

The same definition applies with Y replaced by the Z-group $T_{Y/Z}(\mathcal{M})$.

Under the isomorphisms (2) above, $T_{\underline{\mathrm{Hom}}_{(Z/S)\text{-gr.}}(X,Y)/S}(\mathcal{M})(S')$ corresponds to the $Z_{S'}$-morphisms

$$\varphi : X_{S'} \times_{S'} I_{S'}(\mathcal{M}) \longrightarrow Y_{S'}$$

that are multiplicative in X, that is,

$$\varphi(x_1 x_2, m) = \varphi(x_1, m)\varphi(x_2, m).$$

These in turn correspond to the morphisms of $Z_{S'}$-groups $X_{S'} \to T_{Y/Z}(\mathcal{M})_{S'}$. Hence:

[54] Editor's note: paragraph 4.2.0 has been added; its results are used implicitly in 4.2 and 4.7 of the original.

Proposition 4.2.0.2.

Let X, Y be Z-groups, where Z is over S. There are isomorphisms of S-functors, functorial in $\mathcal{M}$,

$$T_{\underline{\mathrm{Hom}}_{(Z/S)\text{-gr.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{(Z/S)\text{-gr.}}(X, T_{Y/Z}(\mathcal{M})).$$

Taking $Z = S$ gives the following consequence. If Y is a commutative S-group, then so is $T_{Y/S}(\mathcal{M})$, and hence so are $H = \underline{\mathrm{Hom}}_{S\text{-gr.}}(X, Y)$, $\underline{\mathrm{Hom}}_{S\text{-gr.}}(X, T_{Y/S}(\mathcal{M}))$, and $T_{H/S}(\mathcal{M})$.

Corollary 4.2.0.3.

Let X, Y be S-groups. There are isomorphisms of S-functors, functorial in $\mathcal{M}$,

$$T_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{S\text{-gr.}}(X, T_{Y/S}(\mathcal{M})).$$

If Y is commutative, these are moreover isomorphisms of abelian S-groups.

Definition 4.2.0.4.

If Y is an $\mathcal{O}_S$-module, the functor $T_{Y/S}(\mathcal{M})$, respectively $\underline{\mathrm{Lie}}(Y/S, \mathcal{M})$, carries an $\mathcal{O}_S$-module structure induced by that of Y. Equipped with this structure, it will be denoted $T'_{Y/S}(\mathcal{M})$, respectively $\underline{\mathrm{Lie}}'(Y/S, \mathcal{M})$. [55]

Consequently, if X, Y are $\mathcal{O}_S$-modules, then $T'_{Y/S}(\mathcal{M}) = \underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), Y)$. If $H = \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X, Y)$, then $\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X, T'_{Y/S}(\mathcal{M}))$ and $T'_{H/S}(\mathcal{M})$ also carry $\mathcal{O}_S$-module structures.

Corollary 4.2.0.5.

If X, Y are $\mathcal{O}_S$-modules, there are isomorphisms of $\mathcal{O}_S$-modules, functorial in $\mathcal{M}$,

$$T'_{\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X, T'_{Y/S}(\mathcal{M})).$$

Definition 4.2.A.

Let X, L be S-groups, with X acting on L by group automorphisms (cf. I 2.3.5). We define the subfunctor $\underline{Z}^1_S(X, L)$ of $\underline{\mathrm{Hom}}_S(X, L)$ by

$$\begin{aligned}\underline{Z}^1_S(X, L)(S') = \{\, \varphi \in \mathrm{Hom}_{S'}(X_{S'}, L_{S'}) \mid \varphi(x_1x_2) = \varphi(x_1)\big(x_1 \cdot \varphi(x_2)\big), \\ x_1, x_2 \in X(S''),\ S'' \to S' \,\}.\end{aligned}$$

It is called the *functor of crossed homomorphisms* from X to L. [56]

Remark 4.2.B.

a) If L' is another S-group on which X acts by group automorphisms, then

$$\underline{Z}^1_S(X, L \times_S L') \simeq \underline{Z}^1_S(X, L) \times_S \underline{Z}^1_S(X, L').$$

b) If L is an X-$\mathcal{O}_S$-module, then $\underline{Z}^1_S(X, L)$ is the kernel of the differential

$$\partial : \underline{\mathrm{Hom}}_S(X, L) \longrightarrow \underline{\mathrm{Hom}}_S(X^2, L)$$

defined in I 5.1. In particular, $\underline{Z}^1_S(X, L)$ is then an $\mathcal{O}_S$-module.

[55] Editor's note: the prime is used to distinguish the $\mathcal{O}_S$-module structure induced by that of Y from the one that will be obtained from condition E.

[56] Editor's note: the original has been expanded below; in particular, Definition 4.2.A and Remark 4.2.B have been added.

4.2.

Let $u : X \to Y$ be a morphism of S-groups. We describe $L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M})$. First, 3.11.3 gives an isomorphism of S-functors, functorial in $\mathcal{M}$,

$$L^u_{\underline{\mathrm{Hom}}_S(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{Y/S}(X, T_{Y/S}(\mathcal{M})). \tag{†}$$

Since Y is an S-group,

$$T_{Y/S}(\mathcal{M}) = \underline{\mathrm{Lie}}(Y/S, \mathcal{M}) \cdot Y = \underline{\mathrm{Lie}}(Y/S, \mathcal{M})_Y,$$

and one obtains an isomorphism, functorial in $\mathcal{M}$,

$$L^u_{\underline{\mathrm{Hom}}_S(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_S(X, \underline{\mathrm{Lie}}(Y/S, \mathcal{M})).$$

For $S' \to S$, let $u' : X' \to Y'$ be the base change of u. The S-functor[57]

$$\underline{\mathrm{Hom}}_{(Y/S)\text{-gr.}}\big(X, \underline{\mathrm{Lie}}(Y/S, \mathcal{M}) \cdot Y\big)$$

is defined by

$$\begin{aligned}&\underline{\mathrm{Hom}}_{(Y/S)\text{-gr.}}\big(X, \underline{\mathrm{Lie}}(Y/S, \mathcal{M}) \cdot Y\big)(S')\\ &\quad= \mathrm{Hom}_{Y'\text{-gr.}}\big(X', (\underline{\mathrm{Lie}}(Y/S, \mathcal{M}) \cdot Y)_{S'}\big)\\ &\quad= \mathrm{Hom}_{Y'\text{-gr.}}\big(X', \underline{\mathrm{Lie}}(Y'/S', \mathcal{M}) \cdot Y'\big).\end{aligned}$$

Thus (†) induces an isomorphism

$$L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{(Y/S)\text{-gr.}}\big(X, \underline{\mathrm{Lie}}(Y/S, \mathcal{M}) \cdot Y\big). \tag{†'}$$

On the other hand, the composite

$$\mathrm{X} \xrightarrow{u} \mathrm{Y} \xrightarrow{\mathrm{Ad}} \underline{\mathrm{Aut}}_{\mathrm{S\text{-}gr.}}(\underline{\mathrm{Lie}}(\mathrm{Y/S}, \mathscr{M}))$$

Figure 40: The action of X on $\underline{\mathrm{Lie}}(Y/S, \mathcal{M})$.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 27; combined-reader p. 85; printed p. 75.

defines an action of X on $L = \underline{\mathrm{Lie}}(Y/S, \mathcal{M})$ by group automorphisms.

Let $\Phi \in \underline{\mathrm{Hom}}_{Y/S}(X, \underline{\mathrm{Lie}}(Y/S, \mathcal{M}) \cdot Y)$. For $S'' \to S' \to S$ and $x \in X(S'')$, there is a unique expression

$$\Phi(S')(x) = \varphi(S')(x) \cdot u'(x), \qquad \varphi(S')(x) \in \underline{\mathrm{Lie}}(Y'/S', \mathcal{M})(S'').$$

This determines an element $\varphi \in \underline{\mathrm{Hom}}_S(X, \underline{\mathrm{Lie}}(Y/S, \mathcal{M}))$. The map $\Phi(S')$ is a group morphism if and only if, for all $x_1, x_2 \in X(S'')$,

$$\begin{aligned}\varphi(S')(x_1x_2) &= \varphi(S')(x_1)\big(u(x_1)\varphi(S')(x_2)u(x_1)^{-1}\big)\\ &= \varphi(S')(x_1)\big(x_1 \cdot \varphi(S')(x_2)\big),\end{aligned}$$

that is, precisely when $\varphi \in \underline{Z}^1_S(X, \underline{\mathrm{Lie}}(Y/S, \mathcal{M}))$.

[57] Editor's note: this is Definition 4.2.0.1 applied with $Z = Y$, to the Y-groups $u : X \to Y$ and $T_{Y/S}(\mathcal{M}) \to Y$.

Proposition 4.2.

Let $u : X \to Y$ be a morphism of S-groups. There is an isomorphism of S-functors, functorial in $\mathcal{M}$,

$$L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{Z}^1_S\big(X, \underline{\mathrm{Lie}}(Y/S, \mathcal{M})\big).$$

Suppose in addition that Y/S satisfies E. It follows from 4.2.0.3, just as in the proof of 3.11.1, that $\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S$ satisfies E. Thus (cf. 3.5.1)

$$L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M} \oplus \mathcal{N}) \simeq L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}) \\ \times_S L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{N}).$$

This also follows directly from Proposition 4.2 and Remark 4.2.Ba. Consequently, both sides of the isomorphism in Proposition 4.2 carry $\mathcal{O}_S$-module structures obtained from functoriality in $\mathcal{M}$, and that isomorphism respects them.

Proposition 4.2 bis.

Let $u : X \to Y$ be a morphism of S-groups, and suppose that Y/S satisfies E. There is an isomorphism of $\mathcal{O}_S$-modules, functorial in $\mathcal{M}$,

$$L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{Z}^1_S\big(X, \underline{\mathrm{Lie}}(Y/S, \mathcal{M})\big).$$

[58]

When Y/S satisfies E, 3.13.1 also gives, just as in the proof of 3.13.2, an isomorphism functorial in $\mathcal{M}$, for every $u \in \underline{\mathrm{Isom}}_{S\text{-gr.}}(X,Y)$,

$$L^u_{\underline{\mathrm{Isom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}). \tag{*}$$

This yields the following two corollaries.

Corollary 4.2.1.

Let $u : X \to Y$ be a morphism of S-groups. If Y/S satisfies E, there is an isomorphism of $\mathcal{O}_S$-modules, functorial in $\mathcal{M}$,

$$L^u_{\underline{\mathrm{Isom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{Z}^1_S\big(X, \underline{\mathrm{Lie}}(Y/S, \mathcal{M})\big).$$

Corollary 4.2.2.

Let X be an S-group. If X/S satisfies E, there is an isomorphism of $\mathcal{O}_S$-modules, functorial in $\mathcal{M}$,

$$\underline{\mathrm{Lie}}\big(\underline{\mathrm{Aut}}_{S\text{-gr.}}(X)/S, \mathcal{M}\big) \xrightarrow{\sim} \underline{Z}^1_S\big(X, \underline{\mathrm{Lie}}(X/S, \mathcal{M})\big).$$

If Y is commutative, its adjoint action on $L = \underline{\mathrm{Lie}}(Y/S, \mathcal{M})$ is trivial, so $\underline{Z}^1_S(X, L) = \underline{\mathrm{Hom}}_{S\text{-gr.}}(X, L)$. [59]

Corollary 4.2.3.

Let Y be a commutative S-group. There is an isomorphism of S-functors, functorial in $\mathcal{M}$,

$$L^u_{\underline{\mathrm{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{S\text{-gr.}}\big(X, \underline{\mathrm{Lie}}(Y/S, \mathcal{M})\big).$$

[58] Editor's note: the original has been expanded from this point; in particular, Proposition 4.2 bis has been added.

[59] Editor's note: the following sentence has been added.

4.3.

We now consider the case in which X and Y are $\mathcal{O}_S$-modules. Recall (cf. 4.2.0.4) that $T'_{Y/S}(\mathcal{M})$, respectively $\underline{\mathrm{Lie}}'(Y/S,\mathcal{M})$, denotes the functor $T_{Y/S}(\mathcal{M})$, respectively $\underline{\mathrm{Lie}}(Y/S,\mathcal{M})$, equipped with the $\mathcal{O}_S$-module structure induced by that of Y.

When Y/S satisfies E, we shall always write $\underline{\mathrm{Lie}}(Y/S,\mathcal{M})$ for this functor equipped instead with the $\mathcal{O}_S$-module structure defined for every functor satisfying E. By 3.9, the underlying abelian-group structures of $\underline{\mathrm{Lie}}(Y/S,\mathcal{M})$ and $\underline{\mathrm{Lie}}'(Y/S,\mathcal{M})$ coincide, but their module structures need not coincide (see the counterexample in 6.3). For $S' \to S$, $a \in \mathcal{O}(S')$, and $m \in \underline{\mathrm{Lie}}(Y/S,\mathcal{M})(S')$, we write $a \cdot' m$, respectively $a \cdot m$, for the action coming from Y, respectively the action obtained from condition E. We use the same convention for the two actions on $T_{Y/S}(\mathcal{M})$.

There is an isomorphism of $\mathcal{O}_S$-modules

$$T'_{Y/S}(\mathcal{M}) \simeq \underline{\mathrm{Lie}}'(Y/S,\mathcal{M}) \oplus Y.$$

Exactly as for Propositions 4.2 and 4.2 bis, one therefore obtains:

Proposition 4.3.

Let $u : X \to Y$ be a morphism of $\mathcal{O}_S$-modules. There is an isomorphism of S-functors, functorial in $\mathcal{M}$,

$$L^u_{\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}\big(X, \underline{\mathrm{Lie}}'(Y/S,\mathcal{M})\big). \tag{*}$$

If Y/S satisfies E, then $\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X,Y)/S$ satisfies E, and $(*)$ is an isomorphism of $\mathcal{O}_S$-modules when both sides carry the structures obtained from functoriality in $\mathcal{M}$. [60] [61]

Remark 4.3 bis.

Let $u : X \to Y$ be a morphism of $\mathcal{O}_S$-modules.[62] Denote by τ_u the map that sends each $\mathcal{O}_S$-linear morphism $\varphi : X \to \underline{\mathrm{Lie}}'(Y/S,\mathcal{M})$ to

$$u \oplus \varphi : X \longrightarrow T'_{Y/S}(\mathcal{M}) = Y \oplus \underline{\mathrm{Lie}}'(Y/S,\mathcal{M}).$$

The isomorphism of 4.3 belongs to the following commutative diagram, functorial in $\mathcal{M}$.

$$\begin{tikzcd}
\mathrm{L}^u_{\underline{\mathrm{Hom}}_{\mathbf{O}_S\text{-mod.}}(\mathrm{X},\mathrm{Y})/\mathrm{S}}(\mathscr{M}) \arrow[r, "\sim"] \arrow[d, hook] & \underline{\mathrm{Hom}}_{\mathbf{O}_S\text{-mod.}}(\mathrm{X}, \underline{\mathrm{Lie}}'(\mathrm{Y}/\mathrm{S},\mathscr{M})) \arrow[d, hook, "\tau_u"] \\
\mathrm{T}_{\underline{\mathrm{Hom}}_{\mathbf{O}_S\text{-mod.}}(\mathrm{X},\mathrm{Y})/\mathrm{S}}(\mathscr{M}) \arrow[r, "\sim"] & \underline{\mathrm{Hom}}_{\mathbf{O}_S\text{-mod.}}(\mathrm{X}, \mathrm{T}'_{\mathrm{Y}/\mathrm{S}}(\mathscr{M})).
\end{tikzcd}$$

Figure 41: Compatibility of the isomorphism of 4.3 with τ_u.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 30; combined-reader p. 88; printed p. 78.

Moreover, when Y/S satisfies E, 3.13.1 gives, just as in the proof of 3.13.2, for every $u \in \underline{\mathrm{Isom}}_{\mathcal{O}_S\text{-mod.}}(X,Y)$,

$$L^u_{\underline{\mathrm{Isom}}_{\mathcal{O}_S\text{-mod.}}(X,Y)/S}(\mathcal{M}) = L^u_{\underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X,Y)/S}(\mathcal{M}). \tag{*}$$

[60] Editor's note: the following sentence has been added.

[61] Editor's note: on the right-hand side, the scalar action is $(a\varphi)(x) = a\varphi(x)$, where $a\varphi(x)$ denotes the action of a on $\varphi(x) \in \underline{\mathrm{Lie}}(Y/S,\mathcal{M})$. This differs from the action $(a \cdot' \varphi)(x) = a \cdot' \varphi(x) = \varphi(ax)$, but the two agree when $\underline{\mathrm{Lie}}(Y/S,\mathcal{M}) = \underline{\mathrm{Lie}}'(Y/S,\mathcal{M})$.

[62] Editor's note: this remark has been added; it will be useful in 4.5.1.

Corollary 4.3.1.

Let X be an $\mathcal{O}_S$-module satisfying E over S. There is an isomorphism, functorial in $\mathcal{M}$,

$$\underline{\mathrm{Lie}}\big(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(X)/S,\mathcal{M}\big) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}\big(X,\underline{\mathrm{Lie}}'(X/S,\mathcal{M})\big),$$

and this isomorphism respects the $\mathcal{O}_S$-module structures obtained from functoriality in $\mathcal{M}$. In particular, $\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(X)/S$ satisfies E. [63]

Proof. The first assertion follows from $(*)$ and 4.3. For the second, since X/S satisfies E, there are isomorphisms of $\mathcal{O}_S$-modules

$$\underline{\mathrm{Lie}}'(X/S,\mathcal{M}\oplus\mathcal{N}) \simeq \underline{\mathrm{Lie}}'(X/S,\mathcal{M}) \times_S \underline{\mathrm{Lie}}'(X/S,\mathcal{N}).$$

Consequently,

$$\begin{aligned}&\underline{\mathrm{Lie}}\big(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(X)/S,\mathcal{M}\oplus\mathcal{N}\big)\\ &\quad\simeq \underline{\mathrm{Lie}}\big(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(X)/S,\mathcal{M}\big) \times_S \underline{\mathrm{Lie}}\big(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(X)/S,\mathcal{N}\big).\end{aligned}$$

Remark 4.1.1.2a now proves the assertion.

4.3.2.

Before continuing in this direction, let us examine more closely the relations among Y, $\underline{\mathrm{Lie}}(Y/S)$, and $\underline{\mathrm{Lie}}'(Y/S)$. First,

$$\underline{\mathrm{Lie}}(\mathcal{O}_S/S,\mathcal{M}) = \underline{\mathrm{Lie}}'(\mathcal{O}_S/S,\mathcal{M}) = \mathbf{W}(\mathcal{M}), \tag{1}$$

where $\mathbf{W}(\mathcal{M})$ is defined in I 4.6. There is therefore a canonical isomorphism

$$d : \mathcal{O}_S \xrightarrow{\sim} \underline{\mathrm{Lie}}(\mathcal{O}_S/S). \tag{2}$$

Let F be an $\mathcal{O}_S$-module. For every $S_2 \to S_1 \to S$, there is a dihomomorphism

$$\begin{cases} F(S_1) \longrightarrow F(S_2),\\ \mathcal{O}(S_1) \longrightarrow \mathcal{O}(S_2),\end{cases} \tag{3}$$

and hence a morphism of $\mathcal{O}(S_2)$-modules

$$F(S_1) \otimes_{\mathcal{O}(S_1)} \mathcal{O}(S_2) \longrightarrow F(S_2).$$

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In particular, take $S_1 = S'$ and $S_2 = I_{S'}(\mathcal{M})$. One obtains morphisms of $\mathcal{O}(S')$-modules, functorial in $\mathcal{M}$,

$$F(S') \otimes_{\mathcal{O}(S')} T_{\mathcal{O}_S/S}(\mathcal{M})(S') \longrightarrow T'_{F/S}(\mathcal{M})(S').$$

As S' varies, these give morphisms of $\mathcal{O}_S$-modules

$$F \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(\mathcal{M}) \longrightarrow T'_{F/S}(\mathcal{M}), \tag{4}$$

functorial in $\mathcal{M}$. Compatibility with the tangent-bundle projections onto their bases then yields morphisms

$$F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S,\mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}'(F/S,\mathcal{M}), \tag{5}$$

[63] Editor's note: compare editor's note (61). The final sentence and its proof have also been added.

[64] Editor's note: $S' \to S$ has been replaced by $S_2 \to S_1 \to S$, since both S_2 and S_1 must vary below. The original has also been expanded in what follows.

and the following diagram commutes.

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{F} \otimes_{\mathbf{O}_\mathrm{S}} \underline{\mathrm{Lie}}(\mathbf{O}_\mathrm{S}/\mathrm{S}, \mathscr{M}) \arrow[r] \arrow[d] & \mathrm{F} \otimes_{\mathbf{O}_\mathrm{S}} \mathrm{T}_{\mathbf{O}_\mathrm{S}/\mathrm{S}}(\mathscr{M}) \arrow[r] \arrow[d] & \mathrm{F} \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & \underline{\mathrm{Lie}}'(\mathrm{F}/\mathrm{S}, \mathscr{M}) \arrow[r] & \mathrm{T}'_{\mathrm{F}/\mathrm{S}}(\mathscr{M}) \arrow[r] & \mathrm{F} \arrow[r] & 0.
\end{tikzcd}$$

Figure 42: The morphisms (4) and (5) in the split tangent sequences.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 31; combined-reader p. 89; printed p. 79.

The morphisms (5) may be regarded as morphisms of abelian S-groups

$$F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}(F/S, \mathcal{M}). \tag{6}$$

Tensoring F with the isomorphism $d : \mathcal{O}_S \xrightarrow{\sim} \underline{\mathrm{Lie}}(\mathcal{O}_S/S)$ gives, when $\mathcal{M} = \mathcal{O}_S$, a morphism of abelian S-groups

$$F \xrightarrow{\sim} F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S) \longrightarrow \underline{\mathrm{Lie}}(F/S), \tag{7}$$

also denoted $d : F \to \underline{\mathrm{Lie}}(F/S)$.

Remark 4.3.3.

When F/S satisfies E, the morphisms (6) and (7) need not be $\mathcal{O}_S$-linear if both sides are equipped with the module structures obtained from $\mathcal{M}$ by condition E.

For example, let k be a field of characteristic $p > 0$, let $S = \mathrm{Spec}(k)$, and let F be the $\mathcal{O}_S$-module that assigns to every S-scheme T the set $F(T) = \Gamma(T, \mathcal{O}_T)$, with scalars acting through the p-th power:

$$r \cdot f = r^p f \qquad (r \in \mathcal{O}(T),\ f \in F(T)).$$

As an S-functor in groups, $F \simeq \mathbf{G}_{a,S}$. Hence F satisfies E and $\underline{\mathrm{Lie}}(F/S) \simeq \underline{\mathrm{Lie}}(\mathbf{G}_{a,S}/S) \simeq \mathcal{O}_S$. For every $T \to S$, the canonical morphism $d : F(T) \to \mathcal{O}(T)$ is the identity: it respects the abelian-group structure but not the $\mathcal{O}_S$-module structure. [65]

Remark 4.3.4.

The morphisms (4) and (5) can be made explicit as follows. The morphism

$$\Theta : F \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(\mathcal{M}) \longrightarrow T'_{F/S}(\mathcal{M}) = \underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), F)$$

is defined, for $S' \to S$, $\alpha \in \mathcal{O}(I_{S'}(\mathcal{M}))$, and $f : S' \to F$, by

$$\Theta(f \otimes \alpha) = \alpha \cdot (\tau_0 \circ f) = \alpha \cdot (f \circ \rho),$$

where $\tau_0 : F \to T'_{F/S}(\mathcal{M})$ is the zero section and $\rho : I_{S'}(\mathcal{M}) \to S'$ is the structural morphism.

The morphism Θ induces

$$\theta : F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}'(F/S, \mathcal{M}).$$

[65] Editor's note: the original asserted that, when F/S satisfies E, (6), and hence (7), are morphisms of $\mathcal{O}_S$-modules. The counterexample in 6.3 contradicts this. That assertion has been removed, the preceding example inserted, and Definition 4.4 modified accordingly. This does not affect what follows.

This follows from the functoriality in $\mathcal{M}$ noted after (4), and can also be seen directly. Indeed,

$$\underline{\mathrm{Lie}}'(F/S, \mathcal{M})(S') = \{\, \varphi \in \mathrm{Hom}_S(I_{S'}(\mathcal{M}), F) \mid \varphi \circ \varepsilon_{\mathcal{M}} = e \,\},$$

where $e : S' \to F$ is the unit section. On the other hand, $\underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M})(S')$ equals $\Gamma(S', \mathcal{M})$, the kernel of the augmentation

$$\eta : \mathcal{O}(I_{S'}(\mathcal{M})) \longrightarrow \mathcal{O}(S').$$

It remains to see that if $f \in F(S')$ and $\alpha \in \Gamma(S', \mathcal{M})$, then $\alpha \cdot (f \circ \rho) \circ \varepsilon_{\mathcal{M}} = e$.

Apply the dihomomorphism (3) to the S-morphism $\varepsilon_{\mathcal{M}} : S' \to I_{S'}(\mathcal{M})$. The following diagram commutes.

$$\begin{tikzcd}
\mathrm{F}(\mathrm{I}_{\mathrm{S}'}(\mathscr{M})) \arrow[r, "\mathrm{F}(\varepsilon_{\mathscr{M}})"] \arrow[d, "\alpha"'] & \mathrm{F}(\mathrm{S}') \arrow[d, "\eta(\alpha)"] \\
\mathrm{F}(\mathrm{I}_{\mathrm{S}'}(\mathscr{M})) \arrow[r, "\mathrm{F}(\varepsilon_{\mathscr{M}})"] & \mathrm{F}(\mathrm{S}')
\end{tikzcd} .$$

Figure 43: Compatibility of scalar action with $F(\varepsilon_{\mathcal{M}})$.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 32; combined-reader p. 90; printed p. 80.

Thus, for every $\varphi : I_{S'}(\mathcal{M}) \to F$,

$$(\alpha \cdot \varphi) \circ \varepsilon_{\mathcal{M}} = \eta(\alpha) \cdot (\varphi \circ \varepsilon_{\mathcal{M}}),$$

and therefore $(\alpha \cdot \varphi) \circ \varepsilon_{\mathcal{M}} = e$ when $\eta(\alpha) = 0$.

In particular, taking $\mathcal{M} = t\mathcal{O}_S$, one has $\underline{\mathrm{Lie}}(\mathcal{O}_S/S) = t\mathcal{O}_S$, and the morphism $F \to \underline{\mathrm{Lie}}'(F/S)$ is $f \mapsto t \cdot (f \circ \rho)$.[66]

Remark 4.3.5.

Let F be an $\mathcal{O}_S$-module, set

$$E = \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F),$$

and denote by d_F and d_E the $\mathcal{O}_S$-linear morphisms given by 4.3.2(5):

$$d_F : F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}'(F/S, \mathcal{M}),$$
$$d_E : E \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}'(E/S, \mathcal{M}).$$

Remark 4.3.4 gives the following commutative diagram of $\mathcal{O}_S$-linear morphisms.

$$\begin{tikzcd}
\mathrm{E} \otimes_{\mathbf{O}_\mathrm{S}} \underline{\mathrm{Lie}}(\mathbf{O}_\mathrm{S}/\mathrm{S}, \mathscr{M}) \arrow[r, "d_\mathrm{E}"] \arrow[d, equal] & \underline{\mathrm{Lie}}'(\mathrm{E}/\mathrm{S}, \mathscr{M}) \\
\underline{\mathrm{Hom}}_{\mathbf{O}_\mathrm{S}}(\mathrm{F}, \mathrm{F} \otimes_{\mathbf{O}_\mathrm{S}} \underline{\mathrm{Lie}}(\mathbf{O}_\mathrm{S}/\mathrm{S}, \mathscr{M})) \arrow[r, "d_\mathrm{F}"] & \underline{\mathrm{Hom}}_{\mathbf{O}_\mathrm{S}}(\mathrm{F}, \underline{\mathrm{Lie}}'(\mathrm{F}/\mathrm{S}, \mathscr{M})) \arrow[u, "\simeq"', "(*)"]
\end{tikzcd}$$

Figure 44: Compatibility of d_E and d_F.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 32; combined-reader p. 90; printed p. 80.

[66] Editor's note: this remark has been added; it will be useful in 4.6.2.

The right vertical arrow is the isomorphism $(*)$ of 4.3. Thus, by that result, if F/S satisfies E, then E/S satisfies E, and $(*)$ is also an isomorphism of $\mathcal{O}_S$-modules when the right-hand terms are equipped with the module structures obtained from E. [67]

Remark 4.4.0.

In 4.3.2, the morphisms (4) are isomorphisms if and only if the morphisms (5) are. If these equivalent conditions hold, then F/S satisfies E. Indeed, it is enough to verify that

$$\underline{\mathrm{Lie}}'(F/S, \mathcal{M} \oplus \mathcal{N}) \simeq \underline{\mathrm{Lie}}'(F/S, \mathcal{M}) \times_S \underline{\mathrm{Lie}}'(F/S, \mathcal{N}).$$

Under the hypothesis, the horizontal arrows in the following commutative diagram are isomorphisms.

$$\begin{tikzcd}
\mathrm{F} \otimes_{\mathbf{O}_{\mathrm{S}}} \underline{\mathrm{Lie}}(\mathbf{O}_{\mathrm{S}}/\mathrm{S}, \mathscr{M} \oplus \mathscr{N}) \arrow[r, "\sim"] \arrow[d, "\wr"] & \underline{\mathrm{Lie}}'(\mathrm{F}/\mathrm{S}, \mathscr{M} \oplus \mathscr{N}) \arrow[d] \\
\mathrm{F} \otimes_{\mathbf{O}_{\mathrm{S}}} \big(\underline{\mathrm{Lie}}(\mathbf{O}_{\mathrm{S}}/\mathrm{S}, \mathscr{M}) \underset{\mathrm{S}}{\times} \underline{\mathrm{Lie}}(\mathbf{O}_{\mathrm{S}}/\mathrm{S}, \mathscr{N})\big) \arrow[r, "\sim"] & \underline{\mathrm{Lie}}'(\mathrm{F}/\mathrm{S}, \mathscr{M}) \underset{\mathrm{S}}{\times} \underline{\mathrm{Lie}}'(\mathrm{F}/\mathrm{S}, \mathscr{N});
\end{tikzcd}$$

Figure 45: The direct-sum comparison used to verify condition E.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 33; combined-reader p. 91; printed p. 81.

The second vertical arrow is therefore an isomorphism as well, which is exactly condition E for F/S. [68]

Definition 4.4.

An $\mathcal{O}_S$-module F is called *good* if the morphisms

$$F \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(\mathcal{M}) \longrightarrow T_{F/S}(\mathcal{M}),$$

or, equivalently,

$$F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}(F/S, \mathcal{M}),$$

are isomorphisms of abelian S-groups (so that F/S satisfies (E)), and if moreover they respect the $\mathcal{O}_S$-module structures obtained from condition E.

Corollary 4.4.1.

Let F be an $\mathcal{O}_S$-module, and consider the following conditions:[69]

(i) F is a good $\mathcal{O}_S$-module;

(ii) F/S satisfies E, and $d : F \to \underline{\mathrm{Lie}}(F/S)$ is an isomorphism of $\mathcal{O}_S$-modules;

(iii) $\underline{\mathrm{Lie}}(F/S, \mathcal{M}) = \underline{\mathrm{Lie}}'(F/S, \mathcal{M})$.

Then

$$\text{(i)} \iff \text{(ii)} \implies \text{(iii)}.$$

[67] Editor's note: this remark has been added; it will be useful in 4.5 and 4.7.

[68] Editor's note: in view of the changes made in Definition 4.4 (cf. editor's note (65)), this remark has been inserted here; in the original it occurred in the proof of 4.4.1.

[69] Editor's note: the original proved that (i) implies (ii) and (iii); the implication (ii) implies (i) has been added.

Proof. The implication (i) ⇒ (ii) follows from the definition. To prove (ii) ⇒ (i), it remains to show that the morphisms of abelian S-groups, functorial in $\mathcal{M}$,

$$F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Lie}}(F/S, \mathcal{M})$$

are isomorphisms of $\mathcal{O}_S$-modules. Since F/S satisfies E, both sides carry finite direct sums of copies of $\mathcal{O}_S$ to finite products of abelian S-groups. We are therefore reduced to the case $\mathcal{M} = \mathcal{O}_S$, which is precisely the hypothesis.

Finally, (i) ⇒ (iii) follows from the definition and from the fact that the isomorphisms

$$F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Lie}}'(F/S, \mathcal{M})$$

of 4.3.2(5) are $\mathcal{O}_S$-linear.

Examples 4.4.2.

For every quasi-coherent $\mathcal{O}_S$-module $\mathcal{E}$, the $\mathcal{O}_S$-modules $\mathbf{V}(\mathcal{E})$ and $\mathbf{W}(\mathcal{E})$ defined in I 4.6 are good.

Indeed, for every $f : S' \to S$, the morphisms

$$\begin{aligned} \mathbf{V}(\mathcal{E})(S') \otimes_{\mathcal{O}(S')} \mathcal{O}(I_{S'}(\mathcal{M})) &\longrightarrow T_{\mathbf{V}(\mathcal{E})/S}(\mathcal{M})(S'), \\ \mathbf{W}(\mathcal{E})(S') \otimes_{\mathcal{O}(S')} \mathcal{O}(I_{S'}(\mathcal{M})) &\longrightarrow T_{\mathbf{W}(\mathcal{E})/S}(\mathcal{M})(S') \end{aligned}$$

correspond respectively to

$$\begin{aligned} &\mathrm{Hom}_{\mathcal{O}_{S'}\text{-mod.}}\big(f^*(\mathcal{E}), \mathcal{O}_{S'}\big) \otimes_{\mathcal{O}(S')} \Gamma\big(S', D_{\mathcal{O}_{S'}}(\mathcal{M})\big) \\ &\qquad \longrightarrow \mathrm{Hom}_{\mathcal{O}_{S'}\text{-mod.}}\big(f^*(\mathcal{E}), D_{\mathcal{O}_{S'}}(\mathcal{M})\big) \qquad \text{(V)} \\ &\Gamma(S', f^*(\mathcal{E})) \otimes_{\mathcal{O}(S')} \Gamma\big(S', D_{\mathcal{O}_{S'}}(\mathcal{M})\big) \\ &\qquad \longrightarrow \Gamma\big(S', f^*(\mathcal{E}) \otimes_{\mathcal{O}_{S'}} D_{\mathcal{O}_{S'}}(\mathcal{M})\big) \qquad \text{(W)}. \end{aligned}$$

These are isomorphisms because, as an $\mathcal{O}_{S'}$-module, $D_{\mathcal{O}_{S'}}(\mathcal{M})$ is isomorphic to a finite direct sum of copies of $\mathcal{O}_{S'}$. [70]

Proposition 4.5. *Let F be a good $\mathcal{O}_S$-module. Then:*

(i) The S-group $\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)$ satisfies condition (E), and there is a functorial isomorphism

$$\underline{\mathrm{Lie}}\big(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S, \mathcal{M}\big) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}\left(F, \underline{\mathrm{Lie}}(F/S, \mathcal{M})\right),$$

which respects the $\mathcal{O}_S$-module structures obtained from condition (E). In particular, there is an isomorphism of $\mathcal{O}_S$-modules

$$\underline{\mathrm{Lie}}\big(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S\big) \xrightarrow{\sim} \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F).$$

(ii)[71] *Moreover, $\underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F)$ is a good $\mathcal{O}_S$-module.*

Indeed, by 4.4.1, F/S satisfies (E), and

$$\underline{\mathrm{Lie}}(F/S, \mathcal{M}) = \underline{\mathrm{Lie}}'(F/S, \mathcal{M}) \simeq F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathbf{O}_S/S, \mathcal{M}). \tag{1}$$

Assertion (i) now follows from 4.3.1. Put $E = \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F)$. By (1) and Remark 4.3.5, one has the following commutative diagram of morphisms of abelian S-groups.

[70] Editor's note: the preceding proof has been added.

[71] Editor's note: part (ii), an immediate consequence of what precedes, has been added.

$$\begin{tikzcd}
\underline{\mathrm{End}}_{\mathbf{O}_{\mathrm{S}}\text{-mod.}}(\mathrm{F}) \otimes_{\mathbf{O}_{\mathrm{S}}} \underline{\mathrm{Lie}}(\mathbf{O}_{\mathrm{S}}/\mathrm{S}, \mathscr{M}) \arrow[r, "d_{\mathrm{E}}"] \arrow[d, equal] & \underline{\mathrm{Lie}}(\underline{\mathrm{End}}_{\mathbf{O}_{\mathrm{S}}\text{-mod.}}(\mathrm{F})/\mathrm{S}, \mathscr{M}) \\
\underline{\mathrm{Hom}}_{\mathbf{O}_{\mathrm{S}}}(\mathrm{F}, \mathrm{F} \otimes_{\mathbf{O}_{\mathrm{S}}} \underline{\mathrm{Lie}}(\mathbf{O}_{\mathrm{S}}/\mathrm{S}, \mathscr{M})) \arrow[r, "d_{\mathrm{F}}"] & \underline{\mathrm{Hom}}_{\mathbf{O}_{\mathrm{S}}}(\mathrm{F}, \underline{\mathrm{Lie}}(\mathrm{F}/\mathrm{S}, \mathscr{M})) \arrow[u, "\simeq"', "(*)"]
\end{tikzcd}$$

Figure 46: The comparison diagram for the differential d_E.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 34; combined-reader p. 92; printed p. 82.

Here d_F and the vertical isomorphism $(*)$ are isomorphisms of $\mathcal{O}_S$-modules; consequently, so is d_E. This proves (ii).

Scholium 4.5.1. [72] Put (cf. 2.1) $\mathcal{O}_{I_S} = \mathcal{O}_S \oplus t\mathcal{O}_S$, with $t^2 = 0$, and let F be a good $\mathcal{O}_S$-module. For every $S' \to S$, the morphism

$$F(S') \oplus tF(S') = F(S') \otimes_{\mathcal{O}(S')} \mathcal{O}(I_{S'}) \longrightarrow F(I_{S'}) = F(S') \oplus \underline{\mathrm{Lie}}(F/S)(S'),$$

which is the identity on $F(S')$, induces an isomorphism of $\mathcal{O}(S')$-modules

$$tF(S') \xrightarrow{\sim} \underline{\mathrm{Lie}}(F/S)(S').$$

As S' varies, this gives an isomorphism that we may write

$$\underline{\mathrm{Lie}}(F/S) \simeq tF.$$

Thus, for every $S' \to S$, Proposition 4.5 gives a commutative diagram.

$$\begin{tikzcd}
\mathrm{End}_{\mathbf{O}_{\mathrm{S}'}\text{-mod.}}(\mathrm{F}_{\mathrm{S}'}) \arrow[r, "\sim"] & \mathrm{Hom}_{\mathbf{O}_{\mathrm{S}'}\text{-mod.}}(\mathrm{F}_{\mathrm{S}'}, t\,\mathrm{F}_{\mathrm{S}'}) \arrow[r, "\sim"] \arrow[d, hook] & \underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathbf{O}_{\mathrm{S}}\text{-mod.}}(\mathrm{F})/\mathrm{S})(\mathrm{S}') \arrow[d, hook] \\
& \mathrm{Aut}_{\mathbf{O}_{\mathrm{I}_{\mathrm{S}'}}\text{-mod.}}(\mathrm{F}_{\mathrm{I}_{\mathrm{S}'}}) \arrow[r, equal] & \mathrm{T}_{\underline{\mathrm{Aut}}_{\mathbf{O}_{\mathrm{S}}\text{-mod.}}(\mathrm{F})/\mathrm{S}}(\mathrm{S}')
\end{tikzcd}$$

Figure 47: Endomorphisms and automorphisms over the dual numbers.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 35; combined-reader p. 93; printed p. 83.

It follows from 4.3 bis that every

$$X \in \underline{\mathrm{End}}_{\mathcal{O}_{S'}\text{-mod.}}(F_{S'})$$

corresponds to $\mathrm{id} + tX \in \underline{\mathrm{Aut}}_{\mathcal{O}_{I_{S'}}\text{-mod.}}(F_{I_{S'}})$.

Definition 4.6. An S-functor in groups G is called *good* if G/S satisfies condition (E) and $\underline{\mathrm{Lie}}(G/S)$ is a good $\mathcal{O}_S$-module.

[73] Notice that if F is a good $\mathcal{O}_S$-module, then it is a good S-group: indeed, F/S satisfies (E), and $\underline{\mathrm{Lie}}(F/S) \simeq F$ is a good $\mathcal{O}_S$-module.

[72] Editor's note: this scholium, implicit in the original, has been added for use in 4.7.1. Here and below, t, t', and so forth denote square-zero variables; the letter ε is reserved for the unit section of a group.

[73] Editor's note: the following sentence has been added.

Example 4.6.1. *If G is representable, it is good. Indeed, G/S satisfies (E), and $\underline{\mathrm{Lie}}(G/S)$ has the form $V(E)$, hence is good by 4.4.2.*

Lemma 4.6.2. [74] *Let G be an S-functor in groups such that G/S satisfies (E), and put $F = \underline{\mathrm{Lie}}(G/S)$. Then F satisfies (E), and the morphism of abelian groups*

$$d : F \longrightarrow \underline{\mathrm{Lie}}(F/S)$$

respects the $\mathcal{O}_S$-module structures. Consequently, G is good if and only if $F \to \underline{\mathrm{Lie}}(F/S)$ is bijective.

Proof. Suppose that G/S satisfies (E). Let $\mathcal{M}, \mathcal{N}$ be finite free $\mathcal{O}_S$-modules, write

$$F(\mathcal{N}) = \underline{\mathrm{Lie}}(G/S, \mathcal{N}),$$

and let e denote the unit section of G. For every $S' \to S$,

$$F(\mathcal{N})(S') = \{g \in \mathrm{Hom}_S(I_{S'}(\mathcal{N}), G) \mid g \circ \varepsilon_{\mathcal{N}} = e\}.$$

Moreover, $\underline{\mathrm{Lie}}(F(\mathcal{N})/S, \mathcal{M})(S')$ identifies with the set of morphisms

$$\left\{ \Phi \in \mathrm{Hom}_S(I_{S'}(\mathcal{N}) \times_{S'} I_{S'}(\mathcal{M}), G) \,\middle|\, \begin{array}{l} \Phi \circ (\varepsilon_{\mathcal{N}} \times \mathrm{id}_{I_{S'}(\mathcal{M})}) = e \in G(I_{S'}(\mathcal{M})), \\ \Phi \circ (\mathrm{id}_{I_{S'}(\mathcal{N})} \times \varepsilon_{\mathcal{M}}) = e \in G(I_{S'}(\mathcal{N})). \end{array} \right\}.$$

This proves

$$\underline{\mathrm{Lie}}(F(\mathcal{N})/S, \mathcal{M}) \simeq \underline{\mathrm{Lie}}(F(\mathcal{M})/S, \mathcal{N}). \tag{1}$$

Since G/S satisfies (E), it follows that

$$\begin{aligned} \underline{\mathrm{Lie}}(F(\mathcal{N})/S, \mathcal{M}_1 \oplus \mathcal{M}_2) &\simeq \underline{\mathrm{Lie}}(F(\mathcal{M}_1 \oplus \mathcal{M}_2)/S, \mathcal{N}) \\ &\simeq \underline{\mathrm{Lie}}((F(\mathcal{M}_1) \times_S F(\mathcal{M}_2))/S, \mathcal{N}). \end{aligned}$$

By 3.8, the last term is isomorphic to

$$\begin{aligned} &\underline{\mathrm{Lie}}(F(\mathcal{M}_1)/S, \mathcal{N}) \times_S \underline{\mathrm{Lie}}(F(\mathcal{M}_2)/S, \mathcal{N}) \\ &\qquad \simeq \underline{\mathrm{Lie}}(F(\mathcal{N})/S, \mathcal{M}_1) \times_S \underline{\mathrm{Lie}}(F(\mathcal{N})/S, \mathcal{M}_2). \end{aligned}$$

Hence

$$\underline{\mathrm{Lie}}(F(\mathcal{N})/S, \mathcal{M}_1 \oplus \mathcal{M}_2) \simeq \underline{\mathrm{Lie}}(F(\mathcal{N})/S, \mathcal{M}_1) \times_S \underline{\mathrm{Lie}}(F(\mathcal{N})/S, \mathcal{M}_2), \tag{2}$$

so $F(\mathcal{N})/S$ satisfies (E), by Remark 4.1.1.2(a).

It remains to show that the morphism of abelian groups

$$d : F(\mathcal{N}) \longrightarrow \underline{\mathrm{Lie}}(F(\mathcal{N})/S)$$

respects the $\mathcal{O}_S$-module structures. Consider the free $\mathcal{O}_S$-module $\mathcal{M} = t\mathcal{O}_S$. Then

$$\mathcal{O}(I_{S'}(\mathcal{M})) = \mathcal{O}(S')[t]/(t^2) = \mathcal{O}(S') \oplus t\mathcal{O}(S'), \qquad \underline{\mathrm{Lie}}(\mathbf{O}_S/S) \simeq t\mathcal{O}_S.$$

Let $\rho_t : I_S(\mathcal{M}) \to S$ be the structural morphism, corresponding to the injection

$$u_t : \mathcal{O}_S \hookrightarrow D_{\mathcal{O}_S}(\mathcal{M}) = \mathcal{O}_S \oplus t\mathcal{O}_S.$$

[74] Editor's note: this lemma has been added.

Recall (cf. 3.4.2) that for every S-functor X, $F(\mathcal{N})(X)$ is the set of S-morphisms $\phi : I_X(\mathcal{N}) \to G$ such that $\phi \circ \varepsilon_{\mathcal{N}} = e$, and the action of $a \in \mathcal{O}(X)$ is

$$a \cdot \phi = \phi \circ a^*,$$

where a^* is the endomorphism of $I_X(\mathcal{N})$ associated with a; see 2.1.3.

Consequently, by 4.3.4, the composite

$$d : F(\mathcal{N}) \xrightarrow{\sim} F(\mathcal{N}) \otimes_{\mathcal{O}_S} t\mathcal{O}_S \longrightarrow \underline{\mathrm{Lie}}(F(\mathcal{N})/S)$$

is given, for $S' \to S$ and $f \in \mathrm{Hom}_S(S', F(\mathcal{N}))$, by

$$f \longmapsto f \otimes t \longmapsto t \cdot (f \circ \rho_t) = f \circ \rho_t \circ t^*.$$

Here $f \circ \rho_t \in F(\mathcal{N})(I_{S'}(\mathcal{M}))$ is viewed as an S-morphism

$$\phi : I_{I_{S'}(\mathcal{M})}(\mathcal{N}) \longrightarrow G$$

such that $\phi \circ \varepsilon_{\mathcal{N}} = e$. We must prove that d is $\mathcal{O}_S$-linear, or equivalently that

$$d(a \cdot f) = u_t(a) \cdot d(f) \qquad (a \in \mathcal{O}(S')).$$

Viewing $f \in F(\mathcal{N})(S')$ as a morphism $I_{S'}(\mathcal{N}) \to G$, one likewise has $a \cdot f = f \circ a^*$. Functoriality in X of the action of $\mathcal{O}(X)$ on $I_X(\mathcal{N})$ gives the following commutative diagram.

$$\begin{tikzcd}
\mathrm{I}_{\mathrm{S}'}(\mathscr{N}) \arrow[r, "a^*"] & \mathrm{I}_{\mathrm{S}'}(\mathscr{N}) \\
\mathrm{I}_{\mathrm{I}_{\mathrm{S}'}(\mathscr{M})}(\mathscr{N}) \arrow[u, "\rho_t"] \arrow[r, "u_t(a)^*"] & \mathrm{I}_{\mathrm{I}_{\mathrm{S}'}(\mathscr{M})}(\mathscr{N}) \arrow[u, "\rho_t"'] .
\end{tikzcd}$$

Figure 48: Compatibility of scalar action with the structural morphism ρ_t.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 36; combined-reader p. 94; printed p. 84.

Therefore

$$\begin{aligned} d(a \cdot f) &= f \circ a^* \circ \rho_t \circ t^* \\ &= f \circ \rho_t \circ u_t(a)^* \circ t^* \\ &= f \circ \rho_t \circ t^* \circ u_t(a)^* = u_t(a) \cdot d(f). \end{aligned}$$

The penultimate equality follows from the commutativity of $\mathcal{O}$. This completes the proof of Lemma 4.6.2.

Theorem 4.7. *If F is a good $\mathcal{O}_S$-module, then the S-group $\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)$ is good.*

[75] Indeed, by 4.5, $\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S$ satisfies (E), while

$$\underline{\mathrm{Lie}}\big(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S\big) \simeq \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F)$$

is a good $\mathcal{O}_S$-module.

[75] Editor's note: the original has been simplified here by using the additions made in 4.3.1 and 4.5.

4.7.1.

[76] Let G be an S-group and F a good $\mathscr{O}_S$-module. Suppose given a linear representation of G in F, that is (I 4.7.1), a morphism of S-groups

$$\rho : G \longrightarrow \underline{\mathrm{Aut}}_{\mathscr{O}_S\text{-mod.}}(F).$$

If G/S satisfies (E), then 4.1.C and 4.5 give a morphism of $\mathscr{O}_S$-modules, denoted ρ' or $d\rho$,

$$\underline{\mathrm{Lie}}(G/S) \longrightarrow \underline{\mathrm{Lie}}\big(\underline{\mathrm{Aut}}_{\mathscr{O}_S\text{-mod.}}(F)/S\big) \simeq \underline{\mathrm{End}}_{\mathscr{O}_S\text{-mod.}}(F).$$

[77] Furthermore, put $\mathscr{O}_{I_S} = \mathscr{O}_S \oplus t\mathscr{O}_S$, with $t^2 = 0$. It follows from 4.5.1 that, if $S' \to S$ and

$$X \in \underline{\mathrm{Lie}}(G/S)(S') \subset G(I_{S'}),$$

then in $\underline{\mathrm{Aut}}_{\mathscr{O}_{I_{S'}}\text{-mod.}}(F_{I_{S'}})$ one has

$$\rho(X) = \mathrm{id} + t\rho'(X). \tag{*}$$

In other words, for every $S'' \to I_{S'}$ and $f \in F(S'')$,

$$\rho(X)(f) = f + t\rho'(X)(f) \quad \text{in } F(S'').$$

Definition 4.7.2. Let G be a good S-group. Then $\underline{\mathrm{Lie}}(G/S)$ is a good $\mathscr{O}_S$-module, and there is a morphism of S-groups

$$\mathrm{Ad} : G \longrightarrow \underline{\mathrm{Aut}}_{\mathscr{O}_S\text{-mod.}}\big(\underline{\mathrm{Lie}}(G/S)\big).$$

By 4.7.1, it induces a morphism of $\mathscr{O}_S$-modules

$$\mathrm{ad} : \underline{\mathrm{Lie}}(G/S) \longrightarrow \underline{\mathrm{End}}_{\mathscr{O}_S\text{-mod.}}\big(\underline{\mathrm{Lie}}(G/S)\big),$$

or, equivalently, an $\mathscr{O}_S$-bilinear morphism

$$\underline{\mathrm{Lie}}(G/S) \times_S \underline{\mathrm{Lie}}(G/S) \longrightarrow \underline{\mathrm{Lie}}(G/S), \qquad (x, y) \longmapsto [x, y] = \mathrm{ad}(x) \cdot y.$$

Here x, y are arbitrary elements of

$$\underline{\mathrm{Lie}}(G/S)(S') = \underline{\mathrm{Lie}}(G_{S'}/S')(S').$$

If G is commutative, then $[x, y] = 0$.

4.7.3.

The bracket admits the following equivalent description. It is enough to consider sections over S, that is,

$$x, y \in \underline{\mathrm{Lie}}(G/S)(S).$$

There is a canonical isomorphism $I_S \times_S I_S \simeq I_{I_S}$. To avoid confusion, let I and I' denote two copies of I_S, and put

$$\mathscr{O}_I = \mathscr{O}_S[t], \qquad \mathscr{O}_{I'} = \mathscr{O}_S[t'], \qquad t^2 = 0 = (t')^2.$$

[76] Editor's note: the numbering 4.7.1, 4.7.2, and 4.7.3 has been added.
[77] Editor's note: the following sentence has been added.

There is then a commutative square

$$\begin{tikzcd} \mathrm{I} \times \mathrm{I}' \arrow[r] \arrow[d] & \mathrm{I}' \arrow[d] \\ \mathrm{I} \arrow[r] & \mathrm{S} \end{tikzcd} \quad ,$$

Figure 49: The two dual-number structures on $I \times_S I'$.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 37; combined-reader p. 95; printed p. 85.

in which the two arrows from $I \times I'$ identify it with the scheme of dual numbers over I or over I', respectively. Writing $L = \underline{\mathrm{Lie}}(G/S)$, one obtains the following commutative diagram of groups.

$$(1) \qquad \begin{tikzcd} & & 1 \arrow[d] & 1 \arrow[d] & \\ & & \mathrm{L(I)} \arrow[r] \arrow[d] & \mathrm{L(S)} \arrow[r] \arrow[d] & 1 \\ 1 \arrow[r] & \mathrm{L(I')} \arrow[r] \arrow[d] & \mathrm{G(I \times I')} \arrow[r] \arrow[d] & \mathrm{G(I')} \arrow[r] \arrow[d] & 1 \\ 1 \arrow[r] & \mathrm{L(S)} \arrow[r] \arrow[d] & \mathrm{G(I)} \arrow[r] \arrow[d] & \mathrm{G(S)} \arrow[r] \arrow[d] & 1 \\ & 1 & 1 & 1 & . \end{tikzcd}$$

Figure 50: The first exact-sequence diagram used to construct the bracket.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 38; combined-reader p. 96; printed p. 86.

The ninth object in this diagram is $\underline{\mathrm{Lie}}(L/S)(S)$. If G is good, this is $L(S)$, and hence one obtains the next commutative diagram. Its rows and columns are exact sequences of groups; the five groups denoted $L(-)$ are commutative.[78] Under the identifications

$$L(I) = L(S) \oplus tL(S), \qquad L(I') = L(S) \oplus t'L(S),$$

the injections $L(S) \hookrightarrow L(I)$ and $L(S) \hookrightarrow L(I')$ are $u \mapsto tu$ and $u \mapsto t'u$, respectively.

[78] Editor's note: the original has been expanded in what follows.

$$
\begin{tikzcd}
z \in \mathrm{L(S)} \arrow[r, "t"] \arrow[d, "t'"'] & \mathrm{L(I)} \arrow[r] \arrow[d] & \mathrm{L(S)} \ni x \arrow[d] \\
\mathrm{L(I')} \arrow[r] \arrow[d] & \mathrm{G(I \times I')} \arrow[r] \arrow[d] & \mathrm{G(I')} \arrow[d] \\
y \in \mathrm{L(S)} \arrow[r] & \mathrm{G(I)} \arrow[r] & \mathrm{G(S)}
\end{tikzcd} \qquad . \tag{2}
$$

Figure 51: The exact-sequence diagram defining $[x, y]$.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 38; combined-reader p. 96; printed p. 86.

In such a diagram, take x, y as indicated and choose arbitrary lifts $\widetilde{x} \in L(I)$ of x and $y' \in L(I')$ of y. The commutator

$$\widetilde{x}\, y'\, \widetilde{x}^{-1}(y')^{-1} \quad \text{in } G(I \times I')$$

is independent of the chosen lifts and is the image of a uniquely determined element z as shown. One has $z = [x, y]$. [79]

Indeed, use the same letters x and y for their images under the canonical sections $L(S) \to L(I)$ and $L(S) \to L(I')$. Then

$$\widetilde{x} = xu, \qquad y' = yv, \qquad u, v \in L(S) = L(I) \cap L(I').$$

Since $L(I)$ and $L(I')$ are commutative,

$$
\begin{aligned}
\widetilde{x}\, y'\, \widetilde{x}^{-1}(y')^{-1} &= xuyvu^{-1}x^{-1}v^{-1}y^{-1} \\
&= xuyu^{-1}vx^{-1}v^{-1}y^{-1} \\
&= xyx^{-1}y^{-1}.
\end{aligned}
$$

This element maps to the unit in both $G(I)$ and $G(I')$, and therefore comes from the unique z indicated above. Finally, viewing y as an element of $L_{I'}(I')$, and x as an element of $L(S) \subset G(I')$, formula 4.7.1 $(*)$ gives

$$xyx^{-1} = \mathrm{Ad}(x)(y) = (\mathrm{id} + t'\,\mathrm{ad}(x))(y) = y + t'[x, y].$$

Thus $xyx^{-1}y^{-1} \in L(I')$ is the image of $z = [x, y] \in L(S)$.

Two properties are now apparent from this construction:

(i) the bracket is functorial in G. More precisely,

$$G \longmapsto \underline{\mathrm{Lie}}(G/S)$$

is a functor from good S-groups to good $\mathcal{O}_S$-modules equipped with an $\mathcal{O}_S$-bilinear composition law;

(ii) $[x, y] + [y, x] = 0$, since the diagram is symmetric about its main diagonal.[80]

---

[79] Editor's note: the original said in a footnote that the author, at Gabriel's request, acknowledged that the exercise was not immediate, which was precisely why it had not been placed in the text. The details have been supplied here, taking account of the addition in 4.7.1; see also [DG70], §II.4, 4.2.

[80] Editor's note: that is, x and y play symmetric roles. The image of $[y, x]$ in $G(I \times I')$ is the commutator $y'\widetilde{x}(y')^{-1}\widetilde{x}^{-1}$, the inverse of $\widetilde{x}y'\widetilde{x}^{-1}(y')^{-1} = [x, y]$. On the other hand (cf. 4.10 below), it is not known at this point whether $[x, x] = 0$ necessarily holds: if x is represented by $\widetilde{x} \in L(I)$ and by $x' \in L(I')$, it is not a priori clear that $\widetilde{x}$ and x' commute.

Proposition 4.8. *Let F be a good $\mathcal{O}_S$-module. Under the identification*

$$\underline{\mathrm{Lie}}\big(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S\big) = \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F),$$

one has

$$\mathrm{Ad}(g)\cdot Y = g\circ Y\circ g^{-1}, \qquad [X,Y] = X\circ Y - Y\circ X,$$

for every $S'\to S$, $g\in \underline{\mathrm{Aut}}_{\mathcal{O}_{S'}\text{-mod.}}(F_{S'})$, and

$$X,Y\in \underline{\mathrm{Lie}}\big(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S\big)\,(S') = \underline{\mathrm{End}}_{\mathcal{O}_{S'}\text{-mod.}}(F_{S'}).$$

Proof.[81] After base change, we may assume $S'=S$. Put $I=I_S$ and $\mathcal{O}_I=\mathcal{O}_S[t]$, with $t^2=0$. Recall from 4.5.1 that the inclusion

$$i:\underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F)\hookrightarrow \underline{\mathrm{Aut}}_{\mathcal{O}_I\text{-mod.}}(F_I)$$

sends Y to $\mathrm{id}+tY$. By the definition of $\mathrm{Ad}(g)$ in 4.1.A,

$$\mathrm{id}+t\,\mathrm{Ad}(g)(Y) = g\circ(\mathrm{id}+tY)\circ g^{-1} = \mathrm{id}+t(g\circ Y\circ g^{-1}),$$

and therefore $\mathrm{Ad}(g)(Y)=g\circ Y\circ g^{-1}$.

Let I' be a second copy of I_S, and put $\mathcal{O}_{I'}=\mathcal{O}_S[t']$, with $(t')^2=0$. Apply 4.7.3 to

$$G=\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F), \qquad L=\underline{\mathrm{Lie}}(G/S)=\underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F).$$

Here $L=\mathrm{End}(F)$ is the differential identification of Proposition 4.5. The re-edition prints $\mathrm{Aut}(F)$ at this point, but $\mathrm{End}(F)$ is the type-correct object used by the ensuing endomorphism calculation. Identify X with its image under the canonical section $L(S)\hookrightarrow L(I)$. Its image in $G(I\times I')$ is $\mathrm{id}+t'X$, whose inverse is $\mathrm{id}-t'X$. Similarly, Y lifts to $\mathrm{id}+tY$, with inverse $\mathrm{id}-tY$. Their commutator is

$$\begin{aligned}(\mathrm{id}+t'X)\circ(\mathrm{id}+tY)\circ(\mathrm{id}-t'X)\circ(\mathrm{id}-tY)\\ =\mathrm{id}+tt'(X\circ Y-Y\circ X).\end{aligned}$$

This is the image in $G(I\times I')$ of $Z=X\circ Y-Y\circ X\in L(S)$: first Z maps to $tZ\in L(I)$, and then to $\mathrm{id}+t'tZ\in G(I\times I')$. By 4.7.3, $[X,Y]=X\circ Y-Y\circ X$.

Corollary 4.8.1. *Let G be a good S-group and let $x,y,z\in\underline{\mathrm{Lie}}(G/S)(S')$. Then*

$$[x,[y,z]]+[y,[z,x]]+[z,[x,y]]=0.$$

[82] Indeed, since G is good, $\underline{\mathrm{Lie}}(G/S)$ is a good $\mathcal{O}_S$-module; hence Theorem 4.7 shows that

$$\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}\big(\underline{\mathrm{Lie}}(G/S)\big)$$

is a good S-group. The morphism

$$\mathrm{Ad}: G\longrightarrow \underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}\big(\underline{\mathrm{Lie}}(G/S)\big)$$

and the functoriality in 4.7.3(i) yield

$$\mathrm{ad}[x,y]=[\mathrm{ad}\,x,\mathrm{ad}\,y].$$

[81] Editor's note: the original has been expanded and simplified in what follows.
[82] Editor's note: the original has been expanded in what follows.

Together with Proposition 4.8, this gives

$$\operatorname{ad}[x,y] = \operatorname{ad} x \circ \operatorname{ad} y - \operatorname{ad} y \circ \operatorname{ad} x.$$

Applying both sides to z gives the Jacobi identity.

Corollary 4.8.2. *Let G be a good S-group acting linearly on a good $\mathcal{O}_S$-module F; equivalently, let F be a G-$\mathcal{O}_S$-module, with G and F good. Then the linear map*

$$\rho' : \underline{\mathrm{Lie}}(G/S) \longrightarrow \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F)$$

is a representation; that is,

$$\rho'([x,y]) = \rho'(x) \circ \rho'(y) - \rho'(y) \circ \rho'(x).$$

Scholium 4.9. To every good S-group G (for example, every representable one) we have functorially associated a good $\mathcal{O}_S$-module $\underline{\mathrm{Lie}}(G/S)$ equipped with a bilinear map satisfying

$$[x,y] + [y,x] = 0, \qquad [x,[y,z]] + [y,[z,x]] + [z,[x,y]] = 0.$$

We shall call $\underline{\mathrm{Lie}}(G/S)$, with this structure, the *Lie algebra* of G over S. The qualification is that it is not yet known whether it is an $\mathcal{O}_S$-Lie algebra in the strict sense.[83] Every linear representation of G in a good $\mathcal{O}_S$-module F induces a representation of its Lie algebra. In particular, the adjoint representation of G induces the adjoint representation of its Lie algebra.

Definition 4.10. An S-functor in groups G is called *very good* if it is good and $\underline{\mathrm{Lie}}(G/S)$ is an $\mathcal{O}_S$-Lie algebra, that is, if $[x,x] = 0$ identically.

Examples 4.10.1. The following S-groups are very good: $\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)$ for every good $\mathcal{O}_S$-module F (by 4.7 and 4.8); every representable group (as shown below); every good S-group admitting a monomorphism into a very good S-group, in particular every good subgroup functor of a representable group; and every good S-group admitting a faithful linear representation in a good $\mathcal{O}_S$-module, for example every good S-group for which Ad is a monomorphism.

4.11.

Suppose now that G is a group scheme over S. By 4.1.4, $\underline{\mathrm{Lie}}(G/S)(S)$ identifies with the group of infinitesimal automorphisms of G/S invariant under right translation; by 3.14, this is the group of derivations of $\mathcal{O}_G$ over $\mathcal{O}_S$ invariant under right translation. This identification respects the module structure and is an anti-isomorphism of Lie algebras.[84]

To see this, argue as in 4.7.3. Put

$$\mathcal{O}_I = \mathcal{O}_S[t], \qquad \mathcal{O}_{I'} = \mathcal{O}_S[t'], \qquad t^2 = 0 = (t')^2,$$

and let $x \in L(I)$, $y \in L(I')$, where $L = \underline{\mathrm{Lie}}(G/S)$. Left translation λ_x, respectively λ_y, is an S-automorphism of $G_{I\times I'}$ inducing the identity on $G_{I'}$, respectively on G_I. It corresponds to the $\mathcal{O}_S$-automorphism

$$u = \mathrm{id} + t\, d_x, \qquad \text{respectively} \qquad v = \mathrm{id} + t'\, d_y$$

[83] Editor's note: the unresolved point is whether $[x,x] = 0$; see 4.10.

[84] Editor's note: a sign error in the original has been corrected here; cf. [DG70], §II.4, 4.4 and 4.6.

of

$$\mathcal{O}_{G_{I\times I'}} = \mathcal{O}_G[t,t']/(t^2,(t')^2),$$

where d_x, d_y are $\mathcal{O}_S$-derivations of $\mathcal{O}_G$ invariant under right translation. The correspondence between S-automorphisms of $G_{I\times I'}$ and $\mathcal{O}_S$-automorphisms of $\mathcal{O}_{G_{I\times I'}}$ is contravariant. Therefore

$$\lambda_x\lambda_y\lambda_x^{-1}\lambda_y^{-1}$$

corresponds to

$$v^{-1}u^{-1}vu = \mathrm{id} + tt'(d_yd_x - d_xd_y).$$

It follows from 4.7.3 that

$$x \longmapsto -d_x$$

is an isomorphism of Lie algebras; for further details, see [DG70], §II.4, 4.4 and 4.6. The same argument applies to

$$\underline{\mathrm{Lie}}(G/S)(S') = \underline{\mathrm{Lie}}(G_{S'}/S')(S')$$

for every $S' \to S$. Thus one recovers the classical definition.[85]

Scholium 4.11.1. *Via the isomorphism $x \mapsto -d_x$, $\underline{\mathrm{Lie}}(G/S)$ identifies with the functor that assigns to each $S' \to S$ the $\mathcal{O}(S')$-Lie algebra of derivations of $G_{S'}$ relative to S' invariant under right translation.*

Since every representable group is already known to be good by 4.6.1, the preceding discussion gives:

Corollary 4.11.2. *Every representable group is very good.*

Let $\varepsilon : S \to G$ be the unit section of G, and put

$$\omega^1_{G/S} = \varepsilon^*(\Omega^1_{G/S}).$$

Recall from 3.3 that $\underline{\mathrm{Lie}}(G/S)$ is represented by the vector fibration $V(\omega^1_{G/S})$.

Scholium 4.11.3. *Thus to every S-group scheme G we have functorially associated a vector fibration*

$$\mathrm{Lie}(G/S) = V(\omega^1_{G/S})$$

over S, representing the functor $\underline{\mathrm{Lie}}(G/S)$, and hence endowed with the structure of an S-scheme in $\mathcal{O}_S$-Lie algebras. Moreover, this construction commutes with base extension and finite products (cf. 3.4 and 3.8).

Remarks 4.11.4. [86] Let $\pi : G \to S$ be the structural morphism.

(a) The $\mathcal{O}_G$-module $\Omega^1_{G/S}$ is naturally $(G\times_S G)$-equivariant (cf. I, §6). Hence, by I 6.8.1,

$$\Omega^1_{G/S} \simeq \pi^*(\omega^1_{G/S}).$$

For example, $\Omega^1_{G/S}$ is locally free, respectively locally free of finite type, whenever $\omega^1_{G/S}$ is. This holds in particular when S is the spectrum of a field, respectively when S is the spectrum of a field and G is locally of finite type over S.

(b) By I 6.8.2, $\omega^1_{G/S}$ has a canonical G-$\mathcal{O}_S$-module structure, which induces the adjoint action on

$$V(\omega^1_{G/S}) = \mathrm{Lie}(G/S).$$

[85] Editor's note: the original has been expanded in what follows. In particular, the numbering 4.11.1–4.11.8 has been added for later reference.

[86] Editor's note: what follows has been expanded, taking account of the additions made in Expose I, §6.8.

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(c) By EGA I, 5.3.11 and 5.4.6, $\varepsilon : S \to G$ is an immersion; it is a closed immersion if G is separated over S. Thus $\omega^1_{G/S}$ identifies with $\mathcal{I}/\mathcal{I}^2$, where $\mathcal{I}$ is the quasi-coherent ideal defining $\varepsilon(S)$ in an open subscheme $U \subset G$ in which $\varepsilon(S)$ is closed. If $G \to S$ is separated, one may take $U = G$. If $G = \operatorname{Spec} A(G)$ is affine over S, then $\mathcal{I}$ is the augmentation ideal of $A(G)$, namely the kernel of

$$\varepsilon^\natural : A(G) \longrightarrow \mathcal{O}_S$$

(cf. I 4.2).[88]

Remark 4.11.5. The isomorphism $\Omega^1_{G/S} \simeq \pi^*(\omega^1_{G/S})$ identifies the $\mathcal{O}_S$-module $\omega^1_{G/S}$ with the sheaf

$$\pi_*(\Omega^1_{G/S})^G$$

of differentials of G relative to S invariant under right translation: its sections over an open $U \subset S$ are the sections of $\Omega^1_{G/S}$ over $\pi^{-1}(U)$ invariant under right translation (cf. I 6.8.3; compare also $\mathrm{VII_A}$, 2.4).[89]

Notation 4.11.6. Write $\mathscr{L}ie(G/S)$ for the sheaf of sections of the vector fibration $\mathrm{Lie}(G/S) \to S$. It is the $\mathcal{O}_S$-module

$$\mathscr{L}ie(G/S) = (\omega^1_{G/S})^\vee = \mathcal{H}om_{\mathcal{O}_S}(\omega^1_{G/S}, \mathcal{O}_S),$$

dual to $\omega^1_{G/S}$ (cf. EGA II, 1.7.9), and it carries a structure of $\mathcal{O}_S$-Lie algebra.

In general this construction does not commute with base extension. Consequently, the Lie-algebra structure on this module does not determine the structure of an S-scheme in $\mathcal{O}_S$-Lie algebras on $\mathrm{Lie}(G/S)$.[90]

Nevertheless:

Lemma 4.11.7. *Suppose that $\omega^1_{G/S}$ is locally free of finite type. This is the case, in particular, if G is smooth over S (cf. EGA IV_4, 17.2.4), or if S is the spectrum of a field and G is locally of finite type over S. Then*

$$\mathscr{L}ie(G/S)^\vee \simeq (\omega^1_{G/S})^{\vee\vee} \simeq \omega^1_{G/S},$$

and therefore

$$\mathrm{Lie}(G/S) = V(\omega^1_{G/S}) = V\big(\mathscr{L}ie(G/S)^\vee\big) = W\big(\mathscr{L}ie(G/S)\big),$$

where the final equality follows from I 4.6.5.1.

Finally, let $G \to H$ be a monomorphism of functors in groups. Then

$$\mathrm{Lie}(G/S) \longrightarrow \mathrm{Lie}(H/S)$$

[87] Editor's note: for this reason, the linear action of G on $\omega^1_{G/S}$ could be called the pre-adjoint action; by an abuse of terminology it will still be called the adjoint action. A slightly more general construction, used in Expose III (especially III 0.8), is as follows. Suppose that to every $Y \to S$ one assigns an $\mathcal{O}_Y$-module $\mathcal{N}(Y)$, and to every S-morphism $\phi : Z \to Y$ a functorial morphism of $\mathcal{O}_Z$-modules $\mathcal{N}(\phi) : \phi^*\mathcal{N}(Y) \to \mathcal{N}(Z)$. For example, if $\mathcal{J}$ is a quasi-coherent ideal of $\mathcal{O}_S$, one may take $\mathcal{N}(Y) = \mathcal{J}\mathcal{O}_Y$. The action of G on $\omega^1_{G/S}$ then induces a generalized adjoint action of G on the S-functor in abelian groups that assigns to $f : Y \to S$ the group $\mathrm{Hom}_{\mathcal{O}_Y}(f^*\omega^1_{G/S}, \mathcal{N}(Y))$.

[88] Editor's note: the preceding description of $\omega^1_{G/S}$, which is useful later (for example in $\mathrm{VII_A}$, 5.5), has been added.

[89] Editor's note: this description of $\omega^1_{G/S}$ in terms of invariant differentials is not used below.

[90] Editor's note: the reason is that $\mathscr{L}ie(G/S) \simeq (\omega^1_{G/S})^\vee$ need not determine $\omega^1_{G/S}$. For example, let $S = \mathbb{A}^1_k$, where k is a field, and let G be the S-group whose fiber at $s = 0$ is $\mathbb{G}_{a,k}$, while all other fibers are the unit group. Then $\mathscr{L}ie(G/S) = 0$, but $\mathrm{Lie}(G_s/k)(k) = k$.

is also a monomorphism (cf. 3.7). Every vector monomorphism of vector fibrations is a closed immersion.[91] It follows that:

Corollary 4.11.8. *Let $G \hookrightarrow H$ be a monomorphism of S-group schemes.*

(i) The morphism

$$\mathrm{Lie}(G/S) \longrightarrow \mathrm{Lie}(H/S)$$

is a closed immersion; consequently

$$\omega^1_{H/S} \longrightarrow \omega^1_{G/S}$$

is an epimorphism.

(ii) If $\omega^1_{G/S}$ is locally free of finite type, then the corresponding morphism

$$\mathscr{L}ie(G/S) \longrightarrow \mathscr{L}ie(H/S)$$

identifies $\mathscr{L}ie(G/S)$ with an $\mathcal{O}_S$-submodule of $\mathscr{L}ie(H/S)$ that is locally a direct summand.

5. Computation of some Lie algebras

5.1. Examples of Lie algebras: diagonalizable groups

Let $G = D_S(M)$ be a diagonalizable group over S (I 4.4). Since the formation of $\underline{\mathrm{Lie}}(G/S)$ commutes with base extension, it is enough to carry out the construction for $G = D(M)$. We then have

$$G(I_S) = \mathrm{Hom}_{\mathrm{gr}}\big(M, \Gamma(I_S, \mathscr{O}_{I_S})^\times\big) = \mathrm{Hom}_{\mathrm{gr}}\big(M, \Gamma(S, \mathscr{D}_{\mathscr{O}_S})^\times\big).$$

There is a split exact sequence

$$1 \longrightarrow \Gamma(S, \mathscr{O}_S) \longrightarrow \Gamma(S, \mathscr{D}_{\mathscr{O}_S})^\times \longrightarrow \Gamma(S, \mathscr{O}_S)^\times \longrightarrow 1,$$

which shows that $\underline{\mathrm{Lie}}(G)(S)$ identifies with $\mathrm{Hom}_{\mathrm{gr}}(M, \mathscr{O}(S))$, endowed with its evident $\mathscr{O}(S)$-module structure. After base change we therefore obtain:

Proposition 5.1. *There are isomorphisms*

$$\underline{\mathrm{Hom}}_{S\text{-gr}}(M_S, \mathbf{O}_S) \xrightarrow{\sim} \underline{\mathrm{Lie}}(D_S(M)/S)$$

and

$$\mathscr{H}om_{\mathrm{gr}}(\widetilde{M}_S, \mathscr{O}_S) \xrightarrow{\sim} \mathscr{L}ie(D_S(M)/S),$$

where, in the second isomorphism, $\widetilde{M}_S$ denotes the constant sheaf of groups on S defined by M, and $\mathscr{H}om_{\mathrm{gr}}$ denotes the sheaf of homomorphisms of sheaves of groups.

Corollary 5.1.1. *If M is free of finite type—or, as we shall say later, if $D_S(M)$ is a split torus—then, with W as in I 4.6.5,*

$$W\big(\mathscr{L}ie(D_S(M)/S)\big) \xrightarrow{\sim} \underline{\mathrm{Lie}}(D_S(M)/S), \qquad M^\vee \otimes_{\mathbb{Z}} \mathscr{O}_S \xrightarrow{\sim} \mathscr{L}ie(D_S(M)/S).$$

Here $M^\vee$ denotes the dual of the abelian group M. In particular,

$$\mathbf{O}_S \xrightarrow{\sim} \underline{\mathrm{Lie}}(\mathbb{G}_{m,S}/S), \qquad \mathscr{O}_S \xrightarrow{\sim} \mathscr{L}ie(\mathbb{G}_{m,S}/S).$$

[91] Editor's note: let $f : M \to N$ be a morphism of $\mathcal{O}_S$-modules and let $P = \mathrm{Coker}(f)$. If $V(N) \to V(M)$ is a monomorphism, the surjective morphism $\mathrm{Sym}(N) \to \mathrm{Sym}(P)$ factors through $\mathcal{O}_S$; hence $P = 0$.

5.2. Normalizers and centralizers

We first prove a few lemmas. Recall (I 3.1.1) that a sequence

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$$

of $\mathbf{O}_S$-modules is called *exact* if, for every $S' \to S$, the sequence

$$0 \longrightarrow F'(S') \longrightarrow F(S') \longrightarrow F''(S') \longrightarrow 0$$

of $\mathbf{O}(S')$-modules is exact. Likewise, a sequence $1 \to G' \to G \to G'' \to 1$ of S-groups is called exact if the corresponding sequence of groups on S'-points is exact for every $S' \to S$.

Lemma 5.2.1. *Let $1 \to G' \to G \to G'' \to 1$ be an exact sequence of S-groups.* [92]

(i) *The sequences*

$$1 \longrightarrow T_{G'/S}(\mathscr{M}) \longrightarrow T_{G/S}(\mathscr{M}) \longrightarrow T_{G''/S}(\mathscr{M}) \longrightarrow 1$$

and

$$1 \longrightarrow \underline{\mathrm{Lie}}(G'/S, \mathscr{M}) \longrightarrow \underline{\mathrm{Lie}}(G/S, \mathscr{M}) \longrightarrow \underline{\mathrm{Lie}}(G''/S, \mathscr{M}) \longrightarrow 1$$

are exact.

(ii) *Let $1 \to H' \to H \to H'' \to 1$ be a second sequence of groups. It is exact if and only if*

$$1 \longrightarrow G' \times_S H' \longrightarrow G \times_S H \longrightarrow G'' \times_S H'' \longrightarrow 1$$

is exact.

(iii) *If two of G', G, G'' satisfy condition (E), then so does the third.*

(iv) *If $0 \to F' \to F \to F'' \to 0$ is exact and two of the $\mathbf{O}_S$-modules F', F, F'' are good, then so is the third.*

(v) *If two of the S-groups G', G, G'' are good, then so is the third.*

The first part of (i) is immediate, and the second follows from it. Likewise, (ii) is immediate. We prove (iii). For brevity, write $L(\mathscr{M}) = \underline{\mathrm{Lie}}(G, \mathscr{M})$, $L'(\mathscr{M}) = \underline{\mathrm{Lie}}(G', \mathscr{M})$, and similarly for L''. We have the following commutative diagram. [93]

$$\begin{tikzcd}
1 \arrow[r] & \mathrm{L}'(\mathscr{M} \oplus \mathscr{N}) \arrow[r] \arrow[d] & \mathrm{L}(\mathscr{M} \oplus \mathscr{N}) \arrow[r] \arrow[d] & \mathrm{L}''(\mathscr{M} \oplus \mathscr{N}) \arrow[r] \arrow[d] & 1 \\
1 \arrow[r] & \mathrm{L}'(\mathscr{M}) \times \mathrm{L}'(\mathscr{N}) \arrow[r] & \mathrm{L}(\mathscr{M}) \times \mathrm{L}(\mathscr{N}) \arrow[r] & \mathrm{L}''(\mathscr{M}) \times \mathrm{L}''(\mathscr{N}) \arrow[r] & 1
\end{tikzcd}$$

Figure 52: The exact-sequence diagram used to prove Lemma 5.2.1(iii).

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 44; combined-reader p. 102.

The upper row is exact by (i), and the lower row is exact by (i) and (ii). Assertion (iii) follows.

[92] Editor's note: The statement and proof of this lemma have been expanded; this will be useful in 5.3.1.

[93] Editor's note: The proofs of (iii) and (v) have been added.

We prove (iv). There is a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{F}' \otimes_{\mathbf{O}_\mathrm{S}} \mathrm{T}_{\mathbf{O}_\mathrm{S}/\mathrm{S}}(\mathscr{M}) \arrow[r] \arrow[d] & \mathrm{F} \otimes_{\mathbf{O}_\mathrm{S}} \mathrm{T}_{\mathbf{O}_\mathrm{S}/\mathrm{S}}(\mathscr{M}) \arrow[r] \arrow[d] & \mathrm{F}'' \otimes_{\mathbf{O}_\mathrm{S}} \mathrm{T}_{\mathbf{O}_\mathrm{S}/\mathrm{S}}(\mathscr{M}) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathrm{T}_{\mathrm{F}'/\mathrm{S}}(\mathscr{M}) \arrow[r] & \mathrm{T}_{\mathrm{F}/\mathrm{S}}(\mathscr{M}) \arrow[r] & \mathrm{T}_{\mathrm{F}''/\mathrm{S}}(\mathscr{M}) \arrow[r] & 0
\end{tikzcd}$$

Figure 53: The exact-sequence diagram used to prove Lemma 5.2.1(iv).

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 44; combined-reader p. 102.

with exact rows: the first because $T_{\mathbf{O}_S/S}(\mathscr{M})$ is a free, hence flat, $\mathbf{O}_S$-module, and the second by (i). Thus, if two of F', F, F'' are good, so is the third. Finally, (v) follows from (iii) and (iv).

Lemma 5.2.2. *Let G be an S-group, let E, H be two G-objects, and let F be a G-$\mathbf{O}_S$-module.*

(i) *The canonical homomorphism $E^G \times_S H^G \to (E \times_S H)^G$ is an isomorphism.*

(ii) *If F is a good $\mathbf{O}_S$-module, then so is F^G.*

The first assertion is immediate; we prove the second. For every $S' \to S$, there is a commutative diagram

$$\begin{tikzcd}
\mathrm{F}^\mathrm{G}(\mathrm{I}_{\mathrm{S}'}(\mathscr{M})) \arrow[rr, hook] & & \mathrm{F}(\mathrm{I}_{\mathrm{S}'}(\mathscr{M})) \\
\mathrm{F}^\mathrm{G}(\mathrm{S}') \otimes_{\mathbf{O}(\mathrm{S}')} \mathbf{O}(\mathrm{I}_{\mathrm{S}'}(\mathscr{M})) \arrow[r, hook] \arrow[u, "\phi"] & \mathrm{F}(\mathrm{S}') \otimes_{\mathbf{O}(\mathrm{S}')} \mathbf{O}(\mathrm{I}_{\mathrm{S}'}(\mathscr{M})) \arrow[ur, "\simeq\ \phi_1"']
\end{tikzcd}$$

Figure 54: The invariants-and-base-change square in the proof of Lemma 5.2.2.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 45; combined-reader p. 103.

and we must prove that ϕ is bijective. It is plainly injective. To prove surjectivity, let $(t_0, \dots, t_n)$ be a basis of the free $\mathbf{O}(S')$-module $\mathbf{O}(I_{S'}(\mathscr{M}))$, and let

$$u = \sum_{i=0}^{n} f_i \otimes t_i$$

be an element of $F(S') \otimes_{\mathbf{O}(S')} \mathbf{O}(I_{S'}(\mathscr{M}))$ such that $\phi_1(u)$ belongs to $F^G(I_{S'}(\mathscr{M}))$. We show that each f_i belongs to $F^G(S')$.

Let $S'' \to S'$. Via the zero section $\varepsilon_{\mathscr{M}}$, we may regard S'' as lying over $I_{S'}(\mathscr{M})$. Then, for every $g \in G(S'')$,

$$u_{S''} = g \cdot u_{S''} = \sum_{i=0}^{n} g \cdot (f_i)_{S''} \otimes (t_i)_{S''}.$$

Since the $(t_i)_{S''}$ form a basis of

$$\mathbf{O}(I_{S''}(\mathscr{M})) \simeq \mathbf{O}(I_{S'}(\mathscr{M})) \otimes_{\mathbf{O}(S')} \mathbf{O}(S'')$$

over $\mathbf{O}(S'')$, it follows that $g \cdot (f_i)_{S''} = (f_i)_{S''}$, hence $f_i \in F^G(S')$ for all i.

Notation. If E is an S-group and F a sub-S-group, write E/F for the S-functor sending $S' \to S$ to the set $E(S')/F(S')$ of classes $\bar{x} = xF(S')$. If E is commutative, then E/F is a commutative S-group. [94]

Now let G be an S-group and K a sub-S-group, and set $E = \underline{\mathrm{Lie}}(G/S, \mathscr{M})$ and $F = \underline{\mathrm{Lie}}(K/S, \mathscr{M})$. The adjoint action of K on E stabilizes F, and therefore induces an action on E/F. For every $S' \to S$,

$$\begin{aligned}(E/F)^K(S') = \{\bar{x} \in E(S')/F(S') \mid{} & \text{for every } f : S'' \to S' \text{ and } k \in K(S''),\\ & f^*(\bar{x}^{-1}) \operatorname{Ad}(k)(f^*(\bar{x})) \in F(S'')\},\end{aligned}$$

where $f^*(\bar{x})$ denotes the image of $\bar{x}$ in $E(S'')$.

Theorem 5.2.3. *Let G be an S-group and K a sub-S-group. Put (I 2.3.3)*

$$N = \underline{\mathrm{Norm}}_G(K), \qquad Z = \underline{\mathrm{Centr}}_G(K),$$

and let K act on $\underline{\mathrm{Lie}}(G/S, \mathscr{M})$ through the adjoint representation of G.

(i) *If the group law of $\underline{\mathrm{Lie}}(G/S, \mathscr{M})$ is commutative,*[95][96] *then*

$$\frac{\underline{\mathrm{Lie}}(N/S, \mathscr{M})}{\underline{\mathrm{Lie}}(K/S, \mathscr{M})} = \left(\frac{\underline{\mathrm{Lie}}(G/S, \mathscr{M})}{\underline{\mathrm{Lie}}(K/S, \mathscr{M})}\right)^K.$$

(ii) *If the same group law is commutative, then*

$$\underline{\mathrm{Lie}}(Z/S, \mathscr{M}) = \underline{\mathrm{Lie}}(G/S, \mathscr{M})^K.$$

(iii) *If G satisfies (E), then Z satisfies (E); if G and K satisfy (E), then N satisfies (E).*

(iv) *If G is good, then Z is good; if, moreover, G is very good, then Z is very good.*

(v) *If G and K are good, then N is good; if, moreover, G is very good, then N is very good.*

To prove (i) and (ii),[97] we use the following lemma, which follows from the diagram of exact sequences considered in 4.1.B, with G and H interchanged.

Lemma 5.2.3.0. *Let G be an S-group, H a sub-S-group, and $\mathscr{M}$ a free $\mathscr{O}_S$-module of finite type. Then $T_{H/S}(\mathscr{M})$ and $\underline{\mathrm{Lie}}(G/S, \mathscr{M})$ are sub-S-groups of $T_{G/S}(\mathscr{M})$, and*

$$T_{H/S}(\mathscr{M}) \cap \underline{\mathrm{Lie}}(G/S, \mathscr{M}) = \underline{\mathrm{Lie}}(H/S, \mathscr{M}),$$

where

$$T_{H/S}(\mathscr{M}) \cap \underline{\mathrm{Lie}}(G/S, \mathscr{M}) := T_{H/S}(\mathscr{M}) \times_{T_{G/S}(\mathscr{M})} \underline{\mathrm{Lie}}(G/S, \mathscr{M}).$$

[94] Editor's note: These notations have been expanded.

[95] This condition is automatic when G satisfies (E), for example when G is representable.

[96] Editor's note: For an example in which the S-group $\underline{\mathrm{Lie}}(G/S)$ is noncommutative, see 6.3.

[97] Editor's note: The proof has been expanded, and in particular Lemma 5.2.3.0, implicit in the original, has been added.

Since the functors in (i) and (ii) commute with base extension, it is enough to prove the equalities on S-points. Put $H = N$ (respectively, $H = Z$) and let $\alpha = \pm 1$. By the preceding lemma and the definitions of N and Z (I 2.3.3),

$$\begin{aligned} \underline{\mathrm{Lie}}(H/S, \mathscr{M})(S) = \big\{ X \in \underline{\mathrm{Lie}}(G/S, \mathscr{M})(S) \subset G(I_S(\mathscr{M})) \ \big| \\ \text{for every } f : S' \to I_S(\mathscr{M}) \text{ and } u \in K(S'), \\ f^*(X^\alpha)\, u\, f^*(X^{-\alpha})\, u^{-1} \in K(S'), \\ \text{respectively } f^*(X)\, u\, f^*(X^{-1})\, u^{-1} = 1 \big\}. \end{aligned} \tag{*}$$

Here $f^*(X)$ denotes the image of X in $G(S')$.

For brevity, write

$$\mathfrak{g} = \underline{\mathrm{Lie}}(G/S, \mathscr{M}), \quad \mathfrak{k} = \underline{\mathrm{Lie}}(K/S, \mathscr{M}), \quad \mathfrak{n} = \underline{\mathrm{Lie}}(N/S, \mathscr{M}), \quad \mathfrak{z} = \underline{\mathrm{Lie}}(Z/S, \mathscr{M}).$$

If $X \in \underline{\mathrm{Lie}}(H/S, \mathscr{M})(S)$, the equalities (*) hold for every $f : S' \to S$, because f factors through $I_S(\mathscr{M})$ as

$$f = \rho \circ \varepsilon_{\mathscr{M}} \circ f,$$

where $\rho : I_S(\mathscr{M}) \to S$ is the structural morphism and $\varepsilon_{\mathscr{M}} : S \to I_S(\mathscr{M})$ is the zero section. Consequently,

$$\mathfrak{n}(S)/\mathfrak{k}(S) \subset (\mathfrak{g}/\mathfrak{k})^K(S), \qquad \mathfrak{z}(S) \subset \mathfrak{g}^K(S).$$

For the reverse inclusions, assume henceforth that $\mathfrak{g}$ is commutative. Then $\mathfrak{g}^K$ and $(\mathfrak{g}/\mathfrak{k})^K$ are commutative S-groups. Let $X \in \mathfrak{g}(S)$, and write $\bar{X}$ for its image in $(\mathfrak{g}/\mathfrak{k})(S)$. Suppose that $\bar{X} \in (\mathfrak{g}/\mathfrak{k})^K(S)$ (respectively, $X \in \mathfrak{g}^K(S)$); we show that $X \in \mathfrak{n}(S)$ (respectively, $X \in \mathfrak{z}(S)$).

Let $f : S' \to I_S(\mathscr{M})$. Consider the cartesian square

$$\begin{tikzcd} I_{S'}(\mathscr{M}) \arrow[r, "p"] \arrow[d, "\rho'"'] & I_S(\mathscr{M}) \arrow[d, "\rho"] \\ S' \arrow[r, "\rho \circ f"] & S \end{tikzcd}$$

Figure 55: The cartesian square used in the reverse-inclusion argument.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 47; combined-reader p. 105.

and let h be the section of ρ' defined by f. It is enough to show that, for every $v \in K(I_{S'}(\mathscr{M}))$,

$$\left.\begin{aligned} p^*(X^\alpha)\, v\, p^*(X^{-\alpha})\, v^{-1} \in K(I_{S'}(\mathscr{M})), \\ \text{respectively} \qquad p^*(X)\, v\, p^*(X^{-1})\, v^{-1} = 1. \end{aligned}\right\} \tag{**}$$

Indeed, take $v = \rho'^*(u)$ and apply h^* to (**). This gives (*), because $\rho' \circ h = \mathrm{id}_{S'}$ and $p \circ h = f$.

We now prove (**), writing X for $p^*(X)$. Every $v \in K(I_{S'}(\mathscr{M}))$ is uniquely of the form Yk, with $Y \in \underline{\mathrm{Lie}}(K/S, \mathscr{M})(S')$ and $k \in K(S')$. The expression $X^{\alpha} v X^{-\alpha} v^{-1}$ becomes

$$X^{\alpha} Y \bigl(k X^{-\alpha} k^{-1}\bigr) Y^{-1} = X^{\alpha} \operatorname{Ad}(k)(X^{-\alpha}),$$

because $\underline{\mathrm{Lie}}(G/S, \mathscr{M})$ is commutative. This element lies in $\underline{\mathrm{Lie}}(G/S, \mathscr{M})(S')$. By Lemma 5.2.3.0, condition (**) is therefore equivalent, for every $k \in K(S')$, to

$$\begin{cases} \operatorname{Ad}(k)(X) = X, & H = Z, \\ X^{\alpha} \operatorname{Ad}(k)(X^{-\alpha}) \in \underline{\mathrm{Lie}}(K/S, \mathscr{M})(S'), & H = N. \end{cases}$$

For $H = Z$, this follows from $X \in \mathfrak{g}^K(S)$. For $H = N$, the condition may equally be written

$$\operatorname{Ad}(k)(\bar{X}) = \bar{X}, \qquad \operatorname{Ad}(k)(\bar{X}^{-1}) = \bar{X}^{-1}. \tag{*'}$$

The second equality follows from the first because the action of K on $\mathfrak{g}/\mathfrak{k}$ respects its group structure. Thus (**) also follows from $\bar{X} \in (\mathfrak{g}/\mathfrak{k})^K(S)$, proving (i) and (ii).

For (iii)–(v), put

$$\mathfrak{g}(\mathscr{M}) = \underline{\mathrm{Lie}}(G/S, \mathscr{M})$$

and define $\mathfrak{k}(\mathscr{M})$, $\mathfrak{z}(\mathscr{M})$, and $\mathfrak{n}(\mathscr{M})$ similarly. If G/S satisfies (E), then

$$\mathfrak{g}(\mathscr{M} \oplus \mathscr{N}) \simeq \mathfrak{g}(\mathscr{M}) \times_S \mathfrak{g}(\mathscr{N}),$$

and hence, by Lemma 5.2.2(i),

$$\mathfrak{g}(\mathscr{M} \oplus \mathscr{N})^K \simeq \mathfrak{g}(\mathscr{M})^K \times_S \mathfrak{g}(\mathscr{N})^K.$$

It follows that $\mathfrak{z}(\mathscr{M} \oplus \mathscr{N}) \simeq \mathfrak{z}(\mathscr{M}) \times_S \mathfrak{z}(\mathscr{N})$, so Z satisfies (E). If K/S also satisfies (E), then

$$(\mathfrak{g}/\mathfrak{k})(\mathscr{M} \oplus \mathscr{N}) \simeq (\mathfrak{g}/\mathfrak{k})(\mathscr{M}) \times_S (\mathfrak{g}/\mathfrak{k})(\mathscr{N}),$$
$$(\mathfrak{g}/\mathfrak{k})^K(\mathscr{M} \oplus \mathscr{N}) \simeq (\mathfrak{g}/\mathfrak{k})(\mathscr{M})^K \times_S (\mathfrak{g}/\mathfrak{k})(\mathscr{N})^K,$$

and we have the following commutative diagram.

$$\begin{tikzcd}
0 \arrow[r] & \mathfrak{k}(\mathscr{M} \oplus \mathscr{N}) \arrow[r] \arrow[d, "\wr"] & \mathfrak{n}(\mathscr{M} \oplus \mathscr{N}) \arrow[r] \arrow[d] & (\mathfrak{n}/\mathfrak{k})(\mathscr{M} \oplus \mathscr{N}) \arrow[r] \arrow[d, "\wr"] & 0 \\
0 \arrow[r] & \mathfrak{k}(\mathscr{M}) \underset{S}{\times} \mathfrak{k}(\mathscr{N}) \arrow[r] & \mathfrak{n}(\mathscr{M}) \underset{S}{\times} \mathfrak{n}(\mathscr{N}) \arrow[r] & (\mathfrak{n}/\mathfrak{k})(\mathscr{M}) \underset{S}{\times} (\mathfrak{n}/\mathfrak{k})(\mathscr{N}) \arrow[r] & 0
\end{tikzcd}$$

Figure 56: The exact-sequence comparison proving condition (E) for N.

Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 48; combined-reader p. 106.

It follows that

$$\mathfrak{n}(\mathscr{M} \oplus \mathscr{N}) \simeq \mathfrak{n}(\mathscr{M}) \times_S \mathfrak{n}(\mathscr{N}),$$

so N satisfies (E).

Now write $\mathfrak{g} = \mathfrak{g}(\mathscr{O}_S) = \underline{\mathrm{Lie}}(G/S)$, and similarly $\mathfrak{k}, \mathfrak{z}, \mathfrak{n}$. If G is good, then $\mathfrak{g}$ is a good $\mathbf{O}_S$-module, and Lemma 5.2.2(ii) shows that $\mathfrak{z} \simeq \mathfrak{g}^K$ is good; since Z satisfies (E), Z is good. If K is also good, then $\mathfrak{k}$ is good, and Lemmas 5.2.1(iv) and 5.2.2(ii) show that $\mathfrak{g}/\mathfrak{k}$ and $(\mathfrak{g}/\mathfrak{k})^K$ are good. From the exact sequence

$$0 \longrightarrow \mathfrak{k} \longrightarrow \mathfrak{n} \longrightarrow (\mathfrak{g}/\mathfrak{k})^K \longrightarrow 0$$

and Lemma 5.2.1(iv), $\mathfrak{n}$ is good. Finally, if G is very good, so that $[x,x]=0$ identically for $x \in \mathfrak{g}$, then Z and N are plainly very good. This proves (iii)–(v).

Corollary 5.2.3.1. *If the group law of* $\underline{\mathrm{Lie}}(G/S)$ *is commutative, then*

$$\underline{\mathrm{Lie}}\big(\underline{\mathrm{Centr}}(G)/S\big) = \underline{\mathrm{Lie}}(G/S)^G.$$

Corollary 5.2.3.2. *If the group law of* $\underline{\mathrm{Lie}}(G/S)$ *is commutative and* K *is a normal subgroup of* G*, then*

$$(\underline{\mathrm{Lie}}(G/S)/\underline{\mathrm{Lie}}(K/S))^K = \underline{\mathrm{Lie}}(G/S)/\underline{\mathrm{Lie}}(K/S).$$

5.3. Linear representations

Let G be a good S-group acting linearly on a good $\mathbf{O}_S$-module F through

$$\rho : G \longrightarrow \underline{\mathrm{Aut}}_{\mathbf{O}_S\text{-mod}}(F).$$

In 4.7.1 and 4.8.2 we defined the corresponding linear representation

$$\rho' : \underline{\mathrm{Lie}}(G/S) \longrightarrow \underline{\mathrm{End}}_{\mathbf{O}_S\text{-mod}}(F).$$

The sub-S-groups $\underline{\mathrm{Norm}}_G(E)$ and $\underline{\mathrm{Centr}}_G(E)$ are defined for every subset E of F, and in particular for every $\mathbf{O}_S$-submodule E.

Definition 5.3.0. For every $S' \to S$, put

$$\underline{\mathrm{Norm}}_{\underline{\mathrm{Lie}}(G/S)}(E)(S') = \{X \in \underline{\mathrm{Lie}}(G/S)(S') \mid \rho'(X)E_{S'} \subset E_{S'}\},$$
$$\underline{\mathrm{Centr}}_{\underline{\mathrm{Lie}}(G/S)}(E)(S') = \{X \in \underline{\mathrm{Lie}}(G/S)(S') \mid \rho'(X)E_{S'} = 0\}.$$

This construction works for every linear representation of an $\mathbf{O}_S$-Lie algebra in the sense of 4.9, and the two subobjects obtained are $\mathbf{O}_S$-submodules stable under the bracket.

Theorem 5.3.1. *Let* G *be a good* S*-group acting linearly on a good* $\mathbf{O}_S$*-module* F*, and let* E *be an* $\mathbf{O}_S$*-submodule of* F*.*

(i) *One has*

$$\underline{\mathrm{Lie}}\big(\underline{\mathrm{Centr}}_G(E)/S\big) = \underline{\mathrm{Centr}}_{\underline{\mathrm{Lie}}(G/S)}(E),$$

and $\underline{\mathrm{Centr}}_G(E)$ *is a good* S*-group; it is very good if* G *is very good.*

(ii) *Suppose that* E *is a good* $\mathbf{O}_S$*-module. Then*[98]

$$\underline{\mathrm{Lie}}\big(\underline{\mathrm{Norm}}_G(E)/S\big) = \underline{\mathrm{Norm}}_{\underline{\mathrm{Lie}}(G/S)}(E),$$

and $\underline{\mathrm{Norm}}_G(E)$ *is a good* S*-group; it is very good if* G *is very good.*

[98]Editor's note: The original stated the following equality without a hypothesis on E; this has been corrected. See 6.5.1 for counterexamples.

The proof is left to the reader.

5.3.2. Let G be a good S-group. The preceding results apply in particular to the adjoint representation. Let E be a good $\mathbf{O}_S$-submodule of $\underline{\mathrm{Lie}}(G/S)$.[99] It has a centralizer and a normalizer in G, and by 5.3.1 their Lie algebras are the usual centralizer and normalizer of E, computed using the bracket:

$$\underline{\mathrm{Centr}}_{\underline{\mathrm{Lie}}(G/S)}(E)(S') = \{X \in \underline{\mathrm{Lie}}(G/S)(S') \mid [X, E_{S'}] = 0\},$$
$$\underline{\mathrm{Norm}}_{\underline{\mathrm{Lie}}(G/S)}(E)(S') = \{X \in \underline{\mathrm{Lie}}(G/S)(S') \mid [X, E_{S'}] \subset E_{S'}\}.$$

5.3.3. Let K be a sub-S-group of G. Then $\underline{\mathrm{Lie}}(K/S)$ is an $\mathbf{O}_S$-submodule of $\underline{\mathrm{Lie}}(G/S)$. Suppose that it is a good $\mathbf{O}_S$-module,[100] as happens when K is a good S-group. Plainly,

$$\underline{\mathrm{Centr}}_G(K) \subset \underline{\mathrm{Centr}}_G\big(\underline{\mathrm{Lie}}(K/S)\big),$$

$$\underline{\mathrm{Norm}}_G(K) \subset \underline{\mathrm{Norm}}_G\big(\underline{\mathrm{Lie}}(K/S)\big).$$

Hence, by 5.3.1,

$$\underline{\mathrm{Lie}}\big(\underline{\mathrm{Centr}}_G(K)/S\big) \subset \underline{\mathrm{Centr}}_{\underline{\mathrm{Lie}}(G/S)}\big(\underline{\mathrm{Lie}}(K/S)\big),$$
$$\underline{\mathrm{Lie}}\big(\underline{\mathrm{Norm}}_G(K)/S\big) \subset \underline{\mathrm{Norm}}_{\underline{\mathrm{Lie}}(G/S)}\big(\underline{\mathrm{Lie}}(K/S)\big).$$

None of these four inclusions is an equality in general; many examples will appear later. In particular, if K is a normal subgroup of G, then $\underline{\mathrm{Lie}}(K/S)$ is an ideal of $\underline{\mathrm{Lie}}(G/S)$.

6. Miscellaneous remarks

6.1.

The bracket of two infinitesimal automorphisms can be defined for an S-functor X which is not necessarily a group: apply the results of this exposé to $\underline{\mathrm{Aut}}_S(X)$. To obtain a convenient formalism, one is led to assume that X is good, meaning that the $\mathbf{O}_X$-module $T_{X/S}$ is good. If X is an S-group, this agrees with Definition 4.6.

6.2.

There exist functors with infinitesimal endomorphisms that are not automorphisms and hence, a fortiori, do not satisfy condition (E). [101] For a pointed set (E, x_0), let $M_A(E)$ be the free abelian monoid generated by E, and let $M_{A,P}(E, x_0)$ be the quotient of $M_A(E)$ by the equivalence relation generated by $m \sim x_0 + m$. The assignment

$$(E, x_0) \longmapsto M_{A,P}(E, x_0)$$

is left adjoint to the forgetful functor from abelian monoids to pointed sets; $M_{A,P}(E, x_0)$ is the free abelian monoid on the pointed set (E, x_0).

Now let X be the functor assigning to every scheme S the free abelian monoid on $\mathbf{O}(S)$, pointed by zero. Every morphism

$$f : S \longrightarrow I_{\mathbb{Z}} = \mathrm{Spec}(\mathbb{Z}[\varepsilon])$$

[99] Editor's note: This hypothesis has been added; cf. 6.5.1.
[100] Editor's note: This hypothesis has been added; cf. 6.5.1.
[101] Editor's note: The original has been expanded here. In particular, O. Gabber pointed out the correct definition of the free abelian monoid on a pointed set.

corresponds to a square-zero element $u_f \in \mathbf{O}(S)$, and hence defines an endomorphism of $X(S)$ by $x \mapsto x + u_f$, where the sum is taken in $M_{A,P}(\mathbf{O}(S), 0)$. Thus we obtain an endomorphism ϕ of

$$X_{I_{\mathbb{Z}}} = X \times_{\mathbb{Z}} I_{\mathbb{Z}}$$

defined, for $f \in I_{\mathbb{Z}}(S)$ and $x \in X(S)$, by

$$\phi((x, f)) = (x + u_f, f).$$

If $f_0 : S \to I_{\mathbb{Z}}$ is the composite of the structural morphism $S \to \mathrm{Spec}(\mathbb{Z})$ and the zero section, then $u_{f_0} = 0$, so $\phi((x, f_0)) = (x, f_0)$. Thus ϕ induces the identity on X: it is an infinitesimal endomorphism of X which is plainly not an automorphism.

6.3.

There exist modules which are not good. First, the preceding counterexample can be modified slightly.[102] Let $F(S)$ be the free $\mathbf{O}(S)$-module with basis the elements of $\mathbf{O}(S)$. Then F does not satisfy (E) relative to $\mathrm{Spec}(\mathbb{Z})$. Moreover, let G be the $\mathbb{Z}$-group defined by

$$G(S) = \mathrm{GL}_n(R(S)), \qquad n \geq 2,$$

where $R(S)$ is the $\mathbf{O}(S)$-algebra of the abelian group $\mathbf{O}(S)$. Then the $\mathbb{Z}$-group $\underline{\mathrm{Lie}}(G/\mathbb{Z})$ is not commutative.

There are also the following counterexamples.[103] Let $S = \mathrm{Spec}(k)$, where k is a field of characteristic $p > 0$.

(a) Let F be the $\mathbf{O}_S$-module assigning to every S-scheme T the group

$$F(T) = \Gamma(T, \mathscr{O}_T),$$

with its $\mathbf{O}(T)$-module structure defined by $r \cdot f = r^p f$. As an S-functor of groups, F is isomorphic to $\mathbb{G}_{a,S}$. Thus F/S satisfies (E), and

$$\underline{\mathrm{Lie}}(F/S) \simeq \underline{\mathrm{Lie}}(\mathbb{G}_{a,S}/S) \simeq \mathbf{O}_S.$$

For each $T \to S$, the canonical morphism $F \to \underline{\mathrm{Lie}}(F/S)$ is the identity map $F(T) \to \mathbf{O}(T)$. It respects the abelian-group structure but not the module structure, so F is not good (4.4.1).

(b) Let $\alpha_{p,k}$ be the k-functor of groups defined by

$$\alpha_{p,k}(T) = \{x \in \mathbf{O}(T) \mid x^p = 0\}.$$

It is represented by $\mathrm{Spec}(k[X]/(X^p))$, hence is a very good S-group. It also carries an $\mathbf{O}_S$-module structure, but is not a good $\mathbf{O}_S$-module, because the canonical morphism

$$\alpha_{p,k} \longrightarrow \underline{\mathrm{Lie}}(\alpha_{p,k}/k) = \mathbb{G}_{a,k}$$

is not bijective.

[102]Editor's note: The following observation has been added.
[103]Editor's note: Example (a) has been expanded and example (b) has been added.

6.4.

For a group-valued functor G over S, the definitions give

$$(G/S \text{ satisfies (E)}) \Longleftarrow (G \text{ is good}) \Longleftarrow (G \text{ is very good}).$$

It would be interesting to prove or disprove the reverse implications. [104]

6.5.

Let Nil be the $\mathbb{Z}$-functor of groups defined by

$$\operatorname{Nil}(S) = \{x \in \mathbf{O}(S) \mid \text{there exists } n \in \mathbb{N} \text{ with } x^n = 0\}.$$

[105] The functor Nil is very good but not representable. Let Nil^2, $\mathbf{O}_{\mathrm{red}}$, and F be the $\mathbb{Z}$-functors of groups assigning to S, respectively, the ideal $\operatorname{Nil}(S)^2$ and the groups

$$\mathbf{O}_{\mathrm{red}}(S) = \mathbf{O}(S)/\operatorname{Nil}(S), \qquad F(S) = \mathbf{O}(S)/\operatorname{Nil}(S)^2.$$

It is easy to see that

$$\underline{\mathrm{Lie}}(\mathbf{O}_{\mathrm{red}}/\mathbb{Z}) = 0,$$

so the $\mathbf{O}_{\mathbb{Z}}$-module $\mathbf{O}_{\mathrm{red}}$ is not good, although it is a good $\mathbb{Z}$-group. If $\mathscr{M}, \mathscr{N}$ are free $\mathbb{Z}$-modules of finite rank, then

$$\begin{aligned}\operatorname{Nil}^2(I_S(\mathscr{M} \oplus \mathscr{N})) \simeq\ & \operatorname{Nil}^2(S) \oplus \operatorname{Nil}(S) \otimes_{\mathbb{Z}} \mathscr{M} \\ & \oplus \operatorname{Nil}(S) \otimes_{\mathbb{Z}} \mathscr{N},\end{aligned}$$

and hence

$$\begin{aligned}F(I_S(\mathscr{M} \oplus \mathscr{N})) \simeq\ & F(S) \oplus \mathbf{O}_{\mathrm{red}}(S) \otimes_{\mathbb{Z}} \mathscr{M} \\ & \oplus \mathbf{O}_{\mathrm{red}}(S) \otimes_{\mathbb{Z}} \mathscr{N}.\end{aligned}$$

Thus F satisfies condition (E), while

$$\underline{\mathrm{Lie}}(F/\mathbb{Z}) = \mathbf{O}_{\mathrm{red}}.$$

Since $\mathbf{O}_{\mathrm{red}}$ is not a good $\mathbf{O}_{\mathbb{Z}}$-module, F is a $\mathbb{Z}$-group which satisfies (E) but is not good.

6.5.1. We now give the counterexamples announced in 5.3.1–5.3.3. Let S be a scheme, let $F = \mathbf{O}_S^{\oplus 2}$ with the natural action of the good S-group $G = \mathrm{GL}_{2,S}$, and let E be the $\mathbf{O}_S$-submodule of F formed by the pairs (x_1, x_2) for which x_2 is nilpotent. Put

$$N = \underline{\mathrm{Norm}}_G(E).$$

Then

$$\underline{\mathrm{Lie}}(N/S) = \underline{\mathrm{Lie}}(G/S),$$

whereas, for every $S' \to S$,

$$\underline{\mathrm{Norm}}_{\underline{\mathrm{Lie}}(G/S)}(E)(S') = \left\{ \begin{pmatrix} a & b \\ x & c \end{pmatrix} \,\middle|\, a, b, c, x \in \mathbf{O}(S'),\ x \text{ nilpotent} \right\}.$$

[104] Editor's note: A representable G is very good (4.11). For representability criteria see, for example, [MO67]. For automorphisms of an affine algebraic group over an algebraically closed field of characteristic zero, see [HM69].

[105] Editor's note: This paragraph has been added.

Consequently,

$$\underline{\mathrm{Lie}}\big(\underline{\mathrm{Norm}}_G(E)/S\big) \neq \underline{\mathrm{Norm}}_{\underline{\mathrm{Lie}}(G/S)}(E).$$

Taking the semidirect product $G_1 = F \rtimes G$ gives an analogous counterexample in which E is an $\mathbf{O}_S$-submodule of $\underline{\mathrm{Lie}}(G_1/S)$. Moreover, with the notation above,

$$E = \underline{\mathrm{Lie}}(K/S),$$

where K is the subgroup $\mathbf{O}_S \oplus \mathrm{Nil}^2$ of F; that is, $K(S')$ consists of the pairs (x_1, x_2) with $x_2 \in \mathrm{Nil}(S')^2$.

Bibliography

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[106] Editor's note: Additional references cited in this exposé.

Exposé III
Infinitesimal Extensions

by M. Demazure

In this exposé we work in the following general situation. We have a scheme S and a coherent nilpotent ideal $\mathscr{I}$ on S. We denote by S_n the closed subscheme of S defined by the ideal $\mathscr{I}^{n+1}$, for $n \geq 0$. In particular, S_0 is defined by $\mathscr{I}$. Since $\mathscr{I}$ is nilpotent, $S_n = S$ for all sufficiently large n, and the S_i have the same underlying topological space. A typical example is the following: S is the spectrum of a local Artinian ring A, and $\mathscr{I}$ is the ideal defined by the radical of A, so that S_0 is the spectrum of the residue field of A.[0]

In this situation one is given certain data over S_0 and seeks data over S that lift them, that is, data whose base change from S to S_0 recovers the original data. This is done step by step, through the schemes S_n. At each step we shall define the obstructions that arise and classify the resulting solutions when they exist.

The passage from S_n to S_{n+1} may be generalized as follows. We have a scheme S and two ideals $\mathscr{I} \supset \mathscr{J}$ such that $\mathscr{I}\mathscr{J} = 0$. In the preceding case S, $\mathscr{I}$, and $\mathscr{J}$ are respectively

$$S_{n+1}, \qquad \mathscr{I}/\mathscr{I}^{n+2}, \qquad \mathscr{I}^{n+1}/\mathscr{I}^{n+2}.$$

We denote by S_0, respectively $S_{\mathscr{J}}$, the closed subscheme of S defined by $\mathscr{I}$, respectively $\mathscr{J}$, and pose a problem of extending data from $S_{\mathscr{J}}$ to S.

SGA 1, Exposé III, treated the problems of extending morphisms of schemes and of extending schemes. Here we consider the problems of extending group morphisms, group structures, and subgroups.

In §0 we have collected those results of SGA 1, Exposé III, that will be useful to us, putting them into the form best suited to our purpose and sparing the reader constant reference to that exposé.[1] Section 1 collects computations in group cohomology that will be used later and have nothing to do with scheme theory. Sections 2 and 3 treat, respectively, the extension of group morphisms and the extension of group structures. In §4 we briefly recall the proof of a result stated in TDTE IV concerning the extension of subschemes, and apply it to the problem of extending subgroups. For the remainder of the Seminar, only the result of §2 concerning the extension of group morphisms is indispensable.[2]

The idea of reducing problems of infinitesimal extension to the usual cohomological computations in group extensions was suggested by J. Giraud during the oral exposé, whose computations were appreciably more complicated and less transparent. Unfortunately, this

[0] Editor's note: version of October 14, 2024.

[1] Editor's note: the reader may prefer to begin with §1, which is easier and may serve as motivation and a guide to the results of §0.

[2] Editor's note: nevertheless, 3.10 is used in XXIV 1.13. Moreover, 4.37 is used in the proof of IX 3.2 bis and 3.6 bis, although those results also follow from results of Exposé X that do not use 4.37.

method seems to work well only for the first two problems studied, and we have been unable to avoid rather laborious computations for extensions of subgroups.

To simplify the language, we shall call a Y-functor, respectively a Y-scheme, and so on, a functor, respectively a scheme, and so on, equipped with a morphism to the functor Y. This extends the definitions of Exposé I, which concerned only representable Y.

0. Reminders from SGA 1, Exposé III, and miscellaneous remarks

We begin with a general definition.

Definition 0.1. *Let $\mathscr{C}$ be a category, X an object of $\widehat{\mathscr{C}}$, and G a $\widehat{\mathscr{C}}$-group acting on X. We say that X is* formally principal homogeneous[3] *under G if the following equivalent conditions hold:*

(i) *for every object T of $\mathscr{C}$, the set $X(T)$ is either empty or principal homogeneous under $G(T)$;*

(ii) *the morphism of functors*

$$G \times X \longrightarrow X \times X, \qquad (g, x) \longmapsto (gx, x),$$

is an isomorphism.

We shall now put the results of SGA 1, III §5[4] into the form most useful here. Throughout this section we use the following notation. On a scheme S we have two quasi-coherent ideals $\mathscr{I}$ and $\mathscr{J}$ such that

$$\mathscr{I} \supset \mathscr{J}, \qquad \mathscr{I}\mathscr{J} = 0.$$

In particular, $\mathscr{J}^2 = 0$. We denote by S_0, respectively $S_{\mathscr{J}}$, the closed subscheme of S defined by $\mathscr{I}$, respectively $\mathscr{J}$. For every S-functor X, we systematically denote by X_0 and $X_{\mathscr{J}}$ the functors obtained by base change from S to S_0 and $S_{\mathscr{J}}$. We use the same notation for a morphism.

Definition 0.1.1.[5] *Let X be an S-functor. Define a functor X^+ over S by*

$$\mathrm{Hom}_S(Y, X^+) = \mathrm{Hom}_{S_{\mathscr{J}}}(Y_{\mathscr{J}}, X_{\mathscr{J}}) = \mathrm{Hom}_S(Y_{\mathscr{J}}, X)$$

for a variable S-scheme Y. In the notation of Exposé II, §1,

$$X^+ \simeq \prod_{S_{\mathscr{J}}/S} X_{\mathscr{J}}.$$

The identity morphism of $X_{\mathscr{J}}$ defines, by construction, an S-morphism

$$p_X : X \longrightarrow X^+.$$

Explicitly,[6] for every S-scheme Y, the map

$$p_X(Y) : \mathrm{Hom}_S(Y, X) \longrightarrow \mathrm{Hom}_S(Y, X^+) = \mathrm{Hom}_S(Y_{\mathscr{J}}, X)$$

[3]Editor's note: one also calls such an object a *pseudo-torsor*; see EGA IV_4, 16.5.15. The more general notion of a formally homogeneous object, not necessarily principal homogeneous, is defined in Exposé IV, 6.7.1.

[4]Editor's note: see also EGA IV_4, 16.5.14–18.

[5]Editor's note: the numbering 0.1.1 through 0.1.13 has been added to make the definitions and results contained here more visible.

[6]Editor's note: the following sentence has been added, and the next two remarks have been expanded.

is induced by the morphism $Y_{\mathscr{J}} \to Y$.

Remark 0.1.2. If X is an S-group functor, then $X_{\mathscr{J}}$ is an $S_{\mathscr{J}}$-group functor. The defining formula for X^+ therefore equips X^+ with an S-group-functor structure, and the description above shows that

$$p_X : X \longrightarrow X^+$$

is a morphism of S-group functors.

Remark 0.1.3. For every S-group functor Y, one has

$$\mathrm{Hom}_{S\text{-gr}}(Y, X^+) = \mathrm{Hom}_{S_{\mathscr{J}}\text{-gr}}(Y_{\mathscr{J}}, X_{\mathscr{J}}).$$

Indeed, let $f \in \mathrm{Hom}_S(Y, X^+)$ correspond to $f_{\mathscr{J}} \in \mathrm{Hom}_{S_{\mathscr{J}}}(Y_{\mathscr{J}}, X_{\mathscr{J}})$. The condition that f be an S-group morphism is that, for every $T \to S$ and every $y, y' \in Y(T)$,

$$f(y \cdot y') = f(y) \cdot f(y'),$$

which is equivalent to

$$f_{\mathscr{J}}(y_{\mathscr{J}}) \cdot f_{\mathscr{J}}(y'_{\mathscr{J}}) = f_{\mathscr{J}}\big((y \cdot y')_{\mathscr{J}}\big).$$

Since $(y \cdot y')_{\mathscr{J}} = y_{\mathscr{J}} \cdot y'_{\mathscr{J}}$, this is precisely the condition that $f_{\mathscr{J}}$ be a group morphism. Applying this to $Y = X$, we again see that p_X, which corresponds to $\mathrm{id}_{X_{\mathscr{J}}}$, is a morphism of S-group functors.

Return now to the general case and suppose that X is an S-scheme. An S-morphism from a variable S-scheme Y to X^+ is, by definition, an $S_{\mathscr{J}}$-morphism $g_{\mathscr{J}} : Y_{\mathscr{J}} \to X_{\mathscr{J}}$. We define an X^+-functor in abelian groups L_X as follows.

Scholium 0.1.4.[7] If $\pi : Y \to S$ is a morphism of schemes and M is an $\mathscr{O}_S$-module, we denote the inverse image $\pi^* M$ by

$$M \otimes_{\mathscr{O}_S} \mathscr{O}_Y.$$

If $\mathscr{J}$ is an ideal of $\mathscr{O}_S$, we denote by $\mathscr{J}\mathscr{O}_Y$ the ideal of $\mathscr{O}_Y$ that is the image of

$$\pi^*(\mathscr{J}) = \mathscr{J} \otimes_{\mathscr{O}_S} \mathscr{O}_Y \longrightarrow \mathscr{O}_Y.$$

For every morphism of S-schemes $f : Z \to Y$, there is an epimorphism of $\mathscr{O}_Z$-modules

$$f^*(\mathscr{J}\mathscr{O}_Y) \longrightarrow \mathscr{J}\mathscr{O}_Z. \tag{0.1.4}$$

It follows from the commutative diagram below.

$$\begin{tikzcd}
\mathcal{J} \otimes_{\mathscr{O}_S} \mathscr{O}_Y \otimes_{\mathscr{O}_Y} \mathscr{O}_Z \arrow[r, equal] \arrow[d] & \mathcal{J} \otimes_{\mathscr{O}_S} \mathscr{O}_Z \arrow[d] \\
f^*(\mathcal{J}\mathscr{O}_Y) = (\mathcal{J}\mathscr{O}_Y) \otimes_{\mathscr{O}_Y} \mathscr{O}_Z \arrow[r] & \mathcal{J}\mathscr{O}_Z.
\end{tikzcd}$$

Figure 57: The tensor-product diagram yielding the epimorphism (0.1.4).

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 4; combined-reader p. 114.

[7] Editor's note: this scholium has been added.

Definition 0.1.5.[8] Let X be an S-scheme. For every X^+-scheme Y, given by a morphism

$$g_{\mathscr{J}} : Y_{\mathscr{J}} \longrightarrow X_{\mathscr{J}},$$

set

$$\operatorname{Hom}_{X^+}(Y, L_X) = \operatorname{Hom}_{\mathscr{O}_{Y_0}}\big(g_0^*(\Omega^1_{X_0/S_0}), \mathscr{J}\mathscr{O}_Y\big),$$

where $\Omega^1_{X_0/S_0}$ is the module of relative differentials of X_0 over S_0, and $\mathscr{J}\mathscr{O}_Y$ is regarded as an $\mathscr{O}_{Y_0}$-module because it is annihilated by $\mathscr{I}$.

Then L_X is an X^+-functor in abelian groups.[9] Indeed, for every X^+-morphism $f : Z \to Y$, the functor f_0^* and the morphism

$$f_0^*(\mathscr{J}\mathscr{O}_Y) \longrightarrow \mathscr{J}\mathscr{O}_Z$$

of (0.1.4) induce a natural homomorphism $L_X(f)$:

$$\begin{aligned} &\operatorname{Hom}_{\mathscr{O}_{Y_0}}\big(g_0^*(\Omega^1_{X_0/S_0}), \mathscr{J}\mathscr{O}_Y\big) \\ &\qquad \longrightarrow \operatorname{Hom}_{\mathscr{O}_{Z_0}}\big(f_0^*g_0^*(\Omega^1_{X_0/S_0}), f_0^*(\mathscr{J}\mathscr{O}_Y)\big) \\ &\qquad \longrightarrow \operatorname{Hom}_{\mathscr{O}_{Z_0}}\big(f_0^*g_0^*(\Omega^1_{X_0/S_0}), \mathscr{J}\mathscr{O}_Z\big). \end{aligned}$$

Finally, $L_X(f)$ has the following simple local description. The schemes Y and $Y_{\mathscr{J}}$ have the same underlying topological space, as do V and $V_{\mathscr{J}}$ for every open subset V of Y. Let

$$U = \operatorname{Spec}(A), \qquad V = \operatorname{Spec}(B), \qquad W = \operatorname{Spec}(C)$$

be affine opens such that U lies over $\operatorname{Spec}(\Lambda) \subset S$, $g_{\mathscr{J}}(V_{\mathscr{J}}) \subset U$, and $W \subset f^{-1}(V)$. Write I, J for the ideals of Λ corresponding to $\mathscr{I}$, $\mathscr{J}$. Then f, respectively $g_{\mathscr{J}}$, induces a morphism of Λ-algebras

$$\theta : B \longrightarrow C, \qquad \text{respectively} \qquad \varphi : A \longrightarrow B/JB,$$

and clearly $\theta(JB) \subset JC$. On the other hand,

$$m \in \operatorname{Hom}_{\mathscr{O}_{Y_0}}\big(g_0^*(\Omega^1_{X/S}), \mathscr{J}\mathscr{O}_Y\big)$$

induces an element D of

$$\begin{aligned} \operatorname{Hom}_{\mathscr{O}_{V_0}}\big(g_0^*(\Omega^1_{U/S}), \mathscr{J}\mathscr{O}_V\big) &= \operatorname{Hom}_{B/IB}\big(\Omega^1_{A/\Lambda} \otimes_A B/IB, JB\big) \\ &= \operatorname{Der}_\Lambda(A, JB), \end{aligned}$$

and the image of $L_X(f)(m)$ in

$$\begin{aligned} \operatorname{Hom}_{\mathscr{O}_{W_0}}\big(f_0^*g_0^*(\Omega^1_{U/S}), \mathscr{J}\mathscr{O}_Z\big) &= \operatorname{Hom}_{C/IC}\big(\Omega^1_{A/\Lambda} \otimes_A C/IC, JC\big) \\ &= \operatorname{Der}_\Lambda(A, JC) \end{aligned}$$

is simply $\theta \circ D$.

[8] Editor's note: introducing the functor L_X gives an extension, functorial in Y and in X, of SGA 1, III, Proposition 5.1; see 0.1.8 and 0.1.9 below, as well as 0.1.10 and 0.2. Moreover, when X is an S-group functor, L_X gives rise, under certain hypotheses, to an S-group functor L'_X and to an exact sequence $1 \to L'_X \to X \to X^+$; see 0.4, 0.9, and 0.11. These play an essential role in this exposé; see Theorems 2.1, 3.5, and 4.21.

[9] Editor's note: what follows has been added.

Remark 0.1.6.[10] Let $f : X \to W$ be an S-morphism. It induces an S-morphism

$$f^{+} : X^{+} \longrightarrow W^{+}$$

defined as follows. If $g \in \mathrm{Hom}_S(Y, X^{+})$ corresponds to an S-morphism $g_{\mathscr{J}} : Y_{\mathscr{J}} \to X$, then $f^{+}(g)$ is the element $f \circ g_{\mathscr{J}}$ of $\mathrm{Hom}_S(Y, W^{+})$. The following diagram is commutative.

$$\begin{tikzcd}
X \arrow[r, "f"] \arrow[d, "p_X"'] & W \arrow[d, "p_W"] \\
X^{+} \arrow[r, "f^{+}"] & W^{+}
\end{tikzcd} .$$

Figure 58: Naturality of the morphisms p_X and p_W.

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 5; combined-reader p. 115.

Reminders 0.1.7.[11] In this paragraph, given an S-scheme X, we "recall" certain functorial properties of the module of differentials $\Omega^1_{X/S}$ and of the first infinitesimal neighborhood of the diagonal, $\Delta^{(1)}_{X/S}$. These properties follow readily from EGA IV$_4$, §§16.1–16.4, but do not appear there explicitly.

(a) Let us begin by recalling the following facts (cf. EGA II, §§1.2–1.5). Let $g : Y \to X$ be a morphism of schemes, let $\pi : X' \to X$ be an affine X-scheme, and let $\mathscr{B} = \pi_*(\mathscr{O}_{X'})$ be the corresponding quasi-coherent $\mathscr{O}_X$-algebra. Then the Y-scheme $Y \times_X X'$ is affine and corresponds to the quasi-coherent $\mathscr{O}_Y$-algebra $g^*(\mathscr{B})$, and there is a commutative diagram of bijections:

$$\begin{tikzcd}
\mathrm{Hom}_X(Y, X') \arrow[r, "\sim"] \arrow[d, "\wr"'] & \mathrm{Hom}_Y(Y, Y \times_X X') \arrow[d, "\wr"] \\
\mathrm{Hom}_{\mathscr{O}_X\text{-alg.}}(\mathscr{B}, g_*(\mathscr{O}_Y)) \arrow[r, "\sim"] & \mathrm{Hom}_{\mathscr{O}_Y\text{-alg.}}(g^*(\mathscr{B}), \mathscr{O}_Y).
\end{tikzcd}$$

Figure 59: The affine-scheme/algebra bijections after base change by g.

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 5; combined-reader p. 115.

Moreover, these bijections are functorial in the pair (X, X'). More precisely, let W' be an affine W-scheme corresponding to the quasi-coherent $\mathscr{O}_W$-algebra $\mathscr{A}$, and suppose that one has a commutative diagram of morphisms of schemes as follows.

[10] Editor's note: this remark has been expanded.

[11] Editor's note: This paragraph has been added (see also [DG70], §I.4, nos. 1–2).

$$
\begin{tikzcd}
 & \mathrm{X}' \arrow[r, "f'"] \arrow[d] & \mathrm{W}' \arrow[d] \\
\mathrm{Y} \arrow[ur, dashed, "g'"] \arrow[r, "g"] & \mathrm{X} \arrow[r, "f"] & \mathrm{W}
\end{tikzcd}
$$

Figure 60: The scheme morphisms used to express functoriality in (X, X').

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 5; combined-reader p. 115.

Denote by

$$\varphi : \mathscr{A} \longrightarrow f_*(\mathscr{B}), \qquad \varphi^\sharp : f^*(\mathscr{A}) \longrightarrow \mathscr{B}$$

the algebra morphisms associated with f', and by

$$\theta : \mathscr{B} \longrightarrow g_*(\mathscr{O}_\mathrm{Y}), \qquad \theta^\sharp : g^*(\mathscr{B}) \longrightarrow \mathscr{O}_\mathrm{Y}$$

those associated with a variable X-morphism $g' : Y \to X'$. Then the following diagram is commutative, where Y is viewed as a W-scheme through $f \circ g$.

$$
\begin{tikzcd}
\mathrm{Hom}_\mathrm{X}(\mathrm{Y}, \mathrm{X}') \arrow[r, "g' \mapsto f' \circ g'"] \arrow[dd, "\wr"'] & \mathrm{Hom}_\mathrm{W}(\mathrm{Y}, \mathrm{W}') \arrow[dr, "\simeq"] & \\
 & & \mathrm{Hom}_{\mathscr{O}_\mathrm{W}\text{-alg.}}(\mathscr{A}, f_* g_*(\mathscr{O}_\mathrm{Y})) \arrow[d, "\wr"] \\
\mathrm{Hom}_{\mathscr{O}_\mathrm{X}\text{-alg.}}(\mathscr{B}, g_*(\mathscr{O}_\mathrm{Y})) \arrow[urr, "\theta \mapsto f_*(\theta) \circ \phi"] \arrow[rr, "\theta \mapsto \theta \circ \phi^\sharp"] \arrow[d, "\wr"'] & & \mathrm{Hom}_{\mathscr{O}_\mathrm{X}\text{-alg.}}(f^*(\mathscr{A}), g_*(\mathscr{O}_\mathrm{Y})) \arrow[d, "\wr"] \\
\mathrm{Hom}_{\mathscr{O}_\mathrm{Y}\text{-alg.}}(g^*(\mathscr{B}), \mathscr{O}_\mathrm{Y}) \arrow[rr, "\theta^\sharp \mapsto \theta^\sharp \circ g^*(\phi^\sharp)"] & & \mathrm{Hom}_{\mathscr{O}_\mathrm{Y}\text{-alg.}}(g^* f^*(\mathscr{A}), \mathscr{O}_\mathrm{Y}).
\end{tikzcd}
$$

Figure 61: Functoriality of the affine-scheme/algebra correspondences.

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 6; combined-reader p. 116.

(b) Let now X be an S-scheme. Let $\Omega^1_{X/S}$ be the module of differentials of X over S, and let $\Delta^{(1)}_{X/S}$ be the first infinitesimal neighborhood of the diagonal immersion $\delta_X : X \to X \times_S X$. It is a subscheme of $X \times_S X$, in which the diagonal $\Delta_{X/S}$ is a closed subscheme. Write pr^i_X, $i = 1, 2$, for the two projections $X \times_S X \to X$, and write π_X for the restriction of pr^1_X to $\Delta^{(1)}_{X/S}$.

On the one hand, every morphism $f : X \to W$ of S-schemes induces an S-morphism

$$\Delta^{(1)} f : \Delta^{(1)}_{X/S} \longrightarrow \Delta^{(1)}_{W/S}$$

for which the following diagram is commutative.

$$\begin{tikzcd}
X \arrow[r, "\delta_X"] \arrow[d, "f"'] & \Delta^{(1)}_{X/S} \arrow[r] \arrow[d, "\Delta^{(1)} f"] & X \times_S X \arrow[r, "\mathrm{pr}^i_X"] \arrow[d, "f \times f"] & X \arrow[d, "f"] \\
W \arrow[r, "\delta_W"] & \Delta^{(1)}_{W/S} \arrow[r] & W \times_S W \arrow[r, "\mathrm{pr}^i_W"] & W.
\end{tikzcd}$$

Figure 62: Functoriality of the first infinitesimal neighborhood of the diagonal.

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 6; combined-reader p. 116.

On the other hand, through the projection π_X, the scheme $\Delta^{(1)}_{X/S}$ is an affine X-scheme: it is the spectrum of the augmented quasi-coherent $\mathscr{O}_X$-algebra

$$\mathscr{P}^1_{X/S} = \mathscr{O}_X \oplus \Omega^1_{X/S},$$

where $\Omega^1_{X/S}$ is a square-zero ideal. The augmentation is the $\mathscr{O}_X$-algebra morphism

$$\varepsilon_X : \mathscr{P}^1_{X/S} \longrightarrow \mathscr{O}_X$$

that vanishes on $\Omega^1_{X/S}$ and corresponds to the closed immersion $\delta_X : X \hookrightarrow \Delta^{(1)}_{X/S}$. Hence every morphism of S-schemes $f : X \to W$ induces a morphism of augmented $\mathscr{O}_X$-algebras

$$f^*(\mathscr{P}^1_{W/S}) = \mathscr{O}_X \oplus f^*(\Omega^1_{W/S}) \longrightarrow \mathscr{P}^1_{X/S} = \mathscr{O}_X \oplus \Omega^1_{X/S},$$

or, equivalently, a morphism of $\mathscr{O}_X$-modules

$$f_{X/W/S} : f^*(\Omega^1_{W/S}) \longrightarrow \Omega^1_{X/S};$$

see EGA IV_4, (16.4.3.6), and (16.4.18.2) for the notation $f_{X/W/S}$.

Since $\pi_X : \Delta^{(1)}_{X/S} \to X$ is affine, part (a) shows that $\Delta^{(1)} f$ is completely determined by $f_{X/W/S}$. For every X-scheme $g : Y \to X$, the set

$$\mathrm{Hom}_X(Y, \Delta^{(1)}_{X/S}) \simeq \mathrm{Hom}_{\mathscr{O}_Y\text{-alg}}\big(\mathscr{O}_Y \oplus g^*(\Omega^1_{X/S}), \mathscr{O}_Y\big)$$

identifies with a subset of $\mathrm{Hom}_{\mathscr{O}_Y}(g^*(\Omega^1_{X/S}), \mathscr{O}_Y)$, namely

$$\widetilde{\mathrm{Hom}}_{\mathscr{O}_Y}\big(g^*(\Omega^1_{X/S}), \mathscr{O}_Y\big),$$

the subset formed by the $\mathscr{O}_Y$-morphisms $\psi : g^*(\Omega^1_{X/S}) \to \mathscr{O}_Y$ for which $\mathrm{Im}(\psi)$ is a square-zero ideal of $\mathscr{O}_Y$.[12]

Consequently, apply part (a) to the following diagram.

[12] Editor's note: In the proof of SGA 1, III 5.1, the sentence "But the algebra homomorphisms $\mathscr{A} \to \mathscr{O}_Y$ correspond ..." must be corrected accordingly; the remainder of the cited passage is unchanged.

$$\begin{tikzcd}
 & \Delta^{(1)}_{\mathrm{X/S}} \arrow[r, "\Delta^{(1)} f"] \arrow[d] & \Delta^{(1)}_{\mathrm{W/S}} \arrow[d] \\
\mathrm{Y} \arrow[ur, dashed, "g'"] \arrow[r, "g"] & \mathrm{X} \arrow[r, "f"] & \mathrm{W}
\end{tikzcd}$$

Figure 63: The diagram to which the affine functoriality statement is applied.

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 7; combined-reader p. 117.

Taking into account that $\Delta^{(1)}f$ is the restriction of $f \times f$ to $\Delta^{(1)}_{X/S}$, one obtains the following commutative diagram, functorial in the X-scheme $g : Y \to X$.

$$(0.1.7\,(*))\qquad \begin{tikzcd}
\mathrm{Hom}_{\mathrm{X}}(\mathrm{Y}, \mathrm{X} \times_{\mathrm{S}} \mathrm{X}) \arrow[r, "(g,g') \mapsto (f\circ g, f \circ g')"] & \mathrm{Hom}_{\mathrm{W}}(\mathrm{Y}, \Delta^{(1)}_{\mathrm{W/S}}) \\
\mathrm{Hom}_{\mathrm{X}}(\mathrm{Y}, \Delta^{(1)}_{\mathrm{X/S}}) \arrow[u, hook] \arrow[r, "g' \mapsto (\Delta^{(1)} f)\circ g'"] \arrow[d, "\simeq"'] & \mathrm{Hom}_{\mathrm{W}}(\mathrm{Y}, \Delta^{(1)}_{\mathrm{W/S}}) \arrow[u, hook] \arrow[d, "\simeq"'] \\
\mathrm{Hom}^{\square}_{\mathscr{O}_{\mathrm{Y}}}(g^*(\Omega^1_{\mathrm{X/S}}), \mathscr{O}_{\mathrm{Y}}) \arrow[r, "\psi \mapsto \psi \circ g^*(f_{\mathrm{X/W/S}})"] & \mathrm{Hom}^{\square}_{\mathscr{O}_{\mathrm{Y}}}(g^* f^*(\Omega^1_{\mathrm{W/S}}), \mathscr{O}_{\mathrm{Y}}).
\end{tikzcd}$$

Figure 64: The functorial comparison of first-neighborhood morphisms and differential maps, equation 0.1.7 (∗).

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 7; combined-reader p. 117.

Remark 0.1.7.1. Let us conclude this paragraph with the following remark, which will be useful later (cf. 0.1.10). Denote by $\widetilde{L}_X$ the X-functor that associates to every X-scheme $g : Y \to X$ the set

$$\mathrm{Hom}_{\mathscr{O}_Y}\big(g^*(\Omega^1_{X/S}), \mathscr{O}_Y\big),$$

and denote by $\widetilde{L}_f : \widetilde{L}_X \to \widetilde{L}_W$ the morphism of functors defined above: it sends $\psi \in \widetilde{L}_X(Y)$ to $\psi \circ g^*(f_{X/W/S})$. The preceding discussion gives a commutative diagram of functors.

$$\begin{tikzcd}
\mathrm{X} \times_{\mathrm{X}} \mathrm{L}^{\square}_{\mathrm{X}} \arrow[d, "f \times \mathrm{L}^{\square}_{f}"'] & \Delta^{(1)}_{\mathrm{X/S}} \arrow[l, "\sim"'] \arrow[r, hook] \arrow[d, "\Delta^{(1)} f"] & \mathrm{X} \times_{\mathrm{S}} \mathrm{X} \arrow[d, "f \times f"] \\
\mathrm{W} \times_{\mathrm{W}} \mathrm{L}^{\square}_{\mathrm{W}} & \Delta^{(1)}_{\mathrm{W/S}} \arrow[l, "\sim"'] \arrow[r, hook] & \mathrm{W} \times_{\mathrm{S}} \mathrm{W}.
\end{tikzcd}$$

Figure 65: Functoriality of $\widetilde{L}_X$ and the first infinitesimal neighborhood.

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 7; combined-reader p. 117.

Theorem 0.1.8. (SGA 1, III 5.1)[13] *Let Y and X be two S-schemes, let $\mathscr{J}$ be a quasi-coherent square-zero ideal of $\mathscr{O}_Y$, let $Y_{\mathscr{J}}$ be the closed subscheme of Y defined by $\mathscr{J}$, and let $g_{\mathscr{J}} : Y_{\mathscr{J}} \to X$ be an S-morphism.*

(a) *The set $P(g_{\mathscr{J}})$ of S-morphisms $g : Y \to X$ extending $g_{\mathscr{J}}$ is either empty or principal homogeneous under the abelian group*

$$\mathrm{Hom}_{\mathscr{O}_{Y_{\mathscr{J}}}}\big(g^*_{\mathscr{J}}(\Omega^1_{X/S}), \mathscr{J}\big).$$

(b) *Suppose that $i : Y_0 \hookrightarrow Y_{\mathscr{J}}$ is the closed immersion defined by a quasi-coherent ideal $\mathscr{I} \supset \mathscr{J}$ such that $\mathscr{I}\mathscr{J} = 0$, and set $g_0 = g_{\mathscr{J}} \circ i$. Then the preceding abelian group is isomorphic to*

$$\mathrm{Hom}_{\mathscr{O}_{Y_0}}\big(g_0^*(\Omega^1_{X/S}), \mathscr{J}\big).$$

Proof. Part (b) follows at once from part (a). Indeed, since $\mathscr{J}$ is annihilated by $\mathscr{I}$, it may be regarded as an $\mathscr{O}_{Y_0}$-module, and adjunction gives

$$\mathrm{Hom}_{\mathscr{O}_{Y_{\mathscr{J}}}}\big(g^*_{\mathscr{J}}(\Omega^1_{X/S}), \mathscr{J}\big) = \mathrm{Hom}_{\mathscr{O}_{Y_0}}\big(i^* g^*_{\mathscr{J}}(\Omega^1_{X/S}), \mathscr{J}\big).$$

To prove part (a), we may suppose that $P(g_{\mathscr{J}}) \neq \emptyset$, that is, that an S-morphism $g : Y \to X$ extending $g_{\mathscr{J}}$ exists. Write $j : Y_{\mathscr{J}} \hookrightarrow Y$ for the immersion. Then $P(g_{\mathscr{J}})$ is the set of S-morphisms $g' : Y \to X$ such that $g' \circ j = g_{\mathscr{J}}$. Giving such a g' is equivalent to giving an S-morphism

$$h : Y \longrightarrow X \times_S X$$

such that $\mathrm{pr}_1 \circ h = g$ and $h_{\mathscr{J}} = \delta \circ g_{\mathscr{J}}$, where $h_{\mathscr{J}} = h \circ j$ and $\delta : X \hookrightarrow X \times_S X$ is the diagonal immersion.

[13] Editor's note: The statement and proof of SGA 1, III 5.1 are reproduced here, since they are needed below to prove Proposition 0.2; see already the following Corollary 0.1.9. Moreover, $\mathscr{J}$ (respectively $\mathscr{I}$) is another calligraphic form of the letter J (respectively I). Returning to the earlier notation, with $\mathscr{J} \subset \mathscr{I}$ ideals of $\mathscr{O}_S$ such that $\mathscr{I}\mathscr{J} = 0$, this result will be applied with $\mathscr{J} = \mathscr{J}\mathscr{O}_Y$ (respectively $\mathscr{I} = \mathscr{I}\mathscr{O}_Y$).

$$\begin{tikzcd}
\mathrm{X} \times_{\mathrm{S}} \mathrm{X} \arrow[d, "\mathrm{pr}_1"'] & \mathrm{Y}_{\mathfrak{J}} \arrow[l, "h_{\mathfrak{J}} = \delta \circ g_{\mathfrak{J}}"'] \arrow[d, "j"] \\
\mathrm{X} & \mathrm{Y}. \arrow[l, "g"] \arrow[ul, dashed, "h"']
\end{tikzcd}$$

Figure 66: The morphism h determined by two extensions of $g_{\mathscr{J}}$.

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 8; combined-reader p. 118.

Since $h_{\mathscr{J}}$ factors through δ, and since Y lies in the first infinitesimal neighborhood of the immersion $j : Y_{\mathscr{J}} \to Y$ because $\mathscr{J}^2 = 0$, functoriality (EGA IV$_4$, 16.2.2(i)) shows that the desired morphisms h factor uniquely through $\Delta^{(1)}_{X/S}$; see 0.1.7. Put

$$Y' = \Delta^{(1)}_{X/S} \times_X Y, \qquad Y'_{\mathscr{J}} = Y' \times_Y Y_{\mathscr{J}} = \Delta^{(1)}_{X/S} \times_X Y_{\mathscr{J}}.$$

The desired morphisms h are then in bijection with the sections u of $Y' \to Y$ extending the section $u_{\mathscr{J}} = (\delta \circ g_{\mathscr{J}}, \mathrm{id})$ of $Y'_{\mathscr{J}} \to Y_{\mathscr{J}}$. On the other hand, Y' (respectively $Y'_{\mathscr{J}}$) is affine over Y (respectively $Y_{\mathscr{J}}$) and corresponds to the quasi-coherent algebra

$$\mathscr{A} = \mathscr{O}_Y \oplus g^*(\Omega^1_{X/S}), \qquad \text{respectively} \qquad \mathscr{A}_{\mathscr{J}} = \mathscr{A} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_{\mathscr{J}}} = \mathscr{O}_{Y_{\mathscr{J}}} \oplus g^*_{\mathscr{J}}(\Omega^1_{X/S}).$$

Let $\varepsilon : \mathscr{A} \to \mathscr{O}_Y$ be the canonical augmentation, that is, the $\mathscr{O}_Y$-algebra morphism that vanishes on $g^*(\Omega^1_{X/S})$, and define $\varepsilon_{\mathscr{J}} : \mathscr{A}_{\mathscr{J}} \to \mathscr{O}_{Y_{\mathscr{J}}}$ similarly. Then

$$\Gamma(Y'/Y) \simeq \mathrm{Hom}_{\mathscr{O}_Y\text{-alg}}(\mathscr{A}, \mathscr{O}_Y), \qquad \Gamma(Y'_{\mathscr{J}}/Y_{\mathscr{J}}) \simeq \mathrm{Hom}_{\mathscr{O}_{Y_{\mathscr{J}}}\text{-alg}}(\mathscr{A}_{\mathscr{J}}, \mathscr{O}_{Y_{\mathscr{J}}}).$$

Under these isomorphisms, the section $u = (\delta \circ g, \mathrm{id})$ (respectively $u_{\mathscr{J}}$) corresponds to ε (respectively $\varepsilon_{\mathscr{J}}$). Consequently, $P(g_{\mathscr{J}})$ is in bijection with the algebra morphisms $\mathscr{A} \to \mathscr{O}_Y$ that reduce to $\varepsilon_{\mathscr{J}}$; under this bijection, g corresponds to ε.

Put $M = g^*(\Omega^1_{X/S})$. Then $\mathrm{Hom}_{\mathscr{O}_Y\text{-alg}}(\mathscr{A}, \mathscr{O}_Y)$ identifies with the set of $\mathscr{O}_Y$-morphisms $\psi : M \to \mathscr{O}_Y$ whose image is a square-zero ideal. We are interested in those morphisms that induce the zero morphism $M \to \mathscr{O}_{Y_{\mathscr{J}}} = \mathscr{O}_Y/\mathscr{J}$, that is, those that carry M into $\mathscr{J}$. Conversely, since $\mathscr{J}^2 = 0$, every $\mathscr{O}_Y$-morphism $\phi : M \to \mathscr{J}$ comes from a unique algebra morphism $\mathscr{A} \to \mathscr{O}_Y$ reducing to $\varepsilon_{\mathscr{J}}$. Finally,

$$\mathrm{Hom}_{\mathscr{O}_Y}\big(g^*(\Omega^1_{X/S}), \mathscr{J}\big) = \mathrm{Hom}_{\mathscr{O}_{Y_{\mathscr{J}}}}\big(g^*_{\mathscr{J}}(\Omega^1_{X/S}), \mathscr{J}\big)$$

because $\mathscr{J}^2 = 0$, as in the proof of part (b). Thus one obtains a bijection

$$P(g_{\mathscr{J}}) \simeq \mathrm{Hom}_{\mathscr{O}_{Y_{\mathscr{J}}}}\big(g^*_{\mathscr{J}}(\Omega^1_{X/S}), \mathscr{J}\big)$$

under which g corresponds to the zero morphism.

For every

$$m \in \mathrm{Hom}_{\mathscr{O}_{Y_{\mathscr{J}}}}\big(g^*_{\mathscr{J}}(\Omega^1_{X/S}), \mathscr{J}\big),$$

write $m \cdot g$ for the element of $P(g_{\mathscr{J}})$ associated with g and m by the preceding bijection. We have already seen that $0 \cdot g = g$; it remains to prove

$$m' \cdot (m \cdot g) = (m + m') \cdot g. \tag{0.1.8 (*)}$$

This may be checked locally.[14] Indeed, the two preceding morphisms $Y \to X$ induce the same continuous map as g on the underlying topological spaces. It is therefore enough to check that, for every affine open $U = \operatorname{Spec}(A)$ of X lying over an affine open $\operatorname{Spec}(\Lambda)$ of S, and every affine open $V = \operatorname{Spec}(B)$ of $g^{-1}(U)$, they induce the same morphism of Λ-algebras $A \to B$.

Set $J = \Gamma(V, \mathscr{J})$, and let $\varphi, \psi, \eta : A \to B$ be the morphisms induced respectively by g, $m \cdot g$, and $m' \cdot (m \cdot g)$. They agree modulo J. There are unique expressions

$$\psi = \varphi + D, \qquad \eta = \psi + D',$$

where D belongs to

$$\operatorname{Der}_{\varphi}(A, J) = \{\delta \in \operatorname{Hom}_{\Lambda}(A, J) \mid \delta(ab) = \varphi(a)\delta(b) + \varphi(b)\delta(a)\},$$

and D' belongs to $\operatorname{Der}_{\psi}(A, J)$. But $\operatorname{Der}_{\varphi}(A, J) = \operatorname{Der}_{\psi}(A, J)$, since $J^2 = 0$, and both identify with

$$\operatorname{Hom}_{B/J}\big(\Omega^1_{A/\Lambda} \otimes_A B/J, J\big).$$

Under this identification, D corresponds to m and D' to m'. Hence $\eta = \varphi + D + D'$, while $D + D'$ corresponds to $m + m'$, proving (0.1.8 (*)).

Corollary 0.1.9.[15] *Let X be an S-scheme, and retain the notation of 0.1.5. Then X carries a left action of the abelian X^+-group L_X, making X formally principal homogeneous under L_X over X^+. In other words, there is an isomorphism of X^+-functors*

$$L_X \times_{X^+} X \xrightarrow{\sim} X \times_{X^+} X, \qquad (m, x) \longmapsto (x, m \cdot x).$$

Proof. Let $i_0 : X_0 \hookrightarrow X$ be the immersion. Since $X_0 = X \times_S S_0$, one first has

$$i_0^*(\Omega^1_{X/S}) \simeq \Omega^1_{X_0/S_0}$$

(EGA IV, 16.4.5).

Let Y be an X^+-scheme, given by an S-morphism $g_{\mathscr{J}} : Y_{\mathscr{J}} \to X$, and let $g_0 : Y_0 \to X_0$ be obtained by base change. By 0.1.8, if $\operatorname{Hom}_{X^+}(Y, X)$ is nonempty, it is principal homogeneous under

$$\operatorname{Hom}_{\mathscr{O}_{Y_0}}\big(g_0^* i_0^*(\Omega^1_{X/S}), \mathscr{J}\mathscr{O}_Y\big),$$

which identifies with

$$\operatorname{Hom}_{\mathscr{O}_{Y_0}}\big(g_0^*(\Omega^1_{X_0/S_0}), \mathscr{J}\mathscr{O}_Y\big) = L_X(Y).$$

Thus there is a bijection

$$L_X(Y) \times \operatorname{Hom}_{X^+}(Y, X) \xrightarrow{\sim} \operatorname{Hom}_{X^+}(Y, X \times_{X^+} X), \qquad (m, g) \longmapsto (g, m \cdot g).$$

We show that this is functorial in Y. Let $f : Z \to Y$ be a morphism of S-schemes. It is enough to prove that the following diagram is commutative.

[14] Editor's note: The preceding argument is reproduced almost word for word from SGA 1, III 5.1; what follows has been expanded.

[15] Editor's note: This corollary has been added to SGA 1, III 5.1; it proves part (i) of Proposition 0.2 below.

$$\begin{tikzcd}
\mathrm{L_X(Y)} \times \mathrm{Hom_{X^+}(Y,X)} \arrow[r, "\sim"] \arrow[d, "\mathrm{L_X}(f)\times f"'] & \mathrm{Hom_{X^+}(Y, X \times_{X^+} X)} \arrow[d, "f\times f"] \\
\mathrm{L_X(Z)} \times \mathrm{Hom_{X^+}(Z,X)} \arrow[r, "\sim"] & \mathrm{Hom_{X^+}(Z, X \times_{X^+} X)}.
\end{tikzcd}$$

Figure 67: Functoriality in Y of the principal-homogeneous bijection.

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 10; combined-reader p. 120.

If $\mathrm{Hom}_{X^+}(Y,X)=\emptyset$, there is nothing to prove. It is therefore enough to show that, for every S-morphism $g: Y \to X$ extending $g_{\mathscr{J}}$ and every $m \in L_X(Y)$, one has

$$(m \cdot g) \circ f = L_X(f)(m) \cdot (g \circ f). \tag{0.1.9 (*)}$$

These two S-morphisms $Z \to X$ agree on $Z_{\mathscr{J}}$ with $g_{\mathscr{J}} \circ f_{\mathscr{J}}$; in particular, they induce the same continuous map as $g \circ f$ on the underlying topological spaces. It therefore suffices to check the equality on local rings. Let $z \in Z$, put $y = f(z)$, $x = g(y)$, and let A, B, C denote respectively the local rings $\mathscr{O}_{X,x}, \mathscr{O}_{Y,y}, \mathscr{O}_{Z,z}$. Let s be the image of x in S, put $\Lambda = \mathscr{O}_{S,s}$, let J and I be the ideals of Λ corresponding to $\mathscr{J}$ and $\mathscr{I}$, and let $\varphi, \psi : A \to B$ and $\theta : B \to C$ be the Λ-algebra morphisms induced by g, $m \cdot g$, and f, respectively. Then m induces an element D of

$$\mathrm{Hom}_{B/IB}\big(\Omega^1_{A/\Lambda} \otimes_A B/IB, JB\big) = \mathrm{Der}_\Lambda(A, JB),$$

and $\psi = \varphi + D$. Thus $(m \cdot g) \circ f$ corresponds to

$$\theta \circ \psi = \theta \circ \varphi + \theta \circ D.$$

By 0.1.5, $\theta \circ D$ is the image of $L_X(f)(m)$ in

$$\mathrm{Hom}_{C/IC}\big(\Omega^1_{A/\Lambda} \otimes_A C/IC, JC\big) = \mathrm{Der}_\Lambda(A, JC).$$

Consequently, $\theta \circ \varphi + \theta \circ D$ is the image of $L_X(f)(m) \cdot (g \circ f)$. This proves (0.1.9 (*)).

Corollary 0.1.10.[16]

(a) *The functor L_X depends functorially on X. For every S-morphism $f : X \to W$, there is an S-morphism $L_f : L_X \to L_W$ that is a morphism of abelian groups over f^+; that is, the following diagram is commutative,*

[16] Editor's note: The functoriality of L_X in X has been made explicit, in particular in part (b) below.

$$\begin{tikzcd}
\mathrm{L}_\mathrm{X} \arrow[r, "\mathrm{L}_f"] \arrow[d] & \mathrm{L}_\mathrm{W} \arrow[d] \\
\mathrm{X}^+ \arrow[r, "f^+"] & \mathrm{W}^+
\end{tikzcd}$$

Figure 68: The morphism L_f over f^+.

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 11; combined-reader p. 121.

and, for every $Y \to X^+$, the map

$$L_f(Y) : \mathrm{Hom}_{X^+}(Y, L_X) \longrightarrow \mathrm{Hom}_{W^+}(Y, L_W),$$

where Y lies over W^+ through f^+, is a homomorphism of abelian groups.

(b) *Moreover, the following diagram is commutative.*

$$\begin{tikzcd}
\mathrm{L}_\mathrm{X} \times_{\mathrm{X}^+} \mathrm{X} \arrow[r, "\sim"] \arrow[d, "\mathrm{L}_f \times f"'] & \mathrm{X} \times_{\mathrm{X}^+} \mathrm{X} \arrow[d, "f \times f"] \\
\mathrm{L}_\mathrm{W} \times_{\mathrm{W}^+} \mathrm{W} \arrow[r, "\sim"] & \mathrm{W} \times_{\mathrm{W}^+} \mathrm{W}
\end{tikzcd} .$$

Figure 69: Compatibility of L_f with the formally principal homogeneous actions.

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 11; combined-reader p. 121.

Proof. For part (a), L_f is induced by the morphism of $\mathscr{O}_{X_0}$-modules

$$f_{X_0/W_0/S_0} : f_0^*(\Omega^1_{W_0/S_0}) \longrightarrow \Omega^1_{X_0/S_0}$$

(cf. 0.1.7(b)). For every X^+-scheme Y, given by an S-morphism $g_{\mathscr{J}} : Y_{\mathscr{J}} \to X$, one has the following commutative diagram, functorial in Y.

$$\begin{tikzcd}
\mathrm{Hom}_{\mathscr{O}_{\mathrm{Y}_0}}(g_0^*(\Omega^1_{\mathrm{X}_0/\mathrm{S}_0}), \mathcal{J}\mathscr{O}_\mathrm{Y}) \arrow[r, "\mathrm{L}_f(\mathrm{Y})"] \arrow[d] & \mathrm{Hom}_{\mathscr{O}_{\mathrm{Y}_0}}(g_0^* f_0^*(\Omega^1_{\mathrm{W}_0/\mathrm{S}_0}), \mathcal{J}\mathscr{O}_\mathrm{Y}) \arrow[d] \\
\{g_{\mathcal{J}}\} \arrow[r] & \{f_{\mathcal{J}} \circ g_{\mathcal{J}}\}
\end{tikzcd}$$

Figure 70: The homomorphism $L_f(Y)$ on the differential-module descriptions.

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 11; combined-reader p. 121.

Here $L_f(Y)$ is the map

$$\psi \longmapsto \psi \circ g_0^*(f_{X_0/W_0/S_0}),$$

which is indeed a homomorphism of abelian groups.[17]

We now prove part (b). Let Y be an X^+-scheme. If $\mathrm{Hom}_{X^+}(Y, X) = \emptyset$, there is nothing to prove. Thus let $g \in \mathrm{Hom}_{X^+}(Y, X)$. We must show that, for every $m \in L_X(Y)$,

$$f \circ (m \cdot g) = L_f(Y)(m) \cdot (f \circ g). \tag{0.1.10 (*)}$$

With g fixed, $\mathrm{Hom}_X(Y, X \times_{X^+} X)$ is a subset of $\mathrm{Hom}_X(Y, \Delta^{(1)}_{X/S})$, and

$$\begin{aligned} L_X(Y) &= \mathrm{Hom}_{\mathscr{O}_{Y_0}}\big(g_0^*(\Omega^1_{X_0/S_0}), \mathscr{J}\mathscr{O}_Y\big) \\ &= \mathrm{Hom}_{\mathscr{O}_Y}\big(g^*(\Omega^1_{X/S}), \mathscr{J}\mathscr{O}_Y\big), \end{aligned}$$

which is a subset of $\widetilde{L}_X(Y)$; see 0.1.7. Moreover, $L_f(Y)$ is the restriction of $\widetilde{L}_f(Y)$ to $L_X(Y)$. Finally, the bijection

$$L_X(Y) \xrightarrow{\sim} \mathrm{Hom}_X(Y, X \times_{X^+} X), \qquad m \longmapsto (g, m \cdot g),$$

is the inverse of the restriction to $L_X(Y) \subset \widetilde{L}_X(Y)$ of the bijection

$$\mathrm{Hom}_X(Y, \Delta^{(1)}_{X/S}) \xrightarrow{\sim} \{g\} \times \widetilde{L}_X(Y)$$

considered in 0.1.7.1. Consequently, equality (0.1.10 (*)) follows from (0.1.7 (*)). Indeed, let g' be the X-morphism $Y \to \Delta^{(1)}_{X/S}$ defined by $(g, m\cdot g)$. The element of $\widetilde{L}_W(Y)$ corresponding to $(f \circ g, f \circ g')$ is $\widetilde{L}_f(Y)(m) = L_f(Y)(m)$. Thus

$$L_f(m) \cdot (f \circ g) = f \circ (m \cdot g).$$

Lemma 0.1.11. *Let X, X' be two S-schemes. There is a commutative diagram*

$$\begin{tikzcd} \mathrm{L}_{\mathrm{X}} \times_{\mathrm{S}} \mathrm{L}_{\mathrm{X}'} \arrow[r, "\sim"] \arrow[d] & \mathrm{L}_{\mathrm{X} \times_{\mathrm{S}} \mathrm{X}'} \arrow[d] \\ \mathrm{X}^+ \times_{\mathrm{S}} \mathrm{X}'^+ \arrow[r, "\sim"] & (\mathrm{X} \times_{\mathrm{S}} \mathrm{X}')^+ \end{tikzcd} .$$

Figure 71: Compatibility of L_X and X^+ with products.

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 12; combined-reader p. 122.

Proof.[18] For every S-scheme Y,

$$\begin{aligned} \mathrm{Hom}_S(Y, X^+ \times_S X'^+) &= \mathrm{Hom}_S(Y, X^+) \times \mathrm{Hom}_S(Y, X'^+) \\ &\simeq \mathrm{Hom}_S(Y_{\mathscr{J}}, X) \times \mathrm{Hom}_S(Y_{\mathscr{J}}, X') \\ &= \mathrm{Hom}_S\big(Y, (X \times_S X')^+\big). \end{aligned}$$

[17] Editor's note: Observe that L_f depends only on $f_0 : X_0 \to W_0$.

[18] Editor's note: the proof has been expanded.

Thus
$$X^+ \times_S X'^+ \simeq (X \times_S X')^+.$$

Now let Y be a scheme over $X^+ \times_S X'^+$, represented by a morphism
$$h : Y_{\mathscr{J}} \longrightarrow X \times_S X'.$$
Let p, q be the projections from $X \times_S X'$ to X, X', and put $f = p \circ h$, $g = q \circ h$. Since
$$\Omega^1_{(X_0 \times_{S_0} X'_0)/S_0} \simeq p_0^*(\Omega^1_{X_0/S_0}) \oplus q_0^*(\Omega^1_{X'_0/S_0})$$
(EGA IV_4, 16.4.23), there is a natural isomorphism
$$\begin{aligned}\operatorname{Hom}_{\mathscr{O}_{Y_0}}\big(f_0^*(\Omega^1_{X_0/S_0}), \mathscr{J}\mathscr{O}_Y\big) &\times \operatorname{Hom}_{\mathscr{O}_{Y_0}}\big(g_0^*(\Omega^1_{X'_0/S_0}), \mathscr{J}\mathscr{O}_Y\big)\\ &\simeq \operatorname{Hom}_{\mathscr{O}_{Y_0}}\big(h_0^*(\Omega^1_{(X_0 \times_{S_0} X'_0)/S_0}), \mathscr{J}\mathscr{O}_Y\big).\end{aligned}$$
In other words,
$$L_X(Y) \times L_{X'}(Y) \simeq L_{X \times_S X'}(Y).$$

Remark 0.1.12.[19] Let $\mathscr{C}$ be a category stable under fiber products, let S be an object of $\mathscr{C}$, and let T_1, T_2 be two objects over S. For $i = 1, 2$, let L_i and X_i be two objects over T_i, as in the diagram below.

$$\begin{tikzcd}
\mathrm{L}_1 \arrow[dr] & & \mathrm{X}_1 \arrow[dl] & \mathrm{L}_2 \arrow[dr] & & \mathrm{X}_2 \arrow[dl] \\
& \mathrm{T}_1 \arrow[drr] & & & \mathrm{T}_2 \arrow[dll] & \\
& & & \mathrm{S} & & .
\end{tikzcd}$$

Figure 72: The objects L_i, X_i over T_i, and T_i over S.

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 12; combined-reader p. 122.

There is then a natural isomorphism
$$(L_1 \times_{T_1} X_1) \times_S (L_2 \times_{T_2} X_2) \simeq (L_1 \times_S L_2) \times_{T_1 \times_S T_2} (X_1 \times_S X_2).$$
Consequently, the preceding lemma gives:

Corollary 0.1.13. *Let X_1, X_2 be two S-schemes. There is a commutative diagram of isomorphisms:*

$$\begin{tikzcd}
\mathrm{L}_{\mathrm{X}_1 \times_\mathrm{S} \mathrm{X}_2} \times_{(\mathrm{X}_1 \times_\mathrm{S} \mathrm{X}_2)^+} (\mathrm{X}_1 \times_\mathrm{S} \mathrm{X}_2) \arrow[d, "(0.1.11)"', "\simeq"] \arrow[dr, "\simeq"] & \\
(\mathrm{L}_{\mathrm{X}_1} \times_\mathrm{S} \mathrm{L}_{\mathrm{X}_2}) \times_{(\mathrm{X}_1^+ \times_\mathrm{S} \mathrm{X}_2^+)} (\mathrm{X}_1 \times_\mathrm{S} \mathrm{X}_2) \arrow[r, "\sim", "(0.1.12)"'] & (\mathrm{L}_{\mathrm{X}_1} \times_{\mathrm{X}_1^+} \mathrm{X}_1) \times_\mathrm{S} (\mathrm{L}_{\mathrm{X}_2} \times_{\mathrm{X}_2^+} \mathrm{X}_2).
\end{tikzcd}$$

Figure 73: The product isomorphisms of Corollary 0.1.13.

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 13; combined-reader p. 123.

[19] Editor's note: this remark and the corollary that follows have been added.

We can now state:

Proposition 0.2. *For every S-scheme X, one can define a left action of the X^+-abelian group L_X on the X^+-object X, with the following properties:*

(i) *this action makes X formally principal homogeneous under L_X over X^+; that is, the morphism*

$$L_X \times_{X^+} X \longrightarrow X \times_{X^+} X$$

is an isomorphism of X^+-functors;

(ii) *the action is functorial in the S-scheme X: for every S-morphism $f : X \to W$, the following diagram is commutative;*

$$\begin{tikzcd}
L_X \times_{X^+} X \arrow[r] \arrow[d, "L_f \times f"'] & X \arrow[d, "f"] \\
L_W \times_{W^+} W \arrow[r] & W
\end{tikzcd} \quad ;$$

Figure 74: Functoriality of the action in Proposition 0.2(ii).

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 13; combined-reader p. 123.

(iii) *the action commutes with fiber products: for all S-schemes X_1, X_2, the following diagram is commutative.*

$$\begin{tikzcd}
L_{X_1 \times_S X_2} \times_{(X_1 \times_S X_2)^+} (X_1 \times_S X_2) \arrow[r] \arrow[d, "\simeq"'] \arrow[dr, "\simeq"] & X_1 \times_S X_2 \\
(L_{X_1} \times_S L_{X_2}) \times_{(X_1 \times_S X_2)^+} (X_1 \times_S X_2) \arrow[r, "\sim"] & (L_{X_1} \times_{X_1^+} X_1) \times_S (L_{X_2} \times_{X_2^+} X_2). \arrow[u]
\end{tikzcd}$$

Figure 75: Compatibility of the action with fiber products.

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 13; combined-reader p. 123.

Proof.[20] Parts (i) and (ii) follow respectively from Corollaries 0.1.9 and 0.1.10. To prove (iii), write

$$P(X) = L_X \times_{X^+} X$$

for every S-scheme X. Applying (ii) to the projections

$$p_i : X_1 \times_S X_2 \longrightarrow X_i$$

gives the following commutative squares.

[20] Editor's note: the original has been modified and expanded to account for the preceding additions; in particular, those additions incorporate, with details, the content of original page 89.

$$\begin{tikzcd}
\mathrm{P}(\mathrm{X}_1 \times_{\mathrm{S}} \mathrm{X}_2) \arrow[r] \arrow[d, "\mathrm{L}_{p_i} \times p_i"'] & \mathrm{X}_1 \times_{\mathrm{S}} \mathrm{X}_2 \arrow[d, "p_i"] \\
\mathrm{P}(\mathrm{X}_i) \arrow[r] & \mathrm{X}_i
\end{tikzcd}$$

Figure 76: The square induced by each projection p_i.

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 13; combined-reader p. 123.

For $i = 1, 2$, these yield the commutative square

$$\begin{tikzcd}
\mathrm{P}(\mathrm{X}_1 \times_{\mathrm{S}} \mathrm{X}_2) \arrow[r] \arrow[d] & \mathrm{X}_1 \times_{\mathrm{S}} \mathrm{X}_2 \arrow[d, equal] \\
\mathrm{P}(\mathrm{X}_1) \times_{\mathrm{S}} \mathrm{P}(\mathrm{X}_2) \arrow[r] & \mathrm{X}_1 \times_{\mathrm{S}} \mathrm{X}_2.
\end{tikzcd}$$

Figure 77: The product square in the proof of Proposition 0.2(iii).

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 14; combined-reader p. 124.

Combining this square with Corollary 0.1.13 shows that its vertical arrow is an isomorphism and gives precisely the commutative diagram asserted in (iii).

Remark 0.3. Suppose that the X^+-scheme Y is flat over S (cf. SGA 1, IV). We may then write

$$\mathrm{Hom}_{X^+}(Y, L_X) = \mathrm{Hom}_{\mathscr{O}_{Y_0}}\big(g_0^*(\Omega^1_{X_0/S_0}), \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}\big).$$

Remark 0.4. Let $\pi_0 : X_0 \to S_0$ be the structure morphism, and suppose that there is an $\mathscr{O}_{S_0}$-module $\omega^1_{X_0/S_0}$ such that

$$\Omega^1_{X_0/S_0} \simeq \pi_0^*(\omega^1_{X_0/S_0}).$$

(This occurs, in particular, when X_0 is an S_0-group; cf. II 4.11.) Define a functor L'_X over S by

$$\mathrm{Hom}_S(Y, L'_X) = \mathrm{Hom}_{\mathscr{O}_{Y_0}}\big(\omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}, \mathscr{J}\mathscr{O}_Y\big). \tag{0.4.1}$$

Then

$$\mathrm{Hom}_{X^+}(Y, L_X) = \mathrm{Hom}_S(Y, L'_X)$$

for every X^+-scheme Y; in other words,

$$L_X = L'_X \times_S X^+.$$

Since $L_X \times_{X^+} X = L'_X \times_S X$, the action of L_X on X induces an action of L'_X on X, and this action respects $p_X : X \to X^+$.[21] Indeed, if Y is an S-scheme, $h : Y \to X$ is an

[21] Editor's note: what follows has been expanded.

S-morphism, and $m \in L'_X(Y)$, then h and $m \cdot h$ have the same restriction to $Y_{\mathscr{J}}$; that is,

$$p_X(m \cdot h) = p_X(h).$$

Remark 0.5. Keep the hypotheses and notation of 0.4, and suppose in addition that Y is an S-scheme flat over S. Then

$$\operatorname{Hom}_{X^+}(Y, L_X) = \operatorname{Hom}_S(Y, L'_X) = \operatorname{Hom}_{S_0}(Y_0, L_{0X}),$$

where the S_0-functor in abelian groups L_{0X} is defined by the following identity (with T the variable S_0-scheme):

$$\operatorname{Hom}_{S_0}(T, L_{0X}) = \operatorname{Hom}_{\mathscr{O}_T}\big(\omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T\big). \tag{0.5.1}$$

Thus, in the notation of II 1, we have shown that L'_X and $\prod_{S_0/S} L_{0X}$ have the same restriction to the full subcategory of $(\mathbf{Sch})/S$ whose objects are the S-schemes flat over S.

Remark 0.6. Keep the hypotheses and notation of 0.5,[22] and suppose in addition that $\pi_0 : X_0 \to S_0$ has a section ε_0. Then

$$\omega^1_{X_0/S_0} \simeq \varepsilon_0^*(\Omega^1_{X_0/S_0}).$$

First, independently of this last hypothesis, one has

$$\operatorname{Hom}_{S_0}(T, L_{0X}) = \Gamma\Big(T, \mathscr{H}om_{\mathscr{O}_T}\big(\omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T\big)\Big).$$

Suppose now that $\omega^1_{X_0/S_0}$ admits a finite presentation (cf. EGA 0_{I}, 5.2.5), as is in particular the case when X_0 is locally of finite presentation over S_0 (cf. EGA IV_4, 16.4.22). If T is flat over S_0, it then follows from EGA 0_{I}, 6.7.6 that

$$\mathscr{H}om_{\mathscr{O}_T}\big(\omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T\big) \simeq \mathscr{H}om_{\mathscr{O}_{S_0}}\big(\omega^1_{X_0/S_0}, \mathscr{J}\big) \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T,$$

whence

$$\operatorname{Hom}_{S_0}(T, L_{0X}) = \Gamma\Big(T, \mathscr{H}om_{\mathscr{O}_{S_0}}\big(\omega^1_{X_0/S_0}, \mathscr{J}\big) \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T\Big).$$

Using the notation $\mathbf{W}(\,\cdot\,)$ of I 4.6.1, we have therefore proved that, for every S_0-scheme T flat over S_0,

$$\operatorname{Hom}_{S_0}(T, L_{0X}) = \operatorname{Hom}_{S_0}\Big(T, \mathbf{W}\Big(\mathscr{H}om_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J})\Big)\Big).$$

In summary, if $\omega^1_{X_0/S_0}$ admits a finite presentation and one restricts to the category of S-schemes flat over S, then

$$L'_X = \prod_{S_0/S} \mathbf{W}\Big(\mathscr{H}om_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J})\Big), \tag{0.6.1}$$

and $\mathscr{H}om_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J})$ is a quasi-coherent $\mathscr{O}_{S_0}$-module, by EGA I, 9.1.1.

Finally, suppose moreover that $\omega^1_{X_0/S_0}$ is locally free of finite rank, for example that X_0 is smooth over S_0 (in which case it is automatically locally of finite presentation over S_0). Then

$$\mathscr{H}om_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J}) \simeq \underline{\mathrm{Lie}}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}, \tag{0.6.2}$$

[22] Editor's note: the following hypothesis, which was implicit in the original, has been added.

where, by an abuse of language (since X_0 need not be an S_0-group), $\underline{\mathrm{Lie}}(X_0/S_0)$ denotes the dual of the $\mathscr{O}_{S_0}$-module $\omega^1_{X_0/S_0}$.[23]

Proposition 0.2 (and its proof) has two important corollaries.[24]

Corollary 0.7. *Let X be an S-scheme.*

(a) *Every S-endomorphism of X inducing the identity on $X_{\mathscr{J}}$ is an automorphism.*

(b) *There is an exact sequence of groups*

$$0 \longrightarrow \mathrm{Hom}_{\mathscr{O}_{X_0}}(\Omega^1_{X_0/S_0}, \mathscr{J}\mathscr{O}_X) \xrightarrow{i} \mathrm{Aut}_S(X) \longrightarrow \mathrm{Aut}_{S_{\mathscr{J}}}(X_{\mathscr{J}}).$$

(c) *Let $\mathrm{Aut}_S(X)$ act on the first group by transport of structure. Then, for every $u \in \mathrm{Aut}_S(X)$ and $m \in \mathrm{Hom}_{\mathscr{O}_{X_0}}(\Omega^1_{X_0/S_0}, \mathscr{J}\mathscr{O}_X)$,*

$$i(u \cdot m) = u\, i(m)\, u^{-1}.$$

Proof. By 0.2(i), $\mathrm{Hom}_{X^+}(X, X)$ is a principal homogeneous set under $\mathrm{Hom}_{X^+}(X, L_X)$, since it is certainly nonempty: it contains the marked point given by the identity automorphism of X.[25] Consequently, the map $m \mapsto m \cdot \mathrm{id}_X$ induces a bijection

$$\mathrm{Hom}_{\mathscr{O}_{X_0}}(\Omega^1_{X_0/S_0}, \mathscr{J}\mathscr{O}_X) = L_X(X) \xrightarrow{\sim} \mathrm{Hom}_{X^+}(X, X).$$

Let $m \in L_X(X)$, and let $f = m' \cdot \mathrm{id}_X$ be an element of $\mathrm{Hom}_{X^+}(X, X)$. Applying 0.2(ii) to f gives

$$f \circ (m \cdot \mathrm{id}_X) = L_f(X)(m) \cdot f = L_f(X)(m) \cdot (m' \cdot \mathrm{id}_X).$$

On the other hand, because f is an X^+-endomorphism of X, one has $f_{\mathscr{J}} = \mathrm{id}_{X_{\mathscr{J}}}$, hence $f_0 = \mathrm{id}_{X_0}$. Since L_f depends only on f_0 (cf. the editor's note (17) in 0.1.10), it follows that $L_f(X)(m) = m$. Thus the preceding equality becomes

$$(m' \cdot \mathrm{id}_X) \circ (m \cdot \mathrm{id}_X) = m \cdot (m' \cdot \mathrm{id}_X) = (m + m') \cdot \mathrm{id}_X\,.$$

This shows that the bijection $m \mapsto m \cdot \mathrm{id}_X$ carries the group law of $\mathrm{Hom}_{X^+}(X, L_X)$ to composition of X^+-endomorphisms of X.

It follows first that every element of $\mathrm{Hom}_{X^+}(X, X)$ is invertible, proving part (a), and then that there is an exact sequence

$$0 \longrightarrow \mathrm{Hom}_{X^+}(X, L_X) \xrightarrow{i} \mathrm{Aut}_S(X) \longrightarrow \mathrm{Aut}_{S_{\mathscr{J}}}(X_{\mathscr{J}}),$$

which proves part (b).

The morphism i just defined is functorial in X with respect to isomorphisms, because it is defined structurally from the action of L_X on X over X^+, an action that is itself

[23] Editor's note: this is justified by II 3.3 and 4.11. Indeed, the functor $L^{\varepsilon_0}_{X_0/S_0}$, the "tangent space to X_0 over S_0 at ε_0," is represented over S_0 by the vector bundle $\mathbf{V}(\varepsilon_0^*(\Omega^1_{X_0/S_0})) = \mathbf{V}(\omega^1_{X_0/S_0})$. Its sheaf of sections is the dual module $\mathscr{H}om_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{O}_{S_0})$; this is $\underline{\mathrm{Lie}}(X_0/S_0)$ when X_0 is an S_0-group and ε_0 is the identity section.

[24] Editor's note: Corollary 0.7 bis, which was used implicitly in the proof of 0.8, has been added to 0.7. We point out that 0.8–0.12, as well as 0.17, which play an important technical role later in this exposé, are consequences of 0.7 and 0.7 bis.

[25] Editor's note: what follows has been expanded.

functorial in X by Proposition 0.2(ii).[26] Hence every automorphism u of X over S induces, by transport of structure, isomorphisms

$$h : \mathrm{Hom}_{X^+}(X, L_X) \xrightarrow{\sim} \mathrm{Hom}_{X^+}(X, L_X) \quad \text{and} \quad f : \mathrm{Aut}_S(X) \xrightarrow{\sim} \mathrm{Aut}_S(X)$$

such that the following diagram commutes.

$$\begin{tikzcd}
\mathrm{Hom}_{X^+}(\mathrm{X}, \mathrm{L_X}) \arrow[r, "i"] \arrow[d, "h"'] & \mathrm{Aut_S}(\mathrm{X}) \arrow[d, "f"'] \\
\mathrm{Hom}_{X^+}(\mathrm{X}, \mathrm{L_X}) \arrow[r, "i"] & \mathrm{Aut_S}(\mathrm{X})
\end{tikzcd}$$

Figure 78: Functoriality of i under transport of structure.

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 16; combined-reader p. 126; printed p. 116.

Thus $f \circ i = i \circ h$. On the other hand, f is described by the following commutative diagram.

$$\begin{tikzcd}
\mathrm{X} \arrow[r, "a"] \arrow[d, "u"'] & \mathrm{X} \arrow[d, "u"'] \\
\mathrm{X} \arrow[r, "f(a)"] & \mathrm{X}
\end{tikzcd}$$

Figure 79: The automorphism f acts by conjugation with u.

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 16; combined-reader p. 126; printed p. 116.

In other words,

$$f(a) = u \circ a \circ u^{-1} \qquad \text{for every } a \in \mathrm{Aut}_S(X).$$

Writing $i(h(m)) = f(i(m))$ now gives the asserted formula.

Corollary 0.7 bis. *Let X be an S-scheme such that $X_{\mathscr{I}}$ is an $S_{\mathscr{I}}$-monoid. Then L_X has the structure of an S-monoid, and there is a split exact sequence of S-monoids:*

$$\begin{tikzcd}
1 \arrow[r] & \mathrm{L}'_\mathrm{X} \arrow[r, "i"] & \mathrm{L_X} \arrow[r, shift left, "p"] & \mathrm{X}^+ \arrow[l, shift left, "s"] \arrow[r] & 1
\end{tikzcd}$$

Figure 80: The split exact sequence of S-monoids in Corollary 0.7 bis.

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 17; combined-reader p. 127; printed p. 117.

The monoid law induced on L'_X coincides with its abelian-group structure. In particular, if $X_{\mathscr{I}}$ is an $S_{\mathscr{I}}$-group, then L_X is an S-group and is the semidirect product of X^+ and L'_X.

[26]Editor's note: what follows has been expanded.

Proof. Since $X_{\mathscr{J}}$ is an $S_{\mathscr{J}}$-monoid,

$$X^+ = \prod_{S_{\mathscr{J}}/S} X_{\mathscr{J}}$$

is an S-monoid; indeed, $X^+(Y) = X_{\mathscr{J}}(Y_{\mathscr{J}})$ for every $Y \to S$. For every S-scheme Y, let $\widetilde{Y}_{\mathscr{J}}$ denote the affine $Y_{\mathscr{J}}$-scheme corresponding to the quasi-coherent $\mathscr{O}_{Y_{\mathscr{J}}}$-algebra

$$\mathscr{O}_{Y_{\mathscr{J}}} \oplus \mathscr{J}\mathscr{O}_Y,$$

that is, the graded algebra associated with the filtration $\mathscr{O}_Y \supset \mathscr{J}\mathscr{O}_Y$. Then $L_X(Y)$ identifies with $X_{\mathscr{J}}(\widetilde{Y}_{\mathscr{J}})$, and $L'_X(Y)$ identifies with the kernel of the morphism

$$p : X_{\mathscr{J}}(\widetilde{Y}_{\mathscr{J}}) \longrightarrow X_{\mathscr{J}}(Y_{\mathscr{J}})$$

induced by the *zero section* $Y_{\mathscr{J}} \to \widetilde{Y}_{\mathscr{J}}$. Equivalently, that section is induced by the morphism of $\mathscr{O}_{Y_{\mathscr{J}}}$-algebras

$$\mathscr{O}_{\widetilde{Y}_{\mathscr{J}}} \longrightarrow \mathscr{O}_{Y_{\mathscr{J}}}$$

that vanishes on the ideal $\mathscr{J}\mathscr{O}_Y$. Thus, for every $Y \to S$, there is a split exact sequence of monoids, functorial in Y:

$$\begin{tikzcd} 1 \arrow[r] & \mathrm{L}'_{\mathrm{X}}(\mathrm{Y}) \arrow[r, "i"] & \mathrm{L}_{\mathrm{X}}(\mathrm{Y}) \arrow[r, shift left, "p"] & \mathrm{X}^+(\mathrm{Y}) \arrow[l, shift left, "s"] \arrow[r] & 1 . \end{tikzcd}$$

Figure 81: The pointwise split exact sequence for $L_X(Y)$.

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 17; combined-reader p. 127; printed p. 117.

It remains to show that the monoid law induced on L'_X coincides with its abelian-group structure. Let μ be the monoid law of L_X, and let e be its unit section. We must prove that, for every $m, m' \in L'_X(Y)$,

$$\mu(m \cdot e, m' \cdot e) = (m + m') \cdot e.$$

This may be seen in either of two ways. First, repeat the proof of (0.1.10 $(*)$), replacing the morphism $f : X \to W$ there by

$$\mu : L_X \times_S L_X \longrightarrow L_X.$$

Identify $X^+(Y) = X_{\mathscr{J}}(Y_{\mathscr{J}})$ with its image under s in $L_X(Y) = X_{\mathscr{J}}(\widetilde{Y}_{\mathscr{J}})$. For all $g, g' \in X_{\mathscr{J}}(Y_{\mathscr{J}})$ and $m, m' \in L'_X(Y)$, one then obtains

$$\mu(m \cdot g, m' \cdot g') = L_\mu^{(g,g')}(m, m') \cdot \mu(g, g'). \tag{$\star$}$$

Here $L_\mu^{(g,g')}$ denotes the morphism derived from μ at the point (g, g'); in other words, $\widetilde{Y}_{\mathscr{J}}$ lies over $L_X \times_S L_X$ through (g, g'). In particular,

$$\mu(m \cdot e, m' \cdot e) = L_\mu^{(e,e)}(m, m') \cdot e.$$

But

$$L_\mu^{(e,e)}(m, m') = L_{\ell_e}(m') + L_{r_e}(m),$$

where ℓ_e, respectively r_e, denotes left, respectively right, translation by e. Each is the identity map of $X_{\mathscr{J}}$, and therefore $L_\mu^{(e,e)}(m,m') = m + m'$.

Alternatively, one may argue as follows (cf. the proof of [DG70], §II.4, Theorem 3.5). By Lemma 0.1.11, formation of X^+ and of L_X commutes with products, and hence so does formation of L'_X. It follows that the morphism

$$\mu' : L'_X \times L'_X \longrightarrow L'_X$$

induced by μ is a homomorphism for the abelian-group structure. Lemma 3.10 of Exposé II then shows that μ' coincides with the abelian-group law.

0.8.[27] Now let X be an S-scheme such that $X_{\mathscr{J}}$ is an $S_{\mathscr{J}}$-group. Suppose that there is an S-morphism

$$P : X \times_S X \longrightarrow X$$

whose base change

$$P_{\mathscr{J}} : X_{\mathscr{J}} \times_{S_{\mathscr{J}}} X_{\mathscr{J}} \longrightarrow X_{\mathscr{J}}$$

is the group law of $X_{\mathscr{J}}$. An important special case is that in which X is an S-group and P is its group law. We obtain a morphism

$$L_P : L_X \times_S L_X \simeq L_{X \times_S X} \longrightarrow L_X.$$

In fact, this morphism does not depend on P, since it can be computed from the group law $P_{\mathscr{J}}$ of $X_{\mathscr{J}}$, as we now show.[28]

By Proposition 0.2(ii) and (iii), for every $Y \to S$, every $x, x' \in X(Y)$, and every $m, m' \in L'_X(Y)$,

$$\begin{aligned} P(m \cdot x, m' \cdot x') &= P\big((m,m') \cdot (x,x')\big) \\ &= L_P^{(x,x')}(m,m') \cdot \mu(g,g'), \end{aligned}$$

where g, respectively g', is the image of x, respectively x', in $X^+(Y)$. Moreover (cf. the proof of 0.10), $L_P^{(x,x')} = L_\mu^{(g,g')}$, and by 0.7 bis $(\star)$ this is the element of $L'_X(Y)$ defined by the equality

$$L_\mu^{(g,g')}(m,m') \cdot \mu(g,g') = \mu(m \cdot g, m' \cdot g')$$

in $L_X(Y)$.

Write $\times$, rather than μ, for the group law of L_X, and write Ad for the *adjoint action* of X^+ on L'_X. This action factors through X_0 and is induced by the adjoint action of X_0 on $\omega^1_{X_0/S_0}$. Then

$$\begin{aligned} L_\mu^{(g,g')}(m,m') \times g \times g' &= m \times g \times m' \times g' \\ &= \big(m \times \mathrm{Ad}(g)(m')\big) \times g \times g', \end{aligned}$$

and hence

$$L_P^{(x,x')}(m,m') = m \times \mathrm{Ad}(g)(m').$$

We have therefore proved:

Proposition 0.8. *Let $P : X \times_S X \to X$ be an S-morphism such that $P_{\mathscr{J}}$ equips $X_{\mathscr{J}}$ with an $S_{\mathscr{J}}$-group structure. Let $\times$ denote the group law of L'_X, let $(m,x) \mapsto m \cdot x$ denote the morphism $L'_X \times_S X \to X$ defining the action of L'_X on X, and let*

$$\mathrm{Ad} : X^+ \longrightarrow \mathrm{Aut}_{S\text{-gr}}(L'_X)$$

[27] Editor's note: the number 0.8 has been inserted here to mark the return to the original text.
[28] Editor's note: the original has been expanded in what follows.

be the adjoint action of X^+ on L'_X, induced by the adjoint action of X_0 on $\omega^1_{X_0/S_0}$. Then, for every $S' \to S$, every $x, x' \in X(S')$, and every $m, m' \in L'_X(S')$,

$$P(m \cdot x, m' \cdot x') = \big(m \times \mathrm{Ad}(p_X(x))(m')\big) \cdot P(x, x'). \tag{0.8.1}$$

If X is an S-group, we shall write $*$ for its law, e for its unit section, and i for the S-morphism defined by

$$i(m) = m \cdot e$$

for every $S' \to S$ and every $m \in L'_X(S')$.

Corollary 0.9. *Let X be an S-group. Then X^+ is naturally an S-group, and p_X is a morphism of S-groups. Moreover, the S-morphism*

$$i : L'_X \longrightarrow X, \qquad m \longmapsto m \cdot e,$$

is an isomorphism of S-groups from L'_X onto $\mathrm{Ker}(p_X)$, and, for every $S' \to S$, every $x' \in X(S')$, and every $m \in L'_X(S')$,

$$m \cdot x' = (m \cdot e) * x' = i(m) * x'. \tag{0.9.1}$$

The first two assertions were already proved in 0.1.2. Since X is formally principal homogeneous over X^+ under $L_X = L'_X \times_S X^+$, the morphism i is indeed an isomorphism of S-functors from L'_X onto the kernel of p_X. The fact that i is a group morphism and formula (0.9.1) follow from (0.8.1), applied respectively with $x = x' = e$, and with $x = e$, $m' = 1$.

Corollary 0.10. *Let X be an S-group. With the preceding notation, for every $S' \to S$, every $x \in X(S')$, and every $m' \in L'_X(S')$,*

$$x * i(m') * x^{-1} = i\big(\mathrm{Ad}(p_X(x))(m')\big). \tag{0.10.1}$$

This follows from the equality $i(m') * x^{-1} = m' \cdot x^{-1}$ and from (0.8.1), applied with $m = 1$ and $x' = x^{-1}$.

Thus, when X is an S-group, we have explicitly determined the kernel of $X \to X^+$ and the action of the inner automorphisms of X on this kernel. We shall now see that the same can be done for certain, not necessarily representable, S-group functors. One case will be useful to us, namely the functors $\underline{\mathrm{Aut}}$ of Exposé I, 1.7. We state it at once in the next proposition.

Proposition 0.11. *Let E be an S-scheme, and put $X = \underline{\mathrm{Aut}}_S(E)$. The kernel of the morphism of S-group functors*

$$p_X : X \longrightarrow X^+$$

is canonically identified with the commutative S-group functor L'_X defined by

$$\mathrm{Hom}_S(Y, L'_X) = \mathrm{Hom}_{\mathscr{O}_{E_0 \times_{S_0} Y_0}}\left(\Omega^1_{E_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}, \mathscr{J}\mathscr{O}_{E \times_S Y}\right),$$

where Y denotes a variable S-scheme.

Indeed, for a variable S-scheme Y, we have

$$\mathrm{Hom}_S(Y, X) = \mathrm{Aut}_Y(E \times_S Y)$$

and

$$\begin{aligned}\mathrm{Hom}_S(Y, X^+) &= \mathrm{Hom}_S(Y_{\mathscr{J}}, X)\\ &= \mathrm{Aut}_{Y_{\mathscr{J}}}(E \times_S Y_{\mathscr{J}})\\ &= \mathrm{Aut}_{Y_{\mathscr{J}}}\big((E \times_S Y) \times_Y Y_{\mathscr{J}}\big).\end{aligned}$$

Applying 0.7 (b) to the Y-scheme $E \times_S Y$, we obtain an isomorphism of groups

$$\mathrm{Hom}_S(Y, L'_X) \simeq \mathrm{Ker}\big(\mathrm{Hom}_S(Y, X) \longrightarrow \mathrm{Hom}_S(Y, X^+)\big)\,.$$

This isomorphism is readily checked to be functorial in the S-scheme Y. Hence it yields an isomorphism of S-groups

$$L'_X \simeq \mathrm{Ker}(X \longrightarrow X^+),$$

which completes the proof of Proposition 0.11.

Corollary 0.12.[29] *Retain the notation of 0.11: E is an S-scheme and $X = \underline{\mathrm{Aut}}_S(E)$. There is a natural action f of X on L'_X, defined as follows. For every S-scheme Y,*

$$\mathrm{Hom}_S(Y, X) = \mathrm{Aut}_Y(E \times_S Y)$$

and

$$\mathrm{Hom}_S(Y, L'_X) = \mathrm{Hom}_{\mathscr{O}_{E_0 \times_{S_0} Y_0}}\left(\Omega^1_{E_0 \times_{S_0} Y_0 / Y_0}, \mathscr{J}\mathscr{O}_{E \times_S Y}\right).$$

Here

$$\Omega^1_{E_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0} \simeq \Omega^1_{E_0 \times_{S_0} Y_0 / Y_0}$$

(see EGA IV_4, 16.4.5). The first group acts on the second by transport of structure, and this action is functorial in Y. One then has

$$x\, i(m)\, x^{-1} = i\big(f(x)m\big), \tag{0.12.1}$$

for every $Y \to S$, $x \in \mathrm{Hom}_S(Y, X)$, and $m \in \mathrm{Hom}_S(Y, L'_X)$.

Indeed, this follows from 0.7 (c), applied to the Y-scheme $E \times_S Y$.

Reminder 0.13. The direct image of a quasi-coherent module under a morphism of finite presentation is quasi-coherent. Under the same hypotheses, formation of the direct image commutes with flat base change. In the situation

$$\begin{tikzcd}
\mathrm{T} \arrow[d, "f"'] & \mathrm{T}' = \mathrm{T} \times_{\mathrm{S}} \mathrm{S}' \arrow[l, "g'"'] \arrow[d, "f'"] \\
\mathrm{S} & \mathrm{S}' \arrow[l, "g"]
\end{tikzcd}$$

Figure 82: The flat base-change square in Reminder 0.13.

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 20; combined-reader p. 130; printed p. 120.

if f (and hence f') is of finite presentation and g (and hence g') is flat, then for every quasi-coherent $\mathscr{O}_T$-module $\mathscr{F}$,

$$f_*(\mathscr{F}) \otimes_{\mathscr{O}_S} \mathscr{O}_{S'} = f'_*\big(\mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{O}_{S'}\big),$$

or, more elegantly,

$$g^*\big(f_*\mathscr{F}\big) = f'_*\big(g'^*\mathscr{F}\big).$$

[29] Editor's note: "Remark" has been changed to "Corollary".

These two facts hold more generally for a quasi-compact and quasi-separated morphism f; see EGA I, 9.2.1, and EGA III_1, 1.4.15 in the quasi-compact separated case (taking EGA III_2, $\mathrm{Err}_{\mathrm{III}}$ 25 into account), as well as EGA IV_1, 1.7.4 and 1.7.21.

Remark 0.14.[30] Return to the notation of 0.11: let E be an S-scheme, put $X = \underline{\mathrm{Aut}}_S(E)$, and let L'_X be the commutative S-group functor defined by

$$\begin{aligned} L'_X(Y) &= \mathrm{Hom}_{\mathscr{O}_{E_0\times_{S_0}Y_0}}\left(\Omega^1_{E_0/S_0}\otimes_{\mathscr{O}_{S_0}}\mathscr{O}_{Y_0}, \mathscr{J}\mathscr{O}_{E\times_S Y}\right)\\ &= \mathrm{Hom}_{\mathscr{O}_{E\times_S Y}}\left(\Omega^1_{E/S}\otimes_{\mathscr{O}_S}\mathscr{O}_Y, \mathscr{J}\mathscr{O}_{E\times_S Y}\right)\\ &= \Gamma\Big(E\times_S Y, \widetilde{\mathrm{Hom}}_{\mathscr{O}_{E\times_S Y}}\left(\Omega^1_{E/S}\otimes_{\mathscr{O}_S}\mathscr{O}_Y, \mathscr{J}\mathscr{O}_{E\times_S Y}\right)\Big). \end{aligned}$$

Suppose that Y is flat over S. There are then isomorphisms

$$\mathscr{J}\mathscr{O}_{E\times_S Y} \xleftarrow{\sim} (\mathscr{J}\mathscr{O}_E)\otimes_{\mathscr{O}_S}\mathscr{O}_Y \xrightarrow{\sim} (\mathscr{J}\mathscr{O}_E)\otimes_{\mathscr{O}_{S_0}}\mathscr{O}_{Y_0}.$$

Suppose in addition that E is of finite presentation over S. Then $\Omega^1_{E/S}$ is an $\mathscr{O}_E$-module of finite presentation (EGA IV_4, 16.4.22), and therefore, by EGA 0_{I}, 6.7.6,

$$\begin{aligned} &\widetilde{\mathrm{Hom}}_{\mathscr{O}_{E\times_S Y}}\left(\Omega^1_{E/S}\otimes_{\mathscr{O}_S}\mathscr{O}_Y, (\mathscr{J}\mathscr{O}_E)\otimes_{\mathscr{O}_S}\mathscr{O}_Y\right)\\ &\quad\simeq \widetilde{\mathrm{Hom}}_{\mathscr{O}_E}\left(\Omega^1_{E/S}, \mathscr{J}\mathscr{O}_E\right)\otimes_{\mathscr{O}_S}\mathscr{O}_Y. \end{aligned}$$

Let $\pi : E \to S$ and $g : Y \to S$ be the structural morphisms. Apply 0.13 to the diagram

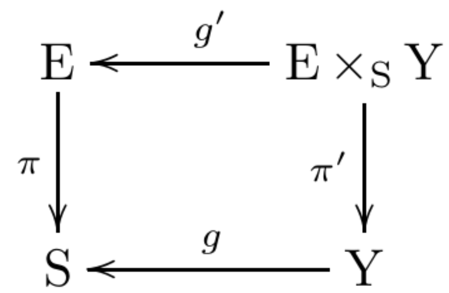

Figure 83: The structural base-change square used in Remark 0.14.

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 21; combined-reader p. 131; printed p. 121.

and to the $\mathscr{O}_E$-module

$$\mathscr{F} = \widetilde{\mathrm{Hom}}_{\mathscr{O}_E}\left(\Omega^1_{E/S}, \mathscr{J}\mathscr{O}_E\right).$$

We obtain

$$\begin{aligned} \Gamma(E\times_S Y, g'^*\mathscr{F}) &= \Gamma(Y, \pi'_* g'^*\mathscr{F})\\ &= \Gamma(Y, g^*\pi_*\mathscr{F})\\ &= W(\pi_*\mathscr{F})(Y). \end{aligned}$$

Thus, if E is of finite presentation over S, we have

$$L'_X = W\Big(\pi_*\widetilde{\mathrm{Hom}}_{\mathscr{O}_E}(\Omega^1_{E/S}, \mathscr{J}\mathscr{O}_E)\Big) \tag{0.14.1}$$

[30] Editor's note: The first part of this remark has been expanded, and a second part has been added.

on the category of S-schemes flat over S. Notice also that the module to which W is applied is quasi-coherent, by EGA I, 9.1.1 and 9.2.1.

Let L_0 be the S_0-functor[31]

$$L_0 = W\Big(\pi_{0*}\widetilde{\mathrm{Hom}}_{\mathscr{O}_{E_0}}(\Omega^1_{E_0/S_0}, \mathscr{J}\mathscr{O}_E)\Big).$$

Returning to the definition of $L'_X(Y)$, and using the isomorphism

$$\mathscr{J}\mathscr{O}_{E\times_S Y} \simeq (\mathscr{J}\mathscr{O}_E)\otimes_{\mathscr{O}_{S_0}}\mathscr{O}_{Y_0},$$

the same reasoning gives

$$L'_X(Y) = L_0(Y_0) = L_0(Y\times_S S_0) = \left(\prod_{S_0/S} L_0\right)(Y).$$

Consequently, on the category of S-schemes flat over S,

$$L'_X = \prod_{S_0/S} W\Big(\pi_{0*}\widetilde{\mathrm{Hom}}_{\mathscr{O}_{E_0}}(\Omega^1_{E_0/S_0}, \mathscr{J}\mathscr{O}_E)\Big).$$

It is not obvious that the action of X on L'_X defined in 0.12 comes from an action of $X_0 = \underline{\mathrm{Aut}}_{S_0}(E_0)$ on L_0. This is nevertheless the case when E is also flat over S.

Indeed, in that case there are canonical isomorphisms

$$\mathscr{J}\mathscr{O}_E \simeq \mathscr{J}\otimes_{\mathscr{O}_S}\mathscr{O}_E \simeq \mathscr{J}\otimes_{\mathscr{O}_{S_0}}\mathscr{O}_{E_0},$$

and hence

$$L_0 \simeq W\Big(\pi_{0*}\widetilde{\mathrm{Hom}}_{\mathscr{O}_{E_0}}\Big(\Omega^1_{E_0/S_0}, \mathscr{J}\otimes_{\mathscr{O}_{S_0}}\mathscr{O}_{E_0}\Big)\Big),$$

and

$$L'_X = \prod_{S_0/S} W\Big(\pi_{0*}\widetilde{\mathrm{Hom}}_{\mathscr{O}_{E_0}}\Big(\Omega^1_{E_0/S_0}, \mathscr{J}\otimes_{\mathscr{O}_{S_0}}\mathscr{O}_{E_0}\Big)\Big). \tag{0.14.2}$$

For every S_0-scheme T, we then have

$$L_0(T) \simeq \mathrm{Hom}_{\mathscr{O}_{E_0\times_{S_0}T}}\Big(\Omega^1_{E_0\times_{S_0}T/T}, \mathscr{J}\otimes_{\mathscr{O}_{S_0}}\mathscr{O}_{E_0\times_{S_0}T}\Big).$$

Moreover,

$$\mathrm{Hom}_{S_0}(T, X_0) = \mathrm{Aut}_T(E_0\times_{S_0}T)$$

acts on $L_0(T)$ by transport of structure, functorially in T. Finally, for every S-scheme Y flat over S, the action by transport of structure of

$$\mathrm{Hom}_S(Y, X) = \mathrm{Aut}_Y(E\times_S Y)$$

on $L'_X(Y) = L_0(Y_0)$ factors through $\mathrm{Aut}_{Y_0}(E_0\times_{S_0}Y_0)$.

Finally, let us extract the following two propositions from SGA 1, Exposé III.

Proposition 0.15. *(SGA 1, Exposé III, 6.8)*[32] *For every affine $S_{\mathscr{J}}$-scheme Y smooth over $S_{\mathscr{J}}$, there exists an S-scheme X, smooth over S, such that $X\times_S S_{\mathscr{J}} \simeq Y$; such an X is unique up to a non-unique isomorphism.*

[31] Editor's note: What follows has been added.

[32] Editor's note: This "global" lifting result uses the "local" lifting result of loc. cit., 4.1, which is stated for S locally Noetherian; see EGA IV_4, 18.1.1 for the general case.

Proposition 0.16. *(SGA 1, Exposé III, 5.5)*[33] *Let X be an S-scheme smooth over S. For every affine S-scheme Y, the canonical map*

$$p_X(Y) : \mathrm{Hom}_S(Y, X) \longrightarrow \mathrm{Hom}_S(Y, X^+) = \mathrm{Hom}_{S_{\mathscr{J}}}(Y_{\mathscr{J}}, X_{\mathscr{J}})$$

is surjective.

Corollary 0.17. *Let E be an affine S-scheme smooth over S, and put $X = \underline{\mathrm{Aut}}_S(E)$. For every affine S-scheme Y, the canonical map*

$$\begin{aligned} \mathrm{Aut}_Y(E \times_S Y) = \mathrm{Hom}_S(Y, X) &\longrightarrow \mathrm{Hom}_S(Y, X^+) \\ &= \mathrm{Aut}_{Y_{\mathscr{J}}}(E_{\mathscr{J}} \times_{S_{\mathscr{J}}} Y_{\mathscr{J}}) \end{aligned}$$

is surjective.

Indeed, $Y \times_S E$ is affine over the affine scheme Y, hence is affine. Applying 0.16, we see that every $S_{\mathscr{J}}$-morphism

$$Y_{\mathscr{J}} \times_{S_{\mathscr{J}}} E_{\mathscr{J}} \longrightarrow E_{\mathscr{J}}$$

extends to an S-morphism

$$Y \times_S E \longrightarrow E.$$

In other words,[34] every $Y_{\mathscr{J}}$-endomorphism of $Y_{\mathscr{J}} \times_{S_{\mathscr{J}}} E_{\mathscr{J}}$ lifts to a Y-endomorphism of $Y \times_S E$. Then 0.7 (a) shows that every $Y_{\mathscr{J}}$-automorphism of $Y_{\mathscr{J}} \times_{S_{\mathscr{J}}} E_{\mathscr{J}}$ lifts to a Y-automorphism of $Y \times_S E$, which is the asserted property.

1. Extensions and Cohomology

1.1.

Let $\mathscr{C}$ be a category stable under fiber products.[35] Let S be an object of $\mathscr{C}$, let G be a representable S-group, and let F be an S-functor in commutative groups on which G acts. The cohomology groups $H^n(G, F)$ were defined in I 5.1. Recall that they are the cohomology groups of a complex denoted $C^*(G, F)$. Writing

$$(G/S)^n = G \times_S \cdots \times_S G \qquad (n \text{ factors}),$$

one has

$$C^n(G, F) = \mathrm{Hom}_S((G/S)^n, F).$$

Since G, and hence each $(G/S)^n$, is representable, one also has

$$C^n(G, F) = F((G/S)^n).$$

It follows from this and from the definition of the coboundary operator that the complex $C^*(G, F)$ depends only on the restriction of F to the full subcategory of $\mathscr{C}/S$ whose objects are the Cartesian powers of G over S. Consequently:

Lemma 1.1.1. *Let $\mathscr{C}$ be a category stable under fiber products, let S be an object of $\mathscr{C}$, and let G be a representable S-group. Denote by $\mathscr{C}(G)$ the full subcategory of $\mathscr{C}/S$ whose objects*

[33] Editor's note: This is an immediate consequence of the definition of "(formally) smooth" adopted in EGA IV_4, 17.3.1 (and 17.1.1).

[34] Editor's note: The original has been modified slightly so that the hypotheses of 0.7 apply exactly.

[35] Editor's note: the hypothesis that $\mathscr{C}$ is stable under fiber products has been added. It holds in the applications in which $\mathscr{C}$ is the category of schemes (**Sch**), or the category of functors $(\mathbf{Sch})^\circ \to (\mathbf{Ens})$. If this hypothesis is omitted, one must instead assume below that the fiber products $G \times_S \cdots \times_S G$ are representable.

are the Cartesian powers of G over S. Let F and F' be two S-functors in commutative groups on which G acts. If F and F' have the same restriction to $\mathscr{C}(G)$, there is a canonical isomorphism

$$H^*(G, F) \xrightarrow{\sim} H^*(G, F').$$

1.1.2. Cohomology and restriction of scalars.[36]

We record another comparison result. Let $T \to S$ be a morphism in $\mathscr{C}$. If F is a T-functor in commutative groups, then the functor obtained by restriction of scalars (cf. Exposé II, 1),

$$F_1 = \prod_{T/S} F,$$

is an S-functor in commutative groups, and there is a morphism of S-functors in groups[37]

$$u : \prod_{T/S} \underline{\mathrm{Aut}}_{T\text{-gr.}}(F) \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(F_1).$$

Now let G be an S-functor in groups, and suppose an action of G_T on F is given by

$$G_T \longrightarrow \underline{\mathrm{Aut}}_{T\text{-gr.}}(F).$$

By the definition of $\prod_{T/S}$, this gives a morphism of S-functors in groups

$$G \longrightarrow \prod_{T/S} \underline{\mathrm{Aut}}_{T\text{-gr.}}(F).$$

Composing with u therefore gives an action of G on $F_1 = \prod_{T/S} F$.[38]

Lemma 1.1.2. *Under the preceding assumptions, there is a canonical isomorphism*

$$H^*\left(G, \prod_{T/S} F\right) \simeq H^*(G_T, F).$$

Indeed, by the definition of cohomology, the standard complexes are canonically isomorphic.

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[36] Editor's note: the heading of this paragraph has been added.

[37] Editor's note: explicitly, let $S' \to S$ and $\alpha \in \underline{\mathrm{Aut}}_{(T\times_S S')\text{-gr.}}(F_{T\times_S S'})$. For every $U \to T \times_S S'$, let $\alpha_U \in \mathrm{Aut}_{\mathrm{gr.}}(F_{T\times_S S'}(U))$ be the element defined by α. Then $u(S')(\alpha)$ is the element β of

$$\underline{\mathrm{Aut}}_{S\text{-gr.}}(F_1)(S') = \underline{\mathrm{Aut}}_{S'\text{-gr.}}\left(\prod_{T\times_S S'/S'} F_{T\times_S S'}\right)$$

such that $\beta_{S''} = \alpha_{T\times_S S''}$ for every $S'' \to S'$.

[38] Editor's note: explicitly, for every $S' \to S$, the action of $G(S')$ on $F_1(S') = F(T \times_S S')$ is given by the action of $G_T(T \times_S S') = G(T \times_S S')$ on $F(T \times_S S')$, together with the group morphism $G(S') \to G(T \times_S S')$ corresponding to the projection $T \times_S S' \to S'$. Equivalently, the action of G_T on F is a morphism $G_T \times_T F \to F$. Applying $\prod_{T/S}$, and writing $G_1 = \prod_{T/S} G_T$, gives, by II 1.2, a morphism $G_1 \times_S F_1 \to F_1$, hence an action of G_1 on F_1. The action of G is obtained through the natural morphism $G \to G_1$, which corresponds by adjunction to id_G.

1.2. Lifting group morphisms.[39]

Following the general principles, we make the following definition.

Definition 1.2.1. *Let*

$$1 \longrightarrow M \xrightarrow{u} E \xrightarrow{v} G$$

be a sequence of morphisms of $\widehat{\mathscr{C}}$-groups. It is called exact *if the following equivalent conditions hold:*

(i) *for every $S \in \operatorname{Ob}\mathscr{C}$, the following sequence of ordinary groups is exact:*

$$1 \longrightarrow M(S) \xrightarrow{u(S)} E(S) \xrightarrow{v(S)} G(S);$$

(ii) *for every object H of $\widehat{\mathscr{C}}$, the following sequence of ordinary groups is exact:*

$$1 \longrightarrow \operatorname{Hom}(H, M) \xrightarrow{u(H)} \operatorname{Hom}(H, E) \xrightarrow{v(H)} \operatorname{Hom}(H, G).$$

Taking $H = G$ in (ii), one sees that the set of sections of v (which need not preserve the group structures) is either empty or principal homogeneous under $\operatorname{Hom}(G, M)$. Suppose that it is nonempty, and let

$$s : G \longrightarrow E$$

be a section of v. For every $S \in \operatorname{Ob}\mathscr{C}$ and every $x \in G(S)$, the element $s(x) \in E(S)$ defines an inner automorphism of E_S that normalizes M_S (more precisely, the image of M_S under u_S), and hence an automorphism of M_S.

Scholium 1.2.1.1. [40] *If M is commutative, one sees set-theoretically that this automorphism is independent of the chosen section and depends only on x, and that it depends multiplicatively on x. In summary, every exact sequence*

$$(E) \qquad 1 \longrightarrow M \xrightarrow{u} E \xrightarrow{v} G$$

in which M is commutative and v admits a section determines a morphism of $\widehat{\mathscr{C}}$-groups

$$G \longrightarrow \underline{\operatorname{Aut}}_{\widehat{\mathscr{C}}\text{-gr.}}(M),$$

called the action of G on M defined by the extension (E).

Definition 1.2.1.2. We saw in I 2.3.7 that v admits a section that is a morphism of $\widehat{\mathscr{C}}$-groups if and only if the extension (E) is isomorphic, as an extension, to the semidirect product of M by G for the preceding action. Such a section of v is called a *section of the extension* (E), or simply a *section of* (E).

If s is a section of (E) and

$$m \in \Gamma(M) \simeq \operatorname{Ker}\bigl(\Gamma(E) \to \Gamma(G)\bigr)$$

(for the definition of Γ, see I 1.2), then the morphism $G \to E$ defined by

$$x \longmapsto u(m)s(x)u(m)^{-1}$$

[39] Editor's note: this heading has been added.

[40] Editor's note: the numbering 1.2.1.1 and 1.2.1.2 has been added to highlight the notions introduced here.

is also a section of (E).[41] It is called the section deduced from s by the inner automorphism defined by m, or by $u(m)$.

Lemma 1.2.2. *Let*

$$(E) \qquad 1 \longrightarrow M \xrightarrow{u} E \xrightarrow{v} G$$

be an exact sequence of $\widehat{\mathscr{C}}$-groups in which M is commutative and v admits a section. Let G act on M in the manner defined by (E).

(i) *The extension (E) canonically defines a class $c(E) \in H^2(G, M)$, whose vanishing is necessary and sufficient for the existence of a section of (E).*

(ii) *If $c(E) = 0$, the set of sections of (E) is principal homogeneous under $Z^1(G, M)$, and the set of sections of (E), modulo the action of the inner automorphisms defined by the elements of $\Gamma(M)$, is principal homogeneous under $H^1(G, M)$.*

(iii) [42] *Let s be a section of (E). The set of conjugates of s by the inner automorphisms defined by $\Gamma(M)$ is in bijection with $\Gamma(M)/H^0(G, M)$.*

The proof is exactly the same as for ordinary groups; starting from a section of v ensures the functoriality of the set-theoretic computations. We briefly indicate the principal steps.

(a) To every section s of v, associate the morphism

$$D_s : G \times G \longrightarrow M$$

defined set-theoretically by

$$u(D_s(x, y)) = s(xy)s(y)^{-1}s(x)^{-1}.$$

The following computation shows that D_s is a 2-cocycle.[43] By the definition of the differential in the standard complex (I 5.1),

$$(\partial^2 D_s)(x, y, z) = \big(s(x)D_s(y, z)s(x)^{-1}\big) \cdot D_s(x, y)^{-1} \cdot D_s(xy, z)^{-1} \cdot D_s(x, yz).$$

Substituting the definition of D_s into this formula gives, without using commutativity,

$$D_s \in Z^2(G, M).$$

(b) If s and s' are two sections of v, there is a morphism $f : G \to M$ such that $s(x) = f(x)s'(x)$. Then

$$\begin{aligned} D_{s'}(x, y) &= f^{-1}(xy)D_s(x, y)s(x)f(y)s(x)^{-1}f(x) \\ &= D_s(x, y) \cdot f^{-1}(xy) \cdot \big(s(x)f(y)s(x)^{-1}\big) \cdot f(x), \end{aligned}$$

and therefore

$$D_{s'} = D_s \cdot \partial^1 f.$$

This shows that the class of D_s in $H^2(G, M)$ is independent of the chosen section s of v; it is the class $c(E)$ of the extension (E).[44]

[41] Editor's note: in the right-hand side, x has been corrected to $s(x)$.

[42] Editor's note: this item has been added as the counterpart of 1.2.4(iii). The assertion is valid for every section of v; see the proof.

[43] Editor's note: the original has been corrected to make it compatible with I 5.1.

[44] Editor's note: the preceding sentence has been added.

(c) Let s and s' be two sections of v, and let $m \in \Gamma(M)$. The equality

$$s(x) = m^{-1}s'(x)m \qquad (x \in G(S),\ S \in \mathrm{Ob}\,\mathscr{C})$$

is equivalent to

$$s(x) = m^{-1}s'(x)m\, s'(x)^{-1}s'(x), \qquad \text{that is,} \qquad s = \partial^0 m \cdot s'.$$

In particular, the stabilizer of s in $\Gamma(M)$ is the subgroup of elements m such that $\partial^0 m = e_M$, namely $H^0(G, M)$. This already proves (iii).

(d) The argument is now familiar.[45] Let s_0 be an arbitrary section of v. There exists a section s, necessarily of the form $s = f \cdot s_0$, that is a group morphism, that is, satisfies $D_s = 0$, if and only if

$$(D_{s_0})^{-1} = \partial^1 f,$$

for some $f : G \to M$, or equivalently if and only if $c(E) = 0$. This proves (i).

In that case, the sections of (E) are precisely the sections $s' = h \cdot s$, where $h : G \to M$ satisfies $\partial^1 h = 0$, that is, $h \in Z^1(G, M)$. Moreover, by (c), two such sections $h_1 \cdot s$ and $h_2 \cdot s$ are conjugate under $\Gamma(M)$ if and only if h_1 and h_2 have the same image in $H^1(G, M)$. This proves (ii).

Still let

$$(E) \qquad 1 \longrightarrow M \xrightarrow{u} E \xrightarrow{v} G$$

be an exact sequence of $\widehat{\mathscr{C}}$-groups with M commutative, and let

$$f : H \longrightarrow G$$

be a morphism of $\widehat{\mathscr{C}}$-groups.

Consider $E_f = H \times_G E$. It is a $\widehat{\mathscr{C}}$-group, and the projection $v_f : E_f \to H$ is a morphism of $\widehat{\mathscr{C}}$-groups; the same holds for $e_f : E_f \to E$. Sending M into E by u and into H by the unit morphism also defines a morphism of $\widehat{\mathscr{C}}$-groups $u_f : M \to E_f$. We have therefore constructed the following commutative diagram of $\widehat{\mathscr{C}}$-groups.

$$\begin{tikzcd}
(\mathrm{E}) & 1 \arrow[r] & \mathrm{M} \arrow[r, "u"] & \mathrm{E} \arrow[r, "v"] & \mathrm{G} \\
(\mathrm{E}_f) & 1 \arrow[r] & \mathrm{M} \arrow[u, "\mathrm{id}"] \arrow[r, "u_f"] & \mathrm{E}_f \arrow[u, "e_f"] \arrow[r, "v_f"] & \mathrm{H} \arrow[u, "f"']
\end{tikzcd}$$

Figure 84: The pullback extension (E_f) and its morphism to (E).

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 27; combined-reader p. 137; printed p. 127.

It follows immediately that:

Lemma 1.2.3.

(i) *The sequence (E_f) is exact.*

[45] Editor's note: what follows has been expanded.

(ii) *The map $s \mapsto e_f \circ s = f'$ gives a bijective correspondence between the morphisms*

$$s : H \longrightarrow E_f$$

such that $v_f \circ s = \mathrm{id}$, that is, the sections of v_f, and the morphisms

$$f' : H \longrightarrow E$$

such that $v \circ f' = f$, that is, the morphisms f' lifting f.

(iii) *Under this correspondence, the sections of (E_f) correspond to the group morphisms f' lifting f.*

Applying Lemma 1.2.2 to the extension (E_f), and taking 1.2.3 into account, gives the following proposition, which formally contains Lemma 1.2.2.

Proposition 1.2.4. *Let*

$$(E) \qquad 1 \longrightarrow M \longrightarrow E \xrightarrow{v} G$$

be an exact sequence of $\widehat{\mathscr{C}}$-groups with M commutative. Let $f : H \to G$ be a morphism of $\widehat{\mathscr{C}}$-groups, and suppose that it lifts to a morphism, not necessarily a group morphism, $f' : H \to E$. Let H act on M by the composite morphism, which is multiplicative and independent of the choice of f',

$$H \xrightarrow{f'} E \xrightarrow{\mathrm{int}} \underline{\mathrm{Aut}}_{\widehat{\mathscr{C}}\text{-gr.}}(M).$$

(i) *The morphism f canonically defines a class $c(f) \in H^2(H, M)$, whose vanishing is necessary and sufficient for the existence of a morphism of $\widehat{\mathscr{C}}$-groups*

$$f' : H \longrightarrow E$$

lifting f.

(ii) *If $c(f) = 0$, the set of morphisms of $\widehat{\mathscr{C}}$-groups f' lifting f, modulo the action of the inner automorphisms defined by elements of $\Gamma(M)$ (that is, by elements $m \in \Gamma(E)$ such that $v(m) = e$), is principal homogeneous under $H^1(H, M)$.*

(iii) *If $f' : H \to E$ is a group morphism lifting f, the set of transforms of f' by the inner automorphisms defined by the elements of $\Gamma(M)$ is isomorphic to*

$$\Gamma(M)/\Gamma(M^H) = \Gamma(M)/H^0(H, M).$$

1.3. Extensions of group laws.

Consider the following situation: one has a morphism in $\widehat{\mathscr{C}}$,

$$(\dagger) \qquad p : X \longrightarrow Y,$$

and a commutative $\widehat{\mathscr{C}}$-group M acting on X, such that X is formally principal homogeneous over Y under M_Y.

If $g : Y \to Z$ is an arbitrary morphism in $\widehat{\mathscr{C}}$, then $g \circ p : X \to Z$ is invariant under M: for every $S \in \mathrm{Ob}\mathscr{C}$, the map $(g \circ p)(S)$ is invariant under the action of $M(S)$ on $X(S)$. Conversely, we shall assume that the following condition holds for $n = 1, 2, 3, 4$:

$(+)_n$: *Every morphism from X^n to M that is invariant under the action of M^n on X^n factors uniquely through $p^n : X^n \to Y^n$, where the superscript n denotes a Cartesian power.*

Lemma 1.3.1.

(i) *If h is a morphism from Y to M, the automorphism u_h of X, defined set-theoretically by $x \mapsto h(p(x)) \cdot x$, preserves the fibers of p and commutes with the operations of M on X.*[46] *Thus, for every $S \in \operatorname{Ob} \mathscr{C}$, $x \in X(S)$, and $m \in M(S)$, one has*

$$p\big(h(p(x)) \cdot x\big) = p(x), \qquad m \cdot h(p(x)) \cdot x = h(p(m \cdot x)) \cdot m \cdot x.$$

(ii) *This construction gives a bijective correspondence between morphisms from Y to M and automorphisms of X that preserve the fibers of p and commute with the operations of M.*

The first assertion is clear, since $p(m \cdot x) = p(x)$ and M is commutative. Conversely, an automorphism u of X preserving the fibers of p can be written set-theoretically as $x \mapsto g(x)x$, where g is a certain morphism from X to M. If u commutes with the operations of M, then g is invariant under M,[47] and the conclusion follows from condition $(+)_1$.

Suppose now, in addition, that a group law on Y and an action of Y on M are given, that is, a morphism of $\widehat{\mathscr{C}}$-groups

$$(\ddagger) \qquad f : Y \longrightarrow \underline{\operatorname{Aut}}_{\widehat{\mathscr{C}}\text{-gr.}}(M).$$

Definition 1.3.2. *A composition law on X,*

$$P : X \times X \longrightarrow X,$$

is called admissible *if it satisfies the following two conditions:*

(i) *P lifts the group law of Y; that is, the following diagram is commutative.*

$$\begin{tikzcd} X \times X \arrow[r, "P"] \arrow[d, "(p,p)"'] & X \arrow[d, "p"] \\ Y \times Y \arrow[r] & Y \end{tikzcd}$$

Figure 85: The compatibility of an admissible composition law with the group law on Y.

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 29; combined-reader p. 139; printed p. 129.

(ii) *For every $S \in \operatorname{Ob} \mathscr{C}$ and all $x, y \in X(S)$, $m, n \in M(S)$, one has*

$$(++) \qquad P(m \cdot x, n \cdot y) = m \cdot f(p(x))(n) \cdot P(x, y).$$

Proposition 1.3.3. *For a group law $*$ on X to be admissible, it is necessary and sufficient that the following four conditions hold:*

(i) *$p : X \to Y$ is a group morphism.*

(ii) *The morphism $i : M \to X$, defined by $i(m) = m \cdot e_X$, is a group isomorphism from M onto* $\ker(p)$*; in other words, set-theoretically,*

$$(m \cdot e_X) * (n \cdot e_X) = (mn) \cdot e_X.$$

[46] Editor's note: here and below, "commutes with p and the operations of M" has been replaced by "preserves the fibers of p and commutes with the operations of M." The equalities that follow have also been added.

[47] Editor's note: indeed, $mg(x)x = mu(x) = u(mx) = g(mx)mx$, and hence $g(mx) = g(x)$.

(iii) *One has* $m \cdot x = (m \cdot e_X) * x = i(m) * x$ *for every* $m \in M(S)$ *and* $x \in X(S)$.

(iv) *The inner automorphisms of* X *act on* $\ker(p)$ *according to the set-theoretic formula*

$$x * i(m) * x^{-1} = i\big(f(p(x))m\big).$$

The proof is immediate.

Lemma 1.3.4.[48] *Let* $h : Y \to M$ *be a morphism and let* u_h *be the automorphism* $x \mapsto h(p(x)) \cdot x$ *of* X *(cf. 1.3.1). Let* P *be an admissible composition law (respectively, an admissible group law) on* X, *and let* P' *be the composition law on* X *obtained from* P *by means of* u_h, *that is,*

$$P'(x, y) = u_h^{-1}\big(P(u_h(x), u_h(y))\big).$$

Then:

(i) P' *is an admissible composition law (respectively, an admissible group law).*

(ii) *For every* $x, y \in X(S)$, *with* $S \in \mathrm{Ob}\,\mathscr{C}$, *set* $v = p(x)$ *and* $w = p(y)$. *Then*

$$P'(x, y) = h(vw)^{-1} \cdot h(v) \cdot f(v)(h(w)) \cdot P(x, y) = (\partial^1 h)(p(x), p(y)) \cdot P(x, y).$$

Proof. One has $u_h^{-1} = u_{h^\vee}$, where $h^\vee : Y \to M$ is defined by $h^\vee(y) = h(y)^{-1}$. By 1.3.2(i) and (ii),

$$P(h(v) \cdot x, h(w) \cdot y) = h(v) \cdot f(v)(h(w)) \cdot P(x, y), \qquad p(P(x, y)) = vw,$$

and hence

$$P'(x, y) = h(vw)^{-1} \cdot h(v) \cdot f(v)(h(w)) \cdot P(x, y) = (\partial^1 h)(p(x), p(y)) \cdot P(x, y).$$

It follows immediately that P' satisfies conditions (i) and (ii) of 1.3.2.

Definition 1.3.5. *Two admissible composition laws obtained from one another by the procedure of 1.3.4 are called* equivalent.[49]

Proposition 1.3.6. *Suppose that an admissible composition law on* X *exists. Then:*

(i) *There exists a canonically determined class* $c \in H^3(Y, M)$, *whose vanishing is necessary and sufficient for the existence of an admissible associative composition law on* X.

(ii) *If* $c = 0$, *the set of admissible associative composition laws on* X *(respectively, the set of their equivalence classes) is principal homogeneous under* $Z^2(Y, M)$ *(respectively,* $H^2(Y, M)$*).*

The proof proceeds in several steps.

(a) Let P be an admissible composition law on X. Since P lifts the associative composition law of Y, there is a unique morphism $a : X^3 \to M$ such that

$$(*) \qquad P(x, P(y, z)) = a(x, y, z) \cdot P(P(x, y), z).$$

[48] Editor's note: the statement and its proof have been expanded with a view to step (e) in the proof of 1.3.6.

[49] Editor's note: this is indeed an equivalence relation, since $u_h^{-1} = u_{h^\vee}$.

Applying conditions 1.3.2(i) and (ii), one sees at once that a is invariant under the action of M^3 on X^3.[50] Condition $(+)_3$ therefore gives:

(1) There exists a unique morphism $DP : Y^3 \to M$ such that

$$P(x, P(y, z)) = DP(p(x), p(y), p(z)) \cdot P(P(x, y), z),$$

and P is associative if and only if $DP = 0$.

(b) Let us compute $P(P(P(x, y), z), t)$ step by step using the preceding formula. Put

$$p(x) = u, \qquad p(y) = v, \qquad p(z) = w, \qquad p(t) = h.$$

One obtains[51] the following pentagonal diagram, in which an arrow $a \xrightarrow{m} b$ means that $b = m \cdot a$.

$$\begin{tikzcd}
& \mathrm{P}(x, \mathrm{P}(y, \mathrm{P}(z, t))) \arrow[dl, "\mathrm{DP}(u,v,wh)"'] \arrow[dr, "f(u)(\mathrm{DP}(v,w,h))"] & \\
\mathrm{P}(\mathrm{P}(x, y), \mathrm{P}(z, t)) \arrow[d, "\mathrm{DP}(uv,w,h)"'] & & \mathrm{P}(x, \mathrm{P}(\mathrm{P}(y, z), t)) \arrow[d, "\mathrm{DP}(u,vw,h)"] \\
\mathrm{P}(\mathrm{P}(\mathrm{P}(x, y), z, t) & & \mathrm{P}(\mathrm{P}(x, \mathrm{P}(y, z)), t) \arrow[ll, "\mathrm{DP}(u,v,w)"']
\end{tikzcd}$$

Figure 86: The pentagonal comparison used to compute the associativity obstruction.

Temporary Loop-1 image; source locator: Polo–Gille Exp3-14oct24.pdf, component-local p. 30; combined-reader p. 140; printed p. 130.

Thus one finds

$$\begin{aligned} DP(u, v, w) \cdot DP(u, vw, h) \cdot f(u)(DP(v, w, h)) \\ \cdot DP(u, v, wh)^{-1} \cdot DP(uv, w, h)^{-1} = e_M, \end{aligned}$$

that is, $\partial^3 DP(u, v, w, h) = e_M$. Moreover, the left-hand side of the preceding formula can be expressed, by means of P and a, as the value at (x, y, z, t) of a certain morphism $X^4 \to M$.

It follows from the uniqueness assertion in $(+)_4$ that $\partial^3 DP$ and e_M, which factor the same morphism, are equal. Hence:

(2) DP is a cocycle; that is, $DP \in Z^3(Y, M)$.

(c) If P and P' are two admissible composition laws on X, there is a unique morphism

$$b : X^2 \longrightarrow M$$

[50] Editor's note: for $x, y, z \in X(S)$, set $u = p(x)$, $v = p(y)$, and $w = p(z)$. Then

$$\begin{aligned} P(mx, P(m'y, m''z)) &= P(mx, m'f(v)(m'')P(y, z)) \\ &= mf(u)(m')f(uv)(m'')P(x, P(y, z)). \end{aligned}$$

On the other hand, since $p(P(x, y)) = uv$, one also has

$$\begin{aligned} P(P(mx, m'y), m''z) &= P(mf(u)(m')P(x, y), m''z) \\ &= mf(u)(m')f(uv)(m'')P(P(x, y), z). \end{aligned}$$

Comparison of these equalities with $(*)$ gives $a(mx, m'y, m''z) = a(x, y, z)$.

[51] Editor's note: the diagram that follows has been added.

such that $P'(x,y) = b(x,y)P(x,y)$. Applying 1.3.2(ii) to P and P', one sees that b is invariant under M^2, and hence, by $(+)_2$:

(3) For every pair (P,P') of admissible composition laws there exists a unique morphism $d(P,P') : Y^2 \to M$ such that

$$P'(x,y) = d(P,P')(p(x),p(y)) \cdot P(x,y),$$

and the set of admissible composition laws is thereby made principal homogeneous under

$$\mathrm{Hom}(Y^2, M) = C^2(Y,M).$$

(d) Under the preceding conditions one has

$$(4) \qquad DP' - DP = \partial^2 d(P,P').$$

(e) The laws P and P' are equivalent if and only if there is a morphism

$$h \in C^1(Y,M) = \mathrm{Hom}(Y,M)$$

such that $d(P,P') = \partial^1 h$. This follows from the definition of equivalence and from 1.3.4(ii).

(f) It remains only to conclude. We seek an associative P', that is, one for which $DP' = e_M$. Now DP is a cocycle whose class in $H^3(Y,M)$ does not depend on the chosen admissible composition law P, by (3) and (4). This class is the required obstruction c. An appropriate P' can be chosen if and only if $c = 0$. Indeed, after choosing an arbitrary P, one must solve, by (1),

$$0 = DP' = DP + \partial^2 d(P,P'),$$

which is possible, by (3) and (4), if and only if $c = 0$. Again by (3) and (4), the set of associative laws P' is principal homogeneous under $Z^2(Y,M)$. By (e), the set of associative laws P' modulo equivalence is principal homogeneous under $H^2(Y,M)$.

2. Infinitesimal extensions of a morphism of group schemes

Retain the notation of §0. Let Y and X be two S-group functors, and let M be the kernel of the group homomorphism $p_X : X \to X^+$. Thus there is an exact sequence of S-group functors

$$1 \longrightarrow M \longrightarrow X \xrightarrow{p_X} X^+.$$

By the definition of X^+, there are isomorphisms

$$\mathrm{Hom}_S(Y, X^+) \xrightarrow{\sim} \mathrm{Hom}_{S_\mathscr{J}}(Y_\mathscr{J}, X_\mathscr{J}),$$
$$\mathrm{Hom}_{S\text{-gr}}(Y, X^+) \xrightarrow{\sim} \mathrm{Hom}_{S_\mathscr{J}\text{-gr}}(Y_\mathscr{J}, X_\mathscr{J}).$$

The map

$$\mathrm{Hom}_S(Y, p_X) : \mathrm{Hom}_S(Y,X) \longrightarrow \mathrm{Hom}_S(Y, X^+)$$

sends an S-morphism $f : Y \to X$ to the S-morphism $f^+ : Y \to X^+$ corresponding under these isomorphisms to the $S_\mathscr{J}$-morphism $f_\mathscr{J} : Y_\mathscr{J} \to X_\mathscr{J}$ obtained from f by base change. If M is commutative, Proposition 1.2.4 applies to this situation.

2.0.[52] In what follows we shall consider the case in which Y is flat over S, and X is an S-group functor of one of the following two types:

[52] Editor's note: The numbering 2.0 has been added for later references.

a X is an S-group scheme;

b $X = \underline{\mathrm{Aut}}_S(E)$, where E is an S-scheme of finite presentation over S.

Write (Flat)$/S$ for the category of S-schemes flat over S. In case a, respectively b, the S-group functor $M = \mathrm{Ker}(X \to X^+)$, its restriction L to (Flat)$/S$, and the actions on M of the inner automorphisms of X were calculated in 0.9, 0.5, and 0.10, respectively in 0.11, 0.14, and 0.12.

More explicitly, in case a, let L_0 be the S_0-functor in commutative groups defined, for every S_0-scheme T_0, by

$$\mathrm{Hom}_{S_0}(T_0, L_0) = \mathrm{Hom}_{\mathscr{O}_{T_0}}\left(\omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0}\right).$$

The group X_0 acts on this functor through its adjoint representation on $\omega^1_{X_0/S_0}$, and

$$L = \prod_{S_0/S} L_0; \qquad L(T) = L_0(T \times_S S_0)$$

for every S-scheme T.

In case b, let $\pi : E \to S$ be the structural morphism. Then L is the abelian-group functor on (Flat)$/S$ defined by

$$\mathrm{Hom}_S(T, L) = \Gamma\Big(T, \pi_* \mathrm{Hom}_{\mathscr{O}_E}(\Omega^1_{E/S}, \mathscr{J}\mathscr{O}_E) \otimes_{\mathscr{O}_S} \mathscr{O}_T\Big),$$

on which X, regarded as a functor on (Flat)$/S$, acts as described in 0.12. In either case there is an exact sequence of group functors on (Flat)$/S$:

$$1 \longrightarrow L \longrightarrow X \longrightarrow X^+. \tag{E}$$

Since Y is flat over S, Lemma 1.1.1 shows that the groups $H^i(Y, M)$ depend only on the restriction L of M to (Flat)$/S$. In case a, where $L = \prod_{S_0/S} L_0$, Lemma 1.1.2 gives isomorphisms

$$H^i(Y, L) \simeq H^i(Y_0, L_0).$$

Taking all this into account, Proposition 1.2.4 yields the following result.[53]

Theorem 2.1. *Let S be a scheme, and let $\mathscr{I} \supset \mathscr{J}$ be two quasi-coherent ideals satisfying $\mathscr{I}\mathscr{J} = 0$, defining the closed subschemes S_0 and $S_{\mathscr{J}}$. Let*

X be an S-group functor of type a or b, with L_0 and L as above;

Y be an S-group scheme flat over S, and $f_{\mathscr{J}} : Y_{\mathscr{J}} \to X_{\mathscr{J}}$ a morphism of $S_{\mathscr{J}}$-groups.

Then:

i The morphism $f_{\mathscr{J}}$ lifts to a morphism of S-groups $Y \to X$ if and only if the following two conditions hold:

i_1 $f_{\mathscr{J}}$ lifts to a morphism of S-functors $Y \to X$. By Proposition 1.2.4 this defines an action of Y on L, independent of the chosen lift. In case a, the resulting action of Y_0 on L_0 comes from $f_0 : Y_0 \to X_0$ and the adjoint action of X_0 on L_0.

[53] Editor's note: The definitions that appeared in the original statement of Theorem 2.1 and occupied most of original page 113 have been moved above.

i_2 *The canonically defined obstruction* $c(f_{\mathscr{J}}) \in H^2(Y, L)$ *vanishes. In case a, this group is isomorphic to* $H^2(Y_0, L_0)$.

(ii) *If the conditions in i hold, the set* $\mathcal{E}$ *of morphisms of* S*-group functors* $Y \to X$ *extending* $f_{\mathscr{J}}$ *is principal homogeneous under* $Z^1(Y, L)$*, isomorphic in case a to* $Z^1(Y_0, L_0)$*. Moreover, the quotient of* $\mathcal{E}$ *by the action of inner automorphisms of* X *defined by sections over* S *that induce the unit section of* $X_{\mathscr{J}}$ *over* $S_{\mathscr{J}}$ *is principal homogeneous under* $H^1(Y, L)$*, isomorphic in case a to* $H^1(Y_0, L_0)$.

(iii) *If* $f : Y \to X$ *is a morphism of* S*-group functors extending* $f_{\mathscr{J}}$*, the set of morphisms* $Y \to X$ *obtained from* f *by those inner automorphisms is isomorphic to*

$$\Gamma(L)/H^0(Y, L),$$

and in case a to $\Gamma(L_0)/H^0(Y_0, L_0)$.

Remark 2.1.1.[54] If $f, f' : Y \to X$ are morphisms of S-group functors extending $f_{\mathscr{J}}$, one obtains a cocycle

$$d(f, f') \in Z^1(Y, L) \quad \big(\simeq Z^1(Y_0, L_0) \text{ in case (a)}\big)$$

such that

$$f' = d(f, f') \cdot f. \tag{*}$$

[55] Write $\bar{d}(f, f')$ for the image of $d(f, f')$ in $H^1(Y, L)$, or equivalently in $H^1(Y_0, L_0)$ in case a.

Remark 2.2. Keep the preceding notation; in particular, Y is flat over S. In case b, formula (0.14.1) identifies L with the restriction to (Flat)$/S$ of

$$\mathbf{W}\Big(\pi_* \operatorname{Hom}_{\mathscr{O}_E}(\Omega^1_{E/S}, \mathscr{J}\mathscr{O}_E)\Big).$$

In case a, suppose in addition that X is locally of finite presentation over S. Formula (0.6.1) then identifies L with the restriction to (Flat)$/S$ of

$$\prod_{S_0/S} \mathbf{W}\Big(\operatorname{Hom}_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J})\Big).$$

In both cases the module to which $\mathbf{W}$ is applied is quasi-coherent, by EGA I, 9.1.1. Suppose further that Y is affine over S.[56] By I, 5.3, one obtains

$$H^i(Y, L) = H^i(Y_0, L_0) = H^i\Big(Y_0, \operatorname{Hom}_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J})\Big) \qquad \text{in case (a)},$$

$$H^i(Y, L) = H^i\Big(Y, \pi_* \operatorname{Hom}_{\mathscr{O}_E}(\Omega^1_{E/S}, \mathscr{J}\mathscr{O}_E)\Big) \qquad \text{in case (b)}.$$

Remark 2.3.

[54] Editor's note: This remark, analogous to 4.5.1, has been added to introduce the notation $d(f, f')$ and $\bar{d}(f, f')$, used in 4.38. Consequently, the part of 2.1(ii) concerning $\mathcal{E}$ itself was also added.

[55] Editor's note: We follow the sign conventions of the original. One could instead write $f' = d(f', f) \cdot f$, but then, in order to retain the equality $d_1(d(i, i')) = d(Y, i'(Y))$ in 4.27 for two immersions $i, i' : Y \hookrightarrow X$, the sign of the class $d(Y, Y')$ introduced in 4.5.1 would have to be changed. This would in turn alter the signs in 4.8, 4.14, and 4.17. We have preferred to retain all the signs of the original, which are correct.

[56] Editor's note: This hypothesis and the reference to I, 5.3 have been added.

(1) By 0.16 and 0.17, condition i_1 is automatic when Y is affine and

$$\begin{cases} X \text{ is smooth over } S, & \text{in case (a)},\\ E \text{ is smooth and affine over } S, & \text{in case (b)}.\end{cases}$$

(2) Under these hypotheses, with Y still flat over S as in 2.0, formulas 2.2(a) and (0.6.2) give in case *a*

$$H^i(Y, L) = H^i(Y_0, L_0) = H^i\Big(Y_0, \operatorname{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big),$$

and in case *b*, by (0.14.2), 1.1.2, and I, 5.3,

$$H^i(Y, L) = H^i\Big(Y_0, \pi_{0*} \operatorname{Hom}_{\mathscr{O}_{E_0}}(\Omega^1_{E_0/S_0}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{E_0})\Big).$$

[57]

We now state several corollaries for a diagonalizable S-group Y (I, 4.4). If S is affine, then, by I, 5.3.3, $H^n(Y, F) = 0$ for every $n > 0$ and every quasi-coherent $\mathscr{O}_S$-module F.

Corollary 2.4. *Let $S_0 \hookrightarrow S$ be a closed subscheme defined by a nilpotent ideal, and let Y be a diagonalizable S-group. Let*

a X be an S-group locally of finite presentation over S, or

b $X = \underline{\mathrm{Aut}}_S(E)$, where E is an S-scheme locally of finite presentation.

If $f : Y \to X$ is a morphism of S-groups whose base change $f_0 : Y_0 \to X_0$ is the unit morphism, then f is the unit morphism.

Indeed, the question is local on S, and in case *b* also local on E. We may therefore suppose that S is affine and, in case *b*, that E is of finite presentation over S. Introduce the closed subschemes $S_n \subset S$ defined by the powers of the ideal defining S_0. This reduces the assertion to the case in which S_0 is defined by a square-zero ideal, where it follows from Theorem 2.1 and Remark 2.2.

If f_0 is not required to be the unit morphism, one has the following statement.

Corollary 2.5. *Let S and S_0 be as in 2.4, with S affine. Let Y be a diagonalizable S-group, X an S-group functor, and $f_0 : Y_0 \to X_0$ a morphism of S_0-group functors.*

i [58]

(ii) *Suppose either that X is a smooth S-group, or that $X = \underline{\mathrm{Aut}}_S(E)$ with E smooth and affine over S. Then f_0 extends to a morphism of S-groups $Y \to X$, and any two extensions are conjugate by an inner automorphism of X defined by a section over S inducing the unit section of X_0 over S_0.*

[57] Editor's note: The preceding material has been added; see Editor's Note 31 in 0.14.

[58] Editor's note: The original stated: "Suppose one of the following two properties holds: *a* X is an S-group locally of finite presentation; *b* $X = \underline{\mathrm{Aut}}_S(E)$, where E is of finite presentation over S. Then f_0 extends to a morphism of S-groups $Y \to X$ if and only if it extends to a morphism of S-functors $Y \to X$." It added that this followed step by step from part (ii) of the theorem. That argument appears insufficient. For example, if $\mathscr{I}^3 = 0$ and $\mathscr{J} = \mathscr{I}^2$, and if $f : Y \to X$ is an S-functor morphism lifting f_0, then f_0 lifts to an $S_{\mathscr{J}}$-group morphism $g_{\mathscr{J}} : Y_{\mathscr{J}} \to X_{\mathscr{J}}$; but must $g_{\mathscr{J}}$ then lift to an S-functor morphism $g : Y \to X$? In any event, assertion i is not used later, where X is always assumed smooth over S.

Introduce S_n as above.[59] For (ii), a scheme smooth over S is necessarily locally of finite presentation. Thus, in case b, a smooth affine E/S is of finite presentation, so the hypothesis in 2.0(b) is indeed satisfied.

Condition i_1 of Theorem 2.1 is automatic by 0.16 and 0.17. Moreover, every section of X_{S_n} over S_n lifts to a section of $X_{S_{n+1}}$ over S_{n+1}: this follows from smoothness in case a, and from 0.17 in case b. Consequently, if f and f' are two lifts of f_0, one may arrange step by step that $f_n = f'_n$ by lifting the inner automorphism supplied by Theorem 2.1(ii). This proves the assertion.

The same argument, together with Remark 2.3, gives:

Corollary 2.6. *Let $\mathscr{I}$ be a nilpotent ideal of S defining S_0, let Y be an affine S-group flat over S, and let X be a smooth S-group.*

i If, for every $n \geq 0$,

$$H^2\Big(Y_0, \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{I}^{n+1}/\mathscr{I}^{n+2}\Big) = 0,$$

then every S_0-group morphism $f_0 : Y_0 \to X_0$ lifts to an S-group morphism $f : Y \to X$.

(ii) *If, for every $n \geq 0$,*

$$H^1\Big(Y_0, \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{I}^{n+1}/\mathscr{I}^{n+2}\Big) = 0,$$

then any two such lifts are conjugate by an inner automorphism of X defined by a section over S inducing the unit section of X_0 over S_0.

Lemma 2.7. *Let S be affine, G an affine S-group, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_S$-module, and $\mathscr{L}$ a locally free $\mathscr{O}_S$-module. Suppose G acts on $\mathscr{F}$ in the sense of Exposé I, hence acts on $\mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{L}$. Let Λ be the ring of S, and let L be the projective Λ-module defining $\mathscr{L}$. Then there is a canonical isomorphism*

$$H^*(G, \mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{L}) \simeq H^*(G, \mathscr{F}) \otimes_\Lambda L.$$

[60]

Let $\mathscr{A}$ be the $\mathscr{O}_S$-algebra $\mathscr{A}(G)$, and consider the complex $\mathscr{C}$ of quasi-coherent $\mathscr{O}_S$-modules

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathscr{F} \otimes \mathscr{A} \longrightarrow \mathscr{F} \otimes \mathscr{A} \otimes \mathscr{A} \longrightarrow \cdots.$$

[61]

By I, 5.3, $H^*(G, \mathscr{F})$, respectively $H^*(G, \mathscr{F} \otimes \mathscr{L})$, is the cohomology of $\Gamma(S, \mathscr{C})$, respectively $\Gamma(S, \mathscr{C} \otimes \mathscr{L})$. Since S is affine, EGA I, 1.3.12 gives

$$\Gamma(S, \mathscr{C} \otimes \mathscr{L}) \simeq \Gamma(S, \mathscr{C}) \otimes_\Lambda L.$$

Because L is projective, hence flat,

$$H^*(\Gamma(S, \mathscr{C}) \otimes_\Lambda L) \simeq H^*(\Gamma(S, \mathscr{C})) \otimes_\Lambda L,$$

which proves the lemma.

[59] Editor's note: What follows has been slightly modified and expanded from the original.

[60] Editor's note: Here $\mathscr{L}$ is given the trivial action of G.

[61] Editor's note: The original proof has been expanded and simplified. It also invoked the isomorphisms $H^n(\mathscr{C} \otimes \mathscr{L}) \simeq H^n(\mathscr{C}) \otimes \mathscr{L}$ and $H_n(\Gamma(-)) \simeq \Gamma(H_n(-))$, where $H^n(\mathscr{C})$ denotes the cohomology sheaf of the complex $\mathscr{C}$.

Using the lemma, Corollary 2.6 becomes:

Corollary 2.8. *Let S be affine and let $\mathscr{I}$ be a nilpotent ideal defining S_0. Suppose every $\mathscr{I}^{n+1}/\mathscr{I}^{n+2}$ is locally free on S_0. Let Y be an affine S-group flat over S, let X be a smooth S-group, and let $f_0 : Y_0 \to X_0$ be an S_0-group morphism.*

i *If $H^2(Y_0, \mathrm{Lie}(X_0/S_0)) = 0$, then f_0 lifts to an S-group morphism $Y \to X$.*

(ii) *If $H^1(Y_0, \mathrm{Lie}(X_0/S_0)) = 0$, then any two such lifts are conjugate by an inner automorphism of X defined by a section over S inducing the unit section of X_0 over S_0.*

Taking $Y = X$ gives:

Corollary 2.9. *Let S and S_0 be as above, and let X be a smooth affine S-group.*

i *If $H^1(X_0, \mathrm{Lie}(X_0/S_0)) = 0$, every endomorphism of X over S that induces the identity on X_0 is the inner automorphism defined by a section over S inducing the unit section of X_0 over S_0.*

(ii) *If $H^2(X_0, \mathrm{Lie}(X_0/S_0)) = 0$, every S_0-automorphism of X_0 extends to an S-automorphism of X.*[62]

Remark 2.10. The assertions involving H^1 have converses by Theorem 2.1. For example, let $S = I_{S_0}$ be the dual-number scheme over S_0 (II, 2.1), and let X be a flat S-group such that every automorphism over S inducing the identity over S_0 is an inner automorphism defined by a section over S inducing the unit section of X_0. Then

$$H^1(X_0, \mathrm{Lie}(X_0/S_0)) = 0.$$

[63]

Corollary 2.11. *Let $S, \mathscr{I}, \mathscr{J}$ be as in 2.1. Let Y be an S-group scheme flat over S, let X be an S-group scheme, and let $f : Y \to X$ be an S-group morphism. The set of morphisms from Y to X obtained from f by conjugation with elements $x \in X(S)$ inducing the unit of $X(S_{\mathscr{J}})$ is isomorphic to the quotient*

$$\mathcal{E} = \mathrm{Hom}_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J}) \big/ \mathrm{Hom}_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J})^{\mathrm{ad}(Y_0)}.$$

Here the second group consists of those $\mathscr{O}_{S_0}$-morphisms $\omega^1_{X_0/S_0} \to \mathscr{J}$ which, after every base change $S' \to S_0$, yield morphisms

$$\omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{S'} \longrightarrow \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{S'}$$

invariant under the action of $Y_0(S')$ on the first factor.

By Theorem 2.1(iii), this set is $\Gamma(L_0)/H^0(Y_0, L_0)$. Now

$$\Gamma(L_0) = \mathrm{Hom}_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J}),$$

and $H^0(Y_0, L_0)$ is precisely $\Gamma(L_0)^{\mathrm{ad}(Y_0)}$ in the sense just stated.

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[62] Editor's note: Let f_0 be an S_0-automorphism of X_0, with inverse g_0. By 2.8(i), lift f_0 and g_0 to S-endomorphisms f and g of X. Then $g \circ f$ and $f \circ g$ induce the identity on X_0, so by 0.7 they are S-automorphisms. Hence f and g are S-automorphisms as well.

[63] Editor's note: This is used in Exposé XXIV, 1.13.

Corollary 2.12. *Under the hypotheses of 2.11, suppose in addition that $\omega^1_{X_0/S_0}$ is locally free of finite rank. Then*

$$\mathcal{E} \simeq \Gamma\Big(S_0, \operatorname{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big) / H^0\Big(Y_0, \operatorname{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big).$$

Indeed,[64] local freeness of finite rank gives

$$\operatorname{Hom}_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J}) \simeq \operatorname{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}.$$

Corollary 2.13. *Suppose in addition that Y_0 is diagonalizable. Then*

$$\mathcal{E} \simeq \Gamma\Big(S_0, \operatorname{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big) / \Gamma\Big(S_0, \operatorname{Lie}(X_0/S_0)^{\operatorname{ad}(Y_0)} \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big),$$

where $\operatorname{Lie}(X_0/S_0)^{\operatorname{ad}(Y_0)}$ *is the factor of the decomposition in I, 4.7.3 corresponding to the zero character of Y_0.*

Indeed, if $Y_0 \simeq D_{S_0}(M)$, the cited result gives a direct-sum decomposition

$$\operatorname{Lie}(X_0/S_0) = \operatorname{Lie}(X_0/S_0)_0 \oplus \bigoplus_{\substack{m\in M\\ m\neq 0}} \operatorname{Lie}(X_0/S_0)_m.$$

Tensoring with $\mathscr{J}$ gives the analogous decomposition for $\operatorname{Lie}(X_0/S_0) \otimes \mathscr{J}$, whence

$$H^0(Y_0, \operatorname{Lie}(X_0/S_0) \otimes \mathscr{J}) \simeq \Gamma\Big(S_0, \operatorname{Lie}(X_0/S_0)_0 \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big).$$

Corollary 2.14. *Suppose in addition that S_0 is affine. Then*

$$\mathcal{E} \simeq \Gamma\Big(S_0, \Big[\operatorname{Lie}(X_0/S_0)/\operatorname{Lie}(X_0/S_0)^{\operatorname{ad}(Y_0)}\Big] \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big).$$

3. Infinitesimal extensions of a group scheme

Still using the notation of §0 ($S, \mathscr{I}, \mathscr{J}$, and so on), let X be an S-scheme, and suppose that $X_{\mathscr{J}}$ is endowed with a group structure. We propose to determine the S-group structures on X that induce the given structure on $X_{\mathscr{J}}$.

From now on, assume that X is flat over S. Let $\mathscr{C}$ be the category of S-schemes flat over S; thus $X \in \operatorname{Ob}\mathscr{C}$. Denote by Y, respectively M, the functor on $\mathscr{C}$ defined by X^+, respectively L'_X. The canonical morphism $p_X : X \to X^+$ defines a morphism in $\widehat{\mathscr{C}}$

$$p : X \longrightarrow Y,$$

and the action of L'_X on X in $\widehat{(\mathrm{Sch})/S}$ defines an action of M on X in $\widehat{\mathscr{C}}$. One checks at once that X is thereby formally principal homogeneous under M_Y over Y (cf. 0.2(i) and 0.4).

The action of X^+ on L'_X defined in 0.8 (denoted Ad there) defines an action, denoted $\widetilde{f}$, of Y on M. On the other hand, by 0.5,

$$\operatorname{Hom}_{\widehat{\mathscr{C}}}(Z, M) \simeq \operatorname{Hom}_{S_0}(Z_0, L_0), \qquad Z \in \operatorname{Ob}\mathscr{C},$$

where L_0 is the functor defined in 0.5.

Lemma 3.1.

[64] Editor's note: The following sentence has been added.

(i) *Condition* $(+)_n$ *of 1.3 holds for every positive integer* n.

(ii) *Let the* S_0*-group* X_0 *act on the* S_0*-functor* L_0 *through its adjoint representation. Then there is a canonical isomorphism*

$$H^*(X_0, L_0) \simeq H^*(Y, M),$$

where the first cohomology is computed in $(\mathrm{Sch})/S_0$*, and the second in* $\mathscr{C}$*.*

Both parts of the lemma follow from the relation

$$\begin{aligned}\mathrm{Hom}_{\widehat{\mathscr{C}}}(Y, M) &\simeq \mathrm{Hom}_{\widehat{(\mathrm{Sch})/S_0}}(X^+ \times_S S_0, L_0)\\ &\simeq \mathrm{Hom}_{S_0}(X_0, L_0)\\ &\simeq \mathrm{Hom}_{\widehat{\mathscr{C}}}(X, M),\end{aligned}$$

which follows at once from the definition of M as $\prod_{S_0/S} L_0$. More generally, the same relation holds with X, Y replaced by X^n, Y^n. Consequently, every morphism $X^n \to M$ factors uniquely through Y^n, which gives $(+)_n$. It also gives the relation $C^*(Y, M) = C^*(X_0, L_0)$, and hence (ii).

We may therefore apply the constructions of 1.3. In particular:

Lemma 3.2. *Let* $P : X \times_S X \to X$ *be a morphism. Then* P *induces the group law of* $X_{\mathscr{J}}$ *if and only if* P *is an admissible composition law on* X *(cf. 1.3.2).*

Indeed, P induces the group law of $X_{\mathscr{J}}$ if and only if it lifts the group law of X^+, or equivalently that of Y. Thus it remains only to show that every morphism P lifting the group law of $X_{\mathscr{J}}$ satisfies identity $((++))$ of 1.3.2(ii), which is exactly what was proved in 0.8.

Proposition 3.3. *Let* S *be a scheme and* S_0 *a closed subscheme defined by a nilpotent ideal. Let* X *be an* S*-scheme that is flat and either quasi-compact or locally of finite presentation over* S*. Let* $P : X \times_S X \to X$ *be a composition law on* X*. Then* P *is a group law if and only if the following two conditions hold:*

(i) P *is associative;*

(ii) P *induces a group law on* $X_0 = X \times_S S_0$*.*

These conditions are clearly necessary. We prove that they are sufficient. Suppose first that $X \to S$ has a section. Since $X(S')$ is then nonempty for every $S' \to S$, it suffices [65] to show that, for every $x \in X(S')$, left and right translation by x are isomorphisms of $X_{S'}$. [66]

[65] Editor's note: The original has been corrected by deleting the inappropriate reference to an exercise of Bourbaki on semigroups (cf. [BAlg], §I.2, Exercises 9–13) and by indicating the role of the left and right translations; see the following editor's note.

[66] Editor's note: Let E be a nonempty set endowed with an associative composition law such that every left translation ℓ_x is bijective, and fix $x_0 \in E$. There is a unique $e \in E$ such that $x_0 e = x_0$; then $x_0 e x = x_0 x$ implies $ex = x$ for every $x \in E$. Moreover, for every x there is a unique x' such that $xx' = e$. Suppose in addition that there is a $b \in E$ for which right translation r_b is injective. Then $xeb = xb$ gives $xe = x$, so e is a unit, and $xx'x = x = xe$ gives $x'x = e$; hence x' is both a left and a right inverse of x, and E is a group.

The hypothesis that r_b be injective is necessary: on any set E, define $xy = y$ for all $x, y \in E$. Every left translation is then the identity (and the law is associative), whereas $r_y(E) = \{y\}$ for every y; thus E is not a group when $|E| > 1$.

We may plainly assume $S' = S$. The translation in question, t, induces on X_0 a translation t_0, which is an automorphism since X_0 is a group. The conclusion follows by flatness (SGA 1 III, 4.2). [67]

Now drop the assumption that X has a section over S, and suppose that there exists $S' \to S$ such that $X_{S'}$ has a section over S'. By what has just been proved, $X_{S'}$ is then an S'-group; let e' be its unit section. The inverse image of e' under $\mathrm{pr}_i : S' \times_S S' \to S'$, $i = 1, 2$, is the unit section of $X_{S''}$ for the group law obtained by pulling $P_{S'}$ back along pr_i. But since P is "defined over S," these two group laws coincide, and so do their unit sections. Hence

$$\mathrm{pr}_1^*(e') = \mathrm{pr}_2^*(e').$$

If $S' \to S$ is a descent morphism (cf. Exposé IV, no. 2), there is a section of X whose base extension is e', and the proof is complete. Since X_X has a section over X, namely the diagonal section, it is now enough to prove that $X \to S$ is a descent morphism. It is flat and surjective, and is either quasi-compact or locally of finite presentation; it is therefore an (fpqc) covering, hence a descent morphism (Exposé IV, no. 6).

Remark. In fact, the hypothesis that $X \to S$ be quasi-compact or locally of finite presentation is superfluous, by the following result, which the reader may prove as an exercise on Exposé IV:

Under the stated hypotheses on S and S_0, if $X \to S$ is flat and $X_0 \to S_0$ is an (fpqc) covering, then $X \to S$ is a descent morphism.

Lemma 3.4. *Two admissible composition laws on X are equivalent (cf. 1.3.5) if and only if one is obtained from the other by an automorphism of X over S that induces the identity on $X_{\mathscr{J}}$.*

Indeed, the morphisms constructed in 1.3.1 are precisely those in the preceding statement, by 0.7. [68]

Taking account of all the preceding results, Proposition 1.3.6 gives:

Theorem 3.5. *Let S be a scheme, and let $\mathscr{I}$ and $\mathscr{J}$ be two ideals on S such that $\mathscr{I} \supset \mathscr{J}$ and $\mathscr{I}\mathscr{J} = 0$. Let S_0 and $S_{\mathscr{J}}$ be the closed subschemes of S that they define. Let X be an S-scheme flat over S (and either locally of finite presentation or quasi-compact over S), and let X_0 and $X_{\mathscr{J}}$ be the schemes obtained by base change. Suppose $X_{\mathscr{J}}$ is endowed with an $S_{\mathscr{J}}$-group structure, and let L_0 denote the S_0-functor in commutative groups defined by*

$$\mathrm{Hom}_{S_0}(T, L_0) = \mathrm{Hom}_{\mathscr{O}_T}\left(\omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T\right),$$

on which X_0 acts through its adjoint representation.

(i) *An S-group structure exists on X that induces the given structure on $X_{\mathscr{J}}$ if and only if the following conditions hold:*

(i_1) *there is a morphism of S-schemes $X \times_S X \to X$ inducing the group law of $X_{\mathscr{J}}$;*

[67] Editor's note: Since X and X_0 have the same underlying topological space and t_0 is an automorphism, t is a homeomorphism, hence an affine morphism; cf. Exposé $\mathrm{VI_B}$, 2.9.1, or EGA IV_4, 18.12.7.1. It therefore suffices to show the following. Let $\mathscr{J}$ be a nilpotent ideal of a ring Λ, and let $\varphi : A \to B$ be a morphism of Λ-algebras, with B flat over Λ, such that $\varphi \otimes_\Lambda (\Lambda/\mathscr{J})$ is bijective. Then φ is bijective. By the "nilpotent Nakayama lemma," φ is surjective; moreover, since B is flat over Λ, one also has $\mathrm{Ker}(\varphi) \otimes_\Lambda (\Lambda/\mathscr{J}) = 0$, whence $\mathrm{Ker}(\varphi) = 0$. Thus φ is bijective.

[68] Editor's note: Indeed, by the proof of 0.7, the S-endomorphisms of X inducing the identity on $X_{\mathscr{J}}$ are the automorphisms $m \cdot \mathrm{id}_X$, where m ranges over $M(X) = \mathrm{Hom}_S(X, M)$. For every $S' \to S$ and every $x \in X(S')$, one has $(m \cdot \mathrm{id}_X)(x) = m(x) \cdot x$. By the proof of 3.1, each $m : X \to M$ factors uniquely through a morphism $h : Y = X^+ \to M$; hence $m \cdot \mathrm{id}_X$ is the automorphism u_h introduced in 1.3.1. The lemma now follows from the definition of equivalence; cf. 1.3.4 and 1.3.5.

(i_2) *a certain obstruction class in* $H^3(X_0, L_0)$*, canonically defined by* X *and the group law of* $X_{\mathscr{J}}$*, is zero.*

(ii) *If the conditions in (i) hold, the set* $\mathcal{E}$ *of group laws on* X *inducing the given law on* $X_{\mathscr{J}}$ *is principal homogeneous under* $Z^2(X_0, L_0)$*. Moreover, the quotient of* $\mathcal{E}$ *by the* S*-automorphisms of* X *inducing the identity on* $X_{\mathscr{J}}$ *is principal homogeneous under* $H^2(X_0, L_0)$*.*

[69] Indeed, by 3.2 every morphism of S-schemes $f : X \times_S X \to X$ inducing the group law of $X_{\mathscr{J}}$ is an admissible composition law on X. By 1.3.6(i), the existence of an associative admissible composition law $P : X \times_S X \to X$ is equivalent to the vanishing of a certain class $c(f) \in H^3(X_0, L_0)$; in that case, 3.3 shows that P is a group law. This proves (i), and (ii) then follows from 3.3 and 1.3.6(ii).

Remark 3.5.1. [70] If μ and μ' are group laws on X inducing the given law on $X_{\mathscr{J}}$, one obtains a cocycle $\delta(\mu, \mu') \in Z^2(X_0, L_0)$. The chosen sign convention is $\mu' = \delta(\mu, \mu') \cdot \mu$; that is, for every $S' \to S$ and $x, y \in X(S')$,

$$\mu'(x, y) = \delta(\mu, \mu')(x_0, y_0) \cdot \mu(x, y).$$

[71]

Write $\bar{\delta}(\mu, \mu')$ for the image of $\delta(\mu, \mu')$ in $H^2(X_0, L_0)$. Finally, if X endowed with μ, respectively μ', is denoted simply by X, respectively X', write $\delta(X, X')$ in place of $\delta(\mu, \mu')$, and similarly for $\bar{\delta}(X, X')$.

Remark 3.6. Let $X_{\mathscr{J}}$ be a smooth affine $S_{\mathscr{J}}$-scheme. By 0.15, up to isomorphism there is a unique S-scheme X, smooth over S, whose reduction modulo $\mathscr{J}$ is $X_{\mathscr{J}}$. If $X_{\mathscr{J}}$ is endowed with an $S_{\mathscr{J}}$-group structure, condition (i_1) is automatic by 0.16. Moreover, 0.6 simplifies the definition of L_0 and gives:

Corollary 3.7. *Let* $S, \mathscr{I}, \mathscr{J}$ *be as in 3.1, and let* $X_{\mathscr{J}}$ *be a smooth affine* $S_{\mathscr{J}}$*-group.*

(i) *The set of smooth* S*-groups whose reduction modulo* $\mathscr{J}$ *is* $X_{\mathscr{J}}$*, up to isomorphism inducing the identity on* $X_{\mathscr{J}}$*, is either empty or principal homogeneous under*

$$H^2\Big(X_0, \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big).$$

(ii) *A smooth* S*-group whose reduction modulo* $\mathscr{J}$ *is* $X_{\mathscr{J}}$ *exists if and only if a certain obstruction in*

$$H^3\Big(X_0, \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big)$$

is zero.

As usual, one obtains the following corollaries.

Corollary 3.8. *Let* S *be a scheme and* S_0 *a closed subscheme defined by a nilpotent ideal* $\mathscr{I}$*. Let* X_0 *be a smooth affine* S_0*-group.*

[69] Editor's note: The following argument has been added.

[70] Editor's note: This remark, analogous to 4.5.1, has been added to introduce the notation $\delta(\mu, \mu')$, or $\delta(X, X')$, used in 4.38. Consequently, the part of 3.5(ii) concerning $\mathcal{E}$ itself has also been added.

[71] Editor's note: We follow the sign conventions of the original in order to have, in 4.38(5), the equality $\partial^1 \bar{d}(X, X') = \bar{\delta}(X, X')$; see also Editor's Note 54.

(i) *If, for every $n \geq 0$,*

$$H^2\Big(X_0, \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{I}^{n+1}/\mathscr{I}^{n+2}\Big) = 0,$$

then any two smooth S-groups whose reduction is X_0 are isomorphic by an isomorphism inducing the identity on X_0.

(ii) *If, for every $n \geq 0$,*

$$H^3\Big(X_0, \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{I}^{n+1}/\mathscr{I}^{n+2}\Big) = 0,$$

then there exists a smooth S-group whose reduction is X_0.

Corollary 3.9. *Let S be affine and S_0 a closed subscheme defined by a nilpotent ideal $\mathscr{I}$. Suppose that every $\mathscr{I}^{n+1}/\mathscr{I}^{n+2}$ is locally free on S_0. Let X_0 be a smooth affine S_0-group.*

(i) *If $H^2(X_0, \mathrm{Lie}(X_0/S_0)) = 0$, then any two smooth S-groups whose reduction is X_0 are isomorphic.*

(ii) *If $H^3(X_0, \mathrm{Lie}(X_0/S_0)) = 0$, then there exists a smooth S-group whose reduction is X_0.*

Corollary 3.10. *Let S_0 be a scheme and let $S = I_{S_0}$ be the dual-number scheme over S_0. Let X_0 be a smooth S_0-group. Then every smooth S-group Y for which Y_0 is S_0-isomorphic to X_0 is S-isomorphic to $X = X_0 \times_{S_0} S$ if and only if*

$$H^2(X_0, \mathrm{Lie}(X_0/S_0)) = 0.$$

[72]

Indeed, by 3.5 the set of classes of such groups Y, up to an S-group isomorphism inducing the identity on X_0, is in bijection with $H^2(X_0, \mathrm{Lie}(X_0/S_0))$. Hence their classes up to an arbitrary S-group isomorphism are in bijection with

$$H^2(X_0, \mathrm{Lie}(X_0/S_0))/\Gamma_0,$$

where

$$\Gamma_0 = \mathrm{Aut}_{S_0\text{-gr.}}(X_0),$$

which acts in the evident way on H^2. The conclusion follows at once. [73]

4. Infinitesimal extensions of closed subgroups

We first state a result valid in an arbitrary abelian category.

Lemma 4.1. *Let*

$$0 \longrightarrow A' \xrightarrow{i} A \xrightarrow{p} A'' \longrightarrow 0$$

be an exact sequence, let $\phi : A' \to Q$ be a morphism, and let $\pi : A'' \to P$ be an epimorphism with kernel C. Let E be the set, up to isomorphism, of quadruples (B, f, g, h) such that the sequence

$$0 \longrightarrow Q \xrightarrow{f} B \xrightarrow{g} P \longrightarrow 0$$

is exact and the following diagram commutes:

[72] Editor's note: This result is used in XXIV, 1.13.

[73] Editor's note: Indeed, $\mathrm{Aut}_{S_0\text{-gr.}}(X_0)$ acts by group automorphisms on the abelian group $H^2(X_0, \mathrm{Lie}(X_0/S_0))$, so the orbit of 0 is the singleton $\{0\}$. Consequently, the quotient set is a singleton if and only if $H^2(X_0, \mathrm{Lie}(X_0/S_0)) = \{0\}$.

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{A}' \arrow[r, "i"] \arrow[d, "\phi"'] & \mathrm{A} \arrow[r, "p"] \arrow[d, "h"'] & \mathrm{A}'' \arrow[r] \arrow[d, "\pi"] & 0 \\
0 \arrow[r] & \mathrm{Q} \arrow[r, "f"] & \mathrm{B} \arrow[r, "g"] & \mathrm{P} \arrow[r] & 0.
\end{tikzcd}$$

(i) *The set E is nonempty if and only if the image in* $\operatorname{Ext}^1(C, Q)$ *of the element A of* $\operatorname{Ext}^1(A'', A')$ *is zero.*

(ii) *Under these conditions, E is a principal homogeneous set under the abelian group* $\operatorname{Hom}(C, Q)$*.*

Introduce the amalgamated sum

$$B' = A \coprod_{A'} Q.$$

There is then a commutative diagram with exact rows: [74]

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{A}' \arrow[r, "i"] \arrow[d, "\phi"'] & \mathrm{A} \arrow[r, "p"] \arrow[d] & \mathrm{A}'' \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathrm{Q} \arrow[r, "j"] & \mathrm{B}' \arrow[r] & \mathrm{A}'' \arrow[r] & 0,
\end{tikzcd}$$

It is clear that the category of solutions to the problem is canonically isomorphic to the category of solutions to the corresponding problem for the sequence

$$0 \longrightarrow Q \longrightarrow B' \longrightarrow A'' \longrightarrow 0$$

and the morphisms id_Q and $\pi : A'' \to P$. [75] In this case, E is in bijection with the set of subobjects N of B' for which $B' \to A''$ induces an isomorphism of N with the kernel C of $A'' \to P$; equivalently, it is the set of morphisms $e : C \to B'$ lifting the canonical morphism $C \to A''$. The abelian group $G = \operatorname{Hom}(C, Q)$ acts on E by $g \cdot e = g + e$, with addition in $\operatorname{Hom}(C, B')$; if $E \neq \varnothing$, this makes E a principal homogeneous set under G.

We deduce the following.

Proposition 4.2. [76] *Let S be a scheme, let $S_{\mathscr{J}}$ be the closed subscheme defined by a quasi-coherent square-zero ideal $\mathscr{J}$, let X be an S-scheme, and let $\mathscr{F}$ be an $\mathscr{O}_X$-module. Put*

$$X_{\mathscr{J}} = X \times_S S_{\mathscr{J}}, \qquad \mathscr{F}_{\mathscr{J}} = \mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{O}_{S_{\mathscr{J}}},$$

and let $\mathscr{G}_{\mathscr{J}} = \mathscr{F}_{\mathscr{J}} / \mathscr{H}_{\mathscr{J}}$ be a quotient module of $\mathscr{F}_{\mathscr{J}}$. Suppose given a morphism of $\mathscr{O}_{X_{\mathscr{J}}}$-modules

$$f : \mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{G}_{\mathscr{J}} \longrightarrow \mathscr{Q}.$$

[74] Editor's note: One has $\operatorname{Coker}(j) = B' \coprod_{Q} 0 = A \coprod_{A'} 0 = A''$, and $\operatorname{Ker}(j) \simeq \operatorname{Ker}(i) = 0$, as one sees by reasoning "as if the category were a category of modules." For a proof entirely in terms of arrows, see for example [Fr64], Theorem 2.5.4 (*).

[75] Editor's note: In what follows, A has been replaced by B', and the end of the argument has been expanded.

[76] Editor's note: The statement has been rewritten so as to fit exactly the application made in 4.3; the proof has also been expanded, following indications supplied by M. Demazure.

Let $\mathcal{E}$ be the sheaf of sets on X defined as follows: for each open subset U of X, $\mathcal{E}(U)$ is the set of quotient modules $\mathscr{G}$ of $\mathscr{F}|_U$ such that $\mathscr{G}/\mathscr{J}\mathscr{G} = \mathscr{G}_{\mathscr{J}}|_U$ and for which there exists an isomorphism

$$h : \mathscr{J}\mathscr{G} \xrightarrow{\sim} \mathscr{Q}|_U$$

making the following diagram commute:

$$\begin{tikzcd}
\mathcal{J} \otimes_{\mathscr{O}_{S_{\mathcal{J}}}} (\mathscr{G}_{\mathcal{J}}|_{U}) \arrow[r, "f|_{U}"] \arrow[d, "\text{can.}"'] & \mathcal{Q}|_{U} \\
\mathcal{J}\mathscr{G} \arrow[ur, "h", "\simeq"'] &
\end{tikzcd}.$$

The isomorphism h is then unique, since $\mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} (\mathscr{G}_{\mathscr{J}}|_U) \to \mathscr{J}\mathscr{G}$ is an epimorphism. Then $\mathcal{E}$ is a sheaf formally principal homogeneous under the sheaf of commutative groups

$$\mathscr{A} = \mathrm{Hom}_{\mathscr{O}_X}(\mathscr{H}_{\mathscr{J}}, \mathscr{Q}) = \mathrm{Hom}_{\mathscr{O}_{X_{\mathscr{J}}}}(\mathscr{H}_{\mathscr{J}}, \mathscr{Q}).$$

Proof. If $\mathcal{E}(U) = \varnothing$, there is nothing to prove; we may therefore suppose that $\mathcal{E}(U)$ contains an element $\widetilde{\mathscr{G}}$. In the following diagram, h is an isomorphism and all the arrows are epimorphisms:

$$\begin{tikzcd}
\mathcal{J} \otimes_{\mathscr{O}_{S_{\mathcal{J}}}} (\mathscr{F}_{\mathcal{J}}|_{U}) \arrow[r] \arrow[d, "\text{can.}"'] & \mathcal{J} \otimes_{\mathscr{O}_{S_{\mathcal{J}}}} (\mathscr{G}_{\mathcal{J}}|_{U}) \arrow[r, "f|_{U}"] \arrow[d, "\text{can.}"'] & \mathcal{Q}|_{U} \\
\mathcal{J}\mathscr{F}|_{U} \arrow[r] & \mathcal{J}\widetilde{\mathscr{G}} \arrow[ur, "h", "\simeq"'] &
\end{tikzcd}.$$

Thus the morphism

$$\mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} (\mathscr{F}_{\mathscr{J}}|_U) \longrightarrow \mathscr{Q}|_U$$

induces a necessarily unique epimorphism $\phi : \mathscr{J}\mathscr{F}|_U \to \mathscr{Q}|_U$. If $\mathscr{G}$ is an $\mathscr{O}_U$-module such that $\mathscr{G}/\mathscr{J}\mathscr{G} = \mathscr{G}_{\mathscr{J}}|_U$ and there is a commutative diagram with exact rows

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{J}\mathscr{F}|_{U} \arrow[r] \arrow[d, "\phi"'] & \mathscr{F}|_{U} \arrow[r] \arrow[d] & \mathscr{F}_{\mathcal{J}}|_{U} \arrow[r] \arrow[d, "\pi"] & 0 \\
0 \arrow[r] & \mathcal{Q}|_{U} \arrow[r] & \mathscr{G} \arrow[r, "p_{\mathcal{J}}"] & \mathscr{G}_{\mathcal{J}}|_{U} \arrow[r] & 0
\end{tikzcd}$$

(where $p_{\mathscr{J}}$ is the projection $\mathscr{G} \to \mathscr{G}/\mathscr{J}\mathscr{G} = \mathscr{G}_{\mathscr{J}}|_U$, so that $\mathscr{Q}|_U = \mathrm{Ker}(p_{\mathscr{J}}) = \mathscr{J}\mathscr{G}$), then $\mathscr{G}$ may be identified with a quotient module of $\mathscr{F}|_U$. Consequently, by 4.1(ii), the set $\mathcal{E}(U)$ is principal homogeneous under the abelian group

$$\mathrm{Hom}_{\mathscr{O}_X}(\mathscr{H}_{\mathscr{J}}, \mathscr{Q})(U) = \mathrm{Hom}_{\mathscr{O}_{X_{\mathscr{J}}}}(\mathscr{H}_{\mathscr{J}}, \mathscr{Q})(U).$$

Proposition 4.3. *(TDTE IV 5.1) Let S be a scheme, let $S_{\mathscr{J}}$ be the closed subscheme defined by a quasi-coherent square-zero ideal $\mathscr{J}$, let X be an S-scheme, and let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-module. Put*

$$X_{\mathscr{J}} = X \times_S S_{\mathscr{J}}, \qquad \mathscr{F}_{\mathscr{J}} = \mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{O}_{S_{\mathscr{J}}}.$$

Let $\mathscr{G}_{\mathscr{J}} = \mathscr{F}_{\mathscr{J}}/\mathscr{H}_{\mathscr{J}}$ be a quasi-coherent quotient module of $\mathscr{F}_{\mathscr{J}}$, flat over $S_{\mathscr{J}}$. For each open subset U of X, let $\mathcal{E}(U)$ be the set of quasi-coherent quotient modules $\mathscr{G}$[77] of $\mathscr{F}|_U$, flat over S, such that $\mathscr{G}/\mathscr{J}\mathscr{G} \simeq \mathscr{G}_{\mathscr{J}}|_U$. Then the $\mathcal{E}(U)$ form a sheaf of sets $\mathcal{E}$ on X, formally principal homogeneous under the sheaf of commutative groups

$$\mathscr{A} = \mathrm{Hom}_{\mathscr{O}_{X_{\mathscr{J}}}}\left(\mathscr{H}_{\mathscr{J}}, \mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{G}_{\mathscr{J}}\right).$$

Proof. Let $\pi : X \to S$ be the structural morphism. Let U be an open subset of X, and let $\mathscr{G}$ be an $\mathscr{O}_U$-module flat over S, such that $\mathscr{G}/\mathscr{J}\mathscr{G} \simeq \mathscr{G}_{\mathscr{J}}|_U$. For every $x \in U$, the stalk $\mathscr{G}_x$ is a flat module over the local ring $\mathscr{O}_{S,s}$, where $s = \pi(x)$. Hence the morphism

$$\mathscr{J}_s \otimes_{\mathscr{O}_{S,s}} (\mathscr{G}/\mathscr{J}\mathscr{G})_x = \mathscr{J}_s \otimes_{\mathscr{O}_{S,s}} \mathscr{G}_x \longrightarrow (\mathscr{J}\mathscr{G})_x$$

is bijective. We therefore have an exact sequence

$$0 \longrightarrow \mathscr{J} \otimes_{\mathscr{O}_S} (\mathscr{G}_{\mathscr{J}}|_U) \longrightarrow \mathscr{G} \longrightarrow \mathscr{G}_{\mathscr{J}}|_U \longrightarrow 0.$$

Since $\mathscr{J} \otimes_{\mathscr{O}_S} (\mathscr{G}_{\mathscr{J}}|_U)$ and $\mathscr{G}_{\mathscr{J}}|_U$ are quasi-coherent $\mathscr{O}_U$-modules, $\mathscr{G}$ is quasi-coherent as well (EGA III, 1.4.17).

Conversely, since $\mathscr{G}_{\mathscr{J}}$ is flat over $S_{\mathscr{J}}$, if $\mathscr{G}$ is a quasi-coherent $\mathscr{O}_U$-module such that $\mathscr{G}/\mathscr{J}\mathscr{G} \simeq \mathscr{G}_{\mathscr{J}}$ and the morphism

$$\mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{G}_{\mathscr{J}} \longrightarrow \mathscr{J}\mathscr{G}$$

is bijective, then $\mathscr{G}$ is flat over S, by the "fundamental criterion of flatness" (SGA 1, IV, 5.5). [78]

Consequently, the set $\mathcal{E}(U)$ considered here is the set considered in 4.2 when f is the identity morphism of $\mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{G}_{\mathscr{J}}$; the conclusion therefore follows from 4.2. ☐

Retain the preceding notation. [79] Let $Y_{\mathscr{J}}$ be a closed subscheme of $X_{\mathscr{J}}$, defined by a quasi-coherent ideal $\mathscr{I}_{Y_{\mathscr{J}}}$, and suppose that $Y_{\mathscr{J}}$ is flat over $S_{\mathscr{J}}$. Applying 4.3 to $\mathscr{F} = \mathscr{O}_X$ and

$$\mathscr{G}_{\mathscr{J}} = \mathscr{O}_{Y_{\mathscr{J}}} = \mathscr{O}_{X_{\mathscr{J}}}/\mathscr{I}_{Y_{\mathscr{J}}},$$

we obtain the following corollary.

Corollary 4.3.1. *Let $S, S_{\mathscr{J}}, \mathscr{J}, X, X_{\mathscr{J}}, Y_{\mathscr{J}}$, and $\mathscr{I}_{Y_{\mathscr{J}}}$ be as above, and suppose that $Y_{\mathscr{J}}$ is flat over $S_{\mathscr{J}}$. Let $\mathscr{A}_{\mathscr{J}}$ be the sheaf of commutative groups*

$$\mathrm{Hom}_{\mathscr{O}_{X_{\mathscr{J}}}}\left(\mathscr{I}_{Y_{\mathscr{J}}}, \mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{O}_{Y_{\mathscr{J}}}\right)$$

[77] Editor's note: To lighten the statement, the hypothesis that $\mathscr{G}$ be quasi-coherent has been added here; the remark that this hypothesis is automatically satisfied has been moved to the proof, which has been expanded accordingly.

[78] Editor's note: See also [BAC], §III.5, Theorem 1.

[79] Editor's note: The discussion that follows has been expanded and Corollary 4.3.1 has been added. Recall also that "pseudo-torsor" is synonymous with "formally principal homogeneous."

on $X_{\mathscr{J}}$, and put $\mathscr{A} = i_*(\mathscr{A}_{\mathscr{J}})$, where $i : X_{\mathscr{J}} \hookrightarrow X$ is the immersion. For each open subset U of X, let $\mathcal{E}(U)$ be the set of closed subschemes Y of U, flat over S, such that

$$Y \times_S S_{\mathscr{J}} = Y_{\mathscr{J}} \cap U.$$

Then $\mathcal{E}$ is an $\mathscr{A}$-pseudo-torsor.

If, moreover, such a Y exists locally—that is, if every $x \in X$ has an open neighborhood U with $\mathcal{E}(U) \neq \varnothing$—then $\mathcal{E}$ is an $\mathscr{A}$-torsor. The $\mathscr{A}$-torsors on X are parametrized by

$$H^1(X, \mathscr{A}) = H^1(X_{\mathscr{J}}, \mathscr{A}_{\mathscr{J}})$$

(see, for example, EGA IV_4, 16.5.15), and $\mathcal{E}$ has a global section—that is, $\mathcal{E}(X) \neq \varnothing$—if and only if its corresponding cohomology class is zero. Thus we obtain:

Corollary 4.4. *Let $S, S_{\mathscr{J}}, \mathscr{J}, X, X_{\mathscr{J}}, Y_{\mathscr{J}}$, and $\mathscr{I}_{Y_{\mathscr{J}}}$ be as above, and suppose that $Y_{\mathscr{J}}$ is flat over $S_{\mathscr{J}}$. Let E be the set of closed subschemes Y of X, flat over S, such that $Y \times_S S_{\mathscr{J}} = Y_{\mathscr{J}}$.*

(i) *The set E is either empty or principal homogeneous under the abelian group*

$$H^0(X, \mathscr{A}) = H^0(X_{\mathscr{J}}, \mathscr{A}_{\mathscr{J}}) = \mathrm{Hom}_{\mathscr{O}_{X_{\mathscr{J}}}}\left(\mathscr{I}_{Y_{\mathscr{J}}}, \mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{O}_{Y_{\mathscr{J}}}\right).$$

(ii) *The set E is nonempty if and only if both of the following conditions hold:*

   (a) *The problem has a solution locally on X.*

   (b) *A certain obstruction vanishes; it lies in*

$$H^1\left(X_{\mathscr{J}}, \mathrm{Hom}_{\mathscr{O}_{X_{\mathscr{J}}}}\left(\mathscr{I}_{Y_{\mathscr{J}}}, \mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{O}_{Y_{\mathscr{J}}}\right)\right).$$

Complement 4.4.1. [80] Retain the notation of 4.4, and suppose that E contains an element Y. Let $\mathscr{I}_Y$ be the ideal of $\mathscr{O}_X$ defining Y, and let $\mathscr{I}_{Y_{\mathscr{J}}}$ be its image in $\mathscr{O}_{X_{\mathscr{J}}}$. As we saw in the proof of 4.2, there is a commutative diagram

$$\begin{tikzcd}
\mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{O}_{X_{\mathscr{J}}} \arrow[r, two heads] \arrow[d] & \mathscr{J}\mathscr{O}_X \arrow[d] \arrow[dl, dashed] \\
\mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{O}_{Y_{\mathscr{J}}} \arrow[r, "\sim"] & \mathscr{J}\mathscr{O}_Y
\end{tikzcd}$$

and hence an epimorphism of $\mathscr{O}_X$-modules

$$\phi : \mathscr{J}\mathscr{O}_X \longrightarrow \mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{O}_{Y_{\mathscr{J}}}.$$

Let $\mathscr{K}$ be its kernel. For every element Y' of E, the morphism $\mathscr{O}_X \to \mathscr{O}_{Y'}$ then factors through $\mathscr{O}_X/\mathscr{K}$, which is the amalgamated sum B' in the proof of Lemma 4.1. If $\mathscr{I}_{Y'}$ denotes the ideal of Y' in $\mathscr{O}_X$, there is a commutative diagram

[80] Editor's note: This complement has been added; it is useful in proving Proposition 4.8(ii).

$$\begin{tikzcd}
 & & 0 \arrow[d] & 0 \arrow[d] & \\
 & & \mathscr{I}_{\mathrm{Y}'}/\mathscr{K} \arrow[r, "\sim"] \arrow[d] & \mathscr{I}_{\mathrm{Y}_{\mathcal{J}}} \arrow[d] & \\
0 \arrow[r] & (\mathcal{J}\mathscr{O}_{\mathrm{X}})/\mathscr{K} \arrow[r] \arrow[d, "\wr"] & \mathscr{O}_{\mathrm{X}}/\mathscr{K} \arrow[r] \arrow[d] & \mathscr{O}_{\mathrm{X}_{\mathcal{J}}} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathcal{J} \otimes_{\mathscr{O}_{\mathrm{S}_{\mathcal{J}}}} \mathscr{O}_{\mathrm{Y}_{\mathcal{J}}} \arrow[r] \arrow[d] & \mathscr{O}_{\mathrm{Y}'} \arrow[r] \arrow[d] & \mathscr{O}_{\mathrm{Y}_{\mathcal{J}}} \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 & .
\end{tikzcd}$$

Replacing X by the closed subscheme defined by $\mathscr{K}$, we are thus reduced to the case $\mathscr{K} = 0$. The datum of Y' is then equivalent to that of the $\mathscr{O}_X$-submodule $\mathscr{I}_{Y'}$ of $\mathscr{O}_X$ which maps bijectively onto $\mathscr{I}_{Y_{\mathscr{J}}}$ under the projection $p : \mathscr{O}_X \to \mathscr{O}_{X_{\mathscr{J}}}$. Let

$$f' : \mathscr{I}_{Y_{\mathscr{J}}} \xrightarrow{\sim} \mathscr{I}_{Y'}, \qquad f : \mathscr{I}_{Y_{\mathscr{J}}} \xrightarrow{\sim} \mathscr{I}_{Y}$$

denote the respective inverse isomorphisms. Then $f' - f$ is an element of

$$\mathrm{Hom}_{\mathscr{O}_{X_{\mathscr{J}}}}\left(\mathscr{I}_{Y_{\mathscr{J}}}, \mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{O}_{Y_{\mathscr{J}}}\right) = \mathrm{Hom}_{\mathscr{O}_{X_{\mathscr{J}}}}(\mathscr{I}_{Y_{\mathscr{J}}}, \mathscr{J}\mathscr{O}_Y),$$

which we denote by $d(Y', Y)$. Notice that $d(Y, Y') = -d(Y', Y)$.

For our fixed Y and variable Y', consider the morphism

$$\mathscr{I}_{Y'} \longrightarrow \mathscr{O}_X \longrightarrow \mathscr{O}_Y = \mathscr{O}_X/\mathscr{I}_Y.$$

Since its composite with $\mathscr{O}_Y \to \mathscr{O}_{Y_{\mathscr{J}}}$ is zero, it takes values in

$$\mathscr{J}\mathscr{O}_Y = \mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{O}_{Y_{\mathscr{J}}}.$$

More precisely, let V be an open subset of X, let x' be a section of $\mathscr{I}_{Y'}$ on V, and let $x_{\mathscr{J}}$ be its image in $\Gamma(V, \mathscr{I}_{Y_{\mathscr{J}}})$. Then

$$x' = f'(x_{\mathscr{J}}) = f(x_{\mathscr{J}}) + (f' - f)(x_{\mathscr{J}}) = f(x_{\mathscr{J}}) + d(Y', Y)(x_{\mathscr{J}}).$$

Consequently, the morphism $\mathscr{I}_{Y'} \to \mathscr{J}\mathscr{O}_Y$ is given by $d(Y', Y)$.

4.5.0.[81] Retain the notation of 4.3.1 and 4.4, and make the following transformations. The quasi-coherent $\mathscr{O}_{X_{\mathscr{J}}}$-module

$$\mathscr{I}_{Y_{\mathscr{J}}}/\mathscr{I}_{Y_{\mathscr{J}}}^2$$

is annihilated by $\mathscr{I}_{Y_{\mathscr{J}}}$, and is therefore the direct image of a quasi-coherent $\mathscr{O}_{Y_{\mathscr{J}}}$-module, denoted by $\mathscr{N}_{Y_{\mathscr{J}}/X_{\mathscr{J}}}$, and called the *conormal sheaf* of $Y_{\mathscr{J}}$ in $X_{\mathscr{J}}$.[82] Since $\mathscr{J} \otimes_{\mathscr{O}_{S_{\mathscr{J}}}} \mathscr{O}_{Y_{\mathscr{J}}}$

[81] Editor's note: The numbering 4.5.0 has been added to mark the return to the original.

[82] Editor's note: The following sentence has been corrected.

is annihilated by $\mathscr{I}_{Y_\mathscr{J}}$, the sheaf of commutative groups $\mathscr{A}_\mathscr{J}$ of 4.3.1 identifies with

$$\begin{aligned}&\mathrm{Hom}_{\mathscr{O}_{Y_\mathscr{J}}}\left(\mathscr{I}_{Y_\mathscr{J}}/\mathscr{I}^2_{Y_\mathscr{J}}, \mathscr{J}\otimes_{\mathscr{O}_{S_\mathscr{J}}}\mathscr{O}_{Y_\mathscr{J}}\right)\\ &\qquad = \mathrm{Hom}_{\mathscr{O}_{Y_\mathscr{J}}}\left(\mathscr{N}_{Y_\mathscr{J}/X_\mathscr{J}}, \mathscr{J}\otimes_{\mathscr{O}_{S_\mathscr{J}}}\mathscr{O}_{Y_\mathscr{J}}\right),\end{aligned}$$

and hence, for every $i \geq 0$,

$$H^i(X_\mathscr{J}, \mathscr{A}_\mathscr{J}) = H^i\Big(Y_\mathscr{J}, \mathrm{Hom}_{\mathscr{O}_{Y_\mathscr{J}}}\left(\mathscr{N}_{Y_\mathscr{J}/X_\mathscr{J}}, \mathscr{J}\otimes_{\mathscr{O}_{S_\mathscr{J}}}\mathscr{O}_{Y_\mathscr{J}}\right)\Big).$$

[83] One may now drop the hypothesis that Y be closed, as follows. First observe that every open subset $U_\mathscr{J}$ of $X_\mathscr{J}$ is obtained by base change from the open subscheme U of X having the same underlying topological space as $U_\mathscr{J}$. Let $Y_\mathscr{J}$ now be a closed subscheme of $U_\mathscr{J}$, flat over $S_\mathscr{J}$, and let $\mathscr{I}_{Y_\mathscr{J}}$ be the quasi-coherent ideal of $\mathscr{O}_{U_\mathscr{J}}$ defining $Y_\mathscr{J}$. If $Y_\mathscr{J}$ lifts to a subscheme Y of X, then Y, having the same underlying topological space as $Y_\mathscr{J}$, is a closed subscheme of U. Consequently, the obstruction to lifting $Y_\mathscr{J}$ to a subscheme, flat over S, of X or of U is the same; it lies in

$$H^1\Big(Y_\mathscr{J}, \mathrm{Hom}_{\mathscr{O}_{Y_\mathscr{J}}}\left(\mathscr{N}_{Y_\mathscr{J}/X_\mathscr{J}}, \mathscr{J}\otimes_{\mathscr{O}_{S_\mathscr{J}}}\mathscr{O}_{Y_\mathscr{J}}\right)\Big).$$

Finally, return to the notation of §0. Let $\mathscr{I}$ be a quasi-coherent ideal of $\mathscr{O}_S$ such that $\mathscr{J}\subset\mathscr{I}$ and $\mathscr{I}\mathscr{J}=0$, and let S_0 be the closed subscheme of $S_\mathscr{J}$ defined by $\mathscr{I}$. For every S-scheme Z, write

$$Z_\mathscr{J} = Z\times_S S_\mathscr{J}, \qquad Z_0 = Z\times_S S_0.$$

Since $\mathscr{J}$ is annihilated by $\mathscr{I}$, the notation of 4.4 gives

$$\mathscr{J}\otimes_{\mathscr{O}_{S_\mathscr{J}}}\mathscr{O}_{Y_\mathscr{J}} = \mathscr{J}\otimes_{\mathscr{O}_{S_0}}\mathscr{O}_{Y_0},$$

and

$$\begin{aligned}&\mathrm{Hom}_{\mathscr{O}_{Y_\mathscr{J}}}\left(\mathscr{N}_{Y_\mathscr{J}/X_\mathscr{J}}, \mathscr{J}\otimes_{\mathscr{O}_{S_\mathscr{J}}}\mathscr{O}_{Y_\mathscr{J}}\right)\\ &\qquad = \mathrm{Hom}_{\mathscr{O}_{Y_0}}\left(\mathscr{N}_{Y_\mathscr{J}/X_\mathscr{J}}\otimes_{\mathscr{O}_{Y_\mathscr{J}}}\mathscr{O}_{Y_0}, \mathscr{J}\otimes_{\mathscr{O}_{S_0}}\mathscr{O}_{Y_0}\right),\end{aligned}$$

and similarly for the other terms. Thus one obtains:

Proposition 4.5. *Let S be a scheme, and let S_0 and $S_\mathscr{J}$ be the closed subschemes defined by quasi-coherent ideals $\mathscr{I}$ and $\mathscr{J}$, where $\mathscr{I}\supset\mathscr{J}$ and $\mathscr{I}\mathscr{J}=0$. Let X be an S-scheme and let $Y_\mathscr{J}$ be a subscheme of $X_\mathscr{J}$, flat over $S_\mathscr{J}$. Define the $\mathscr{O}_{Y_0}$-module*

$$\mathscr{A}_0 = \mathrm{Hom}_{\mathscr{O}_{Y_0}}\left(\mathscr{N}_{Y_\mathscr{J}/X_\mathscr{J}}\otimes_{\mathscr{O}_{Y_\mathscr{J}}}\mathscr{O}_{Y_0}, \mathscr{J}\otimes_{\mathscr{O}_{S_0}}\mathscr{O}_{Y_0}\right).$$

Then:

(i) *A subscheme Y of X, reducing to $Y_\mathscr{J}$ and flat over S, exists if and only if the following two conditions hold:*

(a) *Such a Y exists locally on X.*

(b) *A certain obstruction in $R^1\Gamma(Y_0,\mathscr{A}_0)$ vanishes.*[84]

[83] Editor's note: The discussion that follows has been expanded.

[84] Editor's note: Here $R^1\Gamma(Y_0,\mathscr{A}_0)$ denotes the "coherent" cohomology group $H^1(Y_0,\mathscr{A}_0)$ of the $\mathscr{O}_{Y_0}$-module $\mathscr{A}_0$, in order to distinguish it from the "Hochschild" cohomology groups $H^i(Y_0, M_0)$, where Y_0 is an S_0-group and M_0 an $\mathscr{O}_{S_0}$-module, which will be considered beginning in 4.16.

(ii) *Under these conditions, the set of subschemes Y satisfying the stated requirements is principal homogeneous under the abelian group $\Gamma(Y_0, \mathscr{A}_0)$.*

Remark 4.5.1.[85] It follows from 4.5(ii) that for every pair (Y, Y') of subschemes[86] of X, flat over S and reducing to $Y_{\mathscr{J}}$, there is a "deviation"

$$\begin{aligned} d(Y', Y) &\in \Gamma(Y_0, \mathscr{A}_0) \\ &= \mathrm{Hom}_{\mathscr{O}_{Y_0}}\left(\mathscr{N}_{Y_{\mathscr{J}}/X_{\mathscr{J}}} \otimes_{\mathscr{O}_{Y_{\mathscr{J}}}} \mathscr{O}_{Y_0}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}\right). \end{aligned}$$

With the sign convention adopted in 4.4.1, $d(Y', Y)$ corresponds to the morphism of $\mathscr{O}_X$-modules

$$\mathscr{I}_{Y'} \hookrightarrow \mathscr{O}_X \longrightarrow \mathscr{O}_Y,$$

which takes values in $\mathscr{J}\mathscr{O}_Y \simeq \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}$, and factors through $\mathscr{I}_{Y'_{\mathscr{J}}} = \mathscr{I}_{Y_{\mathscr{J}}}$, then through $\mathscr{N}_{Y_{\mathscr{J}}/X_{\mathscr{J}}}$.

Remark 4.6.[87] *If X is flat over S and $Y_{\mathscr{J}}$ is locally complete intersection in $X_{\mathscr{J}}$, then condition (a) is always satisfied, and every Y flat over S lifting $Y_{\mathscr{J}}$ is locally complete intersection in X. If, moreover, Y_0 is affine, condition (b) is also satisfied.*

Definition 4.6.1. *(cf. SGA 6, VII 1.1) Let B be a commutative ring, let $f : E \to B$ be a B-linear morphism, where E is a free B-module of finite rank d, and let $I = f(E)$. If a basis of E is chosen, f is given by a d-tuple $(f_1, \dots, f_d)$ of elements of B, and I is the ideal generated by the f_i. The* Koszul complex *$K_\bullet(f)$ is the graded B-module $\bigwedge_B^\bullet E$, equipped with the differential of degree -1*

$$x_1 \wedge \cdots \wedge x_i \longmapsto \sum_{j=1}^{i} (-1)^{j-1} f(x_j) x_1 \wedge \cdots \wedge \widehat{x_j} \wedge \cdots \wedge x_i.$$

Thus there is an augmented chain complex, with B/I in degree -1,

$$\cdots \longrightarrow \bigwedge^2 E \longrightarrow E \xrightarrow{f} B \longrightarrow B/I \longrightarrow 0,$$

which is exact in degree 0 by definition, since $I = f(E)$. The morphism f is said to be *regular* if $K_\bullet(f)$ is acyclic in degrees > 0, that is, if the augmented complex above is a resolution of $C = B/I$.

In that case, the proof of SGA 6, VII 1.2(b) shows that the C-modules I^n/I^{n+1}, for $n \in \mathbb{N}$, are free, with I/I^2 of rank d.

Definition 4.6.2. *(cf. SGA 6, VII 1.4) Let X be a scheme, Y a subscheme, and U an open subset of X such that Y is a closed subscheme of U, defined by the quasi-coherent ideal $\mathscr{I}_Y$.*

The subscheme Y is said to be *locally complete intersection* in X if $Y \hookrightarrow X$ is a regular immersion in the sense of SGA 6, VII 1.4; that is, if for every $y \in Y$ there exist an affine open neighborhood V of y in U, a finite free $\mathscr{O}_V$-module $\mathscr{E}$, and a regular morphism $f : \mathscr{E} \to \mathscr{O}_V$ with image $\mathscr{I}_Y|_V$, such that $K_\bullet(f)$ is a resolution of $\mathscr{O}_{Y \cap V}$.

[85] Editor's note: This remark has been placed here; it replaces Remark 4.7 of the original.

[86] Editor's note: "closed subschemes" has been corrected to "subschemes."

[87] Editor's note: We have retained, for reference, Remark 4.6 of the original, which did not include the definition of "locally complete intersection." It is followed by the "correct" definition, taken from SGA 6, VII 1.4 (and replacing that of EGA IV$_4$, 16.9.2), and by proofs of the three assertions made in the remark.

This implies that the immersion $Y \hookrightarrow X$ is locally of finite presentation and, by 4.6.1, that the conormal sheaf

$$\mathscr{N}_{Y/X} = \mathscr{I}_Y / \mathscr{I}_Y^2$$

is a finite locally free $\mathscr{O}_Y$-module.

Lemma 4.6.3.[88] *Let A be a ring, let J be a square-zero ideal of A, put $\overline{A} = A/J$, let B be a flat A-algebra, let E be a free B-module of finite rank, and let $f : E \to B$ be a morphism of B-modules. Suppose that the morphism*

$$g : \overline{E} = E \otimes_A \overline{A} \longrightarrow \overline{B} = B \otimes_A \overline{A}$$

induced by f is regular and that $\overline{B}/g(\overline{E})$ is flat over $\overline{A}$. Then f is regular and $B/f(E)$ is flat over A.

Proof. Put

$$C = B/f(E), \qquad \overline{C} = C \otimes_A \overline{A} = \overline{B}/g(\overline{E}).$$

First, the $\bigwedge_B^i E$ are free B-modules, and hence flat A-modules, since B is flat over A. Since

$$\bigwedge_B^{\bullet} E \otimes_A \overline{A} \simeq \bigwedge_{\overline{B}}^{\bullet} \overline{E},$$

there is an exact sequence of complexes

$$0 \longrightarrow J \otimes_A \bigwedge_B^{\bullet} E \longrightarrow \bigwedge_B^{\bullet} E \longrightarrow \bigwedge_{\overline{B}}^{\bullet} \overline{E} \longrightarrow 0.$$

Moreover, since $J^2 = 0$, for every A-module M one has

$$J \otimes_A M \simeq J \otimes_{\overline{A}} (\overline{A} \otimes_A M).$$

Writing the augmentation arrows as dashed arrows, and denoting the rank of E by d, one obtains the following bicomplex, whose rows are exact.

[88] Editor's note: To prove the results stated in Remark 4.6, Lemmas 4.6.3 and 4.6.4, Proposition 4.6.5, and Remark 4.6.6 have been added.

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & \mathrm{J} \otimes_{\overline{\mathrm{A}}} \bigwedge^d_{\overline{\mathrm{B}}} \overline{\mathrm{E}} \arrow[r] \arrow[d] & \bigwedge^d_{\mathrm{B}} \mathrm{E} \arrow[r] \arrow[d] & \bigwedge^d_{\overline{\mathrm{B}}} \overline{\mathrm{E}} \arrow[r] \arrow[d] & 0 \\
 & \vdots \arrow[d] & \vdots \arrow[d] & \vdots \arrow[d] & \\
0 \arrow[r] & \mathrm{J} \otimes_{\overline{\mathrm{A}}} \overline{\mathrm{E}} \arrow[r] \arrow[d, "\mathrm{id} \otimes g"'] & \mathrm{E} \arrow[r] \arrow[d, "f"] & \overline{\mathrm{E}} \arrow[r] \arrow[d, "g"] & 0 \\
0 \arrow[r] & \mathrm{J} \otimes_{\overline{\mathrm{A}}} \overline{\mathrm{B}} \arrow[r] \arrow[d, dashed] & \mathrm{B} \arrow[r] \arrow[d, dashed] & \overline{\mathrm{B}} \arrow[r] \arrow[d, dashed] & 0 \\
 & \mathrm{J} \otimes_{\overline{\mathrm{A}}} \overline{\mathrm{C}} \arrow[r] \arrow[d] & \mathrm{C} \arrow[r] \arrow[d] & \overline{\mathrm{C}} \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 & .
\end{tikzcd}$$

The right and left columns are also exact, since $K_\bullet(g)$ is a resolution of $\overline{C}$, and $\overline{C}$ is flat over $\overline{A}$. The long exact homology sequence associated with the exact sequence of unaugmented complexes therefore shows that $K_\bullet(f)$ is acyclic in degrees > 0, and in degree 0 gives an exact sequence

$$0 \longrightarrow J \otimes_{\overline{A}} \overline{C} \longrightarrow C \longrightarrow \overline{C} \longrightarrow 0.$$

It follows from the "fundamental criterion of flatness" that C is flat over A (cf. [BAC], §III.5, Theorem 1). □

Lemma 4.6.4.[88] *Let A be a commutative ring, let J be a nilpotent ideal, and let $N \subset M$ be A-modules such that M/N is flat over A. If $x_1, \ldots, x_n$ are elements of N whose images generate the image $\overline{N}$ of N in M/JM, then they generate N.*

Indeed, let N' be the submodule of N generated by the x_i, and put $Q = N/N'$. The morphism $N' \otimes_A (A/J) \to \overline{N}$ is surjective. On the other hand, since M/N is flat over A, the morphism $N \otimes_A (A/J) \to \overline{N}$ is bijective. Thus $Q \otimes_A (A/J) = 0$, whence $Q = 0$ by the "nilpotent Nakayama lemma":

$$Q = JQ = J^2Q = \cdots = 0.$$

We can now prove the following proposition.

Proposition 4.6.5.[88] *Let $S, \mathscr{I}, \mathscr{J}$, and $X, Y_{\mathscr{J}}$ be as in 4.5. Suppose in addition that X is flat over S and that $Y_{\mathscr{J}}$ is locally complete intersection in $X_{\mathscr{J}}$.*

(a) *Condition (a) of 4.5(i) is then satisfied; moreover, every Y flat over S lifting $Y_{\mathscr{J}}$ is locally complete intersection in X.*

(b) *If, in addition, Y_0 is affine, condition (b) of the cited result is also satisfied.*

Proof. The first assertion of (a) follows from Lemma 4.6.3, and the second then follows from Lemma 4.6.4. Moreover, the hypothesis implies (cf. 4.6.2) that $\mathscr{N}_{Y_{\mathscr{J}}/X_{\mathscr{J}}}$ is a finite locally free $\mathscr{O}_{Y_{\mathscr{J}}}$-module. Hence the $\mathscr{O}_{Y_0}$-module

$$\mathscr{A}_0 = \mathrm{Hom}_{\mathscr{O}_{Y_0}}\left(\mathscr{N}_{Y_{\mathscr{J}}/X_{\mathscr{J}}} \otimes_{\mathscr{O}_{Y_{\mathscr{J}}}} \mathscr{O}_{Y_0}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}\right)$$

is quasi-coherent (cf. EGA I, 1.3.12), and therefore $R^1\Gamma(Y_0, \mathscr{A}_0) = 0$ when Y_0 is affine. □

Remark 4.6.6.[88] Let us conclude this paragraph with an example showing that, under the hypotheses of Lemma 4.6.3, a regular sequence $(\overline{g}_1, \overline{g}_2)$ generating the ideal $\overline{I} = g(\overline{E})$ need not lift to a regular sequence in B.

Let k be a field, put $\overline{A} = k[X, Y]$, let $k\varepsilon$ denote the $\overline{A}$-module $\overline{A}/(X, Y)$, so that

$$P\varepsilon = P(0,0)\varepsilon \qquad (P \in \overline{A}),$$

and let $A = \overline{A} \oplus k\varepsilon$, where $J = k\varepsilon$ is a square-zero ideal. Thus $A/J = \overline{A}$.

The algebra $B = A \otimes_k k[Z, T]$ is free, and hence flat, over A, and $\overline{B} = k[X, Y, Z, T]$. Put

$$\overline{g}_1 = XZ - YT, \qquad \overline{g}_2 = XZ - 1.$$

Since $\overline{g}_1$ is irreducible, $\overline{B}/(\overline{g}_1)$ is an integral domain; hence $(\overline{g}_1, \overline{g}_2)$ is a regular sequence in $\overline{B}$, generating the ideal

$$\overline{I} = (XZ - 1, YT - 1).$$

Therefore

$$\overline{C} = \overline{B}/\overline{I} = k[X, Y, X^{-1}, Y^{-1}] = \overline{A}[X^{-1}, Y^{-1}]$$

is a flat $\overline{A}$-algebra, and also a flat A-algebra. But every lift of $\overline{g}_1$ to B has the form

$$XZ - YT + \lambda\varepsilon, \qquad \lambda \in k[Z, T],$$

and therefore annihilates ε.

Source note. The Polo–Gille PDF prints $XY - ZT + \lambda\varepsilon$ for this lift. Since the element being lifted is $\overline{g}_1 = XZ - YT$, the displayed lift above restores that same polynomial before adding $\lambda\varepsilon$.

4.7. Remark 4.7 has been suppressed here and placed in 4.5.1.

Remark 4.8.0.[89] *Let S be a scheme, S' a closed subscheme of S, X an S-scheme, Y an S-subscheme of X, and put $X' = X \times_S S'$ and $Y' = Y \times_S S'$. Then there is a surjective morphism of $\mathscr{O}_{Y'}$-modules*

$$\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \twoheadrightarrow \mathscr{N}_{Y'/X'}.$$

Indeed, after replacing X by a suitable open subscheme, one may suppose that Y is closed and is defined by an ideal $\mathscr{I}_Y$ of $\mathscr{O}_X$. The image of $\mathscr{I}_Y$ in $\mathscr{O}_{X'}$ is then the ideal $\mathscr{I}_{Y'}$ defining Y', and there is a surjective morphism of $\mathscr{O}_{Y'}$-modules

$$\pi : (\mathscr{I}_Y/\mathscr{I}_Y^2) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \twoheadrightarrow \mathscr{I}_{Y'}/\mathscr{I}_{Y'}^2.$$

[89] Editor's note: This remark, used in the proposition that follows, has been inserted here; it appeared in 4.10 of the original.

Suppose in addition that $\mathscr{O}_Y = \mathscr{O}_X/\mathscr{I}_Y$ is flat over $\mathscr{O}_S$. The natural morphism

$$\mathscr{I}_Y \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \longrightarrow \mathscr{I}_{Y'}$$

is then bijective (cf. EGA IV$_2$, 2.1.8).

There is therefore the following commutative diagram with exact rows:

$$\begin{tikzcd}
 & \mathscr{I}^2_{\mathrm{Y}} \otimes_{\mathscr{O}_{\mathrm{X}}} \mathscr{O}_{\mathrm{X}'} \arrow[r] \arrow[d, "\text{surj.}"'] & \mathscr{I}_{\mathrm{Y}} \otimes_{\mathscr{O}_{\mathrm{X}}} \mathscr{O}_{\mathrm{X}'} \arrow[r] \arrow[d, "\wr"] & (\mathscr{I}_{\mathrm{Y}}/\mathscr{I}^2_{\mathrm{Y}}) \otimes_{\mathscr{O}_{\mathrm{Y}}} \mathscr{O}_{\mathrm{Y}'} \arrow[r] \arrow[d, "\pi"', "\text{surj.}"] & 0 \\
0 \arrow[r] & \mathscr{I}^2_{\mathrm{Y}'} \arrow[r] & \mathscr{I}_{\mathrm{Y}'} \arrow[r] & \mathscr{I}_{\mathrm{Y}'}/\mathscr{I}^2_{\mathrm{Y}'} \arrow[r] & 0
\end{tikzcd}$$

It follows from the snake lemma that[90]

$$\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \xrightarrow{\sim} \mathscr{N}_{Y'/X'} \qquad \text{if } Y \text{ is flat over } S. \tag{4.8.0}$$

Proposition 4.8. *Let $S, S_0, S_{\mathscr{J}}$, and $\mathscr{I}, \mathscr{J}$ be as in 4.5.*[91] *Let X be an S-scheme, let Y be a subscheme of X, and let $i : Y \hookrightarrow X$ be the immersion.*

(i) *For every S-morphism $f : T \to X$ such that $f_{\mathscr{J}} : T_{\mathscr{J}} \to X_{\mathscr{J}}$ factors through $Y_{\mathscr{J}}$, one can define an obstruction*

$$c(X, Y, f) \in \operatorname{Hom}_{\mathscr{O}_{T_0}}\big(f_0^*\big(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}\big), \mathscr{J}\mathscr{O}_T\big) \tag{*}$$

whose vanishing is equivalent to the existence of a factorization of f through Y.

(ii) *Let Y' be a second subscheme of X. Suppose that $Y'_{\mathscr{J}} = Y_{\mathscr{J}}$ and that Y and Y' are flat over S. Then there are isomorphisms (cf. 4.8.0)*

$$\mathscr{J}\mathscr{O}_Y \simeq \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0} \simeq \mathscr{J}\mathscr{O}_{Y'}, \qquad \mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_{\mathscr{J}}} \xrightarrow{\sim} \mathscr{N}_{Y_{\mathscr{J}}/X_{\mathscr{J}}},$$

and hence an isomorphism

$$\begin{aligned} u :\ &\operatorname{Hom}_{\mathscr{O}_{Y_0}}\Big(\mathscr{N}_{Y_{\mathscr{J}}/X_{\mathscr{J}}} \otimes_{\mathscr{O}_{Y_{\mathscr{J}}}} \mathscr{O}_{Y_0}, \mathscr{J}\mathscr{O}_Y\Big) \\ &\xrightarrow{\sim} \operatorname{Hom}_{\mathscr{O}_{Y_0}}\big(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}, \mathscr{J}\mathscr{O}_{Y'}\big). \end{aligned}$$

Writing $i' : Y' \hookrightarrow X$ for the canonical immersion and $d(Y, Y')$ for the deviation of 4.5.1, one has[92]

$$c(X, Y, i') = u\big(d(Y, Y')\big). \tag{**}$$

(iii) *The canonical morphism*

$$\mathscr{N}_{Y/X} \xrightarrow{D} i^*(\Omega^1_{X/S})$$

[90] Editor's note: In the original, this was stated in Remark 4.10 under the additional hypothesis that Y' be locally complete intersection in X'. Consequently, that hypothesis also appeared in statements 4.12–4.14; it seems in fact superfluous, and has been deleted from those statements.

[91] Editor's note: The hypothesis that $\mathscr{I}$ be nilpotent has been deleted, since it appears to be superfluous (cf. the proof).

[92] Editor's note: See also 4.27 below.

(cf. SGA 1, II, formula 4.3)[93] *induces a morphism*

$$D_0 : \mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0} \longrightarrow \Omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{X_0}} \mathscr{O}_{Y_0},$$

and hence, for every S-morphism $f : T \to X$ as in (i), a morphism

$$\begin{aligned} v_{f_0} :\ & \mathrm{Hom}_{\mathscr{O}_{T_0}}\Big(f_0^*(\Omega^1_{X_0/S_0}), \mathscr{J}\mathscr{O}_T\Big) \\ & \longrightarrow \mathrm{Hom}_{\mathscr{O}_{T_0}}\big(f_0^*\big(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}\big), \mathscr{J}\mathscr{O}_T\big), \\ & \qquad a \longmapsto a \circ f_0^*(D_0), \end{aligned}$$

where the first group above is $\mathrm{Hom}_{X^+}(T, L_X)$*, cf. 0.1.5. For $a \in \mathrm{Hom}_{X^+}(T, L_X)$, one has*

$$c(X, Y, a \cdot f) - c(X, Y, f) = v_{f_0}(a), \tag{***}$$

where $a \cdot f$ denotes the composite morphism

$$T \xrightarrow{a \times f} L_X \times_{X^+} X \longrightarrow X.$$

We shall prove part (i) of the proposition, leaving it to the reader to (not) verify assertions (ii) and (iii); this verification proceeds by reduction to the affine case and then by comparison of the explicit definitions.[94]

Let us therefore prove (i). The morphism $f : T \to X$ defines a morphism of sheaves of rings[95]

$$\phi : \mathscr{O}_X \longrightarrow f_*(\mathscr{O}_T).$$

Let U be an open subscheme of X in which Y is closed. Since T, respectively $Y_{\mathscr{J}}$, has the same underlying space as $T_{\mathscr{J}}$, respectively Y, the continuous map underlying f sends T into U. Since U is open in X, the morphism ϕ induces a morphism of sheaves of rings

$$\mathscr{O}_U = \mathscr{O}_X|_U \longrightarrow f_*(\mathscr{O}_T),$$

that is, f factors through U.

We may therefore restrict to the case in which Y is closed, hence defined by a sheaf of ideals $\mathscr{I}_Y$. The morphism f factors through Y if and only if the composite

$$\mathscr{I}_Y \longrightarrow \mathscr{O}_X \longrightarrow f_*(\mathscr{O}_T)$$

is zero. Since $f_{\mathscr{J}}$ factors through $Y_{\mathscr{J}}$, the composite

$$\mathscr{I}_{Y_{\mathscr{J}}} \longrightarrow \mathscr{O}_{X_{\mathscr{J}}} \longrightarrow f_*(\mathscr{O}_{T_{\mathscr{J}}})$$

is zero. Consider the following commutative diagram, whose first row is exact:

[93] Editor's note: See also EGA IV$_4$, 16.4.21. Recall that if U is an affine open subset of X such that $Y \cap U$ is defined by the ideal I of $A = \mathscr{O}_X(U)$, if d denotes the differential $A \to \Gamma(U, \Omega^1_{X/S})$, and if $x \in I$, then $D(x + I^2)$ is the element $d(x) \otimes 1$ of $\Gamma(U, \Omega^1_{X/S}) \otimes_A (A/I)$.

[94] Editor's note: These verifications have been carried out below.

[95] Editor's note: On the one hand, the hypothesis that $\mathscr{J}$ be nilpotent—that is, that X_0 have the same underlying topological space as X—has been deleted; on the other hand, the sentence that follows has been expanded.

$$
\begin{tikzcd}
0 \arrow[r] & f_*(\mathcal{J}\mathscr{O}_{\mathrm{T}}) \arrow[r] & f_*(\mathscr{O}_{\mathrm{T}}) \arrow[r] & f_*(\mathscr{O}_{\mathrm{T}_{\mathcal{J}}}) \\
 & & \mathscr{O}_{\mathrm{X}} \arrow[u, "\phi"] \arrow[r] & \mathscr{O}_{\mathrm{X}_{\mathcal{J}}} \arrow[u, "\phi_{\mathcal{J}}"'] \\
 & & \mathscr{I}_{\mathrm{Y}} \arrow[u] \arrow[r] \arrow[uul, dashed, "\phi"] & \mathscr{I}_{\mathrm{Y}_{\mathcal{J}}} \arrow[u]
\end{tikzcd}
$$

It follows that ϕ maps $\mathscr{I}_Y$ into $f_*(\mathscr{J}\mathscr{O}_T)$.[96] Since $\mathscr{J}^2 = 0$, the $\mathscr{O}_X$-module $f_*(\mathscr{J}\mathscr{O}_T)$, with its $\mathscr{O}_X$-module structure induced by ϕ, is annihilated by $\mathscr{I}_Y$. Consequently, ϕ induces a morphism of $\mathscr{O}_X$-modules

$$h : i_*(\mathscr{N}_{Y/X}) = \mathscr{I}_Y/\mathscr{I}_Y^2 \longrightarrow f_*(\mathscr{J}\mathscr{O}_T).$$

On the other hand, one has the following cartesian squares:

$$
\begin{tikzcd}
\mathrm{T}_0 \arrow[r, "f_0"] \arrow[d, "\tau_{\mathrm{T}_0}"] & \mathrm{X}_0 \arrow[d, "\tau_{\mathrm{X}_0}"] & \mathrm{Y}_0 \arrow[l, "i_0"'] \arrow[d, "\tau_{\mathrm{Y}_0}"] \\
\mathrm{T} \arrow[r, "f"] & \mathrm{X} & \mathrm{Y}. \arrow[l, "i"']
\end{tikzcd}
$$

Here $\tau_{T_0}, \tau_{X_0}, \tau_{Y_0}$ are the closed immersions obtained by base change from $S_0 \hookrightarrow S$. Since $\mathscr{J}\mathscr{O}_T$ is a quasi-coherent $\mathscr{O}_T$-module annihilated by $\mathscr{I}$, there is an isomorphism

$$\mathscr{J}\mathscr{O}_T \simeq (\tau_{T_0})_*\tau_{T_0}^*(\mathscr{J}\mathscr{O}_T),$$

and hence

$$f_*(\mathscr{J}\mathscr{O}_T) \simeq (\tau_{X_0})_*(f_0)_*\tau_{T_0}^*(\mathscr{J}\mathscr{O}_T).$$

Thus, by adjunction, h corresponds to a morphism of $\mathscr{O}_{T_0}$-modules

$$h_0 : f_0^*\tau_{X_0}^* i_*(\mathscr{N}_{Y/X}) \longrightarrow \tau_{T_0}^*(\mathscr{J}\mathscr{O}_T).$$

Source notation note. The Polo–Gille prose and this h_0-display print i_{T_0} for the closed immersion labeled τ_{T_0} in the cartesian square and the preceding base-change formulas. We use τ_{T_0} consistently.

Now

$$\tau_{X_0}^* i_*(\mathscr{N}_{Y/X}) \simeq (i_0)_*\tau_{Y_0}^*(\mathscr{N}_{Y/X}) = \mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}.$$

Returning to the customary abuse of notation $\tau_{T_0}^*(\mathscr{J}\mathscr{O}_T) = \mathscr{J}\mathscr{O}_T$, the morphism h_0 identifies with a morphism of $\mathscr{O}_{T_0}$-modules

$$h_0 : f_0^*\big(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}\big) \longrightarrow \mathscr{J}\mathscr{O}_T,$$

which is the required obstruction $c(X, Y, f)$. This proves (i).

[96] Editor's note: The argument that follows has been expanded.

When f is the immersion $i' : Y' \hookrightarrow X$, one sees that $c(X, Y, i')$ comes from the morphism

$$\mathscr{I}_Y \longhookrightarrow \mathscr{O}_X \longrightarrow \mathscr{O}_{Y'},$$

and hence corresponds, by 4.4.1 and 4.5.1, to the class $d(Y, Y')$. This proves (ii).

Source notation note. At this point the Polo–Gille proof prints $i' : Y \hookrightarrow X$, while part (ii) defines i' as the immersion of Y'. The latter domain is used here.

Let us prove (iii). First, $D : \mathscr{N}_{Y/X} \to i^*(\Omega^1_{X/S})$ induces a morphism

$$\begin{aligned} D_0 : \quad \tau^*_{Y_0}(\mathscr{N}_{Y/X}) &\longrightarrow \tau^*_{Y_0} i^*(\Omega^1_{X/S}) \\ &= i_0^* \tau^*_{X_0}(\Omega^1_{X/S}). \end{aligned}$$

Since $X_0 = X \times_S S_0$, one has

$$\tau^*_{X_0}(\Omega^1_{X/S}) \simeq \Omega^1_{X_0/S_0} \qquad \text{(cf. EGA IV}_4\text{, 16.4.5).}$$

This gives the asserted morphism

$$D_0 : \mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0} \longrightarrow \Omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{X_0}} \mathscr{O}_{Y_0}.$$

We shall verify equality $({*}{*}{*})$ after the following remark.

Remark 4.9. The obstruction $c(X, Y, f)$ is computed locally on T. Let $U = \operatorname{Spec}(C)$ be an affine open subset of T lying over an affine open subset $\operatorname{Spec}(A)$ of X, itself lying over an affine open subset $\operatorname{Spec}(\Lambda)$ of S. Let $J \subset I \subset \Lambda$, respectively $I_Y \subset A$, be the ideals corresponding to $\mathscr{J} \subset \mathscr{I}$, respectively $\mathscr{I}_Y$, put $B = A/I_Y$, and let $\phi : A \to C$ be the morphism of Λ-algebras corresponding to $f : T \to X$. Since $f(T_{\mathscr{J}}) \subset Y_{\mathscr{J}}$, one has $\phi(I_Y) \subset JC$, and ϕ therefore induces a morphism of Λ-algebras

$$B \longrightarrow C/JC \longrightarrow C_0 = C/IC.$$

The obstruction $c = c(X, Y, f)$ is then computed by the following commutative diagram:

$$\begin{tikzcd} \mathrm{I_Y} \arrow[r] \arrow[d] & \mathrm{A} \arrow[r, "\phi"] & \mathrm{C} \\ \mathrm{I_Y/I_Y^2} \arrow[r] & \mathrm{I_Y/I_Y^2 \otimes_B C_0} \arrow[r, "c"] & \mathrm{JC} \arrow[u] \end{tikzcd} \quad ,$$

That is, over U, it is the unique element of

$$\operatorname{Hom}_{C_0}\left(I_Y/I_Y^2 \otimes_B C_0, JC\right) = \operatorname{Hom}_{B_0}\left(I_Y/I_Y^2 \otimes_B B_0, JC\right)$$

such that, with the evident notation,

$$c(\overline{x} \otimes_B 1) = \phi(x) \qquad \text{for every } x \in I_Y.$$

[97] We can now complete the proof of 4.8(iii). Equality $({*}{*}{*})$ may be checked locally on T, so we are reduced to the affine situation described above. Let $d_{A/\Lambda} : A \to \Omega^1_{A/\Lambda}$ denote the differential. Over U, the element a corresponds to an element a_U of

$$\begin{aligned} &\operatorname{Hom}_{C_0}\left(\Omega^1_{A_0/\Lambda_0} \otimes_{A_0} C_0, JC\right) \\ &\qquad \simeq \operatorname{Hom}_{B_0}\left(\Omega^1_{A/\Lambda} \otimes_A B_0, JC\right) \simeq \operatorname{Hom}_A(\Omega^1_{A/\Lambda}, JC). \end{aligned}$$

[97]Editor's note: What follows has been added.

On the one hand, $v_{f_0}(a)$ corresponds over U to $a_U \circ D_0$, where D_0 is the morphism of B-modules[98]

$$\begin{aligned} I_Y/I_Y^2 &\longrightarrow \Omega^1_{A/\Lambda} \otimes_A B_0, \\ x + I_Y^2 &\longmapsto d_{A/\Lambda}(x) \otimes 1. \end{aligned}$$

On the other hand (cf. the proofs of 0.1.8 and 0.1.9), the morphism of Λ-algebras $\phi' : A \to C$ corresponding to $a \cdot f$ differs from ϕ by the Λ-derivation $A \to JC$ associated with a_U; that is,

$$\phi' = \phi + a_U \circ d_{A/\Lambda} = \phi + a_U \circ (d_{A/\Lambda} \otimes 1).$$

Consequently, writing $c' = c(X, Y, a \cdot f)$, for every $x \in I_Y$, and writing $\overline{x}$ for its image in I_Y/I_Y^2, one has

$$\begin{aligned} (c' - c)(\overline{x} \otimes 1) &= a_U\big(d_{A/\Lambda}(x) \otimes 1\big) \\ &= (a_U \circ D_0)(\overline{x}) = v_{f_0}(a)(\overline{x}). \end{aligned}$$

This proves that $c' - c = v_{f_0}(a)$.

4.10. Remark 4.10 of the original has been suppressed, since it was made obsolete by the addition of Remark 4.8.0.

4.11. We now propose to study the following situation. Let S, $S_{\mathscr{J}}$, and S_0 be as in 4.8. We have three S-schemes X, X', T, a subscheme Y of X (respectively a subscheme Y' of X'), and morphisms $f : T \to X'$ and $g : X' \to X$.

$$\begin{tikzcd} & \mathrm{Y}' \arrow[d, hook, "i'"'] & \mathrm{Y} \arrow[d, hook, "i"'] \\ \mathrm{T} \arrow[r, "f"] & \mathrm{X}' \arrow[r, "g"] & \mathrm{X} \end{tikzcd} \quad .$$

We suppose that, after reduction modulo $\mathscr{J}$, this diagram can be completed to a commutative diagram

$$\begin{tikzcd} & \mathrm{Y}'_{\mathcal{J}} \arrow[r, dashed] \arrow[d, hook, "i'_{\mathcal{J}}"'] & \mathrm{Y}_{\mathcal{J}} \arrow[d, hook, "i_{\mathcal{J}}"'] \\ \mathrm{T}_{\mathcal{J}} \arrow[ur, dashed] \arrow[r, "f_{\mathcal{J}}"] & \mathrm{X}'_{\mathcal{J}} \arrow[r, "g_{\mathcal{J}}"] & \mathrm{X}_{\mathcal{J}} \end{tikzcd} \quad .$$

By 4.8 we therefore have the obstructions

$$\begin{aligned} c(X, Y, g \circ i') &\in \mathrm{Hom}_{\mathscr{O}_{Y'_0}}\big(i_0'^* g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), \mathscr{J}\mathscr{O}_{Y'}\big), \\ c(X', Y', f) &\in \mathrm{Hom}_{\mathscr{O}_{T_0}}\big(f_0^*(\mathscr{N}_{Y'/X'} \otimes_{\mathscr{O}_{Y'}} \mathscr{O}_{Y'_0}), \mathscr{J}\mathscr{O}_T\big), \\ c(X, Y, g \circ f) &\in \mathrm{Hom}_{\mathscr{O}_{T_0}}\big(f_0^* g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), \mathscr{J}\mathscr{O}_T\big), \end{aligned}$$

[98] Editor's note: Cf. Editor's Note 93.

whose relations we seek to compute. [99]

Lemma 4.12. *Suppose that Y' is flat over S, so that*

$$\mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y'_0} = \mathscr{J}\mathscr{O}_{Y'}.$$

Then:

(i) *There is a natural morphism*

$$\begin{aligned} b_{f_0} : \ &\mathrm{Hom}_{\mathscr{O}_{Y'_0}}\big(i_0'^* g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), \mathscr{J}\mathscr{O}_{Y'}\big) \\ &\longrightarrow \mathrm{Hom}_{\mathscr{O}_{T_0}}\big(f_0^* g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), \mathscr{J}\mathscr{O}_T\big). \end{aligned}$$

(ii) *There is also a natural morphism, functorial in T,* [100]

$$\begin{aligned} a_{g_0}(f_0) : \ &\mathrm{Hom}_{\mathscr{O}_{T_0}}\big(f_0^*(\mathscr{N}_{Y'/X'} \otimes_{\mathscr{O}_{Y'}} \mathscr{O}_{Y'_0}), \mathscr{J}\mathscr{O}_T\big) \\ &\longrightarrow \mathrm{Hom}_{\mathscr{O}_{T_0}}\big(f_0^* g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), \mathscr{J}\mathscr{O}_T\big). \end{aligned}$$

Proof. [101] First note that, with X, X', Y, Y' fixed, giving a T as above is equivalent to giving a morphism $(f, f_{\mathscr{J}}) : T \to X' \times_{X'^+} Y'^+$. Put $Z = X' \times_{X'^+} Y'^+$, and let M and M' be the Z-functors defined, for every $f : T \to Z$, by

$$\begin{aligned} M(T) &= \mathrm{Hom}_{\mathscr{O}_{T_0}}\big(f_0^* g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), \mathscr{J}\mathscr{O}_T\big), \\ M'(T) &= \mathrm{Hom}_{\mathscr{O}_{T_0}}\big(f_0^*(\mathscr{N}_{Y'/X'} \otimes_{\mathscr{O}_{Y'}} \mathscr{O}_{Y'_0}), \mathscr{J}\mathscr{O}_T\big). \end{aligned}$$

[102]

In all cases there is a commutative diagram

$$\begin{tikzcd} f_0^*(\mathcal{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y'_0}) \arrow[r, equal] \arrow[d] & \mathcal{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0} \arrow[d] \\ f_0^*(\mathcal{J}\mathscr{O}_{Y'}) \arrow[r, dashed] & \mathcal{J}\mathscr{O}_T \end{tikzcd}$$

Since Y' is flat over S, the left vertical arrow is an isomorphism, and hence we obtain a morphism of $\mathscr{O}_{T_0}$-modules

$$f_0^*(\mathscr{J}\mathscr{O}_{Y'}) \longrightarrow \mathscr{J}\mathscr{O}_T.$$

[99]Editor's note: Starting with 4.17, this will be applied when X is an S-group, $g : X \times_S X \to X$ is multiplication, Y is a subscheme of X such that $Y_{\mathscr{J}}$ is a subgroup of $X_{\mathscr{J}}$, $Y' = Y \times_S Y$, and the two morphisms $Y^3 \to X^2$ send (y_1, y_2, y_3) respectively to $(y_1 y_2, y_3)$ and $(y_1, y_2 y_3)$. In that case, comparison of the obstructions above will show that the obstruction to Y being a subgroup of X lies in a certain Hochschild cohomology group $H^2(Y_0, N_0)$.

[100]Editor's note: We have removed the hypothesis that Y'_0 be locally complete intersection in X'_0, which is superfluous by 4.8.0. We have also added that $a_{g_0}(f_0)$ is "functorial in T", a fact that plays a crucial role in the proof of 4.17.

[101]Editor's note: The proof has been expanded to make the "functoriality in T" of a_{g_0} explicit.

[102]Editor's note: The situation simplifies from 4.16 onward: we shall restrict to schemes flat over S, Y will be a flat S-group, and $Y' = Y \times_S Y$; this will give S_0-functors N_0 and N'_0.

It induces a morphism of abelian groups

$$\mathrm{Hom}_{\mathscr{O}_{T_0}}\big(f_0^* g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), f_0^*(\mathscr{J}\mathscr{O}_{Y'})\big) \longrightarrow M(T),$$

and, on composing with the morphism

$$M(Y') \longrightarrow \mathrm{Hom}_{\mathscr{O}_{T_0}}\big(f_0^* g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), f_0^*(\mathscr{J}\mathscr{O}_{Y'})\big)$$

induced by f_0^*, we obtain the morphism $b_{f_0} : M(Y') \to M(T)$.

Likewise, in all cases there is a diagram

$$\begin{tikzcd}
g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}) \arrow[r, dashed] \arrow[d] & \mathscr{N}_{Y'/X'} \otimes_{\mathscr{O}_{Y'}} \mathscr{O}_{Y'_0} \arrow[d] \\
g_0^*(\mathscr{N}_{Y_0/X_0}) \arrow[r] & \mathscr{N}_{Y'_0/X'_0}
\end{tikzcd}$$

Since Y' is flat over S, the second vertical arrow is an isomorphism by 4.8.0. We therefore obtain an $\mathscr{O}_{Y'_0}$-morphism

$$i_0'^* g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}) \longrightarrow \mathscr{N}_{Y'/X'} \otimes_{\mathscr{O}_{Y'}} \mathscr{O}_{Y'_0}.$$

This induces a morphism $a_{g_0}(\mathrm{id}_{Y'}) : M'(Y') \to M(Y')$ and, for every $f : T \to Z$, a morphism $a_{g_0}(f) : M'(T) \to M(T)$, such that the following diagram commutes:

$$\begin{tikzcd}
\mathrm{M}'(\mathrm{Y}') \arrow[r, "a_{g_0}(\mathrm{id}_{\mathrm{Y}'_0})"] \arrow[d, "b'_{f_0}"'] & \mathrm{M}(\mathrm{Y}') \arrow[d, "b_{f_0}"] \\
\mathrm{M}'(\mathrm{T}) \arrow[r, "a_{g_0}(f_0)"] & \mathrm{M}(\mathrm{T})
\end{tikzcd}$$

Here b'_{f_0} is defined in the same way as b_{f_0}. Q.E.D.

Remark 4.12.1. [103] *Let M_0 and M'_0 be the Y'_0-functors defined, for every $f : T_0 \to Y'_0$, by*

$$M_0(T) = \mathrm{Hom}_{\mathscr{O}_{T_0}}\big(f_0^* g_0^*(\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0}), \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0}\big),$$
$$M'_0(T) = \mathrm{Hom}_{\mathscr{O}_{T_0}}\big(f_0^*(\mathscr{N}_{Y'/X'} \otimes_{\mathscr{O}_{Y'}} \mathscr{O}_{Y'_0}), \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0}\big).$$

Note at once that $Z_0 = Y'_0$, and that on the category of Z-schemes T flat over S, the functors M and M' coincide respectively with $\prod_{S_0/S} M_0$ and $\prod_{S_0/S} M'_0$. In that case, b_{f_0} is simply the morphism

$$f_0^* : M_0(Y'_0) \longrightarrow M(T_0)$$

induced by f_0. Moreover, for every morphism $u : U \to T$, if $h = f \circ u$, the following diagram commutes:

[103] Editor's note: This remark has been added for use in the proof of 4.17.

$$\begin{tikzcd}
\mathrm{M}'_0(\mathrm{T}_0) \arrow[r, "a_{g_0}(f_0)"] \arrow[d, "u_0^*"'] & \mathrm{M}_0(\mathrm{T}_0) \arrow[d, "u_0^*"] \\
\mathrm{M}'_0(\mathrm{U}_0) \arrow[r, "a_{g_0}(h_0)"] & \mathrm{M}_0(\mathrm{U}_0)
\end{tikzcd}$$

Thus a_{g_0} becomes a morphism of functors

$$\prod_{S_0/S} M'_0 \longrightarrow \prod_{S_0/S} M_0.$$

Proposition 4.13. *Suppose that Y' is flat over S. Then*

$$c(X,Y,g\circ f) = a_{g_0}\big(c(X',Y',f)\big) + b_{f_0}\big(c(X,Y,g\circ i')\big).$$

Since the definitions of the various obstructions and of the morphisms a_{g_0} and b_{f_0} are local, it plainly suffices to verify the formula when all the schemes in question are affine. Write

$$S = \mathrm{Spec}(\Lambda), \qquad S_{\mathscr{J}} = \mathrm{Spec}(\Lambda/\mathscr{J}), \qquad S_0 = \mathrm{Spec}(\Lambda/\mathscr{I}),$$
$$T = \mathrm{Spec}(C), \qquad X = \mathrm{Spec}(A), \qquad Y = \mathrm{Spec}(A/I_Y) = \mathrm{Spec}(B),$$
$$X' = \mathrm{Spec}(A'), \qquad Y' = \mathrm{Spec}(A'/I_{Y'}) = \mathrm{Spec}(B').$$

There is therefore a diagram of rings and ideals [104]

$$\begin{tikzcd}
& \mathrm{B}' & \mathrm{B} \\
\mathrm{C} & \mathrm{A}' \arrow[l, "f"'] \arrow[u, "\pi'"'] & \mathrm{A} \arrow[l, "g"'] \arrow[u, "\pi"'] \\
& \mathrm{I}_{\mathrm{Y}'} \arrow[u, hook] & \mathrm{I}_{\mathrm{Y}} \arrow[u, hook]
\end{tikzcd}$$

Let us study the terms of the formula to be proved. In what follows, for $x \in I_Y$ (respectively $u \in I_{Y'}$), write $\bar{x}$ (respectively $\bar{u}$) for its image in I_Y/I_Y^2 (respectively $I_{Y'}/I_{Y'}^2$). If m belongs to a Λ-module M, write $\bar{m}_0$ for its image in $M_0 = M/\mathscr{I}M$.

We saw that $c = c(X,Y,g\circ f)$ is the unique C_0-morphism

$$I_Y/I_Y^2 \otimes_B C_0 \longrightarrow \mathscr{J}C$$

[104] Editor's note: We retain the notation of the original, writing $f : A' \to C$ and $g : A \to A'$ for the ring homomorphisms corresponding to $f : T \to X'$ and $g : X' \to X$. This explains the formula $c(X,Y,g\circ f)(\bar{x}\otimes 1) = f(g(x))$, for $x \in I_Y$.

such that $c(\bar{x} \otimes 1) = f(g(x))$ for every $x \in I_Y$.

Fix $x \in I_Y$. Since $g_{\mathscr{J}}(Y'_{\mathscr{J}}) \subset Y_{\mathscr{J}}$, one has $g(x) \in I_{Y'} + \mathscr{J} A'$. Write

$$g(x) = x' + \sum_i \lambda_i a'_i, \qquad x' \in I_{Y'}, \quad \lambda_i \in \mathscr{J}, \quad a'_i \in A'.$$

Then

$$c(X, Y, g \circ f)(\bar{x} \otimes 1) = f(g(x)) = f(x') + \sum_i \lambda_i f(a'_i). \tag{1}$$

Consider $a_{g_0}(c(X', Y', f))$. By the definitions just given, it is defined by the following diagram:

$$\begin{tikzcd}
 & \mathrm{I}_{\mathrm{Y}'} \arrow[r, "f"] \arrow[d] & \mathrm{C} \\
\mathrm{I}_{\mathrm{Y}'_0}/\mathrm{I}^2_{\mathrm{Y}'_0} \otimes_{\mathrm{B}'_0} \mathrm{C}_0 & \mathrm{I}_{\mathrm{Y}'}/\mathrm{I}^2_{\mathrm{Y}'} \otimes_{\mathrm{B}'} \mathrm{C}_0 \arrow[l, "\sim"'] \arrow[r, "c(\mathrm{X}',\mathrm{Y}',f)"] & \mathrm{JC} \arrow[u, hook] \\
\mathrm{I}_{\mathrm{Y}_0}/\mathrm{I}^2_{\mathrm{Y}_0} \otimes_{\mathrm{B}_0} \mathrm{C}_0 \arrow[u, "\overline{g}_0"] & \mathrm{I}_{\mathrm{Y}}/\mathrm{I}^2_{\mathrm{Y}} \otimes_{\mathrm{B}} \mathrm{C}_0 \arrow[l, two heads] \arrow[u] \arrow[ur, "a_{g_0}(c(\mathrm{X}',\mathrm{Y}',f))"'] & .
\end{tikzcd}$$

Thus $a_{g_0}(c(X', Y', f))(\bar{x} \otimes 1) = f(u)$, where $u \in I_{Y'}$ has image $\bar{u}$ in $I_{Y'_0}/I^2_{Y'_0}$ satisfying

$$\bar{u}_0 \otimes 1 = \bar{g}_0(\bar{x}_0) \otimes 1 = g_0(\bar{x}_0) \otimes 1.$$

We may therefore take $u = x'$, which gives

$$a_{g_0}(c(X', Y', f))(\bar{x} \otimes 1) = f(x'). \tag{2}$$

Finally consider $b_{f_0}(c(X, Y, g \circ i'))$. By hypothesis, the morphism of Λ_0-algebras $f_0 : A'_0 \to C_0$ factors through B'_0. Since Y' is flat, and hence

$$\mathscr{J} \otimes_{\Lambda_0} B'_0 \xrightarrow{\sim} \mathscr{J} B',$$

we obtain a morphism of B'_0-modules $\psi : \mathscr{J} B' \to \mathscr{J} C$ making the following diagram commute:

$$\begin{tikzcd}
\mathrm{J} \otimes_{\Lambda_0} \mathrm{A}'_0 \arrow[r, "\mathrm{id} \otimes \pi'"] \arrow[d] & \mathrm{J} \otimes_{\Lambda_0} \mathrm{B}'_0 \arrow[r, "\mathrm{id} \otimes f_0"] \arrow[d, "\wr"] & \mathrm{J} \otimes_{\Lambda_0} \mathrm{C}_0 \arrow[d] \\
\mathrm{JA}' \arrow[r, "\pi'"] & \mathrm{JB}' \arrow[r, dashed, "\psi"] & \mathrm{JC}.
\end{tikzcd}$$

Let

$$\varphi : \mathscr{J} B' \otimes_{B'_0} C_0 \longrightarrow \mathscr{J} C$$

be the morphism of C_0-modules induced by ψ. Then, for every $a' \in A'$ and $\lambda \in \mathscr{J}$,

$$\varphi(\lambda \pi'(a') \otimes 1) = \lambda f(a'). \tag{†}$$

The element $b_{f_0}(c(X, Y, g \circ i'))$ is then defined by the commutative diagram

$$\begin{tikzcd}
\mathrm{I_Y} \arrow[rr, "\pi'\circ g"] \arrow[d] & & \mathrm{B'} \\
\mathrm{I_Y/I_Y^2 \otimes_B B'_0} \arrow[rr, "c(\mathrm{X},\mathrm{Y},g\circ i')"] \arrow[d] & & \mathrm{JB'} \arrow[u, hook] \arrow[d] \\
\mathrm{I_Y/I_Y^2 \otimes_B C_0} \arrow[rr] \arrow[drr, "b_{f_0}(c(\mathrm{X},\mathrm{Y},g\circ i'))"'] & & \mathrm{JB' \otimes_{B'_0} C_0} \arrow[d, "\phi"] \\
& & \mathrm{JC}.
\end{tikzcd}$$

Consequently,

$$b_{f_0}(c(X, Y, g \circ i'))(\bar{x} \otimes 1) = \varphi\left(\sum_i \lambda_i \pi'(a'_i) \otimes 1\right) = \sum_i \lambda_i f(a'_i), \tag{3}$$

the last equality following from (†). Comparison of the three explicit formulas (1), (2), and (3) gives the required formula.

Corollary 4.14. *Let Y, Y' be two flat subschemes of X, both reducing to $Y_{\mathscr{J}}$, and suppose that Y_0 is locally complete intersection in X_0. If $f : T \to X$ is an S-morphism such that $f_{\mathscr{J}}$ factors through $Y_{\mathscr{J}} \to X_{\mathscr{J}}$, then*

$$c(X, Y, f) - c(X, Y', f) = b_{f_0}\big(d(Y, Y')\big).$$

Indeed, applying the preceding formula to the diagram

$$\begin{tikzcd}
& \mathrm{Y'} \arrow[d, hook, "i'"'] & \mathrm{Y} \arrow[d, hook, "i"'] \\
\mathrm{T} \arrow[r, "f"] & \mathrm{X} \arrow[r, "\mathrm{id}"] & \mathrm{X}
\end{tikzcd}$$

we find $c(X,Y,f) - c(X,Y',f) = b_{f_0}(c(X,Y,i'))$. Moreover, by 4.8(ii), $c(X,Y,i') = d(Y,Y')$.

Proposition 4.15. *Let X be an S-group smooth over S, and let Y be a sub-S-group flat and locally of finite presentation over S. Then Y is locally complete intersection in X (cf. 4.6.2).*

Proof. [105] We show that the immersion $Y \to X$ is regular in the sense of EGA IV_4, 16.9.2. By EGA IV_4, 19.5.1, this implies that it is regular also in the sense of 4.6.2 (indeed, by loc. cit., the two definitions are equivalent when S is locally noetherian). Thus "regular immersion" will here mean regular in the sense of EGA IV_4, 16.9.2. Since X and Y are flat and locally of finite presentation over S, EGA IV_4, 19.2.4 reduces the assertion to proving that $Y_s \to X_s$ is a regular immersion for every $s \in S$. By EGA IV_4, 19.1.5(ii), it is enough to check this on the geometric fibers of S, and hence when S is the spectrum of an algebraically closed field k.

By Exposé $\mathrm{VI_A}$, 3.2, the quotient X/Y exists and is smooth, the morphism $\pi : X \to X/Y$ is flat, and the following square is cartesian:

$$\begin{tikzcd} Y \arrow[r, "f"] \arrow[d] & X \arrow[d, "\pi"] \\ \bar{e} \arrow[r, "i"] & X/Y \end{tikzcd}$$

Here $\bar{e}$ is the image in X/Y of the unit point of X. By flat base change (cf. EGA IV_4, 19.1.5(iii)), it therefore suffices to show that i is a regular immersion. This is immediate: the noetherian local ring $\mathscr{O}_{X/Y,\bar{e}}$ is smooth (hence regular), so its maximal ideal is generated by a regular sequence.

4.16. [106] Let X be an S-group smooth over S, and denote its group law by $\mu : X \times_S X \to X$. Let $Y_{\mathscr{J}}$ be a sub-$S_{\mathscr{J}}$-group of $X_{\mathscr{J}}$, flat and locally of finite presentation over $S_{\mathscr{J}}$. By 4.15, $Y_{\mathscr{J}}$ is locally complete intersection in $X_{\mathscr{J}}$. Hence, by 4.6.5, every flat [107] S-scheme Y lifting

[105] Editor's note: We have added to the statement the hypothesis that Y be locally of finite presentation over S, and supplied the proof that follows, which is more direct than the one sketched in the original. For completeness, we also spell out the latter. As in the proof above, first reduce to the case $S = \mathrm{Spec}(k)$, where k is algebraically closed. By EGA IV_4, 16.9.10 and 19.3.2, it suffices to see that, for every $y \in Y$, the completion of the local ring $\mathscr{O}_{Y,y}$ is a quotient of a complete noetherian local ring by a regular sequence. By loc. cit., 19.3.3, the set of $y \in Y$ with this property is an open subset $U \subset Y$. Since Y is of finite type over k, it suffices to show that U contains every closed point. Because Y is a k-group, a translation argument reduces this to the completion of $\mathscr{O}_{Y,e}$, that is, to the formal group $\widehat{Y}$ corresponding to Y (cf. Exposé $\mathrm{VII_B}$). Since X is smooth, the affine algebra $\mathscr{A}(\widehat{X})$ is a power series algebra $k[[X_1, \dots, X_n]]$. The Dieudonné structure theorem then shows that $\mathscr{A}(\widehat{Y})$ is isomorphic to a quotient

$$k[[X_1, \dots, X_{r+s}]]/(X_1^{p^{n_1}}, \dots, X_r^{p^{n_r}}),$$

where p is the characteristic exponent of k and $r + s \le n$; cf. Exposé $\mathrm{VII_B}$, Remark 5.5.2(b).

[106] Editor's note: We have reorganized 4.16 by collecting here, on the one hand, the hypotheses stated at the end of 4.15 and, on the other, the definition of the obstruction DY.

[107] Editor's note: The original has been corrected by adding the word "flat".

$Y_{\mathscr{J}}$ is locally complete intersection in X. For such a Y, 4.8.0 gives

$$\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_0} = \mathscr{N}_{Y_0/X_0} = \mathscr{N}_{Y_{\mathscr{J}}/X_{\mathscr{J}}} \otimes_{\mathscr{O}_{Y_{\mathscr{J}}}} \mathscr{O}_{Y_0}. \tag{4.16.1}$$

On the other hand, let $\varepsilon_0 : S_0 \to Y_0$ be the unit section of Y_0, and let $\mathbf{n}_{Y_0/X_0}$ be the quasi-coherent $\mathscr{O}_{S_0}$-module

$$\mathbf{n}_{Y_0/X_0} = \varepsilon_0^*(\mathscr{N}_{Y_0/X_0}).$$

Since Y_0 and X_0 are S_0-groups, it is immediate that $\mathscr{N}_{Y_0/X_0}$ is invariant under, say, left translations by Y_0. Hence [108] it is the inverse image of $\mathbf{n}_{Y_0/X_0}$ under $Y_0 \to S_0$; that is,

$$\mathscr{N}_{Y_0/X_0} = \mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}. \tag{4.16.2}$$

Taking (4.16.1) and (4.16.2) into account, we deduce first from 4.5 that the set of subschemes Y of X, flat over S and lifting $Y_{\mathscr{J}}$, is either empty or principal homogeneous under

$$\mathrm{Hom}_{\mathscr{O}_{Y_0}}\big(\mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}\big). \tag{4.16.3}$$

On the other hand, 4.8(i) shows that, for every such Y and every S-morphism $f : T \to X$ such that $f_{\mathscr{J}} : T_{\mathscr{J}} \to X_{\mathscr{J}}$ factors through $Y_{\mathscr{J}}$, the obstruction $c(X, Y, f)$ to f factoring through Y belongs to

$$\mathrm{Hom}_{\mathscr{O}_{T_0}}\big(\mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0}, \mathscr{J}\mathscr{O}_T\big).$$

If T is also flat over S, this last group is equal to

$$\mathrm{Hom}_{\mathscr{O}_{T_0}}\big(\mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0}, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0}\big).$$

This leads us to introduce the following group functor N_0.

Definition 4.16.1. *Let N_0 be the S_0-functor in commutative groups defined, for every $Z \in \mathrm{Ob}(\mathbf{Sch})_{/S_0}$, by*

$$\mathrm{Hom}_{S_0}(Z, N_0) = \mathrm{Hom}_{\mathscr{O}_Z}\big(\mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_Z, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_Z\big). \tag{*}$$

Then the set of subschemes Y of X, flat over S and lifting $Y_{\mathscr{J}}$, is either empty or principal homogeneous under

$$\mathrm{Hom}_{S_0}(Y_0, N_0) = C^1(Y_0, N_0).$$

For every such Y, consider the diagram

$$\begin{tikzcd} Y \times_S Y \arrow[d, hook, "{(i,i)}"'] & Y \arrow[d, hook, "i"'] \\ X \times_S X \arrow[r, "\mu"] & X \end{tikzcd}$$

[108] Editor's note: See 4.25 below.

and let

$$DY = c\big(X, Y, \mu \circ (i,i)\big)$$

be the obstruction to $\mu \circ (i,i)$ factoring through Y, that is, to Y being stable under the group law of X. By what precedes, DY is an element of

$$N_0(Y_0 \times_{S_0} Y_0) = C^2(Y_0, N_0).$$

Lemma 4.17.[109] *Let X be an S-group smooth over S, and let $Y_{\mathscr{J}}$ be a sub-$S_{\mathscr{J}}$-group of $X_{\mathscr{J}}$, flat and locally of finite presentation over $S_{\mathscr{J}}$. For every subscheme Y of X, flat over S and lifting $Y_{\mathscr{J}}$, consider the obstruction defined in 4.16.1:*

$$DY \in \mathrm{Hom}_{S_0}\big(Y_0 \times_{S_0} Y_0, N_0\big) = C^2(Y_0, N_0).$$

(i) *For Y to be a sub-S-group of X, it is necessary and sufficient that $DY = 0$.*

(ii) *If Y_0 is made to act on N_0, by functoriality, through the inner automorphisms of Y_0, then $DY \in Z^2(Y_0, N_0)$.*

(iii) *If Y and Y' are two subschemes of X, flat over S, lifting $Y_{\mathscr{J}}$ (so that the deviation $d(Y, Y') \in C^1(Y_0, N_0)$ is defined; cf. 4.5.1), then*

$$DY' - DY = \partial^1 d(Y, Y').$$

[110]

We prove these assertions in succession.

4.18. — Proof of 4.17(i). By definition, $DY = 0$ if and only if Y is stable under the group law of X. Thus $DY = 0$ if Y is a subgroup of X. Conversely, if $DY = 0$, then Y is endowed with the induced law μ_Y, which is associative and reduces modulo $\mathscr{J}$ to the group law on $X_{\mathscr{J}}$. Since Y is flat and locally of finite presentation over S, it follows from 3.3 that μ_Y is a group law.

4.19. — Proof of 4.17(ii). We compare the two values of $u = c(X, Y, \mu^2 \circ (i,i,i))$ computed in the following two diagrams (D_j), $j = 1, 2$:

$$(\mathrm{D}_j) \qquad \begin{tikzcd} \mathrm{Y} \times_{\mathrm{S}} \mathrm{Y} \times_{\mathrm{S}} \mathrm{Y} \arrow[d, "{(i,i,i)}"'] & \mathrm{Y} \times_{\mathrm{S}} \mathrm{Y} \arrow[d, hook, "{(i,i)}"'] & \mathrm{Y} \arrow[d, hook, "i"'] \\ \mathrm{X} \times_{\mathrm{S}} \mathrm{X} \times_{\mathrm{S}} \mathrm{X} \arrow[r, "f_j"] & \mathrm{X} \times_{\mathrm{S}} \mathrm{X} \arrow[r, "\mu"] & \mathrm{X} \end{tikzcd}$$

Here $f_1 = (1, \pi)$, $f_2 = (\pi, 1)$, and μ^2 denotes the morphism

$$\mu \circ f_1 = \mu \circ f_2 : X \times_S X \times_S X \longrightarrow X.$$

[111] Put

$$\mu_Y = \mu \circ (i,i), \qquad f_j^Y = f_j \circ (i,i,i), \qquad \mu^{2,Y} = \mu^2 \circ (i,i,i).$$

[109]Editor's note: Sections 4.17 and 4.18 have been modified to take account of the additions made in 4.16.

[110]Editor's note: The original has $DY' - DY = -\partial d(Y, Y')$, but the operator ∂ there is the negative of the differential ∂^1 defined in I, 5.1.

[111]Editor's note: The notation has been slightly modified, and the beginning of the argument has been expanded.

For $j = 1, 2$, let a_j and b_j be the morphisms

$$a_j = a_{\mu_0}\big((f_j^Y)_0\big), \qquad b_j = b_{(f_j^Y)_0},$$

associated by Lemma 4.12 with the pair of morphisms (f_j^Y, μ). Thus

$$\begin{cases} a_j : \operatorname{Hom}_{\mathscr{O}_{Y_0^3}}\Big((f_j^Y)_0^* \mathscr{N}_{Y_0\times Y_0/X_0\times X_0}, \mathscr{J}\mathscr{O}_{Y_0^3}\Big) \longrightarrow \\ \qquad\qquad \operatorname{Hom}_{\mathscr{O}_{Y_0^3}}\Big((\mu^{2,Y})_0^* \mathscr{N}_{Y_0/X_0}, \mathscr{J}\mathscr{O}_{Y_0^3}\Big), \\ b_j : \operatorname{Hom}_{\mathscr{O}_{Y_0^2}}\Big((\mu_Y)_0^* \mathscr{N}_{Y_0/X_0}, \mathscr{J}\mathscr{O}_{Y_0^2}\Big) \longrightarrow \\ \qquad\qquad \operatorname{Hom}_{\mathscr{O}_{Y_0^3}}\Big((\mu^{2,Y})_0^* \mathscr{N}_{Y_0/X_0}, \mathscr{J}\mathscr{O}_{Y_0^3}\Big). \end{cases} \tag{$\dagger$}$$

Since

$$\mathscr{N}_{Y_0\times Y_0/X_0\times X_0} \simeq \operatorname{pr}_1^* \mathscr{N}_{Y_0/X_0} \oplus \operatorname{pr}_2^* \mathscr{N}_{Y_0/X_0}$$

(because X_0 and Y_0 are flat over S_0), and

$$\mathscr{N}_{Y_0/X_0} \simeq \mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0},$$

we have

$$(f_j^Y)_0^* \mathscr{N}_{Y_0\times Y_0/X_0\times X_0} \simeq (\mathbf{n}_{Y_0/X_0} \oplus \mathbf{n}_{Y_0/X_0}) \otimes \mathscr{O}_{Y_0^3},$$

and likewise

$$\begin{aligned} (\mu^{2,Y})_0^* \mathscr{N}_{Y_0/X_0} &\simeq \mathbf{n}_{Y_0/X_0} \otimes \mathscr{O}_{Y_0^3}, \\ (\mu_Y)_0^* \mathscr{N}_{Y_0/X_0} &\simeq \mathbf{n}_{Y_0/X_0} \otimes \mathscr{O}_{Y_0^2}. \end{aligned}$$

Moreover, since Y_0^2 and Y_0^3 are flat over S_0, ($\dagger$) may be rewritten as

$$\begin{cases} a_j : \operatorname{Hom}_{S_0}(Y_0^3, N_0 \oplus N_0) \longrightarrow \operatorname{Hom}_{S_0}(Y_0^3, N_0), \\ b_j : \operatorname{Hom}_{S_0}(Y_0^2, N_0) \longrightarrow \operatorname{Hom}_{S_0}(Y_0^3, N_0). \end{cases} \tag{$\ddagger$}$$

Applying 4.13 twice to $c(X, Y, \mu^{2,Y}) = u$, we obtain

$$\begin{aligned} a_1\big(c(X^2, Y^2, f_1)\big) + b_1\big(c(X, Y, \mu_Y)\big) &= u \\ &= a_2\big(c(X^2, Y^2, f_2)\big) + b_2\big(c(X, Y, \mu_Y)\big). \end{aligned}$$

Now $c(X, Y, \mu_Y) = DY$, and, since $f_1 = (1, \mu)$ and $f_2 = (\mu, 1)$, we have, with the evident notation,

$$c(X^2, Y^2, f_1) = (0, DY), \qquad c(X^2, Y^2, f_2) = (DY, 0).$$

Consequently,

$$u = a_1((0, DY)) + b_1(DY) = a_2((DY, 0)) + b_2(DY).$$

The first point to note is that b_j is simply $\operatorname{Hom}_{S_0}((f_j^Y)_0, N_0)$, that is, the morphism deduced from $(f_j^Y)_0$ by functoriality. The preceding identity therefore becomes

$$\begin{aligned} a_1((0, DY)) - \operatorname{Hom}((\mu, 1), N_0)(DY) + \operatorname{Hom}((1, \mu), N_0)(DY) \\ - a_2((DY, 0)) = 0. \end{aligned}$$

The two middle terms are the $DY(xy, z)$ and $DY(x, yz)$ parts of the formula for the 2-coboundary. It remains only to identify the other two terms.

We first compute the map a_j. It is obtained, by pullback along $(f_j^Y)_0$, from the morphism of $\mathscr{O}_{Y_0^2}$-modules

$$P : \mathbf{n}_{Y_0/X_0} \otimes \mathscr{O}_{Y_0^2} \longrightarrow (\mathbf{n}_{Y_0/X_0} \oplus \mathbf{n}_{Y_0/X_0}) \otimes \mathscr{O}_{Y_0^2}$$

induced by multiplication in Y_0. Consider the vector bundle $V = \mathbf{V}(\mathbf{n}_{Y_0/X_0})$. By duality, P gives

$$\mathbf{V}(P) : V \times_{S_0} V \times_{S_0} Y_0 \times_{S_0} Y_0 \longrightarrow V \times_{S_0} Y_0 \times_{S_0} Y_0,$$

which is expressed set-theoretically by

$$\mathbf{V}(P)(u, v, a, b) = (u + \mathrm{Ad}(a)v, ab, b).$$

[112] This is proved exactly as is the corresponding fact for Lie algebras, that is, for the module $\omega^1_{Y_0/S_0}$. First observe that, by functoriality in Y_0, V is endowed with a group structure in the category of vector bundles over S_0. By the lemma already used for Lie algebras (Exposé II, 3.10), this structure coincides with the group structure underlying its $\mathscr{O}_S$-module structure. Next,

$$\mathbf{V}\big(\mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{Y_0}\big) = \mathbf{V}(\mathscr{N}_{Y_0/X_0})$$

is itself endowed with an S_0-group structure, namely the semidirect product of the structure on V by that on Y_0. It remains only to identify the operations of Y_0 on V to obtain the desired formula.

We now compute the two remaining terms. Consider first $a_1((0, DY))$. It is computed by the following diagram, in which $\mathbf{n}$ denotes $\mathbf{n}_{Y_0/X_0}$:

$$\begin{tikzcd}
\mathfrak{n} \otimes \mathscr{O}_{\mathrm{Y}_0^2} \arrow[r, "\mathrm{P}"] \arrow[d, "(f_1^{\mathrm{Y}})_0^*"] & (\mathfrak{n} + \mathfrak{n}) \otimes \mathscr{O}_{\mathrm{Y}_0^2} \arrow[d, "(f_1^{\mathrm{Y}})_0^*"] \\
\mathfrak{n} \otimes \mathscr{O}_{\mathrm{Y}_0^3} \arrow[dr, "a_1((0,\mathrm{DY}))"'] & (\mathfrak{n} + \mathfrak{n}) \otimes \mathscr{O}_{\mathrm{Y}_0^3} \arrow[d, "(0,\mathrm{DY})"] \\
 & \mathcal{J} \otimes \mathscr{O}_{\mathrm{Y}_0^3}
\end{tikzcd} .$$

Regard the vector bundles defined by these modules as schemes over S_0, and take points with values in an arbitrary scheme. If (u, x, y, z) denotes a point of $\mathbf{V}(\mathscr{J}) \times Y_0^3$, we obtain the following diagram:

[112] Editor's note: The entries a, b have been replaced by ab, b to show that $\mathbf{V}(P)$ is obtained, by pullback to Y_0^2, from the multiplication morphism $V_{Y_0} \times_{S_0} V_{Y_0} \to V_{Y_0}$.

$$\begin{tikzcd}
(\mathrm{Ad}(x)\mathrm{DY}_{y,z}(u), x, yz) & (0+\mathrm{DY}_{y,z}(u), x, yz) \arrow[l, mapsto] \\
& (0+\mathrm{DY}_{y,z}(u), x, y, z) \arrow[u, mapsto] \\
(\mathrm{Ad}(x)\mathrm{DY}_{y,z}(u), x, y, z) \arrow[uu, mapsto] & (u, x, y, z) \arrow[l, mapsto] \arrow[u, mapsto]
\end{tikzcd}\quad .$$

Thus

$$a_1((0, DY))(x,y,z) = \mathrm{Ad}(x)DY(y,z),$$

which is indeed the first term of the coboundary. Likewise, $a_2((DY,0))(x,y,z) = DY(x,y)$, and therefore

$$\begin{aligned} 0 = {} & \mathrm{Ad}(x)DY(y,z) - DY(xy,z) + DY(x,yz) - DY(x,y) \\ = {} & (\partial^2 DY)(x,y,z). \end{aligned}$$

[113]

4.20. — Proof of 4.17(iii). [114] We compare the two values of $v = c(X, Y, \mu \circ (i', i'))$ computed in the following two diagrams:

$$(*) \qquad \begin{tikzcd}
& \mathrm{Y}' \arrow[d, hook, "i'"'] & \mathrm{Y} \arrow[d, hook, "i"'] \\
\mathrm{Y}' \times_{\mathrm{S}} \mathrm{Y}' \arrow[r, "\mu\circ(i',i')"] & \mathrm{X} \arrow[r, equal] & \mathrm{X}
\end{tikzcd}$$

$$(\dagger) \qquad \begin{tikzcd}
& \mathrm{Y} \times_{\mathrm{S}} \mathrm{Y} \arrow[d, hook, "(i,i)"'] & \mathrm{Y} \arrow[d, hook, "i"'] \\
\mathrm{Y}' \times_{\mathrm{S}} \mathrm{Y}' \arrow[r, "(i',i')"] & \mathrm{X} \times_{\mathrm{S}} \mathrm{X} \arrow[r, "\mu"] & \mathrm{X}.
\end{tikzcd}$$

Let $f = \mu \circ (i', i')$. Diagram $(*)$ gives

$$v = DY' + f_0^*\big(c(X, Y, i')\big). \tag{1}$$

Now $Y_0' = Y_0$, and f_0 is the multiplication $Y_0^2 \to Y_0$; hence

$$f_0^*\big(c(X,Y,i')\big)(x_0, y_0) = c(X, Y, i')(x_0 y_0). \tag{2}$$

Put $c = c(X, Y, i')$. Under the identification $N_0' \simeq N_0 \oplus N_0$, the element

$$c(X \times_S X, Y \times_S Y, (i', i'))$$

[113]Editor's note: The signs have been changed to make them compatible with I, 5.1.
[114]Editor's note: The argument that follows has been expanded from the original.

identifies with the pair (c, c). Thus, putting $h = (i', i')$, diagram (†) gives

$$v = h_0^*(DY) + a_{\mu_0}(c, c). \tag{3}$$

Since h_0 is the identity map on Y_0^2, we have $h_0^*(DY) = DY$. Finally, by the preceding calculation of a_{μ_0}, for every $S' \to S$ and every $x_0, y_0 \in Y_0(S_0')$,

$$a_{\mu_0}(c, c)(x_0, y_0) = c(x_0) + \mathrm{Ad}(x_0)\big(c(y_0)\big). \tag{4}$$

It follows that

$$\begin{aligned}(DY' - DY)(x_0, y_0) &= \mathrm{Ad}(x_0)\big(c(X, Y, i')(y_0)\big) - c(X, Y, i')(x_0 y_0) + c(X, Y, i')(x_0) \\ &= \big(\partial^1 c(X, Y, i')\big)(x_0, y_0).\end{aligned} \tag{5}$$

Since $c(X, Y, i') = d(Y, Y')$ (cf. 4.8(iii)), this proves that

$$DY' - DY = \partial^1 d(Y, Y').$$

Theorem 4.21. *Let S be a scheme and let $\mathscr{I}$ and $\mathscr{J}$ be two ideals*[115] *on S, such that $\mathscr{I} \supset \mathscr{J}$ and $\mathscr{I}\mathscr{J} = 0$. Let X be an S-group smooth over S, and let $Y_{\mathscr{J}}$ be a sub-$S_{\mathscr{J}}$-group of $X_{\mathscr{J}}$, flat and locally of finite presentation over $S_{\mathscr{J}}$. Consider the S_0-functor of commutative groups N_0 defined by*

$$\mathrm{Hom}_{S_0}(T, N_0) = \mathrm{Hom}_{\mathscr{O}_T}\Big(\mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T\Big), \qquad T \in \mathrm{Ob}(\mathbf{Sch}/S_0),$$

on which Y_0 acts through the inner automorphisms of X_0.

(i) *For there to exist a sub-S-group of X, flat over S, which reduces to $Y_{\mathscr{J}}$, it is necessary and sufficient that the following two conditions hold:*

  (i_1) *There exists a subscheme Y of X, flat over S, lifting $Y_{\mathscr{J}}$. This condition is automatically satisfied when Y_0 is affine (cf. 4.6.5).*

  (i_2) *A certain canonical obstruction, an element of $H^2(Y_0, N_0)$, vanishes.*

(ii) *If the conditions in (i) hold, the set of sub-S-groups Y of X, flat over S and reducing to $Y_{\mathscr{J}}$, is a principal homogeneous set under $Z^1(Y_0, N_0)$.*[116]

Indeed, condition (i_1) is necessary. Suppose that it holds, and let Y, flat over S, lift $Y_{\mathscr{J}}$. We must find a Y', also flat over S and lifting $Y_{\mathscr{J}}$, such that $DY' = 0$,[117] cf. 4.17(i). By 4.17(iii), this amounts to finding an element $d(Y', Y) \in C^1(Y_0, N_0)$ such that

$$DY = \partial^1 d(Y', Y).$$

[118] Let $c \in H^2(Y_0, N_0)$ be the class represented by DY, which is a cocycle by 4.17(ii). By 4.17(iii), this class does not depend on the choice of Y, and its vanishing is necessary and sufficient for the existence of $d(Y', Y)$ satisfying the preceding equation. This proves (i).

If Y has now been chosen so that $DY = 0$, the equation to be solved becomes $\partial^1 d(Y', Y) = 0$, which proves (ii).

[115] Editor's note: The hypothesis that $\mathscr{I}$ be nilpotent, which appears superfluous, has been removed.

[116] Editor's note: The question whether this set, modulo conjugation by the elements $x \in X(S)$ inducing the identity element of $X(S_{\mathscr{J}})$, is a principal homogeneous set under $H^1(Y_0, N_0)$, occupies 4.23–4.36.

[117] Editor's note: The expression $\partial DY'$ in the original has been corrected to DY'.

[118] Editor's note: See the editor's note to 4.17(iii).

Remark 4.22. Retain the notation of 4.21. By 4.15, Y_0 is locally a complete intersection in X_0; hence $\mathscr{N}_{Y_0/X_0}$ is a finite locally free $\mathscr{O}_{Y_0}$-module, and therefore

$$\mathbf{n}_{Y_0/X_0} = \varepsilon_0^*(\mathscr{N}_{Y_0/X_0})$$

is a finite locally free $\mathscr{O}_{S_0}$-module. Thus, writing

$$\mathbf{n}^\vee_{Y_0/X_0} = \mathrm{Hom}_{\mathscr{O}_{Y_0}}(\mathbf{n}_{Y_0/X_0}, \mathscr{O}_{Y_0}),$$

we have

$$\begin{aligned}&\mathrm{Hom}_{\mathscr{O}_T}\Big(\mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T\Big)\\ &\simeq \mathbf{n}^\vee_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T\end{aligned}$$

for every $T \to S_0$.[119] Consequently, the S_0-functor N_0 is isomorphic to

$$\mathbf{W}\Big(\mathbf{n}^\vee_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big) \simeq \mathbf{W}\Big(\mathrm{Hom}_{\mathscr{O}_{S_0}}(\mathbf{n}_{Y_0/X_0}, \mathscr{J})\Big).$$

It follows that there are isomorphisms[120]

$$\begin{aligned}H^2(Y_0, N_0) &\simeq H^2\Big(Y_0, \mathrm{Hom}_{\mathscr{O}_{S_0}}(\mathbf{n}_{Y_0/X_0}, \mathscr{J})\Big)\\ &\simeq H^2\Big(Y_0, \mathbf{n}^\vee_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big),\\ Z^1(Y_0, N_0) &\simeq Z^1\Big(Y_0, \mathrm{Hom}_{\mathscr{O}_{S_0}}(\mathbf{n}_{Y_0/X_0}, \mathscr{J})\Big)\\ &\simeq Z^1\Big(Y_0, \mathbf{n}^\vee_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big).\end{aligned}$$

4.23. — Still under the hypotheses of 4.21, we shall now study how the set of subgroups Y lifting $Y_{\mathscr{J}}$ behaves under conjugation by sections of X. If x is a section of X over S inducing the unit section of $X_{\mathscr{J}}$, the inner automorphism $\mathrm{Int}(x)$ defined by x sends flat subgroups of X lifting $Y_{\mathscr{J}}$ to flat subgroups of X lifting $Y_{\mathscr{J}}$. Under the conditions of 4.21(ii), the set of these subgroups is a principal homogeneous space under $Z^1(Y_0, N_0)$. We shall see that there is then a subgroup Γ of $B^1(Y_0, N_0)$[121] such that two subgroups of X, flat over S and lifting $Y_{\mathscr{J}}$, are conjugate by an $x \in X(S)$ inducing the unit of $X(S_{\mathscr{J}})$ if and only if their *difference* in $Z^1(Y_0, N_0)$ belongs to Γ. In the best cases we shall show that $\Gamma = B^1(Y_0, N_0)$. Thus the set of flat subgroups of X lifting $Y_{\mathscr{J}}$, modulo conjugation by the elements $x \in X(S)$ inducing the unit of $X(S_{\mathscr{J}})$, is either empty or a principal homogeneous space under $H^1(Y_0, N_0)$ (cf. 4.29 and 4.36).

4.24. — We retain the notation of 4.21. Let Y be a flat subgroup of X whose reduction is $Y_{\mathscr{J}}$. Recall that in 0.5 we introduced the functor L_0^X (respectively L_0^Y) defined, for a variable S_0-scheme T, by the identity

$$\mathrm{Hom}_{S_0}(T, L_0^X) = \mathrm{Hom}_{\mathscr{O}_T}\Big(\omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T, \mathscr{J} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_T\Big),$$

[119] Editor's note: The preceding argument has been expanded. It shows that the following isomorphism is valid without a flatness hypothesis. On the other hand, since 4.16 the discussion has been restricted to S-schemes $f : T \to S$ flat over S, in order to ensure that the group

$$\mathrm{Hom}_{\mathscr{O}_{T_0}}\Big(\mathbf{n}_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{O}_{T_0}, \mathscr{J}\mathscr{O}_T\Big),$$

in which the obstruction $c(X, Y, f)$ lies, coincides with $N_0(T_0)$; see the end of 4.16.

[120] Editor's note: With the notation of I, 5.3, and assuming that Y_0 is affine over S_0.

[121] Editor's note: $Z^1(Y_0, N_0)$ has been replaced by $B^1(Y_0, N_0)$, since the proof shows that Γ is a subgroup of $B^1(Y_0, N_0)$; cf. 4.27–4.29.

and similarly with Y in place of X, as well as the functor

$$L'_X = \prod_{S_0/S} L_0^X.$$

We then have:

Lemma 4.25. *There is a canonical exact sequence of Y_0-$\mathscr{O}_{S_0}$-modules*

$$\mathbf{n}_{Y_0/X_0} \xrightarrow{d} \omega^1_{X_0/S_0} \longrightarrow \omega^1_{Y_0/S_0} \longrightarrow 0 \qquad (+)$$

having the following properties:

(i) *After pullback to Y_0, d gives the morphism $\overline{D}_0$ of 4.8(iii).*

(ii) *If X_0 and Y_0 are smooth over S_0, then d is injective. Since the two modules of differentials are then locally free of finite type, the same is true of $\mathbf{n}_{Y_0/X_0}$, and the sequence is locally split.*

Proof.[122] Let $\pi_0 : Y_0 \to S_0$ denote the structure morphism. By SGA 1 II, formula (4.3) (see also EGA IV$_4$, 16.4.21), there is a canonical exact sequence of $\mathscr{O}_{Y_0}$-modules

$$\mathscr{N}_{Y_0/X_0} \xrightarrow{\overline{D}_0} \Omega^1_{X_0/S_0} \otimes_{\mathscr{O}_{X_0}} \mathscr{O}_{Y_0} \longrightarrow \Omega^1_{Y_0/S_0} \longrightarrow 0. \qquad (\dagger)$$

Since this sequence consists of $(Y_0 \times_S Y_0)$-equivariant modules and morphisms, its pullback (+) by ε_0^* is an exact sequence of Y_0-$\mathscr{O}_{S_0}$-modules, and (†) is the pullback of (+) by π_0^* (cf. Exposé I, Section 6.8). This proves (i).

Suppose in addition that X_0 and Y_0 are smooth over S_0. Then, by SGA 1 II, 4.10 (see also EGA IV$_4$, 17.2.3(i) and 17.2.5), $\overline{D}_0$ is injective and the sequence (†) consists of $\mathscr{O}_{Y_0}$-modules locally free of finite type (and is therefore locally split). By the equivalence of categories I, 6.8.1, d is also injective, and consequently the sequence (+) has the asserted properties.

4.26. —[123] For every S_0-scheme $f : T \to S_0$, the sequence (+) gives an exact sequence of $Y_0(T)$-$\mathscr{O}(T)$-modules

$$\begin{aligned} 0 \longrightarrow \mathrm{Hom}_{\mathscr{O}_T}\left(f^*\omega^1_{Y_0/S_0}, f^*\mathscr{J}\right) &\longrightarrow \mathrm{Hom}_{\mathscr{O}_T}\left(f^*\omega^1_{X_0/S_0}, f^*\mathscr{J}\right) \\ &\longrightarrow \mathrm{Hom}_{\mathscr{O}_T}\left(f^*\mathbf{n}_{Y_0/X_0}, f^*\mathscr{J}\right). \end{aligned}$$

Thus there is an exact sequence of Y_0-$\mathscr{O}_{S_0}$-modules

$$0 \longrightarrow L_0^Y \longrightarrow L_0^X \xrightarrow{d} N_0. \qquad (4.26.1)$$

It gives an exact sequence of complexes of abelian groups

$$0 \longrightarrow C^*(Y_0, L_0^Y) \longrightarrow C^*(Y_0, L_0^X) \xrightarrow{d^*} C^*(Y_0, N_0),$$

and, in particular, the following commutative diagram with exact rows:

[122] Editor's note: The proof has been expanded to take account of the additions made in Exposé I, Section 6.8.

[123] Editor's note: The original has been expanded to make clear that this is an exact sequence of Y_0-$\mathscr{O}_{S_0}$-modules.

$$
\begin{tikzcd}
0 \arrow[r] & \mathrm{C}^0(\mathrm{Y}_0, \mathrm{L}_{0\mathrm{Y}}) \arrow[r] \arrow[d, "\partial"] & \mathrm{C}^0(\mathrm{Y}_0, \mathrm{L}_{0\mathrm{X}}) \arrow[r, "d^0"] \arrow[d, "\partial"] & \mathrm{C}^0(\mathrm{Y}_0, \mathrm{N}_0) \arrow[d, "\partial"] \\
0 \arrow[r] & \mathrm{C}^1(\mathrm{Y}_0, \mathrm{L}_{0\mathrm{Y}}) \arrow[r] & \mathrm{C}^1(\mathrm{Y}_0, \mathrm{L}_{0\mathrm{X}}) \arrow[r, "d^1"] & \mathrm{C}^1(\mathrm{Y}_0, \mathrm{N}_0)
\end{tikzcd} .
$$

Observe that $C^0(Y_0, L_0^Y)$ (respectively $C^0(Y_0, L_0^X)$) is just

$$\mathrm{Hom}_{S_0}(S_0, L_0^Y) = \mathrm{Hom}_S(S, L'_Y)$$

(respectively the analogous group with X in place of Y); by 0.9, it is the group of sections of Y (respectively X) over S inducing the unit section of $X_{\mathscr{J}}$. Also, d^1 is precisely the morphism $v_{i_{Y_0}}$ of 4.8(iii), where $i_{Y_0} : Y_0 \to X_0$ is the canonical immersion.[124]

Lemma 4.27. *Under the conditions of 4.21 for S, $\mathscr{I}$, $\mathscr{J}$, and X, let Y be a subgroup of X, flat over S and lifting $Y_{\mathscr{J}}$. Let $i : Y \hookrightarrow X$ denote the canonical immersion.*[125]

(i) *Let $i' : Y \to X$ be a morphism of S-schemes lifting i_0 (so that i' is also an immersion), let $Y' = i'(Y)$, and let $d(i, i')$ be the element of $C^1(Y_0, L_0^X)$ such that $i' = d(i, i') \cdot i$ (cf. 1.2.4). Then the deviation $d(Y, Y') \in C^1(Y_0, N_0)$ (cf. 4.5.1) is given by*

$$d(Y, Y') = d^1\big(d(i, i')\big).$$

(ii) *Let $x \in C^0(Y_0, L_0^X)$ be a section of X over S inducing the unit section of $X_{\mathscr{J}}$ over $S_{\mathscr{J}}$. Then the deviation $d(Y, \mathrm{Int}(x)Y) \in C^1(Y_0, N_0)$ (cf. 4.5.1) is given by*

$$-d\big(Y, \mathrm{Int}(x)Y\big) = d^1 \partial x = \partial d^0 x.$$

Indeed, Y' is the image of Y under the composite morphism [126]

$$Y \xrightarrow{(d(i,i'),i)} L'_X \times_S X \longrightarrow X,$$

which is denoted $d(i, i') \cdot i$ in 4.8(iii). By loc. cit. and the equality $v_{i_0} = d^1$, we therefore have

$$\begin{aligned} &c\big(X, Y', d(i, i') \cdot i\big) - c(X, Y', i) \\ &\qquad = v_{i_0}\big(d(i, i')\big) = d^1\big(d(i, i')\big). \end{aligned}$$

But $d(i, i') \cdot i = i'$ factors through Y' by definition, so the first term is zero. Moreover, by 4.8(ii),

$$c(X, Y', i) = d(Y', Y) = -d(Y, Y').$$

Hence $d(Y, Y') = d^1(d(i, i'))$, which proves (i).

Now let x be as in (ii). From the formula

$$xyx^{-1} = xyx^{-1}y^{-1}y = (x - \mathrm{Ad}(y)x)y = (-\partial x)(y) \cdot y,$$

we see that Y' is the image of Y under the immersion $i' = (-\partial x) \cdot i_Y$. Part (i) therefore gives

$$-d\big(Y, \mathrm{Int}(x)Y\big) = d^1 \partial x = \partial d^0 x.$$

[124] Editor's note: This follows from the definition of $d : \mathbf{n}_{Y_0/X_0} \to \Omega^1_{X_0/S_0}$ (cf. 4.25) and from the definition of $v_{i_{Y_0}}$ (cf. 4.8).

[125] Editor's note: Part (i) below has been added; it will be useful in 4.35.1 and then in 4.38(4) and (5).

[126] Editor's note: Recall that $L'_X = \prod_{S_0/S} L_0^X$.

Corollary 4.28. *Two subgroups Y and Y' of X, flat over S and lifting $Y_{\mathscr{J}}$, are conjugate by a section of X over S inducing the unit section of $X_{\mathscr{J}}$ if and only if*

$$d(Y,Y') \in \partial d^0 C^0(Y_0, L_0^X) \subset \partial C^0(Y_0, N_0) = B^1(Y_0, N_0).$$

Source note. The French re-edition prints G here; the ambient group throughout this section is X.

Corollary 4.29. *If d^0 is surjective, then Y and Y' as above are conjugate by a section of X over S inducing the unit section of $X_{\mathscr{J}}$ if and only if $d(Y,Y') \in B^1(Y_0, N_0)$.*

Corollary 4.30. *Let Y be as in 4.27. The set of conjugates of Y by sections of X over S inducing the unit section of $X_{\mathscr{J}}$ is isomorphic to*

$$d^1 \partial C^0(Y_0, L_0^X) = C^0(Y_0, L_0^X)/\ker(d^1\partial).$$

Observe now that $C^0(Y_0, L_0^X)/\ker(d^1\partial)$ is computed using only the left-hand square of the commutative diagram in 4.26. In particular, it can also be computed in any diagram of the same type having the same left-hand square. Consider the functor L_0^X/L_0^Y over S_0, defined by

$$\mathrm{Hom}_{S_0}(T, L_0^X/L_0^Y) = \mathrm{Hom}_{S_0}(T, L_0^X)/\mathrm{Hom}_{S_0}(T, L_0^Y).$$

There is a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathrm{C}^0(\mathrm{Y}_0, \mathrm{L}_{0\mathrm{Y}}) & \longrightarrow & \mathrm{C}^0(\mathrm{Y}_0, \mathrm{L}_{0\mathrm{X}}) & \longrightarrow & \mathrm{C}^0(\mathrm{Y}_0, \mathrm{L}_{0\mathrm{X}}/\mathrm{L}_{0\mathrm{Y}}) & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle\partial} & & \downarrow{\scriptstyle\partial} & & \downarrow{\scriptstyle\partial} & & \\ 0 & \longrightarrow & \mathrm{C}^1(\mathrm{Y}_0, \mathrm{L}_{0\mathrm{Y}}) & \longrightarrow & \mathrm{C}^1(\mathrm{Y}_0, \mathrm{L}_{0\mathrm{X}}) & \longrightarrow & \mathrm{C}^1(\mathrm{Y}_0, \mathrm{L}_{0\mathrm{X}}/\mathrm{L}_{0\mathrm{Y}}) & \longrightarrow & 0 \end{array} ,$$

whence, by the preceding observation:

Corollary 4.31. *Let Y be as in 4.27. The set of conjugates of Y by sections of X over S inducing the unit section of $X_{\mathscr{J}}$ is isomorphic to*

$$E = \partial C^0(Y_0, L_0^X/L_0^Y) = C^0(Y_0, L_0^X/L_0^Y)/H^0(Y_0, L_0^X/L_0^Y).$$

Corollary 4.32. *Suppose in addition that S_0 is affine and that $\omega^1_{Y_0/S_0}$ is finite locally free.*[127] *If*

$$\mathscr{F}_0 = \big(\mathrm{Lie}(X_0/S_0)/\mathrm{Lie}(Y_0/S_0)\big) \otimes_{\mathscr{O}_{S_0}} \mathscr{J},$$

then $E = \Gamma(S_0, \mathscr{F}_0)/H^0(Y_0, \mathscr{F}_0)$.

Indeed,[128] since $\omega^1_{Y_0/S_0}$ is finite locally free, as is $\omega^1_{X_0/S_0}$ (because X is assumed smooth over S), 0.6 gives

$$L_0^Y = \mathbf{W}\big(\mathrm{Lie}(Y_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\big), \qquad L_0^X = \mathbf{W}\big(\mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\big).$$

On the other hand, by 4.25 there is an exact sequence of Y_0-$\mathscr{O}_{S_0}$-modules

$$0 \longrightarrow \mathscr{K} \longrightarrow \omega^1_{X_0/S_0} \xrightarrow{\varphi} \omega^1_{Y_0/S_0} \longrightarrow 0, \qquad \mathscr{K} = \ker(\varphi).$$

Since $\omega^1_{Y_0/S_0}$ and $\omega^1_{X_0/S_0}$ are finite locally free, there is a locally split exact sequence

$$0 \longrightarrow \mathrm{Lie}(Y_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J} \longrightarrow \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J} \longrightarrow \mathscr{F}_0 \longrightarrow 0.$$

[127] Editor's note: $\mathrm{Lie}(Y_0/S_0)$ has been corrected to $\omega^1_{Y_0/S_0}$.
[128] Editor's note: The original has been expanded in what follows.

It follows that there is an exact sequence of Y_0-$\mathscr{O}_{S_0}$-modules

$$0 \longrightarrow L_0^Y \longrightarrow L_0^X \longrightarrow \mathbf{W}(\mathscr{F}_0).$$

By the argument used to prove 4.31, we may compute E as the image of the composite map

$$C^0(Y_0, L_0^X) \xrightarrow{\pi} C^0(Y_0, \mathbf{W}(\mathscr{F}_0)) \xrightarrow{\partial} C^1(Y_0, \mathbf{W}(\mathscr{F}_0)).$$

Now π is the map

$$\Gamma\big(S_0, \mathrm{Lie}(X_0/S_0) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\big) \longrightarrow \Gamma(S_0, \mathscr{F}_0).$$

Since S_0 is affine, π is surjective, and this proves the assertion.

Corollary 4.33. *Let $S, \mathscr{I}, \mathscr{J}$, and X be as in 4.21, and let Y be a diagonalizable subgroup of X. Suppose that $\omega^1_{Y_0/S_0}$ is finite locally free and that S_0 is affine.*[129] *The set of subgroups of X conjugate to Y by a section of X over S inducing the unit section of $X_{\mathscr{J}}$ is isomorphic to*

$$E = \Gamma\left(S_0, \frac{\mathrm{Lie}(X_0/S_0)}{\mathrm{Lie}(X_0/S_0)^{\mathrm{ad}(Y_0)}} \otimes_{\Gamma(S_0,\mathscr{O}_{S_0})} \Gamma(S_0, \mathscr{J})\right),$$

that is, to $L_0^X(Y_0)/H^0(Y_0, L_0^X)$.[130]

Indeed, by I, 4.7.3 (cf. 2.13), write

$$\mathrm{Lie}(X_0/S_0) = \mathrm{Lie}(X_0/S_0)^{\mathrm{ad}(Y_0)} \oplus \mathscr{R}.$$

Since Y_0 is commutative,

$$\mathrm{Lie}(Y_0/S_0) \subset \mathrm{Lie}(X_0/S_0)^{\mathrm{ad}(Y_0)},$$

and hence

$$\mathscr{F}_0 = \left(\frac{\mathrm{Lie}(X_0/S_0)^{\mathrm{ad}(Y_0)}}{\mathrm{Lie}(Y_0/S_0)}\right) \otimes_{\mathscr{O}_{S_0}} \mathscr{J} \;\oplus\; \mathscr{R} \otimes_{\mathscr{O}_{S_0}} \mathscr{J},$$

$$\mathscr{F}_0^{\mathrm{ad}(Y_0)} = \left(\frac{\mathrm{Lie}(X_0/S_0)^{\mathrm{ad}(Y_0)}}{\mathrm{Lie}(Y_0/S_0)}\right) \otimes_{\mathscr{O}_{S_0}} \mathscr{J}.$$

By 4.32, therefore, $E \simeq \Gamma(S_0, \mathscr{R} \otimes_{\mathscr{O}_{S_0}} \mathscr{J})$. Returning to the definition of $\mathscr{R}$ completes the proof.

Corollary 4.34. *Let $S, \mathscr{I}, \mathscr{J}$, and X be as in 4.21, and let Y be a diagonalizable subgroup of X. Suppose that $\omega^1_{Y_0/S_0}$ is finite locally free and that S_0 is affine.*[131] *If $x \in X(S)$ induces the unit section of $X_{\mathscr{J}}$ and normalizes Y, then it centralizes Y.*

This follows immediately by comparing the preceding corollary with 2.14. Indeed, 4.33 shows that the elements of $C^0(Y_0, L_0^X)$ which globally preserve Y are the elements of $H^0(Y_0, L_0^X)$, and 2.14 shows that these are precisely the elements which act trivially on the canonical immersion $Y \longrightarrow X$.

4.35. — Return to the general situation of 4.21 and suppose that $Y_{\mathscr{J}}$ is smooth over $S_{\mathscr{J}}$. By 4.25(ii), there is then an exact sequence of Y_0-$\mathscr{O}_{S_0}$-modules

$$0 \longrightarrow \mathrm{Lie}(Y_0/S_0) \longrightarrow \mathrm{Lie}(X_0/S_0) \longrightarrow \mathbf{n}^\vee_{Y_0/X_0} \longrightarrow 0, \tag{*}$$

and these are finite locally free $\mathscr{O}_{S_0}$-modules.

[129] Editor's note: The hypothesis on $\omega^1_{Y_0/S_0}$ has been added, and the hypothesis "S affine" has been replaced by "S_0 affine."

[130] Editor's note: What follows has been added; cf. 4.34.

[131] Editor's note: Cf. Editor's Note 129.

On the other hand, by SGA 1, II, 4.10, every subscheme Y of X lifting $Y_{\mathscr{J}}$ and flat over S is smooth over S.[132] Suppose in addition that S_0 and $Y_{\mathscr{J}}$ are affine. Since $\mathbf{n}^{\vee}_{Y_0/X_0}$ is a locally free $\mathscr{O}_{S_0}$-module, the sequence $(*)$ remains exact after tensoring with $\mathscr{J}$ over $\mathscr{O}_{S_0}$, and then pulling back to Y_0^n. Since the Y_0^n are affine, this gives an exact sequence of complexes of abelian groups

$$0 \longrightarrow C^*(Y_0, L_0^Y) \longrightarrow C^*(Y_0, L_0^X) \xrightarrow{d^*} C^*(Y_0, N_0) \longrightarrow 0,$$

and, in particular, a commutative diagram with exact rows:

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{C}^1(\mathrm{Y}_0,\mathrm{L}_{0\mathrm{Y}}) \arrow[r] \arrow[d, "\partial"] & \mathrm{C}^1(\mathrm{Y}_0,\mathrm{L}_{0\mathrm{X}}) \arrow[r, "d^1"] \arrow[d, "\partial"] & \mathrm{C}^1(\mathrm{Y}_0,\mathrm{N}_0) \arrow[r] \arrow[d, "\partial"] & 0 \\
0 \arrow[r] & \mathrm{C}^2(\mathrm{Y}_0,\mathrm{L}_{0\mathrm{Y}}) \arrow[r] & \mathrm{C}^2(\mathrm{Y}_0,\mathrm{L}_{0\mathrm{X}}) \arrow[r, "d^2"] & \mathrm{C}^2(\mathrm{Y}_0,\mathrm{N}_0) \arrow[r] & 0
\end{tikzcd} .$$

Let now Y and Y' be two subgroups of X lifting $Y_{\mathscr{J}}$ and flat, hence smooth, over S. Since $Y_{\mathscr{J}}$ is affine, 0.15 shows that Y and Y' are isomorphic as schemes lifting $Y_{\mathscr{J}}$; that is, there is an isomorphism of S-schemes

$$f : Y \xrightarrow{\sim} Y'$$

inducing the identity on $Y_{\mathscr{J}}$. On the one hand, by 1.2.4, f defines an element $a \in C^1(Y_0, L_0^X)$ such that

$$f(y) = a(y_0)y$$

for every $y \in Y(S')$, $S' \to S$, and by 4.27(i),

$$d^1(a) = d(Y, Y').$$

Moreover, because Y and Y' are subgroups of X, the last element belongs to $Z^1(Y_0, N_0)$ (cf. 4.21). Thus ∂a is an element of $Z^2(Y_0, L_0^Y)$, whose image $\overline{\partial a}$ in $H^2(Y_0, L_0^Y)$ depends only on the class $\overline{d(Y, Y')} \in H^1(Y_0, N_0)$. This is precisely the definition of the connecting map

$$\partial^1 : H^1(Y_0, N_0) \longrightarrow H^2(Y_0, L_0^Y),$$

so that

$$\partial^1\big(\overline{d(Y, Y')}\big) = \overline{\partial a}.$$

On the other hand, transport the group structure of Y' by f, and let Y_1 be the resulting group, whose underlying scheme is therefore Y. Thus its group law μ_1 is defined, for every $S' \to S$ and $x, y \in Y(S')$, by

$$\mu_1(x, y) = f^{-1}\big(f(x)f(y)\big).$$

By 3.5.1, Y_1 defines a cocycle

$$\delta(Y, Y_1) \in Z^2\big(Y_0, \mathrm{Lie}(Y_0/S_0)\big)$$

[132]Editor's note: What follows, together with Proposition 4.35.1, has been added; it is implicit in the original. Cf. 4.38(5).

such that, for every $S' \to S$ and $x, y \in Y(S')$,

$$\delta(Y, Y_1)(x_0, y_0)xy = \mu_1(x, y) = f^{-1}\big(f(x)f(y)\big).$$

Put $b = \delta(Y, Y_1)$. For every $S' \to S$ and $x, y \in Y(S')$, we have $(b(x_0, y_0)xy)_0 = x_0y_0$. Hence $f(b(x_0, y_0)xy)$ is equal, on the one hand, to

$$a(x_0y_0)b(x_0, y_0)xy,$$

and, on the other hand, to

$$f(x)f(y) = a(x_0)x\,a(y_0)y = a(x_0)\,\mathrm{Ad}(x_0)\big(a(y_0)\big)xy.$$

Comparing these two expressions, we find that $b(x_0, y_0)$ equals

$$\begin{aligned} &a(x_0y_0)^{-1}a(x_0)\,\mathrm{Ad}(x_0)\big(a(y_0)\big) \\ &\quad = \mathrm{Ad}(x_0)\big(a(y_0)\big) - a(x_0y_0) + a(x_0) = (\partial a)(x_0, y_0). \end{aligned}$$

Thus $\delta(Y, Y_1) = \partial a$, and we have proved:

Proposition 4.35.1.[132] *Under the hypotheses of 4.21, suppose in addition that S_0 is affine and that $Y_{\mathscr{J}}$ is smooth over $S_{\mathscr{J}}$ and affine. Let Y, Y' be two subgroups of X lifting $Y_{\mathscr{J}}$ and flat (hence smooth) over S. Let $f : Y \xrightarrow{\sim} Y'$ be an isomorphism of S-schemes inducing the identity on $Y_{\mathscr{J}}$, and let Y_1 denote the group obtained by transporting the group structure of Y' by f. Then*

$$\partial^1\big(\overline{d(Y, Y')}\big) = \overline{\delta(Y, Y_1)}.$$

Proposition 4.36. *Under the hypotheses of 4.21, suppose in addition that $Y_{\mathscr{J}}$ is smooth over $S_{\mathscr{J}}$ and that S_0 is affine. The set of sub-S-groups Y of X, flat (or smooth) over S, reducing to $Y_{\mathscr{J}}$, modulo conjugation by sections of X over S inducing the unit section of $X_{\mathscr{J}}$, is either empty or a principal homogeneous set under the group*

$$H^1\Big(Y_0, \big[\mathrm{Lie}(X_0/S_0)/\,\mathrm{Lie}(Y_0/S_0)\big] \otimes_{\mathscr{O}_{S_0}} \mathscr{J}\Big).$$

It is enough to verify that Corollary 4.29 applies, that is, that

$$d^0 : \mathrm{Hom}_{\mathscr{O}_{S_0}}(\omega^1_{X_0/S_0}, \mathscr{J}) \longrightarrow \mathrm{Hom}_{\mathscr{O}_{S_0}}(\mathbf{n}_{Y_0/X_0}, \mathscr{J})$$

is surjective. This follows from the fact that the sequence (+) of 4.25(ii) is split, since S_0 is affine. [133]

Let us finally state a corollary common to 4.21 and 4.36. In fact, this will be the only form in which we shall subsequently use the general results of this section. [134]

Corollary 4.37. *Let S be a scheme and let S_0 be the closed subscheme defined by a nilpotent ideal $\mathscr{I}$. Let X be an S-group smooth over S, and let Y_0 be a sub-S_0-group of X_0, flat over S_0.*

(i) *If S_0 is affine, Y_0 is smooth over S_0, and if*

$$H^1\Big(Y_0, \big[\mathrm{Lie}(X_0/S_0)/\,\mathrm{Lie}(Y_0/S_0)\big] \otimes_{\mathscr{O}_{S_0}} \mathscr{I}^{n+1}/\mathscr{I}^{n+2}\Big) = 0$$

for every $n \geq 0$, then any two sub-S-groups of X, flat (or smooth) over S and reducing to Y_0, are conjugate by a section of X over S inducing the unit section of X_0.

[133] Editor's note: This also follows from the proof of 4.32.

[134] Editor's note: For example, 4.37 is used in Exposé IX to prove statements 3.2 bis and 3.6 bis.

(ii) *If Y_0 is affine and of finite presentation, and if*[135]

$$H^2\Big(Y_0, \mathbf{n}^\vee_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{I}^{n+1}/\mathscr{I}^{n+2}\Big) = 0$$

for every $n \geq 0$, then there exists a sub-S-group of X, flat over S, reducing to Y_0.

4.38. It remains to relate the three constructions made in this Exposé. To avoid inessential complications, we shall place ourselves in the following situation: S_0 is the spectrum of a field k, S is the spectrum of the dual numbers $D(k)$, G is an S-group smooth over S, and K is a sub-S-group, smooth over S and affine. [136]

Put

$$\mathfrak{g}_0 = \operatorname{Lie} G_0 = \Gamma(S_0, \operatorname{Lie} G_0) = \operatorname{Lie}(G_0/S_0)(S_0), \qquad \mathfrak{k}_0 = \operatorname{Lie} K_0.$$

We have an exact sequence of k-vector spaces [137]

$$0 \longrightarrow \mathfrak{k}_0 \xrightarrow{\ i\ } \mathfrak{g}_0 \xrightarrow{\ d\ } \mathbf{n}^\vee_{K_0/G_0} \longrightarrow 0,$$

which gives rise to an exact cohomology sequence

$$\begin{aligned} 0 \longrightarrow H^0(K_0, \mathfrak{k}_0) &\xrightarrow{\ i^0\ } H^0(K_0, \mathfrak{g}_0) \xrightarrow{\ d^0\ } H^0(K_0, \mathfrak{g}_0/\mathfrak{k}_0) \\ &\xrightarrow{\ \partial^0\ } H^1(K_0, \mathfrak{k}_0) \xrightarrow{\ i^1\ } H^1(K_0, \mathfrak{g}_0) \xrightarrow{\ d^1\ } H^1(K_0, \mathbf{n}^\vee_{K_0/G_0}) \xrightarrow{\ \partial^1\ } H^2(K_0, \mathfrak{k}_0). \end{aligned}$$

These various groups all have a geometric meaning.

a) $H^0(K_0, \mathfrak{k}_0) = \operatorname{Lie} \underline{\operatorname{Centr}}(K_0)$ [138] by II, 5.2.3.

b) $H^0(K_0, \mathfrak{g}_0) = \operatorname{Lie} \underline{\operatorname{Centr}}_{G_0}(K_0)$ [138] (idem).

c) $H^0(K_0, \mathfrak{g}_0/\mathfrak{k}_0) = \operatorname{Lie} \underline{\operatorname{Norm}}_{G_0}(K_0)/\mathfrak{k}_0$[138] (idem).

d)

$$H^1(K_0, \mathfrak{k}_0) = \operatorname{Lie}\big(\underline{\operatorname{Aut}}_{S_0\text{-gr.}}(K_0)\big) \,/\, \operatorname{Im}(\mathfrak{k}_0),$$

where $\operatorname{Im}(\mathfrak{k}_0)$ denotes the image of $\mathfrak{k}_0$ under the morphism $\operatorname{Lie}(\operatorname{Int}_0)$ deduced from

$$\operatorname{Int}_0 : K_0 \longrightarrow \underline{\operatorname{Aut}}_{S_0\text{-gr.}}(K_0).$$

Indeed, it follows from 2.1(ii), applied with $Y = X = K$ and $f_0 = \operatorname{id}_{K_0}$, that $Z^1(K_0, \mathfrak{k}_0)$ is the group of infinitesimal automorphisms of the S_0-group K_0, and that $H^1(K_0, \mathfrak{k}_0)$ is obtained by quotienting by inner infinitesimal automorphisms, that is, by the image of $\mathfrak{k}_0$. Moreover, by II, 4.2.2, we also have

$$Z^1(K_0, \mathfrak{k}_0) = \operatorname{Lie}\big(\underline{\operatorname{Aut}}_{S_0\text{-gr.}}(K_0)\big)\,.$$

[139]

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[135] Editor's note: We have replaced $\operatorname{Hom}_{\mathscr{O}_{S_0}}(\mathbf{n}_{Y_0/X_0}, \mathscr{I}^{n+1}/\mathscr{I}^{n+2})$ by $\mathbf{n}^\vee_{Y_0/X_0} \otimes_{\mathscr{O}_{S_0}} \mathscr{I}^{n+1}/\mathscr{I}^{n+2}$, in accordance with Remark 4.22.

[136] Editor's note: We have slightly modified the original in what follows. In particular, we have replaced X by G and Y by K, and denoted their Lie algebras by $\mathfrak{g}_0$ and $\mathfrak{k}_0$. We have also written $H^i(K_0, \cdot)$ explicitly in place of the abbreviation $H^i(\cdot)$ used in the original.

[137] Editor's note: Equipped with the adjoint action of K_0.

[138] Editor's note: Since the formation of centralizers and normalizers commutes with base change (cf. I, 2.3.3.1), we have written $\underline{\operatorname{Centr}}(K_0)$ in place of $\underline{\operatorname{Centr}}(K)_0$ in the original, and likewise $\underline{\operatorname{Centr}}_{G_0}(K_0)$ and $\underline{\operatorname{Norm}}_{G_0}(K_0)$ in place of $\underline{\operatorname{Centr}}_G(K)_0$ and $\underline{\operatorname{Norm}}_G(K)_0$.

[139] Editor's note: This is the Lie algebra $\operatorname{Der}_k(\mathfrak{k}_0)$ of derivations of $\mathfrak{k}_0$; hence $H^1(K_0, \mathfrak{k}_0)$ is the quotient of $\operatorname{Der}_k(\mathfrak{k}_0)$ by the inner derivations (that is, by the image of $\operatorname{ad} : \mathfrak{k}_0 \to \operatorname{Der}_k(\mathfrak{k}_0)$).

e) By 2.1(ii), $H^1(K_0, \mathfrak{g}_0)$ is the group of deviations between homomorphisms $K \to G$ extending the canonical immersion $i_0 : K_0 \to G_0$, modulo the deviations obtained through the action of the inner automorphisms of G defined by elements of $G(S)$ which give the unit of $G(S_0)$ (that is, elements of $\mathfrak{g}_0$).

f) By 4.36, $H^1(K_0, \mathbf{n}^\vee_{K_0/G_0})$ is the group of deviations between subgroups K' of G extending K_0 and flat over S (hence smooth over S, cf. SGA 1, II, 4.10), modulo the deviations obtained through the action of the inner automorphisms of G constructed as above.

g) By 3.5(ii), $H^2(K_0, \mathfrak{k}_0)$ is the group of deviations between group structures on K extending that of K_0, modulo the S-automorphisms of K inducing the identity on K_0.

We now propose to make explicit the six morphisms of the preceding exact sequence in terms of the geometric interpretation just given.

1) The morphisms i^0 and d^0 are simply those obtained by passing to the Lie algebra (and then, for d^0, to the quotient) in the canonical monomorphisms

$$\underline{\mathrm{Centr}}(K_0) \longrightarrow \underline{\mathrm{Centr}}_{G_0}(K_0) \longrightarrow \underline{\mathrm{Norm}}_{G_0}(K_0).$$

This follows immediately from the definition of the identifications **a)**, **b)**, and **c)**.

2) The morphism ∂^0 is constructed as follows. Let

$$\bar{x} \in \mathrm{Lie}\, \underline{\mathrm{Norm}}_{G_0}(K_0)/\mathfrak{k}_0.$$

Lift it to an element

$$x \in \mathrm{Lie}\, \underline{\mathrm{Norm}}_{G_0}(K_0) \subset \underline{\mathrm{Norm}}_{G}(K)(S).$$

Then $\mathrm{Int}(x)$ defines an automorphism of K inducing the identity on K_0, hence an element of $\mathrm{Lie}\, \underline{\mathrm{Aut}}_{S_0\text{-gr.}}(K_0)$. Denote by $\overline{\mathrm{Int}(x)}$ the image of this element in

$$\mathrm{Lie}\, \underline{\mathrm{Aut}}_{S_0\text{-gr.}}(K_0)/\,\mathrm{Im}(\mathfrak{k}_0).$$

Then

$$\partial^0(\bar{x}) = -\overline{\mathrm{Int}(x)} = \overline{\mathrm{Int}(x^{-1})}. \tag{*}$$

Indeed, let us compute the element of $\mathrm{Lie}\, \underline{\mathrm{Aut}}_{S_0\text{-gr.}}(K_0)$ defined by $\mathrm{Int}(x)$. By definition it corresponds to an element $a \in Z^1(K_0, \mathfrak{k}_0)$ such that

$$xyx^{-1} = a(y_0)y, \qquad y \in K(S'), \quad S' \longrightarrow S.$$

But this can also be written

$$a(y_0) = xyx^{-1}y^{-1} = x - \mathrm{Ad}(y)x = -\partial(x)(y_0),$$

whence $a = -\partial(x)$.

On the other hand,[140] the image under ∂ of

$$x \in \mathrm{Lie}\, \underline{\mathrm{Norm}}_{G_0}(K_0) \subset \mathfrak{g}_0$$

is an element of $Z^1(K_0, \mathfrak{k}_0)$, whose image $\overline{\partial(x)}$ in $H^1(K_0, \mathfrak{k}_0)$ depends only on $\bar{x}$. By definition of the connecting map ∂^0,

$$\partial^0(\bar{x}) = \overline{\partial(x)}.$$

Combined with $a = -\partial(x)$, this proves $(*)$.

3) [141] Let $i : K \to G$ denote the canonical immersion. Let $\bar{u} \in H^1(K_0, \mathfrak{k}_0)$ be the image

[140] Editor's note: We have added the following sentence.

[141] Editor's note: We have expanded the original in what follows, and in $(**)$ we have corrected $u \circ i$ to $i \circ u$.

of an element

$$u \in \operatorname{Lie} \underline{\operatorname{Aut}}_{S_0\text{-gr.}}(K_0) \subset \underline{\operatorname{Aut}}_{S\text{-gr.}}(K).$$

Then

$$i^1(\bar{u}) = \bar{d}(i, i \circ u), \tag{**}$$

where $\bar{d}(i, i \circ u)$ is the class defined in 2.1.1.

Indeed, $\bar{u}$ is the image of an element $v \in Z^1(K_0, \mathfrak{k}_0)$ such that $u(y) = v(y_0)y$, and $i^1(\bar{u})$ is the image in $H^1(K_0, \mathfrak{g}_0)$ of the cocycle $i \circ v \in Z^1(K_0, \mathfrak{g}_0)$. Since i is a group morphism, the equality $u(y) = v(y_0)y$ implies

$$i(u(y)) = i(v(y_0))i(y).$$

It follows that $i \circ v = d(i, i \circ u)$, proving $(**)$.

4) Let $i' : K \to G$ be a group morphism lifting i_0, let $\bar{d}(i, i')$ be the class defined in 2.1.1, and let

$$d\big(K, i'(K)\big) \in C^1(K_0, \mathbf{n}^\vee_{K_0/G_0})$$

be the deviation defined in 4.5.1. By 4.21, $d(K, i'(K))$ belongs to $Z^1(K_0, \mathbf{n}^\vee_{K_0/G_0})$. Denote its image in $H^1(K_0, \mathbf{n}^\vee_{K_0/G_0})$ by $\bar{d}(K, i'(K))$. Then, by 4.27(i),

$$d^1\big(\bar{d}(i, i')\big) = \bar{d}\big(K, i'(K)\big). \tag{†}$$

Source note. In these three adjacent occurrences the French re-edition prints $\mathbf{n}_{K_0/X_0}$, without the dual. The English uses $\mathbf{n}^\vee_{K_0/G_0}$, as required by the exact sequence and cohomology sequence established at the start of 4.38.

5) Finally, let K' be a subgroup of G lifting K_0 and flat, hence smooth, over S. We have assumed that K_0 is affine. We therefore know that K and K' are isomorphic as schemes extending K_0 (cf. 0.15), so there exists an isomorphism of S-schemes

$$f : K \xrightarrow{\sim} K'$$

inducing the identity on K_0. Transport the group structure of K' by f, and let K_1 be the group thereby obtained (whose underlying scheme is thus K); that is, the group law μ_1 of K_1 is defined, for every $S' \to S$ and every $x, y \in K(S')$, by

$$\mu_1(x, y) = f^{-1}\big(f(x)f(y)\big).$$

By 3.5.1,[142] K_1 defines a cocycle $\delta(K, K_1) \in Z^2(K_0, \mathfrak{k}_0)$ such that, for every $S' \to S$ and every $x, y \in K(S')$,

$$\delta(K, K_1)(x_0, y_0)xy = \mu_1(x, y) = f^{-1}\big(f(x)f(y)\big).$$

Then, by 4.35.1,

$$\partial^1 \bar{d}(K, K') = \bar{\delta}(K, K_1). \tag{‡}$$

[142] Editor's note: We have modified the original in what follows, taking account of the additions made in 3.5.1 and 4.35.1.

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[DG70] M. Demazure and P. Gabriel, *Groupes algébriques*, Masson & North-Holland, 1970.

[Fr64] P. Freyd, *Abelian categories*, Harper and Row, 1964.

[143] Editor's note: Additional references cited in this Exposé.

Exposé IV
Topologies and Sheaves

by M. Demazure[*]

This exposé is intended to acquaint the reader with the essentials of the language of topologies and sheaves (without cohomology), a language particularly convenient for questions concerning the formation of quotients, among others.[0]

The first three sections develop the language of forming quotients. The fourth, which is the central part, presents the theory of sheaves, oriented mainly toward its application to questions about quotients; the fifth applies this theory to the formation of quotients in groups and to principal homogeneous bundles. The final section concerns the category of schemes more specifically and defines several useful topologies on that category.

The reader will profitably consult [AS], [MA], [D], and SGA 4; in particular, [D] for the applications of topologies to descent theory, and SGA 4 for questions concerning universes (which are treated especially poorly in this exposé).

1. Universal Effective Epimorphisms

Throughout the remainder of this exposé, a category $\mathscr{C}$ is fixed.

Definition 1.1. *A morphism $u : T \to S$ is called an* epimorphism *if, for every object X, the corresponding map*

$$X(S) = \operatorname{Hom}(S, X) \longrightarrow X(T) = \operatorname{Hom}(T, X)$$

is injective.[1] *It is called a* universal epimorphism *if, for every morphism $S' \to S$, the fiber product $T' = T \times_S S'$ exists and $u' : T' \to S'$ is an epimorphism.*

Definition 1.2. *A diagram of maps of sets*

$$A \xrightarrow{u} B \overset{v_1}{\underset{v_2}{\rightrightarrows}} C$$

is said to be exact *if u is injective and its image consists of those elements $b \in B$ for which $v_1(b) = v_2(b)$. A diagram of the same form in $\mathscr{C}$ is said to be* exact *if, for every object X of $\mathscr{C}$, the corresponding diagram of sets*

$$A(X) \longrightarrow B(X) \rightrightarrows C(X)$$

[*]This text develops the substance of two lectures given by A. Grothendieck, supplementing them at several important points that had been passed over in silence or only briefly touched upon.

[0]Editor's note: version of October 13, 2024.

[1]Editor's note: that is, if u is right-cancellable.

is exact; one then also says that u makes A a kernel *of the pair of arrows* (v_1, v_2). *Dually, a diagram in* $\mathscr{C}$

$$C \overset{v_1}{\underset{v_2}{\rightrightarrows}} B \xrightarrow{u} A$$

is said to be exact if it is exact as a diagram in the opposite category $\mathscr{C}^\circ$; that is, if for every object X of $\mathscr{C}$, the corresponding diagram of sets

$$X(A) \longrightarrow X(B) \rightrightarrows X(C)$$

is exact.[2] *One also says that u makes A a* cokernel *of the pair of arrows* (v_1, v_2).

Definition 1.3. *A morphism $u : T \to S$ is called an* effective epimorphism *if the fiber square $T \times_S T$ exists and the diagram*

$$T \times_S T \overset{\mathrm{pr}_1}{\underset{\mathrm{pr}_2}{\rightrightarrows}} T \xrightarrow{u} S$$

is exact; that is, if u makes S a cokernel of $(\mathrm{pr}_1, \mathrm{pr}_2)$. *It is called a* universal effective epimorphism *if, for every morphism $S' \to S$, the fiber product $T' = T \times_S S'$ exists and the morphism $u' : T' \to S'$ is an effective epimorphism.*

There are evidently implications

$$\begin{array}{ccc}
\text{universal effective epimorphism} & \Longrightarrow & \text{effective epimorphism} \\
\Big\Downarrow & & \Big\Downarrow \\
\text{universal epimorphism} & \Longrightarrow & \text{epimorphism,}
\end{array}$$

but in general no other implication holds.[3]

Definition 1.4.0.[4] We "recall" that a morphism $u : T \to S$ is said to be *squarable* if, for every morphism $S' \to S$, the fiber product $T \times_S S'$ exists.

Lemma 1.4. *Consider morphisms $U \xrightarrow{v} T \xrightarrow{u} S$. Then:*

(a) *if u and v are epimorphisms, then uv is an epimorphism; and if uv is an epimorphism, then u is an epimorphism;*

(b) *if u and v are universal epimorphisms, then uv is a universal epimorphism; and if uv is a universal epimorphism and u is squarable, then u is a universal epimorphism.*

Lemma 1.4 follows immediately from the definitions. We deduce:

Corollary 1.5. *Let $u : X \to Y$ and $u' : X' \to Y'$ be universal epimorphisms, and suppose that $Y \times Y'$ exists. Then $X \times X'$ exists and*

$$u \times u' : X \times X' \longrightarrow Y \times Y'$$

[2]Editor's note: this implies, in particular, that u is an epimorphism.

[3]Editor's note: for example, if $\mathscr{C} = \mathbf{Sch}$ is the category of schemes, it is easy to see that every universal epimorphism is surjective. Let

$$T = \coprod_{p \text{ prime}} \mathrm{Spec}(\mathbf{F}_p), \qquad S = \mathrm{Spec}(\mathbf{Z}).$$

Then the morphism $u : T \to S$ is an epimorphism that is not universal. On the other hand, $T \times_S T$ identifies with T, so the two projections $T \times_S T \to T$ coincide. Since id_T does not descend to a morphism $S \to T$, this shows that u is not an effective epimorphism.

[4]Editor's note: the numbering 1.4.0 has been added for later reference.

is a universal epimorphism.

We also record the following.

Definition 1.6.0.[5] An object S of $\mathscr{C}$ is called *squarable* if its product with every object of $\mathscr{C}$ exists. (If $\mathscr{C}$ has a final object e, this is equivalent to saying that the morphism $S \to e$ is squarable; cf. 1.4.0.)

Lemma 1.6. *Let $u : X \to Y$ be a morphism in $\mathscr{C}_{/S}$. For it to be an epimorphism (respectively, a universal epimorphism, an effective epimorphism, or a universal effective epimorphism), it is sufficient that the corresponding morphism in $\mathscr{C}$ have the same property; it is also necessary if S is assumed to be a squarable object of $\mathscr{C}$.*

The proof is immediate and is left to the reader. The hypothesis that S is squarable is used to interpret the $\mathscr{C}$-morphisms from an object Y of $\mathscr{C}_{/S}$ to an object Z of $\mathscr{C}$ as the $\mathscr{C}_{/S}$-morphisms from Y to $Z \times S$.

Lemma 1.7. *With the notation of 1.4, if u and v are effective epimorphisms and v is a universal epimorphism, then uv is an effective epimorphism.*

To see this, consider the diagram

$$\begin{tikzcd}
S & T \arrow[l, "u"'] & T \times_S T \arrow[l, shift left] \arrow[l, shift right] \\
 & U \arrow[u, "v"] & U \times_S U \arrow[l, shift left] \arrow[l, shift right] \arrow[u, "v \times_S v"'] \\
 & U \times_T U \arrow[u, shift left] \arrow[u, shift right] \arrow[ur] &
\end{tikzcd} \tag{1.7.1}$$

By hypothesis, the first row and the first column are exact; moreover, by 1.5 and 1.6, $v \times_S v$ is an epimorphism, since v is a universal epimorphism. The conclusion follows by an evident diagram chase. Indeed, if an element of $X(U)$ has the same two images in $X(U \times_S U)$, then it has, *a fortiori*, the same two images in $X(U \times_T U)$, and hence comes from an element of $X(T)$, since the first column is exact.

Since the first row is exact, it remains only to verify that the element in question has the same two images in $X(T \times_S T)$. Because $v \times_S v$ is an epimorphism, it is enough to verify that their images in $X(U \times_S U)$ are the same, which is indeed the case.

Proposition 1.8. *Consider morphisms $U \xrightarrow{v} T \xrightarrow{u} S$. If u and v are universal effective epimorphisms, then uv is a universal effective epimorphism. If uv is a universal effective epimorphism and u is squarable, then u is a universal effective epimorphism.*

The first implication follows at once from 1.7. For the second, consider the following diagram (of "bisimplicial" type):

$$\begin{tikzcd}
S & T \arrow[l, "u"'] & T \times_S T \arrow[l, shift left] \arrow[l, shift right] \\
U \arrow[u, "uv"] & U \times_S T \arrow[l] \arrow[u] & U \times_S T \times_S T \arrow[l, shift left] \arrow[l, shift right] \arrow[u] \\
U \times_S U \arrow[u, shift left] \arrow[u, shift right] & U \times_S U \times_S T \arrow[l] \arrow[u, shift left] \arrow[u, shift right] & U \times_S U \times_S T \times_S T. \arrow[l, shift left] \arrow[l, shift right] \arrow[u, shift left] \arrow[u, shift right]
\end{tikzcd} \tag{1.8.1}$$

The first, second, and third columns are exact by the hypothesis that uv is a universal effective epimorphism. The second row is exact because $U \times_S T \to U$ is an effective epimorphism (it has a section over U), and the same is true of the third row, for the same reason. An evident diagram chase then shows that the first row is exact, that is, that u is an effective epimorphism. Since the hypotheses are invariant under an arbitrary base change $S' \to S$, it follows that u is in fact a universal effective epimorphism.

[5] Editor's note: the numbering 1.6.0 has been added for later reference.

Corollary 1.9. *Let $u : X \to Y$ and $u' : X' \to Y'$ be universal effective epimorphisms, and suppose that $Y \times Y'$ exists. Then $X \times X'$ exists and*

$$u \times u' : X \times X' \longrightarrow Y \times Y'$$

is a universal effective epimorphism.

The proof is the same as for 1.5, using the diagram

$$\begin{tikzcd} & Y' & X' \arrow[l, "u'"'] \\ X \arrow[d, "u"'] & X \times Y' \arrow[l] \arrow[u] \arrow[d] & X \times X' \arrow[l] \arrow[u] \arrow[dl, "u \times u'"] \\ Y & Y \times Y' \arrow[l] & . \end{tikzcd} \tag{1.9.1}$$

Corollary 1.10. *Let $u : T \to S$ be a squarable morphism, and let $S' \to S$ be a base-change morphism that is a universal effective epimorphism. Then u is a universal effective epimorphism if and only if*

$$u' : T' = T \times_S S' \longrightarrow S'$$

is one:

$$\begin{tikzcd} T \arrow[d, "u"'] & T' \arrow[l, "v'"'] \arrow[d, "u'"] \\ S & S'. \arrow[l, "v"'] \end{tikzcd} \tag{1.10.1}$$

Only sufficiency requires proof. If u' is a universal effective epimorphism, then so is vu', by 1.8. Since $vu' = uv'$, another application of 1.8 shows that u is a universal effective epimorphism.

Remark 1.11. *The same reasoning shows that in 1.10 one may replace "universal effective epimorphism" by "universal epimorphism," by "universal and effective epimorphism," or simply by "epimorphism" (and in the last case the hypothesis that u be squarable is plainly unnecessary).*

In the proof of 1.8 we used the following result, which deserves to be stated explicitly.

Proposition 1.12. *Let $u : T \to S$ be a morphism admitting a section. Then u is an epimorphism; if $T \times_S T$ exists, it is an effective epimorphism; and if, in addition, u is squarable, it is a universal effective epimorphism.*

The first assertion is contained in 1.4(a), and the third follows at once from the second, so it is enough to establish the second. In fact, there is a stronger conclusion: for every functor $F : \mathscr{C}^\circ \to \mathbf{Set}$, not necessarily representable, the diagram of sets

$$F(S) \longrightarrow F(T) \rightrightarrows F(T \times_S T)$$

is exact. This may be regarded as a special case of the formalism of Čech cohomology (in dimension 0!), which we merely recall here. Assume only that $T \times_S T$ exists, and set

$$\check{H}^0(T/S, F) = \operatorname{Ker}\bigl(F(T) \rightrightarrows F(T \times_S T)\bigr).$$

One may plainly regard $\check{H}^0(T/S, F)$ as a contravariant functor of the variable T in $\mathscr{C}_{/S}$: every S-morphism $T' \to T$ defines a map

$$\check{H}^0(T/S, F) \longrightarrow \check{H}^0(T'/S, F). \tag{+}$$

Fix T, T' in $\mathscr{C}_{/S}$. A familiar calculation shows that, if there exists an S-morphism $T' \to T$, the corresponding map (+) is independent of the choice of that morphism.[6]

Thus $\check{H}^0(T/S, F)$ may be regarded as a functor on the category associated with the set $\mathrm{Ob}(\mathscr{C}_{/S})$, preordered by the relation of domination: T' dominates T if there exists an S-morphism $T' \to T$. In particular, if T and T' are isomorphic in this latter category, that is, if each dominates the other, then (+) is an isomorphism of sets. This applies in particular when T' is the final object of $\mathscr{C}_{/S}$, essentially S itself. The object T always dominates $T' = S$, while the converse holds precisely when T/S has a section. This proves 1.12 in the stronger form announced.

Remark 1.13. *For various applications, the notions introduced and the results stated in this exposé must be developed more generally for a family of morphisms $u_i : T_i \to S$ with the same target, rather than for a single morphism $u : T \to S$. Such a family is called* epimorphic *if, for every object X of $\mathscr{C}$, the corresponding map*

$$X(S) \longrightarrow \prod_i X(T_i)$$

is injective. One similarly introduces the notion of an effective epimorphic family and the universal variants of these notions. Whenever needed below, we shall take for granted that the results of this exposé extend to this more general situation.

Remark 1.14. *Every morphism that is both a monomorphism and an effective epimorphism is an isomorphism.*

Indeed, in the notation of 1.3, the fact that $T \to S$ is a monomorphism implies that the two morphisms

$$\mathrm{pr}_1, \mathrm{pr}_2 : T \times_S T \rightrightarrows T$$

are equal (and are isomorphisms). But a diagram

$$C \overset{v}{\underset{v}{\rightrightarrows}} B \xrightarrow{u} A$$

is exact only if u is an isomorphism, as follows immediately from the definition.[7]

2. Descent Morphisms

Recall the following definitions.

Definition 2.1. *Let $f : S' \to S$ be a morphism such that $S'' = S' \times_S S'$ exists, and let $u' : X' \to S'$ be an object over S'. A* gluing datum *on X'/S', relative to f, is an S''-isomorphism*

$$c : X''_1 \xrightarrow{\sim} X''_2,$$

[6]Editor's note: this is the following argument, communicated by M. Demazure. Let $f, g : T' \to T$, and let $\phi : T' \to T \times_S T$ be the morphism with components f and g, so that $p_1 \circ \phi = f$ and $p_2 \circ \phi = g$. Then $F(\phi) : F(T \times_S T) \to F(T')$ satisfies $F(\phi) \circ F(p_1) = F(f)$ and $F(\phi) \circ F(p_2) = F(g)$. For every $x \in \check{H}^0(T/S, F)$, however, one has $F(p_1)(x) = F(p_2)(x)$. Applying $F(\phi)$ to both sides gives $F(f)(x) = F(g)(x)$, which shows that f and g induce the same map.

[7]Editor's note: a monomorphism that is an epimorphism need not be an isomorphism. For example, in $\mathscr{C} = \mathbf{Sch}$, the morphism

$$\mathrm{Spec}(\mathbf{F}_p) \amalg \mathrm{Spec}(\mathbf{Z}[1/p]) \longrightarrow \mathrm{Spec}(\mathbf{Z})$$

is a monomorphism and a surjective epimorphism, but is not an isomorphism.

where X_i'', for $i = 1, 2$, denotes the inverse image, assumed to exist, of X'/S' under the projection $\mathrm{pr}_i : S'' \to S'$. The gluing datum c is called a descent datum *if it satisfies the "cocycle condition"*

$$\mathrm{pr}_{3,1}^*(c) = \mathrm{pr}_{3,2}^*(c)\ \mathrm{pr}_{2,1}^*(c),$$

where $\mathrm{pr}_{i,j}$, for $1 \leq j < i \leq 3$, are the canonical projections from

$$S''' = S' \times_S S' \times_S S'$$

to S'' (and we now assume that S''' also exists), where $\mathrm{pr}_{i,j}^*(c)$ is the inverse image of c, viewed as an S'''-morphism from X_j''' to X_i''', and where, for each integer k from 1 to 3, X_k''' denotes the inverse image, assumed to exist, of X'/S' under the k-th projection $q_k : S''' \to S'$.

In the second part of the definition we have therefore used customary identifications and abuses of notation,[8] which experience shows to be harmless, but which must plainly be avoided in a rigorous exposition of descent theory—a theory that must itself justify, to some extent, these customary abuses of language. Such a formal exposition ([D]) was written by Giraud, with a view to justifying and making precise SGA 1, Exposé VII, which was never written. For a detailed account of the faithfully flat descent results used constantly in this Seminar, one may consult SGA 1, Exposé VIII.

Continue to let $f : S' \to S$ be a morphism for which $S'' = S' \times_S S'$ exists, and let X be an object over S such that $X' = X \times_S S'$ and $X'' = X \times_S S''$ exist. The inverse images of X' under pr_i, for $i = 1, 2$, then exist and are canonically isomorphic; consequently, X'/S' carries a canonical gluing datum relative to f. When S''' and $X''' = X \times_S S'''$ exist, this is in fact a descent datum. If Y is another object over S satisfying the same conditions as X/S, then for every S-morphism $X \to Y$, the corresponding S'-morphism $X' \to Y'$ is compatible with the canonical gluing data on X' and Y'. In particular, if $S' \to S$ is squarable, then

$$X \longmapsto X' = X \times_S S'$$

is a functor from $\mathscr{C}_{/S}$ to the "category of objects over S' equipped with a descent datum relative to f"—a category whose definition is left to the reader and which is a full subcategory of the "category of objects over S' equipped with a gluing datum relative to f."

With this understood, we make the following definition.

Definition 2.2. *A morphism $f : S' \to S$ is called a* descent morphism *(respectively, an* effective descent morphism*) if f is squarable (that is, $X \times_S S'$ exists for every $X \to S$) and the preceding functor*

$$X \longmapsto X' = X \times_S S'$$

from the category $\mathscr{C}_{/S}$ of objects over S to the category of objects over S' equipped with a descent datum relative to f is fully faithful (respectively, an equivalence of categories).

The first of these notions can be expressed using gluing data alone, and therefore without invoking the triple fiber product S''': f is a descent morphism precisely when it is squarable and $X \mapsto X'$ is a fully faithful functor from $\mathscr{C}_{/S}$ to the category of objects over S' equipped with a gluing datum relative to f. Written out, this means that for any two objects X, Y over S, the following diagram of sets is exact:

$$\mathrm{Hom}_S(X, Y) \longrightarrow \mathrm{Hom}_{S'}(X', Y') \underset{p_2}{\overset{p_1}{\rightrightarrows}} \mathrm{Hom}_{S''}(X'', Y''). \tag{x}$$

[8] Editor's note: for example, we have identified, on the one hand, $\mathrm{pr}_{2,1}^*(X_1'') = \mathrm{pr}_{3,1}^*(X_1'')$, and, on the other hand, $\mathrm{pr}_{2,1}^*(X_2'') = \mathrm{pr}_{3,2}^*(X_2'')$.

Here $p_i(u')$ denotes the inverse image of $u' \in \mathrm{Hom}_{S'}(X', Y')$ under the projection $\mathrm{pr}_i : S'' \to S'$, for $i = 1, 2$. Indeed, by definition, the kernel of the pair (p_1, p_2) is precisely the set of S'-morphisms $X' \to Y'$ compatible with the canonical gluing data.

By the definition of the inverse images Y' and Y'', there are canonical bijections

$$\mathrm{Hom}_{S'}(X', Y') \simeq \mathrm{Hom}_S(X', Y), \qquad \mathrm{Hom}_{S''}(X'', Y'') \simeq \mathrm{Hom}_S(X'', Y).$$

Thus exactness of (x) is equivalent to exactness of

$$\mathrm{Hom}_S(X, Y) \longrightarrow \mathrm{Hom}_S(X', Y) \rightrightarrows \mathrm{Hom}_S(X'', Y), \tag{xx}$$

which is obtained by applying $\mathrm{Hom}_S(-, Y)$ to the diagram in $\mathscr{C}_{/S}$

$$X'' \rightrightarrows X' \longrightarrow X, \tag{xxx}$$

itself obtained from

$$S'' \rightrightarrows S' \longrightarrow S$$

by the base change $X \to S$. In view of 1.6, this proves the first part of the following proposition.

Proposition 2.3. *Let $f : S' \to S$ be a morphism. If f is a universal effective epimorphism, then it is a descent morphism. Conversely, every descent morphism is a universal effective epimorphism if S is a squarable object of $\mathscr{C}$, that is, if its product with every object of $\mathscr{C}$ exists.*

It remains to prove the converse. If f is a descent morphism, we must show that it is a universal effective epimorphism: for every morphism $X \to S$, diagram (xxx) must be exact, or equivalently, its image under $\mathrm{Hom}(-, Z)$ must be an exact diagram of sets for every object Z of $\mathscr{C}$. By hypothesis, $Z \times S$ exists. Let Y be the object of $\mathscr{C}_{/S}$ that it defines. The image of (xxx) under $\mathrm{Hom}(-, Z)$ is then isomorphic to its image under $\mathrm{Hom}_S(-, Y)$, and the latter is exact by the hypothesis on f.

The results concerning descent morphisms may therefore be applied to universal effective epimorphisms. For example:

Proposition 2.4. *Let $f : S' \to S$ be a descent morphism (for example, a universal effective epimorphism). Then:*

(a) *For every S-morphism $u : X \to Y$, the morphism u is an isomorphism (respectively, a monomorphism) if and only if $u' : X' \to Y'$ is one.*

(b) *Let X, Y be two subobjects of S, and let X', Y' be their inverse images in S'. Then X is contained in Y (respectively, is equal to Y) if and only if the same is true of X' and Y'.*

For (a), the definition shows that if u' is an isomorphism in the category of objects with gluing datum, then u is an isomorphism. Every isomorphism of objects over S' that is compatible with gluing data is, however, an isomorphism of objects with gluing datum: its inverse is also compatible with the gluing data. For (b), it remains to show that if X' is contained in Y', that is, if there exists an S'-morphism $X' \to Y'$, then the same is true of X, Y over S. Since $Y' \to S'$, and hence $Y'' \to S''$, is a monomorphism, the morphism $X' \to Y'$ is automatically compatible with the gluing data and therefore comes from an S-morphism $X \to Y$.

The proof applies more generally to two objects X, Y over S, with $Y \to S$ a monomorphism, when one asks whether $X \to S$ factors through Y: it is enough that $X' \to S'$ factor through Y'.

Corollary 2.5. *Let $f : S' \to S$ be a universal effective epimorphism and $g : S \to T$ a morphism such that $S \times_T S$ exists. Suppose that $S'' = S' \times_S S'$ is also a fiber product of S' with itself over T, that is, that the canonical morphism*

$$S' \times_S S' \xrightarrow{\sim} S' \times_T S'$$

is an isomorphism. Then $g : S \to T$ is a monomorphism (and the converse is of course true).

Indeed, consider the cartesian diagram

$$\begin{array}{ccc} S' \times_S S' & \xrightarrow{\sim} & S' \times_T S' \\ \downarrow & & \downarrow f \\ S & \longrightarrow & S \times_T S, \end{array} \tag{2.5.1}$$

where the second horizontal arrow is the diagonal morphism. By 1.9, the second vertical arrow f is a universal effective epimorphism; by hypothesis, the first horizontal arrow is an isomorphism. It follows from 2.4(a), or equally from 2.4(b),[9] that $S \to S \times_T S$ is also an isomorphism, which says precisely that $g : S \to T$ is a monomorphism.

Remark 2.6. *The notions introduced in 2.1 in terms of morphisms between certain inverse limits have an evident expression in terms of the contravariant functors defined by the objects S, S', X' under consideration. Subject to the existence of the fiber products considered in 2.1, there is a one-to-one correspondence between gluing data (respectively, descent data) on X'/S' relative to $S' \to S$, and gluing data (respectively, descent data) for the corresponding objects of*

$$\widehat{\mathscr{C}} = \mathrm{Hom}(\mathscr{C}^\circ, \mathbf{Set}).$$

This makes it possible, if desired, to extend these notions to the case in which no existence hypothesis is imposed on fiber products in $\mathscr{C}$.

Remark 2.7. *The notions introduced in this section generalize to families of morphisms. They can also be presented more intrinsically by means of the notion of a sieve (4.1). For these matters the reader is referred to [D].*

3. Universal Effective Equivalence Relations

3.1. Equivalence Relations: Definitions

Definition 3.1.1. *A $\mathscr{C}$-equivalence relation in an object $X \in \mathrm{Ob}\,\mathscr{C}$ is a representable subfunctor R of the functor $X \times X$ such that, for every $S \in \mathrm{Ob}\,\mathscr{C}$, $R(S)$ is the graph of an equivalence relation on $X(S)$.*

This definition applies in particular to the category $\widehat{\mathscr{C}}$. If X is regarded as an object of $\widehat{\mathscr{C}}$, then a $\widehat{\mathscr{C}}$-equivalence relation in X is simply a subfunctor R of $X \times X$ such that $R(S)$ is the graph of an equivalence relation on $X(S)$, for every object S of $\mathscr{C}$.[10]

If R is a $\mathscr{C}$-equivalence relation in X, let p_i denote the morphism from R to X induced by the projection $\mathrm{pr}_i : X \times X \to X$. Thus there is a diagram

$$p_1, p_2 : R \rightrightarrows X.$$

[9] Editor's note: applied to $f : S' \times_T S' \to S \times_T S = Y$.

[10] Editor's note: the condition is plainly necessary. Conversely, if $R(S)$ is the graph of an equivalence relation for every $S \in \mathrm{Ob}\,\mathscr{C}$, this equivalence relation extends to $R(F)$ for every $F \in \mathrm{Ob}\,\widehat{\mathscr{C}}$: two morphisms $\varphi, \psi : F \to R$ are declared equivalent if, for every $S \in \mathrm{Ob}\,\mathscr{C}$ and every $x \in F(S)$, the elements $\varphi(x)$ and $\psi(x)$ are equivalent in $X(S)$.

Definition 3.1.2. *A morphism $u : X \to Z$ is said to be* compatible with R *if $up_1 = up_2$. An object of $\mathscr{C}$ that is a cokernel of the pair (p_1, p_2) is also called a* quotient object *of X by R, and is denoted X/R.*

Thus one has an exact diagram

$$R \underset{p_2}{\overset{p_1}{\rightrightarrows}} X \xrightarrow{p} X/R,$$

and X/R represents the covariant functor

$$\mathrm{Hom}_{\mathscr{C}}(X/R, Z) = \{\text{morphisms from } X \text{ to } Z \text{ compatible with } R\}.$$

Since quotient objects are taken in $\mathscr{C}$, the quotient X/R is unique whenever it exists.

These definitions extend at once to a $\widehat{\mathscr{C}}$-equivalence relation in X. One should note, however, that the identification of the objects of $\mathscr{C}$ with their images in $\widehat{\mathscr{C}}$ does not commute with the formation of quotients: the quotient X/R of X by R in $\mathscr{C}$ is not *a priori* a quotient of X by R in $\widehat{\mathscr{C}}$. One must therefore avoid identifying $\mathscr{C}$ uncritically with its image in $\widehat{\mathscr{C}}$ in questions involving passage to a quotient.[11]

In what follows, we shall simply say *equivalence relation* to mean a $\widehat{\mathscr{C}}$-equivalence relation; whenever necessary, we shall specify that a $\mathscr{C}$-equivalence relation is intended.[12]

Definition 3.1.3. *If X is an object of $\mathscr{C}$ over S, an* equivalence relation in X over S *is an equivalence relation R in X for which the structure morphism $X \to S$ is compatible with R.*

The canonical monomorphism $R \to X \times X$ then factors through the monomorphism

$$X \times_S X \longrightarrow X \times X$$

and defines an equivalence relation in the object $X \to S$ of $\mathscr{C}_{/S}$. Whenever the quotient X/R exists, it is equipped with a canonical morphism to S, and the corresponding object of $\mathscr{C}_{/S}$ is a quotient of $X \in \mathrm{Ob}\,\mathscr{C}_{/S}$ by the preceding equivalence relation. Conversely, if S is a squarable object of $\mathscr{C}$ and $Y \to S$ is a quotient of X by this equivalence relation in $\mathscr{C}_{/S}$, then Y is a quotient of X by R in $\mathscr{C}$. In any event, we shall never have to consider quotients in $\mathscr{C}_{/S}$ that are not already quotients in $\mathscr{C}$.

Definition 3.1.4. *Let X (respectively X') be an object of $\mathscr{C}$ equipped with an equivalence relation R (respectively R'). A morphism*

$$u : X \longrightarrow X'$$

is said to be compatible with R and R' *if the following equivalent conditions hold:*

(i) *for every $S \in \mathrm{Ob}\,\mathscr{C}$, two points of $X(S)$ congruent modulo $R(S)$ are carried by u to two points of $X'(S)$ congruent modulo $R'(S)$;*

[11] Editor's note: here is an illustration that also previews the sequel of this exposé. Let G be a $\mathscr{C}$-group, let H be a $\mathscr{C}$-subgroup, and let R be the equivalence relation in G defined by $G \times H \to G \times G$, $(g, h) \mapsto (g, gh)$ (cf. 3.2). The functor Q defined by $Q(S) = G(S)/H(S)$ is a quotient in $\widehat{\mathscr{C}}$ (by 4.4.9 applied to the coarsest topology; cf. 4.4.2), but it is not in general the quotient one wants. For example, when $\mathscr{C} = \mathbf{Sch}$, there is an exact sequence of affine group schemes

$$1 \longrightarrow \mu_2 \longrightarrow \mathbf{G}_m \xrightarrow{p} \mathbf{G}_m \longrightarrow 1$$

that identifies $\mathbf{G}_m$ with the quotient $\mathbf{G}_m/\mu_2$. Moreover, because p is finite and locally free, $\mathbf{G}_m$ is the sheaf quotient of $\mathbf{G}_m$ by μ_2 in the larger category of sheaves for the fppf topology; cf. 4.6.6(ii) and 6.3.2. By contrast, the quotient Q in $\widehat{\mathscr{C}}$ is not isomorphic to $\mathbf{G}_m$, since, for example, $Q(\mathbf{Z}) = \{1\}$, whereas $\mathbf{G}_m(\mathbf{Z}) = \{\pm 1\}$. Thus Q is not an fppf sheaf and, *a fortiori*, is not representable.

[12] Editor's note: even for a $\widehat{\mathscr{C}}$-equivalence relation in X, the question of interest is the existence of a quotient in $\mathscr{C}$.

(ii) *there exists a necessarily unique morphism $R \to R'$ making the diagram*

$$\begin{array}{ccc} R & \longrightarrow & R' \\ \downarrow & & \downarrow \\ X \times X & \xrightarrow{u \times u} & X' \times X' \end{array}$$

commute.

By the universal property of X/R, whenever X/R and X'/R' exist there is a unique morphism v making the diagram

$$\begin{array}{ccc} X & \xrightarrow{p} & X/R \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle v} \\ X' & \xrightarrow{p'} & X'/R' \end{array}$$

commute.

Definition 3.1.5. *A subobject (that is, a representable subfunctor) Y of X is said to be* stable under the equivalence relation R *if the following equivalent conditions hold:*

(i) *for every $S \in \operatorname{Ob}\mathscr{C}$, the subset $Y(S)$ of $X(S)$ is stable under $R(S)$;*

(ii) *the two subobjects of R that are the inverse images of Y under p_1 and p_2 are identical.*

An important special case of a stable subobject of X is the following: the quotient X/R exists and Y is the inverse image in X of a subobject of X/R.

Definition 3.1.6. *Let R be an equivalence relation in X, and let $X' \to X$ be a morphism. The equivalence relation R' in X' obtained from the cartesian diagram*

$$\begin{array}{ccc} R' & \longrightarrow & R \\ \downarrow & & \downarrow \\ X' \times X' & \longrightarrow & X \times X \end{array}$$

is called the equivalence relation in X' that is the inverse image *of the equivalence relation R in X. In particular, if X' is a subobject of X, it is called the equivalence relation* induced *on X' by R, and is denoted $R_{X'}$.*

The morphism $X' \to X$ is compatible with R' and R; hence, when the quotients exist, it induces a morphism $X'/R' \to X/R$ (3.1.4). If X' is a subobject of X, we shall see later that in certain cases one can prove that $X'/R' \to X/R$ is a monomorphism and therefore identifies X'/R' with a subobject of X/R. In that situation, the inverse image of this subobject in X is a subobject containing X' and stable under R: the *saturation* of X' with respect to the equivalence relation R.

Proposition 3.1.7. *If the subobject Y of X is stable under R, then for $i = 1, 2$ the following two squares are cartesian:*

$$\begin{array}{ccc} R_Y & \longrightarrow & R \\ {\scriptstyle p_i}\downarrow & & \downarrow{\scriptstyle p_i} \\ Y & \longrightarrow & X. \end{array}$$

The proof is immediate.

3.2. Equivalence Relation Defined by a Group Acting Freely

Definition 3.2.1. *Let X be an object of $\mathscr{C}$, and let H be a $\mathscr{C}$-group acting on X, that is, equipped with a morphism of $\widehat{\mathscr{C}}$-groups*

$$H \longrightarrow \mathrm{Aut}(X).$$

The group H is said to act freely *on X if the following equivalent conditions hold:*

(i) *for every $S \in \mathrm{Ob}\,\mathscr{C}$, the group $H(S)$ acts freely on $X(S)$;*

(ii) *the morphism of functors*

$$H \times X \longrightarrow X \times X$$

defined on sets by $(h, x) \mapsto (hx, x)$ is a monomorphism.

In the preceding definition the group H was assumed to act on X on the left. There is of course an analogous notion when the group acts on the right, that is, when one is given a group morphism from the opposite group H° to $\mathrm{Aut}(X)$.

If H acts freely on X, the image of $H \times X$ under the morphism in (ii) is an equivalence relation in X, called the *equivalence relation defined by the action of H on X*. When the quotient of X by this equivalence relation exists, it is denoted $H\backslash X$ (or X/H when H acts on the right). It represents the following covariant functor: for every object Z of $\mathscr{C}$,

$$\mathrm{Hom}(H\backslash X, Z) = \{\text{morphisms from } X \text{ to } Z \text{ invariant under } H\},$$

where a morphism $f : X \to Z$ is said to be invariant under H if, for every $S \in \mathrm{Ob}\,\mathscr{C}$, the corresponding morphism $X(S) \to Z(S)$ is invariant under the group $H(S)$.

Lemma 3.2.2. *Under the assumptions of 3.2.1, let Y be a subobject of X. The following conditions are equivalent:*

(i) *Y is stable under the equivalence relation defined by H (3.1.5);*

(ii) *for every $S \in \mathrm{Ob}\,\mathscr{C}$, the subset $Y(S)$ of $X(S)$ is stable under $H(S)$;*

(iii) *there exists a necessarily unique morphism f making the diagram*

$$\begin{array}{ccc} H \times Y & \xrightarrow{f} & Y \\ \downarrow & & \downarrow \\ H \times X & \longrightarrow & X \end{array}$$

commute.

Under these conditions, f defines a morphism of $\widehat{\mathscr{C}}$-groups

$$H \longrightarrow \mathrm{Aut}(Y),$$

and the equivalence relation in Y defined by this action of H is precisely the equivalence relation induced on Y by the equivalence relation in X defined by the action of H.

The proof is immediate. There is of course an analogous statement for a right action. The action of H on Y defined above will be called the action induced on Y by the given action of H on X.

Consider now two $\mathscr{C}$-groups H and G, together with a group morphism

$$u : H \longrightarrow G.$$

Then H acts on G by translations (set-theoretically, $hg = u(h)g$), and it acts freely if and only if u is a monomorphism. When it exists, the quotient of G by this action of H is denoted $H\backslash G$. Similarly, one defines a right action of H on G and a quotient G/H. These quotients are functorial with respect to the groups involved. More precisely, one has the following lemma, stated for right quotients.

Lemma 3.2.3. *Let $u : H \to G$ and $u' : H' \to G'$ be monomorphisms of $\mathscr{C}$-groups, and suppose that a morphism of $\mathscr{C}$-groups*

$$f : G \longrightarrow G'$$

is given. The following conditions are equivalent:

(i) *f is compatible with the equivalence relations in G and G' defined by H and H';*

(ii) *for every $S \in \operatorname{Ob}\mathscr{C}$, one has $fu(H(S)) \subseteq u'(H'(S))$;*

(iii) *there exists a necessarily unique multiplicative morphism $g : H \to H'$ such that the diagram*

$$\begin{array}{ccc} H & \xrightarrow{g} & H' \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle u'} \\ G & \xrightarrow{f} & G' \end{array}$$

commutes.

Under these conditions, if the quotients G/H and G'/H' exist, there is a unique morphism $\overline{f}$ making the diagram

$$\begin{array}{ccc} G & \xrightarrow{f} & G' \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle p'} \\ G/H & \xrightarrow{\overline{f}} & G'/H' \end{array}$$

commute.

The first assertion is proved by reduction to the set-theoretic case. The second follows immediately from (i).

The notions introduced above for general equivalence relations could all be translated into the present setting. We record only the following lemma, whose proof is immediate by reduction to the set-theoretic case.

Lemma 3.2.4. *Let $u : H \to G$ be a monomorphism of $\mathscr{C}$-groups, and let G' be a $\mathscr{C}$-subgroup of G. The subobject G' of G is stable under the equivalence relation defined by H if and only if u factors through the canonical monomorphism $G' \to G$. Under this condition, the action of H on G' induced by the action on G defined by u is precisely the action arising from the monomorphism $H \to G'$ through which u factors.*

3.3. Universally Effective Equivalence Relations

Definition 3.3.1. *Let $f : X \to Y$ be a morphism. The* equivalence relation defined by f *in X, denoted $\mathscr{R}(f)$, is the $\widehat{\mathscr{C}}$-equivalence relation in X that is the image of the canonical monomorphism*

$$X \times_Y X \longrightarrow X \times X.$$

Definition 3.3.2. *An equivalence relation R in X is said to be* effective *if:*

(i) *R is representable, that is, it is a $\mathscr{C}$-equivalence relation;*

(ii) *the quotient $Y = X/R$ exists in $\mathscr{C}$;*[13]

(iii) *the diagram*

$$R \overset{p_1}{\underset{p_2}{\rightrightarrows}} X \xrightarrow{p} Y$$

makes R the fiber square of X over Y, that is, R is the equivalence relation defined by p.

Scholium 3.3.2.1.[14] *If R is an effective equivalence relation in X, then p is an effective epimorphism (1.3). If $f : X \to Y$ is an effective epimorphism, then $\mathscr{R}(f)$ is an effective equivalence relation in X, one quotient of which is Y. Thus there is a one-to-one "Galois" correspondence between effective equivalence relations in X and effective quotients of X, that is, equivalence classes of effective epimorphisms with source X.*

Definition 3.3.3. *An equivalence relation R in X is said to be* universally effective *if the quotient $Y = X/R$ exists and, for every $Y' \to Y$, the fiber products $X' = X \times_Y Y'$ and $R' = R \times_Y Y'$ exist and R' is the fiber square of X' over Y'. Equivalently, R is effective and $p : X \to X/R$ is a universal effective epimorphism.*

Scholium 3.3.3.1.[14] *As above, there is a one-to-one correspondence between universally effective equivalence relations in X and universal effective quotients of X.*

Remark 3.3.3.2.[14] Suppose that $\mathscr{C}$ is the category of S-schemes, and write $\mathbf{A}^1_S$ for affine one-space over S. Let $R \subset X \times_S X$ be a universally effective equivalence relation, and let $p : X \to Y$ be its quotient. Then, for every open subset U of Y,

$$\mathscr{O}(U) = \mathrm{Hom}_S(U, \mathbf{A}^1_S)$$

is the set of elements φ of

$$\mathscr{O}(p^{-1}(U)) = \mathrm{Hom}_S(p^{-1}(U), \mathbf{A}^1_S)$$

such that $\varphi \circ \mathrm{pr}_1 = \varphi \circ \mathrm{pr}_2$. In particular, if R is defined by the action of a group H acting freely on the right on X (cf. 3.2.1), then $\mathscr{O}(U)$ is the set of all $\varphi \in \mathscr{O}(p^{-1}(U))$ such that

$$\varphi(xh) = \varphi(x)$$

for every $S' \to S$, $x \in X(S')$, and $h \in H(S')$.

Proposition 3.3.4. *Let R be a universally effective equivalence relation in X. Let $f : X \to Z$ be a morphism compatible with R, hence factoring through a morphism $g : X/R \to Z$. The following conditions are equivalent:*

(i) *g is a monomorphism;*

(ii) *R is the equivalence relation defined by f.*

Indeed, (i) plainly implies (ii), while the converse is precisely 2.5.

Definition 3.3.5. *Let H be a $\mathscr{C}$-group acting freely on X. One says that H acts* effectively, *or that the action of H on X is* effective *(respectively* universally effective*), if the equivalence relation in X defined by the action of H is effective (respectively universally effective).*

[13] Editor's note: the words "in $\mathscr{C}$" have been added.

[14] Editor's note: the numbering 3.3.2.1 (respectively 3.3.3.1) has been added for later reference; Remark 3.3.3.2 has also been added.

3.4. (M)-Effectivity

In practice, universal effective epimorphisms are most often difficult to characterize. Nevertheless, one frequently has at hand a number of morphisms of this kind–in scheme theory, for example, faithfully flat quasi-compact morphisms. This leads to the developments below.

3.4.1. We first state several conditions on a family (M) of morphisms of $\mathscr{C}$:

(a) (M) is stable under base change: every $u : T \to S$ belonging to (M) is squarable (cf. 1.4.0), and for every $S' \to S$, the base change

$$u' : T \times_S S' \longrightarrow S'$$

belongs to (M);

(b) the composite of two elements of (M) belongs to (M);

(c) every isomorphism belongs to (M);

(d) every element of (M) is an effective epimorphism.

Conditions (a) and (b) imply:

(a′) the cartesian product of two elements of (M) belongs to (M). More precisely, let $u : X \to Y$ and $u' : X' \to Y'$ be two S-morphisms belonging to (M). If $Y \times_S Y'$ exists, then $X \times_S X'$ exists and $u \times_S u'$ belongs to (M).

This follows from the diagram

$$\begin{tikzcd}
 & Y' & X' \arrow[l, "u'"'] \\
X \arrow[d, "u"'] & X \times_S Y' \arrow[l] \arrow[u] \arrow[d] & X \times_S X' \arrow[l] \arrow[u] \arrow[dl, "u \times_S u'"] \\
Y & Y \times_S Y' \arrow[l] &
\end{tikzcd}$$

Likewise, (a) and (d) imply:

(d′) every element of (M) is a universal effective epimorphism.

3.4.2. The family (M_0) of universal effective epimorphisms satisfies conditions (a)–(d) of 3.4.1. Indeed, (a), (c), and (d) hold by definition, and (b) follows from 1.8. Henceforth we assume given a family (M) of morphisms of $\mathscr{C}$ satisfying these conditions; our results apply in particular to the family (M_0).

Definition 3.4.3. An equivalence relation R in X is said to be *of type* (M) if it is representable and $p_1 \in (M)$; conditions (b) and (c) then imply that $p_2 \in (M)$.

The relation R is said to be (M)-*effective* if it is effective and the canonical morphism $X \to X/R$ belongs to (M).

A quotient Y of X is said to be (M)-*effective* if the canonical morphism $X \to Y$ belongs to (M).

The following consequences are immediate from the definition.[15]

Proposition 3.4.3.1.

[15] Editor's note: the numbering 3.4.3.1 has been added for later reference, and the proof of part (i) has been expanded.

(i) *An (M)-effective equivalence relation is of type (M) and is universally effective.*

(ii) *An (M)-effective quotient is universally effective (cf. 3.3.3).*

(iii) *The maps $R \mapsto X/R$ and $p \mapsto \mathscr{R}(p)$ establish a one-to-one correspondence between (M)-effective equivalence relations in X and (M)-effective quotients of X.*

(iv) *To be (M_0)-effective is equivalent to being universally effective.*

We prove (i). Since R is (M)-effective, there is a cartesian square

$$\begin{array}{ccc} R & \xrightarrow{p_2} & X \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle p} \\ X & \xrightarrow{p} & X/R, \end{array}$$

and $p \in (M)$. By 3.4.1(a), p_1 and p_2 belong to (M), so R is of type (M).

Set $Y = X/R$, and let $Y' \to Y$ be any morphism. By 3.4.1(a), the fiber products $X' = X \times_Y Y'$ and $R' = R \times_Y Y'$ exist, and the morphisms $X' \to Y'$ and $p_i' : R' \to X'$ belong to (M). Finally, since $R = X \times_Y X$, associativity of the fiber product gives

$$R' = X \times_Y X \times_Y Y' = X' \times_{Y'} X'.$$

This shows that R' is (M)-effective; in particular, R is universally effective. This proves (i), as well as (iv). Parts (ii) and (iii) follow, taking Definition 3.3.2 into account.

3.4.4. Let H be an S-group whose structure morphism belongs to (M). If H acts freely on the S-object X, then it defines an equivalence relation of type (M) in X. Indeed, by (a), the fiber product $H \times_S X$ exists and $\mathrm{pr}_2 : H \times_S X \to X$ belongs to (M). The action of H is said to be (M)-*effective* if the equivalence relation in X defined by this action is (M)-effective.

Proposition 3.4.5 ((M)-effectivity and base change). *Let R be an (M)-effective equivalence relation in X over S, and set $Y = X/R$. Let $S' \to S$ be a base change for which $Y' = Y \times_S S'$ exists. Then $X' = X \times_S S'$ exists, $R' = R \times_S S'$ is an (M)-effective equivalence relation in X' over S', and*

$$X'/R' \simeq (X/R)'.$$

Indeed, the canonical morphisms $X \to Y$ and $R \to Y$ belong to (M); hence, by (a′), X' and R' are representable. By associativity of the fiber product, R' is the equivalence relation in X' defined by the canonical morphism $X' \to Y'$, which belongs to (M). This proves the assertion.

Proposition 3.4.6 ((M)-effectivity and cartesian products). *Let R (respectively R') be an (M)-effective equivalence relation in X (respectively X') over S. If $(X/R) \times_S (X'/R')$ exists, then $X \times_S X'$ exists, $R \times_S R'$ is an (M)-effective equivalence relation in $X \times_S X'$ over S, and*

$$(X \times_S X')/(R \times_S R') \simeq (X/R) \times_S (X'/R').$$

Set $Y = X/R$ and $Y' = X'/R'$. By (a′), the fiber product $X \times_S X'$ exists and the canonical morphism

$$q : X \times_S X' \longrightarrow Y \times_S Y'$$

belongs to (M). The formula

$$(X \times_S X') \times_{(Y \times_S Y')} (X \times_S X') \simeq (X \times_Y X) \times_S (X' \times_{Y'} X')$$

(all unlabelled fiber products being taken over S) shows that $R \times_S R'$ is the equivalence relation in $X \times_S X'$ defined by q, which completes the proof.

3.4.7.[16] Suppose that $\mathscr{C}$ has a final object e, and let $f : G \to G'$ be a morphism of $\mathscr{C}$-groups belonging to (M). By 3.4.1(a), the kernel $\operatorname{Ker}(f)$ is represented by $e \times_{G'} G$, and the morphism $\operatorname{Ker}(f) \to e$ belongs to (M).

On the other hand, the equivalence relation defined by f is the same as that defined by the action of $\operatorname{Ker}(f)$, say on the right, on G: it is the image of the morphism

$$G \times \operatorname{Ker}(f) \longrightarrow G \times G$$

defined on sets by $(g, h) \mapsto (g, gh)$. Consequently, Scholium 3.3.2.1 gives the following corollary.

Corollary 3.4.7.1. *Suppose that $\mathscr{C}$ has a final object e, and let $f : G \to G'$ be a morphism of $\mathscr{C}$-groups belonging to (M). Then the action of $\operatorname{Ker}(f)$ on G is (M)-effective, and G' is the quotient $G/\operatorname{Ker}(f)$.*

3.5. Construction of Quotients by Descent

It frequently happens that one cannot construct a quotient directly, but can do so after a suitable base change. This subsection gives a useful criterion for that situation.

In 2.1 we defined a descent datum on an object X' over S', relative to a morphism $S' \to S$.

Definition 3.5.1. *A descent datum on X', relative to $S' \to S$, is said to be* effective *if X', equipped with this descent datum, is isomorphic to the inverse image over S' of an object X over S, equipped with its canonical descent datum.*

If $S' \to S$ is a descent morphism (2.2), then the object X in the definition is unique up to unique isomorphism. To say that $S' \to S$ is an effective descent morphism (2.2) means that it is a descent morphism and that every descent datum relative to it is effective.

Now consider an equivalence relation R in an object X over S. Let X', X'', and X''' be the inverse images of X over

$$S', \qquad S'' = S' \times_S S', \qquad S''' = S' \times_S S' \times_S S',$$

respectively, and let R', R'', and R''' be the equivalence relations obtained from R by inverse image. Suppose that R' in X' is (M)-effective, and consider the quotient $Y' = X'/R'$, an object over S'. By 3.4.5, its two inverse images over S'' are isomorphic to X''/R''. Thus the S'-object Y' has a canonical gluing datum. Applying in the same way the uniqueness of X'''/R''', one sees that this is in fact a descent datum. (In this argument it has been tacitly assumed that all the displayed fiber products exist. This holds, in particular, if $S' \to S$ is squarable, for example if it is a descent morphism.)

Proposition 3.5.2. *Let R be an equivalence relation in the object X over S, and let $S' \to S$ be a universal effective epimorphism. Assume that every S-morphism whose inverse image over S' belongs to (M) itself belongs to (M). The following conditions are equivalent:*

(i) *R is (M)-effective in X;*

(ii) *R' is (M)-effective in X', and the canonical descent datum on X'/R' is effective.*

[16] Editor's note: this paragraph has been added.

When these conditions hold, the object descended from X'/R' is canonically isomorphic to X/R.

The implication (i)⇒(ii) is simply 3.4.4 translated into the language of descent. Once the converse is proved, the last assertion of the proposition follows from the fact that a universal effective epimorphism is a descent morphism (2.3), and hence the descended object is unique up to unique isomorphism.

We prove (ii)⇒(i). Let $Y' = X'/R'$, and let Y be the descended object. The canonical morphism $p' : X' \to X'/R' = Y'$ is compatible by construction with the descent data: its two inverse images over S'' coincide with the canonical morphism $X'' \to X''/R''$. It is therefore the inverse image over S' of an S-morphism $p : X \to Y$. Since p' belongs to (M), the assumption on $S' \to S$ implies that p also belongs to (M). Moreover, because p' is compatible with the equivalence relation R', the morphism p is compatible with R, again because a universal effective epimorphism is a descent morphism. Hence there is a morphism

$$R \longrightarrow X \times_Y X.$$

To prove that R is (M)-effective and that Y is isomorphic to X/R, it suffices to prove that this morphism is an isomorphism. After base change from S to S', it becomes an isomorphism because R' is effective; it is therefore an isomorphism for the same reason as in 2.4.

Notice that the hypothesis in the proposition is satisfied if (M) is the family (M_0) of universal effective epimorphisms and if $\mathscr{C}$ has fiber products (1.10). We obtain the following.

Corollary 3.5.3. *Suppose that $\mathscr{C}$ has fiber products (fiber products over S would suffice). Let R be an equivalence relation in X over S, and let $S' \to S$ be a universal effective epimorphism. The following conditions are equivalent:*

(i) *R is universally effective in X;*

(ii) *R' is universally effective in X', and the canonical descent datum on X'/R' is effective.*

When these conditions hold, the object descended from X'/R' is canonically isomorphic to X/R.

4. Topologies and Sheaves

The notion of a sieve, and the presentation of the notion of a topology adopted here (4.2.1), which is more intrinsic and in many respects more convenient than the presentation by covering families in [MA], are due to J. Giraud [AS].

4.1. Sieves

Definition 4.1.1. *A* sieve *of the category $\mathscr{C}$ is a subfunctor C of the final functor*

$$e : \mathscr{C}^\circ \longrightarrow \mathbf{Set}.$$

To every sieve C of $\mathscr{C}$ one associates the set $E(C)$ of objects X of $\mathscr{C}$ such that $C(X) \neq \varnothing$, that is, such that the structural morphism $X \to e$ factors through C. Thus

$$\begin{cases} X \in E(C) & \Longleftrightarrow \quad C(X) = e(X) = \{\varnothing\}, \\ X \notin E(C) & \Longleftrightarrow \quad C(X) = \varnothing. \end{cases} \tag{+}$$

The set $E = E(C)$ has the following property:

$$X \in E \text{ and } \operatorname{Hom}(Y, X) \neq \varnothing \quad \Longrightarrow \quad Y \in E. \tag{++}$$

(If the set $\operatorname{Ob}\mathscr{C}$ is endowed with its natural preorder, with Y dominating X when there is an arrow from Y to X, then the sets E satisfying (++) are precisely the subsets of $\operatorname{Ob}\mathscr{C}$ that contain every object dominating one of their elements.)[17]

Conversely, if E is a subset of $\operatorname{Ob}\mathscr{C}$ having property (++), then E can be written uniquely in the form $E(C)$, with C defined by the formulas (+). Thus sieves of $\mathscr{C}$ correspond bijectively to subsets of $\operatorname{Ob}\mathscr{C}$ satisfying (++). By abuse of language, we shall sometimes call the set $E(C)$ itself a sieve of $\mathscr{C}$.

Let C and C' be two sieves of $\mathscr{C}$.[18] Since both are subfunctors of the final functor e, it is equivalent to say that C dominates C' (for the domination relation in $\operatorname{Ob}\widehat{\mathscr{C}}$), that C is a subfunctor of C', or that $E(C) \subset E(C')$. In this case we say that C is *finer* than C'. This is an order relation on the set of sieves of $\mathscr{C}$. Moreover,

$$E(C) \cap E(C') = E(C \times C'),$$

so the set of sieves of $\mathscr{C}$ is filtered for the order relation "is finer than."

Every subset E of $\operatorname{Ob}\widehat{\mathscr{C}}$, and in particular every subset of $\operatorname{Ob}\mathscr{C}$, defines a sieve $C(E)$: the set of $X \in \operatorname{Ob}\mathscr{C}$ such that $F(X) \neq \varnothing$ for at least one $F \in E$ satisfies (++), and hence defines the desired sieve.

This sieve may also be defined as the image of the family of morphisms $\{F \to e \mid F \in E\}$, in the sense of the following definition.

Definition 4.1.2. *Let $\{F_i \to F\}$ be a family of morphisms of $\widehat{\mathscr{C}}$ with common target F. The* image *of this family is the subfunctor of F defined by*

$$S \longmapsto \bigcup_i \operatorname{Im} F_i(S) \subset F(S).$$

Proposition 4.1.3. *Formation of the image commutes with base change. In the preceding notation let I be the image of the family $\{F_i \to F\}$. For every morphism $G \to F$ in $\widehat{\mathscr{C}}$, the image of the family*

$$\{F_i \times_F G \longrightarrow G\}$$

is the subfunctor $I \times_F G$ of G.

Definition 4.1.4.0.[19] *Let C be a sieve of $\mathscr{C}$. If E is a subset of $\operatorname{Ob}\mathscr{C}$ such that $C(E) = C$, then E is called a* base *of C. Every sieve C has a base, for example $E(C)$.*

We now describe $\operatorname{Hom}(C, F)$, where C is a sieve of $\mathscr{C}$ and $F \in \operatorname{Ob}\widehat{\mathscr{C}}$, by means of a base $\{S_i\}$ of C. For every pair (i, j) there is a diagram in $\widehat{\mathscr{C}}$:

$$\begin{array}{ccc} S_i \times S_j & \longrightarrow & S_i \\ \downarrow & & \downarrow \\ S_j & \longrightarrow & e. \end{array}$$

[17] Editor's note: Here "dominating" is meant in the sense of the preorder just mentioned: Y dominates X if there is an arrow $Y \to X$. On the other hand, if X, Y are two subobjects of an object Z, one says (cf. 2.4) that Y contains X if $X \subset Y$. To avoid any ambiguity between these usages, the sequel uses "dominating" in the first case and "containing" in the second.

[18] Editor's note: The following sentence has been expanded.

[19] Editor's note: The numbering 4.1.4.0 has been added to highlight this definition.

It gives a diagram of sets

$$\Gamma(F) = \mathrm{Hom}(e, F) \xrightarrow{\sigma} \prod_i \mathrm{Hom}(S_i, F) \overset{\tau_1}{\underset{\tau_2}{\rightrightarrows}} \prod_{i,j} \mathrm{Hom}(S_i \times S_j, F),$$

with $\tau_1\sigma = \tau_2\sigma$. Hence there is a morphism

$$\mathrm{Hom}(e, F) \longrightarrow \mathrm{Ker}\left(\prod_i \mathrm{Hom}(S_i, F) \rightrightarrows \prod_{i,j} \mathrm{Hom}(S_i \times S_j, F)\right).$$

One checks immediately:

Proposition 4.1.4. *There is an isomorphism, functorial in F,*

$$\mathrm{Hom}(C, F) \xrightarrow{\sim} \mathrm{Ker}\left(\prod_i \mathrm{Hom}(S_i, F) \rightrightarrows \prod_{i,j} \mathrm{Hom}(S_i \times S_j, F)\right),$$

for which the diagram

$$\begin{array}{ccc} \mathrm{Hom}(e, F) & \longrightarrow & \mathrm{Ker}\Big(\prod_i \mathrm{Hom}(S_i, F) \rightrightarrows \prod_{i,j} \mathrm{Hom}(S_i \times S_j, F)\Big) \\ \Big\| & & \uparrow{\scriptstyle\sim} \\ \mathrm{Hom}(e, F) & \longrightarrow & \mathrm{Hom}(C, F) \end{array}$$

is commutative; the bottom arrow is induced by the canonical morphism $C \to e$.

Corollary 4.1.5. *Suppose that the fiber products $S_i \times S_j$ are representable, for example that the S_i are squarable. Then, for every $F \in \mathrm{Ob}\,\widehat{\mathscr{C}}$, there is an isomorphism*

$$\mathrm{Hom}(C, F) \xrightarrow{\sim} \mathrm{Ker}\left(\prod_i F(S_i) \rightrightarrows \prod_{i,j} F(S_i \times S_j)\right).$$

Remark 4.1.6. Let R be a sieve of $\mathscr{C}$, let $\overline{R}$ denote the full subcategory of $\mathscr{C}$ whose object set is $E(R)$, and let

$$i_R : \overline{R} \longrightarrow \mathscr{C}$$

be the inclusion functor. There is an isomorphism, functorial in $F \in \mathrm{Ob}\,\widehat{\mathscr{C}}$,

$$\mathrm{Hom}(R, F) \xrightarrow{\sim} \Gamma(F \circ i_R),$$

for which the diagram

$$\begin{array}{ccc} \mathrm{Hom}(e, F) & \longrightarrow & \mathrm{Hom}(R, F) \\ \downarrow{\scriptstyle\sim} & & \downarrow{\scriptstyle\sim} \\ \Gamma(F) & \longrightarrow & \Gamma(F \circ i_R) \end{array}$$

is commutative; the bottom arrow is induced by i_R.

Definition 4.1.7. *Let $\mathscr{C}$ be a category. A* sieve of the object S *of $\mathscr{C}$ is a sieve of the category $\mathscr{C}_{/S}$.*

A sieve of S is therefore a subobject of S in $\widehat{\mathscr{C}}$. It corresponds canonically to a subset of $\mathrm{Ob}\,\mathscr{C}_{/S}$ that contains the source of every arrow whose target it contains. By abuse of language, such a subset will also be called a sieve of S.

4.2. Topologies: Definitions

Definition 4.2.1. *A* topology *on a category $\mathscr{C}$ consists, for every object S of $\mathscr{C}$, of a set $J(S)$ of sieves of S, called* covering sieves *or* refinements *of S, satisfying the following axioms:*

(T 1) *For every refinement R of S and every morphism $T \to S$, the sieve $R \times_S T$ of T is covering ("stability under base change").*

(T 2) *If R, C are two sieves of S, if R is covering, and if for every $T \in \mathrm{Ob}\,\mathscr{C}$ and every morphism $T \to R$, the sieve $C \times_S T$ of T is covering, then C is a refinement of S.*[20]

(T 3) *If $C \supset R$ are two sieves of S and R is covering, then C is covering.*

(T 4) *For every S, S is a refinement of S.*

These axioms may be reformulated as follows. Suppose that a topology $S \mapsto J(S)$ on $\mathscr{C}$ is given. For every $F \in \mathrm{Ob}\,\widehat{\mathscr{C}}$, let $J(F)$ be the set of subfunctors R of F such that, for every morphism $T \to F$ in $\widehat{\mathscr{C}}$ with T representable, the sieve $R \times_F T$ of T is covering. By (T 1), this notation agrees with the preceding one. An $R \in J(F)$ is again called a *refinement* of F. The preceding axioms immediately imply:

($T'0$) If $F \supset G$ are objects of $\widehat{\mathscr{C}}$, and if $G \times_F S \in J(S)$ for every $S \in \mathrm{Ob}\,\mathscr{C}$ and every morphism $S \to F$, then $G \in J(F)$.

($T'1$) If $G \in J(F)$ and $H \to F$ is a morphism in $\widehat{\mathscr{C}}$, then $G \times_F H \in J(H)$.

($T'2$) If $F \supset G \supset H$, if $G \in J(F)$, and if $H \in J(G)$, then $H \in J(F)$.

($T'3$) If $F \supset G \supset H$ and $H \in J(F)$, then $G \in J(F)$.

($T'4$) For every $F \in \mathrm{Ob}\,\widehat{\mathscr{C}}$, one has $F \in J(F)$.

Conversely, if for every $F \in \mathrm{Ob}\,\widehat{\mathscr{C}}$ one is given a set $J(F)$ of subobjects of F satisfying $T'0$–$T'4$, then $S \mapsto J(S)$ defines a topology on $\mathscr{C}$, and these two constructions are inverse to each other.

Properties $T'1$, $T'2$, and $T'3$ imply the further property[21]:

($T'5$) If G and H are two subobjects of F and $G, H \in J(F)$, then $G \cap H \in J(F)$.

Thus $J(F)$, ordered by $\supset$, is filtered. This observation will be useful below.

4.2.2. The topology defined by J is said to be *finer* than the topology defined by J' if $J(S) \supset J'(S)$ for every $S \in \mathrm{Ob}\,\mathscr{C}$. Equivalently, $J(F) \supset J'(F)$ for every $F \in \mathrm{Ob}\,\widehat{\mathscr{C}}$.

Every set of topologies on $\mathscr{C}$ has a greatest lower bound. More precisely, let I be an index set and, for each $i \in I$, let $S \mapsto J_i(S)$ be a topology on $\mathscr{C}$. Then

$$J(S) = \bigcap_{i \in I} J_i(S)$$

defines a topology, and it is the greatest lower bound of the given set.

In particular, suppose that for every $S \in \mathrm{Ob}\,\mathscr{C}$ a set $E(S)$ of sieves of S is given. The *topology generated by these sets* is the least fine topology for which every element of $E(S)$ is a refinement of S.

[20] Editor's note: In other words, if C is covering "locally with respect to the covering sieve R," then C is covering.

[21] Editor's note: $T'1$ and $T'2$ suffice: $G \cap H = G \times_F H$ belongs to $J(H)$ by $T'1$, and hence to $J(F)$ by $T'2$.

Definition 4.2.3. *Let $\{F_i \to F\}$ be a family of morphisms in $\widehat{\mathscr{C}}$, and let $G \subset F$ be its image (4.1.2). The family is* covering *if $G \in J(F)$. A morphism is covering if the one-member family consisting of that morphism is covering.*

This definition applies in particular to an inclusion: a sieve C of S is covering if and only if the canonical morphism $C \to S$ is covering.

Axioms $T'0$–$T'5$ give the following properties of covering families:

(C 0) If $\{F_i \to F\}$ is a family in $\widehat{\mathscr{C}}$ such that every representable base change $\{F_i \times_F S \to S\}$ is covering, then the original family is covering.

(C 1) If $\{F_i \to F\}$ is covering and $G \to F$ is any morphism, then $\{F_i \times_F G \to G\}$ is covering ("stability under base change").

(C 2) If $\{F_i \to F\}$ is covering and, for every i, $\{F_{ij} \to F_i\}$ is covering, then the composite family $\{F_{ij} \to F\}$ is covering ("stability under composition").

(C 3) If $\{G_j \to F\}$ is covering and $\{F_i \to F\}$ is a family such that for every j there is an i for which $G_j \to F$ factors through $F_i \to F$, then $\{F_i \to F\}$ is covering ("saturation").

(C 4) Every one-member family consisting of an isomorphism is covering.

Notice that (C 2) and (C 3) also imply:

(C 5) If $\{F_i \to F\}$ is a family for which there exists a covering family $\{G_j \to F\}$ such that $\{F_i \times_F G_j \to G_j\}$ is covering for every j, then $\{F_i \to F\}$ is covering ("a locally covering family is covering").

4.2.4. Conversely, let $\mathscr{C}$ be a category having fiber products, and suppose that for every $S \in \mathrm{Ob}\,\mathscr{C}$ one is given a set of families of morphisms in $\mathscr{C}$ with target S, called covering families, satisfying (C 1)–(C 4), and hence also (C 5). Let $J(S)$ be the set of sieves of S having a covering base (or, equivalently by (C 3), all of whose bases are covering). Then $S \mapsto J(S)$ defines a topology on $\mathscr{C}$, and the two preceding constructions are inverse to one another.

In applications it is inconvenient to consider all covering families, since one often has a fairly simple description of a "sufficient" number of them. This leads to the following definitions.

Definition 4.2.5.0.[22] *Let $\mathscr{C}$ be a category, and suppose given, for every $S \in \mathrm{Ob}\,\mathscr{C}$, a set $P(S)$ of families of morphisms in $\mathscr{C}$ with target S. The* topology generated by P *is the least fine topology for which the given families are covering.*

Definition 4.2.5. *A* pretopology *on a category $\mathscr{C}$ consists, for every $S \in \mathrm{Ob}\,\mathscr{C}$, of a set $R(S)$ of families $\{S_i \to S\}$, called* covering for the pretopology*, satisfying:*

(P 1) *For every $\{S_i \to S\} \in R(S)$ and every morphism $T \to S$, the fiber products $S_i \times_S T$ exist and $\{S_i \times_S T \to T\} \in R(T)$.*

(P 2) *If $\{S_i \to S\} \in R(S)$ and, for each i, $\{T_{ij} \to S_i\} \in R(S_i)$, then the composite family $\{T_{ij} \to S\}$ belongs to $R(S)$.*

(P 3) *Every one-member family consisting of an isomorphism is covering.*

[22] Editor's note: This definition, which appeared after 4.2.5 in the original, has been placed here because it will be used in 6.2.1 in a slightly more general setting than that of 4.2.5.

Proposition 4.2.6. *For each S, let $J(S)$ be the set of sieves of S that are covering for the topology generated by the pretopology R. Let $J_R(S)$ be the subset of $J(S)$ consisting of the sieves defined by the families in $R(S)$. Then $J_R(S)$ is cofinal in $J(S)$: every refinement of S contains a sieve defined by a family in $R(S)$.*

For every S, let $J'(S)$ be the set of sieves of S containing a sieve in $J_R(S)$. Clearly $J'(S) \subset J(S)$. To prove $J(S) = J'(S)$, it suffices to prove that the $J'(S)$ define a topology, that is, satisfy (T 1)–(T 4). Axioms (T 1), (T 3), and (T 4) are immediate; only (T 2) remains.[23]

Let $U \in J'(S)$, let C be a sieve of S, and suppose that $C \times_S T \in J'(T)$ for every $T \to U$. We must prove $C \in J'(S)$. By definition, U contains a refinement U' defined by a family $\{S_i \to S\} \in R(S)$. Since (T 3) has already been verified, it is enough to prove $U' \cap C \in J'(S)$; hence we may suppose $U = U'$. By hypothesis, $C \times_S S_i \in J'(S_i)$ for every i. Thus for each i there is a covering family $\{T_{ij} \to S_i\} \in R(S_i)$ such that $T_{ij} \to S_i$ factors through $C \times_S S_i \to S_i$. Consequently $T_{ij} \to S$ factors through $C \to S$. Hence C contains the sieve defined by the composite family $\{T_{ij} \to S\}$, and (P 2) completes the proof.

Axioms (P 1)–(P 3) are those of [MA]. Because pretopologies are so useful in practice, we shall interpret every important result in terms of a pretopology defining the topology under consideration.

Remark 4.2.7. One may introduce a slightly more general notion: for every S, one specifies a set of covering families satisfying (P 1), (P 3), and the conclusion of Proposition 4.2.6. This occurs in particular when the given families satisfy (P 1), (P 3), and (C 5). See [D].

Definition 4.2.8. *Let $\mathscr{C}$ be equipped with a topology, let S be an object of $\mathscr{C}$, and let $P(S')$ be a condition on an argument $S' \in \operatorname{Ob}\mathscr{C}_{/S}$. Suppose that $\operatorname{Hom}(S'', S') \neq \varnothing$ implies $P(S') \Rightarrow P(S'')$. The condition P is said to hold* locally *on S for the topology under consideration if the following equivalent conditions hold:*

(i) *The set of $S' \to S$ for which $P(S')$ holds is a refinement of S.*

(ii) *There is a refinement of S such that $P(S')$ holds for every S' in that refinement.*

(iii) *If the topology is defined by a pretopology, there is a covering family for that pretopology such that $P(S')$ holds for every S' in the family.*

Example 4.2.9. *Let $f : X \to Y$ be an S-morphism. We say that f is* locally an isomorphism *if there is a covering family $\{S_i \to S\}$ such that $f \times_S S_i$ is an isomorphism for every i. Equivalently, there is a refinement R of S such that $X(T) \to Y(T)$ is an isomorphism for every $T \to R$.*

Many other examples of this “local” language will appear below.

4.3. Presheaves, Sheaves, and the Sheaf Associated to a Presheaf

Definition 4.3.1. *Let $\mathscr{C}$ be a category. A* presheaf of sets *on $\mathscr{C}$ is a contravariant functor from $\mathscr{C}$ to the category of sets. The category*

$$\widehat{\mathscr{C}} = \operatorname{Hom}(\mathscr{C}^\circ, \mathbf{Set})$$

is called the category of presheaves *on $\mathscr{C}$. If $\mathscr{C}$ is equipped with a topology, a presheaf P is* separated *(respectively, is a* sheaf*) if, for every $S \in \operatorname{Ob}\mathscr{C}$ and every $R \in J(S)$, the canonical map*

$$P(S) = \operatorname{Hom}(S, P) \longrightarrow \operatorname{Hom}(R, P) \tag{+}$$

[23] Editor's note: Typographical errors in the original have been corrected in what follows.

is injective (respectively, bijective). The full subcategory of $\widehat{\mathscr{C}}$ *whose objects are the sheaves is called the* category of sheaves *and is denoted by* $\widetilde{\mathscr{C}}$.[24]

Proposition 4.3.2. *Let* P *be a separated presheaf (respectively, a sheaf). For every* $H \in \operatorname{Ob}\widehat{\mathscr{C}}$ *and every* $R \in J(H)$*, the canonical map*

$$\operatorname{Hom}(H, P) \longrightarrow \operatorname{Hom}(R, P) \tag{+}$$

is injective (respectively, bijective).

Let P be separated, let H be a presheaf, let $R \in J(H)$, and let $u, v : H \to P$ satisfy $uj = vj$. For every $f : S \to H$, with $S \in \operatorname{Ob}\mathscr{C}$, the sieve $R \times_H S$ is a refinement of S, and $ufj_S = vfj_S$. This is expressed by the diagram

$$\begin{tikzcd}
R \times_H S \arrow[r, "j_S"] \arrow[d, "f_R"'] & S \arrow[d, "f"] & \\
R \arrow[r, "j"] & H \arrow[r, shift left, "u"] \arrow[r, shift right, "v"'] & P.
\end{tikzcd}$$

Since P is separated, it follows that $uf = vf$. This holds for every representable S, so $u = v$.

Now suppose that P is a sheaf. Let $g : R \to P$; we show that it factors through H. For every $f : S \to H$, with $S \in \operatorname{Ob}\mathscr{C}$, the morphism $g \circ f_R : R \times_H S \to P$ factors uniquely through S, thereby defining a morphism $h : S \to P$. By uniqueness, h is plainly functorial in f:

$$\begin{tikzcd}
R \times_H S \arrow[r, "f_R"] \arrow[d, "j_S"'] & R \arrow[r, "g"] \arrow[d, "j"'] & P \\
S \arrow[r, "f"] \arrow[rru, bend right=40, "h"'] & H \arrow[ru, dashed] &
\end{tikzcd}$$

Thus, for every S, we have defined a map $H(S) \to P(S)$ functorial in S, and hence a morphism $H \to P$ with the required properties.

Corollary 4.3.2.1.[25] *Let* R *and* F *be two sheaves. If* R *is a refinement of* F*, then* $R = F$*.*

Indeed, suppose that R is a refinement of F, and write $j : R \hookrightarrow F$ for the inclusion. By 4.3.2, $\operatorname{Hom}(F, R) = \operatorname{End}(R)$; hence there is a $\pi : F \to R$ such that $\pi j = \mathrm{id}_R$. Likewise, $\operatorname{End}(F) = \operatorname{Hom}(R, F)$, and $j\pi j = j$ implies $j\pi = \mathrm{id}_F$. Thus j is an isomorphism.

Proposition 4.3.3 ([AS], 1.3). *Let* $\mathscr{C}$ *be a category and* P *a presheaf on* $\mathscr{C}$*. For every* $S \in \operatorname{Ob}\mathscr{C}$*, let* $J(S)$ *be the set of sieves* R *of* S *such that, for every* $T \to S$*, the map*

$$\operatorname{Hom}(T, P) \longrightarrow \operatorname{Hom}(R \times_S T, P) \tag{+}$$

is injective (respectively, bijective). Then the $J(S)$ *define a topology on* $\mathscr{C}$*; that is, they satisfy (T 1)–(T 4).*

Corollary 4.3.4. *For every* $S \in \operatorname{Ob}\mathscr{C}$*, let* $K(S)$ *be a family of sieves satisfying (T 1), and let* P *be a presheaf on* $\mathscr{C}$*. Then* P *is separated (respectively, a sheaf) for the topology generated by the* $K(S)$ *if and only if, for every* $S \in \operatorname{Ob}\mathscr{C}$ *and every* $R \in K(S)$*, the canonical map*

$$\operatorname{Hom}(S, P) \longrightarrow \operatorname{Hom}(R, P) \tag{+}$$

[24] Editor's note: If Q is a subpresheaf of a separated presheaf P, then Q is separated. Indeed, for every sieve R of S, the composite $Q(S) \hookrightarrow P(S) \hookrightarrow P(R)$ is injective and factors through $Q(S) \to Q(R)$.

[25] Editor's note: This corollary has been added.

is injective (respectively, bijective).

Corollary 4.3.5. *For each $S \in \operatorname{Ob}\mathscr{C}$, let $R(S)$ be a set of families of morphisms in $\mathscr{C}$ with target S, satisfying (P 1), for example a pretopology. Let P be a presheaf on $\mathscr{C}$. Then P is separated (respectively, a sheaf) for the topology generated by R if and only if, for every $S \in \operatorname{Ob}\mathscr{C}$ and every family $\{S_i \to S\} \in R(S)$, the map*

$$P(S) \longrightarrow \prod_i P(S_i)$$

is injective (respectively, the diagram

$$P(S) \longrightarrow \prod_i P(S_i) \rightrightarrows \prod_{i,j} P(S_i \times_S S_j)$$

is exact).

Definition 4.3.6. *The* canonical topology *on a category $\mathscr{C}$ is the finest topology for which every representable functor is a sheaf.*

Corollary 4.3.7. *A sieve R of S is a refinement for the canonical topology if and only if, for every morphism $T \to S$ in $\mathscr{C}$ and every $X \in \operatorname{Ob}\mathscr{C}$, the canonical map*

$$\operatorname{Hom}(T, X) \longrightarrow \operatorname{Hom}(R \times_S T, X)$$

is bijective.

Definition 4.3.8. *A sieve covering for the canonical topology is called a* universal effective epimorphic sieve*.*

Corollary 4.3.9. *A universal effective epimorphic family defines a universal effective epimorphic sieve. Conversely, every squarable family defining a universal effective epimorphic sieve is universal effective epimorphic.*

Return now to a category $\mathscr{C}$ equipped with an arbitrary topology and to the construction of the sheaf associated to a presheaf P. Let S be an object of $\mathscr{C}$. If $R \supset R'$ are two refinements of S, there is a diagram

$$\begin{tikzcd} \operatorname{Hom}(S, P) \arrow[r] \arrow[dr] & \operatorname{Hom}(R, P) \arrow[d] \\ & \operatorname{Hom}(R', P). \end{tikzcd}$$

As already observed, the ordered set $J(S)$ is filtered. Since $S \in J(S)$, there is an evident morphism

$$\operatorname{Hom}(S, P) \longrightarrow \varinjlim_{R \in J(S)} \operatorname{Hom}(R, P).$$

Definition 4.3.10.0.[26] Set

$$\check{H}^0(S, P) = \varinjlim_{R \in J(S)} \operatorname{Hom}(R, P).$$

[26]Editor's note: The numbering 4.3.10.0 has been added for later reference. Moreover, it follows from the definition that if $Q \to P$ is a monomorphism, then the same is true of $LP \to LQ$; thus L "preserves monomorphisms" (see also 4.3.16 for the more general statement that L "commutes with finite inverse limits").

One checks that $\check{H}^0(S,P)$ depends functorially on S, and hence defines a functor LP by

$$\mathrm{Hom}(S, LP) = \check{H}^0(S,P) = \varinjlim_{R\in J(S)} \mathrm{Hom}(R,P). \tag{++}$$

By construction there are morphisms

$$\ell_P : P \longrightarrow LP, \qquad z_R : \mathrm{Hom}(R,P) \longrightarrow \mathrm{Hom}(S, LP).$$

Lemma 4.3.10.

(i) *For every refinement R of S and every $u : R \to P$, the diagram*

$$\begin{tikzcd} P \arrow[r, "\ell_P"] & LP \\ R \arrow[u, "u"] \arrow[r, "i_R"] & S \arrow[u, "z_R(u)"'] \end{tikzcd}$$

is commutative.

(ii) *For every morphism $v : S \to LP$, there exist a refinement R of S and a morphism $u : R \to P$ such that $v = z_R(u)$.*

(iii) *Let Q be a functor and let $u, v : Q \to P$ satisfy $\ell_P u = \ell_P v$. Then the kernel of the pair (u,v) is a refinement of Q.*

(iv) *Let $u : R \to P$ and $u' : R' \to P$. The equality $z_R(u) = z_{R'}(u')$ holds if and only if there is a refinement $R'' \subset R \cap R'$ of S on which u and u' coincide.*

Proof. For (i), we must prove $z_R(u) i_R = \ell_P u$. It is enough to compare the composites of these two morphisms with every morphism $g : T \to R$, where T is representable. Put $f = i_R g$ and $R' = R \times_S T$.[27] We have

$$\begin{tikzcd} P \arrow[r, "\ell_P"] & LP \\ R \arrow[u, "u"] \arrow[r, "i_R"] & S \arrow[u, "z_R(u)"'] \\ R' \arrow[u, "p"] \arrow[r, "i_{R'}", "\sim"'] & T. \arrow[u, "f"'] \arrow[ul, "g"'] \end{tikzcd}$$

By the definition of ℓ_P, $\ell_P u g = z_R(u) f$; this is the special case of the assertion in which i_R is an isomorphism. But $z_R(u) f = z_R(u) i_R g$, proving (i).

Assertions (ii) and (iv) merely express the definition of $\mathrm{Hom}(S, LP)$ as a direct limit.

For (iii), let K be the kernel of (u,v). For every morphism $f : S \to Q$, with S representable, $K \times_Q S$ is a subfunctor of the kernel of $uf, vf : S \to P$. By $T'0$, it is therefore enough to prove the assertion when $Q = S$ is representable. In that case, (ii) and (iv) show that K contains a refinement of S, and hence is itself a refinement of S.

[27] Editor's note: Let $p : R' \to R$ be the projection. Since i_R is a monomorphism, $p = g i_{R'}$. Let g' be the section of $i_{R'}$ defined by g. Then $pg' = g$, hence $pg' i_{R'} = g i_{R'} = p$. This implies $g' i_{R'} = \mathrm{id}_{R'}$, so $i_{R'} : R' \to T$ is an isomorphism with inverse g'.

One finally checks that $P \mapsto LP$ defines a functor

$$L : \widehat{\mathscr{C}} \longrightarrow \widehat{\mathscr{C}},$$

and that $P \mapsto \ell_P$ defines a morphism of functors

$$\ell : \mathrm{Id}_{\widehat{\mathscr{C}}} \longrightarrow L.$$

We now state the essential result.

Proposition 4.3.11.

(i) *For every presheaf P, the presheaf LP is separated and $\ell_P : P \to LP$ is covering (4.2.3).*

(ii) *If P is a sheaf, then $P \to LP$ is an isomorphism.*

(iii) *For every presheaf P and every separated presheaf (respectively, sheaf) F, the map*

$$\mathrm{Hom}(\ell_P, F) : \mathrm{Hom}(LP, F) \longrightarrow \mathrm{Hom}(P, F)$$

is injective (respectively, bijective).

(iv) *If P is separated, then $\ell_P : P \to LP$ is a covering monomorphism (so P is a refinement of LP), and LP is a sheaf.*

Proof. (i) First, $P \to LP$ is covering. Indeed, for every $v : S \to LP$, Lemma 4.3.10(i),(ii) supplies a refinement R of S and a commutative diagram (the map from R to the fiber product is induced by $v' : R \to P$ and $i_R : R \to S$)

$$\begin{tikzcd}
& P \arrow[r, "\ell_P"] & LP \\
& P \times_{LP} S \arrow[u] \arrow[r] & S \arrow[u, "v"'] \\
R \arrow[uur, bend left, "v'"] \arrow[ur] \arrow[urr, bend right, "i_R"'] & &
\end{tikzcd}$$

so $P \times_{LP} S \to S$ is covering. By (C 0), $P \to LP$ is covering.

Now suppose that two morphisms $v_1, v_2 : S \to LP$ induce the same morphism on a refinement R of S. There are refinements R_i, $i = 1, 2$, and morphisms $u_i : R_i \to P$ with $z_{R_i}(u_i) = v_i$. Replacing R by a smaller refinement, we may suppose $R_1 = R_2 = R$. The commutative diagram of Lemma 4.3.10(i) then gives $\ell_P u_1 = \ell_P u_2$. By 4.3.10(iii), u_1 and u_2 coincide on a refinement of R, hence on a refinement of S; by 4.3.10(iv), $z_R(u_1) = z_R(u_2)$. Thus LP is separated.

(ii) This is clear: if P is a sheaf, then $\mathrm{Hom}(S, P) \to \mathrm{Hom}(R, P)$ is already an isomorphism for every refinement R of S.

(iii) Let $u, v : LP \to F$ satisfy $u\ell_P = v\ell_P$. To show $u = v$, it is enough to prove $uf = vf$ for every $f : S \to LP$ with S representable. There are a refinement R of S and a morphism $g : R \to P$ with $f = z_R(g)$. The morphisms uf and vf agree on R, where both equal $u\ell_P g = v\ell_P g$. If F is separated, they therefore agree on S, proving injectivity. If F is a sheaf, the commutative diagram

$$\begin{array}{ccc}
P & \xrightarrow{\ell_P} & LP \\
\big\downarrow & & \big\downarrow \\
F & \xrightarrow{\sim} & LF
\end{array}$$

shows that $\operatorname{Hom}(\ell_P, F)$ is also surjective.

(iv) Suppose that P is separated. To show that $P \to LP$ is a monomorphism, it is enough to consider $u, v : S \to P$, with S representable, satisfying $\ell_P u = \ell_P v$. By 4.3.10(iii), u and v agree on a refinement of S, and therefore agree because P is separated. Thus $P \to LP$ is a monomorphism; since it is covering by (i), P is a refinement of LP.

It remains to show that LP is a sheaf. We already know from (i) that it is separated. Let $S \in \operatorname{Ob}\mathscr{C}$, let R be a refinement of S, and let $h : R \to LP$. We must find $u : S \to LP$ with $ui_R = h$. Since P is a refinement of LP, $R' = P \times_{LP} R$ is a refinement of R, and hence of S. Put $u = z_{R'}(h')$; the situation is

$$\begin{tikzcd} P \arrow[r, "\ell_P"] & LP & \\ R' \arrow[u, "h'"] \arrow[r, "j"] & R \arrow[u, "h"] \arrow[r, "i_R"] & S. \arrow[ul, "u"'] \end{tikzcd}$$

Here the diagonal arrow is $u : S \to LP$. We have $ui_{R'} = \ell_P h' = hj$, hence $ui_R j = hj$. Since R' is a refinement of R and LP is separated, Proposition 4.3.2 gives $ui_R = h$.

Corollary 4.3.12.[28] *Let F be a sheaf and let R be a subobject of F in $\widehat{\mathscr{C}}$. Then R is a separated presheaf, $\ell_R : R \to LR$ is a covering monomorphism, and there is a commutative diagram*

$$\begin{tikzcd} R \arrow[r, "i"] \arrow[dr, "\ell_R"'] & F \\ & LR. \arrow[u, "j"'] \end{tikzcd}$$

Consequently, R is a refinement of F if and only if j is an isomorphism.

We already observed that R is separated and that j is a monomorphism; see editor notes (24) and (26). By Proposition 4.3.11(iv), ℓ_R is a covering monomorphism. Thus, if j is an isomorphism, R is a refinement of F. Conversely, if i is covering, then so is j, and Proposition 4.3.2.1 shows that j is an isomorphism.

Remark 4.3.13. If $J'(S)$ is a cofinal subset of $J(S)$, then

$$\operatorname{Hom}(S, LP) = \varinjlim_{R \in J'(S)} \operatorname{Hom}(R, P).$$

In particular, let $S \mapsto R(S)$ be a pretopology generating the given topology. The functor L can be described using the covering families belonging to $R(S)$. Writing out the preceding formula recovers the construction of [MA].

Let $i : \widetilde{\mathscr{C}} \to \widehat{\mathscr{C}}$ be the inclusion functor. Proposition 4.3.11 gives the following theorem.

Theorem 4.3.14. *There is a unique functor $a : \widehat{\mathscr{C}} \to \widetilde{\mathscr{C}}$ for which the diagram*

$$\begin{tikzcd} \widehat{\mathscr{C}} \arrow[r, "L"] \arrow[d, "a"] & \widehat{\mathscr{C}} \arrow[d, "L"] \\ \widetilde{\mathscr{C}} \arrow[r, "i"] & \widehat{\mathscr{C}} \end{tikzcd}$$

is commutative; that is, $L(L(P))$ is a sheaf for every presheaf P. The functors i and a are adjoint: for every presheaf P and every sheaf F, there is an isomorphism, functorial in P and F,

$$\operatorname{Hom}_{\widehat{\mathscr{C}}}(P, i(F)) \simeq \operatorname{Hom}_{\widehat{\mathscr{C}}}(a(P), F),$$

[28] Editor's note: The statement and proof of this corollary have been expanded.

or simply

$$\mathrm{Hom}(P, F) \simeq \mathrm{Hom}(a(P), F).$$

Definition 4.3.15. *The sheaf $a(P)$ is called the sheaf* associated *to the presheaf P.*

Remark 4.3.16. Since L is constructed using inverse limits and filtered direct limits, it commutes with finite inverse limits.[29]

Moreover, under the identification $L(P\times P) = LP\times LP$, the morphism $\ell_{P\times P}$ is identified with $\ell_P \times \ell_P$. Thus, for example, if P is a presheaf of groups, then LP is canonically a presheaf of groups and the canonical morphism $P \to LP$ is a morphism of groups. The same holds for the functor a. Consequently, if P is a presheaf of groups and F a sheaf of groups, there is an isomorphism

$$\mathrm{Hom}_{\widehat{\mathscr{C}}\text{-gr.}}(P, i(F)) \simeq \mathrm{Hom}_{\widetilde{\mathscr{C}}\text{-gr.}}(a(P), F).$$

See [D] for further details.

4.3.17. Let $\mathscr{V}$ be any category. A *presheaf on $\mathscr{C}$ with values in $\mathscr{V}$* is a contravariant functor from $\mathscr{C}$ to $\mathscr{V}$. Before defining sheaves with values in $\mathscr{V}$, we recall the definition of the inverse limit of a functor. Let $\mathscr{R}$ and $\mathscr{V}$ be categories, and let

$$F : \mathscr{R}^\circ \longrightarrow \mathscr{V}$$

be a contravariant functor. Write $\varprojlim F$ for the object of $\widehat{\mathscr{V}}$ defined by

$$\begin{aligned}(\varprojlim F)(X) &= \mathrm{Hom}_{\widehat{\mathscr{V}}}(X, \varprojlim F)\\ &= \varprojlim_{U\in \mathrm{Ob}\,\mathscr{R}} \mathrm{Hom}_{\mathscr{V}}(F(U), X)\\ &= \mathrm{Hom}(c_X, F),\end{aligned}$$

where X varies over the objects of $\mathscr{V}$, where c_X denotes the contravariant functor from $\mathscr{R}$ to $\mathscr{V}$ that sends every object to X and every arrow to id_X, and where the last Hom is taken in $\mathrm{Hom}(\mathscr{R}^\circ, \mathscr{V})$. If $\mathscr{R}$ has a final object $e_{\mathscr{R}}$, then $\varprojlim F = F(e_{\mathscr{R}})$. If $\mathscr{V} = \mathbf{Set}$, the functor $\varprojlim$ is identified with Γ.

Let S be an object of $\mathscr{C}$ and R a sieve of S. Let $\overline{R}$ be the full subcategory of $\mathscr{C}_{/S}$ whose object set is $E(R)$, and let $i_R : \overline{R} \to \mathscr{C}_{/S}$ be the canonical functor. A presheaf P on $\mathscr{C}$ with values in $\mathscr{V}$ restricts to a functor

$$P_S : (\mathscr{C}_{/S})^\circ \longrightarrow \mathscr{V}.$$

The functor i_R induces a morphism in $\widehat{\mathscr{V}}$:

$$P(S) = P_S(S) = \varprojlim P_S \longrightarrow \varprojlim(P_S \circ i_R).$$

We write

$$\varprojlim_R P = \varprojlim(P_S \circ i_R)$$

for the latter object of $\widehat{\mathscr{V}}$. In view of 4.1.6, Definition 4.3.1 generalizes as follows.

[29] Editor's note: In particular, if K is the kernel of a pair of morphisms of presheaves $u, v : Q \rightrightarrows P$, then LK is the kernel of $Lu, Lv : LQ \rightrightarrows LP$. This will be used in 4.4.5.

Definition 4.3.18. *A presheaf P on $\mathscr{C}$ with values in $\mathscr{V}$ is* separated *(respectively, a* sheaf*) if, for every $S \in \operatorname{Ob}\mathscr{C}$ and every $R \in J(S)$, the canonical morphism in $\widehat{\mathscr{V}}$*

$$P(S) \longrightarrow \varprojlim_{R} P$$

is a monomorphism (respectively, an isomorphism).

When $\mathscr{V}$ is the category **Gr** of groups (or any other category of sets endowed with algebraic structures defined by finite inverse limits), the following notions are equivalent (cf. [D]): a presheaf on $\mathscr{C}$ with values in **Gr** whose underlying presheaf of sets is a sheaf, and a group object in the category of sheaves of sets. With these identifications understood, we shall always regard sheaves with values in a category of sets endowed with algebraic structures defined by finite inverse limits as sheaves of sets equipped, in $\widetilde{\mathscr{C}}$, with the corresponding algebraic structure.

4.4. Exactness Properties of the Category of Sheaves

Theorem 4.4.1.

(i) *Arbitrary inverse limits exist in $\widetilde{\mathscr{C}}$, and "are computed in $\widehat{\mathscr{C}}$"; that is, the inclusion functor*

$$i : \widetilde{\mathscr{C}} \longrightarrow \widehat{\mathscr{C}}$$

commutes with inverse limits. More explicitly, if (X_α) is an inverse system of sheaves, the presheaf

$$\varprojlim_{\alpha} i(X_\alpha) : S \longmapsto \varprojlim_{\alpha} X_\alpha(S)$$

is a sheaf, and

$$i\left(\varprojlim_{\alpha} X_\alpha\right) = \varprojlim_{\alpha} i(X_\alpha).$$

(ii) *Arbitrary direct limits exist in $\widetilde{\mathscr{C}}$. If (X_α) is a direct system of sheaves, then*

$$\varinjlim_{\alpha} X_\alpha = a\left(\varinjlim_{\alpha} i(X_\alpha)\right),$$

where $\varinjlim_\alpha i(X_\alpha)$ is the presheaf direct limit of the $i(X_\alpha)$:

$$\varinjlim_{\alpha} i(X_\alpha) : S \longmapsto \varinjlim_{\alpha} X_\alpha(S).$$

(iii) *The functor*

$$a : \widehat{\mathscr{C}} \longrightarrow \widetilde{\mathscr{C}}$$

commutes with arbitrary direct limits and finite inverse limits.

Assertions (i) and (ii) follow formally from the adjunction formula (4.3.14), and[30] the first assertion of (iii) follows from (ii). The second assertion of (iii) was already noted in 4.3.16.

Scholium 4.4.2. *This theorem makes it possible to use the following method for proving in $\widetilde{\mathscr{C}}$ a statement involving both arbitrary direct limits and finite inverse limits (for example, "every epimorphism is universally effective"; cf. below). One begins by proving the*

[30]Editor's note: the original has been modified here.

corresponding statement in the category of sets, then extends it pointwise to the category of presheaves. One then uses the preceding theorem to pass from the category of presheaves to the category of sheaves. Many examples of this method will occur below (4.4.3, 4.4.6, 4.4.9, and so forth).

Finally, observe that statements concerning the category of presheaves are formally corollaries of the corresponding statements concerning the category of sheaves. It is enough to take the coarsest ("chaotic") topology, namely the topology defined by

$$J(S) = \{S\} \qquad \text{for every } S \in \operatorname{Ob}\mathscr{C};$$

every functor is a sheaf for this topology.

Proposition 4.4.3. *Let $\mathscr{F} = \{F_i \to F\}$ be a family of morphisms of sheaves. The following conditions are equivalent:*

(i) *$\mathscr{F}$ is an epimorphic family;*

(ii) *$\mathscr{F}$ is a universal effective epimorphic family (1.13);*

(iii) *$\mathscr{F}$ is covering (4.2.3);*

(iv) *the sheaf image of $\mathscr{F}$, that is, the sheaf associated with the presheaf image of $\mathscr{F}$ (4.1.2), is F.*

The equivalence of (iii) and (iv) follows from 4.3.12. The remaining equivalences will follow from the next lemmas.

Lemma 4.4.4. *Let $f : X \to Y$ be a monomorphism of sheaves that is also an epimorphism. Then f is an isomorphism.*

The lemma is clear first of all in the category of sets. Let us next prove it in the category of presheaves. Consider the presheaf

$$V : S \longmapsto Y(S) \amalg_{X(S)} Y(S);$$

this is the amalgamated sum of Y and Y over X in the category of presheaves.[31] Let $i_1, i_2 : Y \rightrightarrows V$ be the two coordinate morphisms. If $X \to Y$ is an epimorphism in the category of presheaves, then $i_1 = i_2$. In that case, for every S, the map $X(S) \to Y(S)$ is surjective; since it is also injective, it is bijective, and hence $X \to Y$ is an isomorphism.

Finally, work in the category $\widetilde{\mathscr{C}}$ of sheaves. By 4.4.1(ii), the amalgamated sum in $\widetilde{\mathscr{C}}$ of the sheaves Y and Y over X is the sheaf Z associated with the presheaf V. Consider the diagram

$$X \xrightarrow{f} Y \underset{i_2}{\overset{i_1}{\rightrightarrows}} V \xrightarrow{\tau} Z = a(V).$$

We have $i_1 f = i_2 f$, hence $\tau i_1 f = \tau i_2 f$, and therefore $\tau i_1 = \tau i_2$, since f is an epimorphism in $\widetilde{\mathscr{C}}$. By part (iii) of the following lemma, the presheaf V is separated, so τ is a monomorphism (4.3.11(iv)). Thus $i_1 = i_2$, and we saw above that this implies that $f : X \to Y$ is an isomorphism. This proves Lemma 4.4.4, once it has been verified that V is separated.

Let $R_\varnothing$ denote the presheaf that assigns the empty set to every $S \in \operatorname{Ob}\mathscr{C}$; it is an initial object of $\widehat{\mathscr{C}}$.[32]

Lemma 4.4.5.

[31] Editor's note: the remainder of the proof has been slightly modified.

[32] Editor's note: part (i) of Lemma 4.4.5 has been corrected, and the proofs of all three parts have been given in detail.

(i) *Suppose that* $R_\varnothing \notin J(S)$ *for every* $S \in \operatorname{Ob}\mathscr{C}$. *If* (X_i) *is a family of separated presheaves, then the coproduct presheaf* $\coprod_i X_i$ *is separated.*[33]

(ii) *Consider an equivalence relation in the category of presheaves*

$$X \overset{u}{\underset{v}{\rightrightarrows}} Y,$$

and let $w : Y \to Z$ *be the quotient. If* X *is a sheaf and* Y *is separated, then* Z *is separated.*

(iii) *Consider an amalgamated sum in the category of presheaves, in which* u *and* u' *are monomorphisms:*

$$\begin{array}{ccc} X & \xrightarrow{u} & Y \\ {\scriptstyle u'}\downarrow & & \downarrow \\ Y' & \longrightarrow & V. \end{array}$$

If Y *and* Y' *are separated and* X *is a sheaf, then* V *is separated.*

Proof. For (i), set $X = \coprod_i X_i$. Let $S \in \operatorname{Ob}\mathscr{C}$, let $\tau : R \hookrightarrow S$ be a refinement of S, and let $x_1, x_2 \in X(S)$ satisfy $x_1 \circ \tau = x_2 \circ \tau$. There are indices i, j such that $x_1 \in X_i(S)$ and $x_2 \in X_j(S)$. Since $R \neq R_\varnothing$, there is a morphism $\phi : T \to R$ with $T \in \operatorname{Ob}\mathscr{C}$. Then

$$x_1 \circ \tau \circ \phi = x_2 \circ \tau \circ \phi.$$

Since $X(T)$ is the disjoint union of the $X_k(T)$, it follows that $i = j$. Because X_i is separated and is a subobject of X, the map

$$X_i(S) \longrightarrow X_i(R) \longrightarrow X(R)$$

is injective. Hence $x_1 = x_2$, which proves that X is separated.

We next prove (iii). Let $i : Y \to V$ and $j : Y' \to V$ be the canonical morphisms, and let K be the kernel of the pair

$$Y \times Y' \overset{p}{\underset{q}{\rightrightarrows}} V, \qquad p = i \circ \mathrm{pr}_1, \quad q = j \circ \mathrm{pr}_2 .$$

For $S \in \operatorname{Ob}\mathscr{C}$, the monomorphisms u, u' allow us to identify $X(S)$ with its image in $Y(S)$, respectively in $Y'(S)$. Write $Z(S)$, respectively $Z'(S)$, for the complement. Then $V(S)$ identifies with the disjoint union of $Z(S)$, $Z'(S)$, and $X(S)$. It follows at once that

$$i(S) : Y(S) \longrightarrow V(S), \qquad j(S) : Y'(S) \longrightarrow V(S)$$

are injective and that

$$(u \times u')(S) : X(S) \longrightarrow K(S)$$

is bijective. Consequently i and j are monomorphisms, and $u \times u' : X \to K$ is an isomorphism.

[33] Editor's note: in general, the coproduct of two sheaves F, G is not a sheaf. Indeed, let $S_1, S_2 \in \operatorname{Ob}\mathscr{C}$. Suppose that the coproduct $S = S_1 \coprod S_2$ exists in $\mathscr{C}$, and that the fiber product $S_1 \times_S S_2$ is an initial object $\varnothing$ of $\mathscr{C}$ (cf. I, 1.8). Let R be the sieve of S with basis $\{S_1, S_2\}$. Then $(F \coprod G)(R)$ is the disjoint union of $F(S) \coprod G(S)$ and the sets $F(S_i) \times G(S_j)$, for $i \neq j$; thus $F \coprod G$ is not a sheaf in general. On the other hand, let $\mathscr{C}$ be the category with a single object S and id_S as its only morphism, equipped with the topology defined by $J(S) = \{R_\varnothing, S\}$. Then the only separated presheaves are $R_\varnothing$ and $X = \mathbf{h}_S$ (which is a sheaf), and $X \coprod X$ is not separated.

Set $U = Y \coprod Y'$. For a refinement $\tau : R \hookrightarrow S$, there is a commutative diagram

$$\begin{array}{ccc} U(S) & \xrightarrow{U(\tau)} & U(R) \\ \downarrow & & \downarrow \\ V(S) & \xrightarrow{V(\tau)} & V(R). \end{array}$$

Let $v_1, v_2 \in V(S)$ have the same image in $V(R)$. By the definition of V, they lift to elements $y_1, y_2 \in U(S)$; let z_1, z_2 denote their images in $U(R)$. Then z_1, z_2 have the same image in $V(R)$.

Since $i : Y \to V$ and $j : Y' \to V$ are monomorphisms, the maps $Y(R) \to V(R)$ and $Y'(R) \to V(R)$ are injective. Thus, because Y and Y' are separated, if y_1, y_2 both belong to $Y(S)$, or both belong to $Y'(S)$, then $y_1 = y_2$. Otherwise we may suppose that $y_1 \in Y(S)$ and $y_2 \in Y'(S)$, so that $z_1 \in Y(R)$ and $z_2 \in Y'(R)$. Since $i \circ z_1 = j \circ z_2$, the morphism

$$z_1 \boxtimes z_2 : R \longrightarrow Y \times Y'$$

factors through $K = X$. Moreover, since X is a sheaf, the restriction map identifies $X(S)$ with $X(R)$. Hence there is an $x \in X(S)$ such that

$$u(x) \circ \tau = z_1 = y_1 \circ \tau, \qquad u'(x) \circ \tau = z_2 = y_2 \circ \tau.$$

Since Y and Y' are separated, $u(x) = y_1$ and $u'(x) = y_2$, and therefore $v_1 = v_2$. This proves that V is separated.

Finally, we prove (ii). First observe that the morphism of presheaves

$$X \xrightarrow{u \boxtimes v} K,$$

where K is the kernel of the pair

$$w \circ \mathrm{pr}_i : Y \times Y \rightrightarrows Z,$$

is an isomorphism. Indeed, for every $T \in \operatorname{Ob} \mathscr{C}$, $X(T)$ is an equivalence relation on $Y(T)$, and hence the diagram

$$X(T) \xrightarrow{u \boxtimes v} Y(T) \times Y(T) \overset{w \circ \mathrm{pr}_1}{\underset{w \circ \mathrm{pr}_2}{\rightrightarrows}} Z(T)$$

is exact in the category of sets.

Now let $S \in \operatorname{Ob} \mathscr{C}$, and let $g_1, g_2 : S \to Z$ be two morphisms that agree on a refinement $\tau : R \hookrightarrow S$. By the construction of Z,

$$Z(S) = Y(S)/X(S),$$

so there are morphisms $f_1, f_2 : S \to Y$ such that $wf_i = g_i$. We have $wf_1\tau = wf_2\tau$; by the preceding observation, there is a morphism $\phi : R \to X$ such that

$$u\phi = f_1\tau, \qquad v\phi = f_2\tau.$$

Since X is a sheaf, there is a morphism $\psi : S \to X$ with $\phi = \psi\tau$. Thus, in $Y(R)$,

$$u\psi\tau = f_1\tau, \qquad v\psi\tau = f_2\tau.$$

Because $Y(S) \to Y(R)$ is injective, Y being separated, it follows that $u\psi = f_1$ and $v\psi = f_2$, hence $g_1 = g_2$. Thus Z is separated.

Lemma 4.4.6.[34]

[34] Editor's note: the statement of the lemma and its proof have been given in detail.

(i) *Let $\{F_i \xrightarrow{f_i} F\}$ be a family of morphisms of presheaves, and let $G \subset F$ be its presheaf image. Then the following diagram in $\widehat{\mathscr{C}}$ is exact:*

$$\left(\coprod_i F_i\right) \times_G \left(\coprod_j F_j\right) \underset{\mathrm{pr}_2}{\overset{\mathrm{pr}_1}{\rightrightarrows}} \coprod_i F_i \longrightarrow G.$$

Equivalently, for every presheaf H, the following diagram of sets is exact:

$$\mathrm{Hom}(G, H) \longrightarrow \prod_i \mathrm{Hom}(F_i, H) \underset{q}{\overset{p}{\rightrightarrows}} \prod_{i,j} \mathrm{Hom}(F_i \times_G F_j, H). \tag{*}$$

(ii) *Every covering family of morphisms of sheaves is a universal effective epimorphic family.*

Proof. For (i), let H be a presheaf. The map

$$f^* : \mathrm{Hom}(G, H) \longrightarrow \prod_i \mathrm{Hom}(F_i, H), \qquad \phi \longmapsto (\phi \circ f_i)_i,$$

is injective. Indeed, for each $S \in \mathrm{Ob}\,\mathscr{C}$, the map $\phi(S)$ is determined by the maps $(\phi \circ f_i)(S)$, since the family

$$f_i(S) : F_i(S) \longrightarrow G(S)$$

is surjective. It is clear that the image of f^* is contained in $\ker(p, q)$.

Conversely, let $\phi_i : F_i \to H$ be a family of morphisms such that, for every i, j, the diagram

$$\begin{array}{ccc} F_i \times_G F_j & \longrightarrow & F_j \\ \downarrow & & \downarrow{\scriptstyle \phi_j} \\ F_i & \xrightarrow{\phi_i} & H \end{array}$$

commutes. For every S, the map

$$\coprod_i F_i(S) \longrightarrow H(S)$$

then factors uniquely through a map $\phi(S) : G(S) \to H(S)$. These maps define a morphism $\phi : G \to H$ satisfying $f^*(\phi) = (\phi_i)$. This proves the exactness of $(*)$, and hence (i).

We prove (ii). Since covering families are stable under base change, it is enough to show that every covering family is effective epimorphic. Let $\{F_i \to F\}$ be a covering family of morphisms of sheaves, and let G be its presheaf image. Because the family is covering, the monomorphism $G \hookrightarrow F$ is covering as well; therefore, by 4.3.12,

$$a(G) = F.$$

Moreover, since $G \hookrightarrow F$ is a monomorphism, the fiber products $F_i \times_G F_j$ and $F_i \times_F F_j$ coincide. Part (i) therefore shows that, for every presheaf H, the diagram

$$\mathrm{Hom}(G, H) \longrightarrow \prod_i \mathrm{Hom}(F_i, H) \underset{q}{\overset{p}{\rightrightarrows}} \prod_{i,j} \mathrm{Hom}(F_i \times_F F_j, H) \tag{**}$$

is exact. If, in addition, H is a sheaf, then

$$\mathrm{Hom}(G, H) = \mathrm{Hom}(a(G), H) = \mathrm{Hom}(F, H).$$

Thus $(**)$ shows that $\{F_i \to F\}$ is an effective epimorphic family in the category $\widetilde{\mathscr{C}}$ of sheaves.

Lemma 4.4.7. *Every family of morphisms of sheaves $\{F_i \to F\}$ factors as a covering family $\{F_i \to G\}$ followed by a monomorphism $G \to F$.*

Indeed, it is enough to take G to be the sheaf image of the given family.

Proof of Proposition 4.4.3. We saw in 4.4.6 that (iii) implies (ii), while (ii) evidently implies (i). Finally, let $\{F_i \to F\}$ be an epimorphic family. By Lemma 4.4.7 it factors as a covering family followed by a monomorphism. The latter monomorphism, being dominated[35] by an epimorphic family, is itself an epimorphism; hence it is an isomorphism by 4.4.4.

Remark 4.4.8.[36] Because the presheaf image of the family $\mathscr{F} = \{F_i \to F\}$ is separated, the construction of the associated sheaf shows that the conditions of Proposition 4.4.3 are also equivalent to the following:

(v) for every $S \in \operatorname{Ob}\mathscr{C}$, each $f \in F(S)$ lies locally in the image of $\mathscr{F}$; that is,

(vi) for every $S \in \operatorname{Ob}\mathscr{C}$ and every $f \in F(S)$, the set of morphisms $S' \to S$ such that the image of f in $F(S')$ belongs to the image of one of the $F_i(S')$ is a refinement of S;

(vii) if the topology is defined by a pretopology $S \mapsto \mathscr{R}(S)$, then, for every $S \in \operatorname{Ob}\mathscr{C}$ and every $f \in F(S)$, there is a family $\{S_j \to S\} \in \mathscr{R}(S)$ such that, for each j, the image f_j of f in $F(S_j)$ belongs to the image of one of the $F_i(S_j)$.

Remark 4.4.8 bis. If the sheaf F is representable, the preceding conditions are also equivalent to:

(viii) the set of morphisms $T \to F$, with $T \in \operatorname{Ob}\mathscr{C}$, for which there exist an i and a commutative diagram

$$\begin{array}{ccc} & F_i & \\ & \downarrow & \\ T & \longrightarrow & F \end{array}$$

is a refinement of F.

Indeed, if (viii) holds, the presheaf image of the F_i is dominated by[37] a refinement of F, which implies that the family is covering. Conversely, apply (vi) to $\mathrm{id}_F \in F(F)$.

In pictorial language, this condition says: *locally on F, there is an i for which $F_i \to F$ has a section.* In particular, a morphism

$$G \longrightarrow F,$$

where G is a sheaf and F is a representable sheaf, is covering if and only if it has a section locally on F.

The following lemmas will be useful in $\mathrm{VI_B}$, 8.1 and 8.2.[38]

Lemma 4.4.8.1. *Let $Q \subset P$ be presheaves of groups on $\mathscr{C}$, with Q normal in P. Suppose that P is separated and that Q is a sheaf. Then the presheaf of groups P/Q is separated. Consequently,*

$$a(P/Q) = L(P/Q) = \varinjlim_{R \in J(S)} (P/Q)(R).$$

[35]Editor's note: recall (cf. editor's note 17) that a morphism $g : G \to H$ is said to be *dominated* by a family of morphisms $F_i \to H$ if that family factors through g.

[36]Editor's note: the numbering of the original, which had two paragraphs numbered 4.4.7, has been corrected to agree with later references.

[37]Editor's note: "contains" has been replaced by "is dominated by"; cf. editor's note 17.

[38]Editor's note: Lemmas 4.4.8.1 and 4.4.8.2 have been added.

It is enough to prove the first assertion, since the second follows from it by 4.3.11(iv) and (ii). Let $S \in \operatorname{Ob}\mathscr{C}$, let R be a sieve of S, and let $y \in P(S)/Q(S)$ have identity image in $(P/Q)(R)$. We must show that $y = 1$.

By hypothesis, the morphism $P(S) \to P(R)$ is injective, while $Q(S) \to Q(R)$ is an isomorphism, and the upper row in the following commutative diagram is exact:

$$\begin{array}{ccccccccc} 1 & \longrightarrow & Q(S) & \longrightarrow & P(S) & \longrightarrow & P(S)/Q(S) & \longrightarrow & 1 \\ & & \downarrow{\scriptstyle\sim} & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & Q(R) & \longrightarrow & P(R) & \longrightarrow & (P/Q)(R). & & \end{array}$$

The result follows if the lower row is exact. Let $f : R \to P$ have identity image in $(P/Q)(R)$, and let $\phi : T \to R$ with $T \in \operatorname{Ob}\mathscr{C}$. Then $f \circ \phi$ is the identity in

$$(P/Q)(T) = P(T)/Q(T),$$

that is, $f \circ \phi \in Q(T)$. Hence $\phi \mapsto f \circ \phi$ is a functorial map $R(T) \to Q(T)$, and so defines a morphism of functors $\pi : R \to Q$ such that $\pi(\mathrm{id}_R) = f$. Thus $f \in Q(R)$. This proves the exactness of the lower row and establishes the lemma.

Lemma 4.4.8.2. *Let $H \subset G$ be presheaves of groups on $\mathscr{C}$.*

(i) *If H is normal in G, then $L(H)$ is normal in $L(G)$.*

(ii) *If H is central in G, then $L(H)$ is central in $L(G)$.*

Let $S \in \operatorname{Ob}\mathscr{C}$. We must show (cf. I 2.3.6) that $L(H)(S)$ is normal, respectively central, in $L(G)(S)$. Let

$$h \in L(H)(S), \qquad g \in L(G)(S).$$

There are a sieve R of S and elements $h' \in H(R)$, $g' \in G(R)$ such that

$$h = z_R(h'), \qquad g = z_R(g')$$

in the notation of 4.3.10. Since z_R is a group morphism,

$$ghg^{-1}h^{-1} = z_R(g'h'g'^{-1}h'^{-1}).$$

In case (i),

$$g'h'g'^{-1}h'^{-1} \in H(R),$$

and therefore $ghg^{-1}h^{-1} \in L(H)(S)$. In case (ii), $g'h'g'^{-1}h'^{-1} = 1$, and hence $ghg^{-1}h^{-1} = 1$.

Proposition 4.4.9. *Every equivalence relation in $\widetilde{\mathscr{C}}$ is universally effective (3.3.3). More precisely, let R be a $\widetilde{\mathscr{C}}$-equivalence relation on the sheaf X. Then the sheaf associated with the separated presheaf*

$$i(X)/i(R) : S \longmapsto X(S)/R(S)$$

is a universal effective quotient of X by R.

Let X/R denote the sheaf quotient of X by R, which exists by 4.4.1(ii):

$$X/R = a\big(i(X)/i(R)\big).$$

We must show that $X \to X/R$ is a universal effective epimorphism and that the morphism

$$f : R \longrightarrow X \times_{X/R} X$$

is an isomorphism. The first assertion has already been proved in 4.4.3. As for f, it is obtained by applying the functor a to the morphism

$$i(R) \longrightarrow i(X) \times_{i(X/R)} i(X),$$

or, since $i(X)/i(R)$ is separated by 4.4.5(ii), so that

$$i(X)/i(R) \longrightarrow i(X/R)$$

is a monomorphism, to the canonical morphism

$$i(R) \longrightarrow i(X) \times_{i(X)/i(R)} i(X).$$

Thus we are reduced to proving the same statement in the category of presheaves. But $i(X)/i(R)$ is the presheaf

$$S \longmapsto X(S)/R(S),$$

so the assertion reduces to its set-theoretic analogue, where it is immediate.

Proposition 4.4.10. *Under the hypotheses of 4.4.9, let Y be a subsheaf of X, and write R_Y for the equivalence relation induced on Y by R. Then the canonical morphism (3.1.6)*

$$Y/R_Y \longrightarrow X/R$$

is a monomorphism. It identifies Y/R_Y with a subsheaf of X/R, namely the sheaf image of the composite

$$Y \longrightarrow X \longrightarrow X/R.$$

The morphism of presheaves

$$i(Y)/i(R_Y) = i(Y)/i(R)_{i(Y)} \longrightarrow i(X)/i(R)$$

is a monomorphism. Since the functor a is left exact (4.3.16), it carries monomorphisms to monomorphisms; hence

$$Y/R_Y \longrightarrow X/R$$

is a monomorphism. The final assertion follows from the commutative diagram

$$\begin{array}{ccc} Y & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y/R_Y & \longrightarrow & X/R \end{array}$$

and the fact that $Y \to Y/R_Y$ is covering.

In view of this proposition, we shall always identify Y/R_Y with a subsheaf of X/R.

Proposition 4.4.11. *Let R be a $\widetilde{\mathscr{C}}$-equivalence relation on the sheaf X. For every subsheaf Y of X stable under R, let $Y' = Y/R_Y$, regarded as a subsheaf of $X' = X/R$. Then*

$$Y = Y' \times_{X'} X,$$

and the maps

$$Y \longmapsto Y/R_Y, \qquad Y' \longmapsto Y' \times_{X'} X$$

define a bijective correspondence between the subsheaves Y of X stable under R and the subsheaves Y' of X'.

If Y' is a subsheaf of X', then $Y' \times_{X'} X$ is a subsheaf of X stable under R.[39] If Y' is obtained by passage to the quotient from a subsheaf Y of X, then Y is a subobject of $Y' \times_{X'} X$. It is therefore enough to prove the following: if Y and Y_1 are two subsheaves of X, both stable under R, with Y_1 containing Y, and if their quotients Y/R_Y and Y_1/R_{Y_1} are identical, then $Y = Y_1$.

We may plainly reduce to the case $Y_1 = X$. Write

$$P = i(X)/i(R), \qquad Q = i(Y)/i(R_Y)$$

for the corresponding presheaves. The diagram

$$\begin{array}{ccc} Y & \longrightarrow & Q \\ \big\downarrow & & \big\downarrow \\ X & \longrightarrow & P \end{array}$$

is Cartesian. On the other hand, there is a commutative diagram

$$\begin{array}{ccc} Q & \longrightarrow & a(Q) \\ \big\downarrow & & \big\downarrow{\scriptstyle\sim} \\ P & \longrightarrow & a(P). \end{array}$$

Since $Q \hookrightarrow a(Q)$ is covering by 4.3.11, the monomorphism $Q \hookrightarrow P$ is covering, so Q is a refinement of P. After base change, Y is a refinement of X. Because X and Y are sheaves, 4.3.12 then gives $Y = X$.

4.4.12. In particular, if Y is a subsheaf of X and $Y' = Y/R_Y$, the preceding correspondence defines a subsheaf $\overline{Y}$ of X, stable under R, containing Y, and minimal with these properties. It is called the *saturation* of Y for the equivalence relation R.

4.5. The Case of a Topology Coarser than the Canonical Topology

By 4.3.6 and 4.3.8, the following conditions on a topology $\mathcal{T}$ on $\mathscr{C}$ are equivalent:

(i) $\mathcal{T}$ is coarser than the canonical topology of $\mathscr{C}$;

(ii) every representable presheaf is a sheaf for $\mathcal{T}$;

(iii) every covering sieve for $\mathcal{T}$ is universal effective epimorphic.

If $\mathcal{T}$ is defined by a pretopology $S \mapsto \mathscr{R}(S)$, these conditions are also equivalent to:

(iv) every family belonging to $\mathscr{R}(S)$ is universal effective epimorphic.

When these conditions hold, the canonical functor $\mathscr{C} \to \widehat{\mathscr{C}}$ factors through a functor

$$j_{\mathscr{C}} = j : \mathscr{C} \longrightarrow \widetilde{\mathscr{C}};$$

we also write $j(S) = \widetilde{S}$.[40]

Proposition 4.5.1. *The functor $j : \mathscr{C} \to \widetilde{\mathscr{C}}$ is fully faithful and commutes with arbitrary inverse limits. In particular, it is left exact and therefore preserves algebraic structures defined by finite inverse limits.*

[39] Editor's note: moreover, $(Y' \times_{X'} X)/R = Y'$.

[40] Editor's note: we shall also write $\mathbf{h}_S = \widehat{S}$; cf. the first commutative diagram in 4.5.4.

This follows immediately from the commutative factorization

$$\begin{array}{ccc} & \mathscr{C} & \\ {}^{j}\swarrow & & \searrow{}^{\text{canonical}} \\ \widetilde{\mathscr{C}} & \xrightarrow{i} & \widehat{\mathscr{C}} \end{array}$$

and from 4.4.1(i).

Before giving further properties of j, we must define the topology induced on a category $\mathscr{C}/S$. We no longer suppose here that the given topology is necessarily coarser than the canonical topology. A sieve of an object T in $\mathscr{C}$, together with a morphism $T \to S$, naturally defines a sieve of T in $\mathscr{C}/S$, because the definition of a sieve of T depends only on the category

$$\mathscr{C}/T = (\mathscr{C}/S)/T.$$

For example, if the sieve is defined by a family $\{T_i \to T\}$, its image in $\mathscr{C}/S$ is defined by the same family, now regarded as a family of morphisms in $\mathscr{C}/S$.

Declaring these images of covering sieves to be covering defines a topology on $\mathscr{C}/S$, called the *topology induced* by the given topology. In the terminology of [AS], 2.3, it is the coarsest topology on $\mathscr{C}/S$ for which the canonical functor

$$i_S : \mathscr{C}/S \longrightarrow \mathscr{C}$$

is a comorphism.[41] By definition, the identifications

$$(\mathscr{C}/S)/T = \mathscr{C}/T$$

respect these topologies.

Proposition 4.5.2. *Let S be a representable sheaf on $\mathscr{C}$, and let $F \to S$ be a morphism in $\widehat{\mathscr{C}}$. The presheaf*

$$S' \longmapsto \mathrm{Hom}_S(S', F)$$

on $\mathscr{C}/S$ is separated (respectively, is a sheaf) if and only if F is a separated presheaf (respectively, a sheaf) on $\mathscr{C}$.

For every functor P, one has (I, 1.4.1)

$$\mathrm{Hom}(P, F) = \coprod_{f \in \mathrm{Hom}(P,S)} \mathrm{Hom}_f(P, F).$$

[42] Let $S' \in \mathrm{Ob}(\mathscr{C})$, and let $\eta : C' \to S'$ be a covering sieve. Since S is a sheaf, the map $f \mapsto f \circ \eta$ is a bijection

$$\mathrm{Hom}(S', S) \xrightarrow{\sim} \mathrm{Hom}(C', S).$$

Consequently, the map

$$\mathrm{Hom}_{\widehat{\mathscr{C}}}(S', F) \longrightarrow \mathrm{Hom}_{\widehat{\mathscr{C}}}(C', F)$$

decomposes as the "disjoint union" of the maps

$$\mathrm{Hom}_f(S', F) \longrightarrow \mathrm{Hom}_{f \circ \eta}(C', F).$$

The proposition follows.

[41] Editor's note: that is, for every object $T \to S$ of $\mathscr{C}/S$, every covering sieve R' of $i_S(T \to S) = T$, when regarded as a sieve of the object $(T \to S) \in \mathrm{Ob}(\mathscr{C}/S)$, is covering.

[42] Editor's note: the argument implicit in the original has been expanded below.

Corollary 4.5.3. *If a topology on $\mathscr{C}$ is coarser than the canonical topology, then the topology it induces on $\mathscr{C}/S$ is coarser than the canonical topology of $\mathscr{C}/S$.*

Corollary 4.5.4. *Suppose that the given topology on $\mathscr{C}$ is coarser than the canonical topology. For every $S \in \mathrm{Ob}(\mathscr{C})$, there is an equivalence of categories*

$$\widetilde{\mathscr{C}}_{/\widetilde{S}} \xrightarrow{\sim} \widetilde{\mathscr{C}/S}.$$

The following diagrams commute up to isomorphism; every unlabelled arrow in them is an equivalence:

$$\begin{tikzcd}
\mathscr{C}_{/S} \arrow[r, "(j_{\mathscr{C}})_{/S}"] \arrow[d, equal] & \widetilde{\mathscr{C}}_{/\widetilde{S}} \arrow[r, "(i_{\mathscr{C}})_{/\widetilde{S}}"] \arrow[d, "\wr"] & \widehat{\mathscr{C}}_{/\widehat{S}} \arrow[r, "(a_{\mathscr{C}})_{/\widehat{S}}"] \arrow[d, "\wr"] & \widetilde{\mathscr{C}}_{/\widetilde{S}} \arrow[d, "\wr"] \\
\mathscr{C}_{/S} \arrow[r, "j_{\mathscr{C}/S}"] & \widetilde{\mathscr{C}/S} \arrow[r, "i_{\mathscr{C}/S}"] & \widehat{\mathscr{C}/S} \arrow[r, "a_{\mathscr{C}/S}"] & \widetilde{\mathscr{C}/S}.
\end{tikzcd}$$

If $T \to S$ is an object of $\mathscr{C}/S$, one also has the commutative diagram

$$\begin{tikzcd}
(\widetilde{\mathscr{C}}_{/\widetilde{S}})_{/\widetilde{T}} \arrow[r] \arrow[dd, "\wr"] & (\widetilde{\mathscr{C}/S})_{/\widetilde{T}} \arrow[dr] & \\
& & (\widetilde{\mathscr{C}/S})_{/T}\,. \\
\widetilde{\mathscr{C}}_{/\widetilde{T}} \arrow[r] & \widetilde{\mathscr{C}/T} \arrow[ur, "\sim"] &
\end{tikzcd}$$

The commutativity of the first two squares follows from the definition of the equivalence $\widetilde{\mathscr{C}}_{/\widetilde{S}} \to \widetilde{\mathscr{C}/S}$. To prove the commutativity of the last square, it is enough to observe that the following square commutes:

$$\begin{tikzcd}
\widehat{\mathscr{C}}_{/\widehat{S}} \arrow[r, "L'"] \arrow[d] & \widehat{\mathscr{C}}_{/\widehat{S}} \arrow[d] \\
\widehat{\mathscr{C}/S} \arrow[r, "L''"] & \widehat{\mathscr{C}/S}.
\end{tikzcd}$$

Here L' is the restriction of the functor $L_{\mathscr{C}}$ to $\widehat{\mathscr{C}}_{/\widehat{S}}$, and L'' is the functor $L_{\mathscr{C}/S}$. This is seen immediately by returning to the definition of the functors L, given after 4.3.9. The second diagram is simply the restriction to the categories of sheaves of the corresponding diagram for the categories of presheaves (Exposé I, no. 1), which is commutative.

Scholium 4.5.5. The preceding assertions show that, when the given topology is coarser than the canonical topology, one may identify $\mathscr{C}$ with a full subcategory of $\widetilde{\mathscr{C}}$, itself a full subcategory of $\widehat{\mathscr{C}}$. Under these identifications one may make the customary abuses of language concerning $\mathscr{C} \to \widehat{\mathscr{C}}$, justified by the preceding commutative diagrams. In particular, the first diagram of 4.5.4 shows that the functor a may be used without any special precaution.

In the next subsection we shall see that the identification of $\mathscr{C}$ with a full subcategory of $\widetilde{\mathscr{C}}$, unlike the corresponding identification with $\widehat{\mathscr{C}}$, commutes with the formation of certain direct limits; we shall then explain how this fact is to be used.

From now on, unless expressly stated otherwise, we assume that the given topology is coarser than the canonical topology, and we systematically make the identifications described above.

Proposition 4.5.6. *Let F and G be two sheaves over S, and let $f : F \to G$ be an S-morphism. The following conditions on f are equivalent:*

(i) *f is a monomorphism (respectively, an epimorphism, respectively, an isomorphism) in $\widetilde{\mathscr{C}}$;*

(ii) *f is a monomorphism (respectively, an epimorphism, respectively, an isomorphism) in $\widetilde{\mathscr{C}}_{/S} = \widetilde{\mathscr{C}/S}$.*

For monomorphisms and isomorphisms this is immediate, since it is a question about presheaves. For epimorphisms it follows from the description of epimorphisms as covering morphisms (4.4.3) and from the fact that, by the definition of the induced topology, these are the same in $\widetilde{\mathscr{C}}$ and in $\widetilde{\mathscr{C}/S}$.

Proposition 4.5.7. *Let $f : F \to G$ be a morphism of sheaves. The following conditions are equivalent:*

(i) *f is a monomorphism (respectively, an epimorphism, respectively, an isomorphism);*

(ii) *for every $S \in \mathrm{Ob}(\mathscr{C})$, the morphism $f_S : F_S \to G_S$ is locally a monomorphism (respectively, an epimorphism, respectively, an isomorphism), that is:*

(iii) *for every $S \in \mathrm{Ob}(\mathscr{C})$, the set of morphisms $T \to S$ for which $F_T \to G_T$ is a monomorphism (respectively, an epimorphism, respectively, an isomorphism) is a refinement of S.*

If the given topology is defined by a pretopology $\mathscr{R}$, these conditions are also equivalent to:

(iv) *for every $S \in \mathrm{Ob}(\mathscr{C})$, there is a covering family $\{S_i \to S\} \in \mathscr{R}(S)$ such that, for every i, $F_{S_i} \to G_{S_i}$ is a monomorphism (respectively, an epimorphism, respectively, an isomorphism).*

If $\mathscr{C}$ has a final object e, it is enough in conditions (ii), (iii), and (iv) to take $S = e$.

Clearly (ii), (iii), and (iv) are equivalent. To prove the equivalence of (i) and (ii), as well as the assertion concerning a final object, it is enough to show that (ii) implies (i) and that the properties in question are stable under base change. We prove the latter point first. For monomorphisms and isomorphisms it is immediate, since it is a question about presheaves. For epimorphisms it follows from the fact that every epimorphism of sheaves is universal (4.4.3).

It remains to show that (ii) implies (i). Suppose first that $f_S : F_S \to G_S$ is locally a monomorphism (respectively, an isomorphism). There is then a covering sieve C of S such that f_T is a monomorphism (respectively, an isomorphism) for every $T \to S$ belonging to C. Since an inverse limit of monomorphisms (respectively, isomorphisms) is again one, $\mathrm{Hom}(C, f)$ is a monomorphism (respectively, an isomorphism); see 4.1.4. The commutative diagram

$$\begin{tikzcd} \mathrm{Hom}(C,F) \arrow[r, "\mathrm{Hom}(C,f)"] & \mathrm{Hom}(C,G) \\ F(S) \arrow[u, "\wr"'] \arrow[r, "f(S)"] & G(S) \arrow[u, "\wr"'] \end{tikzcd}$$

therefore shows that $f(S)$ is injective (respectively, bijective).

Finally, suppose that $f : F \to G$ is locally an epimorphism, and let $H \subset G$ be its image. For every $S \to G$, the sheaf $H \times_G S$ is the image of

$$f \times_G S : F \times_G S \longrightarrow S.$$

To prove that f is an epimorphism, it is enough to prove that H is a refinement of G, or equivalently that $H \times_G S$ is a refinement of S for every S. But this holds after each base change $T \to S$ belonging to a refinement of S, since $f \times_G S$ is locally covering. Hence $H \times_G S$ is itself a refinement of S, by Axiom (T 2).

Corollary 4.5.8. *Let F and G be two sheaves over S, and let $f : F \to G$ be an S-morphism. Then f is a monomorphism, respectively an epimorphism, respectively an isomorphism, if and only if it is so locally on S.*

Remark 4.5.9. The proof of the proposition shows that its assertion about monomorphisms (respectively, isomorphisms) remains valid if one assumes only that F is a separated presheaf (respectively, a sheaf) and G is an arbitrary presheaf (respectively, a separated presheaf).

We return temporarily to the case of an arbitrary topology and make the following definition.

Definition 4.5.10. *Let $G \to F$ be a morphism in $\widehat{\mathscr{C}}$. We say that G is a* relative sheaf over *F if, for every F-functor H and every refinement R of H, the canonical map*

$$\mathrm{Hom}_F(H, G) \xrightarrow{\sim} \mathrm{Hom}_F(R, G) \tag{+}$$

is bijective.

Proposition 4.5.2 immediately gives the following generalization.

Proposition 4.5.11. *If F is a sheaf, then G is a relative sheaf over F if and only if G is a sheaf.*

Lemma 4.5.12. *In a sequence*

$$X \longrightarrow T \longrightarrow S,$$

where X, T, S are three objects of $\widehat{\mathscr{C}}$, suppose that X is a relative sheaf over T. Then

$$U = \prod_{T/S} X$$

is a relative sheaf over S.

Indeed, for every S-functor Y, one has

$$\mathrm{Hom}_S(Y, U) = \mathrm{Hom}_T(T \times_S Y, X).$$

If C is a sieve of Y, then $T \times_S C$ is a sieve of $T \times_S Y$, and the conclusion is immediate.

Corollary 4.5.13. *The presheaves* $\mathrm{Hom}_{T/S}(X, Y)$, $\mathrm{Isom}_S(X, Y)$, *and the analogous presheaves are sheaves whenever all the arguments occurring in them are sheaves.*

Indeed, all these presheaves are constructed using fiber products and presheaves $\prod_{T/S} X$ (I, 1.7 and II, 1). It is therefore enough to verify the assertion for a presheaf $\prod_{T/S} X$; in that case it follows from 4.5.11 and 4.5.12.

4.6. Description of the Quotient of a Sheaf by an Equivalence Relation

Recall that the given topology $\mathcal{T}$ is assumed to be coarser than the canonical topology.

Proposition 4.6.1. *Let $R \overset{p_1}{\underset{p_2}{\rightrightarrows}} X$ be a $\widetilde{\mathscr{C}}$-equivalence relation in the sheaf X. Define $F \in \mathrm{Ob}(\widehat{\mathscr{C}})$ as follows: for each $S \in \mathrm{Ob}(\mathscr{C})$,*[43]

$$F(S) = \left\{ \begin{array}{l} \text{sub-}S\text{-sheaves } Z \text{ of } X_S\text{, stable under } R \times S\text{, whose quotient} \\ \text{by } R_Z \text{ is } S\text{, that is, for which } R_Z \rightrightarrows Z \to S \text{ is exact} \end{array} \right\}.$$

[43] Editor's note: $R \times S$ denotes the equivalence relation in $X \times S$ defined by $R \times S_{\mathrm{diag}} \subset X \times X \times S \times S$, and R_Z denotes the equivalence relation that it induces on Z (cf. 3.1.6).

For every sheaf Y, $\mathrm{Hom}(Y,F)$ *is naturally identified with the set*

$$\left\{\begin{array}{l}\text{sub-}Y\text{-sheaves of } X\times Y\text{, stable under } R\times Y\text{,}\\ \text{and having quotient } Y\end{array}\right\}.$$

In particular, the subsheaf $R\subset X\times X$ corresponds to an element $p\in$ $\mathrm{Hom}(X,F)$*, and the exact diagram*

$$R \underset{p_2}{\overset{p_1}{\rightrightarrows}} X \xrightarrow{p} F$$

identifies F with the sheaf quotient X/R.

Indeed, put $Q=X/R$. For every sheaf Y and every morphism $f\in\mathrm{Hom}(Y,Q)$, corresponding to a section $s: Y\hookrightarrow Q\times Y$, consider the diagram

$$\begin{array}{ccccc} R\times Y & \rightrightarrows & X\times Y & \longrightarrow & Q\times Y \\ & & \uparrow & & \uparrow{\scriptstyle s} \\ & & Z & \longrightarrow & Y, \end{array} \qquad (*)$$

in which the right-hand square is cartesian. By 4.4.11, Z is immediately seen to be a sub-Y-sheaf of $X\times Y$, stable under $R\times Y$, with quotient Y. Conversely, every Z of this kind arises from a unique section of $Q\times Y$ over Y. Taking Y first to be representable gives an isomorphism $Q\simeq F$; taking arbitrary Y then gives the asserted description of $\mathrm{Hom}(Y,F)$. Finally, the canonical morphism $X\to Q$ corresponds to the sub-X-sheaf $R\subset X\times X$, which completes the proof.

Corollary 4.6.2. *Let G be any subfunctor of F such that* $\mathrm{Hom}(X,G)\subset\mathrm{Hom}(X,F)$ *contains R. Then the canonical morphism $p: X\to F$ factors through G.*

Since p is covering (4.4.9 and 4.4.3), it follows that G is a refinement of F. In particular, every subsheaf G of F satisfying the preceding condition equals F (4.3.12).

4.6.3. We now consider the case in which X and R are representable. We begin with some terminology. Besides the conditions (a)–(d) introduced in 3.4.1, we shall use the following conditions on a family (M) of morphisms of $\mathscr{C}$; for completeness, we repeat conditions (a)–(c):

(a) (M) is stable under base change.

(b) The composite of two elements of (M) belongs to (M).

(c) Every isomorphism belongs to (M).

$(d_{\mathcal{T}})$ Every element of (M) is covering. [44]

$(e_{\mathcal{T}})$ Let $f: X\to Y$ be a morphism of $\mathscr{C}$. If there is a refinement R of Y such that, for every $Y'\to R$, the morphism $X\times_Y Y'\to Y'$ belongs to (M), then f belongs to (M).

Conditions (a) and (b) imply

(a') the fiber product of two elements of (M) belongs to (M).

On the other hand, by 4.3.9, conditions (a) and $(d_{\mathcal{T}})$ imply

(d') every element of (M) is a universal effective epimorphism.

[44] Editor's note: $\mathcal{T}$ denotes the given topology on $\mathscr{C}$, which is coarser than the canonical topology.

4.6.4. When $\mathscr{C}$ has fiber products, the family of covering morphisms, denoted $(M_{\mathcal{T}})$, satisfies the preceding conditions. Indeed (cf. 4.2.3), (a) follows from (C 1), (b) from (C 2), (c) from (C 4), $(d_{\mathcal{T}})$ from the definition, and $(e_{\mathcal{T}})$ from (C 5). Thus the results established below for a family satisfying these conditions apply in particular to $(M_{\mathcal{T}})$. In particular, $\mathcal{T}$ may be taken to be the canonical topology and (M) to be the family of universal effective epimorphisms.

Lemma 4.6.5. *Let (M) be a family of morphisms satisfying conditions (a)–$(e_{\mathcal{T}})$ above. Let R be a $\mathscr{C}$-equivalence relation of type (M) in $X \in \mathrm{Ob}(\mathscr{C})$. Let $\widetilde{X}$ be the sheaf defined by X, let $\widetilde{R}$ be the $\widetilde{\mathscr{C}}$-equivalence relation in $\widetilde{X}$ defined by R, and let $\widetilde{X}/\widetilde{R}$ be the sheaf quotient. Then R is (M)-effective if and only if $\widetilde{X}/\widetilde{R}$ is representable. When these conditions hold, $\widetilde{X}/\widetilde{R}$ is represented by the quotient X/R.*

Suppose first that R is (M)-effective, and put $Y = X/R$. The canonical morphism $p : X \to Y$ belongs to (M), hence is covering by $(d_{\mathcal{T}})$. The corresponding morphism

$$\widetilde{p} : \widetilde{X} \longrightarrow \widetilde{X}/\widetilde{R}$$

is therefore a universal effective epimorphism in $\widetilde{\mathscr{C}}$ (4.4.3). Consequently it identifies $\widetilde{X}/\widetilde{R}$ with the quotient of $\widetilde{X}$ by the equivalence relation R' defined by $\widetilde{p}$. Since the canonical functor $\mathscr{C} \to \widetilde{\mathscr{C}}$ commutes with fiber products, R' is precisely $\widetilde{R}$, because R is the equivalence relation defined by p.

Conversely, suppose that $\widetilde{X}/\widetilde{R}$ is represented by an object Y of $\mathscr{C}$. Let $p : X \to Y$ be the morphism deduced from the canonical morphism $\widetilde{X} \to \widetilde{X}/\widetilde{R}$; by 4.4.3 it is covering. As before, R is plainly the equivalence relation defined by p. It remains only to show that $p \in (M)$. The cartesian diagram

$$\begin{array}{ccccc} R & \xrightarrow{\sim} & X \times_Y X & \longrightarrow & X \\ & {\scriptstyle p_2} \searrow & \big\downarrow{\scriptstyle \mathrm{pr}_2} & & \big\downarrow{\scriptstyle p} \\ & & X & \xrightarrow{p} & Y \end{array}$$

shows that, after the covering base change p, the morphism p becomes p_2, which belongs to (M). The assertion follows from $(e_{\mathcal{T}})$.

Corollary 4.6.5.1. [45] *Let (M) satisfy conditions (a)–$(e_{\mathcal{T}})$ above, and let $f : G \to G'$ be a morphism of $\mathscr{C}$-groups such that $f \in (M)$. Suppose that* $\mathrm{Ker}(f)$ *is representable (as is the case when $\mathscr{C}$ has a final object e). Then the equivalence relation in G defined by $H = \mathrm{Ker}(f)$ is (M)-effective, and G' represents the sheaf quotient $\widetilde{G}/\widetilde{H}$ for the topology $\mathcal{T}$.*

This follows from 4.6.5 and 3.4.7.1.

We can now state the main result of this subsection.

Theorem 4.6.6. *Let (M) be a family of morphisms satisfying Axioms (a)–$(e_{\mathcal{T}})$ of 4.6.3. Let R be a $\mathscr{C}$-equivalence relation of type (M) (cf. 3.4.3) in an object X of $\mathscr{C}$. Consider the functor $F \in \mathrm{Ob}(\widehat{\mathscr{C}})$ defined by*

$$F(S) = \left\{ \begin{array}{l} \text{sub-}S\text{-sheaves } Z \text{ of } X_S, \text{ stable under } R \times S, \\ \text{whose quotient by } R_Z \text{ is } S \end{array} \right\}.$$

Let F_0 be the subfunctor of F defined as follows: $F_0(S)$ consists of the representable $Z \in F(S)$, that is,

$$F_0(S) = \left\{ \begin{array}{l} \text{sub-}\mathscr{C}/S\text{-objects } Z \text{ of } X_S, \text{ stable under } R \times S, \text{ such that} \\ R_Z \text{ is } (M)\text{-effective with quotient } S, \text{ i.e. such that} \\ Z \to S \text{ belongs to } (M) \text{ and } R_Z \simeq Z \times_S Z \end{array} \right\}.$$

Then:

[45] Editor's note: This corollary has been added.

(i) *The morphism* $p \in \mathrm{Hom}(X, F) = F(X)$ *defined by the subobject* $R \subset X \times X$ *identifies* F *with the sheaf quotient of* X *by* R.

(ii) *The following conditions are equivalent:*

 (a) F *is representable;*

 (b) F_0 *is representable;*

 (c) R *is* (M)*-effective.*

 Under these conditions, $F = F_0 = X/R$.

(iii) *Let* (N) *be a family of morphisms stable under base change, such that for every covering family* $\{S_i \to S\}$ *and every family* $\{T_i \to S_i\}$ *of morphisms in* (N)*, every descent datum on the* T_i *relative to* $\{S_i \to S\}$ *is effective. Suppose that* X *is squarable (cf. 1.6.0) and that* $R \to X \times X$ *belongs to* (N)*. Then* $F_0 = F$.

Proof. Assertion (i) was proved in 4.6.1.

For (ii), we have already proved the equivalence of (a) and (c), as well as the equality $F = X/R$. It remains to prove that either (b) or (c) implies $F_0 = F$. First note, as stated in the theorem, that F_0 is indeed a subfunctor of F: if $S \in \mathrm{Ob}(\mathscr{C})$ and $Z \in F_0(S)$, then $Z \to S$ is squarable, and hence $Z \times_S S' \in F_0(S')$ for every $S' \to S$. Since $R \in F(X)$ belongs to $F_0(X)$, Corollary 4.6.2 shows that (b) implies $F_0 = F$.

Now suppose that (c) holds, and let Q be an object of $\mathscr{C}$ representing X/R. Then $X \to Q$ belongs to (M). For every $S \in \mathrm{Ob}(\mathscr{C})$ and every $Z \in F(S)$, diagram $(*)$ of 4.6.1 gives

$$Z = S \times_{Q \times S} (X \times S).$$

Thus Z is representable and $Z \to S$ belongs to (M), so $Z \in F_0(S)$.

For (iii), let $f \in \mathrm{Hom}(S, F)$ correspond to $Z \in F(S)$. We must show that f factors through F_0, or equivalently that Z is representable. This is immediate when f factors through X, by the following lemma.

Lemma 4.6.7. *Let* $x_0 \in X(S)$*. The image of* x_0 *in* $F(S)$ *corresponds to the subsheaf* $Z \subset X_S$ *defined by the two cartesian squares*

$$\begin{array}{ccccc} X_S & \xrightarrow{\mathrm{id}_{X_S} \times x_0} & X_S \times_S X_S & \longrightarrow & X \times X \\ \uparrow & & \uparrow & & \uparrow \\ Z & \longrightarrow & R_S & \longrightarrow & R. \end{array}$$

This follows at once from the description of the morphism $X \to F$.

Returning to the proof of the theorem, suppose first that f factors through X. Then Z is representable; moreover, since $R \to X \times X$ belongs to (N), so does $Z \to X_S$.

In general, f need not factor through X. Since $X \to F$ is covering (4.4.3), however, 4.4.8(vii) gives a covering family $\{S_i \to S\}$ and, for each i, a morphism $S_i \to X$ making the diagram

$$\begin{array}{ccc} X & \longrightarrow & F \\ \uparrow & & \uparrow{\scriptstyle f} \\ S_i & \longrightarrow & S \end{array}$$

commute. By the preceding argument, the morphism $f_i : S_i \to F$ defined by this diagram belongs to $\mathrm{Hom}(S_i, F_0)$ and corresponds to the subsheaf $Z \times_S S_i \subset X_{S_i}$. The morphism $Z \times_S S_i \to X_{S_i}$ belongs to (N), and the family $\{X_{S_i} \to X_S\}$ is covering. It therefore remains only to prove the following proposition.

Proposition 4.6.8. *Let $\{S_i \to S\}$ be a covering family and let Z be a sheaf over S. Suppose that, for every i, the S_i-functor $Z \times_S S_i$ is represented by an object T_i. Then the family (T_i) carries a canonical descent datum relative to $\{S_i \to S\}$. The sheaf Z is representable if and only if this datum is effective; in that case the descended object represents Z.*

First observe that, by 4.4.3, $\{S_i \to S\}$ is a universal effective epimorphic family in $\widetilde{\mathscr{C}}$, and hence a descent family in $\mathscr{C}$ (2.3). If Z is represented by an object T, then $T \times_S S_i$, regarded as a sheaf, is isomorphic to $Z \times_S S_i$. Thus the descent datum on the T_i is effective, and its uniquely determined descended object is isomorphic to T, hence represents Z.

Conversely, suppose that the canonical descent datum on the T_i is effective, and let T be the descended object. Because $\{S_i \to S\}$ is a descent family in $\widetilde{\mathscr{C}}$, there is an S-morphism $T \to Z$ whose base change to each S_i is the canonical morphism $T_i \to Z \times_S S_i$. This morphism is locally an isomorphism; since both T and Z are sheaves, 4.5.8 implies that it is an isomorphism.

Corollary 4.6.9. *Let R be an (M)-effective equivalence relation in X. For every sheaf F, the map*

$$\mathrm{Hom}(X/R, F) \longrightarrow \mathrm{Hom}(X, F)$$

identifies the first set with the subset of the second consisting of morphisms compatible with R.

Corollary 4.6.10. *Let $\mathcal{T}'$ be a topology coarser than $\mathcal{T}$, for which the morphisms in (M) are covering. Under the hypotheses of 4.6.6(iii), X/R is also the sheaf quotient of X by R for every topology lying between $\mathcal{T}'$ and the canonical topology.*

Remark 4.6.11. If, in the statement of 4.6.6(iii), one assumes in addition that the descended morphism $T \to S$ belongs to (N) whenever the hypotheses there are satisfied, then the inclusion morphisms $Z \hookrightarrow X_S$ also belong to (N). This follows immediately from the construction of Z by descent.

Remark 4.6.12. The implications

$$\text{(c)} \Longrightarrow \text{(b)} \Longrightarrow \text{(a)} \qquad \text{and} \qquad \text{(c)} \Longrightarrow [F_0 = F = X/R]$$

were proved without using the sufficiency direction of Lemma 4.6.5, the only place where condition $(e_{\mathcal{T}})$ enters. They therefore remain valid when (M) satisfies only conditions (a)–$(d_{\mathcal{T}})$. The family of squarable covering morphisms is an example of such a family (M) (compare 4.6.4). For the canonical topology these are precisely the universal effective epimorphisms. Hence:

Corollary 4.6.13. *Let R be a universal effective equivalence relation in X. Then the object X/R of $\mathscr{C}$ is the sheaf quotient of X by R for the canonical topology. It represents the following functor: $(X/R)(S)$ is the set of sub-$\mathscr{C}/S$-objects Z of X_S, stable under $R \times S$, for which the induced equivalence relation is universal effective and has quotient S.*

Similarly, for an arbitrary topology:

Corollary 4.6.14. *Let (M) be the family of squarable covering morphisms. If R is an (M)-effective equivalence relation in X, then the object X/R of $\mathscr{C}$ is the sheaf quotient of X by R, and it represents the functor F_0 of 4.6.6.*

Scholium 4.6.15. We can now make the following clarifications to 4.5.5. In questions involving inverse limits alone—fiber products, algebraic structures, and so forth—the results of Exposé I and 4.5.5 allow us to identify $\mathscr{C}$, without distinction, with a full subcategory of either $\widetilde{\mathscr{C}}$ or $\widehat{\mathscr{C}}$. The same is not true for questions that mix inverse and direct limits.

Whenever both inverse and direct limits occur, and especially when passing to quotients (for example, when putting a group structure on the quotient of a group by an invariant subgroup), we shall regard the given category as embedded in the category of sheaves. Thus, if R is a $\mathscr{C}$-equivalence relation in an object X of $\mathscr{C}$, the notation X/R will mean the sheaf quotient of X by R, formerly denoted $j(X)/j(R)$, and, when this sheaf is representable, also the object that represents it. The preceding results show that, in the most important cases, a quotient in $\mathscr{C}$ is also a quotient in the category of sheaves. In any event, we shall not use the notation X/R for a quotient in $\mathscr{C}$ that does not coincide with the quotient in $\widetilde{\mathscr{C}}$—for instance, one that is not universal. This modifies the definitions of no. 3.

To study a problem of this kind, one therefore works first in the category of sheaves, where all the usual results hold (cf. no. 4.4), and then specializes the resulting statements to the original category, using the results of the present subsection and, when available, criteria for the effectivity of descent. Examples of this method will appear in the following subsections.

4.7. Use of Effectivity Criteria: Isomorphism Theorem

In this subsection we give an example of the use of effectivity criteria. The initial data are a topology $\mathcal{T}$ on $\mathscr{C}$, still assumed coarser than the canonical topology; a family (M) of morphisms of $\mathscr{C}$ satisfying Axioms (a)–$(e_{\mathcal{T}})$ of 4.6.3; and a family (N) of morphisms of $\mathscr{C}$ which may satisfy the following axioms:

(a) (N) is stable under base change;

$(f_{\mathcal{T}})$ morphisms in (N) descend for the given topology. More precisely, for every $S \in \mathrm{Ob}(\mathscr{C})$, every covering family $\{S_i \to S\}$, and every family $\{T_i \to S_i\}$ of morphisms in (N), every descent datum on the T_i relative to $\{S_i \to S\}$ is effective; if T denotes the descended object, then $T \to S$ belongs to (N).

Since every element of (M) is covering (condition 4.6.3 $(d_{\mathcal{T}})$), Axiom $(f_{\mathcal{T}})$ implies the following axiom:[46]

(f_M) if $Y \to X$ belongs to (N) and $X \to X_1$ belongs to (M), every descent datum on Y relative to $X \to X_1$ is effective; if Y_1 denotes the descended object, then $Y_1 \to X_1$ belongs to (N).

Here is an example, to be treated later: $\mathscr{C}$ is the category of schemes, $\mathcal{T}$ is the faithfully flat quasi-compact topology, (M) is the family of faithfully flat quasi-compact morphisms, and (N) is the family of closed immersions, or the family of quasi-compact immersions.[47]

Taking 4.6.11 into account, the principal result of 4.6.6 may be recalled as follows.

Proposition 4.7.1. *Let X be a squarable object of $\mathscr{C}$, and let R be an equivalence relation of type (M) in X. Suppose that $R \to X \times X$ belongs to (N), where (N) satisfies (a) and $(f_{\mathcal{T}})$. Then the sheaf quotient X/R is given by*

$$\begin{aligned}(X/R)(S) = \{Z \subseteq X_S \mid\ & Z \text{ is an } S\text{-subobject stable under } R \times S,\\ & Z \to X_S \text{ belongs to } (N), \quad Z \to S \text{ is covering or belongs to } (M),\\ & R_Z \simeq Z \times_S Z\}.\end{aligned}$$

We also have the following result.

[46]Editor's note: Axiom (f_M), which originally appeared before Proposition 4.7.2, has been moved here.
[47]Editor's note: cf. §6.4; see also VI_A, 5.3.1.

Proposition 4.7.2. *Let $X \in \mathrm{Ob}(\mathscr{C})$, let R be an (M)-effective equivalence relation in X, and let (N) be a family of morphisms satisfying (a) and (f_M). For every subobject Y of X which is stable under R and for which $Y \to X$ belongs to (N), the equivalence relation R_Y induced in Y by R is (M)-effective. Moreover, the quotient*

$$Y/R_Y = Y'$$

is a subobject of $X' = X/R$, and $Y' \to X'$ belongs to (N).

The map $Y \mapsto Y'$ is a bijection from the set of subobjects Y of X which are stable under R and for which $Y \to X$ belongs to (N), onto the set of subobjects Y' of X' for which $Y' \to X'$ belongs to (N). Its inverse is

$$Y' \longmapsto Y' \times_{X'} X.$$

Indeed, since R is (M)-effective, the morphism $X \to X'$ belongs to (M). Let Y' be a subobject of X' such that the canonical morphism $Y' \to X'$ belongs to (N). Then $Y = Y' \times_{X'} X$ is a subobject of X stable under R, and the morphisms $Y \to X$ and $Y \to Y'$ belong respectively to (N) and (M), since both families are stable under base change. Let R_Y denote the equivalence relation induced in Y by R. By 4.4.11, the sheaf quotient Y/R_Y is represented by Y'; hence 4.6.5 shows that R_Y is (M)-effective.

Conversely, let Y be a subobject of X, stable under R, such that $Y \to X$ belongs to (N). Since Y is stable under R, its two inverse images in

$$R = X \times_{X'} X$$

are identical, and Y is equipped with a descent datum relative to $X \to X'$. The required conclusion follows from Axiom (f_M) for the family (N).

Corollary 4.7.3. *Let $X \in \mathrm{Ob}(\mathscr{C})$, and let R be an (M)-effective equivalence relation in X. Suppose in addition that $R \to X \times X$ belongs to (N), where (N) satisfies (a) and $(f_{\mathcal{T}})$. Then, for every Y as in 4.7.2, $R_Y \to Y \times Y$ also belongs to (N), and consequently 4.7.1 gives*

$$\begin{aligned}(Y/R_Y)(S) = \{Z \subseteq Y_S \mid {} & Z \text{ is an } S\text{-subobject stable under } R_Y \times S,\\ & Z \to Y_S \text{ belongs to } (N) \text{ (hence so does } Z \to X_S),\\ & Z \to S \text{ is covering,} \quad R_Z \simeq Z \times_S Z\,\}.\end{aligned}$$

5. Passage to the Quotient and Algebraic Structures

5.1. Principal Homogeneous Bundles

Definition 5.1.0.[48] We recall (III 0.1) that an object X equipped with a right action of H is said to be *formally principal homogeneous*[49] under H if the canonical morphism of functors

$$X \times H \longrightarrow X \times X, \qquad (x, h) \longmapsto (x, xh),$$

is an isomorphism. Equivalently (cf. *loc. cit.*), for every $S \in \mathrm{Ob}\,\mathscr{C}$, the set $X(S)$ is formally principal homogeneous under $H(S)$: it is either empty or principal homogeneous under

[48] Editor's note: the numbering 5.1.0 has been introduced for later reference.

[49] Editor's note: one also says "pseudo-torsor"; cf. EGA IV_4, 16.5.15. The more general notion of a formally homogeneous object (not necessarily principal homogeneous) is defined in Addendum 6.7.1 at the end of this exposé.

$H(S)$. In particular, when H acts on itself by right translations, it is formally principal homogeneous under itself.

Definition 5.1.1. An object X with an action of H is said to be *trivial* if it is isomorphic, as an H-object, to H with its translation action.

Proposition 5.1.2. *If X is formally principal homogeneous under H, there is an isomorphism*

$$\Gamma(X) \xrightarrow{\sim} \mathrm{Isom}_{H\text{-obj.}}(H, X)$$

of principal homogeneous sets under $\Gamma(H)$.

To a section x of X, associate the morphism $H \to X$ defined on sets by $h \mapsto xh$. The assertion follows immediately by reduction to the set-theoretic case.

Corollary 5.1.3. *There is an isomorphism of H-objects*

$$X \xrightarrow{\sim} \mathrm{Isom}_{H\text{-obj.}}(H, X).$$

Corollary 5.1.4. *An object with a group action is trivial if and only if it is formally principal homogeneous and has a section.*

Definition 5.1.5. Let $\mathscr{C}$ be a category equipped with a topology. An S-object X with an action of the S-group H is called a *principal homogeneous bundle under H* if it is locally trivial, that is, if the following equivalent conditions hold:[50]

(i) the collection of morphisms $T \to S$ for which the functor $X \times_S T$ is trivial under $H \times_S T$ is a refinement of S;

(ii) for a pretopology defining the given topology, there is a covering family $\{S_i \to S\}$ such that, for every i, the S_i-functor $X \times_S S_i$ is trivial under the S_i-group functor $H \times_S S_i$, equivalently, it has a section over S_i.

Proposition 5.1.6. *Let $\mathscr{C}$ be a category equipped with a topology $\mathcal{T}$, and let (M) be a family of morphisms of $\mathscr{C}$ satisfying axioms (a)–($e_{\mathcal{T}}$) of 4.6.3. Let H be an S-group whose structure morphism $H \to S$ belongs to (M), and let P be an S-object with an action of H. The following conditions are equivalent:*

(i) *P is a principal homogeneous bundle under H (Definition 5.1.5);*

(ii) *P is formally principal homogeneous under H, and its structure morphism $P \to S$ belongs to (M);*

(iii) *there exists a morphism $S' \to S$ belonging to (M) such that, after base change from S to S', P becomes trivial; that is, $P \times_S S'$ is trivial under $H \times_S S'$;*

(iv) *H acts freely and (M)-effectively on P, and the quotient P/H is isomorphic to S.*

First, (ii) and (iv) are equivalent. In either case $P \to S$ belongs to (M), hence is squarable, which ensures that the fiber products $H \times_S P$ and $P \times_S P$ are representable. Condition (ii) plainly implies (iii): one may take $S' = P$, since formal principal homogeneity implies that $P \times_S P$ is trivial under $H \times_S P$ by 5.1.4, its diagonal section providing a section. Condition (iii) plainly implies (i), because $\{S' \to S\}$ is a covering family by axiom ($d_{\mathcal{T}}$).

It remains to prove (i)⇒(ii). The morphism of sheaves

$$P \times_S H \longrightarrow P \times_S P$$

[50] Editor's note: in this case one also says that X is an H-torsor.

is locally an isomorphism and hence is an isomorphism (4.5.8); thus P is formally principal homogeneous. The structure morphism $P \to S$ is locally isomorphic to the structure morphism $H \to S$, which belongs to (M). It therefore belongs to (M) by axiom $(e_{\mathcal{T}})$.

The equivalence between (i) and (iv) admits the following generalization.

Proposition 5.1.7. *Under the same assumptions on $\mathscr{C}$ and (M), let H be an S-group and let X be an S-object on which H acts on the right. Suppose that $H \to S$ belongs to (M). The following conditions are equivalent:*

(i) *H acts freely and (M)-effectively on X;*

(ii) *there exists an S-morphism $p : X \to Y$, compatible with the equivalence relation on X defined by the action of H, such that the induced action of $H \times_S Y$ on X over Y makes X a principal homogeneous bundle under H_Y over Y.*

Under these conditions, p identifies Y with the quotient X/H.

If $p : X \to Y$ is compatible with the action of H, then the induced action of $H \times_S Y$ on X over Y defines the same equivalence relation on X as the original action of H, by the isomorphism

$$H_Y \times_Y X \xrightarrow{\sim} H \times_S X.$$

The proposition follows from this observation and the preceding equivalence (iv)⇔(i).

Corollary 5.1.7.1.[51] *Let $\mathscr{C}$ be a category having a final object and fiber products, and let it be equipped with a topology $\mathcal{T}$ coarser than the canonical topology. Let $f : G \to H$ be a morphism of $\mathscr{C}$-groups, put $K = \mathrm{Ker}(f)$, and suppose that f is covering for $\mathcal{T}$. Then H represents the sheaf quotient G/K, and f is a K_H-torsor. Equivalently, one may say that G is a K-torsor over H.*

Indeed, because f is covering, it is a universal effective epimorphism (4.4.3). Hence, by 3.3.3.1, H is the quotient of G by the equivalence relation

$$\mathscr{R}(f) = G \times_H G.$$

Moreover, the morphism

$$G \times K \longrightarrow G \times_H G, \qquad (g, k) \longmapsto (g, gk),$$

is an isomorphism of $K_G = G \times_H K_H$-objects; its inverse is $(g, g') \mapsto (g, g^{-1}g')$. Thus $\mathscr{R}(f)$ is the equivalence relation defined by K. Since $f : G \to H$ is covering, it is a K_H-torsor by 5.1.6(ii), or directly by Definition 5.1.5(ii).

We can now sharpen Theorem 4.6.6 in the case of a quotient by a group action.

Proposition 5.1.8. *Under the assumptions of 5.1.7, let F_0 denote the functor over S defined as follows: for each $S' \to S$, $F_0(S')$ is the set of representable S'-subfunctors Z of $X \times_S S'$ that are stable under $H \times_S S'$ and are principal homogeneous bundles under this S'-group for the induced action (3.2.2).*

(i) *The following conditions are equivalent:*

(a) *the action of H on X is (M)-effective and free;*[52]

(b) *F_0 is representable.*

[51] Editor's note: this special case has been added as a corollary; it will be used several times in the following exposés.

[52] Editor's note: the words "and free" have been added.

Under these conditions, $F_0 = X/H$*.*

(ii) *Let* (N) *be a family of morphisms stable under base change, such that for every covering family* $\{S'_i \to S'\}$ *and every family of morphisms* $\{T_i \to S'_i\}$ *in* (N)*, every descent datum on the* T_i *relative to* $\{S'_i \to S'\}$ *is effective. Suppose that* $X \times_S H \to X \times_S X$ *belongs to* (N)*, and that* X *is squarable. Then the element* $p \in \mathrm{Hom}(X, F_0)$ *corresponding to the subobject* $X \times_S H$ *of* $X \times_S X$ *identifies* F_0 *with the sheaf quotient* X/H*.*

5.2. Group Structures and Passage to the Quotient

In this subsection we are concerned with the algebraic structures that can be placed on the quotient (G/H) of a group by a subgroup. We first work in the category of sheaves on $\mathscr{C}$ for an arbitrary topology. Taking the canonical topology and using 4.5.12 will then give results concerning universal effective passage to the quotient in $\mathscr{C}$.

Proposition 5.2.1. *Let* $u : H \to G$ *be a monomorphism of sheaves of groups. There is a unique* G*-object structure on the quotient sheaf* G/H *for which the canonical morphism*

$$p : G \longrightarrow G/H$$

is a morphism of G*-objects. This structure is functorial in the pair* (G, H)*: given a commutative diagram*

$$\begin{array}{ccc} H & \longrightarrow & G \\ \downarrow & & \downarrow f \\ H' & \longrightarrow & G', \end{array}$$

the morphism $f : G/H \to G'/H'$ *of 3.2.3 is compatible with the morphism* f *between the acting groups.*

Indeed, G/H is the sheaf associated with the presheaf

$$i(G)/i(H) : S \longmapsto G(S)/H(S).$$

Since the functor a is left exact, it carries objects with a group action to objects with a group action. The presheaf $i(G)/i(H)$ is an $i(G)$-object, and therefore

$$G/H = a\big(i(G)/i(H)\big)$$

is an $a(i(G)) = G$-object. This construction plainly has all the stated properties.

Corollary 5.2.2. *Let* $u : H \to G$ *be a monomorphism of* $\mathscr{C}$*-groups, and suppose that the action of* H *on* G *is universally effective. There is a unique* G*-object structure on the quotient object* $G/H \in \mathrm{Ob}\,\mathscr{C}$ *for which* $p : G \to G/H$ *is a morphism of objects with an action. In the sense of the preceding proposition, this structure is functorial in the pair* (H, G)*, with* H *acting universally effectively on* G*.*

Proposition 5.2.3. *Let* $u : H \to G$ *be a monomorphism of sheaves of groups that identifies* H *with an invariant subgroup sheaf of* G*. There is a unique sheaf-of-groups structure on the quotient sheaf* G/H *for which the canonical morphism* $p : G \to G/H$ *is a group morphism. This structure is functorial in the pair* (H, G)*, with* H *invariant.*

The proof is similar to that of 5.2.1.

Corollary 5.2.4. *Let* $u : H \to G$ *be a monomorphism of* $\mathscr{C}$*-groups that identifies* H *with an invariant subgroup of* G*. Suppose that the action of* H *on* G *is universally effective. There* *is a unique group structure on the quotient object $G/H \in \mathrm{Ob}\,\mathscr{C}$ for which the canonical morphism $G \to G/H$ is a group morphism. This structure is functorial in the pair (H, G), with H invariant and acting universally effectively.*

The group structure of G/H can be characterized more explicitly.

Proposition 5.2.5. *Under the assumptions of 5.2.4, let K be a $\mathscr{C}$-group and let $f : G \to K$ be a morphism. The following conditions are equivalent:*

(i) *f is a group morphism compatible with the equivalence relation defined by H;*

(ii) *f is a group morphism whose restriction $H \to K$ is the trivial morphism;*

(iii) *f factors through a group morphism $G/H \to K$.*

In particular, there is an isomorphism, functorial in the group K,

$$\mathrm{Hom}_{\mathscr{C}\text{-gr.}}(G/H, K) \xrightarrow{\sim} \big\{f \in \mathrm{Hom}_{\mathscr{C}\text{-gr.}}(G, K) \mid f \circ u = e\big\}.$$

The equivalence of (i) and (ii) is proved set-theoretically, and (iii)⇒(ii) is immediate. The equivalence of (iii) and (ii) follows from the formula

$$\mathrm{Hom}(G/H, K) \simeq \mathrm{Hom}\big(i(G)/i(H), K\big)$$

and the definition of the group structure on G/H.

Remark 5.2.6. In the preceding situation, if the kernel of f is exactly H, then the morphism $G/H \to K$ through which f factors is a monomorphism. This follows at once from 3.3.4.

For sheaves of groups, 4.4.11 can be sharpened as follows.

Proposition 5.2.7. *Let G be a sheaf of groups and H an invariant subgroup sheaf. For every subgroup sheaf K of G containing H, let $K' = K/H$, regarded as a subgroup of $G' = G/H$. Then*

$$K = K' \times_{G'} G,$$

and the maps

$$K \longmapsto K/H, \qquad K' \longmapsto K' \times_{G'} G$$

give a bijection between the subgroup sheaves of G containing H and the subgroup sheaves of G'. Under this correspondence, the invariant subgroup sheaves of G and of G' correspond.

The first assertion follows readily from 4.4.11 and 3.2.4. It remains to show that K is invariant in G if and only if K' is invariant in G'. If K is invariant in G, then the presheaf $i(K)/i(H)$ is invariant in $i(G)/i(H)$; by the usual argument, the same is true of the associated sheaves. Conversely, if K' is invariant in G', then the fiber product $K' \times_{G'} G$ is invariant in G, as is immediate.

Now let L be an arbitrary subgroup sheaf of G. Denote by $\overline{L}$ the saturation of L for the equivalence relation defined by H; we also write $\overline{L} = L \cdot H$.

Proposition 5.2.8. *Under the preceding assumptions, $L \cdot H$ is a subgroup sheaf of G containing H, and the image of L in G/H is identified with*

$$(L \cdot H)/H \simeq L/(H \cap L).$$

Indeed, let L' denote the image sheaf of L in G/H. Under the correspondence of the preceding proposition, it is the subgroup sheaf of G/H corresponding to $L \cdot H$. Since $L \to L'$ is covering, and hence a universal effective epimorphism of sheaves, it follows from 4.4.9 that L' is identified with the quotient of L by the kernel of $L \to L'$, which is plainly $H \cap L$.

Finally, consider a sheaf of groups G, a subgroup sheaf K, and an invariant subgroup sheaf H of K. We first define a right action of the sheaf of groups $H\backslash K$ $(= K/H)$ on G/H. The group K acts on G by right translations. Since H is invariant in K, this action is compatible with the equivalence relation defined by the action of H, and therefore induces an action of K on G/H, that is, a morphism

$$K^{\circ} \longrightarrow \mathrm{Aut}(G/H)$$

from the group opposite to K. Since the target is a sheaf (4.5.13) and this morphism is trivial on H, it factors through K/H and defines the desired action. The right and left actions of G on itself commute; hence the actions of G and K/H on G/H commute.

Proposition 5.2.9. *Under the preceding assumptions, K/H acts freely on the right on G/H, and there is a canonical isomorphism of G-objects*

$$(G/H)/(K/H) \simeq G/K.$$

If K is invariant in G, in which case K/H is invariant in G/H by 5.2.7, this isomorphism respects the group structures on both sides.

There is an isomorphism of presheaves

$$\big(i(G)/i(H)\big)/\big(i(K)/i(H)\big) \xleftarrow{\sim} i(G)/i(K),$$

which respects the $i(G)$-object structures. Applying the functor a to this relation gives the asserted result.

Corollary 5.2.10. *Let G be a $\mathscr{C}$-group, K a $\mathscr{C}$-subgroup of G, and H an invariant $\mathscr{C}$-subgroup of K. Let (M) be a family of morphisms of $\mathscr{C}$ satisfying axioms (a)–($e_{\mathcal{T}}$). Suppose that the right action of H on G (respectively, on K) is (M)-effective. Then K/H acts naturally and freely on the right on G/H, and this action commutes with that of G. The following conditions are equivalent:*

(i) *the action of K on G is (M)-effective;*

(ii) *the action of K/H on G/H is (M)-effective.*

Under these conditions, there is an isomorphism of objects[53] with a G-action

$$(G/H)/(K/H) \simeq G/K.$$

5.3. Use of Effectivity Criteria: Noether's Theorem

[54] Let $\mathscr{C}$, $\mathcal{T}$, and (M) be as usual. Let (N) be a family of morphisms satisfying axioms (a) and (f_M) of 4.7. Combining 5.2.7 and 4.7.2 gives the following result.

Proposition 5.3.1. *Let G be a $\mathscr{C}$-group, and let H be an invariant $\mathscr{C}$-subgroup of G that acts (M)-effectively on G. For every $\mathscr{C}$-subgroup K of G containing H such that $K \to G$ belongs to (N), the action of H on K is (M)-effective, and the quotient $K/H = K'$ is a $\mathscr{C}$-subgroup of $G/H = G'$ such that $K' \to G'$ belongs to (N). The map $K \mapsto K'$ is a bijection*

[53] Editor's note: of $\mathscr{C}$.

[54] Editor's note: in fact, the isomorphisms established in 5.2.8–5.2.10 deserve to be called "Noether isomorphism theorems."

between the set of $\mathscr{C}$-subgroups K of G that contain H and for which $K \to G$ belongs to (N), and the set of $\mathscr{C}$-subgroups K' of G' for which $K' \to G'$ belongs to (N). Its inverse is

$$K' \longmapsto K \times_{G'} G.$$

Under this correspondence, the invariant subgroups of G and of G' correspond.

Corollary 5.3.2. *If $H \to G$ belongs to (N), then $\mathscr{C}$ has a final object e, and the unit section $e \to G/H$ belongs to (N).*[55]

6. Topologies in the Category of Schemes

6.1. The Zariski Topology

This is the topology generated by the following pretopology: a family of morphisms $\{S_i \to S\}$ is covering if each morphism is an open immersion and the union of the images of the S_i is all of S. It is denoted by (Zar).

Definition 6.1.1. *A sheaf for the Zariski topology is also called a* functor of local nature*: it is a contravariant functor from* **Sch** *to* **Set** *such that, for every scheme S and every covering of S by open subschemes S_i, the diagram*

$$F(S) \longrightarrow \prod_i F(S_i) \rightrightarrows \prod_{i,j} F(S_i \cap S_j)$$

is exact.

In particular, a functor of local nature carries direct sums to products. Since every representable functor is a sheaf, the Zariski topology is coarser than the canonical topology.

As a matter of terminology, whenever we say "local" or "locally" without further qualification, the reference is to the Zariski topology and hence to the usual meaning of these words.

6.2. A Procedure for Constructing Topologies

Proposition 6.2.1. *Let $\mathscr{C}$ be a category, let $\mathscr{C}'$ be a full subcategory, and let P be a set of families of morphisms in $\mathscr{C}$ with common target, stable under base change and composition (that is, satisfying axioms (P 1) and (P 2) of 4.2.5). Let P' be a set of families of morphisms in $\mathscr{C}'$ that contains every singleton family consisting of an identity isomorphism. Equip $\mathscr{C}$ with the topology generated by P and P' (cf. 4.2.5.0), and suppose that the following three conditions hold:*

(a) *If $\{S_i \to S\} \in P'$, so that $S_i, S \in \operatorname{Ob}\mathscr{C}'$, and $T \to S$ is a morphism in $\mathscr{C}'$, then the fiber products $S_i \times_S T$, formed in $\mathscr{C}$, exist and the family $\{S_i \times_S T \to T\}$ belongs to P'; in particular, $S_i \times_S T \in \operatorname{Ob}\mathscr{C}'$.*

    This condition implies that P' is stable under base change in $\mathscr{C}'$, but is not equivalent to that assertion: it also requires the inclusion functor $\mathscr{C}' \hookrightarrow \mathscr{C}$ to commute with certain fiber products.

(b) *For every $S \in \operatorname{Ob}\mathscr{C}$, there exists a family $\{S_i \to S\} \in P$ with $S_i \in \operatorname{Ob}\mathscr{C}'$ for every i.*

[55] Editor's note: this follows by applying the proposition with $K = H$.

(c) *In a situation*

$$\begin{tikzcd} & S_{ijk} \arrow[d, "(P')"'] \\ S_i \arrow[d, "(P')"'] & S_{ij} \arrow[l, "(P)"'] \\ S & \end{tikzcd} \tag{6.2.1.c}$$

where S, S_i, S_{ij}, S_{ijk} are objects of $\mathscr{C}'$, where $\{S_i \to S\} \in P'$, $\{S_{ij} \to S_i\} \in P$ for each i, and $\{S_{ijk} \to S_{ij}\} \in P'$ for each pair ij, there exists a family $\{T_n \to S\} \in P'$, and for each n a multi-index (ijk), fitting into a commutative diagram

$$\begin{tikzcd} S_{ijk} \arrow[dr] & T_n \arrow[l] \arrow[d] \\ & S. \end{tikzcd} \tag{6.2.1.d}$$

Under these hypotheses, a sieve R of an object $S \in \operatorname{Ob}\mathscr{C}$ is covering if and only if there is a composite family

$$\begin{tikzcd} R \arrow[d] & S_{ij} \arrow[l, dashed] \arrow[d, "(P')"'] \\ S & S_i \arrow[l, "(P)"'] \end{tikzcd} \tag{6.2.1.e}$$

where $S_i, S_{ij} \in \operatorname{Ob}\mathscr{C}'$, $\{S_i \to S\} \in P$, and $\{S_{ij} \to S_i\} \in P'$ for every i, such that the resulting morphisms $S_{ij} \to S$ factor through R. Equivalently, the sieve generated by this composite family is contained in R.

Proof. Families belonging to P and P' are covering, and therefore so is a composite of such families, by (C 2). A sieve of the displayed form is thus covering because it contains a covering sieve.

Conversely, it is enough to show that the sieves of this form constitute a topology, that is, to verify axioms (T 1)–(T 4) of 4.2.1.

Axiom (T 4). Let $S \in \operatorname{Ob}\mathscr{C}$. By (b), there exists a family $\{S_i \to S\} \in P$ with $S_i \in \operatorname{Ob}\mathscr{C}'$. The singleton families $\{S_i \xrightarrow{\mathrm{id}_{S_i}} S_i\}$ belong to P' by hypothesis. Hence the maximal sieve S of S has the required form:

$$\begin{tikzcd} S_i \arrow[r] \arrow[d, "(P')"', "\mathrm{id}"] & S \arrow[d, "\mathrm{id}_S"] \\ S_i \arrow[r, "(P)"] & S. \end{tikzcd} \tag{6.2.1.T4}$$

Axiom (T 3). This is evident.

Axiom (T 2). Let R be a sieve of S of the required form, and let C be a sieve[56] of S such that, for every $T \in \operatorname{Ob}\mathscr{C}$ and every morphism $T \to S$ factoring through R, the sieve $C \times_S T$ of T has the required form. Since $S_{ij} \to S$ factors through R, the sieve

$$C_{ij} = C \times_S S_{ij}$$

[56] Editor's note: the original has been corrected here.

of S_{ij} has the required form; it is represented schematically by

$$
\begin{tikzcd}
 & S_{ij} \arrow[ddl] \arrow[d, "(P')"'] & C_{ij} \arrow[l] \arrow[dd] \\
 & S_i \arrow[d, "(P)"'] & \\
R \arrow[r] & S & C. \arrow[l]
\end{tikzcd}
\tag{6.2.1.T2a}
$$

Consequently, for every pair ij, there is a diagram of the form

$$
\begin{tikzcd}
S_{ijkl} \arrow[d, "(P')"'] \arrow[ddr] & \\
S_{ijk} \arrow[d, "(P)"'] & \\
S_{ij} & C_{ij}. \arrow[l]
\end{tikzcd}
\tag{6.2.1.T2b}
$$

We have therefore produced a composite family

$$S_{ijkl} \xrightarrow{(P')} S_{ijk} \xrightarrow{(P)} S_{ij} \xrightarrow{(P')} S_i \xrightarrow{(P)} S \tag{6.2.1.T2c}$$

belonging to $P \circ P' \circ P \circ P'$, factoring through C, and having every object other than S in $\mathscr{C}'$.

Apply condition (c) to each family $\{S_{ijkl} \to S_i\}$. For every i we obtain a family $\{T_{in} \to S_i\} \in P'$ such that $T_{in} \to S$ factors through one of the S_{ijkl}, and hence through C:

$$
\begin{tikzcd}
T_{in} \arrow[r] \arrow[d, "(P')"'] & S_{ijkl} \arrow[d] \\
S_i \arrow[d, "(P)"'] & C \arrow[dl] \\
S &
\end{tikzcd}
\tag{6.2.1.T2d}
$$

Thus the sieve C of S has the required form, completing the verification of (T 2).

Axiom (T 1). Let R be a sieve of S of the prescribed form and let $T \to S$ be a morphism in $\mathscr{C}$. We show that the sieve $R \times_S T$ of T again has the prescribed form. Consider the commutative configuration

$$
\begin{tikzcd}
 & S_{ij} \arrow[ddl] \arrow[d, "(P')"'] & & U_{ikj} \arrow[ll] \arrow[d, "(P')"'] \\
 & S_i \arrow[d, "(P)"'] & T_i \arrow[l] \arrow[d, "(P)"'] & U_{ik} \arrow[l, "(P)"'] \arrow[dl, "(P)"] \\
R \arrow[r] & S & T \arrow[l] & .
\end{tikzcd}
\tag{6.2.1.T1a}
$$

Put $T_i = S_i \times_S T$. The family $\{T_i \to T\}$ belongs to P, by (P 1). By (b), choose families $\{U_{ik} \to T_i\} \in P$ with $U_{ik} \in \operatorname{Ob} \mathscr{C}'$. By (P 2), $\{U_{ik} \to T\} \in P$. Condition (a) shows that

$$U_{ik} \times_{S_i} S_{ij} = U_{ikj}$$

is an object of $\mathscr{C}'$ and that, for each ik, $\{U_{ikj} \to U_{ik}\} \in P'$. The commutative diagram

$$
\begin{tikzcd}
R \arrow[d] & U_{ikj} \arrow[l] \arrow[d, "(P')"'] & \\
S & T \arrow[l] & U_{ik} \arrow[l, "(P)"']
\end{tikzcd}
\tag{6.2.1.T1b}
$$

shows that the morphisms $U_{ikj} \to T$ factor through the sieve $T \times_S R$ of T. This sieve therefore has the required form, and the proof is complete.

Corollary 6.2.2. *Let $S \in \operatorname{Ob} \mathscr{C}'$, and let R be a sieve of S. Then R is covering if and only if there exists a family $\{T_i \to S\} \in P'$ that factors through R.*

Such a sieve is plainly covering. Conversely, apply condition (c) with the family $\{S_i \to S\}$ reduced to the identity isomorphism of S; the proposition then shows that every covering sieve has the stated form.

Corollary 6.2.3. *A presheaf F on $\mathscr{C}$ is separated (respectively, is a sheaf) if and only if the map*

$$F(S) \longrightarrow \prod_i F(S_i)$$

is injective (respectively, the diagram

$$F(S) \longrightarrow \prod_i F(S_i) \rightrightarrows \prod_{i,j} F(S_i \times_S S_j)$$

is exact) in each of the following two cases:

(i) *$\{S_i \to S\} \in P$;*

(ii) *$S, S_i \in \operatorname{Ob} \mathscr{C}'$ and $\{S_i \to S\} \in P'$.*

The conditions are necessary because the families in question are covering. Let C be the sieve of S that is the image of a composite family

$$\{S_{ij} \xrightarrow{(P')} S_i \xrightarrow{(P)} S\}.$$

A straightforward diagram chase shows that the conditions of the corollary imply that

$$\operatorname{Hom}(S, F) \xrightarrow{g} \operatorname{Hom}(C, F)$$

is injective, respectively bijective. Every covering sieve R of S contains a sieve C of this form, and there is a commutative diagram

$$\begin{tikzcd} \operatorname{Hom}(S,F) \arrow[r, "f"] \arrow[dr, "g"'] & \operatorname{Hom}(R,F) \arrow[d, "h"] \\ & \operatorname{Hom}(C,F). \end{tikzcd} \tag{6.2.3.1}$$

Since g is injective, so is f, and hence F is separated. Moreover, R refines C, so h is also injective. If g is bijective, then f is bijective as well, and F is a sheaf.

Remark 6.2.4. *The preceding corollary does not follow from 4.3.5, since P' is not stable under base extension.*

Remark 6.2.5. *Condition (c) holds in particular when:*

(i) *P' is stable under composition;*

(ii) *whenever $\{S_j \to S\}$ is a family of morphisms in $\mathscr{C}'$ belonging to P, it contains a subfamily belonging to P'.*

6.3. Application to the Category of Schemes

Take $\mathscr{C}$ to be the category of schemes, $\mathscr{C}'$ the full subcategory of affine schemes, and P the set of surjective families of open immersions. We shall consider the following sets P'_i:

P'_1 : finite surjective families of flat morphisms;

P'_2 : finite surjective families of flat, finitely presented, quasi-finite morphisms;

P'_3 : finite surjective families of étale morphisms;

P'_4 : finite surjective families of finite étale morphisms.

The description of P'_2 is accompanied by the following clarification.[57]

For each P'_i, except P'_4, the conditions of Proposition 6.2.1 hold. Condition (c) follows from 6.2.5: since an affine scheme is quasi-compact, every family of morphisms in $\mathscr{C}'$ belonging to P contains a finite subfamily that still belongs to P, and hence belongs to P'_i for $i = 1, 2, 3$. The topology T_i generated by P and P'_i is denoted and named as follows:[57]

T_1	= (fpqc)	the faithfully flat quasi-compact topology,
T_2	= (fppf)	the faithfully flat topology (locally of finite presentation),
T_3	= (ét)	the étale topology,
T_4	= (étf)	the finite étale topology.

Since $P'_1 \supset P'_2 \supset P'_3 \supset P'_4$, one has

$$(\text{fpqc}) \geq (\text{fppf}) \geq (\text{ét}) \geq (\text{étf}) \geq (\text{Zar}).$$

Proposition 6.3.1.

(i) *A sieve R of S is covering for T_i, $1 \leq i \leq 3$, if and only if there exist a covering (S_p) of S by affine open subschemes and, for every p, a family $\{S_{pq} \to S_p\} \in P'_i$, with all S_{pq} affine, such that every morphism $S_{pq} \to S$ factors through R.* [58]

(ii) *A presheaf F on* **Sch** *is a sheaf for* (fpqc) *(respectively,* (fppf), (ét), *or* (étf)*) if and only if:*

(a) *F is a sheaf for* (Zar)*, that is, a functor of local nature;*

(b) *for every faithfully flat (respectively, faithfully flat, finitely presented and quasi-finite; surjective étale; or surjective finite étale) morphism $T \to S$, with T and S affine, the diagram*

$$F(S) \longrightarrow F(T) \rightrightarrows F(T \times_S T)$$

is exact.

(iii) *The topologies T_i, for $i = 1, 2, 3, 4$, are coarser than the canonical topology.*

(iv) *Every surjective family of flat open morphisms (respectively, flat morphisms locally of finite presentation; étale morphisms; or finite étale morphisms) is covering for* (fpqc) *(respectively,* (fppf), (ét), *or* (étf)*).*

[57] Editor's note: let P''_2 be the set of finite surjective families of morphisms in $\mathscr{C}'$ that are flat and of finite presentation. By Proposition 6.3.1 below, the topology T_2 generated by P and P'_2 coincides with the topology generated by P and P''_2. This follows from the results of EGA IV$_4$, §17.16 on quasi-sections; see the proof of 6.3.1. We recall the following special case of EGA IV$_4$, 17.16.2. Let S be affine and let $f : X \to S$ be a surjective flat morphism locally of finite presentation. Then there exist a faithfully flat, finitely presented, quasi-finite morphism $S' \to S$, with S' affine, and an S-morphism $S' \to X$.

[58] Editor's note: every family $\{S_{pq} \to S_p\} \in P'_i$ is finite. Hence $S'_p = \coprod_q S_{pq}$ is affine, and the family may be replaced by the morphism $S'_p \to S_p$, which still belongs to P'_i.

(v) *Every finite surjective family of flat quasi-compact morphisms is covering for* (fpqc). *In particular, every faithfully flat quasi-compact morphism is covering for* (fpqc).

Proof. Part (i) follows from 6.2.1 and part (ii) from 6.2.3, taking into account that a sheaf for the Zariski topology carries direct sums to products. Every representable functor is a sheaf for (Zar) and satisfies condition (ii)(b), by SGA 1, VIII, 5.3. Thus T_1 is coarser than the canonical topology, which proves (iii).

We prove (iv). Let $\{S_i \to S\}$ be a family as in the statement. By covering S with affine open subschemes, we reduce immediately to the case in which S is affine.[59]

First suppose that the morphisms $S_i \to S$ are flat and open (respectively, étale). Cover each S_i by affine open subschemes S_{ij}. Since the morphisms are open, the images T_{ij} of the S_{ij} form an open covering of S. The affine scheme S is quasi-compact, so finitely many T_{ij}, indexed by a finite set E, cover it. Then

$$S' = \coprod_{(i,j)\in E} S_{ij}$$

is affine, and $S' \to S$ belongs to P_1', respectively P_3', and is therefore covering. Since it factors through the given family, that family is covering.

In the (étf) case, every S_i is finite over S and hence affine. In the preceding argument one may take the singleton covering $\{S_i\}$ of S_i, which gives $S' \to S \in P_4'$.

Now suppose that the morphisms $f_i : S_i \to S$ are flat and locally of finite presentation. For every $s \in S$, the proof of EGA IV_4, 17.16.2 supplies an affine subscheme $X(s)$ of some S_i, with $s \in f_i(X(s))$, such that the restriction

$$g_i : X(s) \longrightarrow S$$

is flat, finitely presented, and quasi-finite. Its image $g_i(X(s))$ is an open neighborhood $U(s)$ of s, by EGA IV_2, 2.4.6. Since S is affine, finitely many such opens $U(s_j)$, $j = 1, \dots, n$, cover S. Consequently,

$$X' = \coprod_{j=1}^{n} X(s_j)$$

is affine and $X' \to S$ is surjective, flat, finitely presented, and quasi-finite, so it belongs to P_2'.[60] This completes the proof of (iv).

We prove (v). Let $\{S_i \to S\}$ be a finite surjective family of flat quasi-compact morphisms, and cover S by affine open subschemes T_j. The schemes

$$S_{ij} = T_j \times_S S_i$$

are quasi-compact, and therefore admit finite affine open coverings T_{ijk}. Each $T_{ijk} \to T_j$ is flat, and the family $\{T_{ijk} \to T_j\}$ is finite and surjective, hence covering for T_1. By composition, $\{T_{ijk} \to S\}$ is covering as well. It factors through the given family, which is therefore covering:

$$\begin{tikzcd} S_i \arrow[d] & S_{ij} \arrow[l] \arrow[d] & T_{ijk} \arrow[l] \arrow[dl] \\ S & T_j \arrow[l] & \end{tikzcd} \tag{6.3.1.v}$$

[59]Editor's note: the argument that follows has been simplified by using the fact that S is now assumed affine.

[60]Editor's note: this shows that if P_2'' denotes the set of finite surjective families of flat, finitely presented morphisms in $\mathscr{C}'$, then the topology generated by P and P_2'' equals T_2. Moreover, in the notation used at the beginning of the proof of (iv), if each S_i is covered by affine opens of finite presentation over S, then the resulting morphism $S' \to S$ belongs to P_2''.

Corollary 6.3.2. *Let (M_i) be the following families of morphisms:*

(M_1) : faithfully flat quasi-compact morphisms;

(M_2) : faithfully flat morphisms locally of finite presentation;

(M_3) : surjective étale morphisms;

(M_4) : surjective finite étale morphisms.

The family (M_i) satisfies axioms (a), (b), (c), (d_{T_i}), and (e_{T_i}) of 4.6.3. [61]

Conditions (a), (b), and (c) are classical (see EGA and SGA 1, passim).[62] By 6.3.1(iv),(v), (M_i) satisfies (d_{T_i}). It remains to verify (e_{T_i}). It is enough to verify (e_{T_1}), which implies all the others; this follows from SGA 1, VIII, §§4–5.

Corollary 6.3.3. *Let X be a scheme and let R be an equivalence relation in X of type (M_i). Then R is (M_i)-effective if and only if the sheaf quotient of X by R for T_i is representable; in that case it is represented by the quotient X/R.*

This is 4.6.5.

6.4. Effectivity Conditions

We now seek families (N) of morphisms satisfying axiom (f_T) of 4.7. Observe first that (f_{T_1}) implies (f_{T_i}), so it is enough to consider the (fpqc) topology.

Lemma 6.4.1. *The following families of morphisms satisfy axiom (f_{T_1}) of 4.7, that is, they "descend by* (fpqc)*":*

(N) : open immersions;

(N') : closed immersions;

(N'') : quasi-compact immersions.

By 6.3.1(ii), it is enough to verify that these families descend for the Zariski topology and along a faithfully flat quasi-compact morphism. The first assertion is clear. For (N), the second is SGA 1, VIII, 4.4; for (N'), it is loc. cit., 1.9; for (N''), argue as in loc. cit., 5.5, using the two preceding results.

Corollary 6.4.2. *The same result holds for quasi-compact open immersions.*

These results make the general results of 4.7.1, 4.7.2, 5.1.8, 5.3.1, and so on applicable to the present situation. We state the first as an example.

Corollary 6.4.3. (= 4.7.1 + 4.6.10). *Let X be a scheme and R an equivalence relation in X. Suppose that $R \to X$ is faithfully flat and quasi-compact, and that $R \to X \times X$ is a closed immersion (respectively, open, quasi-compact, or quasi-compact open). Then the sheaf quotient X/R is the same for the* (fpqc) *topology as for the canonical topology. For every scheme S,*

$$(X/R)(S) = \left\{ \begin{array}{l} \text{closed (respectively, open, retrocompact, or open retrocompact)} \\ \text{subschemes } Z \text{ of } X_S, \text{ stable under } R \times S, \text{ such that} \\ Z \to S \text{ is faithfully flat and quasi-compact and} \\ R_Z \rightrightarrows Z \to S \text{ is exact} \end{array} \right\}. \tag{6.4.3.1}$$

[63]

[61] Editor's note: the original has been corrected by adding surjectivity to (M_3) and (M_4); it is automatically satisfied in the other two cases.

[62] Editor's note: see EGA I, 6.6.4 for "quasi-compact"; EGA II, 6.1.5 for "finite"; EGA IV_1, 1.6.2 for "locally of finite presentation"; EGA IV_2, 2.2.13 for "faithfully flat"; and EGA IV_4, 17.3.3 for "étale".

[63] Editor's note: recall that a subscheme Z of a scheme T is called *retrocompact* if the immersion $Z \hookrightarrow T$ is quasi-compact; see EGA 0_{III}, 9.1.1.

6.5. Principal Homogeneous Bundles

We merely record the terminology:

topology	principal homogeneous bundles
(fpqc)	principal homogeneous bundles (without qualifier)
(ét)	quasi-isotrivial principal homogeneous bundles
(étf)	locally isotrivial principal homogeneous bundles
(Zar)	locally trivial principal homogeneous bundles.

(6.5.1)

6.6. Other Topologies

Other topologies on the category of schemes are sometimes useful. One is the *global finite étale topology* (étfg), generated by the pretopology whose covering families are the surjective families of finite étale morphisms. It is not finer than the Zariski topology. The corresponding principal homogeneous bundles are called *isotrivial*.

The position of this topology, and of the chaotic topology, in the lattice is as follows:

$$\begin{tikzcd}
 & \text{(canonical)} \arrow[d] & \\
 & \text{(fpqc)} \arrow[d] & \\
 & \text{(fppf)} \arrow[d] & \\
 & \text{(ét)} \arrow[d] & \\
 & \text{(étf)} \arrow[dl] \arrow[dr] & \\
\text{(Zar)} \arrow[dr] & & \text{(étfg)} \arrow[dl] \\
 & \text{(chaotic)} &
\end{tikzcd} \tag{6.6.1}$$

[64]

6.7. Homogeneous Spaces

Let[65] G be an S-group scheme, let X be an S-scheme equipped with a left action of G, and let

$$\Phi : G \times_S X \longrightarrow X \times_S X$$

be the morphism of S-schemes defined on points by

$$(g, x) \longmapsto (gx, x).$$

Recall (cf. 5.1.0 and III.0.1) that X is called a *formally principal homogeneous space* under G if the following equivalent conditions hold:

(i) for every $T \to S$, the set $X(T)$ is empty or is a principal homogeneous space under $G(T)$;

[64] Editor's note: recall (cf. 4.4.2) that the chaotic topology is the coarsest topology, defined by $J(S) = \{S\}$ for every $S \in \operatorname{Ob} \mathscr{C}$.

[65] Editor's note: the numbering that follows has been added.

(ii) Φ is an isomorphism of S-functors;

(iii) Φ is an isomorphism of S-schemes.

The equivalence (i)$\Longleftrightarrow$(ii) is clear, and (ii)$\Longleftrightarrow$(iii) holds because $\mathscr{C} = (\mathbf{Sch}/S)$ is a full subcategory of $\widehat{\mathscr{C}}$.

The notion of a *formally homogeneous space*, not necessarily principal homogeneous, is obtained by requiring Φ to be an epimorphism in the category of sheaves for an appropriate topology $\mathcal{T}$. Indeed, requiring Φ to be an epimorphism of S-functors is equivalent to requiring that, for every $T \to S$, the set $X(T)$ be empty or homogeneous (not necessarily principal homogeneous) under $G(T)$. This is too restrictive, as the following simple example shows.

Let

$$S = \operatorname{Spec} R, \qquad G = \mathbf{G}_{m,R}, \qquad X = \mathbf{G}_{m,R},$$

and let G act on X by $t \cdot x = t^2 x$. Then Φ is surjective, finite, and étale, and hence is an epimorphism in the category of sheaves for the (étf) topology (and, *a fortiori*, an epimorphism of S-schemes). On the other hand, the points 1 and -1 of $X(R)$ are not conjugate under any element of $G(R)$. Thus

$$G(R) \times X(R) \longrightarrow X(R) \times X(R)$$

is not surjective.[66] We are therefore led to the following definition.

Definition 6.7.1. *Let G be an S-group, let X be an S-scheme equipped with an action of G, and let $\mathcal{T}$ be a topology on $(\mathbf{Sch}/S)$ coarser than the canonical topology. The scheme X is called a* formally homogeneous space under G, *relative to $\mathcal{T}$, if the following equivalent conditions hold:*

(i) *the morphism $\Phi : G \times_S X \to X \times_S X$ is an epimorphism in the category of sheaves for $\mathcal{T}$;*

(ii) *for every $T \to S$ and every $x, y \in X(T)$, there exist a $\mathcal{T}$-covering morphism $T' \to T$ and an element $g \in G(T')$ such that $y_{T'} = g \cdot x_{T'}$.*

Remark 6.7.2. *Condition (i) implies in particular that Φ is a universal effective epimorphism in $(\mathbf{Sch}/S)$, by 4.4.3. As is readily seen, it follows that Φ is surjective; cf. 1.3, Editor's note (3).*

Proposition and Definition 6.7.3. [67] *Let G be an S-group, let X be an S-scheme equipped with an action of G, and let $\mathcal{T}$ be a topology on $(\mathbf{Sch}/S)$ coarser than the canonical topology. The following conditions are equivalent:*

(i) *X satisfies both of the following conditions:*

  (1) *the morphism $\Phi : G \times_S X \to X \times_S X$ is covering, that is, X is a formally homogeneous G-space;*

[66] Editor's note: this difficulty plainly comes from the following fact. Let $\mathscr{C}'$ be a full subcategory of $\widehat{\mathscr{C}}$ containing $\mathscr{C}$, for example the category $\widetilde{\mathscr{C}}_{\mathcal{T}}$ of sheaves on $\mathscr{C}$ for a topology $\mathcal{T}$ coarser than the canonical topology. If $f : X \to Y$ is a morphism in $\mathscr{C}$, then the implications

$$f \text{ epimorphic in } \widehat{\mathscr{C}} \Longrightarrow f \text{ epimorphic in } \mathscr{C}' \Longrightarrow f \text{ epimorphic in } \mathscr{C}$$

are generally strict.

[67] Editor's note: cf. [Ray70], Definition VI.1.1.

(2) *the structural morphism $X \to S$ is also covering, that is, locally for $\mathcal{T}$ it has a section (cf. 4.4.8 bis, (b)).*

(ii) *Locally on S for the topology $\mathcal{T}$, the scheme X, with its G-action, is isomorphic to the quotient sheaf of G by a subgroup scheme H, for $\mathcal{T}$. In other words, there is a covering family $\{S_i \to S\}$ such that every $X \times_S S_i$ represents the sheaf quotient of $G \times_S S_i$ by some subgroup scheme H_i.*

*Under these conditions, X is called a G-*homogeneous space*, relative to $\mathcal{T}$.*

Proof. Assume (ii). Put

$$G_i = G \times_S S_i, \qquad X_i = X \times_S S_i.$$

The scheme X_i has a section over S_i, namely the composite of the unit section $\varepsilon_i : S_i \to G_i$ with the projection $\pi_i : G_i \to X_i = G_i/H_i$. It follows that $X \to S$ is covering.

Moreover, π_i is covering, so $\pi_i \times \pi_i$ is covering as well, by 4.2.3 (C 1),(C 2). There is a commutative diagram[68]

$$\begin{array}{ccc} G_i \times_{S_i} X_i & \xrightarrow{\Phi_i} & X_i \times_{S_i} X_i \\ {\scriptstyle \mathrm{id}\times\pi_i}\big\uparrow & & \big\uparrow{\scriptstyle \pi_i\times\pi_i} \\ G_i \times_{S_i} G_i & \underset{\sim}{\xrightarrow{\Psi_i}} & G_i \times_{S_i} G_i, \end{array} \tag{6.7.3.1}$$

where Φ_i is obtained from Φ by the base change $S_i \to S$, and Ψ_i is the isomorphism defined on points by

$$(g, g') \longmapsto (gg', g).$$

The composite $(\pi_i \times \pi_i) \circ \Psi_i$ is covering; hence Φ_i is covering by 4.2.3 (C 3). Thus Φ is locally covering and is therefore covering, by 4.2.3 (C 5). This proves (ii)$\Longrightarrow$(i).

Conversely, assume (i), and first suppose in addition that the structural morphism $X \to S$ has a section σ. By EGA I, 5.3.13, σ is an immersion. Define $H = G \times_X S$ by the following diagram, whose two squares are cartesian:

$$\begin{array}{ccccc} H & \longrightarrow & G & \xrightarrow{\mathrm{id}_G \boxtimes \sigma} & G \times_S X \\ \big\downarrow & & {\scriptstyle \pi}\big\downarrow & & {\scriptstyle \Phi}\big\downarrow \\ S & \xrightarrow{\sigma} & X & \xrightarrow{\mathrm{id}_X \boxtimes \sigma} & X \times_S X. \end{array} \tag{6.7.3.2}$$

Here π, $\mathrm{id}_G \boxtimes \sigma$, and $\mathrm{id}_X \boxtimes \sigma$ are the morphisms defined on points, for $T \to S$, $g \in G(T)$, and $x \in X(T)$, by

$$\pi(g) = g \cdot \sigma_T, \qquad (\mathrm{id}_G \boxtimes \sigma)(g) = (g, \sigma_T), \qquad (\mathrm{id}_X \boxtimes \sigma)(x) = (x, \sigma_T).$$

Then π is covering, and H is a subgroup scheme of G representing the stabilizer $\mathrm{Stab}_G(\sigma)$ of σ (cf. I, 2.3.3). Thus, for every $T \to S$,

$$H(T) = \{g \in G(T) \mid g \cdot \sigma_T = \sigma_T\}.$$

Let G/H denote the presheaf

$$T \longmapsto G(T)/H(T),$$

[68] Editor's note: the Polo–Gille re-edition prints $G_i \times_{S_i} X_i$ at the upper right. We write $X_i \times_{S_i} X_i$, since Φ_i, as the base change of $\Phi : G \times_S X \to X \times_S X$, and $\pi_i \times \pi_i$ both have this codomain.

and let $a(G/H)$ be its associated sheaf for $\mathcal{T}$. The preceding construction gives a commutative diagram of presheaves equipped with a G-action:

$$\begin{tikzcd} G \arrow[r, "\pi"] \arrow[d] & X \\ G/H \arrow[ur, "\bar{\pi}"'] & \end{tikzcd} \tag{6.7.3.3}$$

where $\bar{\pi}$ is a monomorphism. Since π is covering, $\bar{\pi}$ is covering as well. By 4.3.12, $\bar{\pi}$ therefore induces an isomorphism

$$a(G/H) \xrightarrow{\sim} X.$$

We have proved: if X is a G-homogeneous space such that $X \to S$ has a section σ, then X represents the sheaf quotient G/H, where $H = G \times_X S$ is the stabilizer of σ.

In general, by hypothesis there is a covering family $\{S_i \to S\}$ such that every morphism

$$X_i = X \times_S S_i \longrightarrow S_i$$

has a section σ_i. Put $G_i = G \times_S S_i$. The morphism

$$\Phi_i : G_i \times_{S_i} X_i \longrightarrow X_i \times_{S_i} X_i$$

obtained from Φ by base change along $S_i \to S$ is still covering.

The preceding argument then gives

$$X_i \simeq G_i/H_i,$$

where H_i is the stabilizer of σ_i in G_i. This completes the proof of (i)$\Longrightarrow$(ii).

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[69] Editor's note: this reference has been added.